# EXHALE: Update Log (stage 3, from 2026-09-23)

## Contents


1. The Ritz residuals of the preconditioned spectrum (2026-09-23) . . . 9
2. Two records the campaign left open, closed (2026-09-23) . . . 11
3. Two statements of the source that were not true, corrected (2026-09-23) . . . 11
4. Four more source comments that were not true, corrected (2026-09-23) . . . 12
5. One equivalent-width operator for the observation and every model (2026-09-23) . . . 13
6. The stage export of the stationary alternation (2026-09-23) . . . 14
7. Three defects the family C audit found, fixed, and two small ones (2026-09-23) . . . 14
8. The regression check of section 7, a cell writer, and corrections from the spectral and On stall tests (2026-09-24) . . . 16
9. Three defects the transient measurements found, repaired (2026-09-24) . . . 17
10. The elemental flux records ride on the Riemann face mass flux; a row writer for the helium element (2026-09-24) . . . 18
11. The molecular K_zz = 1e9 ladder certified across the observed equivalent width (2026-09-24) . . . 19
12. The regression of sections 9 and 10, and the validation of the crossing (2026-09-24) . . . 20
13. The width of the line on the molecular crossing, and the catalog summary brought up to date (2026-09-24) 22
14. The user's decisions of 2026-09-24 on the open items, and what each gave . . . 23
15. The grid of the crossing (decision item 3) and the regression of sections 13 and 14 (2026-09-24) . . . 25
16. The user's decisions of 2026-09-24 on the four remaining items . . . 25
17. The atomic-data audit of 2026-09-24, applied . . . 26
18. The H2-front energy imbalance, the H3+ non-LTE interpolant, and the descent below He/H 1.45 (2026-09-24) . . . 28
19. The second round of the atomic-data audit (2026-09-24/25) . . . 28
20. The molecular K_zz = 1e9 family under the audited physics (2026-09-25) . . . 30
21. The code-size appendix moves to this log (2026-09-25) . . . 31
22. The originals supplied on 2026-09-25, read and applied . . . 31
23. The Markdown documents move to md/ (2026-09-25) . . . 33
24. The last items of the MoCHII comparison, and the molecular crossing under the corrected physics (2026-09-25/26) . . . 34
25. The helium-hydrogen charge exchange from its originals (2026-09-26) . . . 36
26. A trial state no longer stops the run at an unclosed lower ghost (2026-09-26) . . . 37
27. The molecular family under the corrected physics: catalog publication and the width of the line (2026-09-26) . . . 38
28. The catalog summary brought to the tree's state, and why the molecular winds do not certify (2026-09-26) 38
29. Errors fixed and models solved first: the plan of 2026-09-26 . . . 39
    29.1 An OpenMP data race in the metal sweep (fixed) . . . 39

29.2 Fix batch (errors confirmed in the source) . . . 40
29.3 The interdiffusion enthalpy flux (new physics, serious) . . . 40
29.4 He+ + H non-radiative charge exchange near threshold . . . 40
29.5 Goldens, fixture, suites . . . 41
29.6 Models solved . . . 41
29.7 A progress-control defect of the composition relaxation . . . 41
29.8 Incidents of this series . . . 42
29.9 The lower-ghost H2 closure stalled with a zero move (fixed) . . . 42
29.10 The _adv post-process made second order . . . 43
29.11 The lower face at the reservoir level . . . 43
29.12 A restartable state after every outer pass . . . 44
29.13 The unsolved molecular cases by continuation (state of 2026-09-28 16:30) . . . 45
30. The base contact held through each steady solve, and the Intel build (2026-09-28/29) . . . 46
30.1 OpenMP thread count: measured, and the default team . . . 46
30.2 The Intel compiler . . . 46
30.3 Why the molecular continuations alternated for 60 to 80 passes . . . 47
30.4 The side of the contact is held through each steady solve . . . 48
30.5 An output argument that was never set . . . 48
30.6 Runs . . . 48
31. The code audit of 2026-09-29: findings checked and applied . . . 49
32. The follow-up review of the audit corrections (2026-09-29) . . . 50
33. The state of every pass published as one generation (audit F6) . . . 51
34. The enthalpy the transported carriers carry (audit F1), and the regression comparison of residual columns (2026-09-29) . . . 53
35. Why the mapped atomic He/H 9.7 seeds at reduced XUV do not solve (2026-09-29, open) . . . 55
35.1 The review of the account (2026-09-29) . . . 57
36. The face-flux budget export and the discrete equilibrium tests of the base (2026-09-29) . . . 57
37. One discrete equilibrium at the base (2026-09-29) . . . 59
38. The atomic He/H 9.7 cases: the seed projection, and the element enthalpy flux as the remaining obstacle (2026-09-29) . . . 60
39. The review of the 2026-09-29 handoff: state identity of the continuation factor, the projected seed, the output precision; the continuation runs (2026-09-29/30) . . . 61
40. Why the atomic He/H 9.7 continuation fails: two mechanisms, and the heat conduction the models omit (2026-09-30, open) . . . 62
41. Heat conduction with the conductivity of the mixture, and its continuation factor (2026-09-30) . . . 64
42. Pseudo-transient steps accepted by the pseudo-time control (2026-09-30) . . . 66
43. The continuation chains, first three hours, and their step control (2026-09-30) . . . 67
44. The outer under-relaxation sets the pace of the chains: omega = 1 (2026-09-30) . . . 67

45. The first state with heat conduction: a conductive thermosphere below an exobase at 1.5 R_p (2026-09-30, open) . . . 68
46. Heat conduction in the fiducial-XUV cases (2026-09-30, running) . . . 69
47. The photochemical 0.10 case on the full equation (2026-09-30) . . . 69
48. The fiducial-XUV atomic case with conduction (2026-09-30) . . . 69
49. The advected temperature correction keeps the transport terms (2026-09-30) . . . 70
50. A floor of the energy row near convergence (2026-09-30; located in section 53, explained and removed in section 58) . . . 71
51. The molecular photochemical fiducial case with conduction (2026-09-30; EW corrected by section 54) . 72
52. The molecular scalar He/H 1.45 case with conduction (2026-09-30) . . . 72
53. Where the energy-row floor of section 50 comes from (2026-09-30; the cause is in section 58) . . . 73
54. The review of the 2026-09-30 handoff: findings checked (2026-09-30) . . . 75
55. One failure contract for the advection-derived product (2026-10-01) . . . 76
56. Continuation factors written exactly; legacy F7.5 tokens read as intervals (2026-10-01) . . . 77
57. The H2 conductivity without overflow (2026-10-01) . . . 78
58. The energy-row floor was a rounding staircase of the absorbed fraction (2026-10-01) . . . 78
59. One (1 - eˆ-x)/x for the code, and the other small-value forms (2026-10-01) . . . 79
60. The combined build and the tests of sections 55-59 (2026-10-01) . . . 80
61. The column energy solve accepted on its residual; the refused rows carry the solution (2026-10-01) . . . 80
62. The Codex review of the 2026-10-01 summary, applied (2026-10-01) . . . 81
63. The reviewed part of the LHS 1140 b EW worktree, taken into the tree (2026-10-03) . . . 82
64. R19 is the thermal rate of Esposito, Coppola & De Fazio (2015) (2026-10-03) . . . 85
65. The Codex review of sections 63-64, checked (2026-10-03) . . . 85
66. The LHS 1140 b EW plan started: phases 0 and 1 (2026-10-03) . . . 86
67. Phase 2: carried ionization stages certify on LHS 1140 b (2026-10-03) . . . 88
68. Phase 3: the He/H 2.09 reference with conduction and carried ionization stages (2026-10-03) . . . 89
69. The 10.2-30 eV photoelectron partition of Furlanetto & Stoever (2010), a default-off option (2026-10-03) 89
70. The conduction bath below the level is the profile's temperature (2026-10-03) . . . 91
71. One H2 gate for both products; recombination photoelectrons partitioned as on the stellar path (2026-10-03) 91
72. The enthalpy of the carried ionization stages; the movement-bound rule (2026-10-03) . . . 92
73. The He/H 2.09 reference on the merged binary (2026-10-03) . . . 93
74. Photochem/wind elemental-flux closure on the new physics (2026-10-03) . . . 93
75. The EW depends on the matching-level temperature: the energy coupling to Photochem is required (2026-10-03) . . . 94
76. The wind writes the heat it conducts into the lower atmosphere (2026-10-03) . . . 95
77. Column energy balance under the conducted flux; the energy-coupled closure (2026-10-03) . . . 95
78. The published V-T rates in the column energy balance (2026-10-03) . . . 97
79. The EW of the energy-coupled state; the NH3 nu2 + H2 rate as an option and its bracket (2026-10-03) . 99

80. The NH3 nu2 + H2 bracket re-closed: both cases converge (2026-10-04) . . . 100
81. He 2^3S transport: the metastable carried as a fourth helium state (2026-10-04) . . . 102
82. The advected product steps each element on the velocity of its own nuclei (2026-10-04) . . . 105
83. The EW evaluate reloads the exact restart (2026-10-04) . . . 107
84. Phase 3 of the LHS 1140 b EW plan accepted; TO_BE_DONE (AH) closed (2026-10-04) . . . 108
85. Phase 4 of the LHS 1140 b EW plan: outer domain, outer boundary, transit-weighted shares, Roche geometry (2026-10-04) . . . 109
86. Phase 5 of the LHS 1140 b EW plan, first step: the EW falls with He/H; the column energy balance limits the low side (2026-10-04) . . . 111
87. LHS 1140 b He 10830: the width-matching velocity kernel in the EW chain (2026-10-04) . . . 112
88. The stationary evaluate certifies the stored pair (2026-10-04) . . . 114
89. The column energy balance with absorption by the overlying column; the thin limit flags instead of refusing; phase 5 at He/H 1.5 and 1.2 (2026-10-04) . . . 117
90. Phase 5 closed: the EW has no crossing in He/H; the thin check uses the escape of emitted photons; the model-top pinning averages a 2-cycle (2026-10-04) . . . 121
91. LHS 1140 b: why the He 10830 EW rises as He/H falls, dEW decomposed by field and by balance term (2026-10-04) . . . 123
92. Phase 6 of the LHS 1140 b EW plan: numerical and observational validation at He/H 0.65 (2026-10-04) 125
93. Closure acceptance: the conducted heat flux has its own residual; Codex's resume and solve-exit refusals reviewed (2026-10-04) . . . 127
94. The three phase-6 problems fixed: the mapper's base cells, the He 2^3S carrier row, the certificate of a written state (2026-10-04) . . . 128
94.1 The mapper seeds the first refined cell from the reservoir row . . . 128
94.2 The He 2^3S carrier row at the first gated cell . . . 129
94.3 The certificate of a written state and of its reload . . . 131
94.4 Tests (gfortran tree build, md5 7a48413c1ec482c0305cb69c08a976a9) . . . 131
95. Route B: the GJ 699 proxy spectrum puts the He 10830 EW three times above the observation; the escape boundary condition's hydrogen carrier split made self-consistent (2026-10-04) . . . 132
96. The mapper's base-row check takes the consumer's tolerance, 1e-6 (2026-10-04) . . . 135
97. Code audit of 2026-10-04: a non-finite restart center is refused; a zero conducted flux can close (2026-10-04) . . . 136
98. Route B step 2: the observed He 10830 EW needs 1.33-1.39 times the GJ 1132 ionizing XUV at He/H 2.09 (2026-10-04) . . . 136
99. Route B step 3: the observed He 10830 EW fixes F_EUV(100-911) at 12.9-13.7 erg cm^-2 s^-1 over deep He/H 0.65-5 (2026-10-04) . . . 138
100. LHS 1140 b: heat conduction in the wind makes the mass-loss rate depend on composition (2026-10-04) 140
101. Route A: deep He/H 2.09 to 1e4 on the GJ 1132 spectrum; the He 10830 EW passes a minimum and turns up but stays below the observation (2026-10-04) . . . 142

102. The EUV that matches the observed EW, deep He/H 0.2 to 1e4; the exchange inside the warmed column solved as a transfer option (2026-10-05) . . . 145
103. The physics-and-algorithms document audited against the source (2026-10-05) . . . 147
104. The user documents audited against the source (2026-10-05) . . . 148
105. The lower-ghost chemistry of LHS 1140 b at He/H 2e4: the reservoir H2 partition solved with its own ionization, and the molecular rows normalized by the turnover of their element (2026-10-06) . . . 150
105.1 The reservoir H2 partition was pinned at the ionization of the previous state . . . 150
105.2 The rows were scaled by a bound that places trace ions at the whole of their element . . . 150
105.3 Tests . . . 151
105.4 The actual cases . . . 152
106. A failed conduction stage is a refused and retaken step; the handoff-free base-level conflict warns; stale source comments corrected (2026-10-06) . . . 153
106.1 A failed Crank-Nicolson stage refuses the attempted step . . . 153
106.2 Base BC: pressure beside a density key without a handoff . . . 154
106.3 Stale source comments (each checked in the code first) . . . 154
106.4 The atomic acceptance scale (measured, not changed) . . . 155
107. The helium carry of `map_state_to_grid.py --reservoir` holds the hydrogen-nucleus count of every row (2026-10-06) . . . 155
107.1 Cause . . . 155
107.2 Fix (`src/utils/map_state_to_grid.py`) . . . 156
107.3 Tests (`src/tests/state_mapper/`) . . . 156
107.4 The real seeds (MEASURED) . . . 156
108. The EUV that matches the observed EW reaches He/H 1e5 (2026-10-06) . . . 157
109. The equivalent width and the width-matching kernel read the planet-rotation column (user decision 2026-10-06) . . . 157
110. Resonant H+/H and He+/He coefficients in the collisional diagnostic; HITRAN band coefficients and two comments of the conducted-heat column corrected (2026-10-07) . . . 158
111. The recomputed NH3/CH4 band coefficients barely move the EW-matched state (2026-10-07) . . . 160
112. The acceptance of the ionization sweep measures every kinetic row against the turnover of its element group; constraint rows are judged in their own units (item (AM), 2026-10-07) . . . 161
112.1 What was wrong . . . 161
112.2 Change (a): the explicit row layout (no number moves) . . . 161
112.3 Change (b): the acceptance correction . . . 162
112.4 Measurement (control build against the (b) build) . . . 163
112.5 Newly refused states, and what they are . . . 164
112.6 Q5: slow rows inside fast groups (measurement and decision) . . . 164
112.7 Open . . . 165
113. The carrier source runs the oxygen cycle only with the oxygen chemistry on (2026-10-07) . . . 165

114. The oxygen-guard fix on the energy-coupled LHS 1140 b states: the He/H 2.09 line falls by a factor 2.5 (2026-10-07) . . . 166
115. The element operator settles on the pressure gradient of the mixture, and its non-resonant ion-neutral pairs are the polarization form alone (2026-10-07) . . . 167
116. The carried ionization stages are scaled by their gross channels; the molecular solve scale groups as the acceptance does; a resolution floor and an H3+ own-row criterion in the acceptance (item (AO) D1-D4, 2026-10-07) . . . 171
116.1 D4 (defect): the carried proton and helium stages carry their gross production and loss . . . 171
116.2 D1: the molecular solve scale sums the kinetic rows of each group . . . 172
116.3 D2: the resolution floor of the acceptance . . . 173
116.4 D3: H3+ against its own gross rate . . . 175
116.5 Suites, frozen binary, documents . . . 175
116.6 Open . . . 176
117. The energy-coupled LHS 1140 b states with every correction to date (binary bd2b93d3): the He 10830 line collapses at He/H 0.65 (2026-10-08) . . . 176
118. The element simplex of the ionization acceptance is bounded at the resolution of a fraction; the mirror root is refused, a candidate outside the simplex is projected and judged again, and the atomic sweep re-solves a cell without a root (item (AO) (iii), 2026-10-08) . . . 177
118.1 The defect . . . 177
118.2 The fix (`src/modules/radiation/ionization_equilibrium.f90`) . . . 178
118.3 Tests . . . 179
118.4 Regression cases (control -> final) . . . 179
118.5 Frozen binary, documents, open . . . 180
119. One description of the matching surface: the carrier base face open against the stated ghost partition, the ghost hydrogen census and one ghost temperature from the profile, Photochem's top at p_match, and the wind's H and H2 fluxes handed over separately (item (AN) items 2-4, user decisions 2026-10-07; 2026-10-08) . . . 181
119.1 What changed . . . 181
119.2 Tests (RED on the source of section 118, GREEN now) . . . 182
119.3 Regression cases reached (gfortran build, control = the live source before this section, md5 e2e1ca99) . . . 183
119.4 Frozen binary . . . 183
119.5 The three LHS 1140 b states: the open face does not certify . . . 183
120. The cell solve stops when its Newton correction resolves every unknown relative to its own magnitude; the hydrogen-only sweep faces the same judge (item (AP), user decision 2026-10-08) . . . 184
120.1 The stopping test . . . 184
120.2 The hydrogen-only sweep is judged like the others . . . 185
120.3 Tests (RED: the solver of section 119; GREEN: this section) . . . 186

120.4 Regression cases (control -> this section) . . . 186
120.5 The three LHS 1140 b k00 stationary solves . . . 187
120.6 Documents . . . 188
121. Why the open-face LHS 1140 b winds were refused: an unbounded reconstruction of the trace ionization stages, and an outer alternation that oscillates between a helium-retaining and a helium-depleted wind (item (AO) D5 and section 119.5, 2026-10-08) . . . 188
121.1 The refusing rows: a trace stage drained through its own outflow face . . . 188
121.2 The fix (`Reconstruct_scalar`, WENO3 branch) . . . 189
121.3 The trajectory: the outer alternation oscillates between two winds . . . 189
121.4 Regression cases (section 120 -> this section) . . . 190
121.5 What is left, and the proposal . . . 191
121.6 Documents . . . 191
122. The three LHS 1140 b structural closures with binary a76da494 and the damped outer alternation: He/H 2.09 certifies with the matching surface continuous; route B and He/H 0.65 stop at k = 0 (2026-10-08) 191
122.1 He/H 2.09 (`phase3/energy_coupling_nlte_structural_a76da494`) . . . 192
122.2 Route B (He/H 0.65, s = 1.3) and He/H 0.65 (`phase5`) . . . 193
122.3 Open . . . 193
123. Energy continuity at the open matching surface measured; the WENO3 floor of the primitive variables tested (2026-10-08) . . . 193
124. The Photochem adapter runs on its stated step budget and retries a stalled steady-state pass once; the He/H 2.09 open-face closure resumed at k = 5 and converged there (user decisions 2026-10-08; `md/k05_chemistry_failure_20261008.md`) . . . 194
125. Trial admissibility of the stationary residual, recovery of the outer alternation to the last converged wind, and the WENO3 primitive floor at the domain scale (plan rev2 step 2, user approval 2026-10-09) . . . 196
125.1 Trial admissibility (`steady_newton.f90`, `UW_conversions.f90`, `charge_exchange.f90`, `constrained_chemical_equilibrium.f90`) . . . 197
125.2 Failed-wind recovery (`EXHALE_main.f90`, `steady_wind_with_element_diffusion`) . . . 198
125.3 The WENO3 primitive floor at the domain scale (`Reconstruction.f90`, `hydrodynamic_rows_body.inc`) . . . 199
125.4 Checkpoint: regression matrix (gfortran, OMP_NUM_THREADS = 1, no golden refresh) . . . 199
125.5 The route-B and He/H 0.65 k00 entries (closure environment, omega 0.5, bound 0.04 held, 9-12 threads) . . . 202
125.6 Also corrected . . . 203
126. An open base face starts from a seed that agrees with its ghost: the base cells take the ghost hydrogen partition, and the carriers are relaxed at the entry wind before the first solve (change 4 of step 2, user approval 2026-10-09; `md/cold_base_cell_diagnosis_20261009.md`) . . . 203
126.1 What was implemented . . . 204
126.2 Tests . . . 204

126.3 Pass 1, cells 1-60 (the diagnosis's control repeated) . . . 204
126.4 Full alternations from k00 (closure environment, omega 0.5, bound 0.04 held, 9 threads, budget 60 passes) . . . 205
126.5 Matrix . . . 206
127. The marching stops on the face mass flux and the state change, not on the cell-centered du; the local WENO3 floor is restored; secondary ionization is on from step 0 (2026-10-09/10, user decisions) . . . 206
127.1 Changes (working tree; frozen binary `LHS1140b/models/.P2/EXHALE_5ed2b073.x`, manifest `BINARY_MANIFEST_5ed2b073943e.txt`, ifx; gfortran tree build md5 7a1ff33f) . . . 206
127.2 Measurements behind the thresholds (wasp_full, local floor unless stated, secondary ionization off, continued from a loaded state) . . . 207
127.3 Regression matrix (gfortran 7a1ff33f, one thread, `tmp/step2/regm/fin`), against the step-2 matrix of 7fe0b827 (`regm/new`: same local floor, staged secondary ionization) . . . 207
127.4 Outer ghost condition: design note only . . . 208
127.5 Exit status of a failed stationary solve; change 4 (b) diagnosed (2026-10-10, user decisions) . . . 208
127.6 The WASP regression cases move their outer face past the sonic point (user decision 2026-10-10) 209
128. The column conduction as an exact excess over the climate column; the latent and eddy heat of the column measured (2026-10-10; TO_BE_DONE (AR) 3(ii), (AU)) . . . 210
129. One stellar geometry at the matching face, set by the wind's input (user decision 2026-10-10) . . . 210
130. Eddy heat transport in the reference climate and the column; with K_zz = 1e9 no reference climate exists (user decision 2026-10-10, option (b)) . . . 211
131. The JFNK hand-off reads the face mass-flux spread (user decision 2026-10-10) . . . 212
132. The H3+ LTE cooling function is MiZATeP (Mizus et al. 2017); the Miller et al. 2013 fit is removed (plan rev4 step B) . . . 212
133. Turbulent dissipation heating does not make the chemistry's K_zz = 1e9 a heat diffusivity; recorded, not implemented (review, user directive 2026-10-10) . . . 213
134. The chemistry is handed the hydrogen-NUCLEUS flux, not the component-1 mass flux (plan rev4 step C)214
135. Step D: documents brought to the code; four defects the documentation review found (2026-10-10) . . . 215
136. Checkpoint of plan rev4 steps A-D: frozen binary and regression matrix (2026-10-10) . . . 215
137. The six long-failing test rows were stale expectations, not code defects; two rows made stale by section 127.5 (2026-10-10) . . . 216
138. A Roche domain extended past the potential ridge: the cold IC, the outer ghost and the absorbing columns assumed a potential rising outward (plan rev4 step E, 2026-10-11) . . . 217
139. The internal energy of cold, highly supersonic expanding cells from the advected adiabat (dual-energy method) (plan rev4 step E, 2026-10-11) . . . 218
140. Step-E inputs moved into the tree; documents brought to sections 138-139 (plan rev4 step E, 2026-10-11)219
141. Stale folders deleted (user decision, 2026-10-11) . . . 220
142. The regression matrix measured across sections 138-139 (2026-10-11/12) . . . 221
143. WASP-121 b run of record: the spectrum Huang et al. computed with (user decision, 2026-10-12) . . . 222

144. A subsonic outer face: no further treatment; its Mdot labeled not quantitative (user decision, 2026-10-12) 222
145. Diagnosis: the base-cell mass row refusal is the known split between the solver merit and the certification scale (2026-10-12) . . . . . 223
146. Section 145 corrected after the Codex review: the refused Yan state is the marching fallback (2026-10-12) 223
147. Second Codex review: the F1-F3 plan corrected (rev2) (2026-10-12) . . . . . 224
148. Error-testing material deleted from the source and the test set (2026-10-16) . . . . . 224
149. Uncalled routines, unbuilt sources and work-item option names removed from the source (2026-10-17) 229

**Code-size summary: EXHALE vs. the original ATES (2026-06-07; updated 2026-10-05)** **232**

**Fortran source inherited unchanged from ATES** **243**

This log continues `md/Update_EXHALE_stage2.md` from 2026-09-23. The earlier logs are kept under their own names:

- `docs/Update_EXHALE_stage0.{tex,pdf}`: the early phase (the ATES fork up to 2026-08-15).
- `md/Update_EXHALE_stage1.md` (typeset `docs/Update_EXHALE_stage1.{tex,pdf}`): sections 1-171, 2026-06 to 2026-09-05; "Update_EXHALE section N" in the source comments and the older memos points into that file.
- `md/Update_EXHALE_stage2.md` (typeset `docs/Update_EXHALE_stage2.{tex,pdf}`): sections 1-17, 2026-09-05 to 2026-09-22.

The entries of 2026-09-23 were first written into the stage 2 log as its sections 18 to 24 and moved here the same day as sections 1 to 7; a record of that day that cites "`md/Update_EXHALE_stage2.md` section 18" to "section 24" means section 1 to section 7 of this file (the references known at the move were rewritten).
Each entry below is one dated section: what changed, why, which files, the check that was run and its result. Section numbers restart at 1. The plan the entries execute is `md/PLAN_20260923_rev2.md`.

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## 1. The Ritz residuals of the preconditioned spectrum (2026-09-23)

The spectrum hook `EXHALE_PRECOND_SPECTRUM=1` printed Ritz values with no accuracy of their own (`md/ISSUES_20260909.md` item (D8), plan section 16). It now prints the residual `||A y - theta y||` of each.
**The residual.** For a Ritz pair `(theta, y = V_k s)` with `||s|| = 1` the Arnoldi relation `A V_k = V_k H_k + h(k+1,k) v(k+1) e_kˆT` gives `A y - theta y = h(k+1,k) s_k v(k+1)`, so the residual is `|h(k+1,k)| |s_k|`. `ritz_values_of_a_matrix_free_operator` now hands back `h(k+1,k)`, and the routine prints that residual and the residual over the value's magnitude for the six smallest values of every compression. That form assumes a linear operator and an orthonormal basis, and the finite-difference action is linear only to its truncation, so for the three smallest values of the whole operator the residual is also measured by applying the operator once more to `y`, once to each part of a complex pair. The operator action is now one routine, `action_of_the_preconditioned_operator`, called by the recursion and by the measured residual alike; the complex-pair packing of dgeev is undone in `coefficients_of_the_ritz_vector`. Every existing `[diag 15]` line is printed as before, and the residuals are new lines.

**Verification.** On the stored state of `atomic_scalar_gj1132x0.01_kzz1e9/HeH9.7` (`g0002_20260919T004843Z_f7485b14`) with `EXHALE_TRUST_REGION=1`, `EXHALE_ELEM_DIAG=1`, `EXHALE_PRECOND_SPECTRUM=1` and `EXHALE_JFNK_MAXIT=1`, a 17-second run: the 30 `[diag 15]` lines of the binary before the change are byte-identical after it, so extracting the action moved nothing in the recursion; and the Arnoldi estimate and the measured residual agree to five or six digits for all three measured pairs, the complex one included. MEASURED. The hook runs only under `EXHALE_PRECOND_SPECTRUM=1` inside the trust-region step, which no case of the regression matrix enables, so the matrix was not rerun. Two unused local declarations of the same file (`islot`; `tv`, `net`) were removed on the way; they carried no value.

**What it shows.** At 200 products that spectrum is the one phase 9 of the campaign quoted (largest 9.99385E-01, smallest 2.52128E-04, ratio 3.96381E+03, reproduced to every digit), and its small end is not resolved: `h(k+1,k) = 0.168501`, and the residual over the magnitude is 12.8, 3.58, 1.86, 1.86, 1.09 and 1.09 for the six smallest values, every one larger than one.

**The whole space.** `n_ritz_step` became `EXHALE_RITZ_STEPS=<k>` (default 200, capped at the number of unknowns, below 2 refused). At k = 1500, the number of unknowns of this operator, the Hessenberg is similar to the operator, and the same fixture took 136 s. MEASURED: smallest 5.60717E-06, ratio 1.78560E+05, 43 values below 1E-4 and 319 below 1E-2, against 0 and 6 at 200 products. So "nothing below 1E-4" is wrong on that checkpoint, and "there is no near-null mode of the preconditioned operator" is refuted. What the recursion finds are small eigenvalues of the operator the Krylov cycle sees: the hydrodynamic block alone (three unknowns a cell; `nvar_jac = 3 + nspec_row`, with no species rows on this route), scaled and preconditioned. [Corrected later on 2026-09-23 by `md/lhs1140b_spectral_checks_20260923.md`: this read "and shifted by the pseudo-time diagonal"; the shift is in the band preconditioner only, the hook being handed `idtau_leg = 0`.] They are not established as time scales of the atmosphere, and they say nothing about the alternated composition-hydrodynamics map. [Wording corrected 2026-09-23 after the review of `md/PLAN_20260923.md`: this paragraph read "it has a cluster of slow modes".] Below about 1E-5 even the full recursion does not resolve them: the residual measured on the operator is 9.72E-06, 9.88E-06 and 9.56E-06 for the three smallest values (the first two one conjugate pair), larger than the pair's magnitude and 0.96 of the third's, so none is resolved from zero at the production probe length; the floor was later measured to be the rounding of the forward difference (it falls as arcˆ-0.97 with the probe length, spectral checks memo). What the measurement bounds rigorously is the smallest singular value of the action as applied: on the normalized Ritz vector its norm is at most |theta| + ||r|| = 1.5E-05. Whether the 1E-5 floor of the residual is truncation, rounding or nonadditivity of the finite-difference action is not decided by these runs; `measure_the_additivity_of_the_jacobian_action` is the narrower test. The vectors (the real and imaginary parts of the pair, and the third) carry 53 to 61 per cent in the mass rows and peak in the wind around cells 93 to 169, not at the base, but since they are not resolved their location is not established either. The documents that quoted the 200-product numbers carry a dated correction beside them. What stands is the direct measurement they were offered beside: on the alternated route the Krylov cycle reaches its tolerance in one to four products.

## 2. Two records the campaign left open, closed (2026-09-23)

**`EXHALE_RESIDUAL` now writes the configuration it measured under.** The route stopped at `src/EXHALE_main.f90:1335`, before `write_setup_report` and `write_resolved_config` further down, so `EXHALE_setup.out` was left empty and `EXHALE_resolved.out` absent. It now writes both before its stop, as the direct steady route already did before its own. MEASURED on the stored state of `atomic_scalar_gj1132x0.01_kzz1e9/HeH9.7`: 62 and 48 lines written where there were none, the printed residual byte-identical (9.806145E-01, 8.723527E-06, 1.071147E+00), and the resolved file identical to a normal run's but for two added lines, the base ghost count and its electrons as solved, present because this route evaluates the residual, and so solves the ghost, before it writes.

**The conversion now asserts the charge.** `molecular_seed_from_atomic_state` computed the positive charge density before and after the 2 H -> H2 transfer and never compared the two. It now does, cell by cell, against `inventory_feasible_slack` (eight units in the last place) times the charge density, the bound `element_inventory` already uses for two expressions of the same nuclei; a neutral cell must come back exactly. A breach stops the run, and the largest change is printed in units of `eps * n_chg`. A relative change was not used because a neutral cell carries no charge and the ratio is not defined there. The `molecular_seed` suite asserts it in each of its three modes: 29 of 29 pass, and the change is 0.00 units in every mode, so in memory the transfer conserves the charge bitwise, and the 1.5 units the M2 conversion audit measured were its file round trip, as that audit said. The failure path was read, not exercised.

## 3. Two statements of the source that were not true, corrected (2026-09-23)

Found while checking the review of `md/PLAN_20260923.md` against the code; the plan's revision is `md/PLAN_20260923_rev1.md`. Neither change moves a number: one is a comment and one is the text of a printed line.

**The certificate printed a withdrawn reason.** Its report says why the species rows below `cert_regime_wind_r` = 1.20 are reported and do not gate, and for the band 1.10 to 1.20 it printed that the row's discretization, 4.6e-4 at the binding radius, "stands above any tolerance one could set there". The comment above `cert_regime_wind_r` said the same, that a 1e-5 gate would ask the residual to fall below the error of its own discrete equation. That inference was withdrawn on 2026-09-11 (`md/code_status_20260910.md` section 3.3): the algebraic residual of the discrete equations is not bounded below by their truncation error, and the same routine took a wind H2 row to 2.4e-13 on a held background. The gating decision is unchanged; its stated reason is now the one that document gives, that no state of this code has come near 1e-5 there and that the terms of the row are conserved only to 2e-3 (`certification.f90`, the comment of the two regimes and the report block).

**The absent-carrier threshold claimed a round-off argument.** The comment on `carrier_absent_fraction` (`diffusive_photochemistry.f90`) said that 1e-20 of the free reservoir is "four decades below the round-off of a double against the reservoir, so nothing at or under it is a resolved density at all", and that such a carrier carries nothing any reported quantity depends on. A small term vanishing from a rounded sum does not make it unresolved: stored as its own double, 4e-17 stays 4e-17 with a relative spacing of 1.5e-16, and its production, loss and transport stay resolved with it. The comment now says the threshold is a choice and not a round-off limit, and that the test reads the abundance

alone, not the production or the incoming flux that could raise it. The threshold itself is unchanged.
Verified: the build compiles (`EXHALE.x` md5 `69233669123c368b2d58aebc132eda85`; the printed string moved it from `67646328ad476a976a7953aba65bdffc`), and one `stationary evaluate` of a stored molecular state (the pass-6 state of `kzz1e9/HeH0.55` from the 2026-09-21 alternation audit) prints the corrected block and reproduces to four digits the verdict that audit measured on it under its own binary (carrier balance H2 9.722E-04 at cell 500, elemental transport He/H 7.353E-04 at cell 336). The line counts of both files are unchanged. The regression matrix was not rerun: the printed text enters no output file the matrix compares, and a comment enters no object code.

## 4. Four more source comments that were not true, corrected (2026-09-23)

Found while checking the second review of the LHS 1140 b plan (`md/PLAN_20260923_review1.md`) against the code; the plan's revision is `md/PLAN_20260923_rev2.md`. All four are comments in `diffusive_photochemistry.f90`; the line count of the file is unchanged and the rebuilt `EXHALE.x` has the same md5, `69233669123c368b2d58aebc132eda85`, so no object code moved.

- **The boundary section of the module header said "OUTER: zero gradient, as the element operator does"**, and that a run whose carriers are not negligible there is flagged in `output/Oxygen_chemistry.txt`. The carrier operator imposes zero diffusive, eddy and settling flux on the outer face (`Jf(N) = 0`), which with a settling drift is not a zero gradient, and `Oxygen_chemistry.txt` is written only by oxygen-chemistry runs and flags no carrier. The header now states the flux condition and points to `carrier_face_coefficients`.
- **`carrier_face_coefficients` justified that closure by the outer cells being collisionless**, quoting `Kn_bulk` = 1.4-2.4 at the domain top and an exobase at 18.8-25.4 R_p. Those numbers come from runs made before the 2026-09-13 well-balanced re-solve. MEASURED with `src/utils/collisional_validity.py` on the current certified states and on the `kzz1e9/HeH0.55` audit state: `Kn_bulk` at the top cell is 0.055-0.142, the exobase lies above the domain, and the largest species value in the critical region is 0.41-0.59. The comment now says the outer cells are transitional and that the zero face flux rests on the other reason it gives, the measured inward settling flux of a ghost-built face. `md/collisional_validity.md` carries the same measurement as a dated note.
- **The row-scale floor was said to be converged by construction** ("a density that small cannot change any observable, so a row below it IS converged", and "every term of such a row is round-off"). The floor is the rate that would move a carrier by 1e-20 of its element in one signal-crossing time, a scale that a row is judged against; it proves nothing about the row.
- **The absence test was said to separate round-off from a real row**. It reads the abundance alone, and a carrier far below the threshold can carry a real, resolved production; the comment now says that the test removes such a cell from the carrier gate without asking whether production or incoming flux could raise it, and that this is a choice.

**Stale (2026-10-07):** the Knudsen numbers, `Kn_bulk = 0.1` radii and exobase radii in this passage were computed by `src/utils/collisional_validity.py` before its resonant H+ + H and He+ + He momentum-transfer coefficients (`md/Update_EXHALE_stage3.md` section 110). On the 102 stored LHS 1140 b states the change moves the `Kn_bulk =`

0.1 radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.

Outside the source, the comment on the 2.26 km/s overlay shift in LHS1140b/make_memo_figures.py said the equivalent width is unaffected. The equivalent widths are computed from unshifted arrays, so the shift enters none of them; applied to a model it would move the fixed-window EW by 1.2e-5 relative (MEASURED). The comment now says both.

Verified: make rebuilt EXHALE.x to the same md5; the Python file parses. Nothing was run beyond that, since no executable statement changed.

## 5. One equivalent-width operator for the observation and every model (2026-09-23)

The user's decision on md/PLAN_20260923_rev2.md section 4: the observed statistic stays as it is (1.1084293096 +/- 0.0294774727 %A, working target 1.108 +/- 0.030), and every model is put through the same operator. A model spectrum, already convolved to the instrument resolution, is interpolated linearly onto the observation's own vacuum samples inside 10832.60 to 10834.20 A (44 samples, 10832.622 to 10834.175 A) and integrated with their trapezoid weights. Before, each model was integrated with a masked trapezoid on its own grid inside the nominal window, so the two integrals started and stopped at different wavelengths.

**One module, six readers.** LHS1140b/he10830_equivalent_width.py holds the window, the vacuum/air ratio, the observation, and the operator (model_equivalent_width, transit_file_equivalent_width, observed_equivalent_width). The six places that computed an EW each on its own now call it: LHS1140b/models/status.py (the results table and, since this change, the EW column of the provenance table, which it used to copy from each run's REPRODUCE.md), LHS1140b/make_memo_figures.py (the unbroadened and the broadened curves, and the observed value, whose variable and printed label no longer call the released spectrum a digitization), LHS1140b/models/compare_archive.py, LHS1140b/models/write_reproduce.py (whose line also said the window was in air; it is in vacuum), and the inline EW of LHS1140b/models/run_case.sh. The unused regular expression that read the EW out of REPRODUCE.md was removed.

**What moved, MEASURED.** Over the 177 EXHALE spectra of the catalog the change is at most 1.45e-4 %A, relative -1.1e-3 to +0.9e-3; 31 spectra move in the fourth decimal, 15 of the 83 certified rows of the catalog table among them (for example the certified molecular K_zz = 1e9 He/H 2.13 and 9.7 read 1.5604 and 2.4060 where they read 1.5603 and 2.4061), and one crossing moves (K_zz = 0: 3.1011 to 3.1010). Broadened curves and p-winds curves move more, up to about 0.6 per cent, because their wings reach the window edges that the old model integral kept and the observation's samples do not (for example the matched-kernel EW of the well-mixed He/H 0.4 solution, 1.067 to 1.065). LHS1140b/MODELS.md sections 7 and 8 were regenerated by models/status.py --write; docs/lhs1140b_catalog_state.tex carries the new EWs, a paragraph on the operator and the recomputed comparison of its section 4 (1.374869 to 1.356745 %A, 0.60 standard deviations), rebuilt to seven pages and checked; the memo docs/lhs1140b_exhale_vs_pwinds.tex and its figures are brought to the same operator in a separate pass (md/lhs1140b_ew_operator_memo_changes_20260923.md).

## 6. The stage export of the stationary alternation (2026-09-23)

EXHALE_STAGE_EXPORT=1 (default off) writes, for every pass of the alternation, the state the driver holds at four named stages: hydro_entry (before the hydrodynamic solve; from the second pass on, the state the previous composition update handed over), hydro_return (the accepted solve after the ionization sweep that follows it), certification (the joint test, with the three residual rows), and composition_return (after the element and carrier updates and the rebuild of p and T at the fixed conserved state). Each goes to output/stage_state_p<pass>_<stage>.txt with every row, ghosts included: r, rho, v, p, T, heat, cool, the particle and electron densities, H I, H II, He I, He II, He III, He(2^3S), H2, H2+, H3+, HeH+, the conserved variables, and at the certification the residual rows. At hydro_entry and at the certification the driver also asks conservation_budget for the next assembly under the stage's name (conservation_budget_request_next, new), so the energy row's terms of that state are exported without counting assemblies, which probes and trials also consume. It is the instrument md/PLAN_20260923_rev2.md P1.5 and P1.6 (b) call for (the four named-stage captures of review R4).

The writer calls nothing that sets a module variable. MEASURED on one pass from the certified molecular_scalar_gj1132_kzz1e9/HeH2.13 g0008 state: the data rows of Hydro_ioniz.txt and Ion_species.txt are bitwise the same with the export off (twice), with it on, and with the previous binary 69233669123c368b2d58aebc132eda85; the log differs only by the export lines. The two exported budget files carry their stage names. Binary EXHALE.x md5 b017fe42407c980068e112a8ca3d71d3, copied to LHS1140b/models/EXHALE_b017fe42.x with its manifest. The regression matrix was not rerun: every new call in the driver is guarded by the key, and the budget module gained a subroutine and a header line that are reached only on a request, so with the key off no executed statement of any route changed. The bitwise check above covers the stationary alternation; the matrix's cases, which march, were not run.

## 7. Three defects the family C audit found, fixed, and two small ones (2026-09-23)

The audit of the 13-decade H2 hole in molecular_photochem_gj1132_kzzprofile/HeH9 (md/lhs1140b_family_c_h2_slab_20260923.md, item P1.7 of md/PLAN_20260923_rev2.md) traced it to the molecular seed and found why no solve removed it. Each finding was checked in the source before it was changed.

**The element relaxation threw away its own kept steps** (binary_element_diffusion.f90, relax_element_composition). Its retry budget counts every rejected step of the relaxation, and when it ran out the routine returned element_relaxation_failed and restored the entry composition, even after 24 to 31 steps had been kept. The module's outcome table, and the driver that reads it, reserve that outcome for a relaxation that kept no step; the driver then refuses the pass and skips the carrier relaxation (EXHALE_main.f90, the outer_element_update_refused branch). This is how the carrier operator never ran on the family C case in four solves, and how the 2026-09-18 rung at He/H 1.411 ended at its second pass ("took 28 steps and found no admissible 29th", READ, L34c section 4). A relaxation that has kept steps now ends as a partial advance (element_relaxation_step_budget) with the kept steps handed back, and says so once in the log; one that kept none is still refused with its entry composition. The retry budget still counts every rejection. Test: two new rows of src/tests/element_operator/ on a test knob (element_refuse_after_accepted_for_test, -1 in every

production run) that lets three steps through and refuses the rest: the relaxation reports the partial advance, three steps, and a moved composition; with no step let through it reports the refusal and returns the entry composition bitwise. Every row of the suite passes.

**A sweep could rewrite a transported H2** (`ionization_equilibrium.f90`, the extraction of the molecular partition). With the carriers transported the sweep is handed H2 (`x_h2_fixed`) and solves the rest against it; a root meets the imposed value through its own row, but a non-root acceptance (class 4) or a clamp onto the element budget (class 3) returned another x(4), and the extraction wrote that back as the cell's H2. MEASURED on the seed's intermediate state of the family C case: the returned H2 differed from the handed one at exactly the 21 non-root cells, by up to 82 per cent; with the fix it differs nowhere by more than 7.8e-16. The imposed value is now written and the atomic hydrogen takes the remainder of the budget.

**The molecular seed read a root off cells the sweep had not solved** (`molecular_seed_from_atomic_state.f90`, the `local` mode). The seed takes the root of each cell's own H2 row at the ions of one sweep; at cells where that sweep adopted a non-root with every ion at zero, the root was 4e-17 of the hydrogen, thirteen decades below the neighbouring cells, and that was the hole. The sweep now records the acceptance class of each cell (`ieq_cell_acceptance_class`, written and never read inside the sweep), and the seed reads no root off a class-4 or class-6 cell: such a cell takes the root of its nearest solved neighbours, interpolated in log x2 over r, and the smaller of that and the fit. MEASURED on the same reproducer: the 21 cells go from 4e-17 (and from up to 6.8e-4 where the rising slab was) to a smooth 1.8e-4 to 2.1e-3 between the solved cells 428 and 450; outside them the seed moves by at most 8.2e-4 relative, through the H2 fix above. Two further changes the audit proposed were tried and not kept: iterating sweep and root to a common fixed point did not converge in 30 iterations and drove the base layer to its unshielded branch (x2 = 6.9e-4 at 974 K), because a cell's shielding column is the H2 above it and the iteration is not a local one; and adopting the root in place of the smaller of root and fit follows from the same iteration and was withdrawn with it. The `molecular_seed` suite passes 29 of 29.

**Smaller.** The comment on `nH_free` in `diffusive_photochemistry.f90` said it excludes the hydrogen of the ion stages and molecular ions; it is the cell's hydrogen less the HeH+ nucleus (`carrier_element_totals`). And `carrier_row_terms.txt` was written with `ES14.7`, which drops the `E` of a three-digit exponent (`1.0332735-100`); it is now `ES15.7E3`.

**Not changed, reported.** Two provenance attributions in the state index of the family C case are wrong (the audit memo, D4): the certificate of g0002 belongs to a later solve, and g0001's manifest names another binary than its own file header. The index is written by the catalog tools and was not edited by hand.

Verified: the element-operator and molecular-seed suites pass; the H2 fix and the seed on the family C reproducers as above; the stationary route with `EXHALE_STAGE_EXPORT` unchanged. `EXHALE.x` md5 `45190de9ba3752eb720a728d54d3a1e3`. The one regression case that runs the H2 carrier operator, `mol_carrier`, was run with the binary before and after the element and H2 fixes (section 8).

## 8. The regression check of section 7, a cell writer, and corrections from the spectral and On stall tests (2026-09-24)

**The one regression case that runs the H2 carrier operator.** `mol_carrier` was run from its input with `OMP_NUM_THREADS=1 EXHALE_MAXSTEPS=12000` twice, once with the binary before the fixes of section 7 (`EXHALE_b017fe42.x`) and once after (`EXHALE_af853fe7.x`), in copies (`LHS1140b/models/.P1/d2d3_mol_carrier/`). MEASURED: the two `Hydro_ioniz.txt` differ by at most 1.3e-11 relative and the two `Ion_species.txt` by at most 9.4e-13; neither run met a non-root cell or the retry budget of the element relaxation, so the difference is the H2 now written exactly at the imposed value where a root had met it to the solver's tolerance. Below 0.1 per cent, it needs no golden refresh.

**A cell writer on the `EXHALE_NEWTON_TEST` route** (`EXHALE_main.f90`). The route printed only the row maxima of the solver's vector F(Y), so where it differed from the `EXHALE_RESIDUAL` route the cell was not known (ISSUES D5). It now writes `output/newton_residual_profile.txt` with the columns of `output/residual_profile.txt` (its temperature column is 0: the route forms no temperature array). MEASURED on `molecular_scalar_gj1132_kzz1e9/HeH2.13` g0008 with `EXHALE_RELOAD_EQ=0`: the two routes differ in cells 1 and 2 alone, where the base face mass flux differs by 9 per cent, and nowhere else in 500 cells; the mass-row maxima (5.10e-09 against 1.37e-09) are that base difference.

**From the checks of the preconditioned spectrum** (`md/lhs1140b_spectral_checks_20260923.md`):

- the cost line of the additivity measurement printed the run's cumulative count of inner cell solves where it meant this measurement's own (`nt_calls`, now `nt_calls_here`, `steady_newton.f90`);
- the comment that the Newton model's products reuse the frozen WENO3 weights is true on the line-search route and not on the trust-region route, where the merit evaluation sets `weno_mode = 0` and nothing restores it before the leg and the spectral hook; the comment now says so, and which of the two the leg should use is left open;
- the base boundary's comment that the states "sit decades from" the thresholds of the wind window now records a state that sits one decade away and a probe that crosses the switch (`base_boundary.f90`);
- the records that called the spectrum's operator "shifted by the pseudo-time diagonal" (ISSUES D8, stage 3 section 1, the p2_base note, the plan rev2 section 3.6) were corrected: the shift is in the band preconditioner only.

**From the On stall test** (`md/lhs1140b_on_stall_end_to_end_20260923.md`):

- `Ionization transport: True` was refused with `Coupled carrier solve: True` but not with `On stall`, which hands the same registry the same unknowns once the alternation stalls; it is now refused with either (`input_read.f90`);
- the comment on `EXHALE_JFNK_MAXIT` said a named cap makes the run take one stationary solve; it forbids a second entry into the stationary route and caps the solve of every pass inside the one entry, which the test measured (`EXHALE_main.f90`); the plan rev2 carried the same error and is corrected. The handover itself fails: the conditions are met and the transition begins, and the re-evaluation of the snapshot's residual stops the run on the face mass flux guard, the carrier balance also reading a background the snapshot does not hold. That is a change of the solver's control flow and is reported for a decision, not fixed.

**The update log.** The entries of 2026-09-23 were moved from the stage 2 log (its sections 18 to 24) into this file on the user's instruction, and `src/utils/update_log_to_tex.py` now takes the stage as its argument (`2` or `3`; the code-size appendix stays with stage 2). `docs/Update_EXHALE_stage2.tex` was regenerated for sections 1 to 17 (310 pages, no error or undefined reference) and this log's twin built (7 pages at the time).
Verified: the element-operator and molecular-seed suites pass; the builds compile. `EXHALE.x` md5 `46d5b9f4c0a62c1ab0cdf9bb165839ef`.

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## 9. Three defects the transient measurements found, repaired (2026-09-24)

The P4 measurements of the plan (`md/lhs1140b_p4_transients_20260923.md`, whose worker reported them and left them in place) found three defects; each is repaired and its path checked.

**The default CFL number was a single-precision literal.** `real*8 :: CFL = 0.6` stored 0.6000000238 (`parameters.f90`); every run that took the default marched at a step 3.97e-8 longer than the one it declared (MEASURED: the first step on a certified molecular state is 0.54736376538957865 s with the default and 0.54736374363929763 s with `CFL: 0.6`). Now `0.6d0`. The same kind of literal set the Voronov (1997) collisional-ionization coefficients of all seventeen rows of `Cool_coeff.f90` (11.26 stored as 11.2600002), and the physics probe `voronov_carbon_ionization.f90` mirrored it on purpose to compare exactly; both are now double-precision literals, and the probe passes at its 1e-12 tolerance (MEASURED). The rounding was below 1e-7 relative, four decades under the precision of the fits; a declared constant is still the constant it declares.

**`EXHALE_UPDATE_MAP` could stop before its last requested step.** The diagnostic switched off the flux-spread stop, the residual gate and the JFNK hand-off, but not the step-change stop or the stall stop, and on a stationary state a step of a quarter of the CFL step already moves the state by less than `dtu_th`: a six-step map ended after three blocks. While the map runs, every stop flag is now cleared after the stop tests of each step (`EXHALE_main.f90`), so the map ends on its own step count and the staged secondary ionization cannot change the physics between two of its steps; the stop line says "the update map wrote its N requested step(s)". MEASURED on a copy of the failing run without `Force start`: six blocks, dt from 6.6296e-05 to 6.6296e-11 of the code unit.

**`EXHALE_DUMP_IC=1` wrote placeholders as the lower ghost rows.** `load_IC` fills the primitive ghost rows with cell 1's state until the first `Apply_BC`, which writes the ghosts of the conserved array only; the dump wrote the primitive array. It now takes the ghost rows of the primitive array the boundary derivation held (`bc_W_hold`). MEASURED: the two lower ghost rows read rho = 1.302e14 and 1.242e14, T = 233.0998 and 226.0275 K, against the placeholder 4.857e13 and 546.06 K before, and the file's 233.1004 and 226.0281 K (the 2.5e-6 is the ghost composition the boundary re-derives from the loaded column).

The P4 memo now carries its command record (section 7, which its worker had not written) and the repairs beside each defect.

## 10. The elemental flux records ride on the Riemann face mass flux; a row writer for the helium element (2026-09-24)

**What was wrong.** The element transport operator is handed the face mass flux of the Riemann solve (`face_mass_flux_of_state`), the one the mass row differences. Two records of it were not: the base budget entry (`element_base_flux`) and `output/element_flux_profile.txt` built their face mass flux as the face mean of the cell-centred density and velocity. Near the base the cell-centred velocity carries a collocated two-cell odd-even mode (project record P44), so on the certified `molecular_scalar_gj1132_kzz1e9/HeH0.55` state the file read 4 pi r^2 F = -1.07e9, +1.16e8, -3.8e7 and +5.9e7 g/s at faces 1 to 4, while the Riemann flux is 3.934813e7 g/s through every face (MEASURED). The "standing base sound wave" the comments named was that mode, not a wave of the solution.
**What the records are now** (`binary_element_diffusion.f90`, `certification.f90`, `EXHALE_main.f90`):

- `write_element_flux_profile` takes the face mass flux of the state it describes and is called by the certification of that state, with the flux the element row was measured on, whenever `EXHALE_DIFFUSION_CHECK=1` or a lower-atmosphere profile is in use; the element step no longer writes it. Before, the file held the profile of the last relaxation, taken before the last hydrodynamic solve, and a stationary evaluate never wrote it.
- The base budget entry uses the flux of its own evaluation: the relaxation's `Frho_in`, or on the marching path the step flux `F0/6 + F1/6 + 2 F2/3` of the three SSP-RK3 stages (the flux whose divergence is the step's density change), read once per step; with neither, it says "not measured".
- `face_mass_flux_cgs` converts a code face flux with n0 mu v0 and the face mean of msum.
- The velocity is no longer an argument of `element_diffusion_step` or `relax_element_composition` (nothing read it).
- `composition_residual` and `trace_composition_residual` reduced their norm with `max`, whose result with a NaN is compiler dependent; a non-finite row now sets the norm to huge.
- `element_transport_residual`, documented as side-effect free, added its face reconstructions to the run's face counters; it now restores them.

**The helium row, term by term.** `EXHALE_ELEMENT_ROW_TERMS=1` (default off) writes `output/element_row_terms.txt` from the certification's own evaluation: the diffusive and advective terms of each face, the residual, the floor, the scale and the measure, with the face F_rho, X_face and J. Its measure agrees with the certification's full-precision anchor to the sixteenth digit.
**What it measured** (`md/lhs1140b_element_row_terms_20260924.md`): on the certified He/H 0.55 and 1.8 states the helium flux through r < 1.10 varies by 2.2e-3 and 7.8e-5 of its median, nearly all of it the step across cell 2 (the first solved cell, row 6.1e-5 and 2.3e-5, ungated); every other cell below r = 1.2 stands at or below 1.5e-7 and 2.7e-7. At cell 2 the residual would change the helium density in 9e9 and 3e10 s, against an advective cell time of 2e6 s and a diffusive one of 47 s. The comment in `certification.f90` that gave "element fluxes not conserved, spread 9.8" as the reason for not gating r < 1.10 had measured the face-mean artifact; it and the printed line now say the band is reported by decision, and the gating itself is unchanged. The Phase-E statement that the LHS 1140 b overlap window is structurally empty rested on the same quantity and is marked stale where it stands (`md/phase_e_flux_closure_design.md`,

src/utils/element_flux_closure.py, md/certification_tolerance_anchoring_20260910.md); it was not re-measured. src/utils/collisional_validity.py and md/collisional_validity.md used the same wording for the base cells; corrected (the computation, whose velocity scale binds four decades below its threshold there, is unchanged).

**Checks.** The solved state does not move: an evaluate of the certified He/H 1.45 state by the new binary writes Hydro_ioniz.txt, Ion_species.txt and both _adv files identical to the 46d5b9f4 evaluate apart from the provenance line, CERTIFIED, and its element_flux_profile.txt carries 7.0353671e7 g/s from face 1 outward (MEASURED; the worker measured the same on the He/H 0.55 and 1.8 states). Suites: element operator 65 of 65 (its base record rows assert the new definition and fail against the old objects by 1.54 and 1.31), diffusion_tests.x 35 of 35, certification 123 PASS, steady species rows 202, species face flux 18 (the last two by the worker). The dead velocity arrays of diffusion_tests.f90 and certification_contexts.f90 are removed, and the test T6, which claimed a uniform mixture "under an arbitrary velocity field" that the operator never read, now says what it tests.

## 11. The molecular K_zz = 1e9 ladder certified across the observed equivalent width (2026-09-24)

Items P1.1 to P1.3 of the plan, run under the binary EXHALE_46d5b9f4.x (the tree of section 8), every rung a stationary alternation from a certified parent mapped onto the new reservoir with the pressure held (map_state_to_grid.py --ic --reservoir He/H, the base mixing ratio from the same nucleus fraction f), 24 passes and a 3 h ceiling allowed. Run directories under LHS1140b/models/.P1/; driver .P1/make_rung.sh.

| He/H | parent | passes | wall of the passes | outcome | EW [%A] | log10 Mdot |
|---|---|---|---|---|---|---|
| 0.55 | the 2026-09-21 audit state (pilot) | 12, then 4 | 4.30 h, then 0.07 h | certified in the continuation | 0.1015 | 7.59 |
| 1.8 | certified 2.13 (descent pilot) | 12, then 3 | 1.04 h, then 0.04 h | certified in the continuation | 1.3584 | 7.88 |
| 1.45 | certified 1.8 | 17 | 1.15 h | certified | 1.0964 | 7.85 |
| 1.6 | certified 1.8 | 21 | 1.22 h | certified | 1.2155 | 7.86 |
| 1.52 | certified 1.45 | 14 | 0.88 h | certified | 1.1532 | 7.86 |
| 0.689 | certified 0.55 | 24 (budget) | 2.85 h | not certified: elemental row 9.9e-4, still falling | 0.274 (uncertified state) | 7.67 |

MEASURED. Every certified state was re-entered four times in a chain of Restart intent: stationary evaluate, each entry reading the state the one before wrote: CERTIFIED at every entry, the largest relative change of any field after four entries 1.2e-12 (the 1.8 chain; 3.5e-13 or less for the others). The EW is the catalog's measurement operator (section 5) on the spectrum EXHALE_transit.py synthesized from the published state. The two adjacent certified rungs

1.45 and 1.52 straddle the observed 1.108 +/- 0.030 %A, and `models/status.py` puts the crossing of the log-log chord at He/H = 1.4643. The interval is 0.0205 dex, below the plan's 0.03 dex floor, so the refinement stopped there.

**Connectivity** (`.P1/connectivity/compare_rungs.py`, figure `docs/figures/lhs1140b_kzz1e9_molecular_rungs.pdf`): from 1.45 to 2.13 the temperature, velocity, density, H II and He 2^3S profiles, the mass-loss rate (7.847 to 7.902), the outer Mach number (0.515 to 0.508) and the H2 front (x(H2) = 1e-2 at 1.200 to 1.161 R_p) move monotonically, and neighbor differences scale with the step (about 3 per cent in T for 0.02 dex). The 0.55 state is a different wind: x(H2) above 0.5 to 1.13 R_p and near 1e-2 in the far wind, a peak temperature of 3400 K against 5200 K and more, He 2^3S thirty times lower. The interval between 0.55 and 1.45 is not sampled (the 0.689 rung is its first point), so whether the two are one branch, and whether a second crossing lies between them, is not established. The 0.689 rung's last state (uncertified) is intermediate: x(H2) above 0.5 to 1.10 R_p as at 0.55, but 1.5e-4 at 10 R_p, near the 1.45 value.

**The line shape** (`docs/figures/lhs1140b_kzz1e9_molecular_ew.pdf`): the crossing is of the integrated EW alone. At He/H = 1.45 the model red line is 4.07 per cent deep, 3.3 times the depth Cherubim et al. (2026) fit (1.24 +0.22/-0.23 per cent), and correspondingly narrower; the molecular solutions share the line-shape disagreement of the unbroadened atomic spectra.

**Published** (`models/run_case.sh` with `SEED=` the certified run, which solves nothing new from a certified entry, evaluates, synthesizes the spectrum and writes `REPRODUCE.md`): `molecular_scalar_gj1132_kzz1e9/HeH0.55` g0007 and the new cases `HeH1.45`, `HeH1.52`, `HeH1.6`, `HeH1.8` (g0002 each), added to `models/make_models.py`. The new cases' `input.inp` is the certified `HeH2.13` input with the He/H changed, the input the states were solved under; `make_models.py` itself would write `Restart intent: stationary equilibrate` for a molecular case, which the older `HeH0.083` carries and the certified cases do not (a difference of route, reported for a decision). The catalog inventory reads 157 state indexes, 100 certified. `LHS1140b/MODELS.md` sections 7 and 8 and `models/CASE_INVENTORY.{md,json}` were rewritten by `status.py --write`; `docs/lhs1140b_catalog_state.tex` carries the new rows (section 13) and `docs/lhs1140b_exhale_vs_pwinds.tex` its corrected molecular paragraph and campaign table (both rebuilt, no error, pages checked). A verification binary copied into `models/` had become the "newest EXHALE_.x" that `status.py` takes as the binary of record; it was moved to the run directory that used it.

**The physical record of the straddling states** (plan section 3.1, with the writer of section 10, `.P1/element_row_terms/record_straddle.txt`): the H2 carrier row is worst inside the gated wind (5.16e-6 and 7.90e-6), and the helium element row is worst at cell 2, ungated (2.04e-5 and 1.27e-4), the next worst below r = 1.2 being 1.07e-6 (cell 6) and 5.4e-6 (cell 3), both in the first cells above the base; at cell 2 the residual would change the helium density in 2.9e10 and 4.6e9 s, against an advective cell time of 1.5e6 s, and the region below 1.2 passes the helium flux on to within 9.3e-5 of its throughput.

**P3** (`md/lhs1140b_p3_low_xuv_20260923.md`, run 2026-09-23 under `EXHALE_69233669.x`): the atomic He/H 9.7 branch certifies at 0.0675, 0.0662 and 0.065 of the fiducial XUV (EW 0.0239, 0.0172, 0.0132 %A) and refuses at 0.06 and 0.0624 on the elemental row; each certified rung re-entered four times, CERTIFIED throughout.

## 12. The regression of sections 9 and 10, and the validation of the crossing (2026-09-24)

**The matrix** (`make check`, binary `ac232b34`, the tree with section 9; sixteen cases run, against the goldens of 2026-09-20). Verdict REGRESSION FAIL, for three reasons, MEASURED and attributed with the binary of the day before

(`EXHALE_46d5b9f4.x`, the tree without section 9) run on copies of the cases:

1. **The CFL literal of section 9.** The control reproduces the goldens (`mol_lyman_werner` identical, `mol_carrier` within 1.3e-11), so every movement is today's, and in these marching cases the only change that acts is the step 3.97e-8 shorter. The state files move by at most 9.1e-5 (`wasp_he23off`; 1.6e-5 or less in the rest), below the 1e-3 tolerance. What exceeds it is the `adv_mass_row` column of `Hydro_ioniz_adv.txt`, the mass-row measure the post-processing writes per row, in the four molecular 12000-step snapshots that are far from stationary (du near 0.8): 1.2e-3 to 4.3e-3 relative (`mol_lyman_werner`, `mol_diffusion`, `mol_sec_ion`, `mol_carrier`), and 0 against 5.6e-15 in one row of `wasp_full_newton`. `hydrostatic_column` moves one ghost-row ion density from 4.1e-185 to 1.2e-170 (the control is identical to its golden). None of these is a change of a physical quantity above the tolerance.
2. **`oxygen_chemistry` stops before its first step**, with the control as well: "ghost composition is not a fixed point", the map applied 20 times and still moving by 5.7e-7 against an accuracy of 1e-11, after the constrained molecular roots of the cold start were clamped onto the simplex. The golden (2026-09-20 13:53) predates the ghost fixed point (commit of 2026-09-20 16:18); the case has not run since. The refusal is the boundary's designed behavior, so what to do with it is reported for a decision, not changed.
3. Two files of `iontrans_metals` have no reference (MISS, not a failure).

**Section 10 does not move them.** The five cases that exercise the changed paths (element operator and its records, the certification's element block) were re-run with the tree of section 10 (`76c3ce17`) against the outputs of the matrix: `mol_diffusion`, `lower_profile`, `iontrans_metals`, `mol_carrier`, `hydrostatic_column`, REGRESSION PASS, all byte-identical (tolerance 0). The other cases run without element diffusion and do not reach those paths.

**The goldens were not refreshed.** The refresh of the sixteen cases that section 9 moves was refused by the session's permission check; the outputs of the matrix stand in the case directories and are what a refresh would snapshot.

**The validation of the crossing of section 11** (plan order 12):

- A reverse continuation: He/H 1.52 reached downward from the certified 1.6 certifies (pass 13) and agrees with the 1.52 reached upward from 1.45 to 5.3e-6 in every hydrodynamic field and 1.4e-4 in every species, EW 1.15317 against 1.15316 %A (`LHS1140b/models/.P1/reverse_HeH1.52_from1.6/`).
- A refined grid (`LHS1140b/models/.P1/refine_grid/`): the two straddling states on 1000 cells (the base block and every other spacing halved, the outer radius unchanged) certify (passes 16, 15) and give EW 1.0640 and 1.1196 %A (against 1.0964 and 1.1532 on 500 cells), log10 Mdot 7.84 and 7.85. They still straddle 1.108; the chord moves from He/H 1.4643 to 1.505. The discretization shift, 3 per cent in EW, is of the size of the observation's working error; a 2000-cell pair is running to read the order.
- The physical record of section 11 and the operator of section 5.
- The domain test waits for a grid option (a decision).

## 13. The width of the line on the molecular crossing, and the catalog summary brought up to date (2026-09-24)

**The broadening the line needs** (user instruction: after the EW, match the width with an assumed turbulent Gaussian as `docs/lhs1140b_exhale_vs_pwinds.tex` does). `LHS1140b/models/.P1/connectivity/broaden_molecular.py` restates the memo's operation (the excess absorption of the instrument-convolved spectrum convolved with a Gaussian of FWHM f in velocity, the three-Gaussian extractor of `he_line_metrics.py`, the EW operator of section 5, f matched where the red-pair FWHM reaches the measured 0.841 A) and first reproduces the memo on its atomic well-mixed He/H 0.40 state: f = 22.22 km/s, depth 1.229, blue 0.161, ratio 7.65, EW 1.0645 (the memo prints 1.065). MEASURED on the certified molecular K_zz = 1e9 states:

| He/H | cells | EW, f = 0 | matched f [km/s] | red [%] | blue [%] | red/blue | EW at f |
|---|---|---|---|---|---|---|---|
| 0.55 | 500 | 0.1015 | 22.32 | 0.113 | 0.014 | 7.89 | 0.0981 |
| 1.45 | 500 | 1.0964 | 22.21 | 1.224 | 0.160 | 7.64 | 1.0600 |
| 1.52 | 500 | 1.1532 | 22.20 | 1.287 | 0.169 | 7.63 | 1.1149 |
| 1.6 | 500 | 1.2155 | 22.19 | 1.357 | 0.178 | 7.61 | 1.1752 |
| 1.8 | 500 | 1.3584 | 22.16 | 1.516 | 0.200 | 7.58 | 1.3135 |
| 2.13 | 500 | 1.5604 | 22.12 | 1.741 | 0.231 | 7.55 | 1.5087 |
| 9.7 | 500 | 2.4060 | 21.80 | 2.684 | 0.359 | 7.47 | 2.3264 |
| 1.45 | 1000 | 1.0640 | 22.21 | 1.188 | 0.155 | 7.65 | 1.0288 |
| 1.52 | 1000 | 1.1196 | 22.19 | 1.250 | 0.164 | 7.64 | 1.0825 |
| observed | | 1.108 +/- 0.030 | | 1.24 +0.22/-0.23 | 0.25 +0.14/-0.12 | 6.7 +12.7/-3.1 | 1.108 +/- 0.030 |

At He/H 1.45 the width is reached at 22.21 km/s (sigma 9.43 km/s) and the depth, the blue component and the ratio then fall inside the observed 1 sigma intervals without being tuned. The kernel does not depend on composition or chemistry (22.12 to 22.32 over 0.55 to 2.13; the memo's atomic families 22.20 to 22.40). The Gaussian wings carry 3.3 per cent of the EW out of the window, so the broadened EW crosses the working target 1.108 at He/H 1.5112 between the certified 1.45 and 1.52 on the catalog grid; on 1000 cells the broadened EW of 1.52 is 1.0825, and the chord extended past 1.52 reaches the target at 1.55 (DERIVED, an extrapolation). On the advection-corrected profiles weighted by n(2 3S) r over r, c_s = 4.33 km/s in the line-forming region of both straddling states, so the kernel is Mach 2.2 (sigma), 3.1 (sqrt 2 sigma, the v_turb of Lampon et al. and p-winds) and 3.8 (sqrt 3 sigma). Figure `docs/figures/lhs1140b_kzz1e9_molecular_broadened.pdf`.

**A correction to the memo found on the way.** The memo's c_s of the line-forming region of the well-mixed He/H 0.55 state, 4.1 km/s, did not follow from its own T = 2499 K and mu = 1.94; recomputed with the memo's weighting (n(2 3S) r integrated over r on the `_adv` profiles, which reproduces T = 2498 K and mu = 1.93) it is 4.21 km/s, and the Mach numbers of the matched kernel move from 2.3, 3.3, 4.0 to 2.2, 3.2, 3.9 (summary and section of the memo corrected, marked in place).

**The catalog summary.** Renamed on the user's instruction to `docs/lhs1140b_catalog_state.tex`

(no date in the name; it was lhs1140b_catalog_state_20260922.tex, and the references in md/du_stop_vs_stationary_20260920.md, md/session_handoff_20260920.md and the stage 2 and 3 logs now point to the new name; the stage 2 twin Update_EXHALE_stage2.{tex,pdf} still carries the old name and was not regenerated). Brought up to date rather than annotated: the counts (157 indexes, 100 certified; 103 models, 91 certified), the table by group, the table of certified models (eight new rows) and of uncertified ones (the molecular 0.55 row out, the He/H 9.7 rungs at 0.06 and 0.0624 in), the mechanisms (and the count of refusing rows, which the old text stated wrongly: the energy row, not the mass row, appeared most often), the operator-difference statement re-measured over the 91 certified rows (still at most 1.5e-4 %A, 15 rows in the fourth decimal), and a new section 5 on the molecular crossing, its validation and the broadening. Rebuilt (11 pages, no error), every changed page checked.

## 14. The user's decisions of 2026-09-24 on the open items, and what each gave

The list of section 6 of md/lhs1140b_plan_rev2_execution_20260924.md, reordered by what it changes, was put to the user; every item was decided "as recommended". Items 2 and 3 (the rungs below He/H 1.45, and the grid of the catalog) are still running and are recorded in the next section.

**Item 1, the width of the line.** The added Gaussian of section 13 is an empirical parameter of the comparison, fixed by the measured width alone; every composition read from the full line is stated as conditional on it (docs/lhs1140b_catalog_state.tex section 5, the execution record).

**Item 4, which cells gate the species rows: compared, not changed.** A default-off diagnostic, EXHALE_CERT_GATE_DIAGNOSTIC=1 (certification.f90), prints for every carrier and element row what the verdict would be if every cell from cell 3 outward were gated against the wind tolerance (1e-5), cells 1 and 2 reported. MEASURED on all 99 ladder cases re-evaluated with it (md/lhs1140b_gating_diagnostic_20260924.md): every stored verdict reproduces; under the candidate rule 38 of the 91 certified cases stay certified and 53 would be refused, 52 of them by the helium element row in the first cells above the base (cell 3 in 45 cases, up to 9.1e-2) and one by the H2 row (molecular kzz1e9/HeH9.7, 4.76e-5 at r = 1.127); no uncertified case would become certified. In 27 atomic K_zz cases the element row exceeds 1e-5 in 76 to 205 cells up to r = 1.09 to 1.19 (on the atomic kzz1e9 He/H 1.50 and 1.60 states 1.5e-3 and 1.8e-3 at cell 3, 184 and 182 cells out to r = 1.16). The premise of the item, that below r = 1.2 only the first solved cell stands above 1e-5, holds on the six certified molecular K_zz = 1e9 states and not on the atomic ladders. The gate was not changed. The same worker's compiler warnings showed two truncations in the certification report (System_HeH_mol_metals printed as System_HeH_mol_metal, the carrier row's scale text cut at 76 characters); both fields were widened (log text only). The diagnostic shares the gate's reduction with the verdict (species_row_distance_over_gated_cells); the certification suite passes 143 of 143, and with the variable unset every output is byte-identical to the build before it (MEASURED by the worker on four states).

**Item 6, a grid option that keeps the interior.** Outer shells [r_face,cells]: <r_face> <cells> (K33d of md/input_schema.md, default off): the grid is built as before, and <cells> cells are appended beyond its outer face, their widths in geometric progression with the ratio that puts the last face exactly on r_face; every constructed center up to the first outer ghost, every face, width and volume and the window indices are bitwise those without the key (new test grid_and_gates/outer_shells_base_identity, 40 of 40), N and everything sized on it follow the extended grid, and a state written with shells carries them in its # grid line, so it loads only into a run with the same shells.

With the key absent every output is byte-identical to the control build (MEASURED on an LHS 1140 b evaluate and on `hydrostatic_column`).

**Item 5, the domain test (P1.4 (c)) and the outer closure** (`md/lhs1140b_outer_shells_domain_test_20260924.md`). The straddling molecular states on the 500-cell grid plus 43 shells to 60 R_p (spacing ratio continuous to 4e-4) both certify (13 and 10 passes). MEASURED: the wind turns transonic, with a sonic point at 40.3 R_p (Mach 0.51 at the old 29 R_p edge becomes 0.77); the EW falls by 1.74 per cent at both compositions (1.0964 to 1.0773, 1.1532 to 1.1330), log10 Mdot moves by 2e-4, and the profiles move by at most 0.13 per cent inside 2 R_p and by up to 20 to 30 per cent at the old edge. The EW crossing moves from He/H 1.4643 to 1.4886; with the matched kernel (unchanged, 22.2 km/s) both extended states lie below 1.108. The H2 carrier's last-cell defect stays two decades below the gate on both domains (5.6e-7 at 60 R_p): the zero diffusive outer face does not decide these states; the subsonic outflow boundary at 30 R_p holds the outer wind on a subsonic branch and acts through the hydrodynamics (tentative).

**Item 7, the composition update holding the pressure.** `Composition update holds: energy|pressure` (K15j, default `energy`): with `pressure`, the composition update of the alternation keeps rho, v and p of every cell, re-closes the eliminated species at the temperature of the new composition and rebuilds the energy from the caloric equation of state (`rebuild_energy_at_held_pressure`, `EXHALE_main.f90`), printing the energy it adds. MEASURED (`md/lhs1140b_composition_update_pressure_20260924.md`): it removes what it was built for, the base pressure step (cells 1 to 3 held exactly; the base face mass flux handed to the next solve 4.46e-8 against the wind's 4.44e-8, where the default hands -2.84e-5), and the next hydrodynamic solve still fails in all three tests: on the well-mixed He/H 0.083 and 0.55 cases on the energy row at the H2 front (the carrier update raises the cooling there by 31 to 44 per cent at the fixed wind, under either value), and on the K_zz = 1e9 rung 1.8 to 1.6, where the default recovers and certifies at pass 21, it fails from pass 2 on and never recovers. The obstacle behind the base step is the front's energy balance under a composition change at fixed wind. The default is unchanged; the option stays, default off, with its negative result recorded. With the key absent every output is byte-identical to the control build (MEASURED on the P1.6 entry and on `hydrostatic_column`).

**Item 8, one route for every molecular catalog case.** `make_models.py` wrote `Restart intent: stationary equilibrate` for a molecular case; the certified molecular cases carry `stationary`, and were solved under it from an atomic-to-molecular conversion as well as from molecular continuations (He/H 2.13 and 9.7, runs of 2026-09-19, READ). So instead of a rule by the kind of seed (the recommendation), every molecular case now takes `stationary`: `make_models.py` writes it, with `Coupled carrier solve: False` and the base-grid comment line the certified cases carry, and its output now equals the `input.inp` of every molecular K_zz = 1e9 and well-mixed case byte for byte (the two photochemical molecular cases differ only in the order of one line); `molecular_scalar_gj1132_kzz1e9/HeH0.083/input.inp`, the one case with `equilibrate`, was changed to match.

**Item 9, the `On stall` handover:** deferred.

**Item 10, the goldens.** Refreshed for the sixteen cases the matrix of section 12 ran (all but `oxygen_chemistry`), from the outputs of that matrix; the previous goldens are kept in `backup/regression/golden_pre_cfl_20260924/`.

**Item 11, `oxygen_chemistry`.** The case now starts from its own golden of 2026-09-20 (the 1000-step snapshot of the cold start, kept in its `IC/`) and marches 1000 steps from there (`Load IC? True`, `Restart intent: relaxation`; its `README.md` records why); the load passes the ghost fixed point and the run ends at du = 20.9, log10 Mdot 9.39

(MEASURED with the control build). Its golden was snapshotted from that run.

**The smaller items.** The Phase-E overlap window is entry (AH) of `md/TO_BE_DONE.md`, deferred to the next time the LHS 1140 b profile runs are handled. `docs/Update_EXHALE_stage2.{tex,pdf}` was regenerated for the edit of its section 17 (310 pages, no error). `src/utils/map_state_to_grid.py` wrote `nan` as the center shift in its `# mapped:` line between grids of different cell counts; it now says the shift is not defined there.

## 15. The grid of the crossing (decision item 3) and the regression of sections 13 and 14 (2026-09-24)

**Three grids, three compositions.** The certified molecular K_zz = 1e9 states at He/H 1.45, 1.52 and 1.6 were mapped onto grids of 1000 and 2000 cells (the base block and every other spacing halved at each step, the outer radius unchanged; target grids from cold-start dumps) and solved again with the rung environment; all six certify (1000 cells: 16, 15, 15 passes; 2000 cells: 13, 13, 19). MEASURED through the catalog's evaluation, synthesis and EW operator (`LHS1140b/models/.P1/refine_grid/`, `.P1/connectivity/grid_convergence.txt`):

| He/H | EW, 500 cells | 1000 cells | 2000 cells | continuum (DERIVED) | log10 Mdot 500 / 1000 / 2000 |
|---|---|---|---|---|---|
| 1.45 | 1.0964 | 1.0640 | 1.0475 | 1.030 | 7.85 / 7.84 / 7.84 |
| 1.52 | 1.1532 | 1.1196 | 1.1025 | 1.085 | 7.86 / 7.85 / 7.85 |
| 1.6 | 1.2155 | 1.1804 | 1.1625 | 1.144 | 7.86 / 7.86 / 7.86 |

The EW converges at first order (successive-difference ratio 1.96, p = 0.97, at every composition), and the continuum column is the Richardson extrapolation at that order. The crossing of the working target 1.108 moves from He/H 1.4643 (500 cells) to 1.5054 (1000), 1.5273 (2000) and 1.552 (continuum); from the broadened line of section 13, from 1.5112 to 1.5547, 1.5780 and about 1.60 (the last just above the 1.6 rung); every other value is interpolated between certified rungs. The matched kernel does not move with the resolution (22.18 to 22.21 km/s on all nine states). The 500-cell catalog overestimates the EW by about 6 per cent, twice the observation's working error; a formally second-order scheme converging at first order in the EW points to a first-order error in the line-forming region, whose source is not established. Combined with the domain factor of section 14 (DERIVED, the two measured separately), the crossing stands near 1.58 from the EW and near 1.63 from the broadened line. Which grid the catalog should use is left to the user with these numbers.

**The regression of sections 13 and 14.** After all the code of this series had landed (the diagnostic of item 4, the grid option of item 6, the composition option of item 7, the widened report fields, and the mapper record), the tree was rebuilt (`EXHALE.x` md5 `2df491cb020e8200be247e2469964a63`, no warning) and the whole matrix run against the goldens refreshed in section 14: REGRESSION PASS, all seventeen cases byte-identical (66 files; the two `_adv` files of `iontrans_metals` have no reference, as that route writes none), `oxygen_chemistry` included on its warm start.

## 16. The user's decisions of 2026-09-24 on the four remaining items

Put to the user after sections 14 and 15; decided the same day.

**The catalog's grid: 500 cells, the compositions quoted from the grid continuum.** The catalog keeps its 500-cell grid (every certified state, the chains and the ladders stay as they are); a composition read from it is quoted as the continuum value of section 15, with the difference between the 500-cell and the continuum value as the grid uncertainty of every catalog number: from the EW, He/H 1.55 (the 500-cell crossing is 1.46, 0.025 dex lower, because the 500-cell EW is about 6 per cent high), from the broadened line about 1.60 (1.51 on 500 cells); with the domain factor of section 14, 1.58 and 1.63 (DERIVED, the two effects measured separately). `docs/lhs1140b_catalog_state.tex` section 5 states the rule.

**The gating candidate: not adopted.** The species rows keep their current gate; the diagnostic `EXHALE_CERT_GATE_DIAGNOSTIC=1` stays as a default-off report.

**The pressure-holding composition update: kept, default off.** `Composition update holds: pressure` stays in the code with its negative result (section 14, item 7); the default remains `energy`.

**The energy imbalance at the H2 front: explored.** The obstacle item 7 exposed, the energy row at the H2 front failing after a composition change at fixed wind, is being diagnosed (`LHS1140b/models/.P1/h2_front_energy/`); its result is recorded in a later section.

## 17. The atomic-data audit of 2026-09-24, applied

`md/EXHALE_ATOMIC_DATA_AUDIT_2026-09-24.md` (an audit of the atomic data against the sibling code MoCHII) was checked item by item in the code and, where it could be obtained, in the original source, and applied; the four memos carry every verdict, source, size and command: `md/atomic_data_audit_applied_core_20260924.md` (the H/He cooling core), `md/atomic_data_audit_metal_tables_20260924.md` (the CHIANTI metal tables), `md/charge_exchange_detailed_balance_20260924.md` (the charge-exchange table) and `md/atomic_data_audit_applied_docs_20260924.md` (comments, unreachable code, the detailed-balance test). Every item the audit raised was found to be real; the corrections are physics corrections and are on by default.

**The H/He cooling core.**

- Recombination cooling of every ion is k T alpha (3/2 + dln alpha/dln T) on the coefficient the balance itself uses (B1, B7, B12). Hui & Gnedin (1997) print the He III fit with a factor 8 where the hydrogenic scaling gives 2: He III removed 3.05 kT per capture at 1e4 K and now 0.76 kT. The relation reproduces Hummer (1994) Table 1 to 0.1 per cent.
- The post-process temperature root and `eval_cool` now call one cell assembly, `radiative_cooling_of_cell` (B2); the two coolings differed by a factor 1.04-1.60 on the `wasp_full` post-process and now agree to 1.3e-14.
- **The 2^3S -> 1^1S electron-impact de-excitation (B3)**, the detailed-balance reverse of q13, is in every place the triplet balance is written, with the 19.82 eV returned to the gas. Oklopcic & Hirata (2018) leave it out as small against 2^3S -> 2^1S, 2^1P; that holds at 1e4 K (ratio 0.055) and not below: 0.15 at 5e3 K, 0.57 at 3e3 K, 2.8 at 2e3 K (MEASURED), the temperatures of the LHS 1140 b line-forming region.
- Voronov (1997) fits take Voronov's fitted dE again (B4: Mg II x1/1.47, C I x1/1.10 at 5e3 K); n = 2 recombination is the balance's case-B coefficient split with Draine's 2s fraction (B6, +2.6 per cent at 1e4 K); He I, He II (A3) and H I (B8) collisional-excitation cooling are CHIANTI 11.0.2 sums with the observed level energies (x1.11,

x1.29 and x1.11 at 1e4, 1e4 and 5e3 K), H I on one rate set shared with the n = 2 model; single-precision literals are gone from the four files (B10); `use_2lev_cool`, `lambda_2level`, `melem_Z`, `cool_FeII_func` and the other unreachable wrappers are removed (B11).

**The metal line cooling (A2).** Mg II took the CHIANTI `.scups` scaling energy (4.27 eV) as its transition energy where the observed g-weighted 4.4300 eV belongs, 1.15 times too high at 1e4 K and 1.39 at 5e3 K; it is refitted with observed energies and every bound channel (0.2 per cent). Ca II and Mg I carried the resonance line alone; they now take (T, n_e) statistical-equilibrium tables with every channel (at n_e = 1e9 cm^-3 the old coefficient was 1.7-4 times low for Ca II and 5.8-480 times low for Mg I over 1e4-3e3 K). Fe I stays on its Boltzmann (LTE-limit) table, now stated as an upper limit at most 2.0 times the low-density value; the data for a density-resolved Fe I solve are not available here.

**Charge exchange.** The reverse rows of Huang et al. (2023) Table 4 depart from detailed balance with internal partition functions by factors 0.18 to 2.9. Each measured or computed direction is kept as the original paper gives it, and the other is computed from it by detailed balance with NIST partition functions; four reverse rows are not carried (A2, A10, C4, C6), their forward channel ending in an excited or radiative product; three transcriptions are corrected to the originals (A12 Si2+ + H: 1.23e-9 and the sign of its exponent; C5: Zhao et al. 2004 eq. 19 in full; A15: Stancil et al. 1998 Table 1 directly). Ten pairs whose originals could not be obtained (mostly Physical Review A) stay as printed and are marked KNOWN_BLOCKED in the new test `src/tests/charge_exchange_detailed_balance`; among them N + H+ (its endothermic direction falls with T) and Na + H+ (both directions below 1e-60 cm^3 s^-1, so no Na-H charge exchange acts).

**Other corrections of the same series.** Comments that contradicted the code or the source (C1, C4 to C9, C11) are corrected; `rk_R21_H_Hep_cx`, `rk_R22_Hp_He_cx` and the three unreachable Lyman-Werner fits are removed after a call-graph search; the Lyman-Werner warning of `ionization_equilibrium.f90` and `write_output.f90` no longer cites a DB96 fit the band share has not used since 2026-09-06 (`NH2_db96_max` removed); `src/tests/a2_m1` compiles again against a built tree and its saved output was regenerated (the 0 K photolysis thresholds; `md/a2_reaction_audit.md` carries a correction block); the `Makefile` flag stamp now hashes the first line of `$(FC) --version`, whose conda-forge package build number `-dumpfullversion` drops: the previous `build/` held 35 objects of gcc 16.2.0-4 and 92 of 16.2.0-5 linked into one binary.

**Checks.** The tree was rebuilt from `make distclean` (`EXHALE.x` md5 `551dae520985d63c9ea8b15c21106589`, no warning; copied to `LHS1140b/models/EXHALE_551dae52.x` with `BINARY_MANIFEST_551dae520985.txt`); `physics_probe`, `charge_exchange_detailed_balance` (65 pass, 10 known blocked) and `charge_exchange_rows` pass on it. The regression matrix and the remaining suites are recorded in the next sections.

**Effect (MEASURED by the workers, on builds of the same physics).** `wasp_full`: log10 Mdot -2e-4 (cooling core), +4e-4 (charge exchange), -2.7e-3 (metal tables, a du-stopped case, so below its path spread); Fe III x0.48 at 1.2 R_p. LHS 1140 b: the He 10830 EW of the certified molecular states falls by 14.9 to 16.7 per cent on evaluation, almost all from B3, and by 20 per cent after one stationary re-solve (He/H 1.45: 1.0964 -> 0.8775 %A; 1.52: 1.1532 -> 0.9180), because n(2^3S)/n(He) falls by 41-58 per cent at 5-10 R_p, where n_e q31g is 3-100 times A31. No certificate of the catalog holds under the corrected physics. **Every LHS 1140 b number of this log before this section, and of `docs/lhs1140b_catalog_state.tex`, is of the physics before the audit.**

## 18. The H2-front energy imbalance, the H3+ non-LTE interpolant, and the descent below He/H 1.45 (2026-09-24)

**The H2 front (decision of section 16).** Diagnosed with the tree binary of section 15 and no change to the source (`md/lhs1140b_h2_front_energy_imbalance_20260924.md`, figure `docs/figures/lhs1140b_h2_front_energy_imbalance.pdf`). MEASURED: across the composition update one term of the front's energy row moves, the H3+ infrared cooling (99.6 per cent of the cooling there): on the well-mixed He/H 0.083 update H2 at the front rises 3.2 per cent, H3+ answers with +8.6 per cent (+17.5 with the pressure held), and the cooling of the spike cell rises by 31 (44) per cent against a heating that rises by 1 per cent. The code path is consistent (the next solve assembles the relaxation's cooling to 1e-7 and 3e-5). About two thirds of the jump came from the interpolant of the H3+ non-LTE factor (below). With the composition frozen a hydrodynamic root exists and the trust-region solve reaches it (||R|| 1.5e-7 and 1.6e-8), where the line search cannot take its first step; the well-mixed He/H 0.083 alternation still does not converge with the pressure held and the trust region (6 passes, the front walking outward, Mdot -26 per cent). The trust region inside the alternation is a solver-control change and was not made.

**The H3+ non-LTE factor is now a power law between the columns of Miller et al. (2013) Table 6** (`h3p_cooling.f90`, `s_table`). The emission per H3+ ion can grow at most as fast as the collision rate that populates the emitting levels, so the local index d ln s / d ln n(H2) lies between 0 and 1, and the table's node-to-node indices do (0.00 to 0.89). The interpolant linear in s over log n gave 12.6 just above the 1e8 cm^-3 column and 0.17 just below it at 2000 K (DERIVED), and over r = 1.25-1.40 R_p of the well-mixed He/H 0.083 state an H3+ cooling 1.8 to 5.6 times the power law's. In T the interpolant stays linear; its form there is not fixed by any bound and is written at the code with the row ratios. The correction is on by default and is in the binary of section 17.

**The descent below He/H 1.45 (decision item 2 of section 14)** ran under the P1.2 rules on `EXHALE_46d5b9f4.x` (the physics before section 17; `LHS1140b/models/.P1/descend_from_1.45_record.txt`): 1.152 from 1.45 not certified (18 passes, 3 h); 1.2924 from 1.45 certified (16 passes); 1.152 from 1.2924 not certified (24 passes, the H2 carrier row 3.1e-4 at cell 247); 1.2202 from 1.2924 not certified (24 passes, the He/H element row 8.6e-5 at cell 234 and falling by about 10 per cent a pass); the driver stopped on two consecutive failed refinements. The interval 0.55 to 1.29 stays unsampled, and the one new certified state is of the physics before section 17.

**The molecular family under the corrected physics (user decision of 2026-09-24: this family first).** The certified He/H 1.45, 1.52, 1.6, 1.8 and 2.13 states are being re-solved from themselves with `EXHALE_551dae52.x` (`LHS1140b/models/.P1/newphys/`, the rung environment, 24 passes, 4 h); their result, the new crossing and what follows for the rest of the catalog are recorded in a later section.

## 19. The second round of the atomic-data audit (2026-09-24/25)

The audit's author rechecked the applied tree from MoCHII (`~/MoCHII/MoCHII_v1.00/md/EXHALE_AUDIT_RECHECK_2026-09-24.md` and `..._SECOND_RECHECK_2026-09-24.md`): every item of section 17 was confirmed as applied correctly, and the open ones were listed. Each was verified in the code and the original source before it was applied; memos `md/atomic_data_audit_round2_core_20260924.md`, `md/charge_exchange_detailed_balance_20260924.md`

section 10 and `md/chianti_descaling_spline_20260924.md`.

**Recombination and the helium network.**

- The ground-state capture coefficient alpha_1 is the Milne relation on the photoionization cross section the transfer uses, for H I, He I and He II (0.998 of Hummer 1994 for H I, 1.006-1.011 of Hummer & Storey 1998 for He I); case B = Badnell alpha_A minus that alpha_1 (the Mao & Kaastra fit, 2.7 per cent low for He I at 1e4 K, is removed).
- The Oklopcic & Hirata (2018) alpha_1 is the capture to 1^1S alone; with the singlet/triplet coupling off the excited-singlet captures were missing. Case B is now split into triplet and singlet captures with Hummer & Storey (1998) Table 5 in both configurations.
- **H ground captures escape where the gas is thin** (new key `H_rec_escape`, default True, K14d2 of `md/input_schema.md`): alpha_HII = alpha_B + y_HI alpha_1 with the local escape weight of the He II construction, case A in the thin outer wind and case B in the thick base; `False` is case B everywhere. A correction of the case-B assumption where it does not hold, hence on by default.
- The He III recombination cascade (ground capture, direct capture to n = 2, He II Ly-alpha, the two-photon continuum of Nussbaumer & Schmutz 1984, the 2s-2p mixing of He II) is absorbed on the spot under `He_rec_coupling`.
- The collisional feed of 2^3S through the higher triplets (feed/q13 = 0.018, 0.101, 0.328 at 5e3, 1e4, 2e4 K, CHIANTI 11.0.2) is added to q13 wherever the triplet balance reads it, and so are the 2^3S -> n^1L (n >= 3) excitations.
- A(2s -> 1s) = 8.2249 s^-1 (Nussbaumer & Schmutz 1984; was 8.26); the 2s-2p l-mixing of H by protons, He+ and He2+ from Pengelly & Seaton (1964) with their own cut-off rule (the Lamb shift or fine structure for 2s, not the Debye radius); the 2^3S electron-impact ionization as Black (1981) eq. 12 prints it; `coll_rate_prefactor` = 8.629132e-6 from CODATA, one parameter; the 2^3S triplet cooling is a CHIANTI sum (Black's 13179 K was his fitted threshold, not an energy); the (T, n_e) cooling tables above their top density continue with the last slope clipped to [-1, 0]; q31g is recorded in the diagnostic states (state 28 of `stage_channels.txt`).

**Energy terms that were missing.** The metal recombination energy (k T alpha (3/2 + dln alpha/dln T)) and the metal collisional-ionization energy (potential times the Voronov rate) are in the cooling; the charge-exchange energy defects heat the gas (new heat channel 20, `heat_charge_exchange`: per reaction the defect less the energy radiated by an excited product, product states from the sources read; the ledger closes to 1.4e-8 eV an event). On `wasp_full` the charge-exchange heat is 18 per cent of the heating at the base, 3.9 at 1.1 R_p, 0.4 at 1.3 R_p; the base share is mostly Fe + H+, whose product state could not be read (Rutherford & Vroom 1972) and is taken as ground, an upper bound.

**Charge exchange.** N + H+ / N+ + H are now Kingdon & Ferland (1996) Table 1 (Butler & Dalgarno 1979) with the reverse by detailed balance (the printed Huang rows were 69 and 1.5e7 times low at 1e4 K); Na + H+, Na+ + H and K + H+ are not carried (no reading of the printed Table 4 form gives a physical rate; the originals, Dutta et al. 2001 and Watanabe et al. 2002, could not be obtained); D26 and D30 are derived by detailed balance from the source directions; the gate now also requires every carried rate to obey the threshold law (205 pass, 6 known blocked, 2 known defects).

**CHIANTI descaling.** CHIANTI prescribes the natural cubic spline through the `.scups` points (`descale_scups.pro`,

CHIANTI Technical Reports 4 and 13); the scripts of `cooling_data/` used the not-a-knot spline (as ChiantiPy). Every CHIANTI-based coefficient was regenerated with the natural spline: He I cooling moves by -2 to +0.5 per cent over 3e3-3e4 K (Upsilon(1^1S-2^1P) by -16 per cent at 1e4 K), H I by +-0.7 per cent, O I fine-structure Upsilon by +1.3 to +6.7 per cent, most others below 1 per cent.

**Other repairs of the series.** The Lyman-Werner band share the H2O/OH continua see is now summed cell by cell along the path at each cell's own (T, n_H) (it was one uniform slab at the local T: the band's photon ledger was 18 per cent short on the warm-started `oxygen_chemistry` state; only `Oxygen chemistry: True` runs reach it); six test suites were repaired (`md/test_suite_repairs_20260924.md`).

**Build and checks.** Tree rebuilt from `make distclean`: `EXHALE.x` md5 `b059a4be0459d65fdc3dff9894ba5d2f`, no warning, copied to `LHS1140b/models/EXHALE_b059a4be.x` with `BINARY_MANIFEST_b059a4be0459.txt`. All 42 suites run on it (MEASURED): 38 pass; `grid_and_gates` fails its three `stationary_evaluate_*` rows on the pinned fixture `backup/regression/wasp_full_newton/IC/` (solved under the old physics; a re-solve under this physics certifies, `info = 0`, ||R|| 2.3e-9, and waits in `LHS1140b/models/.P1/suite_repairs/wfn_IC_r2/` for the golden refresh); `residual_determinism` fails its known closure-spread row (1.17e4, the solver item open since 2026-09-18); the two `steady_*` suites need the log of a finished steady solve as input.

**Effect (MEASURED by the workers).** Of the core items alone: `wasp_full` log10 Mdot 13.35 -> 13.32, the H ionization front outward (x(H+) 0.36 lower at 1.27 R_p), T up to +11 per cent; `mol_metals` -0.004. Of the charge-exchange items: `wasp_full` +4.2e-3, N II down by up to 1.8 dex. On the LHS 1140 b He/H 1.45 state the core items lower the He 10830 EW by 7 per cent on evaluation, mostly through the triplet/singlet split of case B (0.041 %A), `H_rec_escape` (0.020) and the Milne alpha_1 (0.011).

**The LHS 1140 b molecular family between the two rounds** (binary `EXHALE_551dae52.x`, `LHS1140b/models/.P1/newphys/`): re-solved from the certified states, no rung certified in 24 passes (the H2 carrier row at the front falling by 0.72-0.86 a pass), then all five certified in 7-11 more passes. Their EWs (MEASURED, evaluate + synthesis + the EW operator): He/H 1.45 0.9813, 1.52 1.0185, 1.6 1.0587, 1.8 1.1500, 2.13 1.2765 %A; the 1.108 crossing near He/H 1.70. These are of the physics of section 17 only; the states under this section's physics are being solved from them.

## 20. The molecular K_zz = 1e9 family under the audited physics (2026-09-25)

The user's decision of 2026-09-24 was to re-solve this family first. The five states He/H 1.45, 1.52, 1.6, 1.8 and 2.13 were re-solved from the states certified under the first round (section 19) with `EXHALE_b059a4be.x` (`LHS1140b/models/.P1/newphys/HeH<x>_r2/`, the rung environment, `resolve.sh`): all five certified in 13 passes (the H2 carrier row at the front falling from 0.5 to 6e-6 - 9e-6). Each was then evaluated, synthesized and measured with the EW operator (`evaluate_ew.sh`, `eval_r2/`; every evaluation certified). MEASURED:

| He/H | EW, audit round 2 | round 1 | before the audit | log10 Mdot, round 2 |
|---|---|---|---|---|
| 1.45 | 0.9387 | 0.9813 | 1.0964 | 7.899 |
| 1.52 | 0.9776 | 1.0185 | 1.1532 | 7.906 |
| 1.6 | 1.0198 | 1.0587 | 1.2155 | 7.913 |

| He/H | EW, audit round 2 | round 1 | before the audit | log10 Mdot, round 2 |
|---|---|---|---|---|
| 1.8 | 1.1159 | 1.1500 | 1.3584 | 7.927 |
| 2.13 | 1.2497 | 1.2765 | 1.5604 | 7.945 |

The 1.108 crossing of the log-log chord moves from He/H 1.4643 (before the audit) to **1.7833, between the certified 1.6 and 1.8**, on the 500-cell grid; the EW falls by 14 to 20 per cent at fixed He/H and log10 Mdot rises by about 0.05. The grid continuum factor and the domain factor of sections 15 and 14 were measured under the old physics and are not re-measured here. Figure `docs/figures/lhs1140b_kzz1e9_audit_stages.pdf` (`.P1/newphys/compare_physics_stages.py`): the EW against He/H at the three stages, the line at He/H 1.8, and T, v, n(2^3S), x(H+) and x(H2) at He/H 1.8. At He/H 1.8 (MEASURED on the _adv profiles, before the audit / round 1 / round 2): the line keeps its shape and loses depth; the peak temperature is 5439 / 5315 / 5600 K; the velocity at the outer edge 1.21 / 1.40 / 1.36 km/s; x(H+) at 20 R_p 0.059 / 0.056 / 0.050 (the ground captures escape in the thin far wind); the metastable density at its peak (1.35-1.38 R_p) 33.3 / 33.3 / 31.8 cm^-3, and its radial integral over 1-1.5, 1.5-3 and 3-30 R_p 11.1 / 10.9 / 10.2, 28.8 / 28.3 / 27.6 and 28.4 / 24.2 / 23.4 cm^-3 R_p. The loss is in the cool outer wind beyond 3 R_p (-18 per cent), where the superelastic 2^3S -> 1^1S collisions outrun the radiative decay, and the EW falls by 18 per cent with it.

## 21. The code-size appendix moves to this log (2026-09-25)

On the user's instruction the hand-written code-size appendix `docs/Update_EXHALE_appendix.tex` is now input by this log's typeset twin instead of the stage 2 twin (`src/utils/update_log_to_tex.py`, `STAGES`); `docs/Update_EXHALE_stage2.{tex,pdf}` were regenerated without it. Its references to "this log" now name the stage 2 log where they meant it, and it states that its numbers (measured 2026-09-22) do not yet include the code of stage 3; they are re-measured with `src/utils/codesize.py` when the atomic-data series is closed.

## 22. The originals supplied on 2026-09-25, read and applied

The user supplied the papers the two audit rounds could not obtain; each was read and the code checked against it (memos `md/charge_exchange_detailed_balance_20260924.md` sections 11 and 11.8, `md/atomic_sources_hhe_20260925.md`).

**Charge exchange.**

- Na + H+ (Dutta et al. 2001, non-radiative, product H(n=2), endothermic by 1.682 eV) and Na, K + H+ (Watanabe et al. 2002, radiative): neither paper gives a fit and Huang et al.'s Table 4 forms are in neither; the rate is now the Maxwellian average of Dutta's cross sections (their own rate figure violates the threshold law below about 5000 K) plus Watanabe's radiative values: 8.4e-12 at 1e4 K against 4.4e-171 as coded before. The reverse is not carried (it starts from H(n=2)). K + H+ from Watanabe.
- Fe + H+, O+, N+ (Rutherford & Vroom 1972 Table I): the 300-1200 K values in log-log, held above 1200 K; the measured cross sections start at 8 eV, so every wind rate is an extrapolation, bracketed by 1.4e-9 and 4.0e-9 at

1e4 K (DERIVED; the held value is the source's own extrapolation to 6 per cent; Pequignot & Aldrovandi 1986 infer 2.7e-9). The product state is not determined in the paper; the near-resonance reading (Fe+ z4F/z4D, 1.18 eV of heat an event instead of the full 5.70 eV defect) is the one Pequignot & Aldrovandi's Table 3 also takes; the range is stated at the code.

- S + H+ (Zhao et al. 2005 Table III), Si + He+ (Satta et al. 2013 eq. 8, Table 1), S + C+ (Chenel et al. 2010 Table 2), Na + Ca+ (Kwolek et al. 2019, measured at 0.015-0.09 eV, the Langevin value kept as an upper bound) replace the Table 4 transcriptions; the reverses of channels whose forward ends in an excited product (Fe rows, Si + He+, Na + Ca+) are not carried. C + Si+ (D1/D2) stays blocked: the paper Table 4 cites for it does not contain it. The detailed-balance gate: 194 pass, 0 fail, 1 known blocked.
- Effect: `wasp_full` log10 Mdot -3e-4; the charge-exchange heat at the base falls from 18 to 6.7 per cent of the heating; Na I D2 transit depth 1.169 -> 1.156 per cent.

**Hydrogen and helium.**

- The He I collision strengths of the metastable network (q13, q31g, q31a, q31b) were exponential fits the ATES code carried, not coefficients of Lampon et al. (2020), who take the values from Bray et al. (2000). Bray tabulates 5.6e3-5.6e5 K; the fits passed through the first four table values and were an unsupported extrapolation below 5.6e3 K (1.10-1.16 times the new strengths at 1.5e3 K). Now: Bray's table inside its range (PCHIP in log-log; one misprint of Table 2 corrected), the CHIANTI he_1 temperature dependence joined continuously outside it; the feed through the higher triplets, the 2^3S cooling, the 2^3S -> n^1L excitations and the He I cooling sums were rebuilt on Bray. On the LHS 1140 b He/H 1.8 state the He 10830 EW rises by 2.3 per cent on evaluation.
- H 2s-2p mixing by electrons: the coded rate (Janev et al. 2003, from Chibisov 1969) was ten times below Seaton's (1955) first-order dipole constant; now Seaton (1955) eq. (55), approximation V (5.78e-5 cm^3 s^-1 at 1e4 K against his Table 1 5.7e-5; Osterbrock & Ferland Table 4.10 agree).
- A(2s -> 1s) = 8.22461 s^-1 (H) and 526.823 s^-1 (He II), Drake (1986) eq. (27) (section 19 wrote 8.2249, the Nussbaumer & Schmutz 1984 value it replaced).
- The case-B share of recombinations into H 2s is now Pengelly's (1964, Table I) hydrogenic fraction, the one the He II branch already used; the quadratic in T that section 17 (item B6) called "Draine's 2s fraction" is Christie et al.'s (2013) fit, and Draine (2011) gives only Tables 14.2 and 14.3.
- The atomic-mode (no 2^3S) He I case-B photons are now sent at their own exits (584 A, 19.8 eV, two-photon) split by the network's rates, instead of one averaged energy (3.10 instead of 4.12 eV of heat a capture at high density; z = 0.966 and 0.660 at low and high density against Draine's 0.96 and 0.67).
- Voronov (1997): all 20 rows equal the published Table I; its stated range (1 eV-20 keV) is written at the code with what the fit does below it. Anderson et al. (2000) confirm the CHIANTI H I strengths to 2.5 per cent at 0.5-5 eV; Yan & Babb (2023) published values equal the arXiv ones; Dere et al. (1997) and Del Zanna et al. (2015) say nothing about the descaling spline, which rests on the CHIANTI IDL code.

**Build.** Tree rebuilt from `make distclean`: `EXHALE.x` md5 `50aa18d79bcd25f219b5b59baa31b8b8`, no warning, copied to `LHS1140b/models/EXHALE_50aa18d7.x` with its manifest. The manifests of 2026-09-24/25 listed files as changed that were not (the header lines of the previous manifest matched the file-name search); the three lists were rebuilt from

the md5 lines of consecutive manifests. The refined-grid runs of the round-2 physics that were still running (He/H 1.6 on 2000 cells, 2.13 on 1000 and 2000) were stopped as superseded; the molecular family, the regression fixture, the regression matrix and the suites are being re-run on this binary.

**The texts the corrections made stale** (`md/stale_texts_20260925.md`): the physics document (`docs/physics_overview/*.tex`, the PDF rebuilt, 70 pages), `docs/transmission_spectrum.{tex,md,pdf}` and `md/atomic_data_EXHALE_vs_MoCHII.md` were brought to the code. The transit post-processor built the n = 2 populations of the Balmer lines with Christie et al.'s rates while claiming to mirror the solver; it now uses the solver's rates, one transcribed function each (agreement with the Fortran 3.1e-13, MEASURED), and keeps only its own Ly-alpha and Balmer fields (which the profiles do not carry). H-alpha peak depth of HD 209458 b -1.73 per cent, of `wasp_full` +0.14 per cent (MEASURED), almost all from the proton l-mixing, which lowers n(2s)/n(2) at 1.3-3 R_p from 0.53-0.69 to 0.26-0.34. The n = 2 threshold cross section 1.4e-17 cm^2 is Seaton (1959) at the n = 2 edge (1.378e-17 DERIVED), not Osterbrock & Ferland.

## 23. The Markdown documents move to md/ (2026-09-25)

On the user's instruction the 350 Markdown files directly under `docs/` moved to the new top-level `md/`; `docs/` keeps the LaTeX and PDF documents, the figures and the dated audit directories, whose 13 Markdown files stayed with the files they describe. Done with `mv`, not `git mv`: git sees the 35 tracked files as deleted under `docs/` and new under `md/`. Paths that followed:

- `.gitignore`: the rule that keeps the development record off the public repository (`docs/*.md` with 35 exceptions) is now `md/*.md` with the same 35 exceptions under `md/`.
- 73 relative links of 24 moved files into `docs/` subdirectories now read `../docs/...`.
- 3079 references `docs/<name>.md` in 416 maintained files (sources and their comments, tests, scripts, `README.md`, `README_HOWTO.md`, the `md/` files themselves, the `docs/` LaTeX, the regression README and scripts) now read `md/<name>.md`; four of them are strings the binary prints (the setup report, one lower-atmosphere message, one test message), so the tree binary changed (`EXHALE.x` md5 `b30e56f63335431dcb22b9fe3f80f948`, no warning); the state files the regression compares carry none of them.
- `src/utils/update_log_to_tex.py` reads the logs from `md/`.
- Left as they were: run records and products (inputs, outputs, state files, `REPRODUCE.md`, logs, the `.P1` run areas, archived regression snapshots), whose text records the paths of their day and whose bytes other records hash; and the one string `LHS1140b/models/make_models.py` writes into the 94 stored catalog inputs, so that it still reproduces them byte for byte.
- The workspace `CLAUDE.md` and the session memory point to `md/`.
- The 17 `docs/` documents whose text names these paths were rebuilt on the user's instruction (the two log twins regenerated first): every build without error or undefined reference, the new paths in the PDFs.

## 24. The last items of the MoCHII comparison, and the molecular crossing under the corrected physics (2026-09-25/26)

`md/EXHALE_MOCHII_ATOMIC_COMPARISON_2026-09-25.md` (a side-by-side of the atomic processes of EXHALE and MoCHII, written from the MoCHII side) found the recombination construction, the recombination cooling, the He 2^3S network and the H 2s-2p mixing now the same in both codes to four digits at 1e4 K, and listed five open items for EXHALE; each was verified and applied (memos `md/charge_exchange_detailed_balance_20260924.md` section 12 and `md/photoionization_584_auger_20260925.md`).

- **N2+ + H -> N+ + H+** (Barragan et al. 2006 Table 2, 9.70e-10 cm^3 s^-1 at 1e4 K, the product N+ 2s2p^3 3D0, so no reverse; heat 4.57 eV an event; `metals.inp: cx_N2p_H <scale>`, default 1). On `wasp_full` N III falls by up to 5.4 decades.
- **He2+ + H -> He+ + H+ + photon** (Kingdon & Ferland 1996 Table 1, 1.0e-14, radiative, no reverse and no heat), in every system that carries He III; He III falls by 0.69 dex at the `wasp_full` base. The original of the rate (West, Lane & Cohen 1982, Phys. Rev. A 26, 3164) could not be obtained.
- **Density-dependent detailed balance.** Where the source names the product level, the reverse is the thermal reverse times the population-weighted factor of the metastable levels at the local (T, n_e) (five-level statistical equilibrium of N I, S II, C I from CHIANTI 11.0.2), and the heat of the forward channel removes the forbidden-line energy a metastable product radiates at low density; elsewhere the thermal reverse is exact at every density. `wasp_full` (n_e 8e8-3.5e9) moves by less than 0.2 per cent. A 0/0 below 60 K in the metastable Boltzmann weights (one LHS 1140 b base cell sits at 40.8 K) was guarded.
- **He I 584 A.** The line is resonantly scattered by He0 inside the cell, and each scattering ends in absorption by H I (or H2, or a metal ion), in conversion to 2^1S through the 2.06 um branch (1/912) and the existing two-photon exit, or in escape (Wood, Mathis & Ercolano 2004 sect. 4.1, eqs. 15-17, generalized to every continuum absorber of the cell and to escape); their eq. 17 carries He/H = 0.1 in its coefficient.
- **Inner-shell photoionization.** Each subshell absorption heats with the photoelectron plus the Auger energy, the fluorescence and the further ionization energy are not heat, and the neutrals of C, N, O, Mg, Si, Ca and Fe are carried straight to X++ where the vacancy autoionizes (Kaastra & Mewe 1993 Tables 2 and 3; Ca II 3s is treated as radiative, since the measured energies forbid its autoionization). At 98 eV the electrons of a Mg II absorption receive 0.333 of the photon energy instead of 0.847.

**Build.** `EXHALE.x` md5 `f4da21f29579d87370efec2a4220f4c4`, no warning, copied to `LHS1140b/models/EXHALE_f4da21f2.x` with its manifest. On LHS 1140 b the two He items move the He 10830 EW by +0.05 per cent (He/H 1.8), but a state of the previous physics does not certify under them, so the molecular states are re-solved on this binary.

**The molecular K_zz = 1e9 family under the physics of section 22** (`EXHALE_50aa18d7.x`; `.P1/newphys/`, `.P1/refine_grid_r3/`). He/H 1.8 and 2.13 re-solved from the round-2 states; 1.6 and 1.52 as rungs from 1.8 (1.45 failed its rung: the H2 carrier row held at 1e-1 through four passes). MEASURED, every evaluation certified except the one marked:

| He/H | EW, 500 cells | 1000 cells | 2000 cells | continuum (DERIVED) |
|---|---|---|---|---|
| 1.52 | 1.0038 | | | |
| 1.6 | 1.0475 | 1.0187 | 1.0039 | 0.988 |
| 1.8 | 1.1468 | 1.1162 | 1.1003 (a) | 1.083 |
| 2.13 | 1.2853 | 1.2516 | 1.2343 | 1.216 |
| crossing of 1.108 | 1.721 | 1.783 | 1.819 | 1.860 |

(a) the re-evaluation of the certified state refuses the base mass row by a factor 1.9 (1.4e-8 against 7.4e-9); its EW is quoted for reference.

The EW converges at first order with the grid (successive-difference ratios 1.93-1.95, p = 0.95-0.97), as under the physics before the audit, and the crossing moves from He/H 1.46 (500 cells) and 1.55 (continuum) before the audit to 1.72 and 1.86. The 60 R_p domain factor of section 14 was measured before the audit and is not applied here.

**The documents brought to the code** (`md/user_manual_update_20260926.md`): `docs/EXHALE_user_manual.tex` (rebuilt, 103 pages, no error or undefined reference, every changed page looked at), `README.md` and `README_HOWTO.md` now describe the input keys (`H_rec_escape`, `He_rec_coupling`, the `metals.inp` keys `cx_N2p_H`, `cx_O2p_H`, `cx_full`), the 24 columns of `Heating_breakdown.txt` (charge-exchange heat in column 24), the `stage_channels.txt` layout and the atomic data of sections 17-24. Found in the code and fixed: `H_rec_escape` was parsed but missing from the list of recognized keys (a spurious "unrecognized input line" warning, and a second statement not refused); the missing-`metals.inp` message promised built-in abundances that are zero; comments said `cx_full` toggles the He-H pair and that O2+/N2+ + H are dropped; the `stage_channels.txt` header omitted the metal charge-exchange channels 27-32. `md/input_schema.md` K14e (still "Koskinen 2013") and its `metals.inp` table (no group-E keys) were brought up to date. Not changed: `H_rec_escape` and `He_rec_coupling`, like the other physics switches of their group, are neither echoed in the setup report nor restart option tokens.

**Checks of the binary of this section** (`EXHALE_f4da21f2.x`). The suites: `physics_probe` did not compile one new probe (`inner_shell_photoabsorption_ledger` called `cx_set_cell` without the electron density the density-dependent reverses added; the two items were written at the same time); fixed, and the suite passes. `grid_and_gates` fails its fixture rows plus `outer_iteration_ending_is_the_stagnation_one` and `restart_lower_ghost_is_the_derived_boundary` (7.9e-14 against 1e-14), all on the pinned fixture of the previous physics; `residual_determinism` keeps its known row; the two `steady_*` suites need a solve log. The regression matrix ran all 17 cases; every one differs from the goldens by the physics (`wasp_full` log10 Mdot and the base inflow: 214 inflow cells against 257). **The user held the golden refresh (2026-09-26).** The stationary re-solve of the fixture from the previous one did not converge (info = 2: the energy row 3.1e-2 at cell 406, the closure of `System_HeH_TR_metals` 4.6e-3 at cell 340), while the matrix case `wasp_full_newton` (marching, then Newton) certified (info = 0, ||R|| 4.4e-9); that output is the candidate fixture. The tree was rebuilt after the comment and message corrections of this section (`EXHALE.x` md5 `0a7532d5087f14df41142fb499b68240`; it differs from `EXHALE_f4da21f2.x` in printed strings and comments only).

**The molecular states on `EXHALE_f4da21f2.x`** (`.P1/newphys/HeH<x>_r4`, re-solved from the certified states of the

previous physics; certified in 9, 9, 8 and 7 passes; every evaluation certified). He 10830 EW, 500 cells: He/H 1.52 1.0047, 1.6 1.0483, 1.8 1.1477, 2.13 1.2862 (+0.06-0.09 per cent against the table above); the 1.108 crossing 1.7194 (1.7213 before). The refined grids were not re-solved on this binary; the crossing's grid factor above stands within the size of this change.

## 25. The helium-hydrogen charge exchange from its originals (2026-09-26)

The He2+ + H original the user supplied (West, Lane & Cohen 1982) and an analysis made on the MoCHII side of the He+ + H exchange (`~/MoCHII/MoCHII_v1.00/md/heii_h_charge_exchange/`) were checked against the published papers and applied (`md/charge_exchange_detailed_balance_20260924.md` section 12.9). Every rate below is DERIVED from the cited cross sections unless marked READ.

- **He2+ + H -> He+(1s) + H+ + photon.** The 1.0e-14 cmˆ3 sˆ-1 of Kingdon & Ferland (1996) is not a result of any calculation: it is the value they set for every reaction without a significant channel (READ, their sect. 2). Now the Maxwellian average of the West et al. (1982) radiative cross sections (Figs. 4-6, with the resolved resonances): 2.0e-13 at 1e3 K, 1.65e-13 at 1e4 K, 1.57e-13 at 3e4 K, 16-20 times the old value; a calculation above 1e3 K, an extrapolation below.
- **He+ + H -> He + H+, non-radiative.** The Kingdon & Ferland fit reproduces the singlet-conditional cross sections of Zygelman et al. (1989) without the 1/4 of the spin statistics, and those cross sections are themselves hundreds of times those of Kimura et al. (1993) and Loreau et al. (2014). Now the Loreau et al. (2014) H+ + He cross section by microscopic reversibility (weights 4:1, threshold 13.65 eV/u, zero below 15 eV/u): 4.4e-17 at 1e4 K instead of 9.0e-15, 1.5 per cent of the radiative channel; the band at the code (up to 2.9x at 1e4 K from the threshold treatment; 0.4-2.5x above 3e4 K). The reverse He + H+ (B1) follows by detailed balance (5.1e-22 at 1e4 K instead of 1.0e-19); the heat of an event is unchanged.
- **He+ + H, radiative.** Stancil, Lepp & Dalgarno (1998) Table 1 row 19 prints 1.20e-15 (READ); the code carried the 1.25e-15 of Glover & Jappsen (2007). Now 1.20e-15.

Effect (MEASURED): `wasp_full` log10 Mdot -0.002 (inside the marching spread), He III down to 0.94 dex at the base, He II and He 2ˆ3S +48 and +52 per cent at 1.1 R_p, the charge-exchange share of the heating 5.6 -> 0.9 per cent at 1.05 R_p. **LHS 1140 b He/H 1.8 (evaluation of the certified state): the old He+ + H exchange supplied up to 10.7 per cent of the heating; without it the peak temperature falls from about 5600 to 5150 K, He II and He 2ˆ3S rise by 14 and 15 per cent at 1.5 R_p, and the He 10830 EW rises from 1.1477 to 1.1818 %A (+3.0 per cent).** The certificate does not hold under the corrected rates (energy row 8.4e-2), so the molecular states are re-solved on the binary of this section (`EXHALE.x` md5 `59c6be36820b8ba5c958c294ad9e6f02`, copied to `LHS1140b/models/EXHALE_59c6be36.x` with its manifest).

**The molecular states re-solved on `EXHALE_59c6be36.x`** (`.P1/newphys/HeH<x>_r5`, from the certified states of section 24): He/H 1.6, 1.8 and 2.13 certified in 25, 23 and 21 passes, every evaluation certified. He 10830 EW, 500 cells (MEASURED): 1.6 1.0577, 1.8 1.1550, 2.13 1.2902 (+0.9, +0.6 and +0.3 per cent against section 24; the +3.0 per cent of the evaluation at fixed wind is mostly given back by the solved wind); the 1.108 crossing moves from 1.7194 to 1.7026 (DERIVED). He/H 1.52 stopped in its second pass: a JFNK trial iterate with ||R|| near 2 left a lower

ghost whose base H2 handoff partition did not close in 30 passes (last move 2.0e-6 against 1e-10), and the guard of `ionization_equilibrium.f90` (a boundary nothing stands behind is refused) ends the run with `error stop` even on a trial state. Whether a trial should instead be refused and the step shortened is a solver-control question put to the user.

## 26. A trial state no longer stops the run at an unclosed lower ghost (2026-09-26)

User decision: as recommended (`md/ghost_closure_trial_refusal_20260926.md`). The guard of `ionization_equilibrium.f90` that stops the run when the lower ghost's base H2 handoff partition does not close against its own ionization balance (tolerance 1e-10, 30 passes) now distinguishes a state a caller can refuse from one it cannot: the stationary JFNK's trial evaluations (`eval_residual` of `steady_newton.f90` when it screens admissibility: line search, trust region, Levenberg-Marquardt and pseudo-time trials, the finite-difference probes) and the relaxation's trial chemistry (`equilibrate_chemistry_at_fixed_conserved_state`) declare it before the sweep, and there an unclosed ghost refuses the trial (logged as `CANDIDATE REFUSED`, counted in the ledger, refusal code 7, closure reason 5), the solver then shortening the step as for its other refusals; an accepted state, the state written, the marching sweep and every other path keep the stop. The same rule now covers the ghost-composition fixed point one step later. MEASURED: the He/H 1.52 re-solve of section 25 passes the trial that stopped it (the lambda = 1 trial of JFNK solve 3 iteration 7; the halved trials close) and certifies at pass 24; an evaluate of He/H 1.6 and a copy of `mol_carrier` are byte-identical to the build before (`EXHALE_59c6be36.x`) in every data line; the suites pass, the two failing ones (`residual_determinism`, `grid_and_gates` fixture rows) failing identically on the build before. `EXHALE.x` md5 `7db5afe9b1d80ce757fcc07d1b499330`, copied to `LHS1140b/models/EXHALE_7db5afe9.x` with its manifest.

The He/H 1.52 state on this physics: EW 1.0149 %A (certified evaluation); with 1.6, 1.8 and 2.13 of section 25 the 500-cell ladder is 1.0149, 1.0577, 1.1550, 1.2902 and the crossing of 1.108 stays at 1.7026.

**The grid of the crossing on this physics** (`.P1/refine_grid_r5/`, binary `EXHALE_59c6be36.x`, the 1000-cell He/H 1.8 and 2.13 continued on `EXHALE_7db5afe9.x`, identical physics; every state and evaluation certified). He 10830 EW, MEASURED; continuum DERIVED (Richardson at the measured order):

| He/H | 500 cells | 1000 cells | 2000 cells | continuum |
|---|---|---|---|---|
| 1.6 | 1.0577 | 1.0297 | 1.0150 | 0.999 |
| 1.8 | 1.1550 | 1.1249 | 1.1094 | 1.093 |
| 2.13 | 1.2902 | 1.2571 | 1.2403 | 1.223 |
| crossing of 1.108 | 1.703 | 1.764 | 1.797 | 1.838 |

First order again (successive-difference ratios 1.91-1.96, p = 0.93-0.97). Against the physics before the audit (1.464 on 500 cells, 1.552 in the continuum) the crossing has moved to 1.703 and 1.838; the continuum value is interpolated between the certified continuum points of 1.8 and 2.13.

## 27. The molecular family under the corrected physics: catalog publication and the width of the line (2026-09-26)

**Catalog.** The certified He/H 1.52, 1.6, 1.8 and 2.13 states of sections 25 and 26 were published into `LHS1140b/models/molecular_scalar_gj1132_kzz1e9/` (`run_case.sh`, `SEED=` the certified run, `EXHALE_7db5afe9.x`): every re- measurement "original claim REPRODUCED", every evaluation certified; EW 1.0149, 1.0577, 1.1550, 1.2902 %A, log10 Mdot 7.89, 7.90, 7.91, 7.93. The catalog's He/H 0.55, 1.45 and 9.7 molecular states are of the physics before the audit and were not re-solved.

**The added Gaussian** (`.P1/connectivity/broaden_molecular_final.py`, the operations of section 13; output `broaden_molecular_final.txt`, figure `docs/figures/lhs1140b_kzz1e9_molecular_broadened_final.pdf`). The memo check (atomic well-mixed He/H 0.40) reproduces f = 22.22 km/s. MEASURED: the matched kernel is 22.10-22.19 km/s FWHM on every state and grid (sigma 9.39-9.42 km/s), and at it the red depth, blue depth and ratio of the He/H 1.8 state lie inside the observed 1 sigma (red 1.289, blue 0.170, ratio 7.60). The broadened EW (MEASURED on each grid, continuum DERIVED):

| He/H | 500 cells | 1000 cells | 2000 cells | continuum |
|---|---|---|---|---|
| 1.6 | 1.0227 | 0.9955 | 0.9813 | 0.966 |
| 1.8 | 1.1167 | 1.0876 | 1.0726 | 1.057 |
| 2.13 | 1.2474 | 1.2155 | 1.1992 | 1.182 |
| crossing of 1.108 | 1.781 | 1.851 | 1.890 | 1.933 |

As turbulence the kernel is supersonic: on the He/H 1.6 and 1.8 states the line-forming region (n(2^3S) r weighting) has T = 2650 and 2756 K, mu = 1.94 and 2.03, c_s = 4.33 km/s, so sigma = 9.4 km/s is Mach 2.2 as a line-of-sight dispersion, 3.1 as sqrt(2) sigma and 3.8 as an isotropic speed, as before the audit. From the EW alone the crossing is 1.703 (500 cells) and 1.838 (continuum); from the broadened line 1.781 and 1.933.

## 28. The catalog summary brought to the tree's state, and why the molecular winds do not certify (2026-09-26)

`docs/lhs1140b_catalog_state.tex` (13 pages, rebuilt, pages 5 to 8 checked by eye):

- The certified table carries the molecular K_zz = 1e9 rows the catalog now publishes, READ from each case's `run.log` and `models/status.py`: 1.52, 1.6, 1.8 and 2.13 under `EXHALE_7db5afe9.x` (note a); 1.45 under `EXHALE_b059a4be.x`, the second round of corrections, EW 0.9387 (note b; under `EXHALE_50aa18d7.x` neither its re-solve nor its rung from 1.52 certified within the time ceiling, and it was not attempted under the binary of record); 9.7 (note c, below). The rows of the table before this entry were the pre-audit values (1.6: EW 1.2155, now 1.0577).
- A new subsection of section 3, "Why the molecular winds do not certify", assembled from `md/lhs1140b_stationary_L34c_20260918.md`, `md/lhs1140b_alternation_audit_20260921.md`, `md/lhs1140b_coupled_trial_20260921.md` and `md/TO_BE_DONE.md` L22: the composition half of the

partitioned route refuses (the H2 carrier row in all six cases), by two measured mechanisms (the composition update takes the wind out of solution in the well-mixed cases; a drift under a saturated movement bound in `kzz1e9/HeH0.083`); the coupled route fails in its Krylov solve; the open item is a preconditioner that holds the species rows. Every one of those measurements predates the atomic-data corrections and none was repeated under `EXHALE_7db5afe9.x`.

- The two molecular cases that published no state (`molecular_scalar_gj1132_wellmixed/HeH2.13`, `molecular_photochem_gj1132_kzzprofile/HeH2.09`) are now named.

Found and fixed: `LHS1140b/models/status.py` printed the `not_solved.md` reason beside `molecular_scalar_gj1132_kzz1e9/HeH9.7`, which is certified. The certified generation `g0008_20260920T052941Z_c3c466a4` is the 2026-09-20 evaluation of the state written on 2026-09-19; the `not_solved.md` is that of a re-solve made later the same day (`g0006_20260920T000506Z_5b2061aa`, `composition_refusal`, the H2 carrier row 5.13e-4 at cell 262 after 40 passes), which did not replace it. A certified row with a `not_solved.md` beside it now reads "a later re-solve, not the state shown: ..." (MEASURED on the table output). `MODELS.md` was not rewritten.

## 29. Errors fixed and models solved first: the plan of 2026-09-26

User instruction (2026-09-26): re-checking solutions that move little against the observational error is not the priority; fixing errors and solving the models that do not solve is. Plan `md/PLAN_20260926.md`; the user approved its recommendations (goldens and fixture refreshed; the composition-side solver work reopened if continuation does not close the cases; `diffusion_check` taken out of the model count). New physics only where it can change results seriously; the rest is recorded, not applied.

### 29.1 An OpenMP data race in the metal sweep (fixed)

`meg_g02` (the inner-shell Auger share of neutral photoionization to X++, added with binary f4da21f2 on 2026-09-25) was missing from the private list of the He + metals cell sweep (`ionization_equilibrium.f90`, then ~2137, a `default(shared)` loop), so threads overwrote each other's value between the write and the two reads. MEASURED: route-B stationary restart of `wasp_full_newton` failed at 8 threads (composition elimination never contracts, `info = 2`) and certified at 1 thread; two 8-thread runs of one input differed before the solve started (`md/metal_sweep_openmp_race_20260926.md`). Every metals-on run at more than one thread since f4da21f2 was affected; the regression matrix (1 thread) and every metals-on catalog certificate (last published 2026-09-20) were not. Fix: `meg_g02` private, and all four cell loops of `ioniz_eq` converted to `default(none)` with explicit shared lists; an audit of every parallel region of `radiation/` and `lower_atmosphere/` found one more, formal, race (two table-ready flags read without synchronization; now `omp atomic read/write seq_cst`). New suite `src/tests/thread_count_identity/` (200 steps of `wasp_full` at 1 and 8 threads, data lines bitwise): 7db5afe9 FAIL (3.2e-3 Hydro, 5.4e-2 Ion), fixed binary PASS. The r0 case (route B, 8 threads) now certifies, `info = 0`, ||R|| 6.50e-9. No 1-thread run moved.

### 29.2 Fix batch (errors confirmed in the source)

F1 free-free Gaunt factor: van Hoof et al. (2014) eq. (21) uses the infinite-mass Rydberg (READ, their sect. 2), so Ry/k = 157887.51 K, not 157807 (the Hui & Gnedin 1997 constant, kept in that fit); F2 the carrier diffusivity read by outputs and diagnostics was zero on a stationary path (MEASURED D_H2 = 0 at every cell; now 7.9e6 - 8.2e9 cm^2/s); F3 the determinism replay now passes u to `cell_row_scales`; F4 pressure formed before the composition refresh at 3 of 9 sites (coupled-block entry, after `solve_steady_*`, trial end; the other 6 verified not stale); F5 the CO column, dissociation rate and shielding factor joined the step checkpoint; F6 an inert `Coupled carrier solve` is reported; F7 five wrong comments; F8 the `_adv` headers carry every provenance field; F10 seven callerless `Cool_coeff` wrappers removed and the Voronov H/He fit defined once (rates bitwise unchanged). Effect on the regression inputs (single thread): F1 alone, at most 1.4e-4 relative (`wasp_full` velocity), log10 Mdot unchanged to 4 digits.

### 29.3 The interdiffusion enthalpy flux (new physics, serious)

The energy equation of a diffusing mixture carries q_d = sum_i h_i J_i (Cook 2009, Phys. Fluids 21, 055109, eqs. 11-13; Schunk 1977, Rev. Geophys. Space Phys. 15, 429, eqs. 15b, 16, 20c; READ). EXHALE moved He relative to H and carried no such term in either energy equation. MEASURED size on LHS 1140 b He_diffusion states: 13-26 % of the local energy budget at 1.1-5 R_p (`md/interdiffusion_enthalpy_flux_20260926.md`). Implemented in both routes (binary model: q_d = J_He (h_He - h_H), sensible enthalpies with the electrons each ion carries; formation energy verified to be carried already); key `Interdiffusion enthalpy flux:` default True with `He_diffusion` (the term belongs to the equation), False only to match published models that omit it (Koskinen 2013, 2022; Yelle 2004); restart token `interdiff_enth` (older states restart under `Restart option change: interdiff_enth`). Suite `src/tests/interdiffusion_enthalpy_flux/` 18/18. Effect on the molecular He/H 1.8 state (24 + 24 passes, carrier row 1.6e-5 at the end, not yet certified): T -166 K median over 1.3-2 R_p (largest -236 K, -4.4 %), He/H -3.1 %, the n(2^3S) column -1.3 %, log10 Mdot +0.002.

### 29.4 He+ + H non-radiative charge exchange near threshold

Section 25 took the Loreau et al. (2014) H+ + He cross section forward by reciprocity with sigma_r = 0 below 15 eV/u, where their calculation starts. That closes an open channel: the forward (exothermic) rate then fell as exp(-E/kT) at low T (5e-34 at 300 K). Adopted, as in MoCHII (the user's decision relayed by the MoCHII session), sigma_r(x) = sigma_r(15) sqrt((x - x_th)/(15 - x_th)) between x_th = 13.655 eV/u and 15 eV/u, i.e. sigma_f v constant at low energy (ion-induced-dipole capture with a constant transfer probability; an assumption no calculation tests). Rate: 9.64e-17 at 1e4 K (was 4.41e-17), 9.56e-17 at 1e3 K (was 1.5e-21); 3.3 % of the radiative channel at 1e4 K. Generator `cooling_data/heii_h_charge_exchange_rate.py` (reproduces the old table to 0.4 % with the old cut).

The radiative channel, the user asked, is one coefficient in both codes: Stancil, Lepp & Dalgarno (1998) Table 1 row (19), 1.20e-15 (T/300)^0.25, the Zygelman et al. (1989) Table I DIRECT column times the 1/4 of the singlet approach (READ; 1.17e-15 at 300 K from the original). MoCHII moved from the Glover & Jappsen (2007) R26 1.25e-15, which cites the same data and cannot be reproduced from it (2026-09-26); EXHALE unchanged. A digitization of Courtney et al. (2021) Fig. 3 agrees with this fit within 2 % over 1e3 - 1e4 K and lies 1.9x above it at 100 K; a fit to their tabulated

rate needs the data file (University of Georgia Molecular Opacity Project, not reachable from here).

### 29.5 Goldens, fixture, suites

Goldens re-snapshotted from the tree binary 4e275f1d (the physics of 7db5afe9) after a full matrix (all 17 cases differed from the round-3 goldens, as expected); backup `golden_pre_round45_20260926`. Fixture `wasp_full_newton/IC` replaced by the 1-thread route-B certified state of 7db5afe9 (backup `IC_pre_round45_20260926`). Every suite on the integrated binary: all pass except grid_and_gates (3 rows: the fixture is not a root of the fixed physics, to be re-made with the final binary), residual_determinism (known row), the two suites that need run logs.
Closing the series (2026-09-27): the matrix on the final binary 6844f188 (sections 29.1-29.4, 29.7, 29.9) differs from the goldens of 7db5afe9 in every case beyond 1e-3, largest in the helium columns of the lower cells (3.7 % at row 1 of `wasp_full`, the He+ + H rate at low T) and in the molecular cases (the ghost start, the progress control); goldens refreshed from it (backup `golden_pre_final_20260927`). Fixture re-solved on 6844f188 at 8 threads (`info = 0`, ||R|| 4.11e-9; backup `IC_pre_final_20260927`); `grid_and_gates` 345/345. Every suite passes except the known `residual_determinism` row and the two suites that need run logs.

### 29.6 Models solved

MEASURED, `.P1/s1_remeasure/`, `.P1/s2_continuation/`: every failure of the catalog's unsolved molecular and atomic cases stands under 7db5afe9 (six passes each); continuation then solved:

- `molecular_scalar_gj1132_kzz1e9/HeH1.45`: rung from the certified 1.52, certified in 13 passes.
- `molecular_scalar_gj1132_wellmixed/HeH2.13`, 1.8, 1.4: the well-mixed molecular cases differ from the K_zz = 1e9 ones ONLY by `He_diffusion` off (their `base.inp` carries `Kzz_base 1.0e9`, which the carrier transport reads). From the certified kzz1e9/2.13 state, He/H set uniform (new `map_state_to_grid.py --uniform`) and diffusion off: certified in 10 passes; then rungs 1.8 (10 passes) and 1.4 (14 passes) with `make_rung.sh` (`UNIFORM=1`). The direct re-solve of the catalog's well-mixed 2.13 state is refused by the progress control at pass 5, so the continuation is what solves it.
- `molecular_photochem_gj1132_kzzprofile/HeH9`: on the final binary of the series (2dd9a865, face at the level), mapped from its 40-pass state on 6844f188 (`.P1/s2_continuation/photo9_f`, row 7.4e-5 falling 0.77 a pass) and continued 40 passes: certified (H2 carrier row 9.2e-6 at cell 217), `.P1/s2_continuation/photo9_v4` (2026-09-27). Its earlier stops were the lower-ghost closure defect of section 29.9.
- Ionization-stage transport on the molecular 1.8 state stalls on the He++ row under the progress control; the cause is the control (29.7).

### 29.7 A progress-control defect of the composition relaxation

Diagnosed (`.P1/iontr_diag/`): each pass of the carrier relaxation ends on its particle-count cap, the state moves a fixed amount a pass, the relative gated measure stays flat because residual and scale fall together, and the outer control reads that as no progress, halves the cap and ends the loop. With the cap held (diagnostic override) the same state certified in 14 passes.

Fixed (EXHALE_main.f90 progress control, binary d4b571e8): a pass whose consumed carrier update ended on its movement bound, whose solve converged and whose joint distance did not rise by more than 1.1x is not judged by the relative measures and does not halve the bound; the bound doubles after such a pass up to 0.5 and halves when a bound-limited update raises the distance of a converged wind or precedes the first failed solve; omega follows the same rule and no longer drops on hydro entry failures; EXHALE_CARRIER_TRUST_HOLD still freezes the bound. Also fixed: the scale x_ref of the carrier fixed-point test was the H2 maximum alone, so with transported stages and no molecules it sat at its 1e-30 floor and the test could never pass (logged displacement ~1.7e28, MEASURED on iontrans_metals). MEASURED: the hot-Uranus stagnation fixture certifies at pass 42 (it stagnated; the suite row that needs that ending now holds the bound); iontrans_metals accepts at pass 17 instead of 48, the two certified states within 7.6e-5; the molecular He/H 1.8 state with transported stages certifies in 14 passes from a starting bound of 0.5 (EXHALE_CARRIER_TRUST=0.5) but not from the default 1e-2, where the hydro solve fails at pass 2 under the old control and the new one alike; with the interdiffusion flux on, that state's hydro solve fails on entry and a slow mode on the He front (0.86-0.93 a pass) remains. REGRESSION on kzz1e9/HeH0.083: a growing bound makes the H2 row at the outer cells 498-500 overshoot (up to 1.5e-1) and the bound saws; neither control certifies that case (the old one stalls on the elemental row at cell 338). mol_carrier bitwise unchanged (the outer loop is not entered). Open: the starting bound for transported stages, the H2 outer-cell overshoot, the hydro entry failures with the interdiffusion flux.

### 29.8 Incidents of this series

- A sed insertion with an empty line address put a three-line comment before every line of src/tests/boundary_state/molecular_seed_of_an_atomic_state.sh; the 153 blocks were removed and the file checked by diff against the afternoon copy (identical except the intended line).
- The tree binary was rebuilt while a regression matrix was running on it; the matrix was stopped and is re-run after the series.

### 29.9 The lower-ghost H2 closure stalled with a zero move (fixed)

The molecular photochemical-column cases (He/H 9 and 2.09) stopped at passes 1-3 with "the base handoff partition of a lower ghost did not close": the closure's own move exactly zero and its residual 9-41 times the 1e-10 tolerance. Diagnosed (.P1/ghost_stop/): each closure pass after the first started the molecular solve from the previous pass's accepted state, whose x(4) held the OLD imposed H2 partition; the pinned row x(4) - x_h2_fix then entered hybrd1 carrying the whole secant move, and hybrd1 (xtol = sqrt(eps), diag = 1) can return a move below 2 xtol ||x|| ~ 3e-8 untouched with info = 1 (hybrd.f90: first delta = pnorm, halved on the rejected step, then delta <= xtol xnorm). The write-back then imposed the value, the next pass's map saw no move, and 28 identical solves followed (27 of 27 traced events; 41 of 41 refusals and stops in the S1 logs, residuals 8.2e-10 - 2.8e-8, all below that floor). Fix: impose_transported_fractions sets every unknown a transport solve owns (H2 partition, oxygen carriers, ionization stages) to its imposed value before both the molecular and the atomic hybrd1 solves; the stop report prints the pin error. MEASURED with the fix (worker build): 0 of 127092 ghost closures unclosed, pin error exactly 0, no refusal and no stop; the He/H 9 case then fails on its H2 carrier row (0.88), a separate problem. Binary 6844f188. Every molecular base-handoff regression case is reached (ghost composition moves by 1e-12 - 1e-9).

### 29.10 The _adv post-process made second order

User decision (2026-09-27): the catalog quotes the He 10830 EW of the _adv post-process. That post-process was a first-order upwind recursion in r (backward Euler of dx/dr = R/v with the rates at r_j and the residence time at r_{j-1}), the main source of the first-order grid convergence of the EW (`md/ew_grid_order_20260926.md`). Now the variable-step BDF2, y_j - a1 y_{j-1} + a2 y_{j-2} = g h_j F(y_j, r_j) with w = h_j/h_{j-1}, a1 = (1+w)^2/(1+2w), a2 = w^2/(1+2w), g = (1+w)/(1+2w) (derived at `post_process_adv.f90`, coefficients as eq. (1.4) of Li & Liao, arXiv:2201.00527; L-stable, so a population that equilibrates within a cell does not ring as under the trapezoid rule), for every composition system and the energy equation, rates and 1/v at the same point; backward Euler on the first step, after a refused cell, for w > 1 + sqrt(2), and as a retake of a step with a negative population or no temperature root (0 retakes on the nine states). MEASURED on the nine certified grid states (He/H 1.6, 1.8, 2.13 on 500, 1000, 2000 cells), with the state held fixed: the post-process share of the EW error falls from +3.0 % / +1.0 % (p = 1.00) to below 3e-4, and the _adv 2^3S column converges at p = 1.98 - 2.14 (1.00 - 1.01 before); a closed-form column test gives p = 1.989. The EW itself changes by -3.9 % (500 cells) to -0.97 % (2000 cells) at He/H 1.6 (1.05692 -> 1.01571 on 500 cells) and still converges at first order: what remains is the solved state's first-order base closure (being fixed) and the transit integration (p ~ 1.4, opposite sign).

Also fixed: the h div(rho v) term of the _adv energy equation was built from differences of the cell-centred rho v r^2, which carries the odd-even pattern of the cell-centred velocity at the base (1.8e12 to 3.4e12 between neighbours while the face mass flux changed by 1e-10), giving 805 / 4008 / 579 K at the first corrected cell on 500 / 1000 / 2000 cells; it is now the state's own mass row, the face-flux divergence. Below about 1.13 R_p the molecule-free post-process lacks the molecular heating and H3+ cooling that set the run's energy balance there; by the user's decision (2026-09-27) the run T is kept there with status 3 "unsupported" wherever the omitted heating and cooling exceed the carried ones (196 / 389 / 776 cells on 500 / 1000 / 2000 cells, up to 1.134 - 1.137 R_p; EW change at most 1.2e-4 relative); the user manual's status table says so. Binary ca651768. Suites: adv_static_limit 53, transit_state 12, transit_census 21, physics_probe 2199, thread_count_identity 9, all pass. Every _adv golden moves (5 short cases: T up to 5.9 %, n(2^3S) up to 9.4 %). The catalog EWs predate this change by -1 % to -4 %. Found, not changed: `thermal_damkohler_number` is the net rate |H - Cl/u, near zero at a balanced cell, not the relaxation rate its comment describes. Docs: the transit tool now says a stationary SOLVE writes no _adv products and its evaluation does; the project CLAUDE.md says the matrix compares the _adv pair as well.

### 29.11 The lower face at the reservoir level

The reservoir (the `base.inp` handoff or the Photochem profile's matching level) is stated at r = 1, but the domain's lower face sat at r_edg(0) = 1 + dr/2, and `base_boundary.f90` carried the reservoir pressure up to that face along the reservoir's own isentrope: a half-cell slab of 226 K gas whose thickness scales with dr lay between the level and the domain (MEASURED: the lower layer offset by -2.2 % / -1.15 % in density on 500 / 1000 cells, `md/ew_grid_order_20260926.md` section 3). Now the face is the level in every grid mode (Uniform, Stretched, Mixed: `define_grid.f90`, r(0) = 2 - r(1), refusals if a ghost reaches r <= 0 or the face is not 1), the face state is the stated reservoir with no transport along any isentrope, the ghosts hold the reservoir continued below the level, and the Pho-

tochem matching level is the face on every grid; boundary model id `..._face_at_level_v4`. MEASURED on the molecular He/H 1.8 state, 500 / 1000 / 2000 cells, all certified: the pressure gap between the interior at the level and the reservoir falls from 2.93 / 1.44 / 0.70 % to 0.114 / 0.067 / 0.038 %; the lower-layer density offset falls 20-40x; the wind quantities stay first order with errors 1.6-1.9x smaller (log10 Mdot 7.9147 / 7.9113 / 7.9094 -> 7.9116 / 7.9098 / 7.9086; `_adv` EW 1.1548 / 1.1248 / 1.1091 -> 1.1482 / 1.1214 / 1.1074 %A, measured before the `_adv` change of 29.10). The remaining first-order error is likely the thermal layer at the contact, which the grid does not resolve (cell-1 T 800 / 695 / 604 K against the 226 K reservoir, 13 % lower at every halving; not shown to drive the wind). Two other first-order terms were found and left: the element-diffusion Dirichlet He/H is imposed at the cell-1 center, dr/2 above the level (`binary_element_diffusion.f90` 946-959; the He/H change near the base is 1.5e-4), and the conduction bath at the ghost center r(0) (`viscous_conduction.f90` 485; conduction off in these runs). Every cell center moves by dr/2, so the grid guard refuses every state written before: such a state loads only as a seed made with `map_state_to_grid.py --ic` onto the new grid (a 1-step cold run gives the target grid). The regression fixtures and goldens are migrated in the same series (MEASURED, 2026-09-27: every fixture of the default matrix and every fixture a suite reads mapped with `map_state_to_grid.py --ic` onto a 1-step cold-run grid; `wasp_full_newton/IC` re-solved and certified, info = 0, ||R|| 1.43e-9, and `iontrans_metals/IC` a copy of it; the `hp_*` seeds, whose recipe from `mol_base_handoff` is now refused by the `carrier` restart token, mapped from their previous pairs with the seeds kept (x(H II) 0 and 1e-12); `jfnk_hd189`, a named case outside the matrix whose legacy pair has no `# columns` header, not migrated; the two suites that read catalog states now map them into their own scratch; every suite passes, and the standing `residual_determinism` row now passes (5.9e-2 against < 1); goldens refreshed from the matrix run on 2dd9a865, backup `golden_pre_facelevel_20260927`); the LHS 1140 b catalog states stay records of the previous boundary (their EWs move by -0.6 % at 500 cells, below the observational error). Binary 2dd9a865.

### 29.12 A restartable state after every outer pass

The stationary route wrote its state only when the run ended, so a run stopped from outside (a wall-clock `timeout`) lost every pass it had made: MEASURED on 2026-09-28, three LHS 1140 b continuations lost 35, 13 and 9 passes. Now each outer pass that does not end the iteration writes the state the next pass starts from, `output/Hydro_ioniz_last_pass.txt` and `output/Ion_species_last_pass.txt`, overwritten each pass, by the same writer and headers as the final state files, rebuilt from (u, f_sp) in the order of `rebuild_state_from_checkpoint` with every side effect on the caloric mixture undone; both files carry `certified=F cert_reason=pass_snapshot_p<n>` and a `# pass_snapshot` line (the movement bound and omega of that pass, which a restart does not inherit: pass them by `EXHALE_CARRIER_TRUST` / `EXHALE_DIFF_OMEGA`), so a pair mixed from two passes is refused by `load_IC` and no tool reads a snapshot as certified. Each file is written to `<name>.part` and renamed. MEASURED: 0.03 - 0.05 s a pass on 500 cells (passes of 90 - 135 s); a run killed during pass 3 left the complete pass-2 pair, which restarted and continued as the uninterrupted run did; the final state files of a normal run are bitwise those of the previous binary. `LHS1140b/models/.P1/s2_continuation/config_rung.sh` continues from the pair when the parent has no final state. Binary a7255a03. The regression matrix on it, strict (`REGRESSION_REL_TOL=0`): all 17 cases byte-identical to the goldens of 2dd9a865 (MEASURED).

### 29.13 The unsolved molecular cases by continuation (state of 2026-09-28 16:30)

All runs under `LHS1140b/models/.P1/s2_continuation/` (scripts `config_rung.sh`, `make_rung.sh` in `.P1/`, `chain.sh`, `facelevel_continue.sh`; env as the catalog: `EXHALE_PTC_DTAU0=1.0`, 8 threads). Certified states (MEASURED, `certified=T`, the JFNK ||R|| of the last solve 6.5e-9 - 2.9e-8):

| family | He/H | passes | binary | note |
|---|---|---|---|---|
| kzz1e9 | 1.45 | 14 | 7db5afe9 | rung from 1.52 |
| kzz1e9 | 1.30 | 5 (after 24) | 7db5afe9 | |
| kzz1e9 | 1.223 | 9 (after 40 + 40) | 2dd9a865 | moved to the face-at-level grid |
| kzz1e9 | 1.186 | 24 (after 30) | 2dd9a865 | |
| kzz1e9 | 1.168 | 20 | 2dd9a865 | |
| kzz1e9 | 1.15 | 46 | a7255a03 | carrier bound held at 1e-2 |
| wellmixed | 2.13, 1.8, 1.4, 1.1, 0.85, 0.7603, 0.68, 0.6116 | 10 - 21 each | 7db5afe9 | path finding, previous grid |
| wellmixed | 0.6116, 0.55, 0.4 | 16, 18, 17 | 2dd9a865 | on the face-at-level grid |
| photochem kzzprofile | 9 | 41 | 2dd9a865 | after 40 passes on 6844f188 |

Not solved yet: kzz1e9 below 1.15 (1.12 running, gated row falling to 1e-4 then flat at 1.3e-4 over passes 14-16); wellmixed 0.3347 and below; photochem kzzprofile 2.09 (gated row oscillating 0.08 - 0.54, hydro info = 2 on most passes, about 40 min a pass); kzz1e9 0.083 and the wellmixed 0.083 are the far ends of the two chains.
What was learned on the way (MEASURED):

- The well-mixed molecular cases are the kzz1e9 ones with `He_diffusion` off (their `base.inp` carries `Kzz_base 1.0e9`, which the carrier transport reads); a kzz1e9 state with He/H set uniform (`map_state_to_grid.py --uniform`) certifies with diffusion off in 10 passes, which is how the family was entered.
- Runs stopped by a wall-clock `timeout` lost every pass (35, 13 and 9 passes), which section 29.12 fixed for every later run; and `config_rung.sh` ran under `set -e`, so a run that timed out never wrote its `EXIT.txt` and a waiting continuation never started (fixed).
- The growing carrier bound of section 29.7 drives a sawtooth in the H2 cases below He/H ~1.2 (kzz1e9 1.15: the gated row between 5e-3 and 2e-1 over 35 passes with hydro info alternating 0/2) and in the well-mixed 0.3347 rung (between 2e-4 and 1e-2, the refusing cell moving between the front near cell 357 and the outer cell 500). Holding the bound at 1e-2 (`EXHALE_CARRIER_TRUST_HOLD=1e-2`) certified kzz1e9 1.15 in 46 passes, and 1.12 then fell 2.6x a pass to 1e-4; it did not stop the well-mixed 0.3347 oscillation (1e-3 - 2.7e-3 over 19 passes). The rule that grows the bound is therefore not right for the H2 carrier; an open item.

Documents brought to the code of this series (2026-09-28): the user manual (input keys `Interdiffusion enthalpy flux`, `Outer shells`, `Composition update holds`, `H2 double ionization`, `H2 neutral dissociation` and the retired `Oxygen transport` added from `known_keys`; the `_last_pass` output files; the face at the level; the BDF2 post-process; the progress control; the thread-count note; stale statements corrected, among them the tokens a `Restart option change:` line may not name, four not six), the physics and algorithms document (the inter-diffusion enthalpy flux in the energy equations, the lower face, the grid constructions, the He+ + H pair, Ry/k, the post-process step, the progress control and the snapshot; stale statements corrected, among them the default of `H2 double ionization`, `chung80`, and the claim that every extension defaults to off) and `README.md`. Two source comments corrected (`base_boundary.f90` model id, `steady_residual.f90` energy row scale; comments only), and `src/tests/interdiffusion_enthalpy_flux/run.sh` made executable: `make test` had skipped it silently.

## 30. The base contact held through each steady solve, and the Intel build (2026-09-28/29)

### 30.1 OpenMP thread count: measured, and the default team

MEASURED on lart2 (2 x Xeon Gold 6154, 36 cores, 72 hardware threads), `EXHALE_a7255a03.x` (gfortran 16.2, -O3), one run at a time; A is `wasp_full` marching for 300 steps, B one stationary outer pass of the well-mixed molecular He/H 0.3347 state (500 cells, not converged). Records: `LHS1140b/models/.P1/omp_scaling_lart2/`.

| threads | A [s] | A speed-up | B [s] | B speed-up |
|---|---|---|---|---|
| 1 | 201.7 | 1.00 | 1535.6 | 1.00 |
| 8 | 47.4 | 4.26 | 234.8 | 6.54 |
| 18 | 34.7 | 5.81 | 126.2 | 12.17 |
| 36 | 32.2 | 6.26 | 87.8 | 17.49 |
| 72 | 41.8 | 4.83 | 89.7 | 17.12 |

The output is bitwise the same at every thread count. The second hardware thread of a core adds nothing (72 against 36), so the default team when `OMP_NUM_THREADS` is unset is now the number of physical cores the process may run on (`init.f90`, `physical_core_count`: the processors of `Cpus_allowed_list` grouped by their `thread_siblings_list`; where Linux does not provide the lists, every processor counts), in place of min(cores, 16), a 2026-07 marching measurement. Checked on lart2: 36 with nothing set, 4 under `taskset -c 0-3`, 1 under `0,36`, 5 with `OMP_NUM_THREADS=5`. `README_HOWTO.md` updated. The rung scripts take the team from `NTHREADS` (default 8).

### 30.2 The Intel compiler

lart2 has no classic ifort: in oneAPI 2026.1.1 and 2025.3.1 `ifort` is a link to `ifx`. The ifx build refused `diffusive_photochemistry.f90`: the derived types `carrier_module_state`, `carrier_checkpoint` and `thermochemical_state` were given the PUBLIC attribute twice (the module's `public ::` list and the `type, public ::` statement), which the standard forbids and gfortran accepted; the `, public` of the type statements was removed. The regression case `mol_carrier` stayed bitwise identical (REGRESSION_REL_TOL=0). The Intel build now records the oneAPI compiler and MKL libraries in its rpath (`Makefile`), so it runs under the `env -i` of the run scripts.

MEASURED on lart2 against the gfortran build of the same source (cases as in 30.1; B normalized by the closure evaluations, because the three builds take 27, 67 and 38 JFNK iterations from the same state):

| threads | A gfortran / ifx [s] | B gfortran / ifx [ms per closure evaluation] |
|---|---|---|
| 1 | 201.4 / 104.6 | 1.81 / 0.99 |
| 8 | 47.3 / 27.1 | 0.275 / 0.147 |
| 18 | 34.6 / 20.5 | 0.151 / 0.078 |
| 36 | 34.5 / 19.2 | 0.107 / 0.054 |

ifx is 1.7 to 2.0 times faster; `-march=native` gains gfortran at most 10 per cent. The hydrodynamic columns agree within 3.4e-9 and the species within 1.3e-4 (trace neutral metals) of the gfortran output. By the user's instruction every calculation started on lart3 and lart4 from 2026-09-28 uses the ifx build (`EXHALE_b037d586.x`, manifest `BINARY_MANIFEST_b037d5869f76.txt`); the goldens stay gfortran snapshots.

### 30.3 Why the molecular continuations alternated for 60 to 80 passes

The well-mixed He/H 0.3169 rung (`.P1/s2_continuation/wm8_0.3169`) ran 80 passes with the hydrodynamic solve converging on every other pass and the carrier update reversing itself each pass (cosine of consecutive x(H2) increments -0.990 to -1.000, amplitude 6.3e-3 at the H2 front). The stage exports show that the wind itself alternated between two base states: the converged one with the reservoir owning the level (lower ghosts at 232 K) and the failed one with the interior owning it (ghosts at 1200-1212 K). The upwind side of the base contact is decided at every evaluation from the spread d_window of the wind-window mass flux (reservoir where d_window <= 2e-3) and otherwise from the sign of the matched face velocity (`base_boundary.f90`). MEASURED over three stalled runs:

| run | solves that converged returned | solves that failed returned |
|---|---|---|
| wm8 0.3169, passes 60-79 | d_window 1.355e-4, ghost 232 K | d_window 2.000e-3 to 2.25e-3, ghost 1200-1212 K |
| wm6 0.3347, passes 47-60 | 1.33e-4, 232 K | 2.00e-3 to 2.35e-3, 1222-1236 K |
| kzz7 1.09, passes 30-43 | 8.4e-5, 234 K | 2.00e-3 to 4.07e-3, 1136-1413 K |

Several failed solves stopped at d_window = 2.000e-3 to four digits: the iterate was held at the threshold, where the residual steps. The composition update of the default contract moves the base pressure, the first steps of the next solve send a mass-flux pulse through the window, d_window crosses 2e-3 and the base goes over to the interior side; the root is on the reservoir side, across the step. The code comments stated that the threshold stands far from every operating point; that holds for written states and not for the iterates of a solve, and the comments now say so. The rule that grows the carrier bound (TO_BE_DONE (g)) harmed the H2 cases for the same reason, as far as these records show: a larger composition step makes a larger pulse.

Tests from the pass-80 state of wm8_0.3169 (ifx, carrier bound held at 3e-3, MEASURED):

- T1, the reservoir side held for every solve by a measurement setting (`EXHALE_BASE_WIND_SPREAD=1e3`): certified in 6 passes, every solve info = 0, ghosts at 232 K throughout; continued under the unmodified rule it certified

again in 2 passes (wm9_0.3169_spread_check6), the two states within 1e-3 of each other.

- T2, Composition update holds: pressure alone: the solves of passes 2-4 failed on the reservoir side (the line-search stagnation recorded in md/lhs1140b_h2_front_energy_imbalance_20260924.md) and the run stopped on its no-progress rule. Not a remedy here.

### 30.4 The side of the contact is held through each steady solve

base_boundary.f90 / EXHALE_main.f90. Before each steady solve of the stationary route the side of the contact is fixed by the first evaluation of the solve (its entry state) and held for every residual and Jacobian evaluation inside it (hold_base_contact_direction_from_entry, release_base_contact_direction). After the solve the rule is applied to the returned state (base_contact_reverses_in_state); where a converged solve returns a state whose rule selects the other side, one more solve from it holds that side. A root is therefore accepted only on the side its own rule selects, as before; only the path to it changes. The marching route and every evaluation outside a steady solve read the rule at every evaluation. EXHALE_BASE_CONTACT_PER_EVALUATION=1 restores the old reading inside the solves (a control). Each solve logs the side it held, the number of evaluations, and at how many of them the rule read the other side. Documented in docs/physics_overview/boundaries_initial.tex.
MEASURED (gfortran build d9047b32, the pass-80 state of wm8_0.3169):

- H1, the code as changed: certified in 6 passes with the same worst rows as T1 (3.41e-4, 7.37e-4, 6.44e-4, 7.71e-5, 2.37e-5, 1.65e-6); in passes 2 to 5 the rule read the interior side at 822 of 2702, 850 of 3014, 259 of 2741 and 109 of 2341 evaluations, in pass 6 at none.
- H0, the same build with EXHALE_BASE_CONTACT_PER_EVALUATION=1: passes 1-6 alternated info = 0, 2, 0, 2, 0, 2, the old cycle.

Regression (the full default matrix of 17 cases, gfortran d9047b32 against the goldens of 2dd9a865, MEASURED 2026-09-29): 16 cases bitwise identical in all four files; wasp_full_newton, the one case whose steady solve met the contact threshold, agrees within 1.7e-9 of every column's own maximum. Its first solve held the interior side (the rule read the reservoir side at 5365 of 5968 evaluations inside it), the returned state's rule read the reservoir side, and the second solve, held there, returned the old root. The harness flagged one entry of Hydro_ioniz_adv.txt, the rounding-level residual column adv_mass_row (0 against 3.1e-15 in a column whose largest value is 7e-3), as a relative difference of 1; it is not a difference of the state. Goldens are not refreshed (the user's decision: later).

### 30.5 An output argument that was never set

hydrodynamic_rows_body.inc: hydrostatic_equilibrium_of_each_cell_fv declared pL_asm and pR_asm INTENT(OUT) and never set them (ifx warning 6843); the caller forms both itself before any use, so no value was wrong. The two arguments were removed from the routine and its one call.

### 30.6 Runs

- chain_verified.sh (new, .P1/s2_continuation/): chain.sh with every certified rung continued for up to 6 passes under a second environment (CHECK_ENV), the rung counting only if that continuation certifies too.

- Well-mixed chain `wm10` from the re-certified 0.3169 state (ifx, T1's setting, every rung verified under the unmodified rule): 0.3 (9 passes), 0.26 (21), 0.22 (24), 0.19 (22) certified and verified by 09:00 on 2026-09-29; continuing to 0.083.
- K_zz = 1e9 chain `kzz8` (ifx, carrier bound held at 3e-3): 1.09 (32 passes), 1.075 and 1.06 (the first 1.06 failed), 1.03, 1.0, 0.97, 0.94, 0.91, 0.88, 0.85, 0.8, 0.75, 0.7, 0.65 certified by 08:50 on 2026-09-29.
- photochem kzzprofile He/H 2.09: `photo209_v5` (default control, binary 2dd9a865) certified at pass 72; its launcher failed after the run because `config_rung.sh` was edited under it from another host (its EXIT.txt was written by hand, with the reason).

## 31. The code audit of 2026-09-29: findings checked and applied

`md/CODE_AUDIT_20260929.md` (Codex, with its probes in `docs/audit_20260905/code_audit_20260929/`). Each finding was checked against the source before anything was changed. The audit document was made ASCII (curly quotes and dashes); its disposition section records the same.

- **F2, the admissibility predicate accepted infinities: FIXED.** `UW_conversions.f90`, `positive_density_and_internal_energy`: +Inf density or energy compares true against zero and passed. Every conserved component and the internal energy must now be finite. The audit's probe (`guard_probe.f90`, linked against the fixed objects) now reads finite accepted, +Inf energy refused, +Inf density refused, NaN refused. A finite state takes the same branch as before, so no output moves.
- **F3, a steady route wrote a state without certifying it: FIXED.** The direct `EXHALE_PTC=1` route wrote whatever certificate the solve had left; it now calls the same final certification as the stationary restart (`certify_the_state_to_be_written`, one routine for both: the claim, the boundary rebuilt from the written composition, the residual assembled from that state, the certification evaluated on it). And a solve whose root stays on the other side of its own contact rule after the second solve (section 30.4) now returns info = 2 instead of 0. The restart route calls the same statements in the same order as before.
- **F4, the transit tool replaced non-finite transmission by 1 or 0: FIXED.** `EXHALE_transit.py`, `_checked_transmission`: a non-finite value or a departure from [0, 1] beyond 1e-12 stops the synthesis and names the first wavelength; the rounding of the Ly-alpha disk average is still clipped (the only line that was clipped before), and He 10830, H-alpha and H-beta are checked without clipping. The comment that exp(-tau) overflows in a saturated wing was wrong (it underflows) and is gone. MEASURED on the molecular K_zz = 1e9 He/H 2.13 catalog output: every number of the five `tpm_*` files is identical before and after.
- **F5, the equivalent-width operator: FIXED.** `LHS1140b/he10830_equivalent_width.py`: the model curve is refused when it is not one-dimensional, not of one length, not finite (a NaN wavelength used to pass) or not strictly ordered, and a transit file with a non-finite column or a non-positive continuum; the observational error now uses the trapezoidal weights of the estimator itself. MEASURED: EW 1.1084293096 %A unchanged, error 0.0294775 -> 0.0288956 %A (0.030 -> 0.029 to two figures). The error is ours, computed from the released spectrum under independent samples, not a number of Cherubim et al. (2026).
- **F7, a changed link library did not relink: FIXED.** `Makefile`: a second stamp whose name hashes the link

command (compiler, flags, `LDLIBS`); the executables depend on it. MEASURED: `make -q` with an unchanged command returns 0, with `LDLIBS=-l_exhale_missing_library` it returns 1, and `make -n` then shows the link alone, no compilation.

- **F8, the formation-energy diagnostic proposed a double count: CORRECTED (text).** `src/utils/formation_energy_flux_diagnostic.py` said the sensible-energy equation lacks -div(F_form) and a chemical source; in the sensible-energy form the chemical conversion is already the reaction heating and cooling (h nu - I for a photoionization), and adding either term counts it twice. The header now states the correct balance and what the script measures (the chemical share of the energy ledger); no computation changed.
- **F1, the sensible enthalpy carried by the diffusing carriers is not in the energy equation: CONFIRMED, MEASURED, OPEN.** The carrier transport (`diffusive_photochemistry.f90`) moves H2 relative to the gas and returns no energy flux; `interdiffusion_enthalpy_active` requires He diffusion and covers the elements only, so an H2 counterflow at zero hydrogen flux carries q = (h_H2 - 2 h_H) Phi_H2 that nothing represents. DERIVED on two certified states (well-mixed He/H 0.19 and K_zz = 1e9 He/H 0.65), from the face diffusive fluxes of the carrier row terms and h_H2 - 2 h_H = k (u_rv - 5T/2) with a rigid rotor and a harmonic vibrator (0.4 per cent from the code's own H2 caloric at 1000 K): |div q| is 1 to 3 per cent of the heating over the molecular layer (median 1.8 per cent over $1.001 < r < 1.03$), 8 to 11 per cent at the base cell, and as large as the net heating or larger where heating and cooling cancel; the integral of |div q| is 1.4 to 1.5 per cent of the integrated heating, while the SIGNED integral vanishes (-1e-18 of it), as the closed diffusive boundary of the carrier operator requires (script, inputs with their md5 and the numbers: `docs/audit_20260905/code_audit_20260929_F1/`). It can move the temperature of the H2 layer; the implementation (the carrier face fluxes into the energy rows of the steady residual and the marching update, the compensating atomic flux from the element constraints, tests) is the next item.
- **F6, the pass snapshot is two renames, not one publication: OPEN, a decision.** A reader-interface and data-layout change (a state generation with a manifest and one published pointer); left for the user's decision.
- Sections 11 to 13 of the audit (the wind-only certification scope, the abundance mask of absent carriers, trace caloric approximations, general EOS Riemann tests, the outer diffusive closure, parallel tests, optimization order) are statements of scope and proposals, recorded in `md/TO_BE_DONE.md`; none is a defect to fix in place.

## 32. The follow-up review of the audit corrections (2026-09-29)

`md/CODE_AUDIT_20260929_review1.md` (Codex; probes in `docs/audit_20260905/code_audit_20260929_recheck/`) confirmed F2, F4, F5 and F7 and the substance of F3 and F8, and named four remaining points. All four were checked in the source and applied.

- **R1, the direct steady route stopped with status 0 whatever the final certificate said.** It now ends with `certification_stop_uncertified`, the exit contract of the other steady routes: a stationary claim the final certificate refuses exits 2 after the state and the resolved configuration are written; an unfinished snapshot exits 0.
- **R2, the formation-energy diagnostic described more than it computes.** It computes the advective part u_form

v only; the header now says so, gives the stationary identity with the relative transport term, lists what a difference contains (omitted relative transport, the species residual of the saved state, missing species, the flux reconstruction), and states the total energy balance of an open, gravitating wind with its work and boundary terms instead of "only radiation".

- **R3, the admissibility test could overflow before rejecting.** The sign of rho e is now taken as the sign of 2 rho E - (rho v)^2 on the components divided by the largest of them, so nothing can overflow (`admissible_conserved_cell`, one definition used by the stage test and by the positivity repair, whose own cell test had accepted infinities as well). Under -ffpe-trap=overflow,invalid the 1e160 cell of the review is admitted (rho e = 5e159) with no trap, and its negative-energy variant, a tiny-density large-momentum cell and a negative-energy ordinary cell are refused. The three marching stages now test the interior before `Apply_BC` builds the boundary on it: `Apply_BC` reads the physical cells and writes only the ghosts, and neither the test nor the repair reads the ghosts of the trial, so the decisions are unchanged and an inadmissible interior no longer reaches the boundary and EOS routines.
- **R4, the transmission messages quoted metres as Angstrom.** The wavelength arrays of `EXHALE_transit.py` are in metres; the messages now quote 1e10 times them, and the function states its unit.
- **The F1 estimate** is now kept as a script with its inputs and the signed and absolute integrals (section 31 corrected: the quoted 1.5 per cent was the integral of |div q|; the signed integral vanishes).

Checks of this section (MEASURED): the overflow probe above; the audit's admissibility probe (finite accepted, +Inf and NaN refused); the six regression cases that march or take the stationary restart route (`hp_zero_seed`, `hp_trace_seed`, `hp_front`, `hydrostatic_column`, `oxygen_chemistry`, `iontrans_metals`) bitwise identical to their goldens with REGRESSION_REL_TOL=0 (gfortran `EXHALE.x` 67c12cd1); the transmission function on a NaN and on a rounding excursion. The production ifx build of this source is `EXHALE_64954795.x` (manifest `BINARY_MANIFEST_64954795cb2a.txt`), built with `make -B` on lart2 because `/nfs/mocafe` is mounted with a 30-minute attribute cache there and a plain `make` did not see the sources edited on lart4 (it relinked the old objects; the md5 stayed that of the previous build).

Two findings of the runs of this morning, recorded here and in `md/TO_BE_DONE.md` (m): the well-mixed chain (`wm10`, the contact held) certified 0.3 to 0.1 and stopped at 0.09, where every hydrodynamic solve converged and the H2 carrier row of the OUTERMOST cell (500, r = 30 R_p) grew slowly pass after pass (0.09: 1.2e-3 to 2.5e-3; 0.09487: 3.5e-3 to 4.0e-3) with every carrier relaxation ending where the carriers stopped moving, the symptom the 2026-09-24 memo recorded at He/H 0.083 and a candidate for the zero diffusive carrier flux at the outer face (audit section 12.3); and the four reduced-XUV atomic He/H 9.7 cases fail their first solves with the contact held, so their obstacle is not the contact threshold.

## 33. The state of every pass published as one generation (audit F6)

Audit finding F6 (`md/CODE_AUDIT_20260929.md` section 8, `md/CODE_AUDIT_20260929_review1.md` section 9): the stationary route wrote the state of every outer pass as `output/Hydro_ioniz_last_pass.txt` and `output/Ion_species_last_pass.txt`, each half through its own `.part` file and its own rename. An interruption

between the two renames left halves of two passes, and the pass number of cert_reason=pass_snapshot_p<n> did not tell two runs apart (two runs both write pass 1), so a mixed pair from two runs with compatible metadata was not excluded. Changed with the user's approval.

What changed (src/modules/files_IO/write_output.f90, src/modules/files_IO/load_IC.f90, src/modules/functions/utilities.f90, src/EXHALE_main.f90):

- **A run identity.** establish_run_identity, called once after init, forms run_identity = yyyymmddThhmmss.sss_c<count> from date_and_time (to the millisecond) and the 64-bit system_clock count, both standard Fortran; characters outside [A-Za-z0-9._-] are replaced.
- **One generation for each pass.** publish_pass_state_generation (it replaces write_pass_state_pair) writes, with the writers and header lines the pair had, Hydro_ioniz.txt, Ion_species.txt and manifest.txt into output/pass_state/.<gen>.tmp/, with <gen> = r<run_identity>_p<pass>. The manifest states, one key=value a line, state_id, run_identity, pass, previous_state_id, the two file names with their sizes in bytes (INQUIRE size=), certified=F and cert_reason=pass_snapshot_p<pass>.
- **The order of publication.** (1) the three files are written and closed in the temporary directory; (2) rename(2) of that directory onto output/pass_state/<gen>; (3) output/pass_state/current.part is written with <gen>, closed and renamed onto output/pass_state/current, the one rename that publishes; (4) the generation current named before is kept, and the one it had replaced (its manifest's previous_state_id) is removed file by file under its explicit name and then remove(3)d as an empty directory. Directories are made with a C mkdir binding (mode 0755) and removed with C remove, beside the existing rename binding; no shell command and no pattern is used. A name read from a file is removed only if it starts with r and has only [A-Za-z0-9._-] characters.
- **An identity in both halves.** The coupling line of both halves of a generation ends in state_id=<gen> (write_coupling_state_header takes it as an optional argument; no other file carries it). load_IC reads it with the certification pair (parse_certification_claim) and adopt_certification_claim refuses a pair whose halves state two different ids, and a pair in which one half states an id and the other none. A pair in which neither states one (every final state, every older file) loads exactly as before; a pair that does prints the generation.
- **A measurement key.** EXHALE_PASS_STATE_STOP_AT=<step>[:<pass>] stops the run with stop 3 right after step 1, 2 or 3 of the publication of that pass (of every pass if none is named). Unset: no effect.

What changes for the user: the files moved from output/{Hydro_ioniz,Ion_species}_last_pass.txt to output/pass_state/<gen>/{Hydro_ioniz,Ion_species}.txt, and the generation to take is the one output/pass_state/current names; the old names are no longer written. The final Hydro_ioniz.txt / Ion_species.txt of a run are unchanged (no state_id). An interruption before step 2 leaves a .<gen>.tmp directory, one between steps 2 and 3 a complete generation no pointer names; neither is read and neither is removed by a later run. The continuation script of the LHS 1140 b rungs is being executed by running jobs, so it was not edited in place: the new version is LHS1140b/models/.P1/s2_continuation/config_rung.sh.new, to be moved onto config_rung.sh when no process uses it. Its entry precedence is (a) the parent's final pair, (b) the generation current names, read once and taken only if its three files exist and the pointer, the manifest's state_id and both headers' state_id agree,

otherwise refused, (c) a legacy *_last_pass.txt pair with the checks it had. run_check.sh clears output/*.txt of a case before a run and leaves output/pass_state/, which holds at most two generations and is not compared. src/tests/steady_selfconsistent_residual/run.sh never read the pair. Documents updated: README.md, README_HOWTO.md, md/input_schema.md appendix D, docs/EXHALE_user_manual.tex (section "pass_state/", label sec:passstate, and the key in the table of environment variables), docs/physics_overview/steady_state.tex, boundaries_initial.tex, postprocessing.tex; both PDFs rebuilt without errors or undefined references.
Checks (MEASURED 2026-09-29 on lart4, gfortran 16.2, binary built in its own object directory build_f6, md5 b8cf7a21f2fa1149b1d9131cafbd44a0; the tree also held other uncommitted edits in progress at that moment):

- **Publication and retention.** iontrans_metals inputs and IC in a scratch directory, EXHALE_OUTER_PASSES=4, 4 threads, exit 0: passes 1, 2 and 3 published; afterwards current named ..._p3, ..._p2 was kept, ..._p1 was removed, no .tmp or current.part was left; each manifest's byte sizes (94446, 448089) equal the files, and the manifest and both headers state the same state_id. The final pair carries no state_id. The regression run of iontrans_metals (below) published 14 passes and left exactly ..._p13 and ..._p14, current naming ..._p14.
- **Interruption.** Three runs started within 30 ms of one another, each on a directory holding the generations of the first run, with EXHALE_PASS_STATE_STOP_AT=k:2, all exiting with status 3, got three different identities (...228.498_c...163735, ...228.527_c...351728, ...228.522_c...722562). k = 1: current named the run's complete p1; .<p2>.tmp/ held its three files. k = 2: current named p1; p2 was complete and unnamed. k = 3: current named the complete p2; p1 and the previous run's p3 were still present (step 4 not reached). At every point current named a generation whose three files existed and whose manifest and headers stated its name. The previous run's p2 had been removed at the publication of the new p1, across the two runs.
- **load_IC.** Hydro of p2 with Ion of p3: refused, exit 1, both ids printed. Hydro of p3 with a final Ion_species.txt, and the reverse: refused, exit 1. Both halves of p3: loaded, exit 0, "the state is the pass state generation ...". The legacy certified IC pair of the case (no state_id) loaded as before (the runs above).
- **config_rung.sh.new**, with BIN=/bin/true on dummy parents: a final pair is taken first even with generations beside it; a parent with only generations gets the pair current names (byte-identical copies, LAUNCH.txt seed "LAST PUBLISHED PASS (pass_snapshot_p3, output/pass_state//)"); a manifest naming another id, a generation missing Ion_species.txt, and a pointer to an absent generation are refused (exit 1); a legacy *_last_pass.txt pair is taken; a legacy pair of two passes and an empty parent are refused.
- **Final outputs.** REGRESSION_EXE=<that binary> REGRESSION_REL_TOL=0 run_check.sh check iontrans_metals hp_front: all six compared files data identical (bitwise) to their goldens, REGRESSION PASS.

## 34. The enthalpy the transported carriers carry (audit F1), and the regression comparison of residual columns (2026-09-29)

**The physics.** A molecular carrier (H2; OH, H2O, CO with the oxygen chemistry) is moved by the carrier operator relative to the rest of its own elements: its diffusive, eddy and settling number flux Phi_c moves nuclei that the element fluxes do not, and the non-carrier species of the same elements move the other way, nucleus for nucleus. Cook (2009,

eqs. 11-13) then puts q_car = sum_c Phi_c [h_c - sum_el nu_{c,el} hbar_el] into the energy equation, h_c the enthalpy of one carrier particle and hbar_el the enthalpy of the non-carrier species of element el per nucleus, every particle with the electrons its charge gave up, as in `component_specific_enthalpies`; an H2 molecule rising through atomic hydrogen at zero hydrogen flux carries k (u_rv - 5T/2), about -1.5 kT at 1000 K. Before this section the energy equation carried the element part only (section 29.3), which does not see a carrier moving within its element.

**The code.**

- `diffusive_photochemistry.f90`: `carrier_enthalpy_face_flux` (the flux at every face from the carrier operator's own coefficients and face rule, `carrier_face_coefficients` and `carrier_face_flux`, with the temperature and the particle count of the state passed, so it follows the iterate of a steady solve), `carrier_enthalpy_divergence_of_state` (cells 1 ... N on the carrier rows' geometry; both ends of the carrier column are closed, so the column sum is zero) and `carrier_enthalpy_active`. Writes no module state.
- The stationary energy row (`steady_residual.f90`, `assemble_residual`) adds the divergence into `Sidf`, the one record of the enthalpy of relative transport. The residual module cannot use the carrier module (which uses it), so the carrier module registers the procedure in a pointer of the residual module when the carrier set is decided (`carrier_set_init`); a run without the term leaves it null.
- The marching step (`EXHALE_main.f90`) subtracts dt div q_car after an accepted carrier step, from the fluxes of the composition that step returned (the step's own fluxes to first order in dt, exactly at a fixed point, so the two routes solve one energy equation).
- One key for both terms, `Interdiffusion enthalpy flux:` (default True): it now also acts with `Molecular carrier transport: True`. A new restart token `carrier_enth` (absent in older files, read as F), so a restart from a state written without the term names it on a `Restart option change:` line.
- Not covered: the ionization stages of `Ionization transport` (their face flux contains the element's advection; the relative part is not formed), recorded in `md/TO_BE_DONE.md` (j).

**Checks (MEASURED).**

- New suite `src/tests/carrier_enthalpy_flux/` (10 rows, all PASS) on a synthetic HI/H2 column at zero hydrogen flux: both end faces carry nothing; the column sum of V div q is 4e-17 of the sum of |V div q|; the face flux equals Phi_H2 k (u_rv - 5T/2), Phi_H2 recomputed from the operator's coefficients, to 3e-15; the key off and a run without carrier transport give exactly zero.
- The certified well-mixed He/H 0.19 state (`wm10_0.19_check`) reloaded with the term (`LHS1140b/models/.P1/audit_F1/wm019_F1`, gfortran b029a1e8): the code's divergence, from the conservation budget of pass 1, is a median 1.84 per cent of the heating over 1.001 < r/R_p < 1.03 (the independent estimate of section 31: 1.83), 1.6 per cent in absolute integral (1.5), zero in signed integral, largest 9.9 per cent at r = 7.2 R_p. That state is not stationary with the term: the gated carrier row stood at 7.1e-3 on the first pass and fell by about 18 per cent a pass (1.7e-4 at pass 20), every hydrodynamic solve converging. So every molecular state certified before this section is a state of the previous energy equation.

**The regression comparison of residual columns** (user decision of 2026-09-29). `backup/regression/compare_within_tolerance.py` measured every entry against itself, so a residual column that

is rounding at a solution read 0 against 3.1e-15 as a relative difference of 1 (`adv_mass_row` of `wasp_full_newton`, section 30.4). A column whose header name ends in `_row` is now measured against the largest value of its golden column; every other column keeps the value-by-value measure (a state column spans decades, and a column-wide scale would hide a change of its small values). The flagged file now reads within 2.5e-9.

**What the term changes (MEASURED).** The He/H 0.19 reload above certified again after 35 passes (`certified_in_wind`). Against the state certified without the term: T higher by 8 to 19 K in the H2 layer (r < 1.01 R_p), lower by a median 4.8 K over 1.3-3 R_p and by at most 24 K (r = 4.0 R_p, 1774 -> 1749 K); the wind-window mass flux -0.55 per cent; the He 2^3S density higher by a median 1.9 per cent over 1.05-5 R_p, at most 6.3 per cent. The equivalent width was not measured (the stationary route writes no `_adv` pair); a change of the order of the He 2^3S change, about 2 per cent, is DERIVED, against an observational error of 2.6 per cent. Figure: `LHS1140b/models/.P1/audit_F1/wm019_before_after_F1.pdf`.

**The regression matrix and the goldens (MEASURED 2026-09-29).** The full matrix on the F1 + F6 source: twelve cases bitwise or within 1e-3; the four cases with `Molecular carrier transport: True` differ as F1 makes them (`mol_carrier`, `hp_zero_seed`, `hp_trace_seed`, `hp_front`); and `oxygen_chemistry` wrote nothing, because its IC predates the `carrier_enth` token: its input now names `Restart option change: carrier_enth`, and it is then bitwise its golden, since in that 1000-step cold transient the carrier interval is never covered (1049 of 1049 step attempts) and the carriers never move, so F1 adds nothing by construction. The goldens of the four changed cases were re-snapshotted (the previous ones kept in `backup/regression/golden_pre_F1_20260929/`), with `iontrans_metals` so that its reference directory is current, and the four re-checked bitwise PASS.

## 35. Why the mapped atomic He/H 9.7 seeds at reduced XUV do not solve (2026-09-29, open)

The four atomic He/H 9.7 cases at 0.01, 0.06 and 0.0624 of the fiducial XUV (`LHS1140b/models/.P1/atomic97/`) were moved from the catalog states onto the face-at-level grid (`facelevel_continue.sh`, section 29.11) and restarted stationary. The hydrodynamic solve ended `info=2` on every pass. Measured on the 0.06 case, `LHS1140b/models/.P1/atomic97/diag_R/` (one directory for each measurement, `Restart intent: stationary evaluate` unless named otherwise):

| run | state and binary | residual norm | largest row |
|---|---|---|---|
| D1 | catalog state, previous grid, `EXHALE_69233669.x` (the binary that wrote it) | 5.6e-7 | the catalog value |
| D2 | the same state, `EXHALE_7db5afe9.x` (2026-09-26, previous grid) | 5.7e-2 | energy, r = 1.072 |
| D4 / D5 | mapped seed, `EXHALE_6dd54955.x`, element enthalpy flux off / on | 1.88 / 1.88 | mass, cell 13 (all of it below r = 1.03) |

| run | state and binary | residual norm | largest row |
|---|---|---|---|
| D6 | mapped seed, `EXHALE_a7255a03.x` (the first face-at-level binary) | 1.88 | as D4 |
| D9 | D4 with v below r = 1.003 set to the wind flux / (rho r^2) | 1.86 | mass, cell 13 |
| D10 | D4 with ln rho, ln p and T below r = 1.03 interpolated as cubics | 1.82 | mass, cell 110 (where the two interpolations meet) |
| D11 | D4 with 50, 110 or 200 cells rebuilt in the discrete well-balanced equilibrium | 1.54 | mass, cell 6 |
| D13 / D14 | D11 (110 cells) with cell 1 anchored to the reservoir pressure, without / with the smooth velocity of D9 | 1.54 / 1.18 | mass, cell 6 / cell 1 |

What this shows, and what it does not:

- The seed is a wind: its mass-flux spread above r = 1.03 is 0.32 per cent, the same as the catalog state's. The spread of 0.42 to 0.48 in the logs of the stationary restarts is what the first JFNK solve left (27 to 42 non-monotone accepts), not the seed.
- CORRECTED after the review (35.1): the two bullets written here first read D9 to D14 from the maximum over all cells and excluded the interpolation of the lower layer and the reconstruction. That was wrong. The element enthalpy flux is not the direct cause of the fixed-state mass maximum (D4 = D5 in the mass row), and nothing more is excluded.
- The heating of the lower layer at fixed T is higher by about 8 per cent under the face-at-level binary than under `EXHALE_7db5afe9.x` at the same radius (1.139e-9 against 1.053e-9 erg cm^-3 s^-1 at r = 1.010); the increase is the charge-exchange channel (35.1), while cooling is below 2 per cent of heating from the base to 1.1 R_p in both.
- A marching relaxation does not settle the seed: 5000 steps take the mass-flux spread from 0.003 to 3.8 and back to 1.9, and 41000 steps (D8) leave it at 1.09 to 1.10 from step 15000 on. That relaxation ran its first stage with PLM (`Restart intent: relaxation` enters the two-stage PLM -> WENO3 marching), while the state and every stationary solve above are WENO3.
- `Restart intent: relaxation` does not relax a seed whose spread is below the Newton switch (1e-2): a restart arms the hand-off at load (`du_newton_armed = do_load_IC`), and the 0.06 seed was handed to JFNK at marching step 2.

The two 0.01 cases were already refused in the catalog with the same signature (flat mass flux, mass and energy rows of order 1 in the first cells); the 0.06 and 0.0624 cases acquired it with the grid and code change. The whole account,

with the next steps considered, is `md/atomic_heh97_reduced_xuv_20260929.md`. The four runs are stopped (their `EXIT.txt` say why). The same failure, a first JFNK stalled at ||R|| 1.32 on the mass row of cell 2, stopped the F1 resolve of the mapped `m7_HeH1.45` state; that He/H is instead being reached from `kzz_1.223_v4b`, a state solved on the face-at-level grid (`LHS1140b/models/.P1/audit_F1/launch_F1_kzz_1.223_to_1.45.sh`).

### 35.1 The review of the account (2026-09-29)

`md/atomic_heh97_reduced_xuv_20260929_review.md` (its retained evidence is the directory `atomic_heh97_review_20260929/` of `docs/audit_20260905/`: a diagnostic linked to the production objects, built with bounds checks, reproducing the production face fluxes exactly) was checked here and taken over:

- The continuity defect is a numerical face flux: on the reconstructed traces of the mapped seed the Roe pressure-jump term (-Delta p_WB / 2c, Delta p_WB about 1e-7 of p at Mach 4e-8) reverses the mass flux at face 12 and removes most of it at face 13, and at face 0, whose two traces differ by 4.9e-4 of the pressure, carries about 1e4 times the wind flux inward. The low-Mach velocity-jump correction scales Delta v and leaves this (1.876).
- A lower layer projected into the reconstruction's discrete equilibrium with the momentum set to the wind flux takes cell 13 from 1.876 to 1.3e-4; cell 1 stays at 1.18 and the join at cell 111 rises to 1.50 (READ from the retained results). The interior is repaired; the base boundary and the join are not.
- The base boundary continues cell 1 along a hydrostatic isentrope, the reconstruction balances each cell with a constant-density equilibrium, and `Rec_BC` keeps the interior trace of face 0: three constructions, not one discrete equilibrium. The comment in `base_boundary.f90` (`characteristic_base_face_state`) that claimed the boundary exactly balanced in any hydrostatic atmosphere at any resolution was corrected (comment only; no binary change beyond line numbers).
- The heating difference is the nonradiative He+ + H -> He + H+ channel, 8.785e-11 against zero at r = 1.010 (`Heating_breakdown.txt` of D2 and D4, MEASURED here as well): the rate of section 25, 9.6e-17 cmˆ3 sˆ-1 at 1057 K against 2.8e-21 before, whose threshold continuation carries a band of 1.5e-5 to 3.6 times at 1e3 K. A deliberate physics change, not a defect; a sensitivity test moves the forward rate, its reverse and the heat together. Heating cannot make the fixed-state continuity defect.
- The first JFNK solve cut the certification distance from 7.96e10 to 1.19e9 and the mass row from 1.88 to 0.099 while spoiling the outer wind; a failed multi-scale solve, not by itself evidence against the non-monotone acceptance.

The next steps in the review's order (a face-flux budget diagnostic, discrete equilibrium tests of the boundary with the reconstruction, one discrete equilibrium at the base, a seed that satisfies the lower-layer equations, continuation, and only then the nonlinear formulation) are listed in `md/atomic_heh97_reduced_xuv_20260929.md` section 5; none is started, and the base boundary change needs its design agreed first.

## 36. The face-flux budget export and the discrete equilibrium tests of the base (2026-09-29)

Steps 1 and 2 of `md/atomic_heh97_reduced_xuv_20260929_review.md` section 8. No equation, flux, source or boundary changed.

**The face-flux budget export (schema 3).** `EXHALE_CONSERVATION_BUDGET=<n>` (`src/modules/time_step/conservation_budget.f90`, called from `assemble_residual` in `steady_residual.f90`) now appends to the 42 columns of schema 2, for the upper face of each row: the reconstructed left and right traces (rho, v, p) the flux was formed from, `wb_dp_eq` and the two departures, the pressure jump Delta p_WB the Roe flux uses, the central mass transport of the physical flux, the pressure-jump part of the numerical mass flux (production `Num_flux` with the jump minus the same call with the jump set to zero: an attribution, not a flux), and the recomputed numerical mass flux; for each cell, the absolute and normalized continuity residual (the scale the stationary solve divides by) and the signed cumulative mass budget. A header block states the lower boundary of the assembly: the reservoir, the interior state continued to the face along cell 1's isentrope (recomputed from the inputs the boundary used and checked bitwise against the values it stored), the characteristic face velocity, the two traces of face 0 and the numerical base flux. Columns that a branch does not define are NaN, with the reason in the header. The flux branch counters the extra `Num_flux` calls would move are saved and restored. MEASURED: on the review's 0.06 input the new columns reproduce the review's table (face 0: +3.28224e-4, 9995.84 times the wind flux; faces 12 and 13: pressure-jump parts -4.05157e-8 and -2.59041e-8 against central transport 3.28e-8), the continuity rows telescope from the exported face fluxes with a difference of 0, and the state files are the same with the export on and off (data identical; only the provenance time differs); `hydrostatic_column` with the export off is bitwise the binary before the change. The base block of that input shows the two interior pressures of face 0 that do not agree: the isentropic continuation the boundary compares with the reservoir, 0.999877 p0, and the reconstructed interior trace the flux sees, 0.999509 p0.

**The discrete equilibrium tests.** Two measurement programs in `src/tests/grid_and_gates/` (`base_reservoir_discrete_equilibrium.f90`, `low_mach_face_mass_flux.f90`, a shared module `base_equilibrium_columns.f90`, input `base_reservoir_equilibrium.inp`; `run.sh` names `base_reservoir_equilibrium` and `low_mach_face_flux`), each on a ladder of three lower cell widths (2e-4 x 50, 1e-4 x 100, 5e-5 x 200 R_p), production `Apply_BC`, `Reconstruct`, `Num_flux` and `assemble_residual` with no sources. Columns: `a_iso`, `a_warm` built exactly in the reconstruction's discrete constant-density equilibrium (wb_dp_eq = 0) and anchored to the reservoir pressure at face 0; `b_iso` an analytic isothermal column; `c_isen` an analytic isentrope. MEASURED (16 PASS, 37 FAIL, the failures being the findings; the README marks both tests RED by design):

- The well-balanced interior keeps its discrete equilibrium to rounding (interior momentum rows 1.2e-14 and 4.6e-14 of their scale on `a_iso`, `a_warm`). The base does not: on `a_iso` the face-0 flux is -1.46e-4 of (rho c)_res, v_b = -2.6e-4 c_res, the ghost-0 / cell-1 mismatch is wb_dp_eq(0) = 2.9e-4 p, and the cell-1 momentum row is 3.1e-3 of its scale.
- These are truncation errors: the face fluxes converge at order 2.0, the cell-1 momentum row at order 1.0, on every column.
- Against a manufactured wind at Mach M they grow as 1/M: on `a_iso` the face-0 flux misses the wind by -0.136 at M = 1e-3, -14.6 at 1e-5 and -1.47e4 at 1e-8 (1.47e4, 3.61e3, 8.93e2 on the three grids), while the interior faces 2-10 stay within 5.3e-4 at every M. On `b_iso` (continuum but not discrete balance) the interior faces are off by 1.03 at M = 1e-8 as well.

Interpretation (not measured as such): two mechanisms act at face 0. The outgoing relation takes p_i from the isentropic continuation of cell 1, which differs from cell 1's own constant-density equilibrium at the face by about (1/2 gamma)(dr/2H)^2 (4.3e-4 here) and drives v_b; and the reconstructed right trace departs from p_res because WENO3 forms its departure from ghost 0, which lies on the reservoir isentrope and not on cell 1's equilibrium. Refinement cannot bring the base flux to 1 per cent of a Mach 1e-8 wind (a base cell about 300 times narrower than the finest tested would be needed); the boundary has to share the reconstruction's discrete equilibrium (review step 3), and a seed has to be projected onto it (step 4).

Also corrected: the header of `hydrostatic_residual.f90` named a SAVEd `tol = 1.0` in `define_grid` as the reason one grid is built in one process; `define_grid` now sets `tol` before each search, and the reason is the operator's module arrays allocated at the first N. The grid-and-gates README count ("Twenty-one tests") and its list of names were stale and were corrected.

## 37. One discrete equilibrium at the base (2026-09-29)

Step 3 of `md/atomic_heh97_reduced_xuv_20260929_review.md` section 8. Under `Well balanced: True` the base condition reads the interior pressure at the level face as cell 1's own constant-density equilibrium, p_i = P_dn(1) = p_1 + rho_1 (phi_c(1) - phi_i(0)), the pressure the reconstruction extrapolates cell 1 to (`base_boundary.f90`, `interior_state_at_level_face`), in place of cell 1's hydrostatic isentrope. The two ghosts continue the same equilibrium below the level, P_up(0) = p_b and P_up(-1) = P_dn(0), at the reservoir-side temperature (`base_ghost_averages`). Once the contact rule has chosen the face density rho_b, the face velocity carries the interior's mass flux at that density (v_b = v_i rho_i/rho_b + (p_b - p_i)/(rho_i c_i)): a stationary face with flow through it has no contact. The reservoir still states the pressure and entropy of inflowing gas; the equilibrium shared is mechanical, not thermal (Kaeppeli & Mishra 2016, A&A 587, A94, section 2.1). Without the option the previous construction is unchanged. The model identity is `..._face_at_level_discrete_equilibrium_v5` with the option and the unchanged `..._face_at_level_v4` without it (`base_boundary_model_id`).

MEASURED (the tests of section 36): on columns built in the discrete equilibrium, v_b, the face-0 flux and the base momentum rows went from 2.6e-4 c, 1.46e-4 rho c and 3.1e-3 of the weight to below 1e-14; the base face carries a manufactured wind of Mach 1e-3 to 1e-9 to 2.1e-3 (isothermal column) and 2.5e-3 (warming column), first order in the cell width, where the previous boundary missed it by 1.47e4 at Mach 1e-8. Carrying the flux at rho_1 instead of rho_b would have left 2.8 and 6.7 per cent; a temperature contact at the level with flux through it is not a stationary flow and was moved to a static-only test column. Analytic columns (continuum, not discrete, balance) now show their O((dr/H)^2) offset at the base face (order 2.0). grid_and_gates: 361/37 -> 393/11 PASS/FAIL, the eleven all the analytic isothermal column. On the mapped atomic 0.06 seed the cell-1 row went 1.378 -> 1.206 and the face-0 flux 1.0e4 -> 2.2e4 times the wind: the seed's cell 1 lies 5.5e-4 of p off the discrete equilibrium, which the isentrope had partly offset; the seed must be projected (section 38). Two continuation states certified under v4 (K_zz = 1e9 and well-mixed He/H 0.55, with the carrier enthalpy flux) re-certified under v5 in 2 and 4 passes, starting at ||R|| 1.14 and 1.13. Also corrected: the v2 identity in `md/input_schema.md`, the v3 literal in `src/tests/boundary_state/ghost_composition_fixed_point.sh`, the missing `carrier_enth` token in `molecular_seed_of_an_atomic_state.sh`.

## 38. The atomic He/H 9.7 cases: the seed projection, and the element enthalpy flux as the remaining obstacle (2026-09-29)

**The seed projection.** Measurement key `EXHALE_SEED_HYDROSTATIC_CELLS=K` (`src/modules/init/init.f90`, `seed_layer_in_discrete_equilibrium`, off when unset, requires `Well balanced: True`): after load_IC, cell 1 is set so that its constant-density equilibrium reaches the reservoir pressure at the level face, cells 2 to K so that wb_dp_eq of faces 1 to K-1 vanishes, every cell above K (upper ghosts included) is scaled by the one factor that balances face K, all with T and the mass fractions of every cell held; the momentum of cells 1 to K carries the mass flux of cell K+1 and above K the velocity is divided by the density factor. MEASURED on the mapped 0.06 seed under v5 (`LHS1140b/models/.P1/atomic97/diag_R/P<K>_v5_projected`): mass and momentum rows 1.88 and 5.0e-3 -> 0.133 and 3.2e-5 (K = 50), 3.6e-2 and 8.5e-6 (110), 3.7e-3 and 7.6e-7 (200 and 300); deeper than local Mach 1e-3 the neglected momentum flux shows (1.5e-5 at K = 400). The energy row stays at 1.05: it is not projected.

**What the energy row is.** From the budget export of the projected seed (cell, dF_energy, heat - cool, Sidf): cell 50, 8.6e-6, 9.4e-6, +1.44e-5; cell 200, 4.3e-6, 4.6e-6, +5.4e-6; cell 2, 5.0e-6, 3.0e-5, -5.4e-4. Without Sidf the energy row would be about 8 per cent of the heating from cell 3 up; the element interdiffusion enthalpy flux (section 29.3), absent when the seeds were solved, is 1.1 to 1.5 times the heating over 1.0 - 1.15 R_p. It is physical: helium stays while hydrogen escapes, and the escaping hydrogen carries (5/2) kT per particle relative to the barycentric velocity, an enthalpy flux of the order of the wind's own. Cell 2's spike is the helium diffusive flux through face 1, 2.8 times that of the faces above, because cell 2's helium fraction is about 1e-6 off the diffusive steady state against the reservoir composition of cell 1; the seed's composition is not in its element steady state (the catalog refused these states on the elemental He/H row). With the term off, cell 1 still reads 0.94: the heat of the base cell, which the previous boundary's spurious face fluxes had been carrying away.

**Solves.** From the projected seeds with the term on (`*_v5proj`, `EXHALE_710ab9ca.x`), the hydrodynamic solve returned info = 2 on every pass with the mass and momentum rows unchanged, the energy row grew 1.09 -> 1.40 over the element steps, and 0.06, 0.0624 and the photochemical 0.01 were refused by the progress control at pass 4 (the scalar 0.01 was stopped in its second pass). With the term off (`*_v5proj_enthoff`), the same seeds solve: hydro info = 0 on pass 1 in 83 s, ||R|| 5.1e-8 (0.06) and 4.8e-8 (0.0624); the elemental He/H row, 2.2e-2 and 1.1e-2 at pass 1, moves between 5.7e-3 and 1.4e-2 over passes 2 to 5, then falls by about a factor 2 a pass, and both states CERTIFY at pass 24 (He/H row 8.5e-6 and 9.9e-6 against 1e-5; run logs of `*_v5proj_enthoff`). A marching test of the projected 0.06 seed (20000 steps, term on) settles at a mass-flux spread of 1.09 to 1.10, as before the boundary change.

**The continuation factor.** Measurement key `EXHALE_INTERDIFF_ENTH_SCALE=s`, 0 <= s <= 1, 1 when unset (`binary_element_diffusion.f90`, `interdiffusion_enthalpy_divergence`, the one place both the stationary row and the marching update take the term from): the divergence is multiplied by s, so a state of the full equation (s = 1) can be reached from one solved without the term in steps. A state at 0 < s < 1 is a step of that continuation, not a prediction.

## 39. The review of the 2026-09-29 handoff: state identity of the continuation factor, the projected seed, the output precision; the continuation runs (2026-09-29/30)

The review is `md/session_handoff_code_review_20260929.md` (evidence in `md/session_handoff_code_review_20260929_evidence/`). Every finding was checked against the code before anything was changed; the verdicts:

| item | verdict |
|---|---|
| R1 continuation factor absent from the state identity | CONFIRMED: `EXHALE_INTERDIFF_ENTH_SCALE` changes the energy equation, the `interdif` |
| R2 certificate carried onto a transformed state | CONFIRMED (the projection kept `ic_certified`; so did a restart that changes a named optio |
| R3 list-directed output loses precision | CONFIRMED with ifx 2026.1: 15 significant digits, up to 12 ulp on reload (gfortran: 0 ulp) |
| R4 projection without admissibility check | CONFIRMED on the review's synthetic grid (negative density and pressure returned) |
| R5 terminal success of the continuation scripts | CONFIRMED in `adaptive_enthalpy_scale_chain.sh` and in `xuv_descent_chain.sh` (th |
| Q1 element enthalpy budget over cells 2..N | not a defect: cell 1's composition is steady, so the reservoir supplies through face 0 the elemen |
| Q2 expected low-Mach limits reported as failures | improvement adopted |

**Checks** (ifx and gfortran builds of the edited tree; only the paths the edits touch). `src/tests/interdiffusion_enthalpy_flux/run.sh`: 18 PASS. Gate tests `low_mach_face_flux output_state restart_round_trip restart_option_change restart_intent base_reservoir_equilibrium`: 120 PASS, 11 XFAIL, 0 FAIL. IC dumps (ifx) of scratch copies of the certified 0.06 term-off state: plain `certified=T certified_in_wind interdiff_enth=F`; projected (K = 244) `certified=F projected_seed`; s = 0.125 with the term named `certified=F option_change_seed interdiff_enth=0.12500`. Dump, reload, dump: physical rows differ by at most 1 ulp in rho, v, p, 6 ulp in T and 5 ulp in the species (the unit conversions and T from p, not the format); ghost velocity 1e-8 (the boundary rebuilds the ghosts, as the reader does). The review's own probe, linked against the new objects: the projection is refused at cell 2 and nothing is adopted. The regression matrix was not run (user instruction of 2026-09-29).

**What the correction leaves as it was.** The binaries in use by the running chains (`LHS1140b/models/.P1/atomic97/EXHALE_acb07e93.x`) predate R1: every intermediate rung they write at 0 < s < 1 states `interdiff_enth=T` (`*_adNN_s<value>/`, the value in the directory name and in `LAUNCH.txt`); those states are continuation steps and are not published. State files written before 2026-09-29 by ifx builds carry 15-digit values; they are left as they are.

**The continuation runs** (binary `EXHALE_acb07e93.x`, ifx; lart3 and lart4):

- Element enthalpy flux factor from the certified term-off states, `adaptive_enthalpy_scale_chain.sh`: s = 0.125 CERTIFIES at both 0.06 and 0.0624 (19 and 20 passes); from there a step of 0.1125 fails (hydro info = 2, energy row 0.27 to 1.8), and the chain at 0.06 advances in steps of about 0.02 to 0.03 (s = 0.153, 0.174 converged; 0.195, 0.206 failed).
- XUV descent at s = 0 from the certified 0.06 term-off state, `xuv_descent_chain.sh` (new): 0.06 -> 0.05 fails (hydro info = 2 all 24 passes; the JFNK starts at ||R|| 0.18 with the energy row of the outer wind, r = 15, and stagnates near 1e-2), the chain retries at q = 0.913.

- The photochemical case: the catalog's certified 0.10 state (previous grid, HLLC) mapped onto the face-at-level grid and projected (K = 230; the density moved by 1.5e-3, against 5.2e-2 on the unsolved 0.01 seed), solved with ROE and the term off: hydro info = 2 on all 9 passes, refused; loaded energy row 0.51 at cell 1 and 0.45 at r = 13. The same run with HLLC loads to four digits the same residual (energy 0.5075 at cell 1, first JFNK ||R|| 0.4455 against 0.4457), so the flux choice does not carry the mismatch; it was stopped.
- The scalar 0.01 case from its projected seed with the term off (the run of section 38, lart3): hydro info = 2 on every pass; its mass and energy rows fell from 6.1e-2 / 1.1e-1 at pass 1 to 5.35e-4 / 6.97e-4 by pass 17 and stay there (passes 17 to 19: the hydrodynamic solve is not accepted), while the element row falls (1.9e-4 at pass 19).

## 40. Why the atomic He/H 9.7 continuation fails: two mechanisms, and the heat conduction the models omit (2026-09-30, open)

Diagnosis of the failed rungs of section 39, read from their run logs (`LHS1140b/models/.P1/atomic97/*_adNN_*`, `*_xd01_*`); every number MEASURED.

**A. The alternation between the hydrodynamic solve and the element step diverges** (0.06, s = 0.174 to 0.2375). Each hydrodynamic solve converges (info = 0, quadratic in its last iterations, lam = 1, the linear solve at its tolerance in 1 to 6 products), but the elemental He/H row grows from pass to pass and the under-relaxation of the element step does not stop it; the hydrodynamic solve fails only once the composition has moved far:

| s = 0.195, pass | hydro start ||R|| | hydro | He/H row | omega |
|---|---|---|---|---|
| 1 | 0.092 | info = 0 | 9.0e-3 | 0.5 |
| 2 | 0.17 | info = 0 | 2.3e-2 | 0.5 |
| 3 | 0.32 | info = 0 | 1.4e-1 | 0.25 |
| 4 | 0.72 | info = 0 | 3.8e-1 | 0.125 |
| 5 | 0.39 | info = 2 | 3.5e-1 | 0.125 |

The rung at s = 0.174 that the chain accepted ended at pass 14 with the He/H row at 0.51 (hydro info = 0): the acceptance on `hydro info=0` alone (section 39, R5) carried an unconverged composition into every later rung. The last states of the 0.06 chain that are converged are s = 0.125 (certified) and s = 0.153 (He/H row 1.05e-4, not certified); the chain stopped at 01:07 with its step below 0.005.

**B. The stationary solve does not move from its start** (0.0624 from s = 0.125 to 0.2375; the XUV step 0.06 -> 0.05 at s = 0). The composition converges at the held hydrodynamics (He/H row 9.9e-6 -> 2.7e-8 over 12 passes), but every pass starts the solve at the same ||R|| (0.2671; 0.183) and ends there. Within a solve the pseudo-time steps are accepted at lam = 1 while dtau doubles; near dtau ~ 2e3 a step overshoots (||R|| 0.06 -> 0.24 -> 0.28, the ionization root of 1 to 2 cells outside the simplex in the trial evaluations), dtau is cut, ||R|| returns to 0.053 while the fixed-scale merit rises, and after 40 iterations without improvement of either functional the solve stops and returns its best judged iterate, which is the entry state (every move raises the continuity row against its 3e-12 tolerance). The linear model is not the obstacle (the cycle reaches its tolerance in one product); the path the pseudo-transient has to take is longer than the solve's stagnation window.

**The energy equation of these models omits heat conduction** (`cond=F`, the default). Evaluated on the stored states with the conductivity the code carries (`viscous_conduction.f90`: Watson, Donahue & Walker 1981 neutral atomic hydrogen, 4.45e4 (T/1000 K)^0.7 erg cm^-1 s^-1 K^-1), |(1/r^2) d/dr(r^2 kappa dT/dr)| against the local heating, and the conductive flux F_c against the advected enthalpy flux of the wind

(`LHS1140b/models/.P1/atomic97/conduction_vs_heating.py <Hydro_ioniz.txt>`, from the columns of the state file and R0 of its grid line):

| state | r = 1.001 | 1.01 | 1.05 | 1.1 | 1.2 | 1.5 | 2.0 R_p |
|---|---|---|---|---|---|---|---|
| atomic scalar 0.06 He/H 9.7 (section 38, certified) | 99 | 55 | 5.2 | 51 | 42 | 22 | 17 |
| atomic photochem 0.10 He/H 9.7 (catalog) | 67 | 31 | 3.6 | 30 | 23 | 9.0 | 7.1 |
| atomic scalar 0.33 He/H 9.7 (catalog) | 15 | 8.0 | 2.5 | 7.9 | 6.8 | 2.9 | 1.2 |
| atomic scalar 1.0 He/H 9.7 (catalog) | 4.1 | 2.1 | 0.8 | 2.3 | 1.9 | 1.0 | 0.34 |
| atomic scalar 1.0 He/H 0.55 (catalog) | 0.8 | 1.1 | 0.5 | 0.5 | 0.6 | 0.4 | 0.06 |
| molecular photochem He/H 9 (catalog) | 73 | 4.0 | 1.6 | 1.5 | 0.02 | 1.1 | 0.45 |

(|conduction| / heating.) On the 0.06 state F_c is 1.5 to 3.5 erg cmˆ-2 sˆ-1 from the base to 1.3 R_p against an advected enthalpy flux of 0.014 to 0.15; on the fiducial atomic He/H 9.7 state 2 to 3.4 against 0.3 to 2.8. The lower layers of the LHS 1140 b catalog are therefore solved without a term of the energy equation that is of order one and, at reduced XUV, one to two orders above the heating. *Likely* this is also why the reduced-XUV models are so stiff: without conduction the temperature of a layer whose wind runs at Mach 1e-8 is set by the heating divided by the mass flux, and the element enthalpy flux, of the order of the wind's own, moves it by a large factor (section 38). Not established: whether the temperature structure with conduction differs by as much as the ratio suggests; the published models that include conduction (Koskinen et al. 2013a, 2022; Yelle 2004) state it as small for their hot-Jupiter cases, whose base wind is 2.5 km/s (Koskinen 2022) and not Mach 1e-8.

**Conductivity of a helium-dominated gas.** The code applies the atomic hydrogen value whatever the composi-

tion. Yelle (2004), eqs. 10-11, forms the neutral conductivity of a mixture from the constituents' values (Banks & Kockarts 1973) with a Wilke-type rule; at He/H = 9.7 the helium value decides it. Banks & Kockarts (1973) is not in `references/`; the value is not adopted until it is read in the original.

**Test.** From the certified 0.06 term-off state, `Conduction: True` switched on at once (binary `EXHALE_a1c14b9c.x`, tree source of section 39; `*_cond_enthoff`, `*_cond_enthon` with the element enthalpy flux on at s = 1): the loaded ||R|| is 1.00 and 1.84 (the conduction term alone), and the first stationary solve of each STAGNATED after 90 iterations and returned the entry state. A state with conduction is not reached from one without it in one solve either; both runs were stopped after their first pass.

Open, for the user: whether conduction becomes part of the LHS 1140 b model (it changes every model of the catalog), with which conductivity for the helium-rich mixtures, and by which route the states with it are reached (a continuation factor on kappa as on the element enthalpy flux, or time marching with the implicit conduction update).

## 41. Heat conduction with the conductivity of the mixture, and its continuation factor (2026-09-30)

User decision of 2026-09-30: heat conduction enters the LHS 1140 b model, reached by continuation in the conductivity; the no-conduction runs were stopped (the 0.0624 s chain had ended at s = 0.181, the XUV descent was stopped at x = 0.0573, rung 4; the scalar 0.01 run had been refused at pass 20).

**The conductivity** (`viscous_conduction.f90`, section 2, Eq. 4'), read in the published versions:

```
kappa = (1/n) [ n_e 1.2e-6 T^2.5 + n_HI 379 T^0.69 + n_HeI 299 T^0.69 ]   erg cm^-1 s^-1 K^-1
```

Salz et al. (2015, A&A 576, A21), section 3.4.2, Eqs. 24-26 (electron-ion, Spitzer; neutral hydrogen; the number-weighted sum), and Sutton et al. (2015, JGR 120, 6884), Eq. 5 (the helium coefficient 299 T^0.69 in the same weighted form); both quote Banks & Kockarts (1973), which was not available. n counts every particle (electrons, atoms, ions, metals); the ions add nothing of their own (Yelle 2004, Eq. 9, gives 7.4e-8 T^2.5/sqrt(mu), below 1e-2 of kappa_H at 1e4 K). The previous coefficient, Watson et al. (1981) 4.45e4 (T/1000 K)^0.7 (verified in Erkaev et al. 2016, Eq. A6), was the neutral hydrogen value applied to any composition; the hydrogen term now is Salz Eq. 25, within 2 per cent of it from 300 to 7000 K (1.013 at 300 K, 1.0006 at 1000 K, 0.981 at 7000 K). On the 0.06 state the mixture coefficient is 0.81 of kappa_H where helium is 90 per cent of the particles. The viscosity keeps the hydrogen value (5); it is not used by any LHS 1140 b case. None of the three sources gives an H2 coefficient; the H2 term is n_H2 k_H2(T) with k_H2 the tabulated conductivity of Incropera et al. (2007, Fundamentals of Heat and Mass Transfer, 6th ed., Table A.4), 100 to 2000 K, linear in (ln T, ln k) and continued outside with the exponent of the end interval (an extrapolation), `h2_conductivity` (the user supplied the book, 2026-09-30; Holman, Heat Transfer, 10th ed., Fig. 1-4, shows H2 and He only up to 500 C). The same table gives helium 0.152 and 0.354 W m^-1 K^-1 at 300 and 1000 K, against 299 T^0.69 = 1.53e4 and 3.51e4 erg cm^-1 s^-1 K^-1: the helium coefficient of Sutton et al. agrees to 1 per cent. (A first version of this entry stopped molecular runs with conduction; that stop is removed.) The table was first interpolated in (ln T, ln k); at the user's request (2026-09-30) it is replaced by a fitted function, kappa_H2 = A T^s [1 + c E(theta_v/T)], E(x) = x^2 e^x/(e^x - 1)^2, A = 272.523, s = 0.735652, c = 0.370446, theta_v = 5987 K (the v = 0 -> 1 spacing of the Roueff et al. 2019 ladder the code carries): the Eucken form, fitted over 200-2000 K, largest deviation

2.5 per cent (200 K), rms 1.05 per cent; valid 200-2000 K, an extrapolation outside (+7.6 per cent at 150 K, +20 per cent at 100 K where rotation freezes out). Other forms measured on the same table: a power law 6 per cent, Sutherland 2 to 12 per cent, a cubic in ln T 1 per cent (untrustworthy outside). The fit has a continuous derivative (the interpolation kinked at 25 nodes); above 2000 K it lies 3 and 6 per cent below the end-slope continuation at 3000 and 4000 K, neither being a measurement. Linked probe: 200/300/1000/2000 K give 1.025, 0.989, 1.013, 0.986 of the table. No run with molecules and conduction has been made with either form. Yelle (2004) Eq. 10, the Wilke-type mixture rule, was not used: as printed it is not dimensionally the Wilke form (a factor 1/r in the sum), *likely* a typesetting error.

**Code.** `thermal_conductivity(Tcell, f_sp, kap)` forms the number fractions from the composition of the same cells (calc_ne, calc_ntot: the equation of state's own counts), and the composition is passed down `thermal_conduction_coeffs`, `thermal_conduction_source`, `viscous_conduction_sources` (steady residual) and `viscous_conduction_step` (marching); the test driver `src/tests/energy_update/conduction_floor_tests.f90` passes a neutral-hydrogen composition. Continuation factor `EXHALE_CONDUCTION_SCALE = s` (`conduction_scale()`), 1 when unset; the `cond` token of the options field carries s wherever s /= 1, so a restart across a change of s names `cond` (the rule of section 39, R1); the input echo and the setup report state the coefficient and the factor.

**Checks.** gfortran and ifx builds; `src/tests/energy_update/run.sh`: 54 PASS, 0 FAIL. `Restart intent: stationary evaluate` of the certified 0.06 term-off state with `Conduction: True` and the budget export: the conduction term of the energy row (Sene, viscosity off) over the heating agrees with an independent evaluation of Eq. 4' from the written columns to five digits at every radius tested (-46.04 at 1.003, +41.40 at 1.10, -34.07 at 1.20, +12.42 at 3.0 R_p).

**Continuation, first attempt** (`.P1/atomic97/factor_continuation_chain.sh`, new: a rung is accepted only when certified, the chain ends successfully only at s = 1 certified with exit 0, every rung names its token; binary `EXHALE_0c5924c7.x`): from the certified term-off 0.06 and 0.0624 states, the first rung s = 0.005 returns its entry state from every hydrodynamic solve (hydro info = 2, ||R|| 1.000 at the energy row of cell 1). The row of cell 1 is dominated by the conduction through the base face, where the operator holds the reservoir's temperature (Dirichlet, `conduction_base_level_T`) and the no-conduction states carry a large temperature jump across half a 2e-4 R_p cell; since the row is measured against its own largest term, its relative residual stays at 1 for any s. Both chains were stopped at pass 6 of rung 1. A smaller factor cannot help; the obstacle is mechanism B of section 40.

**Mechanism B measured** (`.P1/atomic97/diag_R/B1_jump_scan_cond0.005`, `EXHALE_RESID_JUMP_SCAN=1` at the entry state of that rung): along the first Krylov direction the residual is smooth from 2 arcs down to 3e-5 arcs of the probe step; below that the only steps are rounding (the largest moves its row by 4e-7 of it, 0.6 to 1.7 times the arithmetic bound epsilon x face state / width). A discontinuous residual is therefore not what rejects the steps of lam 1e-6 at dtau 8e-5. At such dtau the step is dY ~ -dtau F, the explicit pseudo-time step, and its rejection means the merit (the scaled 2-norm of F) rises along the physical relaxation direction. *Likely* this is the obstacle: the pseudo-transient continuation is globalized by a merit line search, which can refuse a relaxation that is stable in time but not monotone in that norm. Kelley & Keyes (1998, SIAM J. Numer. Anal. 35, 508), abstract: "Standard globalization strategies such as line search or trust region methods often stagnate at local minima. Ψtc succeeds in many of these cases by taking advantage of the underlying PDE structure of the problem"; their pseudo-time step is set by the residual reduction (SER, their Eq. 1.5),

and they report a nonmonotonic steady-state residual norm at the outset of the iterations for iterates far from the steady state. Changing the acceptance of the solve is a solver change and is held for the user (the stage 2 solver program is closed, 2026-09-10).

## 42. Pseudo-transient steps accepted by the pseudo-time control (2026-09-30)

User decision of 2026-09-30: the acceptance of the pseudo-transient solve may be changed (the stage 2 solver program is reopened for this item).

**What was changed** (`steady_newton.f90`, `solve_steady_jfnk`; measurement key, off by default, so no solve without it moves). `EXHALE_PTC_SER=1|2`: a trial of the line search is accepted when it is admissible (the existing `admissible` test of `eval_residual`: positive, finite, a chemical root in every cell) and no cell's density or total energy moves by more than `EXHALE_PTC_SER_MAX_CHANGE` (default 0.3) of itself; otherwise lam is halved. No merit descent is asked. The returns to the best iterate after 20 iterations without improvement, the stagnation stop after 40, the return after a run of damped steps, and the damped Gauss-Newton escape replacing an accepted step are not taken; the solve ends at its gates, at its iteration cap or after 12 consecutive iterations without an admissible trial, and hands back its LAST iterate unless the best one met the gates (its iterates are a relaxation path, not a search). The pseudo-time: `=1`, SER (Kelley & Keyes 1998, Eq. 1.5), dtau <- dtau min(2, max(1/2, f2/f2_try)); `=2`, doubled on a full step; both times lam on a shortened step. The outer alternation adopts the returned state whatever info is (`EXHALE_main`, after `solve_steady_jfnk`), so progress carries over.

**Measured** (`.P1/atomic97/diag_R/SER/`, one outer pass, cap 400 iterations, binaries `EXHALE_d0d5bf27.x` (=1) and `EXHALE_9d0b8d4c.x` (=2); every fixture returned its entry state under the merit line search):

| fixture | merit line search | SER=1 | SER=2 |
|---|---|---|---|
| T1: 0.06, conduction factor 0.005 (section 41 rung 1) | entry state, | | R |
| T3: 0.0624, no conduction, element enthalpy factor 0.125 -> 0.2375 (section 40, B) | entry state, | | R |
| T2: 0.06, conduction factor 1 at once | entry state | | |

Under `=2` every step of T1 is taken at lam = 1 with dtau doubling from 2 to 4.2e6; T3's ||R|| rises from 0.06 to 1.2 at iterations 10 to 16 (a nonmonotone transient the merit line search refused) and then converges quadratically. `=1` keeps dtau near its entry value because the merit falls by about a per cent per step. `=2` is the one used from here. Pass 1 of T3 leaves the elemental He/H row at 0.23: mechanism A (the diverging hydro / element alternation, section 40) is a separate question, which the continuation chains meet with their certified-only acceptance.

**Continuation restarted** (`.P1/atomic97/launch_conduction_then_enthalpy.sh`, lart3, `CHAIN_ENV=EXHALE_PTC_SER=2`): from the certified term-off 0.06 and 0.0624 states, conduction by

EXHALE_CONDUCTION_SCALE (S0 = 0.005), then, from the certified full-conduction state, the element enthalpy flux by EXHALE_INTERDIFF_ENTH_SCALE (S0 = 0.05); logs <case>_ser2_cond_chain.log, <case>_ser2_cond_interdiff_enth_chain.log.

## 43. The continuation chains, first three hours, and their step control (2026-09-30)

With EXHALE_PTC_SER=2 (section 42) every conduction rung below s ~ 0.14 certified in 7 to 29 passes (5 to 15 minutes): 0.06 up to s = 0.0673, 0.0624 up to 0.137, the photochemical 0.10 up to 0.0673 (its term-off state had certified first, in 9 passes, where the merit line search had refused it). Two defects of the chain script were measured and fixed (.P1/atomic97/factor_continuation_chain.sh):

- the step ratio grew as q^1.5 on every success, so 0.06 jumped from 0.0673 to 0.698 (x10.4) and the photochemical case to 0.32; a rung that fails under EXHALE_PTC_SER=2 runs to the solve's iteration cap (3000, the stagnation stop being off there), 2 h 20 min for the 0.06 rung (hydro info = 2, elemental row 0.89). Now the ratio is capped (Q_CAP, 2) and every solve of a chain carries EXHALE_JFNK_MAXIT=400 (the certified solves take 22 to 28 iterations);
- 0.0624 was refused twice at 30 passes with the elemental row at 1.35e-5 and 1.07e-5 against 1e-5, i.e. still converging; a refused rung whose hydrodynamic solve converged and whose elemental row is below NEAR (1e-4) is now continued once at the same factor from its own state before the step is shortened. The factor at which the alternation of 0.0624 diverges (mechanism A) is between 0.137 (certified) and 0.225 (refused, elemental row 7.8e-2 at pass 5). The chains were stopped at 06:35 and restarted from their last certified rungs (S_START) by one wrapper for all three cases, .P1/atomic97/launch_full_equation_from.sh (conduction to 1, then the element enthalpy flux, then for 0.06 and the photochemical case the XUV descent to 0.01).

The cond and interdiff_enth tokens carry the factor in F7.5, five decimals: two factors that differ below 1e-5 carry one token.

## 44. The outer under-relaxation sets the pace of the chains: omega = 1 (2026-09-30)

In every certified conduction rung the elemental He/H row falls, after a transient of 5 to 8 passes, by 0.505 per pass (three rungs tabulated: photochemical 0.32 -> 0.453, scalar 0.0624 0.194 -> 0.306, scalar 0.06 0.190 -> 0.247), which is 1 - omega at the default omega = 0.5 of the element step (EXHALE_main, comp_omega): the undamped alternation contracts by about 0.01 per pass there, and the damping alone makes the 21 to 22 passes of a rung. The existing key EXHALE_DIFF_OMEGA starts omega at the value given (the halving on a pass without progress stays). Three known rungs rerun at omega = 1, everything else as the chains (.P1/atomic97/diag_R/omega1/):

| rung | omega = 0.5 (chain) | omega = 1 |
|---|---|---|
| photochemical 0.32 -> 0.453 | certified, 22 passes, 1 h 53 min | certified, 6 passes, 40 min |
| scalar 0.06 0.190 -> 0.247 | certified, 21 passes | certified, 6 passes, 51 min |
| photochemical 0.453 -> 0.761 | refused at pass 5 (He/H row 0.099, mechanism A) | certified, 7 passes, 48 min (one pass at hydro info = 2 halved c |

At omega = 1 the He/H row falls by about ten a pass (6.5e-2, 3.2e-2, 3.8e-3, 3.4e-4, 2.8e-5, 2.3e-6). *Likely* part of mechanism A (section 40) was the damping itself: an element step held back by half while the hydrodynamic solve answers in full. The chains were stopped at 13:09 and restarted with `EXHALE_DIFF_OMEGA=1.0` added to their `CHAIN_ENV` (`launch_full_equation_from.sh`) from the highest certified states: photochemical 0.761 (the omega = 1 rung), scalar 0.0624 0.410, scalar 0.06 0.300. The default of `comp_omega` is not changed.

## 45. The first state with heat conduction: a conductive thermosphere below an exobase at 1.5 R_p (2026-09-30, open)

The photochemical K_zz(p) He/H 9.7 case at 0.10 of the fiducial XUV reached conduction factor 1 (the equation: `Conduction: True`, mixture kappa of section 41, element interdiffusion enthalpy flux still off) certified at 13:27 (`.P1/atomic97/atomic_photochem_gj1132x0.10_kzzprofile_HeH9.7_ser2_cond01_s1`, 14 passes, 17 min, from the omega = 1 rung at 0.761). Against the certified state of the same case without conduction (`..._v5proj_enthoff_ser2`), MEASURED (`.P1/atomic97/compare_conduction_on_off.py`, figure `.P1/atomic97/diag_R/cond_on_off_photochem_x010.png`):

| r [R_p] | T [K] off -> on | n [m_H cm^-3] off -> on | v [cm/s] off -> on |
|---|---|---|---|
| 1.01 | 1041 -> 494 | 6.4e12 -> 4.4e12 | 4.0e-2 -> 2.1e-4 |
| 1.10 | 3463 -> 940 | 7.2e10 -> 1.2e9 | 3.0 -> 0.64 |
| 1.50 | 7246 -> 933 | 1.2e9 -> 2.0e6 | 94 -> 210 |
| 3.0 | 3182 -> 900 | 2.8e7 -> 1.3e3 | 1.0e3 -> 7.9e4 |
| 10 | 740 -> 719 | 2.9e5 -> 19 | 8.7e3 -> 4.9e5 |

The thermosphere is nearly isothermal at 900-940 K from 1.05 to 5 R_p (the XUV heat conducted down to the 226 K reservoir), in place of a 7290 K peak at 1.43 R_p; the mass-loss rate falls from 7.3e6 to 2.7e4 g/s (median over r >= 1.1 R_p, x0.004); the He(2^3S) density falls by decades above 1.1 R_p. `src/utils/collisional_validity.py` on the two states: without conduction the domain is subsonic (max Mach 0.07) with Kn_bulk 0.40 at the outer boundary; WITH conduction the exobase (Kn_bulk = 1) is at 1.52 R_p, Kn_bulk = 0.1 at 1.30 R_p, the critical point at 8.8 R_p, and Kn reaches 859 in the acceleration region: the hydrodynamic Mdot (log 4.41) is UNVALIDATED, and Jeans escape through the exobase (T = 932 K, lambda_J = 17) gives 9.8 g/s. *Likely* the low-XUV LHS 1140 b atmosphere with conduction is a conduction-limited thermosphere in the Jeans regime, the structure of the Earth's thermosphere, and not a hydrodynamic wind; the continuum solution above ~1.3 R_p is outside its validity. Not established: the fiducial-XUV cases with conduction (not yet solved), and the element enthalpy flux, whose continuation from this state is running.

**Stale (2026-10-07):** the Knudsen numbers, `Kn_bulk = 0.1` radii and exobase radii in this passage were computed by `src/utils/collisional_validity.py` before its resonant H+ + H and He+ + He momentum-transfer coefficients (`md/Update_EXHALE_stage3.md` section 110). On the 102 stored LHS 1140 b states the change moves the `Kn_bulk = 0.1` radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.

## 46. Heat conduction in the fiducial-XUV cases (2026-09-30, running)

User decision of 2026-09-30: beside the low-XUV chains, switch conduction on in the fiducial-XUV cases the observations are compared with, to decide the direction of the catalog. Started 14:16 on lart4 (`.P1/fiducial_conduction/launch_fiducial_conduction.sh`, binary `EXHALE_e37b8e27.x` with the H2 conductivity fit, `EXHALE_PTC_SER=2`, cap 400, omega = 1):

- `molecular_scalar_gj1132_kzz1e9/HeH1.45` (the He/H at which the model EW crosses the observed 1.108, section 14 of this log) and `molecular_photochem_gj1132_kzzprofile/HeH9`: certified 2026-09-29 on the face-at-level grid with the element and carrier enthalpy fluxes on (boundary model v4; loaded by the v5 binary as a seed); conduction by its factor from 0.005 to 1 (`factor_continuation_chain.sh`; rungs and logs in `.P1/atomic97/`).
- `atomic_scalar_gj1132_kzz1e9/HeH9.7` (previous grid, 2026-09-20): mapped, 230-cell projection, term-off solve, then conduction and the element enthalpy flux by `launch_full_equation_from.sh` (whose fixed binary `EXHALE_9d0b8d4c.x` differs from `e37b8e27` only in the H2 conductivity, absent in an atomic run). Entry inputs carry the spectrum file by absolute path (the catalog's `../../../sed/` does not resolve from `.P1/`).

## 47. The photochemical 0.10 case on the full equation (2026-09-30)

From the certified conduction state of section 45, the element interdiffusion enthalpy flux by its factor (0.05 -> 1, `interdiff_enth` chain, omega = 1): rungs 0.2, 0.4, 0.8 and 1 certified in 10 to 15 passes each; the full equation (conduction and the element enthalpy flux, factor unset) certified at 14:22 (`.P1/atomic97/atomic_photochem_gj1132x0.10_kzzprofile_HeH9.7_ser2_cond_interdiff_enth06_s1`).
Against the conduction-only state, MEASURED (`compare_conduction_on_off.py`): Mdot 2.711e4 -> 2.725e4 g/s (+0.5 per cent), T within 0.5 K at every radius tested (943.4 K at 1.2 R_p), n and v within 0.5 per cent; `collisional_validity.py`: exobase 1.518 R_p, critical point 8.81 R_p, Kn up to 855 in the acceleration region, Jeans escape 9.88 g/s; HYDRODYNAMIC RESULT UNVALIDATED as in section 45. With conduction on, the element enthalpy flux, which without it was 1.1 to 2 times the heating of the lower layer (section 38) and the hardest term to switch on, moves the solution by less than a per cent: conduction carries the energy of that layer. The XUV descent 0.10 -> 0.01 on the full equation started from this state (`atomic_photochem_gj1132_kzzprofile_HeH9.7_full_xuv_descent.log`).
**Stale (2026-10-07):** the Knudsen numbers, `Kn_bulk = 0.1` radii and exobase radii in this passage were computed by `src/utils/collisional_validity.py` before its resonant H+ + H and He+ + He momentum-transfer coefficients (`md/Update_EXHALE_stage3.md` section 110). On the 102 stored LHS 1140 b states the change moves the `Kn_bulk = 0.1` radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.

## 48. The fiducial-XUV atomic case with conduction (2026-09-30)

`atomic_scalar_gj1132_kzz1e9/HeH9.7` (fiducial XUV), mapped from the previous grid and projected (section 46): the term-off solve certified in 2 passes; conduction by its factor 0.005 -

> 1 certified in 9 rungs of 1 to 2 passes each, 14 minutes, reaching the equation at 14:34 (`.P1/atomic97/atomic_scalar_gj1132_kzz1e9_HeH9.7_fid_ser2_cond09_s1`; the element enthalpy flux continuation follows). Against the term-off state, MEASURED (`compare_conduction_on_off.py`, figure `.P1/atomic97/diag_R/cond_on_off_atomic_scalar_fiducial.png`; `collisional_validity.py`):

| | no conduction | conduction |
|---|---|---|
| T max | 7826 K at 1.434 R_p | 6342 K at 1.419 R_p |
| T at 1.05 / 1.5 / 3.0 R_p | 2433 / 7784 / 4365 K | 2798 / 6298 / 3741 K |
| Mdot (median r >= 1.1) | 7.57e7 g/s | 5.25e7 g/s (x0.69) |
| critical point | none in domain (max Mach 0.508) | none (max Mach 0.51) |
| Kn_bulk at the outer boundary | 0.060 | 0.066 |

At the fiducial XUV conduction redistributes the heat (the lower layer warms, the peak cools by 19 per cent) and lowers the mass-loss rate by 31 per cent; the solution stays a subsonic continuum wind with the exobase above the domain, UNVALIDATED on the same grounds as before (Kn 0.7-1.1 in the acceleration region, no critical point in the domain). At 0.10 of that XUV (section 45) the same term turns the wind into a conductive thermosphere with an exobase at 1.5 R_p. The effect on the He 10830 equivalent width is not yet computed.

**Stale (2026-10-07):** the Knudsen numbers, `Kn_bulk = 0.1` radii and exobase radii in this passage were computed by `src/utils/collisional_validity.py` before its resonant H+ + H and He+ + He momentum-transfer coefficients (`md/Update_EXHALE_stage3.md` section 110). On the 102 stored LHS 1140 b states the change moves the `Kn_bulk = 0.1` radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.

## 49. The advected temperature correction keeps the transport terms (2026-09-30)

**Defect.** The advection-corrected profiles (`post_process_adv`, the `*_adv.txt` files the He 10830 synthesis reads) re-derive the temperature from the steady internal-energy equation (E), div(u v) + p div v = heating - cooling, which holds neither heat conduction nor the enthalpy flux of the element and carrier fluxes. On a run that solved them the corrected temperature is then the answer to another equation: MEASURED on the fiducial atomic case on the full equation (section 48), 7939 K against the run's 5489 K at 1.2 R_p and 8479 K against 6259 K at 1.5 R_p.

**Fix** (user decision 2026-09-30: the consistent form, conduction of the corrected temperature). `T_equation` takes heating = photoheating + `heat_extra` + `cond_diag` T (new `teq_state` fields, zero for a run without the terms: the residual is unchanged); `post_process_adv` fills them from the conduction triplets of the run's operator (`thermal_conduction_coeffs`, now public) and the enthalpy flux divergence of the element (`interdiffusion_enthalpy_divergence_of_state`) and carrier (`carrier_enthalpy_divergence`) fluxes. Conduction couples each cell to both neighbors, so after the marching sweep (cell below at its corrected value, cell above at the pass's profile) the corrected cells are solved together by a Newton iteration on the column (`energy_column_newton`: banded Jacobian, two sub- and one super-diagonal, by one-sided differences; the conductivity and the enthalpy flux divergence re-evaluated at each iterate; steps capped at 30 per cent; converged at a relative step of 1e-10; a solve that

fails returns the marching profile and is reported). Cells that keep the run's temperature are fixed values of the solve.

**Checks.** gfortran and ifx builds; `src/tests/adv_static_limit`: 53 PASS, 0 FAIL; `src/tests/energy_update`: 54 PASS (section 41). A run without the terms (the fiducial term-off state): the column solve does not run, the solved state is bitwise unchanged, and the `*_adv` rows move by at most 1.4e-11 relative (the ifx build re-associates the heating sum; below the standing 0.1 per cent rule). The full-equation fiducial state: the column solve converges in 4 to 7 iterations each pass (relative step 5e-11, 502 cells); the corrected temperature follows the run's where conduction carries the energy (1129/5507/6346 K against 1127/5489/6259 K at 1.01/1.2/1.5 R_p) and departs from it beyond 3 R_p, where the non-equilibrium composition changes the balance (figure `.P1/atomic97/diag_R/adv_transport_terms_fiducial.png`).

**He 10830 EW** (WINERED kernel, the catalog's operator, `.P1/fiducial_conduction/ew_of_state.sh`; observed 1.108 +- 0.029):

| state | adv | solution (no advection correction) |
|---|---|---|
| term-off (no conduction, no element enthalpy flux) | 1.573 | 12.99 |
| full equation, adv without the transport terms | 0.946 | 7.52 |
| full equation, adv with them | **0.649** | 7.52 |

On this case heat conduction and the element enthalpy flux lower the model EW from 1.57 to 0.65 (x0.41); the He/H at which the model EW crosses the observed value (1.5 in the catalog, section 14) will move. The previous EW of every conduction run is superseded by this operator.

## 50. A floor of the energy row near convergence (2026-09-30; located in section 53, explained and removed in section 58)

Two chains slow down on rungs whose hydrodynamic solve ends at the 400-iteration cap (info = 1): the 0.0624 conduction rung at factor 0.751 (He/H row 1e-7, energy row 1e-5 to 5e-5 against its tolerance 1e-6 for twelve passes of 140 s each; the rung then CERTIFIED at pass 15, 15:44, when one pass's state fell inside every tolerance) and the first XUV descent rung of the photochemical case (0.10 -> 0.0833, passes of 900-1000 s, running). The rungs are not stuck but their certification rests on a fluctuation. In both the solve reaches ||R|| ~ 5e-5 within 25 iterations and then stays between 5e-6 and 5e-5 for the rest of the cap, at dtau = 1e14, with every step accepted.

Tried: `EXHALE_PTC_SER_RNORM` (default 1e-3), the pseudo-time acceptance of section 42 only while ||R|| is above the switch and the merit line search below it (binary `EXHALE_1543a8f2.x`). The two fixtures of section 42 still certify in 22 and 28 iterations (T1 ||R|| 3.5e-7, T3 1.5e-7). On the 0.0624 rung 0.579 -> 0.751 (`.P1/atomic97/diag_R/ser_phase/P1_*`) the transient is crossed as before (||R|| 1.9 -> 5.7e-5 in iterations 19 to 24) and the merit phase then wanders at the same 1e-5 to 5e-5 for 376 iterations, the Krylov forcing term pinned at its ceiling 0.1 (the residual not falling, the Eisenstat-Walker rule does not tighten it). The stall is therefore not the acceptance rule. *Likely* the energy row carries a floor of that size on this state (candidates: the tolerance of the composition elimination inside the residual, an ionization root switching branch); not measured. The test was stopped after pass 1. The switch is kept (it changes nothing on the fixtures and is the standard terminal phase), with this record that it did

not cure the stall.

## 51. The molecular photochemical fiducial case with conduction (2026-09-30; EW corrected by section 54)

`molecular_photochem_gj1132_kzzprofile/HeH9` (fiducial XUV; the molecular network with the element and carrier enthalpy fluxes, section 46) reached conduction factor 1 certified at 16:22 in 9 rungs of 10 to 15 passes (`.P1/atomic97/molecular_photochem_gj1132_kzzprofile_HeH9_cond09_s1`; binary `EXHALE_e37b8e27.x`, the first molecular run with the H2 conductivity). Against its catalog state (no conduction; boundary model v4, re-measured by the current binary, whose evaluate refuses the old certificate, exit 2, as expected across the boundary change), with the advected profiles of section 49 (binary `EXHALE_1543a8f2.x`) and the WINERED operator, MEASURED:

| | no conduction (catalog) | conduction |
|---|---|---|
| T max | 7202 K at 1.525 R_p | 6194 K at 1.441 R_p |
| T at 1.001 / 1.1 / 1.5 / 3 R_p | 985 / 2878 / 7197 / 4350 K | 444 / 3788 / 6168 / 3732 K |
| Mdot (median r >= 1.1) | 9.78e7 g/s | 5.95e7 g/s (x0.61) |
| He 10830 EW, adv | 1.836 | **1.103** |
| He 10830 EW, solution | 14.81 | 8.56 |
| critical point / Kn at the outer boundary | none (Mach 0.57) / 0.059 | none (Mach 0.53) / 0.065 |

(Figure `.P1/atomic97/diag_R/cond_on_off_molecular_photochem_fiducial.png`.) The observed EW is 1.108 +- 0.029 (`he10830_equivalent_width.py`). **Correction (section 54):** in this evaluation all ten column energy solves of the advected profile were rejected (`marching profile kept: T`, `.P1/fiducial_conduction/ew_M2_s1/eval.log`). The 1.103 is therefore the EW of the retained marching profile, not of a converged corrected-energy profile, and its agreement with the observation is not established. With the atomic fiducial case (section 49, 1.57 -> 0.65) conduction lowers the model EW by a factor 0.4 to 0.6 at the fiducial XUV, so the catalog's inference of the He/H that reproduces the observation (the crossing near 1.5 of section 14, molecular scalar family) no longer holds. Where the crossing now lies is undetermined: this case and the molecular scalar He/H 1.45 case of section 52 belong to different families. Both hydrodynamic results stay UNVALIDATED on the grounds of section 48 (no critical point in the domain, Kn up to 1 in the acceleration region).

## 52. The molecular scalar He/H 1.45 case with conduction (2026-09-30)

`molecular_scalar_gj1132_kzz1e9/HeH1.45` (the He/H at which the catalog's model EW crossed the observed value, section 14) reached conduction factor 1 certified at 16:39 (9 rungs of 12 to 20 passes; `.P1/atomic97/molecular_scalar_gj1132_kzz1e9_HeH1.45_cond09_s1`). MEASURED as in section 51 (advected profiles of section 49, binary `EXHALE_1543a8f2.x`; figure `.P1/atomic97/diag_R/cond_on_off_molecular_scalar_HeH1.45.png`):

| | no conduction (catalog state, re-measured) | conduction |
|---|---|---|
| He 10830 EW, adv | 0.864 | **0.153** |
| He 10830 EW, solution | 9.17 | 2.15 |
| T max | 5036 K at 1.465 R_p | 3711 K at 1.316 R_p |
| T at 1.001 / 1.01 R_p | 1432 / 1453 K | 481 / 993 K |
| Mdot (median r >= 1.1) | 7.75e7 g/s | 2.92e7 g/s (x0.38) |
| critical point / Kn at the outer boundary | none (Mach 0.57) / 0.090 | none (Mach 0.65) / 0.172 |

The catalog carries EW 1.0964 for this case (section 14); re-measured here by the current binary with the advected profiles of section 49 it gives 0.864. Which part of that difference is the operator (the transport terms of the advected energy equation) and which is the re-evaluation by the current boundary (v5) was not separated. With conduction the EW falls by a factor 0.18, to 0.14 of the observed 1.108 +- 0.029, while the molecular photochemical He/H 9 case falls onto it (section 51): with conduction the He/H that reproduces the observation is higher than the catalog's 1.5, between 1.45 and 9 on these two families (not the same family; not yet bracketed on one).

## 53. Where the energy-row floor of section 50 comes from (2026-09-30; the cause is in section 58)

The floor of section 50 (hydrodynamic solves ending at the iteration cap with ||R|| 1e-5 to 1e-4 against the energy tolerance 1e-6) was measured on the 0.06 conduction rung at factor 0.773561 (IC: pass state r20260930T172935.418_c1790756975418473_p22 of `atomic_scalar_gj1132x0.06_kzz1e9_HeH9.7_ser2_cond08_s0.773561`; runs in `.P1/atomic97/diag_R/floor_*`, binaries `EXHALE_9d0b8d4c.x` and, for the new diagnostics below, `EXHALE_dfb51387.x`). Every run starts at the same ||R|| = 3.882e-5, energy row of cell 296.

**The row.** At cells 290 to 312 (r = 1.75 to 2.06 R_p) the energy row is the difference of the advective divergence and the conduction source, each ~1e-10 in the row's units, with heating ~4e-12; the residual is +-1e-15 to 4e-15, alternating in sign from cell to cell, i.e. 1e-5 to 4e-5 of the row's scale. The temperature there is 748 K and flat: the relative first difference of T is 4.3e-5 to 5.3e-5 per cell and the relative second difference 1.4e-6 to 2.1e-6 per cell. The H ionization crosses x(H+) = 0.5 between cells 306 and 307.

**Excluded, by measurement.**

| candidate | test | result |
|---|---|---|
| acceptance rule | `EXHALE_PTC_SER_RNORM` (section 50) | same floor |
| scaling of the linear model | `EXHALE_JUDGED_ROWS`, `EXHALE_JUDGED_MERIT` | cap at 1.5e-5 / 5.3e-5 |
| probe length of the Jacobian action | `EXHALE_JV_PROBE_ARC` = 10, 100; `EXHALE_JV_COLUMN_SCALE=1` | 40 iterations, best 8.2e-6 / |
| frozen WENO3 weights in the model | new `EXHALE_JV_FRESH_WEIGHTS=1` (products with the weights recomputed) | first 9 iterations identical |
| tolerance of the per-cell ionization solve | `EXHALE_IEQ_TOL` = 1e-12, 1e-14 | start |
| sonic point, Roe entropy fix | Mach number of the band | 0.005 to 0.013: subsonic, |

The additivity test (`EXHALE_JV_ADDITIVITY=1`) puts its defect in cells 1 to 32 (mass and momentum rows; along directions off the Krylov basis the second difference does not scale with the arc at cell 1), not in the band. That non-smoothness at the base is recorded here and not pursued.

**Located.** `EXHALE_LINEAR_ROWS=1` now prints, cell by cell over a band (`EXHALE_LINEAR_ROWS_CELLS=lo,hi`, default 290,312), the energy row F, the model's F + J dY, the row actually found at Y + dY, their ratio, and the difference quotient (F(Y + t dY) - F)/t for t = 0.1 to 1e5. (The "linear residual left over the row of F" printed before is the column maximum in raw units, which the base cells set; it says nothing about this band.) On the first iteration:

- cells 290 to 304 and 308 to 312: actual change / predicted = 0.90 to 1.13, the row falls by ~10 per step;
- cells 305, 306, 307: 1.91, 2.05, 1.41; the row changes sign at the same magnitude (306: +1.42e-15 -> -1.49e-15) and does so again on the next iteration (ratios 2.03, 4.62, 0.43).

The difference quotient along dY is constant within 10 to 20 per cent for t >= 10 and at t = 0.1, and doubles at t = 1 in cells 305 to 307 only (306: -1.28, **-2.91**, -1.59, -1.39, -1.43, -1.42, -1.44 e-15): the row carries a deterministic jump of 0.6 to 1.5e-15 between states 1e-12 apart, i.e. 1e-5 of its scale. Heating and cooling in cell 306 jump by 3e-11 (relative) at the same t, and the H I / H II fractions of cells 306 and 307 by 1e-11 to 2e-11, ten times their neighbors', with a relative step of ~1e-12 in every cell.

**The source.** With `EXHALE_EQ_MEASURE_TRACE=1` every residual evaluation stops the composition elimination after 1 pass, on a particle-count increment of 6e-12 at cell 307 (the closure target is 1e-8). Forcing more passes (`EXHALE_RESID_EQ_TOL=1e-15`, `EXHALE_RESID_SC_MAX=12`), the increment at cell 307 stays at 3.0e-11 for all 12 passes: the elimination does not converge there. `EXHALE_IEQ_REPORT_CELL=307` shows x(H+) of that cell alternating between two values on successive passes,

```
5.030758860692102E-01   5.030758860599341E-01   (difference 9.3e-12)
```

each returned with a reaction residual of 1e-16 and solver info 1; with the cell tolerance at 1e-15 the two values are the same to the last digit and the solver returns info 4 (no further progress). *Likely* the elimination alternates between two nearby states. It is not established that the cell residual is flat (section 54): the printed residual is the normalized one, whose hydrogen turnover bound includes the Penning term with the total He density, and successive passes change the coefficients of the cell problem (T, n_e, attenuation), so the two x values can be roots of two slightly different problems.

**The amplification.** An indeterminacy of 1e-11 in the composition is an indeterminacy of 1e-11 in the particle count and in T = p/((n_tot + n_e) k). Where the row is the balance of advection and conduction on a profile whose relative curvature is 1.7e-6 per cell, that moves the conduction divergence by 1e-11/1.7e-6 = 6e-6 of itself, which is the row's scale. MEASURED jumps 0.6 to 1.5e-15 against a scale of 1e-10 agree with this. Newton cannot bring the row below the jump; the solve therefore wanders at 1e-5 to 1e-4 and a rung certifies only when one pass lands inside the tolerance (section 50).

**The same place in the other chains.** The worst energy cell of the last 40 iterations against the x(H+) = 0.5 crossing:

| rung | worst energy cells (count of 40) | crossing between cells |
|---|---|---|
| 0.0624, factor 0.913 | 297 (30), 290, 296, 292 | 296 / 297 |

| rung | worst energy cells (count of 40) | crossing between cells |
|---|---|---|
| photochemical descent, x = 0.0634 | 287 (20), 284, 282, 299 | 288 / 289 |
| 0.06, factor 0.774 | 301 (10), 303, 308, 307 | 306 / 307 |

**Not decided, for the user.** (i) Condition the cell ionization residual at x(H+) ~ 0.5, so that the composition is determined to rounding there. (ii) The energy row's certification scale in a conduction-dominated layer: it is the largest of the row's terms, and the conduction term is itself the difference of two face fluxes that are, by the first and second differences above, ~30 times larger; measured against the face fluxes the same jump would be ~2e-7 of scale (an estimate from those differences, not measured). Neither was changed. Remedy (ii) is withdrawn (section 54): a larger denominator admits a 30 times larger absolute imbalance and does not make the physical residual smaller.

**Also seen.** `EXHALE_RESID_SC_TRACE=8` from a second seed (the column's composition shifted by five cells) converges at fixed Y to a different composition (max relative difference 4.8 at cell 266, ‖R‖ 1.78 against 3.9e-5). The shifted seed changes the elemental totals too (hydrogen at cell 144 by 37 per cent, section 54), so the two runs solve different elemental problems. They do not show multiple fixed points of one problem.

**Code.** `EXHALE_JV_FRESH_WEIGHTS=1` (steady_newton.f90, measurement only) and the band and ladder printout of `EXHALE_LINEAR_ROWS` (one residual evaluation per ladder length) were added; with neither key set the run is unchanged (iteration 1 of the diagnostic run reproduces ‖R‖ = 1.945e-5). The comment of `residual_norms` (steady_residual.f90) said the two regions' rows are made dimensionless by different physics; `residual_row_scale` has no region switch, and the comment now says so. The chains were not touched.

## 54. The review of the 2026-09-30 handoff: findings checked (2026-09-30)

The review is `md/session_handoff_20260930_review.md` (source tree 60a90fb). Each finding was checked against the code or the saved products before being accepted. The plan that carries them out is `md/PLAN_20260930.md`, revised as `md/PLAN_20260930_rev1.md` after its review `md/PLAN_20260930_review.md`, and as `md/PLAN_20260930_rev2.md` after `md/PLAN_20260930_rev1_review.md`; `md/PLAN_20260930_rev3.md` is revision 2 re-checked against the code and the stored runs (the ionization-transport route's failed 09-23 attempt on this planet, the chains untouched by the derived-state exit code, He 2^3S transport expected). No code was changed at this step; the statements of sections 51 and 53 and of the handoff that the review showed to be wrong were corrected in place.

| review item | checked |
|---|---|
| 4.3 column rejection not propagated | `ew_M2_s1/eval.log`: `marching profile kept: T` on 10 of 10 passes; `ew2_fid_atomic_ful` |
| 4.1 transport on the input composition | `transport_terms_of_profile` passes `f_sp_in` to `thermal_conduction_coeffs`, the elemen |
| 4.2 step-only convergence | exit on `dmax < 1e-10`; no final residual evaluation |
| 4.4 fixed ten outer passes | `do k = 1,10` with no convergence test |
| 10.1 metal re-solve leaves an outdated n_e | `calc_ne` before the metal re-solve; `calc_mmw` and `p_out` after it use that n_e (only `pp_metals=` |
| 5 factor serialization | `write(opt_value,'(F7.5)')`: 0.773561 and 0.773564 give one token |
| 7 EW launcher | `ew_of_state.sh` defaults to `EXHALE_e37b8e27.x`, continues after a failed evaluation or transit, |

| review item | checked |
|---|---|
| 10.3 wrapper exit | launch_full_equation_from.sh ends exit 0 after the XUV descent |
| 8 control bytes | factor_continuation_chain.sh line 56 has literal 0x01 and 0x02 in place of \1, \2 |
| 10.2 H2 conductivity overflow | x*x*exp(x)/(exp(x)-1)**2 guarded only at x > 700; exp(x)**2 overflows above x ~ 355, i.e. |
| 2.1 label | report_accepted_cell_state prints x(min(4,nx)) as x(H2 nuclei); in the atomic He 2^3S b |
| 1 shifted seed | comp_b(k,:) = f_sp(min(N,k+5),:) moves elemental totals |
| 2 rounding-floor interpretation | the printed residual is normalized_reaction_residual; passes change T, n_e and attenuation |
| 2.2 composition stopping measure omits transport | increment_of_what_the_residual_reads reads particle count, H2 share, net radiative source |
| 3 face-flux certification scale | - |
| 6 crossing inference across families | - |

## 55. One failure contract for the advection-derived product (2026-10-01)

Plan md/PLAN_20260930_rev3.md A1, A2 and A6; the defect is finding 4.3 of section 54 (a rejected column energy solve left its rows labeled corrected and the evaluation exited 0). Implemented by the worker and checked on the diff and the runs.

**The column solve** (energy_column_newton, post_process_adv.f90) returns an outcome: *rejected* with its reason (linear solve, nonfinite, non-positive T, metal temperature band, iteration cap) and the triggering cell, or *step_passed_unverified* (the step test passed; no residual is formed after the last step, so it is not called converged). A rejected column keeps the marching profile and writes its WHOLE unknown set adv_failed (decision D2(i): no schema change; the ghost rows are written failed but not counted). For a metal-band rejection the log prints the triggering cell's energy row term by term in erg cm^-3 s^-1, at the column's T and at the run's T (R = q0 fv/(mup mum dr_step), the scaling of T_equation).

**The derived-state record.** post_process_adv returns adv_derived_state (chemistry, column outcome and reason, physical-cell counts, outer = unverified: the outer loop is still ten fixed passes, n_pp_passes); both callers print it and write_output writes it as # adv_derived_state: ... into both *_adv headers (schema and columns unchanged; the regression comparison excludes # lines). The evaluate route exits 7 when the record is rejected (2 stays the certification refusal of the input state, written first; 7 is used by nothing else in src/ or the shell consumers). read_adv_validity (exhale_transit_lib.py) refuses a row when adv_T_status /= 0 OR adv_comp_status = 2, returns refused_reason and derived_state (None = unknown); EXHALE_transit.py prints the record and refuses a rejected adv spectrum unless EXHALE_TRANSIT_DIAGNOSTIC=1, which writes the refusal into the spectrum metadata. ew_of_state.sh requires the binary, a fresh work directory, checks every step, records the md5 and state_id of both state files, the binary md5 and the EXHALE_* overrides, judges solution and adv separately, and states the EW in percent-angstrom (the former mA? was wrong). run_case.sh accepts 7 as "solved state certified, adv product refused": the state is published with its certification, the refusal and the verdict line go into run.json (adv_product, adv_derived_state), REPRODUCE.md and the last log line, the transit pass is skipped (the transit tool itself exits 7 on a rejected record) and earlier tpm_* products of the case are moved to superseded_case_products/ so that none can be read as this run's line measure; an unknown exit still fails the pass. Launcher tests (LHS1140b/models/tests/):

evaluate_entry.sh 25/25 (the five exit-7 checks fail on the former tools), runner_records.sh 18/18 after a fixture defect of its own was fixed (its synthetic tree lacked he10830_equivalent_width.py and the observation CSV that status.py imports; 14/18 on the former fixture), and the other five launcher suites unchanged.

**A6.** After every composition update (the H/He sweep, the metal re-solve and its failure branch) n_tot, n_e, mmw and the caloric quantities are refreshed, then xion, then heating, the energy solve and p_out. wasp_full with pp_metals 2: max |p - (n_tot + n_e) k T|/p of the written composition 4.4e-8 -> 7.8e-16; the _adv T column moves by 1.0e-6. The xion refresh acts in every mode: on the fiducial atomic full-equation state the outer passes now reach their fixed point (column iterations 20..4 -> 20..1 over the ten passes) and the _adv T moves by 2.9e-10. **Separate defect found and fixed:** the pp_metals = 2 metal re-solve never set ieq_cell%T_K, so its charge exchange ran at the temperature of the last equilibrium cell and the cx_require_cell guard aborted every pp_metals = 2 run (reproduced on wasp_full; one line).

**The M2 rejection explained.** ew_M2_s1 (molecular photochemical fiducial, section 51): reason metal_T_band at cell 437 (r = 10.35 R_p) on 10 of 10 passes: the column's T 2445 K is 2.007 times the run's 1218 K. The row at 2445 K closes to 9.5e-12 of its largest term (compression work 5.6e-13 = photoheating 3.7e-13 + conduction 1.6e-13; radiative cooling 1e-16), and with that cell alone put back at 1218 K the conduction from its neighbors (7.9e-10) leaves a residual equal to the largest term. So 2445 K is a root of the corrected, conduction-coupled equation and only the band refuses it; the band is a guard of the cell-by-cell Brent solve against spurious hot metal-cooling roots and has no stated validity for the column solve. Whether the column is to be accepted on its residual instead (plan B2) is held for the user; the band is unchanged and the case stays rejected (EW 1.1032 diagnostic; the solution spectrum 8.563 is a result).

**Checks.** src/tests/adv_static_limit: 53 -> 63 PASS, 0 FAIL, with the gfortran and the ifx binary (rows K: a stored certified state evaluated plain and with each of the five injected rejections, EXHALE_ADV_TEST_REJECT=<reason>, measurement-only; rows L: the reader on composition-only failure, retained/unsupported/not-evaluated kept apart, an absent record read as unknown). Evaluate-only reruns in .P1/fiducial_conduction/verify_w1/: M2 exit 7 with the record above; fiducial atomic exit 0, step_passed_unverified, EW 0.6491 unchanged. Known and left as it is: a gfortran runtime error exits 2, the code of the certification refusal (TO_BE_DONE (o)).

## 56. Continuation factors written exactly; legacy F7.5 tokens read as intervals (2026-10-01)

Finding 5 of section 54 (plan A3). load_IC.f90 writes a factor s /= 1 of cond and interdiff_enth as ES23.16E3 (factor_token_text, adjustl; s = 1 stays T), and the writer, reader, comparison and absent-token values share one length, 32 (previously 8/16/16/8); with both factors the options field is 296 of its 1024 characters, and a field or token that would not fit is refused, never cut. A legacy F7.5 token (factor_token_kind, these two tokens only) states the interval [v - 5e-6, v + 5e-6]: a requested factor inside it loads the state as a seed with the claim dropped (legacy_factor_token, one printed line, a # legacy_factor_token history line keeping the original text); a factor outside it is a physics change under the existing rule (named on Restart option change:, or refused). Malformed values are refused even when named. The run log (conduction_scale, interdiffusion_enthalpy_scale) and the setup report print the factor ES13.6 and the report names a migration. gfortran and ifx write 0.773561, 1e-6, 1 - 1e-12 and 0 as the same token text and each reloads the other's state at the same factor. Test restart_factor (grid_and_gates 25): 38/38 PASS with both compilers (15 of the first 27 FAIL on 60a90fb); the gate tests of section 39 unchanged

(gfortran 119 PASS, 11 XFAIL, 0 FAIL; the 16 ifx FAILs are the fixtures' 1-ulp compiler difference, present before). `grid_and_gates/run.sh` uses `-module` and `-qmkl` for ifx (before, ifx could not link the Fortran drivers). Round trip: rho, v, p bitwise (gfortran) or 1 ulp (ifx), T and species 1-3 ulp, the loader's own arithmetic. States written on 2026-09-29/30 with F7.5 tokens (every chain rung) keep them and load only as seeds.

## 57. The H2 conductivity without overflow (2026-10-01)

Finding 10.2 of section 54 (plan A4). `h2_conductivity` evaluated E(x) = x^2 e^x/(e^x - 1)^2 (x = 5987 K/T) as written: (e^x - 1)^2 overflows above x ~ 355 (T < 16.9 K), where the form returned 0 in place of E < 1e-148, and x^2 e^x above x = 696.69, so 8.553 K <= T < 8.593 K gave NaN, and `thermal_conductivity` added n_H2 x NaN even where n_H2 = 0. (The review and section 54 had said "NaN above x ~ 355"; the NaN interval is the narrow one.) E is now `einstein_heat_capacity`: x^2 e^-x/(1 - e^-x)^2 for x >= 0.05, the series 1 - x^2/12 + x^4/240 - x^6/6048 below (truncation < 2.3e-16; the two agree at the switch to 6.7e-16), 0 above x = 700; `h2_conductivity` stops on a non-positive or non-finite T; the H2 term is formed only where n_H2 > 0, in the same summation order, so an atomic mixture is bitwise unchanged. Fit and range (200-2000 K) unchanged. `src/tests/energy_update`: 54 -> 69 PASS (8.57 K = 1.3235866213750098e3, 200/748/2000 K unchanged to 1e-14, the 1e6 K series limit, continuity, n_H2 = 0 mixture bitwise atomic). Stationary evaluate of the molecular scalar He/H 1.45 conduction state, HEAD against HEAD + fix (ifx): `Hydro_ioniz` and `Ion_species` bitwise; `*_adv` at rounding (T 7e-14).

## 58. The energy-row floor was a rounding staircase of the absorbed fraction (2026-10-01)

Plan C1-C3, on the state of section 53 (`diag_R/floor_jvadd`). Keys added, default off: `EXHALE_IEQ_DUMP_CELL=<j>` writes every He 2^3S-branch solve of cell j to `ieq_cell_dump.txt` (tags, T, densities, every rate coefficient, start and returned vectors, dimensional rows, turnover bounds, normalized residual, the forward-difference Jacobian) and the cell's photoionization rates at perturbed entry states to `ieq_cell_rate_probe.txt`; `EXHALE_RESID_SC_TRACE_SEED=shift|h2|partition` selects seed B of the two-seed trace (`partition`: the elemental inventory and the charge closure kept, a fraction of H I and He I moved into H II and He II). A5: `report_accepted_cell_state` names unknown 4 by the system (He 2^3S in the atomic triplet branch). With every key unset the outputs equal the HEAD build's.

**Verdict: hypothesis (b) of section 53, and the cause.** At cell 307 the elimination alternated between x(H+) = 0.50307588606921 and 0.50307588605993 (1.8e-11 apart). The dumped rows are reproduced from the dumped coefficients in binary64 (146 records, to 1.8e-17 of the turnover bound). Each returned x is the 50-digit root of ITS OWN pass's coefficients to 1 ulp and 1.4e-11 (normalized) from the other's; the binary64 residual is monotone across 41 ulps, not flat. The coefficients differ because P_HI, the cell's photoionization rate, is a staircase function of the cell's own entry n(H I): steps of 4-5e-11 relative every 1.7e-11 in n(H I). Cause: `absorbed_fraction_per_unit_depth` (`utilities.f90`) took the closed form (1 - e^-d)/d for d >= 1e-8, whose rounding is a staircase with relative steps eps/d (1.5e-9 at this cell's d = 1.5e-7). The smooth composition map contracts (-0.073 per pass, measured with the partition seed), but its fixed point lay on a step, so a period-2 orbit resulted, and that orbit was the floor: two seeds end on it in opposite phase, ||R_B - R_A|| = 1.7e-5. The "flat residual" of section 53 is withdrawn.

**Fix.** The Taylor series through d^8 (Horner) for 1e-8 <= d < 1e-2 (truncation < 3e-25), the closed form above (error <= 2.2e-14); the old comment's truncation figure for the three-term series at 1e-8 was 4e-25 and is 4.2e-26. Kahan's (1 - u)/(-ln u) does not work: ifx -O3 -xHost folds ln(exp(-d)) to -d and the output was bitwise the closed form's. Reach: every XUV photoionization and photoheating rate (`photoionization_field_at_cell_H` and `_HHe` through `cell_mean_attenuation`, the a_tau = 0 branch) and the H2O/OH FUV band rates. Effect on that state (max over physical cells, new against the control): x(H+) 7.8e-11, T 1.3e-11, rho 1.6e-10, v 2.2e-6 at the base cell only (below 1e-9 elsewhere), the mean rho v r^2 over r >= 1.1 R_p 2.7e-10.

**Result.** floor_jvadd with the chain settings (`EXHALE_PTC_SER=2`, cap 400, omega 1, factor 0.773561): ONE outer pass, 2 JFNK steps, info = 0, CERTIFIED, energy row 1.16e-7 (tolerance 1e-6), 1.3 s; the HEAD build with the same settings: 400 iterations, info = 1, energy row 3.56e-5, still 9.0e-6 after three passes, never certified. Test `absorbed_fraction_switch` (physics_probe 28): 14 assertions (quad-precision error on each side, continuity at both switches, mpmath at 1.5e-7, 1e-3, 0.1, 1, 10, no staircase over 41 points at 1.5e-7), all pass with gfortran and ifx at -O0; the former form fails 5. Runs in `diag_R/w4_*`; figures in `diag_R/w4_figures/`.

Consequence for the chains (stopped 2026-10-01, section 55 of the handoff): their slow rungs were this staircase, not the acceptance rule or the stopping measure of the elimination (plan C4, which stays a comment defect of that measure, not a cure).

## 59. One (1 - e^-x)/x for the code, and the other small-value forms (2026-10-01)

`one_minus_exp_over_x` (`utilities.f90`: the series through x^8 below 1e-2, the closed form above, error <= 2.2e-14) is now the only evaluation of the quotient. Evaluated through it, each with its own name, limits and meaning: `absorbed_fraction_per_unit_depth` (section 58), `absorbed_fraction_over_tau` and `absorbed_photon_fraction` (`util_ion_eq.f90`: the case-A/B ground capture weights and the on-the-spot absorption of recombination radiation; former switch 1e-4 with errors 2.2e-12 and 4e-14), the thin branch of `line_escape_probability_one_face` (`Cool_coeff.f90`, the metal and helium fine-structure line escape; former closed form from 1e-8, error up to 2e-8), and the Ly-alpha Sobolev escape `beta_sob` (`lya_rt.f90`), which returned 0 in place of 1 for tau_S below ~1e-16 and at tau_S = 0 (a limit defect; reached only with excited hydrogen on, which the LHS 1140 b runs do not have). Then the three forms that lose accuracy where their value is small: the stimulated-emission factor 1 - e^-Ek/T of `line_center_opacity_lte` and the absorbed Lyman-continuum fraction of `excited_hydrogen.f90` as y (1 - e^-y)/y (limit y), and the lower incomplete gamma functions of the l-mixing rate, `lower_gamma_2` and `lower_gamma_3`, as their power series in Horner form below 0.25 (through x^12) and 0.5 (through x^17), closed form above, errors <= 1.1e-14 (the former three-term series switched at 1e-2 with jumps of 6.7e-8 and 8.4e-8 there). `fine_structure_escape_probability` takes its thin reference in quadruple precision (its double reference was the cancelling form). Test `one_minus_exp_over_x_sites` (physics_probe 46): 87 assertions (mpmath values, continuity at every switch, no staircase over 41-point scans, the limits), all pass with gfortran and ifx at -O0; the former code fails 36. Evaluate-only: floor_jvadd moves <= 5.7e-14 beyond section 58; the molecular photochemical s1 state <= 1.2e-10 (H II, O II at a ghost cell), CERTIFIED before and after (mass 2.7e-9, energy 2.3e-8); the last three forms change neither run (excited hydrogen off; y of order 0.1-1 on the closed-form side).

## 60. The combined build and the tests of sections 55-59 (2026-10-01)

The tree with every edit of sections 55-59 built with gfortran (md5 a68670605f8d55c3a84e8c5ce4e04ca8) and ifx (fe48f7b774aac4490bd00334644fca6b); the suites the edits touch, run once on that tree: `energy_update` 69 PASS, 0 FAIL; `grid_and_gates` 423 PASS, 12 XFAIL, 0 FAIL (every program passed); `adv_static_limit` 63 PASS, 0 FAIL with the gfortran binary and 63/63 with the ifx binary; `physics_probe` 2295 PASS, 0 FAIL. After section 61 the tree is rebuilt (`EXHALE.x` aacba6fd93e4a547902ae732ad0b9bb2, `EXHALE_ifx.x` bff027c3c4ab416a429b30b5b70e9a00) and `adv_static_limit` (62 PASS, rows K with five reasons), `energy_update` and `evaluate_entry.sh` rerun on it (counts in section 61). The launcher suites of `LHS1140b/models/tests/` were run by the worker of section 55 (`evaluate_entry.sh` 25/25, `runner_records.sh` 18/18, five others unchanged). The regression matrix was not run and no golden was touched (user instruction of 2026-09-29; the measured effects are stated with each change, the largest 2.2e-6 in v at the base cell of the diagnostic state, section 58). The workers' intermediate binaries (`EXHALE_w*.x`, `build_w*/`, `diag_R/w4_bin/`) are superseded by these two.

## 61. The column energy solve accepted on its residual; the refused rows carry the solution (2026-10-01)

User decision of 2026-10-01 ("잔차"): the conduction-coupled column of the advected profile is judged on its residual, not on the metal temperature band (TO_BE_DONE (p), plan rev3 B2). Implemented by the worker of section 55 and checked on the diff, the tests and the three evaluate-only runs.

**Acceptance.** After the iteration stops (step 1e-10 or 100 iterations, a stopping rule only) the transport terms are formed again at the final profile, and every unknown's energy row R = q0 fv/(mup mum dr_step) must satisfy |R| <= atol_E + cert_tol_energy S_E, with S_E the largest of the grouped terms (div(rho e v) as one term, p div v, heating, cooling, the conduction divergence, the element and the carrier enthalpy divergences) and atol_E = 10 eps times the parts the row is assembled from (the grouped terms, the two advective halves, the three conduction face terms). MEASURED: the stagnated floor is 0.73 to 0.99 eps of that sum on the three conduction states, while the grouped sum alone under-states it by 1e3 to 1e5, the conduction divergence being a small difference of three large face terms. Today the absolute term never decides: the achieved |R| is 1e4 below rtol S_E. The [0.5, 2] T_in band is removed from the column (it guards the cell-by-cell Brent solve, which keeps it, and had refused a root, section 55). Outcome `converged_on_residual`; reason `residual_above_tolerance` with the worst cell's term table; record fields `column_worst_cell`, `column_worst_residual`, `column_worst_residual_rel`.

**Cells without a marching root.** Cells whose marching Brent step finds no root (its bracket [0.05, 4] T_in) are column unknowns by default and the residual decides (`cells_marching_unrooted` in the record; `EXHALE_ADV_COLUMN_EXCLUDE_UNROOTED=1` restores the old set, measurement only). On the photochemical s1 state three cells at 24.5-28.5 R_p have their roots at 4.0-4.3 T_in; with them in the column every pass solves the same set (the 354/351 alternation is gone).

**Refused rows carry the solution.** The status-3 rows of that product (1.0-1.09 R_p) were T-unsupported but composition-corrected, and their corrected n(He 2^3S) was 70 to 5600 times the run's: the molecule-free advection systems omit the He(2^3S) + H2 destruction that the solution's balance includes. A composition class (formed with the

particle-count class on the state handed in, no threshold) now refuses a cell whose H+, He+ or He(2^3S) loses faster to H2 (R10 + R13, R17 + R23, `ioniz_HeI23S_H2`, from `mol_rates`) than to the carried channels; the class is applied inside the passes, so the cascade takes the run's composition as its history there; the metal re-solve of `pp_metals = 2` skips refused cells; and every refused row is written with the run's exact numbers (species and metal stages when the composition field is refused; T when the T field is 1, 3 or 4; p, heat, cool when both are refused): the restored fields equal the solution's columns to 0 relative on the three states; a row with corrected T and refused composition is not the solution as a whole. The class is a threshold rule on the input state and does not bound the omitted H2 contribution. The transit census still counts them as refused (173 rows to 1.089 R_p, photochemical; 246 rows to 1.322 R_p, scalar He/H 1.45).

**Results** (evaluate-only, `ew_of_state.sh`, tree ifx binary; EW in percent-angstrom, adv / solution):

| state | column | unknowns (unrooted) | worst \|R\|/S_E | EW adv | EW solution |
|---|---|---|---|---|---|
| fiducial atomic full-equation | converged | 500 (0) | 2.8e-11 | 0.6491 | 7.5179 |
| molecular photochemical He/H 9, conduction | converged | 329 (3) | 1.6e-11 | **0.7705** | 8.5630 |
| molecular scalar He/H 1.45, conduction | converged | 256 (0) | 2.9e-12 | **0.1408** | 2.1471 |

All three exit 0 with the COLUMN accepted; the derived state as a whole stays `unverified`: the transport coefficients are formed on the input composition (plan B1), the outer iteration is the fixed ten passes (B3) and the momentum balance is not measured (B4). The photochemical adv EW of an ACCEPTED corrected profile is 0.7705 against 1.1032 of the refused marching profile (section 51, now superseded) and the observed 1.108 +- 0.029: 30 per cent low. Its He 10830 line-center optical depth peaks at 0.59 at b = 1.2 R_p (the solution's at 1.45), 85 per cent of it from 1.1-3 R_p, where T_adv/T_run is 1.00-1.04 and n(He 2^3S) is 0.93 to 0.60 of the solution's (the advected, non-equilibrium population); T_adv reaches 2 to 4.4 T_run beyond 10 R_p, which carries under 1 per cent of tau. The correction of the scalar He/H 1.45 product moved from 0.1525 to 0.1408 (-7.7 per cent) because its composition class reaches 1.32 R_p. Tests: `adv_static_limit` 62 PASS (63 before: rows K have five reasons now), `evaluate_entry.sh` 25/25.

## 62. The Codex review of the 2026-10-01 summary, applied (2026-10-01)

`md/code_improvements_20261001.md` summarizes sections 53-61; Codex reviewed it in place (its comments R1-R8 stay in that file). Its statements were checked against the tree before anything was changed.

- R2, R5, R8 (wording): "exact root" is agreement to floating-point accuracy (0.65 and 1.64 ulp on the last two records; the two x(H+) are 9.2e-12 apart, 1.8e-11 relative); refusal of an `_adv` row is per field (composition, or T), so a row with corrected T and refused composition is not the solution as a whole, and the H2 class is a threshold rule that does not bound the omitted contribution; "profiles move by <= 1e-10" was false as a general statement: on the diagnostic state the maximum relative change over physical cells is x(H+) 7.8e-11, T 1.3e-11, rho 1.6e-10, v 2.2e-6 (the base cell only), the mean rho v r^2 over r >= 1.1 R_p 2.7e-10. The summary, sections 58, 60 and 61 and the handoff now say so.
- R3 (HIGH, correct): what section 61 accepts is the column subproblem with lagged coefficients (the transport coefficients on the input composition, the fine-structure escape probabilities of the start of the pass); the full-state energy residual on one derived composition (plan B1), the outer iteration (B3) and the momentum

balance (B4) are not tested. The stale comments of `post_process_adv.f90` (the derived-state block) and of `adv_derived_state_verdict` (`exhale_transit_lib.py`), which still said the column residual is untested, now list what is and is not tested.

- R4 (MEDIUM, correct): `EXHALE_transit.py` continued on an `unknown` derived-state verdict (no record) with a warning; it now exits 7 unless `EXHALE_TRANSIT_DIAGNOSTIC=1`, as for a rejected record, and the metadata carries `adv_derived_state_unknown` and the override. `ew_of_state.sh` passes the key on an unknown record in its diagnostic mode; the manual and the READMEs say so. adv_static_limit rows M (the rows-K product with its record stripped): refused without the key, a curve with the key stating the unknown record; 64 PASS (62 + 2). Left as it is: `run_case.sh` calls the transit without the key, so a product written by a binary older than the record would fail its case; the current binary always writes the record.
- R6 (correct): the two bitwise rows of `absorbed_fraction_switch` (H2O and OH band rates against the XUV cell mean) fail under ifx -O3 -xHost by 1 ulp because the optimizer reassociates the test's reference product; they now assert 2 ulps (`..._to_2ulp`, 4.5e-16) with a DIAGNOSTIC line of the bitwise-unequal counts: 14/14 with the suite's -O0 flags, with gfortran -O3 (0 unequal) and with ifx -O3 -xHost (552 and 405 unequal of 1506, max 2.2e-16). A second copy of the function would still be caught (its difference is the closed form's 1e-9 to 1e-14 error below d = 1e-2).
- R7: the saved EWs reintegrate to 0.6490803760 / 7.5179432737, 0.7705246807 / 8.5629850578, 0.1407893262 / 2.1470782882 (adv / solution); recorded in the summary.

## 63. The reviewed part of the LHS 1140 b EW worktree, taken into the tree (2026-10-03)

`md/lhs1140b_ew_codex_review_20261003.md` reviews the changes an autonomous Codex run made in the detached worktree `../LHS1140b_EW_20261001/code` (same base, 3152b74). Its sections 3.1 and 3.2 (accept; fix first) and 4 (a defect of this tree) are applied here; section 3.3 (molecular-ion transport, joint species-flux modules, metastable balance diagnostic, HeH+ Newton rows) is not. Each item was re-derived against this tree's code, not copied.

**Physics corrected.**

- Recombination photons absorbed by H2 (`util_ion_eq.f90` `recombination_radiation_absorbed`, `ionization_equilibrium.f90`): the whole H2 share made H2+ and heated with one threshold. It is now split into the four final states of `h2_photo_channels` (M H2+, S dissociative ionization, D double ionization, N neutral dissociation) in the ratio of the channel cross sections the stellar field uses (`h2_photoabsorption_cross_sections`, shared with `set_energy_vectors`), total absorption unchanged, heat = photoelectron(s) + fragment kinetic energy of the channel. The capture kinetic energy of a continuum channel goes to every final state (energy conservation; the Codex version gave it to the ionizing ones only). Channel shares at the default switches, MEASURED: S 1.9-6.3% at 19.8-31.1 eV; S 11.4% and N 1.1% at 40.8 eV (He II Ly-alpha, inside the neutral window the old comment said it was outside of); S 17.0% and D 0.37% at 54.4 eV.
- `h2_channel_threshold` of N: 33 -> 32 eV, the lower continuity anchor where `frac_H2_neutral_dissociation` turns on (32-33 eV absorptions had a cross section and no energy

recipient). No current recombination channel lies in 32-33 eV.

- The recombination-coupling condition at the two sweep sites is `use_h_rec_escape .or. (use_he_rec_coupling .and. thereis_He)`; the old form could read the `_hrc` arrays unset in a helium-free run.
- Carrier enthalpy flux (`diffusive_photochemistry.f90` `carrier_enthalpy_face_flux`): the recoil set of the energy row is now the set `carrier_write_back` rescales. Before, H+ took part of the hydrogen recoil although a carried H+ (`Ionization transport: True`) is written from its own row, and HeH+ took part although the write-back never moves it. Physics document `docs/physics_overview/molecules_metals.tex` updated.
- Class 6 of the ionization sweep: a constrained projection that fails `ionization_fractions_physical` keeps the entry composition and records the element-budget violation (it was accepted as a root); the class-6 residual is that of the retained state (also stated in `steady_newton.f90`). `impose_transported_fractions` is applied before the admissibility test and at write-back. The decay-rate diagnostic releases the He II / He III fixed flags as well.
- Jeans escape flux of `collisional_validity.py`: n vbar/4 (1+lambda)eˆ-lambda; the old prefactor was 2/sqrt(pi) = 1.128 too large. The planet radius and mass of a run are read from its `EXHALE_resolved.out` by one rule in `examples/exhale_io.py` `load_run` (the lower-atmosphere profile and `base.inp` overwrite R0); `collisional_validity.py` and `formation_energy_flux_diagnostic.py` use it. He mass 3.9715 m_H (the table's), not 4.

**Restart and output.**

- Exact code-unit restart: `conserved_state.txt` (new `conserved_state_restart.f90`) beside every 'eq' write and every pass-state generation, read as `conserved_state_IC.txt` when it belongs to the IC pair (checksums, grid, species, boundary). A state that cannot be written exactly is not written (warning, existing file removed); a file that does not match, an option-change restart and a seed run use the dimensional pair with a notice. Exit statuses unchanged. README, manual (`conserved_state.txt` subsection), `README_HOWTO.md` pass-state restore, `md/input_schema.md`, GUI copy, mapper refusal and `element_flux_closure.py` (post-processing step = `Restart intent: stationary evaluate`) follow.
- Species densities re-formed after `Apply_BC` at the six output and evaluation sites (the Newton residual's order); ghost output rows formed from the conserved ghost cells by the tree's own routines (`boundary_gas_state_from_conserved`); `write_resolved_config` after the certification on both branches of the loaded-restart route.
- Comments corrected: subsonic outflow ghost (one incoming characteristic), ion-neutral friction (Langevin friction independent of T, rigid core as Tˆ(1/2); `1/D = 1/D_pol + 1/D_hs` stated as an interpolation, up to a factor 2 in friction at the crossover), the Banks & Kockarts diameter (2.99 A, not 2.7 A; also `md/binary_diffusion_design.md`), mol_rates R14 and R19 sources (coefficients unchanged), `n_car_max` derived from the species table.

**Not taken, with reason.** The element eddy-flux outputs (no caller once the row dumps of 3.3 are left out); the `spherical_flux_divergence` refactor (stops on a NaN trial; the loop stays); the Codex wording that weakened two

correct statements of `collisional_validity`.

**Tests.** Four builds from one source (`3152b74`; all changes; the enthalpy recoil set only; the recombination channels and chemistry only), run single-threaded on copies of `hp_front` (100 steps) and `carrier_model_a_newton` (stationary restart), MEASURED largest relative change over physical cells against the control build:

| case | change | largest relative change |
|---|---|---|
| hp_front | enthalpy recoil set | v 3.0e-7, HeH+ 1.2e-8 |
| hp_front | recombination channels | H2+ 1.1e-4, H3+ 7.0e-6, HeH+ 8.8e-7, cool 7.5e-7 |
| carrier_model_a_newton | enthalpy recoil set | HeH+ 1.6e-4, cool 1.5e-4, T 7.0e-5, rho 4.2e-5 |
| carrier_model_a_newton | recombination channels | 0 (`He_rec_coupling: False`) |

Both are snapshots, not converged states, so these are the changes of one evaluation and not of a solution. Full profiles (T, v, rho, H2, H+, H2+, H3+, HeH+, He 2^3S) overlaid and inspected (session scratch `cmp/cmp_*.pdf`).

`make test` on the merged tree (`EXHALE.x` 60bd2895135134813116b4b0ac63fcaa): 44 of 48 entries PASS; `steady_completion_flag` and `steady_selfconsistent_residual` fail without `EXHALE_STEADY_RUNLOGS` (they read the logs of a steady run and have no default; not a code result); `boundary_state` and `physics_probe` failed on the test defects below and pass after them (`physics_probe` 2295 PASS, 0 FAIL). `carrier_enthalpy_flux` gains section (e): a column with H+ (a third of the atomic hydrogen) and HeH+ (1e-3), checked face by face against Phi_H2 [h_H2 - 2 hbar_H] with H+ local and with H+ carried; both rows pass (1.8e-15) and both FAIL on the 3152b74 code (1.2e-3 from HeH+ alone; 1.11 with H+ carried, the old flux being about twice the right one there). Three test defects found and corrected: `contact_law_at_the_base.f90` called `thermal_conduction_source` without the composition argument added in 60a90fb (compile error); `helium_584_resonance_line.f90` called `absorb_recombination_channel` without the new channel-split argument; `boundary_of_the_installed_composition.sh` asserted a bitwise zero face-state gap, which the density re-formation after `Apply_BC` moves to rounding (MEASURED 1.6e-15; the entry-composition order it guards against leaves 7.5e-7), now a 1e-13 relative bound. The exact restart was exercised on `wasp_he23off` inputs: two write-reload round trips bitwise in u and f_sp; a negative ghost fraction is not written and the run exits 0; stale and mismatched files are ignored with a notice. Not run: molecular-seed, pass-state generation, the closure driver, GUI.

**Open, for the user.** R19 (HeH+ + H -> H2+ + He): 9.1e-10 cm^3 s^-1 is a room-temperature measurement; De Fazio et al. (2014) Table 2 gives 1.54e-9 (1000 K) and 1.65e-9 (2000 K), Esposito (2015) via Sil et al. (2025) 1.82e-9 exp(-121/T); not changed. Esposito, Coppola & De Fazio (2015) was then read in its accepted manuscript (the published version could not be obtained): a Boltzmann average over all 178 rovibrational states of HeH+, 10-15000 K, with the authors' own polynomial fit (1.52e-9 at 1000 K, a maximum 2.00e-9 near 6000 K; 1-6% from De Fazio up to 2000 K; the Sil Arrhenius form is 9-11% below it at 4000-8000 K). The R19 comment of `mol_rates.f90` now states this. The constant is kept because it is the Koskinen et al. (2022) Table 1 entry the Model A comparison uses; the replacement is the user's decision (decided the same day: section 64). `python/paper_data.py` still reads the input.inp planet radius.

## 64. R19 is the thermal rate of Esposito, Coppola & De Fazio (2015) (2026-10-03)

HeH+ + H -> H2+ + He (`mol_rates.f90` `rk_R19_HeHp_H`) was the Koskinen et al. (2022) Table 1 constant 9.1e-10 cmˆ3 sˆ-1, a measurement near 300 K (Karpas et al. 1979) held at every wind temperature. By the user's decision of 2026-10-03 it is now the thermal rate of Esposito, Coppola & De Fazio (2015, J. Phys. Chem. A 119, 12615), read in the accepted manuscript (the published version could not be obtained): close-coupling cross sections for v = 0, j = 0-5 and quasi-classical trajectories for the other states, Boltzmann-averaged over all 178 rovibrational states of HeH+, in the authors' fit k = sum_{i=0}ˆ{6} a_i (log10 T)ˆi (max error 2.5%; their eq 4 prints the upper limit 7 with seven coefficients, and the seven reproduce their Fig. 6). T is held to the authors' 10-15000 K (the polynomial is negative below about 6 K and above about 3e4 K). The function now takes T; its two callers (`System_HeH_mol.f90` `set_mol_coeffs`, `molecular_reaction_heat.f90`) pass the temperature they already pass to the other rates.

MEASURED values of the built function: 1.088e-9 (300 K), 1.524e-9 (1000 K), 1.753e-9 (2000 K), 1.997e-9 (5354 K), 2.004e-9 (6000 K), held 1.571e-9 above 15000 K and 5.25e-11 below 10 K; 1-6% from De Fazio (2014) Table 2 at 100-2000 K, within 3% of Garcia Munoz (2025) Table A.6 at 500-5000 K. The Sil et al. (2025) Arrhenius form of the same rate is not used (9-11% below it at 4000-8000 K).

Effect, MEASURED against the build of section 63 (same source but R19), single-threaded, largest relative change over physical cells; full profiles of T, v, rho, H2, H+, H2+, H3+, HeH+ and He+ overlaid and inspected (session scratch `cmp/r19_*.pdf`):

| case | HeH+ | H2+ | H3+ | H2 | T, v, rho |
|---|---|---|---|---|---|
| hp_front (100 steps) | -46% | +68% (outer wind) | -18% | +1.4% | <= 5.8e-7 |
| carrier_model_a_newton | -35% | +0.07% | -0.02% | +0.02% | <= 3.7e-5 |

`docs/physics_overview/molecules_metals.tex` gains the R19 paragraph and the two references; `docs/koskinen2022_model_a_comparison.tex` now says that its "matched reactions" row holds for the runs of that document and that the code departs from Table 1 in R12, R15, R17, R19 and R20.

`make test` on this tree (`EXHALE.x` 3d27cc49f507582c6cdc9683e6aedca7): 46 of 48 entries PASS (5176 PASS lines, 11 XFAIL); the two that fail are `steady_completion_flag` and `steady_selfconsistent_residual`, which need `EXHALE_STEADY_RUNLOGS` (section 63).

## 65. The Codex review of sections 63-64, checked (2026-10-03)

Codex reviewed sections 63-64 and the documents of the day in `md/claude_code_physics_review_20261003.md` (with `_checks.json`, `_timescales.json`, `_profiles.pdf`, `_carrier_test.txt`, `_r19_probe.txt`) and changed five files (`examples/exhale_io.py`, the two md documents, `src/utils/collisional_validity.py` and the scattering assessment). Each change was checked against the code.

- `examples/exhale_io.py` `load_run`: `EXHALE_resolved.out` is now read from the parent of the output directory, not from the directory of the input file. Correct: EXHALE writes the resolved file in the run directory and

the outputs in its `output/`, so an input file from elsewhere could pair an output with another run's radius and mass (Codex reproduced a factor 2 in Mdot/M_p with a foreign file). Exercised on `lower_profile` with and without an input file: both read that run's own resolved file.

- `md/binary_diffusion_design.md`, `md/collisional_validity.md`, `src/utils/collisional_validity.py`: HeH+ is stated not to be a friction carrier of the element operator (true: `binary_element_diffusion.f90` leaves it out of both carrier lists), and the "omitted channels are conservative" statement is limited to the retained coefficients. Both correct.
- `md/lhs1140b_ew_codex_scattering_assessment_20261003.md`: Codex added a qualification. Its substance was right: the lifetime had been compared with bulk advection only. The diffusion and drift times are now MEASURED on the same state (HeH+ diffusion by Blanc's law from the pair coefficients of `collisional_validity.py`, K_zz = 1e9, settling in gravity and the ambipolar field eE = -kT d ln(n_e T)/dr plus gradient drift, n_e the production charge budget): at the He 2^3S column peak t_chem = 11 s against t_adv 2.8e5, t_diff 4.9e7 and t_drift 1.2e7 s, so Da stays 2.5e4, also with every Coulomb partner dropped (D 30 times larger; t_drift 3.9e5 s). Advection is the shortest transport time except in 210 cells of the slow base layer below 1.18 R_p (diffusion shorter; 0.4% of the radial He 2^3S column). The qualification is replaced by these numbers and the overstated sentences are corrected. Two Codex follow-ups in the same review file then showed that the ambipolar field adds to gravity in 197 of the 500 cells (a field-free drift is not an upper bound; eE/mg down to -0.92 with the metal-ion electrons, -0.49 without), that the neutral-only row is a sensitivity and not a bound on the pair law, that the opening claim was too categorical, and that the first field-inclusive table still carried field-free drift times and a wrong count ("advection shortest in all but one cell", which had missed the cells where diffusion is shortest). All were reproduced and corrected; the assessment states the field, the margin at the peak (D 43 times larger for drift to reach advection, 1.1e6 times for it to reach chemistry), the local scope of the claim, and that the column shares are radial, not EW weights. Da at the column peak is 2.5e4 throughout.

Points of the review that stand as open items: the recombination photoelectrons of `util_ion_eq.f90` heat locally with no secondary ionization or excitation, where the stellar path partitions the photoelectron energy (stated at the code site; at 54.4 eV an H2 channel leaves a photoelectron above the secondary-ionization threshold; effect not measured); and the thermal R19 rate assumes HeH+ internal states in equilibrium, which chemical relaxation does not show.

## 66. The LHS 1140 b EW plan started: phases 0 and 1 (2026-10-03)

The plan `md/PLAN_20261003_lhs1140b_ew.md` was approved by the user with all five decisions of its section 3. Runs are in `LHS1140b/models/.P2/ew_fit/`; the binary is `LHS1140b/models/.P2/EXHALE_f036290f.x` (ifx, clean export of dd4a591, manifest `.P2/BINARY_MANIFEST_f036290f65fd.txt`).

**Phase 0** (`.P2/ew_fit/NOTES_phase0.md`).

- Entry `molecular_photochem_gj1132_kzzprofile_HeH2.09.entry/` from the certified generation g0001 (2026-09-29) of that case; its input differs from the He/H 9 entry in the planet name and He/H only.
- Evaluate-only on f036290f: both the HeH2.09 state and the He/H 9 conduction state are NOT CERTIFIED. HeH2.09: cell-1 rows O(1), from the base being rebuilt as `discrete_equilibrium_v5` from a state written under

v4 (the He/H 9 entry was in the same position and certified through its conduction chain). He/H 9 conduction (v5 already): carrier H2 0.838, energy 2.6e-2, from the equation changes of sections 63-64.

- The stored EWs reproduce bitwise on their own binaries: 0.7705246807 (He/H 9 conduction, `verify_w1d/EXHALE_w1d.x`) and 1.5008683274 (HeH2.09, `EXHALE_6dd54955.x`). On f036290f (diagnostic, refused state) they are 0.7705418877 and 1.4601886445.
- Conduction continuation of HeH2.09 started on lart4 15:33 KST (`.P2/ew_fit/factor_continuation_chain.sh`, a copy writing into `.P2/ew_fit`, with `config_rung.sh` requiring BIN); rung 1, s = 0.005, certified in 21 passes.

**Phase 1** (`.P2/ew_fit/phase1/NOTES_phase1.md`; all on the He/H 9 conduction state, a solution of the w1d equations, each number with its binary).

- Where the line forms (EW with n(2^3S) zeroed beyond r_cut): on the local-balance solution state (EW 8.563) 17% inside 5 R_p and 18% beyond 13 R_p; on the advected product of the same state (EW 0.7705, MEASURED here, `phase1/ew_adv/truncation_adv.txt`) 17% inside 2 R_p, 55% inside 5 R_p, 77% inside 8 R_p, 95% inside 13 R_p. Kn_bulk = 0.1 at 6.97 R_p. The solution and advected EWs of one state differ by a factor 11: carried ionization is a first-order effect on the EW.
- He 2^3S local balance (solution state): first-order EW change +0.013 to +0.019 percent-angstrom (2.2-3.3 times the numerical target 0.0058).
- Ion-neutral drift: 0.3-1.1 of the element velocity at 3-30 R_p (H+/H I and He+/He I; the resonant charge-exchange cross section used in one bracket is not verified in a published source). The one-velocity stage flux is not valid there.
- Stage enthalpy (`TO_BE_DONE.md` (j)): up to 2e-2 of the energy-row scale; required whenever ionization transport is on.
- Recombination photoelectrons above 10.2 eV deposited as heat: below 5e-5 of the heating, but up to 43 times the energy-row tolerance at 1.0-1.5 R_p; < 2.3e-7 of the row scale where the line forms.
- Energy coupling to Photochem: the wind conducts 4.25 erg cm^-2 s^-1 into its base (35% of the heat it absorbs; the enthalpy flux there is 120 times smaller). The Photochem column at the level is an isotherm at the clima skin temperature with no energy equation; an LTE thin estimate gives a warming of 120-240 K within 0.2-0.4 H below the level. The one-way coupling is not justified for this planet.

**Stale (2026-10-07):** the Knudsen numbers, `Kn_bulk = 0.1` radii and exobase radii in this passage were computed by `src/utils/collisional_validity.py` before its resonant H+ + H and He+ + He momentum-transfer coefficients (`md/Update_EXHALE_stage3.md` section 110). On the 102 stored LHS 1140 b states the change moves the `Kn_bulk = 0.1` radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.

**Corrected in the tree.** The ENERGY comment of `recombination_radiation_absorbed` (`util_ion_eq.f90`) listed He II Ly-alpha on H I (27.2 eV) among the channels above the 30 eV threshold and omitted the He III ground capture on He I, H2 and the metal ions; it now lists them. The comment of `conduction_base_level_T` called the isentropic continuation the lower atmosphere's own temperature; it now states that this holds only for an adiabatic lower atmosphere and gives the measured 3.5 K offset from the profile isotherm.

The catalog publisher `.P1/publish_rows/publish_solved.sh` wrote its EW (percent-angstrom) under the key `he10830_red_pair_equivalent_width_mA`; the key is now `..._percent_angstrom`. The 59 run.json records already published keep the old key (held for the user).

**Held for the user.** The bath temperature itself (profile vs isentrope), the 10.2-30 eV electron partition (Xu & McCray 1991 not in `references/`), the energy coupling to Photochem, and the ion-neutral drift: decisions, see the report of this date.

## 67. Phase 2: carried ionization stages certify on LHS 1140 b (2026-10-03)

`.P2/ew_fit/phase2/NOTES_phase2.md`. The failed 2026-09-23 attempt (`.P1/iontrans_atomic_kzz1e9_HeH2.13`: He++ carrier row 2.0e-2 at cells 416-419, omega halved to 0.125) was re-run from its own state, mapped onto the face-at-level grid, on EXHALE_f036290f.x, with the 09-23 conditions (conduction off, `Interdiffusion enthalpy flux: False`). With `EXHALE_DIFF_OMEGA=1.0` it certifies at outer pass 11 (`phase2/C_p40`, `certified=T`; checked here): He++ row 8.6e-6, H+ 2.3e-8, He+ 5.2e-7, He/H 6.8e-9. `EXHALE_PTC_SER` does not change the outcome; omega = 0.5 refuses (the outermost element row halves each pass while the hydro solve fails on the cell-500 momentum row).

- The 09-23 failure was the state and the solver of that binary, not the He++ equation: the row's terms show the locally balanced x(He III), steep outward, advected with no source (measure about half of d ln x per cell over 7-29 R_p). Beyond about 3 R_p the He III recombination time exceeds the flow time, so the stage is frozen, 20-50 times below the local root.
- Physical effect of carrying the stages (no conduction): x(H II) at 11 R_p 0.34 -> 0.042, n_e at 29 R_p about 10 times lower, outer T 460 -> 1175 K, edge velocity 1.2 -> 1.9 km/s, log Mdot +0.012 dex; unchanged inside about 1.5 R_p.
- EW of the certified state (MEASURED here, `phase2/ew/C_p40`): solution spectrum 1.0483 percent-angstrom (status result), advected product 1.0182 (conditional). The two agree within 3% once the stages are carried, against a factor 11 between the solution and advected EWs of a local-balance state (section 66): the bracketing-on-adv, accepting-on- solution route of the plan's decision 2 is consistent.
- Open (proposed, not coded): the element operator's outer face carries zero diffusive flux (`binary_element_diffusion.f90` 219-228, 3245-3250), a wall condition at an open outflow; at the top D_12 = 3.7e13 cm^2/s and Pe = 2.8, so it makes a 1-2 cell layer (X_He -0.30% at cell 500). An outflow condition (gradient by the zero-gradient ghost, settling and thermal drift kept with the cell-N coefficients) is proposed as a default-off option, in phase 4. With the enthalpy flux on the hydro solve fails on the cell-500 momentum row from pass 7 (`phase2/G_from_C_enthon`, stopped): phase 4.
- Comment corrected: `carrier_row_terms_write` (`diffusive_photochemistry.f90`) described two printed scales; it prints one, `rowdump_scale`, as its file header says, and the helium-stage production and loss columns are the positive and negative parts of the net source (checked at `carrier_source`).

## 68. Phase 3: the He/H 2.09 reference with conduction and carried ionization stages (2026-10-03)

- Conduction continuation from the certified HeH2.09 entry (`.P2/ew_fit/molecular_photochem_gj1132_kzzprofile_HeH2.09`): nine rungs s = 0.005 -> 1, each certified (10-37 passes), 15:33-17:39 KST; final state `..._HeH2.09_cond09_s1`. EW of that state (local ionization balance): advected product 0.6255, solution spectrum 7.6754 percent-angstrom.
- Ionization transport switched on (`iontrans`, omega 1): refused at pass 12 (`..._cond_iontrans01`), the H2 carrier row at cells 217-240 oscillating 0.08-0.18. Diagnosis (`phase3/iontrans_diag/NOTES.md`): a limit cycle of the progress control, not of the equations. The relaxation of the carried stages stopped every pass on the particle-count movement bound in the outer wind (5.7-13.8 R_p), where the carried stages differ most from the local ones (x(H II) 0.35 against 0.040 at 11 R_p); the wind solve then responds (dT/T 6e-4 to 2e-3 at cells 217-246), and the nearly cancelling H2 cycle at 1.20-1.33 R_p (production and loss 8.6 cmˆ-3 sˆ-1, 1.5e4 times the transport divergence) amplifies that into the row (measured: 2.8e-4 after the composition update, 3.7e-2 after the next wind solve at fixed carriers); the bound rule halves on that rise and doubles on its fall, so the relaxation never reaches the carrier fixed point. Once it does, every run contracts by 0.62-0.63 per pass.
- With `EXHALE_CARRIER_TRUST_HOLD=0.32` (or `EXHALE_CARRIER_TRUST=0.32`), omega 1, the same parent certifies in 24-25 passes; continuing the refused pass-12 state with the original controls certifies at pass 33. The three certified states agree to |d ln n| <= 4e-7 (checked here: `certified=T`, `iontrans=T cond=T`).
- Carried stages against the local ones (He/H 2.09, conduction on): log Mdot 7.819 -> 7.805; x(H II) at 11 R_p 0.35 -> 0.040; outer x(He II) about 0.8 -> 0.2; edge T 469 -> 1113 K, edge v 1.5 -> 3.2 km/s; at the H2 front (1.15-1.17 R_p) He II x1.8, H2 x0.44.
- EW of the certified state `phase3/iontrans_diag/B2_hold0.32` (MEASURED here, `phase3/ew_B2_iontrans`): solution spectrum 0.5830, advected product 0.5680 percent-angstrom (2.6% apart). Observed 1.1084 +- 0.0289.
- Not yet in this state's equations: the stage enthalpy of `TO_BE_DONE.md` (j) (section 66: up to 2e-2 of the energy-row scale with ionization transport), the ion-neutral drift (held for the user), He 2ˆ3S transport (section 66: +0.013 to +0.019 on a local-balance state), the 10-30 eV photoelectron partition (being implemented), the energy coupling to Photochem (held for the user).
- Proposed, not coded: the progress rule `bound_step_harmed` (`EXHALE_main.f90` near 7848) should count a row rise as harm only if the same row, evaluated at the composition the relaxation returned (before the next wind solve), also rose; or should not shrink the bound when the bound-limited cells and the cells that set the joint distance differ. Section 29.13's open item ("the rule that grows the bound is not right for the H2 carrier") now has its mechanism.

## 69. The 10.2-30 eV photoelectron partition of Furlanetto & Stoever (2010), a default-off option (2026-10-03)

Photoelectrons of 10.2-30 eV were deposited entirely as heat (`E_sec_ion`, the bottom of the Dalgarno, Yan & Liu 1999 tables), although above 10.2 eV they excite and ionize H I and He I. New

key `Low-energy electron partition: Furlanetto2010` (default none, the previous behavior); notes `LHS1140b/models/.P2/ew_fit/sec_partition/NOTES.md`.

- Source: Furlanetto & Stoever 2010, MNRAS 404, 1869 (published PDF in `references/`), and the authors' electronic tables (paper footnote 9; the archive `elec_interp.tar.gz` is no longer on the authors' site, the copy used is the Wayback Machine snapshot of 2017-11-02, SHA-1 equal to the archive's CDX digest; `references/Furlanetto_2010_tables/`). Rows 10.0-32.2 eV at all 14 ionized fractions are carried as parameter arrays in the generated `src/modules/radiation/low_energy_electron_tables.f90` (generator `src/utils/furlanetto2010_tables_to_fortran.py`, which checks the archive digest and the bytes of every table file). The physics is in `low_energy_electron_degradation.f90`; one routine decides for every bin which of the three bands (above 30 eV, 10.2-30 eV, below 10.2 eV) a photoelectron falls in, for the stellar and the recombination paths alike.
- Interpolation as the authors' `elec_interp.c` (linear in E and in x), except two stated differences: everything below 10.2 eV is heat (the paper's section 7), and x is clamped to the table range [1e-4, 0.999]. Against the C code at 3042 (E, x) points: largest differences 2.6e-7 (f_heat), 3.4e-7 (n_Lya), 7.7e-8 (n_ion,HI). Energy closure: the tables themselves close to 2.7e-5; the code's partition to 3.6e-5 on rows and 9.8e-5 between rows. Xu & McCray 1991 (ApJ 375, 190, the ADS scan of the published article) Figs. 4-5 agree qualitatively.
- Accounting: heat f_heat E0; n_ion,HI and n_ion,HeI added to the existing secondary-ionization rates (n_ion,HeII is zero below 30 eV); the excitation energy leaves as line radiation, as the existing partition treats its non-heat, non-ionization share (Ly-alpha trapping in thick H I is not followed: the heat is a lower bound there). Branching between H I and He I renormalized to the cell's composition by the existing rule from the tables' He/H 0.087. Cells with n(H2) > 0 keep the heat-only treatment (no source covers H2 at 10-30 eV).
- Restart contract: a token `lowE_partition` on the `# options` line, so a state is not silently restarted under the other setting; old states read as F. Every state file now carries `lowE_partition=F` when off.
- Default path: control and measured builds give identical numerical outputs on `wasp_he23off` (20 steps) and `hp_front` (100 steps), with and without `Secondary_ionization: Immediate`; physics_probe 2295 PASS; the new suite `src/tests/low_energy_electron_degradation` 14 PASS (rerun here).
- Discontinuity at 30 eV where the source changes (atomic gas): Delta f_heat -0.011 (x = 1e-4), -0.006 (1e-2), -0.002 (0.1), 0 (0.5, 0.9); not smoothed (no source).
- Option on: `wasp_he23off` heating -0.67%, P_HI +0.63%, T -0.41% at the ionization front (1.03-1.05 R_p). On the molecular LHS 1140 b and hot-Uranus states every cell has n(H2) > 0, so the main solve is unchanged, while the advected post-process, which rebuilds an H/He gas without molecules, applies it everywhere (He/H 9 conduction state: adv T -2.9%, heat -12%, He 2^3S +3.8%). This inconsistency of the gate is held for the user. `md/input_schema.md` K14c's "E0 > 40 eV" corrected to 30 eV.
- Not changed, reported: photoelectrons above 30 eV on the recombination path (He III ground-capture channels) are still deposited as heat, where the stellar path partitions them.

## 70. The conduction bath below the level is the profile's temperature (2026-10-03)

User decision. `base_reservoir_temperature_at` (`base_boundary.f90`), the bath temperature the conduction operator holds at the base ghost (`conduction_base_level_T`), continued the reservoir's hydrostatic isentrope below the level even where a lower-atmosphere profile is handed over; on the LHS 1140 b photochem profiles the column there is an isotherm at the clima skin temperature, and the isentrope put the bath 3.5 K above it (188.89 against 185.42 K), lowering the conductive flux into the level by about 4.6% at a fixed cell-1 temperature (section 66). With a profile it now returns the profile's temperature at that radius, through a new accessor `lap_value_at_radius` (`lower_atmosphere_profile.f90`, linear in r between the bracketing levels, nearer end outside); without a profile, the isentrope as before. Comment of `conduction_base_level_T` updated.

- Checked on the certified He/H 2.09 state with conduction and carried stages (`phase3/iontrans_diag/B2_hold0.32`), evaluate-only, scratch ifx build of the tree (md5 3c236561) against EXHALE_f036290f.x (`.P2/ew_fit/bath_T/`): the only row that moves is the energy row of cell 1 (1.6e-8 -> 4.4e-2 of its scale, refused); every other row equal to the digits printed. The state is to be re-solved with the fixes now in progress, then the matching-level temperature sensitivity (plan section 8).
- Runs without a profile, and the base ghost state of the advection (its own isentrope in `base_ghost_averages`), are unchanged.

## 71. One H2 gate for both products; recombination photoelectrons partitioned as on the stellar path (2026-10-03)

Notes: `LHS1140b/models/.P2/ew_fit/sec_partition2/NOTES.md`.

- **H2 gate (user decision).** The 10.2-30 eV partition of section 69 is gated by n(H2) > 0 in one place (`low_energy_electron_partition`); every path now passes the SOLVED state's H2 density (the advected post-process passes `nmol_eq(:,1)`, the same column its `_adv` files carry). On the He/H 9 conduction state (500 of 500 cells with H2) option on and off now give identical solved and advected products in every cell (before: the advected product differed in 329 cells); on the atomic `wasp_he23off` the option acts as before (advected heat -0.67%, x(H+) -4.6%).
- **Recombination photoelectrons (defect).** They were deposited whole as heat; above `E_sec_ion` the stellar path partitions the same energies. Each recombination photoelectron is now passed, at its fixed representative energy, through the stellar path's bands and partition (`photoelectron_partition_at_energy`, the same bracket and interpolation as `photoelectron_partition_on_grid`, 0-1.2e-16 apart), under the stellar path's switch (`use_sec_ion .and. sec_ion_active`, the STAGED default included): f_heat of it is heat, its H I, He I and H2 ionizations go to dP_HI, dP_HeI, dP_H2 (dissociative one per 22 H2+), the rest leaves as line radiation. Channels concerned: He III ground capture on H I, He I, H2 and metal ions with thresholds below 24.4 eV, and He II Ly-alpha on metal ions below 10.8 eV (Na I, Ca I, Mg I, Fe I; previously missing from the comment). Inner-shell metal absorptions partition the photoelectron and the Auger electron separately, as the stellar path does. With the 10.2-30 eV option on, the recombination path now uses the He I ground singlet as the He I target,

as the stellar path does (it used the summed He I).

- Tests: physics_probe 2499 PASS, 0 FAIL (new driver `recombination_photoelectron_partition.f90`, 204 PASS: energy closure 1.2e-15, partition-whole equals heat-only to 4.4e-16, gate closed bitwise with solved H2); `low_energy_electron_degradation` 14 PASS; rerun here on the merged tree.
- Effects (control vs measured, OMP 1): `wasp_full` and `hp_front` with the default staged coupling: numerically identical (the coupling is not yet switched on in these snapshots); with `Secondary_ionization: Immediate`: wasp_full heat -5.3e-5, T -2.7e-5, x(H+) +3.4e-5 at 1.036-1.041 R_p (the He recombination heating channel itself -0.23%); hp_front below 1e-7. The certified He/H 2.09 transport state: wind energy row 8.4e-16 -> 1.3e-11, gated H+ row 1.6e-8 -> 3.3e-8, far below tolerance.
- Not traced: on the He/H 9 conduction state (option off) one cell, 118 (1.035 R_p, where n_e falls off), moves He 2^3S -1.2% and He III +1.4%; every other cell below 1.1e-4.
- Limit: each channel is partitioned at its representative energy, not averaged over the capture continuum.

## 72. The enthalpy of the carried ionization stages; the movement-bound rule (2026-10-03)

Notes: `LHS1140b/models/.P2/ew_fit/phase3/stage_enthalpy/NOTES.md`; work in a worktree, merged into the tree by a three-way merge (the two option tokens added at the same time, `lowE_partition` and `stage_enth`, are both in the vocabulary, n_opt = 25, every token array checked at 25 entries).

- **Stage enthalpy (`TO_BE_DONE.md` (j), now fixed for the stages).** The stage flux F_k = x_k N_el + E_k, E_k = -n_el K dx_k/dr, moves the stage relative to its element by E_k; by Cook (2009, eqs. 11-13) it carries q_st = sum_el sum_k E_k [h_k - hbar_el], hbar_el the enthalpy per nucleus of the species `carrier_write_back` rescales (hydrogen: H I, H2+, H3+; helium: He I with its 2^3S; HeH+ never). With every heavy particle at the same sensible enthalpy the exchange moves only the electrons': q_st = h_e sum_k Z_k E_k, h_e = gamma/(gamma-1) kT, exact for helium and corrected for the H2+/H3+ share of hydrogen. Routine `ionization_stage_enthalpy_face_flux`; the recoil set is formed once (`recoiling_nucleus_enthalpies`) for this and the molecular term (`carrier_enthalpy_face_flux` renamed `molecular_carrier_enthalpy_face_flux`); `carrier_enthalpy_divergence_of_state` returns the sum, so the stationary residual, the marching update and the advected column energy solve take it from one routine. Under `Interdiffusion enthalpy flux` (default on), only with `Ionization transport: True`; state token `stage_enth`. Validity at the code site: zero current, one temperature, no stage drift beyond E_k (the ion-neutral drift and its enthalpy are not modeled), no Dufour term.
- Tests: new `src/tests/ionization_stage_enthalpy_flux/` 13 PASS (closed ends exactly 0, column sum 1.1e-16, manufactured q 7.7e-16, H2+/H3+ recoil 2.3e-15, zero for uniform fractions, iontrans off and key off); `carrier_enthalpy_flux` 13 PASS, `ionization_stage_flux` 45 PASS; an independent recomputation from the B2 state files agrees with the production Sidf difference to 1.8e-13 of its maximum. Default path (`wasp_he23off`) identical; `hp_front` (iontrans on) moves by up to 7.9e-6, identical with the key off, i.e. the new term.
- On the certified He/H 2.09 transport state (B2): the term is up to 1.7e-3 of the energy-row scale (1.05-1.2 R_p),

above 1e-6 in 369 of 500 cells, about 12 times smaller than the phase-1 estimate on a local-balance state. Resolved with it (`R_B2_stage_hold0.32`): certified in 11 passes; T -6.2e-5, v -2.3e-4, x(H+) +1.5e-4, He 2^3S +4.1e-4 at most; log Mdot 7.80532 -> 7.80523; EW solution 0.5829 (was 0.5830), advected 0.5679.

- **Movement-bound rule** (`bound_step_harmed`, `EXHALE_main.f90`): a rise of the joint distance counts as harm of the step only if the update moved the cell that carries the joint distance by at least half of the largest movement (`prog_cell`, `update_count_move`); otherwise the rise is the wind's response there, logged as such, and the bound grows. In the refused run of section 68 the update had moved those cells by 0.002-0.17 of the largest movement. Without information the old rule applies. Verified: the He/H 2.09 iontrans switch without a trust hold now certifies in 28 passes (refused at 12 before; the same solution as B2 to |d ln n| 4.7e-7); the atomic phase-2 route 11 passes and the conduction rung s = 0.32 -> 0.64 28 passes, both bitwise equal to the control build.
- Every iontrans state certified before this section (B2, B3, C, C2) solved the previous energy equation. The docs (`md/input_schema.md` K44, the manual's `Interdiffusion enthalpy flux` entry, `docs/physics_overview/`) are updated; `TO_BE_DONE.md` (j) is marked fixed for the stages.
- `make test` on the merged tree: 45 of 48 entries PASS. Besides the two steady suites that need `EXHALE_STEADY_RUNLOGS`, `grid_and_gates` fails `stationary_evaluate_*`: the stored certified state of `backup/regression/wasp_full_newton`, evaluated with the secondary ionization armed, now moves OIII by 6.0e-5 and is refused. With the control build of section 71 (before the recombination photoelectron partition) the same test passes (3.8e-10, certified, exit 0): the stored state is a solution of the previous equations. Its refresh belongs with the deferred refreshes.
- Merged ifx binary for the next runs: `LHS1140b/models/.P2/EXHALE_6af10ba4.x` (manifest `.P2/BINARY_MANIFEST_6af10ba4*.txt`). The He/H 2.09 reference is being re-solved on it from B2 (`..._HeH2.09_ref_merged01`, no trust hold, `Restart option change: stage_enth`).

## 73. The He/H 2.09 reference on the merged binary (2026-10-03)

Re-solved from B2 on `EXHALE_6af10ba4.x` (sections 66-72 in it), no trust hold, `Restart option change: stage_enth`: certified in 21 outer passes (`.P2/ew_fit/molecular_photochem_gj1132_kzzprofile_HeH2.09_ref_merged01`, checked `certified=T cert_reason=certified_in_wind`). EW (`phase3/ew_ref_merged01`): solution spectrum 0.5793, advected product 0.5643 percent-angstrom (B2: 0.5830 / 0.5680); observed 1.1084 +- 0.0289. Next: the matching-level temperature sensitivity (plan section 8) and the Photochem/wind elemental-flux closure on this physics (plan phase 3 step 4).

## 74. Photochem/wind elemental-flux closure on the new physics (2026-10-03)

`LHS1140b/models/.P2/ew_fit/phase3/closure/NOTES.md`; driver unchanged (`element_flux_closure.py` passes the reference's keys and env; checked in the running process environment), binary 6af10ba4, Photochem 0.9.0 with climate on, omega 0.5, tolerance 5%.

- Iteration 0 (the reference's profile, trial H 2.566e7, He 5.335e7 g/s from the no-conduction closure): measured H 2.1209e7, He 4.2318e7 g/s (overlap window, spread 2e-7): eps 0.21 / 0.26. With conduction and carried stages

the wind takes 17% less H and 21% less He.

- Converged at k = 2: eps 0.026 / 0.033 below 5%, window spread < 1e-7; three wind solves certified (4-6 passes each), log Mdot 7.800.
- The measured fluxes do not respond to the trial: over the iterations they moved by 1e-5 while the trial moved by 15-18%; eps fell only by the damping. The fixed point is the measured flux (H 2.12095e7, He 4.23172e7 g/s); the accepted trial is 2.6% / 3.3% above it (one undamped step closes it, for phase 6).
- At the 1 microbar match: He/H -1.7e-5, T unchanged, q_H2 +1.4e-5; the column above changes by at most 0.16%; the wind by < 1e-5 (He 2^3S -4.5e-4 near the H2 front). EW of k02: solution 0.5793, advected 0.5643, equal to the reference to the four digits recorded.
- Not done: the undamped step; the Kn, Mach and HeH+ Da profiles of the final state; the reproduction of the 2026-09-16 profile by the present adapter.

## 75. The EW depends on the matching-level temperature: the energy coupling to Photochem is required (2026-10-03)

`LHS1140b/models/.P2/ew_fit/phase3/T0_sensitivity/NOTES.md`. On the He/H 2.09 reference (section 73) the lower-atmosphere profile was raised by Delta T at and above the match and tapered below it with the column response phase 1 solved (all-band shape, 1/e 0.19 H, for 120 K; rotational-band shape, 1/e 0.41 H, for 240 K), levels inserted every 0.02 H to 3 H, r re-integrated hydrostatically (the match rises by 2.6e-4 and 1.1e-3 R_p, so R0 changes), composition and K_zz held at the reference values; headers marked as diagnostic modifications. The reference state was mapped onto each new grid (`map_state_to_grid.py --ic`) and re-solved with the reference controls; both certified (24 and 21 passes).

| Delta T [K] | EW solution / advected [percent-angstrom] | log10 Mdot | base conductive flux [erg cm^-2 s^-1] |
|---|---|---|---|
| 0 | 0.5793 / 0.5643 | 7.8029 | 10.0 |
| 120 | 0.6998 / 0.6855 | 7.8720 | 9.53 |
| 240 | 0.8098 / 0.7964 | 7.9261 | 8.43 |

- dEW/dT0 is about 1.0e-3 percent-angstrom per K: a 6 K change of the matching-level temperature is the numerical target. The EW rises through Mdot (+17%, +33%) and a denser outer flow (n(2^3S) +17/+32% at 3 R_p, +40/+80% at 20 R_p), not through the base layer.
- The reference conducts 10.0 erg cm^-2 s^-1 into its base (0.52 of the absorbed heating), not the 4.25 measured on the He/H 9 state (section 66); the column's actual warming under this flux was not solved and is likely larger than the imposed range.
- By plan section 8 the two-way energy coupling is required. Not separated: the effect of the raised R0 from that of the raised T0; the warmed column's composition was held fixed.

## 76. The wind writes the heat it conducts into the lower atmosphere (2026-10-03)

Phase 1 defect 2 (LHS1140b/models/.P2/ew_fit/phase1/NOTES_phase1.md section 5): the conductive flux through the base face carried 35-52 % of the wind's absorbed heat and was written nowhere; it was recovered from the budget's Sene.

- viscous_conduction.f90: heat_conducted_into_lower_atmosphere(Tcell, f_sp) forms the base face's own term of row 1 of thermal_conduction_coeffs, q = -kappa_{1/2}(T_1 - T_b)/(r_1 - r_0), T_b the bath at the ghost centre r(0) (conduction_base_level_T), converted with p0 v0 = n0 mu v0^3, sign: **positive into the lower atmosphere**. It is held apart from conduction_base_heat_flux, which follows the last residual evaluation (a perturbed state inside Newton).
- certification.f90 calls it on the state it certifies (the evaluation made on the state about to be written, as for the elemental flux windows); gates nothing.
- write_setup_report.f90, EXHALE_resolved.out with Conduction on: base_conduction_state (unmeasured before the wind exists, then measured), base_conductive_flux_cgs, base_conduction_T_bath_K, base_conduction_T_cell1_K. One place, the record the closure driver already reads. Documented in md/input_schema.md and the user manual (EXHALE_resolved.out section).
- Binary LHS1140b/models/.P2/EXHALE_20021893.x (ifx, md5 2002189393ed2fa9012355fc8d2cd3fb, manifest .P2/BINARY_MANIFEST_2002189393ed.txt; it differs from 6af10ba4 in these three files only).

**Checks.**

- A stationary evaluate of a copy of the He/H 2.09 reference (ref_merged01) writes 10.01598 erg cm^-2 s^-1, T_bath 185.022 K, T_1 339.796 K: the 10.0160 of routes S and K of section 75 (MEASURED; agreement to the 4 decimals that note gives). Closure k02: 10.01605.
- src/tests/boundary_state/run.sh contact_law (gfortran private build): five new assertions (the recorded flux is the base face term reversed times p0 v0 to 1e-14; the two temperatures; unmeasured without conduction) and the existing 23 all PASS. The test now sets p0, which it did not (input_read sets p0 = n0 mu v0^2 at every run).
- Not run: the regression matrix and the other suites. The change adds a measurement that writes module variables only and two output lines; nothing the solve reads changes.

## 77. Column energy balance under the conducted flux; the energy-coupled closure (2026-10-03)

Plan section 8: the two-way energy coupling. Notes: LHS1140b/models/.P2/ew_fit/phase3/energy_coupling/NOTES.md.
**Column.** src/utils/conducted_heat_column_balance.py solves, below p_match and down to where the response is below 1e-3 K and the bottom flux below 1e-6 of F (or the tropopause),

```
(1/r^2) d/dz (r^2 kappa dT/dz) = C(T, p) - C(T0(p), p),
kappa dT/dz = F at p_match, T = T0 at the bottom,
```

in s = -ln p at fixed pressure (layer mass p/g fixed, thickness following T), 0.01 in ln p, conservative finite volumes, Newton. kappa is the mixture conductivity of `viscous_conduction.f90`; C the optically thin band emission of CH4, NH3, H2O, C2H2, CO, CO2 from the climate model's own k-distributions; only C(T) - C(T0) enters, so F = 0 returns the climate T bit for bit. Rotational bands in LTE (NH3 rotation-inversion lines A <= 4.8 sˆ-1 against a pressure-broadening rate of 1.4e4 sˆ-1 at 1 microbar, HITRAN). A vibrational band is a two-level system, net cooling eps (e_LTE

- a), eps = C10/(C10 + A10), a = e_LTE(T0)/2 (the upward intensity of the isothermal stratosphere), but only when its V-T rates for both He and H2 are published; otherwise it stays in LTE and is flagged. Treatments: `nlte`, `nlte_partial` (incomplete rate sets used with the partners that are published, a lower bound on that band's emission), `lte`, `rotational`. The thin assumption is refused when more than 10 % of a bin's emission excess (bins carrying > 1 % of the total) would be absorbed by the column above (escape exp(-1.66 tau_bar)).

**Published inputs** (published versions, read):

- CH4 dyad + H2: Halthore, Allen & De Cola 1994 (1994ApJ...424L..61H), Eq. (5): C20 = 4.315e6 p[torr] exp(-40/Tˆ(1/3)); one measured temperature (~296 K), slope from pure CH4 150-500 K.
- CH4 nu4 A10 = 2.12 sˆ-1: Yelle 1991 (1991ApJ...383..380Y) Table 2.
- Band intensities, 296 K, HITRAN line sums (hitran.org, 2026-10-03): NH3 rotation-inversion 1.718e-17, NH3 nu2 2.247e-17 (Weaver & Mumma 1984, 1984ApJ...276..782W, Table 3: 2.2e-17), CH4 dyad 5.195e-18 (Yelle 1991: 5.42e-18 for nu4) cmˆ-1/(molecule cmˆ-2).
- NOT obtainable (publisher refuses this host): CH4 + He (Yardley, Fertig & Moore 1970; Siddles, Wilson & Simpson 1994; Menard-Bourcin et al. 2005) and NH3 nu2 + He (Hovis & Moore 1978, 1980); no NH3 + H2 measurement found. On LHS 1140 b therefore `nlte` = `lte`. (Superseded by section 78: the papers were obtained, and `nlte` now carries them.)

**k-table checks** (MEASURED): band strength of the k-tables / HITRAN sum at 296 K: NH3 rotation 0.987, NH3 nu2 1.005, CH4 dyad 1.006. k-table LTE emission of a fundamental / two-level h c nu0 A10 g1 exp(-h c nu0/kT): CH4 dyad 1.04, 1.03, 1.06 and NH3 nu2 1.11, 1.13, 1.20 at 185, 300, 500 K (the bin-mean product sigma_bar x Bbar and the inversion structure of nu2).

**Tests** `src/tests/conducted_heat_column/run.sh`: 16 PASS (F = 0 identity bit for bit; manufactured column with constant kappa and linear cooling against the analytic exponential, 7.8e-7, order 2.000; budget: emission excess equals F to 3e-16 in three treatments; collision limit of the two-level band equals LTE to 6e-9 K; optically thick column refused).

**Adapter.** `photochem_to_lower_profile.py --conducted-heat-flux F` (default 0: profile and base.inp byte-identical, MEASURED against closure k02 at its own arguments): levels inserted every 0.05 in ln p over 8 H below the match in the column handed to the chemistry; chemistry -> column balance with its composition -> chemistry re-solved on the new T from its previous solution (`reinitialize_to_new_climate_PT`, the escape boundary condition re-imposed, the grid re-pinned) -> until T_match moves < 1 K -> profile with the column solution's nodes inserted where the rise exceeds 1e-3 K, T of the column solution, n and rho from p/(kT), r re-integrated hydrostati-

cally. Header keys `conducted_heat_flux_cgs`, `column_energy_treatment`, `column_T_match_K`; they enter the solution_id. Record `conducted_heat_column.txt`. `lower_profile_schema.py` writes the keys only when present.

**F = 10 erg cmˆ-2 sˆ-1 on the k02 column** (MEASURED):

| treatment | T_match [K] | 1/e depth [H] |
|---|---|---|
| nlte (= LTE) | 337.5 | 0.070 |
| nlte_partial | 345.6 | 0.073 |
| rotational | 556.3 | 0.137 |

NH3 nu2 61 %, CH4 dyad 22 %, NH3 rotation 14 % of the flux (LTE). Emission-weighted tau above the match <= 0.039; up to 7.9 % of a band's excess absorbed above. The bounds 337-556 K differ by 36 times the 6 K target: the NH3 nu2 and CH4 + He rates decide the answer.

**Closure.** `element_flux_closure.py`, config `energy_coupling` (default false; `energy_coupling_omega` 0.5, `T_match_tol` 6 K): the wind's `base_conductive_flux_cgs` damped into the next column's `--conducted-heat-flux`; history columns F_cond_trial, F_cond_meas, T_match; convergence also needs |T_match(k) - T_match(k-1)| < T_match_tol; `--fcond0` or the seed run's record starts it. Also fixed: the `--print-config-template` output was not valid JSON (an escaped quote in a non-raw string). Running in `phase3/energy_coupling/` (lart4, 12 threads, nlte = LTE: the lower bound of T_match), seeded from closure k02.

## 78. The published V-T rates in the column energy balance (2026-10-03)

The five papers section 77 could not obtain were supplied and read in their published versions (`ExoAtmosphere/references/`; tables checked on the page images). `conducted_heat_column_balance.py`, section PUBLISHED RATES, every rate cited at its code site:

| band, partner | rate [cmˆ3 sˆ-1] | process | T range | source |
|---|---|---|---|---|
| CH4 dyad + He | 1.7e-14 (295) ... 1.7e-15 (140), 7 points | coupled nu2/nu4 pair -> ground, relaxation rate | 140-295 K | Siddles, Wilson & Simpson 1994, Chem. Phys. 188, 99, Table 5 |
| CH4 dyad + H2 | 1.44e-13 (193), 3.68e-13 (296) | nu4 (= nu2) -> ground, elementary V-T | 193-296 K | Menard-Bourcin et al. 2005, J. Phys. Chem. A 109, 3111, Table 4 |
| NH3 nu2 + He | 1.1e-13 (198), 2.8e-13 (293), 4.1e-13 (398) | nu2 -> ground V-T,R, relaxation rate | 198-398 K | Hovis & Moore 1980, J. Chem. Phys. 72, 2397, Table I (293 K: Hovis & Moore 1978, J. Chem. Phys. 69, 4947, Table I) |

| band, partner | rate [cm^3 s^-1] | process | T range | source |
|---|---|---|---|---|
| NH3 nu2 + H2 | not measured; ESTIMATE = He rate x k(H2)/k(4He) of the CD4 bending modes (27.5 at 295 K, 36 at 240 K, 49 at 190 K) | - | - | analog: Siddles et al. 1994 Tables 1-2 |

- A relaxation rate (k10 + k01) is turned into k10 by the population factor 1 + sum (g_u/g_0) exp(-E_u/kT) (<= 3 per cent in the measured ranges).
- Inside the measured range: ln k interpolated linearly in T^(-1/3) (every published value reproduced). Outside it: the Landau-Teller law continued from the nearest point, slope fitted to all points of the pair (b = -54.6, -40.8, -37.5 K^(1/3) for CH4-He, CH4-H2, NH3-He; Halthore et al. 1994 used -40 for CH4). The column spans 185-560 K, so NH3-He is extrapolated 160 K above its data and CH4-H2 260 K.
- The analog argument for NH3-H2: the H2 enhancement is V-R transfer into the widely spaced H2 rotational levels, set by the quantum removed (CD4 bend 996 cm^-1 against NH3 nu2 950 cm^-1); the CH4 dyad gives 22-27. Uncertain by a factor of a few.
- A10 from the HITRAN lines with the rotational levels thermalized, T-dependent (NH3 nu2 13.2 s^-1, CH4 dyad 1.96 s^-1 at 185 K; published 15 and 2.12).
- Treatments redefined: `nlte` uses the measured and the estimated rates, `nlte_partial` the measured ones only.
- Cross-checks: Menard-Bourcin CH4-He 1.69e-14 (296 K) against Siddles 1.7e-14; at 193 K 1.4 times Siddles. Yardley, Fertig & Moore 1970 Table I, p tau = 2.3 (He) and 0.12 (H2) microsecond atm at room temperature, i.e. 1.75e-14 and 3.4e-13.

**Tests** (`src/tests/conducted_heat_column`): 33 PASS. New: every tabulated published value reproduced to 1e-12; the NH3-H2 estimate is the He rate times the CD4 ratio; the Landau-Teller continuation is continuous at the ends of the data; `nlte` treats the two bands that carry the flux; the collision limit equals LTE.

**Adapter alone at F = 10.016** (k02 configuration, MEASURED):

| treatment | T_match [K] | 1/e depth [H] | NH3 nu2 / CH4 dyad / NH3 rotation |
|---|---|---|---|
| all LTE | 337.50 | 0.070 | 61 / 22 / 14 % |
| nlte | 353.24 | 0.077 | 62 / 16 / 17 % |
| nlte_partial (NH3-H2 left out) | 380.23 | 0.091 | 38 / 28 / 26 % |

eps at the match: NH3 nu2 0.78 (0.32 without H2), CH4 dyad 0.60. nlte - LTE = +15.7 K, above the 6 K target. The NH3-H2 estimate moves T_match by 27 K, 4.5 times the target: it is the largest remaining uncertainty.

**Closures.** The all-LTE closure (`phase3/energy_coupling/`) was stopped at k = 2: the rates entered the module while

it ran, so its nlte would have changed meaning mid-record. Its k00-k01 stand as the all-LTE record (T_match 337.7, 335.8 K; log10 Mdot 7.880, 7.870 against 7.800 without the coupling). The closure with the published rates runs in phase3/energy_coupling_nlte/ (lart4, 12 threads), seeded as before.

## 79. The EW of the energy-coupled state; the NH3 nu2 + H2 rate as an option and its bracket (2026-10-03)

**The coupled closure converged** (phase3/energy_coupling_nlte/, section 78) at k = 2: eps_H 0.015, eps_He 0.043 (tol 0.05), T_match moved 1.15 K (tol 6 K). Its EW, by the reference operator (ew_of_state.sh, EW_DIAGNOSTIC=1, binary 6af10ba4, R = 68000, solution and adv), is energy_coupling_nlte/ew_k02/EW.txt.

**--nh3-h2-rate-analog** (photochem_to_lower_profile.py; module conducted_heat_column_balance.py, NH3_H2_RATE_ANALOGS, ColumnResponse(nh3_h2_rate_analog=...)). The unmeasured NH3 nu2 + H2 de-excitation rate is the measured NH3 nu2 + He rate times k(H2)/k(He) of an analog:

- cd4_bending (default): CD4 bending modes, Siddles et al. 1994 Tables 1-2, 27.5 (295 K) to 49 (190 K);
- water_bending_overtone: H2O 2nu2 at 295 K, k(H2) = 2.9e-12 cm^3 s^-1 (Zittel & Masturzo 1991, J. Chem. Phys. 95, 8005, Table I, H2(16)O) over k(He) = 3.2e-13 (Hovis & Moore 1980, Table III, H2(18)O, 296 K) = 9.06, held constant in T (the H2 rate exists at 295 K only). Zittel & Masturzo 1989 Table III has no He value. Both table values checked on the published text. Used by treatment nlte only. The profile header and conducted_heat_column.txt record the analog (nh3_h2_rate_analog) for the nlte treatment; the EXHALE reader ignores the key. The solution_id does not include it (it includes column_T_match, which differs). Documented in README_HOWTO.md with --column-energy-treatment.

**Tests** (src/tests/conducted_heat_column): 41 PASS. New: the water estimate equals the He rate times 2.9e-12/3.2e-13 at 185, 295 and 560 K; the default column solution equals the explicit cd4_bending one bit for bit; the water analog lies between CD4 and no H2 term in T_match; the analog does not enter nlte_partial; an unknown analog is refused. RED before the change (module reconstructed without the option): 29 PASS, then a TypeError at the first new assertion. The default nlte, nlte_partial and lte column solutions of the test column are bit-identical to the module before the change (T_match 352.358, 380.122, 336.988 K).

**Bracket** (phase3/energy_coupling_bracket/, NOTES.md there). Two single wind solves at the coupled state's fluxes, not new closures: the closure driver with --kmax 1, the k02 trial elemental fluxes, F_cond = the 9.6264 k02 measured, seeded from k02's output through map_state_to_grid.py --ic, the closure's binary, keys and environment; only the adapter option changed.

| case | T_match [K] | F_cond trial / measured [erg cm^-2 s^-1] | log10 Mdot [g/s] | EW solution [%A] | EW adv [%A] | eps_H / eps_He |
|---|---|---|---|---|---|---|
| uncoupled reference | 185.0 | - / 10.016 | 7.8029 | 0.5793 | 0.5643 | - |

| case | T_match [K] | F_cond trial / measured [erg cm^-2 s^-1] | log10 Mdot [g/s] | EW solution [%A] | EW adv [%A] | eps_H / eps_He |
|---|---|---|---|---|---|---|
| coupled, CD4 analog (converged) | 349.88 | 9.7267 / 9.6264 | 7.8854 | 0.7255 | 0.7114 | 0.015 / 0.043 |
| water 2nu2 analog (single solve) | 358.01 | 9.6264 / 9.6227 | 7.8928 | 0.7401 | 0.7262 | 0.023 / 0.063 |
| no NH3 + H2 term (single solve) | 375.29 | 9.6264 / 9.5429 | 7.9041 | 0.7629 | 0.7491 | 0.035 / 0.092 |

(MEASURED; log10 Mdot at cell N-20 recomputed from the states, the history's 7.890 for k02 being the two-decimal post-processing print; the reference's 10.016 is the section 76 evaluate.)

- The energy coupling raises the EW by 0.146 %A (+25 per cent), 25 times the numerical target 0.0058, and Mdot by 21 per cent. The EW stays below the observed 1.108 +- 0.029.
- The NH3 nu2 + H2 estimate: water analog +0.015, no H2 term +0.037 %A against CD4, 2.5 and 6.4 times the target. It remains the largest single uncertainty of the coupled EW.
- The neglected re-iteration: the conducted flux is closed (out/in -0.04 and -0.9 per cent, about 1 K of T_match); the He flux is not (eps_He 0.063 and 0.092). Its effect on the EW was not measured.
- Figures: `energy_coupling_bracket/fig_coupled_profiles{,_inner}.png`, `fig_bracket_profiles{,_inner}.png`, `fig_bracket_differences.png`. The coupled wind is 18-37 per cent denser beyond 1.1 R_p, its temperature peak about 80 K higher, and n(He 2^3S) up to 1.5 times higher at 30 R_p.

## 80. The NH3 nu2 + H2 bracket re-closed: both cases converge (2026-10-04)

The two bracket cases of section 79 were continued from their single wind solves as full energy-coupled closures, identical to `phase3/energy_coupling_nlte/` (binary 20021893, Photochem arguments, `exhale_env`, keys, Resid tol 4e-6, 12 OpenMP threads, omega 0.5, omega_E 0.5, tol 0.05, T_match_tol 6 K, kmax 12) except the column option: `phase3/energy_coupling_water_analog/` (`--nh3-h2-rate-analog water_bending_overtone`) and `phase3/energy_coupling_no_h2/` (`--column-energy-treatment nlte_partial`). Each was seeded from its bracket `k00/output`, with first trial elemental fluxes and F_cond equal to the driver's own damped update from that k00 (trial + 0.5 (measured - trial); F_cond 9.624561 and 9.584647 erg cm^-2 s^-1). Both converged at k = 1 (two wind solves each, info 0, about 2-3 min each). EW by the operator of section 79 (`ew_of_state.sh`, `EW_DIAGNOSTIC=1`, binary 6af10ba4). Comparison: `phase3/energy_coupling_bracket_converged/` (compare.py, compare.txt, NOTES.md).

| case | k | T_match [K] | F_cond trial / measured [erg cmˆ-2 sˆ-1] | log10 Mdot [g/s] | EW solution [%A] | EW adv [%A] | eps_H / eps_He |
|---|---|---|---|---|---|---|---|
| CD4 analog, converged | 2 | 349.88 | 9.7267 / 9.6264 | 7.8854 | 0.7255 | 0.7114 | 0.015 / 0.043 |
| water 2nu2 analog, converged | 1 | 357.97 | 9.6235 / 9.6221 | 7.8928 | 0.7400 | 0.7261 | 0.006 / 0.016 |
| no NH3 + H2 term, converged | 1 | 374.62 | 9.5744 / 9.5692 | 7.9048 | 0.7643 | 0.7505 | 0.009 / 0.024 |
| water 2nu2 analog, single solve (section 79) | 0 | 358.01 | 9.6264 / 9.6227 | 7.8928 | 0.7401 | 0.7262 | 0.023 / 0.063 |
| no NH3 + H2 term, single solve (section 79) | 0 | 375.29 | 9.6264 / 9.5429 | 7.9041 | 0.7629 | 0.7491 | 0.035 / 0.092 |

(MEASURED; log10 Mdot = 4 pi rho v rˆ2 at cell N-20 of the written state.)

- Re-closing changes the single-solve values by: water T_match -0.03 K, log10 Mdot -0.00004, EW -0.0001 (solution and adv); no H2 term -0.67 K, +0.0007, +0.0014. The re-iteration left open in section 79 moves the EW by at most 0.0014 %A, a quarter of the numerical target 0.0058.
- The bracket against the converged CD4 state: water +0.0145 / +0.0147, no H2 term +0.0388 / +0.0391 %A (2.5 and 6.7 times the target); T_match +8.1 and +24.7 K.
- The water k01 state: the solve certified it as written (mass row 2.17e-12 against 3.0e-12), but the evaluate of the reloaded state on 6af10ba4 refuses it on the mass row alone (3.316e-12 against 3.0e-12 at cell 253, r = 1.377 R_p; evaluate exit 2), so its EW products carry the label `diagnostic`. The EW is 1e-4 from the certified single solve of the same case. The cause of the reload difference was not investigated.
- Figures `energy_coupling_bracket_converged/fig_converged_profiles{,_inner}.png`: against CD4 the water and no-H2 winds are warmer below 1.1 R_p (up to +11, +29 K), cooler at 1.19 R_p (-11, -32 K), warmer beyond (+13, +33 K near 2.2 R_p; +3, +8 K at 30 R_p); n(He 2ˆ3S) is 0.96 and 0.89 times the CD4 value just outside the front at 1.15 R_p and 1.04 and 1.10 times at 30 R_p.

## 81. He 2^3S transport: the metastable carried as a fourth helium state (2026-10-04)

Plan `md/PLAN_20261003_lhs1140b_ew.md` phase 3 step 3 (decision 4); design `md/He23S_transport_design_20261003.md`, approved with the key name `He 2^3S transport` and `Ionization transport: True` required. Phase 1 (`LHS1140b/models/.P2/ew_fit/phase1/NOTES_phase1.md` section 1) had measured that the local balance of the level is exact inside 2 R_p and 1-4 % off at 3-30 R_p, where the 10830 line forms.

**What the option does.** `He 2^3S transport: True` (default off) makes x3 = n(2^3S)/n(He nuclei) carrier `ic_HeTR` (index 8) of the transport-chemistry operator. It rides on the helium nucleus flux with the stage flux of `ionization_stage_transport.f90` equation (1), its source is the level balance `tr_triplet_row` (row 4 atomic, row 8 molecular with the H2 quench) at the trial densities, the singlet closes the three carried helium states, and every sweep is handed the transported value (`x_hetr_fix` replaces the level row). It is the steady advection balance of Oklopcic & Hirata (2018, Eq. 15) and Lampon et al. (2020, Eq. 11) with the stage eddy term.

**Files.**

- `parameters.f90` (`he23s_transport`; the stale comment of `ionization_transport`, which said H+ alone and "requires Molecular carrier transport", corrected to the code: three stages, atomic gas allowed); `input_read.f90` (key, refusals without `Include He23S` or `Ionization transport`); `load_IC.f90` (token `he23strans`, the 26th, not a layout token, `F` when absent; coupling tag and notes); `utilities.f90` (`he23strans=T` on the coupling line); `write_setup_report.f90` (setup text; `he23s_transport` in `EXHALE_resolved.out` and the parse dump only when on).
- `diffusive_photochemistry.f90`: index, name `He2^3S`, set and registration, the unknown, lookups, `carrier_is_nucleus_fraction` (the three stages and the level: the carriers written per element nucleus on the nucleus flux; `carrier_is_ionization_stage` keeps naming the three stages), the face state, the nucleus sum with nk = 3, the helium availability without the reservation of the level, the singlet closure with the trial level, the source row and its ledger, the FD guard and the projection on the three-dimensional simplex, the write-back of the level from its own row, the headroom of the level (the helium nucleus density: `element_box_side` refuses an excited level), and `helium_metastable_excitation_energy_divergence` (a measurement, item 3 below).
- `binary_element_diffusion.f90` (`n_car_max = n_bsp`), `ion_cell_state.f90` (`x_hetr_fixed`, `x_hetr_row`, `x_hetr_fix`), `ion_residual_core.f90` (the imposed level row), `ionization_equilibrium.f90` (flags, seed, release in the rate matrix, OpenMP shared list), `constrained_chemical_equilibrium.f90` (fixed species, seed, debug record), `attempted_step.f90` (hash), `certification.f90`, `ionization_stage_transport.f90` (header).
- Tests: new `src/tests/helium_metastable_transport/` (run.sh and driver); `src/tests/certification/certification_contexts.f90` (below).

**Decisions taken in the implementation.**

1. Entry name `carrier balance He2^3S` (no blank: the carrier name is one whitespace-delimited field of the row-term record).

2. Certification: `carrier balance He2^3S` at `cert_tol_carrier_wind`; `level balance He 2^3S` not applicable under the key; `ionization stage nucleus sum He` with nk = 3.
3. The excitation-energy flux of the level relative to He I is NOT a term of the energy row, contrary to design section 4 ("added if it exceeds 1e-3"): the 19.82 eV are chemical energy, booked where the level's reactions run (q13 cooling, q31g heating, Penning heat) at the transported population, so adding the flux would count them twice. It is printed by every certification as a reported-only line. MEASURED on the validation state: max |div q_xl/(energy row's largest term) = 3.0e-7 over the column (cell 198, 1.14 R_p), 6.6e-9 at r >= 1.2 R_p; written at the code site.

**Tests** (gfortran private build; all MEASURED).

- `helium_metastable_transport`: 22 assertions, all PASS (RED before: the driver does not compile against the entry text, which has no `ic_HeTR`). The level is the solved last carrier on the nucleus flux; the available helium reserves HeH+ alone (1e-14); a partition between the old reserved bound and the new simplex is left unchanged, one outside is returned to it; the write-back keeps H and He nuclei and mass to 1.1e-16 and writes the level from its row; the ledger sprod - sloss = src holds for the level in molecular and atomic gas; at the closed-form root of the level balance the row reads 7.4e-17 of its terms; d src(He II)/d n3 is the analytic g3 - g1 + n_e (b3 - b1) to 1e-6. Manufactured three-state helium column, 250/500/1000: 2^3S row 7.29e-7, 9.68e-8, 1.25e-8 (observed order 2.935), extrapolated 7.5e-8 at the anchoring spacing 1.8447e-3 R_p and 5.0e-7 at the gating spacing 3.5215e-3 R_p (gate 1e-5); nucleus-sum identity 4.9e-16 against the nk = 3 bound 1.5e-14.
- The 22 existing suites whose drivers link the changed modules (adv_static_limit, carrier_boundary_jacobian, carrier_constraint_attribution, carrier_enthalpy_flux, carrier_helium_inventory, carrier_outer_boundary, carrier_reference_scales, carrier_retry, carrier_returned_state_acceptance, certification, coupled_block_handover, coupled_block_jacobian, element_operator, grid_and_gates, interdiffusion_enthalpy_flux, ionization_imposed_fractions, ionization_stage_enthalpy_flux, ionization_stage_flux, krylov_and_dogleg, species_masses, stage_row_balance, steady_species_rows): all pass except two items. (a) `certification_contexts` assumed an even report capacity (its last entry's tolerance and verdict were hard-coded to the helium entry); the capacity grew by one carrier and became odd, so the expectation is now taken from the same parity as the name and measure (test defect; 145 PASS after). (b) `grid_and_gates/restart_intent_and_metadata.sh` fails three `stationary_evaluate_*` assertions (a rebuilt O III of 6.0e-5 against 1e-8, the stored certified fixture re-certified F, exit 2) with the tree's own `EXHALE.x` built before this change (md5 60fa14bf) and identically with the new build: pre-existing, the stored fixture is no longer certified under the current equations; not fixed (it needs the fixture re-made).
- Default off, one stored case: the EW evaluate (`ew_of_state.sh`, `EW_DIAGNOSTIC=1`) of `energy_coupling_nlte/k02` with the new binary writes `Hydro_ioniz`, `Ion_species` and their `_adv` pair with data rows bitwise identical to the reference evaluate made with 6af10ba4 (`energy_coupling_nlte/ew_k02`); the headers differ only by the run time and the new token `he23strans=F`; EW 0.7255 / 0.7114 % A as before. (The first build of this evaluate stopped in the all-carrier headroom loop on the level, `element_box_side`; that is the headroom fix above, made before any result was taken.)

- Restart contract with the binary: off -> on accepted with the token named and refused unnamed; on -> off likewise; the key without `Ionization transport` or without `Include He23S` refused at startup.

**Binary.** `LHS1140b/models/.P2/EXHALE_4c0e8603.x` (ifx 2026.1.0, md5 4c0e86033617eca84914079aa61a8edf; manifest `.P2/BINARY_MANIFEST_4c0e86033617.txt`, source copy `.P2/build_he23s_ifx/`): the source of 20021893 (the tree still held its changes, checked file by file against that manifest) plus this section.

**Validation** (`LHS1140b/models/.P2/ew_fit/phase3/he23s_transport/`). `solve/` restarts the converged `energy_coupling_nlte/k02` state with `He 2^3S transport: True` and `Restart option change: he23strans`, the closure's keys and environment (Resid tol 4.0e-6, 12 threads, Well balanced, stationary intent, Newton, Secondary_ionization Immediate). It is CERTIFIED in 10 outer passes; the binding row of every pass was `carrier balance He2^3S` at cell 217 (r = 1.20 R_p), 5.8e-2 on pass 1 and 8.59e-6 on pass 10 (gate 1e-5); below the gate the row reads 1.48e-4 at cell 1 (reported). EWs with the same operator and binary 4c0e8603 for both states (the transported state's input carries the new key, which 6af10ba4 does not read):

| | local balance (k02) | transported |
|---|---|---|
| EW solution [% A] | 0.7255 | 0.7334 |
| EW adv [% A] | 0.7114 | 0.7114 |
| log10 Mdot [g/s], cell N-20 | 7.885432 | 7.885431 |
| n(2^3S) / local, 3.02 R_p | 1 | 0.99951 |
| 10.03 R_p | 1 | 1.03947 |
| 29.03 R_p (cell N) | 1 | 1.16845 |
| level entry | `level balance He 2^3S` 9.5e-21 (READ, k02/run.log) | `carrier balance He2^3S` 8.59e-6 |

(MEASURED except where marked.) n(2^3S) moves by less than 4e-5 inside 2 R_p, by -0.45 % at most near 5.7 R_p, and by +18 % at most near 25 R_p; T moves by at most 3.1e-6 and v by 1.1e-5 relatively. The solution EW rises by 0.0079 % A (+1.1 %), 1.4 times the numerical target 0.0058. The phase-1 estimate (+0.013 to +0.019 % A) was made on a different state (He/H 9 deep, EW 8.56 % A), so the two are not directly comparable; the shape is the same (negative at 3-7 R_p, positive beyond), the outer rise is larger here (+17 % at 29 R_p against +4 % there). The adv product is unchanged (0.7114): it rebuilds x3 along streamlines from the hydrodynamic state, which moved by 1e-5. Against it, the Lagrangian x3 of `post_process_adv` on the transported state lies 1-4 % below the transported x3 at 1.5-10 R_p and 1.8 % below at 29 R_p; it reads 3.2-3.7 % below the local balance at 2-8 R_p, where the local balance is exact, which appears to come from the adv product's own H/He partition on its molecule-free background (module header of `post_process_adv.f90`) rather than from the level's transport; not investigated further. [Traced in section 82: the adv product advected every element on the bulk velocity; corrected there.] Elemental fluxes moved by -7e-7 (H) and -1.8e-6 (He) relatively and the base conducted heat by +2e-7, far inside the 5 % closure tolerance (READ, `closure.json`), so the Photochem closure was not re-run. Figure `he23s_profile_comparison.png` (n(2^3S) of the two states and the adv product; ratios to the local balance).

**Not verified.** The option in an atomic gas end to end (the unit tests cover the atomic rows; the validation run is molecular); the marching route with the option (the advected register of the level and the conversion of its advected

mass fraction into a fraction of the helium nuclei belong to the marching step, which no run of this section took); the inner ghost value of x3 is the sweep's local root as for the stages (not re-examined).

**Records.** `md/input_schema.md` (K15f2, K44), `docs/EXHALE_user_manual.tex` (the new key; the `Ionization transport` entry rewritten to the code: three stages, atomic gas allowed, `Coupled carrier solve` refused, which the old text had the other way round), `README_HOWTO.md`, `docs/physics_overview/{ionization,molecules_metals,summary_table}.tex` (the stale "fifth carrier on the mass flux" paragraph rewritten), `md/He23S_transport_design_20261003.md` section 11.

## 82. The advected product steps each element on the velocity of its own nuclei (2026-10-04)

**Finding.** Section 81 left open why the `_adv` product of the He 2^3S-transported reference (`LHS1140b/models/.P2/ew_fit/phase3/he23s_transport/`) reads n(2^3S) 3.2-3.7 % below the solution at 2-8 R_p, where the level is in local balance. The level is not the cause: n(2^3S)/n(He+) of the adv product agrees with the solution's to 0.1 % there (MEASURED), and the 2^3S deficit follows a He+ deficit (He+ 0.963-0.970 of the solution, H+ 1.056-1.062). The stages are transport-dominated at those radii (the He+ carrier row of the solution balances its net source against the stage-flux divergence, `solve/output/carrier_row_terms.txt`), so their values depend on the transport law, and the two products used different ones:

- the solution carries every stage on its element's nucleus flux (`ionization_stage_transport.f90`, equation (1)), i.e. on v_H and v_He;
- the adv product stepped every row with c1 = g h/v, the mass-weighted velocity (`post_process_adv.f90` `composition_step`, the old line `adv_cell%c1 = step_width(jc, bdf2)*R0/(v(jc)*v0)`; the adv systems `System_implicit_adv_{H,HeH,HeH_TR}.f90` used that one c1 for every row). With `He_diffusion` on and the strong separation of this state (He/H 2.09 at the base, 0.59 at 29 R_p) the nuclei do not move with v: v_He/v = 0.82-0.97 and v_H/v = 2.1-1.08 at 1.3-4 R_p (MEASURED from `element_flux_profile.txt`). The stage eddy term of (1), which the marching step also lacks, is small: K ldln x/drl/v_el = 0.035 (He+) at 1.3 R_p, 0.005 at 1.5 R_p, below 0.001 beyond 1.75 R_p (K = 1e9 cm^2 s^-1). Directions agree: the adv helium rode too fast (less ionized), the adv hydrogen too slow (more ionized). **Verdict: a physics defect of the adv product whenever element diffusion is on**; the fractions it steps are fractions of one element's nuclei, which move with that element's velocity.

**Fix.** With element continuity div(n_el v_el) = 0 the steady stage balance div(x n_el v_el) = S is v_el dx/dr = S/n_el.

- `post_process_adv.f90`: v_He = v + J/(rho Y) and v_H = v - J/(rho (1 - Y)) per cell, J the diffusive helium mass flux of the state (`helium_diffusive_face_flux`, the flux the run's element row is the divergence of; mean of the two faces), Y the helium mass fraction (`element_mass_fractions`), rho from `mixture_mass_sum`; both equal v without `He_diffusion`. `composition_step` sets c1 = g h/v_H and the new c1_he = g h/v_He. The validity conditions use them: inflow of either element refuses the cell (condition (i)), and the Damkohler number of condition (ii) takes each element's rates with its own residence time. The omitted stage eddy term and its measured size are written at the code site.

- ion_cell_state.f90 (adv_rates%c1_he, the step statement), System_implicit_adv_HeH.f90 and System_implicit_adv_HeH_TR.f90 (the helium rows on c1_he), System_implicit_adv_H.f90 (comment), charge_exchange.f90 (he_h_cx_fvec_adv takes c1 and c1_he: the H row on c1, the He rows on c1_he).

**Tests.** New suite src/tests/adv_element_velocity/ (driver element_velocity_rate_weights.f90): on both adv layouts, with (c1, c1_he) = (c, c) and (c, 2c) the rate part of the hydrogen row does not move and every helium row doubles, charge-exchange pair included; with (2c, c) the reverse. 9 assertions PASS on the private build (relative change 0 against 1e-12); RED on the entry text (compile error: c1_he not a member of adv_rates). src/tests/charge_exchange_rows/advection_pair_reservoir.f90 sets c1_he. Suites whose drivers use the changed modules, private gfortran build: adv_static_limit 64, charge_exchange_rows 257, charge_exchange_detailed_balance 298, attempted_step 70, carrier_constraint_attribution 10, carrier_helium_inventory 15, carrier_reference_scales 16, carrier_retry 156, constrained_network_layout 8, ionization_imposed_fractions 48, stage_row_balance 10, physics_probe 2499 PASS; 0 FAIL. Without element diffusion the change is the identity: the evaluate of the stored backup/regression/wasp_full_newton state (he_diffusion F) with a gfortran control build of the entry text and the gfortran build of the change writes Hydro_ioniz_adv.txt and Ion_species_adv.txt with data rows bitwise identical (504 rows, 503 corrected cells).

**Effect on the reference** (he23s_transport/ew_solve_advvel/, binary LHS1140b/models/.P2/EXHALE_8cc7b546.x, ifx, md5 8cc7b5462cc2390eefed1bb3658e307f, manifest BINARY_MANIFEST_8cc7b5462cc2.txt; the source of 4c0e8603 plus this section). MEASURED, adv product against the solution:

| r [R_p] | n(2^3S) bulk v | n(2^3S) element v | He+ bulk / element | H+ bulk / element |
|---|---|---|---|---|
| 2.00 | 0.9680 | 1.0122 | 0.9668 / 1.0121 | 1.0480 / 0.9604 |
| 3.02 | 0.9635 | 1.0066 | 0.9629 / 1.0082 | 1.0615 / 0.9679 |
| 4.98 | 0.9662 | 1.0028 | 0.9666 / 1.0063 | 1.0604 / 0.9726 |
| 7.95 | 0.9686 | 1.0014 | 0.9698 / 1.0054 | 1.0576 / 0.9750 |
| 20.10 | 0.9728 | 1.0009 | 0.9743 / 1.0045 | 1.0524 / 0.9779 |

adv EW 0.7114 -> 0.7369 % A (+0.0255, 4.4 times the numerical target 0.0058); the solution EW 0.7334 does not depend on the adv product. The remaining offset starts at the first corrected cell (1.157 R_p), at the He ionization front, where He+ rises by a factor 17 in seven cells and the first adv step jumps by +15 % in He+ against the solution (it did so with the bulk velocity as well, +14 %); the frozen-in stages carry it outward. It is a difference between two discretizations of one equation across a front resolved by a few cells (upwind limited face fluxes against a backward-Euler start of the BDF2 march), not a missing term; it was not reduced further. Figure he23s_transport/acceptance/fig_adv_vs_solution.png.

**Superseded.** Every adv EW of a run with He_diffusion: True made before this section was stepped on the bulk velocity: the adv columns of sections 79-81 (0.7114, 0.7261, 0.7505, 0.5643 and the bracket's) and every adv number of the LHS 1140 b catalog with element diffusion. They are not recomputed here.

## 83. The EW evaluate reloads the exact restart (2026-10-04)

**Finding.** Section 80 recorded that the water-analog state (`phase3/energy_coupling_water_analog/k01`), certified when written (mass row 2.17e-12 against 3.0e-12 at cell 255), was refused by the EW evaluate (`ew_k01/`: mass row 3.316e-12 at cell 253, exit 2, EW labels `diagnostic`). The evaluate did not load the exact restart: `LHS1140b/models/.P1/fiducial_conduction/ew_of_state.sh` staged only the dimensional pair, `Hydro_ioniz.txt` and `Ion_species.txt` as the `_IC` files, and never `output/conserved_state.txt`, which section 63 writes beside every 'eq' write and which `init.f90` reads as `output/conserved_state_IC.txt`. The run therefore rebuilt (rho, rho v, E) from 16-digit rho, v, p with the energy re-formed from p through the caloric equation of state, and the mass row in the wind, which sits at the rounding level of the face flux difference there, moved past its 3.0e-12 floor at one cell. A defect of the script, not of the code. The `generation=absent` of `EW.txt` is not related: the `state_id` field is written only on a pass-state generation, and a final 'eq' write carries none (the script's header now says so).

**Fix.** `ew_of_state.sh` stages `conserved_state.txt` as `conserved_state_IC.txt` when the state has one, records its md5 (`exact_state`) and whether the evaluate read it (`exact_state_read`, from the reader's own message or its IGNORED notice) in `EW.txt`. `element_flux_closure.py` already staged it (section 63). `LHS1140b/models/.P2/ew_fit/phase1/run_after.sh` stages the pair only; it is the record of a finished measurement and is left as it ran.

**Verification** (MEASURED; same binaries as before).

| state | binary | evaluate | certification on reload | EW solution / adv [% A] |
|---|---|---|---|---|
| water k01, text reload (`ew_k01/`) | 6af10ba4 | exit 2 | REFUSED, mass 3.316e-12 / 3.0e-12, cell 253 | 0.74001762 / 0.72605621 (diagnostic) |
| water k01, exact (`ew_k01_exact/`) | 6af10ba4 | exit 0 | CERTIFIED, mass 1.348e-11 / 1.6e-11 (rounding anchor), cell 207 | 0.74001762 / 0.72605621 (result / conditional) |
| reference, text reload (`he23s_transport/ew_solve/`) | 4c0e8603 | exit 0 | CERTIFIED, mass 2.83e-12 / 3.2e-12, cell 250 | 0.73339944 / 0.71142279 |
| reference, exact (`he23s_transport/ew_solve_exact/`) | 4c0e8603 | exit 0 | CERTIFIED, mass 2.00e-12 / 3.0e-12, cell 252 | 0.73339944 / 0.71142279 |

With the exact state the physical cells of u are restored bitwise (the evaluate's own `conserved_state.txt` against the solve's: U rows identical except the two lower ghosts, which the boundary rebuilds). The species fractions are not bitwise: the stationary evaluate takes one composition sweep from the loaded composition (its log: chemical residual 1.6e-16, particle count 4.5e-16), which moves f_sp by at most 4.4e-13 relative (text reload: 2.2e-12), and the mass-row verdict cell moves with it (the solve wrote cell 255 at 2.17e-12; the exact reload binds at cell 207). The mass row in

the wind is a rounding-level quantity against a floor of the same order; that is the certification's design (the rounding anchor of each cell), not changed here.

## 84. Phase 3 of the LHS 1140 b EW plan accepted; TO_BE_DONE (AH) closed (2026-10-04)

Plan `md/PLAN_20261003_lhs1140b_ew.md` phase 3 steps 4-6 on the reference `LHS1140b/models/.P2/ew_fit/phase3/he23s_transport/solve/` (section 81; binary 4c0e8603). Record and figures: `he23s_transport/acceptance/` (`NOTES.md`, `acceptance.py`, `acceptance_numbers.txt`, five figures, all inspected).

**Step 4, the elemental-flux closure and (AH).** The closure measures on `element_flux_profile.txt`, whose face fluxes are the Riemann face mass flux of the certified state (`write_element_flux_profile`, called from `certification.f90` with `Frho_cert`), reduced over the overlap window by the code (`lower_profile_F_*` of `EXHALE_resolved.out`). Re-measured from the profile itself on the reference: window 1.00058-1.01892 R_p, 82 faces (not empty); F_H 2.32443e7 g/s, F_He 5.35726e7 g/s, spreads 3.8e-9 and 3.1e-9 (the face-mean measure of Phase E gave 60-700 %); the fluxes are constant through the whole column to the 8 printed digits. Against the Photochem trial fluxes of the closure's last iterate (2.28992e7, 5.12443e7): 1.48 % (H) and 4.35 % (He), inside the 5 % tolerance, so the closure's 5 % agreement holds on the Riemann measure. TO_BE_DONE (AH) closed with these numbers; the older `atomic_photochem_*_kzzprofile` rungs it named were not re-evaluated (superseded by the molecular closure).

**Step 5, spectrum and EW.** Solution 0.7334 % A (decision 2's acceptance basis), adv 0.7369 (section 82); observed 1.1084 +- 0.0289, so the He/H 2.09 reference lies 0.375 % A (13 sigma, independent-sample error) below the observation with the full equation set. The line is twice as deep and three times narrower than observed (peak 2.52 % against 1.28 %, FWHM 0.24 A against 0.76 A on a 3-sample running mean of the observation), and the observed peak sits 0.15 A redward of the model's.

**Step 6, acceptance.**

- Certified wind under unchanged tolerances (`solve/run.log`, final state): mass 3.36e-12 / 4.3e-12 (rounding-anchored, cell 242), momentum 2.3e-11, energy 1.5e-8, carriers H2 1.7e-7, H+ 4.0e-9, He+ 6.3e-8, He++ 1.7e-7, He2^3S 8.59e-6 (gate 1e-5, r >= 1.2 R_p), nucleus sums 2.2e-16 / 4.3e-16, elemental transport 2.7e-12. Re-certified on the exact reload.
- Photochem at steady state: `reached_steady_state T`, iteration 2; the profile is the k02 one byte for byte.
- Flux agreement over the stated overlap region: above.
- EW, line profile and full profiles against the observation: figures `fig_line_profile.png`, `fig_profiles.png`, `fig_adv_vs_solution.png`.
- Kn, Mach, HeH+ Damkohler (`fig_validity.png`): Kn_bulk = 0.1 at 5.69 R_p and 1 at 17.21 R_p; sonic point 25.7 R_p, Mach 1.08 at the boundary; HeH+ Da (code rates R16, R18, R19; advection and eddy times) 3.4e6 at the n(2^3S) peak (1.40 R_p), 94 at 5 R_p, 3.4 at 10 R_p, 0.12 at 29 R_p; Da < 10 in 82 cells holding 2.0 % of the radial He 2^3S column.

**Stale (2026-10-07):** the Knudsen numbers, `Kn_bulk = 0.1` radii and exobase radii in this passage were computed by `src/utils/collisional_validity.py` before its resonant H+ + H and He+ + He momentum-transfer coefficients

(md/Update_EXHALE_stage3.md section 110). On the 102 stored LHS 1140 b states the change moves the Kn_bulk = 0.1 radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.

**Not verified.** The outer region (beyond Kn_bulk 0.1, where most of the line forms; exobase inside the domain) is phase 4; the ion-neutral drift and the NH3 nu2 + H2 bracket are not in the EW; the HeH+ Damkohler number leaves out molecular diffusion.

**Carried into phase 4.** State he23s_transport/solve/; binary for new calculations .P2/EXHALE_8cc7b546.x (4c0e8603 plus section 82, which changes only the adv product: the evaluate of the reference writes Hydro_ioniz.txt, Ion_species.txt and Heating_breakdown.txt with data rows bitwise identical on 4c0e8603 and 8cc7b546, MEASURED; a full solve on 8cc7b546 was not run).

## 85. Phase 4 of the LHS 1140 b EW plan: outer domain, outer boundary, transit-weighted shares, Roche geometry (2026-10-04)

Plan md/PLAN_20261003_lhs1140b_ew.md phase 4 on the phase-3 reference (section 84). Record, scripts and figures (all inspected): LHS1140b/models/.P2/ew_fit/phase4/NOTES.md. Binary .P2/EXHALE_8cc7b546.x (ifx), the reference's keys and environment, Photochem closure not re-run. EW in percent-angstrom (% A); numerical target 0.0058. All MEASURED unless marked READ.

**Roche geometry** (code unit R0 = 0.16253525 R_J): Hill radius 38.01 R0, L1 37.61, volume-equivalent Roche-lobe radius (Eggleton) 26.72 (nominal R_p: 39.18, 38.77, 27.54). The reference runs Domain mode: Spherical. EXHALE_transit.py ends the impact parameters at R_*/R_p = 12.93.

**EW against the outer radius.** Same keys; base block kept; for 15 and 20 the cell count that keeps the reference's stretch ratio (widths at shared radii within 7e-4); beyond 30, Outer shells [r_face,cells]: 37.6 15 (face at L1, interior bit for bit the reference's); seeds by map_state_to_grid.py --ic. All certified (21-23 passes); every final pass bound by carrier balance He2^3S at cell 217.

| outer radius | grid | EW sol / adv | log10 Mdot | sonic point | Kn = 0.1 | Kn = 1 |
|---|---|---|---|---|---|---|
| 15 | 458, face 14.65 | 0.7565 / 0.7602 | 7.885528 | none (Mach 0.24 at edge) | 5.97 | not reached |
| 20 | 476, face 19.52 | 0.7044 / 0.7078 | 7.885798 | none (Mach 0.57) | 5.62 | not reached |
| 30 (reference) | 500, face 29.27 | 0.7334 / 0.7369 | 7.885456 | 25.69 | 5.69 | 17.21 |
| 37.6 | 515, face 37.60 | 0.7360 / 0.7396 | 7.885425 | 25.73 | 5.70 | 17.29 |

**Stale (2026-10-07):** the Knudsen numbers, Kn_bulk = 0.1 radii and exobase radii in this passage were computed by src/utils/collisional_validity.py before its resonant H+ + H and He+ + He momentum-transfer coefficients (md/Update_EXHALE_stage3.md section 110). On the 102 stored LHS 1140 b states the change moves the Kn_bulk =

`0.1` radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.

Elemental fluxes move by <= 1e-3 relative and the conducted heat by <= 2.5e-4 against the reference, all inside the 5 % closure tolerance (the neglected re-closure). With the face past the sonic point the extension to L1 changes the EW by +0.0026 (0.45 of the target); faces inside it (15, 20) put the flow on another subsonic branch beyond 5 R_p and change the EW by +3.2 % and -4.0 % (not monotonic).

**Outer boundary condition.** The code has one rule for each outer closure and no key or environment variable selecting another (READ): hydrostatic free-outflow ghost (the edge is supersonic here, Mach 1.08); zero diffusive flux for conduction (1.2e-3 to 3.0e-3 of the advected enthalpy flux at the last interior faces), for the element operator (a one-cell layer, X_He -0.12 % at face 499) and for the carriers (eddy/advective 1.7e-8 to 2.8e-7). No admissible alternative exists to run; the last two cells carry 1.5e-4 of the EW, so the element wall of TO_BE_DONE (m) / section 67 cannot move the EW of this reference by more than 1e-6 relative (the proposed outflow option is not needed here and was not implemented). The cell-500 momentum row of section 67: the momentum row is largest at the last cell on every domain, 2.3e-11 (30) and 5.5e-11 (37.6) against 1e-8; it does not affect this reference.

**Transit-weighted shares** (n(2^3S) truncated beyond R in the transit synthesis of the solution state, EW differenced): beyond Kn = 0.1 (5.69 R0) 39.8 % (radial column 5.3 %); beyond R_*/R_p 4.5 %; beyond Kn = 1 (17.21) 1.09 %; beyond the sonic point 0.094 %; beyond the volume-equivalent Roche radius 0.050 %.

**Stale (2026-10-07):** the Knudsen numbers, `Kn_bulk = 0.1` radii and exobase radii in this passage were computed by `src/utils/collisional_validity.py` before its resonant H+ + H and He+ + He momentum-transfer coefficients (`md/Update_EXHALE_stage3.md` section 110). On the 102 stored LHS 1140 b states the change moves the `Kn_bulk = 0.1` radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.

**Roche geometry of the transit.** `EXHALE_transit.py` has no He 10830 triaxial branch; `phase4/roche/roche_he10830.py` restates the tool's triplet transit with the state on Roche equipotentials (`roche_recon.ReconMap`); spherical mode reproduces the tool to 2.7e-7 in transmission (EW 0.733390 against 0.733399). Option (b) on the reference: 0.7247 (-0.0087, -1.2 %). The reference re-solved in the Roche potential (`Domain mode: Roche`, certified in 19 passes, sonic point 22.35): 0.7240 with the spherical transit and 0.7156 (-0.0178, -2.4 %) with the Roche reconstruction, the self-consistent pair. Option (c): Kn < 0.1 alone gives 0.4412, a bracket [0.441, 0.733] 50 times the target. Recommendation (for decision 3, not applied): keep the 30 R_p spherical domain for the scan and state the tidal geometry as a measured interval [Roche-consistent, spherical] at the two models bracketing the crossing; the full comparison is NOTES section 6.

**Corrected on the way.** `EXHALE_transit.py`, triaxial branch (`triaxial_depth`, off by default): the impact parameters ran to the outermost grid radius while the depth was normalized to the annulus up to R_ib = min(r_N, R_*/R_p), so for r_N > R_*/R_p the mean transmission of the wider annulus was applied to the narrower one. The impact parameters now end at R_ib. Spherical products data-identical before and after on the reference (He 10830, H-alpha, Ly-alpha); the triaxial H-alpha line-center depth of the reference moves 0.001 -> 0.004 %; unchanged by construction when r_N <= R_*/R_p (checked with R_* = 1 R_sun). No stored product used the branch.

**Not verified.** Re-closure of the new domains; refinement of the shells and of the 15/20 grids; the transit tool's central-

transit, uniform-disk geometry against the real impact parameter; wind direction along the potential gradient in the reconstruction; any domain beyond L1.

## 86. Phase 5 of the LHS 1140 b EW plan, first step: the EW falls with He/H; the column energy balance limits the low side (2026-10-04)

Plan `md/PLAN_20261003_lhs1140b_ew.md` phase 5 step 2 on the two neighbors of the phase-3 reference (section 84). Record, script and figures (both inspected): `LHS1140b/models/.P2/ew_fit/phase5/NOTES.md`. Binary `.P2/EXHALE_8cc7b546.x` (ifx), lart4, 12 threads. No code changed. EW in percent-angstrom (% A), observed 1.1084 +- 0.0289. All MEASURED.

**Route.** Each He/H runs the energy-coupled elemental-flux closure of the reference (`energy_coupling_nlte/closure.json` with only `--abundances He=<v>`, the binary and the template changed; tol 0.05, T_match tol 6 K, omega 0.5, kmax 12) with `He 2^3S transport: True` (resolved on in every k00), seeded by continuation in He/H from the certified reference state: the driver maps it onto the new reservoir (`map_state_to_grid.py --ic --reservoir He/H`); no molecular state is rescaled.

| deep He/H | closure | T_match [K] | F_cond [erg cm^-2 s^-1] | log10 Mdot | EW sol / adv | red peak, FWHM | Kn_bulk 0.1 / 1 / sonic [R_p] |
|---|---|---|---|---|---|---|---|
| 1.5 | refused at k00 by the column energy balance (below) | | | | | | |
| 1.8 | converged k = 3, certified | 359.5 | 11.65 | 7.9359 | 0.7975 / 0.8015 | 2.77 %, 0.270 A | 6.11 / 18.93 / 26.26 |
| 2.09 | reference | 349.9 | 9.63 | 7.8854 | 0.7334 / 0.7369 | 2.52 %, 0.273 A | 5.69 / 17.21 / 25.69 |
| 3.0 | stopped at k02 (below); not converged | 340.5 (k02) | 6.73 | 7.8190 | 0.6873 / 0.6900 (k02, diagnostic); 0.6977 / 0.7004 (k01) | 2.28 %, 0.282 A | 5.03 / 14.55 / 24.50 |

**Stale (2026-10-07):** the Knudsen numbers, `Kn_bulk = 0.1` radii and exobase radii in this passage were computed by `src/utils/collisional_validity.py` before its resonant H+ + H and He+ + He momentum-transfer coefficients (`md/Update_EXHALE_stage3.md` section 110). On the 102 stored LHS 1140 b states the change moves the `Kn_bulk = 0.1` radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.

(Observation: red peak 1.28 %, FWHM 0.80 A on a 3-sample running mean; the widths are those of `phase5/compare.py`, which gives 0.273 A on the reference where section 84 quoted 0.24 A from another reading.)

**Direction and slope.** The EW falls as He/H rises: dEW/d(He/H) = -0.221 % A per unit He/H on 1.8-2.09 (both accepted) and -0.051 on 2.09-3.0 (diagnostic k02; -0.039 with k01). At lower He/H the wind is cooler (T_max 4795, 4960, 5352 K) and denser, and n(He 2^3S) is larger beyond about 3 R_p. The line deepens at a nearly fixed width (0.27-0.28 A against 0.80 A observed). The crossing of 1.108 by extrapolation alone: He/H 0.39 (linear), 0.87 (linear in ln He/H), 0.98-1.20 (quadratics); all below 1.5.

**Validity limit of the column treatment (He/H 1.5).** The column energy balance under the conducted flux (section 77) refuses its own optically thin treatment at He/H 1.5 (`ThinLayerRefusal`, `HeH1.5/k00/adapter.log`): the column above absorbs 10.8 % of the emission excess of the CH4 nu4+nu2 bin 1275-1350 cm^-1 and 10.3 % of the NH3 rotational bin 100-160 cm^-1, against the 10 % limit. Band maxima of that fraction (CH4 dyad / NH3 rotational): He/H 3.0 5.4-5.8 / 5.5-5.6 %; 2.09 7.7 / 7.5 %; 1.8 8.5-8.9 / 8.4-8.5 %; 1.5 10.8 / 10.3 % (refused). The fraction grows as He/H falls (lower mean molecular weight, higher CH4 and NH3 mixing ratios). Every extrapolated crossing lies below the He/H at which the treatment holds, so bracketing 1.108 needs radiative transfer in the column bands (absorption by the overlying column) in place of the optically thin limit; the limit itself was not changed. SUPERSEDED (section 90): these fractions used the flux transmission exp(-1.66 tau_bar) where the escape of locally emitted photons is E_2-type. With the correct escape (adapter alone, the same final arguments) the band maxima CH4 dyad / NH3 rotational are: 3.0 (k02) 16.1 / 13.1 %; 2.09 20.3 / 16.5 %; 1.8 (k03) 22.2 / 18.1 %; 1.5 (thin k01) 24.6 / 20.1 %. Every state of this section is past the 10 % limit.

**He/H 3.0 refused at k02 on the mass row at its rounding floor.** The wind solve certified its final state (cell 246, r = 1.333 R_p: 3.686e-12 against 3.7e-12, distance 0.987); the closure's stationary evaluate of that exact state measures 3.936e-12 (1.054) and exits 2, deterministically (12 and 8 threads give the same, scratch re-evaluation); the evaluate refreshes the composition by one sweep (up to 3.3e-6 relative), and its written state re-evaluates at 3.686e-12 and certifies. The certification verdict of the solve is therefore not reproduced by the evaluate of the exact restart at a cell whose tolerance is its rounding anchor: a reproducibility defect of the mass-row floor, reported, not changed (shared acceptance interface). The closure was not converged at k02 in any case (eps_He 0.070, T_match moved 10.5 K). The branch was stopped as the plan prescribes; states kept.

**Not verified.** A converged He/H 3.0 closure; any model below He/H 1.8; the NH3 nu2 + H2 bracket, the drift and the Roche interval at the new He/H; the 0.24 / 0.273 A FWHM difference on the reference.

## 87. LHS 1140 b He 10830: the width-matching velocity kernel in the EW chain (2026-10-04)

User decision 2026-10-04: the line width is treated by convolution with an added Gaussian velocity kernel whose FWHM f is adjusted, for each model, so that the model line reaches the measured width, by the convention of `docs/lhs1140b_exhale_vs_pwinds.tex` Sect. `sec:broadening`; the crossing EW is the window EW of the model convolved with its own f. Record, script, numbers and figure (inspected): `LHS1140b/models/.P2/ew_fit/phase5/nonthermal_kernel/NOTES.md`. All MEASURED; EW in percent-angstrom, observed 1.1084 +- 0.0289.

**What the kernel is.** A Gaussian line-of-sight velocity distribution standing for the velocity field of the outflow that

the 1-D model does not contain; not a computed physics term. Convention: the excess (max T - T)/max T of the R = 68,000 instrument-convolved curve (column 3 of `tpm_He10830.txt`, `T_instr`; corrected 2026-10-05: this entry first said "instrument- and rotation-convolved", but column 3 carries no planet rotation; for LHS 1140 b the rotation column gives an EW equal to 1e-6 %A) convolved with a Gaussian of FWHM f in velocity (`gaussian_filter1d`, reflecting boundary); depth, width and red/blue read by the three-Gaussian extractor used on the data; f solved so that the blended red-pair FWHM is the measured 0.841 A; EW by the window operator (vacuum 10832.60-10834.20 A, observation samples, trapezoid weights). The width audit's sigma = 9.33 km/s is kept as a reference value only.

**Code.** `LHS1140b/he10830_equivalent_width.py`: `broaden_excess`, `broadened_line_metrics`, `width_matching_fwhm_kms` (moved unchanged from `LHS1140b/make_memo_figures.py`, which now imports them), and `width_matched_product` / CLI `--width-matched` (writes `tpm_He10830_broadened.txt` with the record in its header). `LHS1140b/models/.P1/fiducial_conduction/ew_of_state.sh`: option `EW_NONTHERMAL_KERNEL=width_match` (or a FWHM in km/s) adds a `nonthermal_kernel` line and, on each product line, `kernel: fwhm_kms= sigma_kms= EW_kernel= EW_nokernel= red= blue= red_blue= fwhm_A=`; `EW=` stays the unbroadened EW. Default unset: unchanged. `EXHALE_transit.py` and `exhale_transit_lib.py` unchanged.

**Tests.** Kernel off: EW.txt of HeH1.8/k03 identical to the stored one (up to the state-directory prefix), `tpm_He10830.txt` data bitwise equal; in the kernel runs all six unbroadened products equal the stored ones bitwise. The kernel conserves the absorption summed over the grid to 1.6e-16-5.4e-16 relative (the trapezoid moves by -1.0e-4 because the broadened blue wing reaches the grid end). The moved functions return bitwise the numbers of the replaced definitions. The window EW falls by 3.31-3.32 %: the red pair loses 4.0 % through the window edges, the blue component leaks 0.005-0.006 in.

**Results (solution; adv in the NOTES).**

| He/H | f [km/s] (sigma) | red [%] | blue [%] | red/blue | FWHM [A] | EW no kernel / kernel |
|---|---|---|---|---|---|---|
| 1.8 | 22.05 (9.36) | 0.890 | 0.117 | 7.60 | 0.841 | 0.7975 / 0.7710 |
| 2.09 | 22.02 (9.35) | 0.818 | 0.108 | 7.60 | 0.841 | 0.7334 / 0.7091 |
| 3.0 (k01, unconverged closure) | 21.92 (9.31) | 0.779 | 0.103 | 7.56 | 0.841 | 0.6977 / 0.6746 |
| observed | | 1.254 | 0.198 | 6.35 | 0.841 | 1.1084 |

f is 21.9-22.1 km/s at every He/H (22.22 on the 2026-09 solutions of the memo). At that f the shape matches and the depth is 0.62-0.71 of the observed: the kernel lowers the window EW by 3.3 %, moving every model further below 1.108. Cross-checks (chi^2 on the 129 samples, v_wind 2.26 km/s): free f with the EW free 15.7-17.4 km/s (chi^2 147-241); with the window EW held at 1.1084 by an amplitude scale, 22.3-22.4 km/s (chi^2 18.7-18.8), close to the width-matching f.

**Not verified.** The uncertainty of f from the measured width; the blue depth near the reflecting grid end; the memo and its figures were not regenerated (`make_memo_figures.py` changed only by the import, not run).

## 88. The stationary evaluate certifies the stored pair (2026-10-04)

**Finding.** Section 86 recorded that the He/H 3.0 k02 state (`LHS1140b/models/.P2/ew_fit/phase5/HeH3.0/k02`) was certified by its wind solve (mass row at cell 246, r = 1.333 R_p: 3.686e-12 against the rounding-anchored 3.7e-12, distance 0.987) and refused by the closure's `Restart intent: stationary evaluate` of the same exact state (3.936e-12, distance 1.054, exit 2), while the state that evaluate wrote re-certified. The evaluate took one composition sweep from the stored composition BEFORE it certified, and certified, wrote and exited on the swept state. The verdict was therefore not a function of the stored state. MEASURED (binary `EXHALE_99a76cb0.x`, below) the sweep moves the physical composition very little: max |dfl/f over species carrying at least 1e-12 of a cell's mass 4.88e-12 (cell 499, C I), max over cells of sum_k |df_k| 1.51e-16, particle count 5.2e-16. The 3.3e-6 relative change in `Ion_species.txt` quoted in section 86 sits in the two lower ghost rows (row 2, He 2^3S, r = 0.99990 R_p), which the boundary derives; the physical rows of the text files differ by less than 1e-8. The mass row of the wind is a rounding-level quantity judged against a rounding-anchored tolerance, and a change of the caloric map at the 1e-16 level moves it by 7 % at cell 246.
**Rule adopted.** The certificate a file carries is the one measured on the state written in it, and the evaluate writes the state it loaded.

- `stationary_state_of_the_loaded_restart` (`EXHALE_main.f90`) certifies the STORED pair first: the conserved state and the composition exactly as loaded, with the heating and cooling of that composition in the field it attenuates. They come from one sweep with the composition of the cells above the two lower ghosts held (`ioniz_eq_sources_of_held_composition`, `ionization_equilibrium.f90`): the rates are built from the entry composition (`xuv_field_block_cells` = 0, `xuv_self_field_passes` = 1, so every rate is a function of that composition and T), heat and cool are contracted with the entry densities, the interior composition is returned bit for bit (asserted; the run stops otherwise), and the two lower ghosts, which no file stores, are solved and derived by the boundary as on every route. That certification is the verdict: the `certified=` pair of every file written, the exit status (0, 2 a refused claim, 7 a rejected derived product) and `stored pair verdict` in the log.
- `evaluate` writes `Hydro_ioniz.txt`, `Ion_species.txt`, `conserved_state.txt`, the channel breakdowns and the `_adv` pair from the stored pair (the `_adv` pair carries the stored pair's certificate as `derived_from` provenance), then takes one composition sweep from the loaded state as it was before the held sweep, lower ghosts included (`composition_sweep_movement_of_the_stored_pair`), and reports its movement and its certification (`record_verdict = .false.` in `certification_evaluate`, which then sets no file pair, no exit status, no resolved-configuration record and writes no row-term or flux-profile file), with the two certifications printed entry beside entry (`certification_reports_side_by_side`). The swept composition is never written.
- `Restart intent: stationary` (the solve) makes the same stored-pair certification (the answer to the file's claim) and starts the solve from the swept state, entered at the loaded state as before.
- The solve routes already certify what they write: the stationary restart, the direct steady route (`certify_the_state_to_be_written`) and the final marching write assemble the residual from (u, f_sp, heat,

cool) and write those same arrays. The heat and cool are those of the last sweep, taken at its entry temperature (the documented one-sweep lag); on k02 the stored-pair evaluate, whose sources are the stored composition's at the state's own temperature, reads the energy row at 2.06661e-8 (cell 58) where the solve's final certification read 2.067e-8 (cell 58).

- Found while checking this: the periodic marching write (every 1000 steps) wrote the run's last certified pair, measured at an earlier monitor step on another state. It now writes `certified=F cert_reason=marching_snapshot_s<step>` and restores the run's pair after the write, as the pass-state snapshot already did. The final write is unaffected.
- `ew_of_state.sh` records the written state's pair as `written_state` (was `work_state`).

**Binary.** `LHS1140b/models/.P2/EXHALE_99a76cb0.x`, md5 99a76cb05e56ad3a6ee80e6c433814fd, ifx 2026.1.0, built from `.P2/build_storedpair_ifx/`; manifest `.P2/BINARY_MANIFEST_99a76cb05e56.txt`. To be used for the phase 5-6 acceptance of the EW plan. Against `BINARY_MANIFEST_8cc7b5462cc2.txt` exactly three sources differ (`EXHALE_main.f90`, `certification.f90`, `ionization_equilibrium.f90`), and the tree matched that manifest before the change.

**Verification** (MEASURED, lart4, 12 OpenMP threads unless stated).

| state | certification | binding cell | measure / tol | distance | exit |
|---|---|---|---|---|---|
| HeH3.0 k02, solve's final (run.log, 8cc7b546) | solve | 246 | 3.686e-12 / 3.7e-12 | 0.987 | |
| HeH3.0 k02, old evaluate (8cc7b546) | after sweep | 246 | 3.936e-12 / 3.7e-12 | 1.054 | 2 |
| HeH3.0 k02, new evaluate, 12 and 8 threads | stored pair (verdict) | 246 | 3.686e-12 / 3.7e-12 | 0.987 | 0 |
| same run | after sweep (reported) | 246 | 3.936e-12 / 3.7e-12 | 1.054 | |
| reference `phase3/he23s_transport/solve`, solve's final | solve | 242 | 3.360e-12 / 4.3e-12 | 0.790 | |
| reference, new evaluate | stored pair (verdict) | 242 | 3.360e-12 / 4.3e-12 | 0.790 | 0 |
| same run | after sweep (reported) | 252 | 2.001e-12 / 3.0e-12 | 0.667 | |

The stored-pair verdict reproduces the solve's own final certification at both states, and the after-sweep reading reproduces the old evaluate entry by entry (largest measures, cells, binding cells); of the volume averages only the energy

row's moves, 8.886e-10 (old) against 8.865e-10, not traced further (the after-sweep sweep runs after another sweep has filled the frozen background and the boundary caches). Every other entry of k02 is within its tolerance on both. The k02 evaluate writes the stored state: `conserved_state.txt` U and F rows of the physical cells and upper ghosts bitwise equal to the input, `Ion_species.txt` physical rows equal, `Hydro_ioniz.txt` p and T within 1.9e-15 relative of the solve's text (re-derived from u), heat within 1.6e-14 and cool within 7.4e-13 (the sources of the stored composition at its own temperature against those of the solve's last sweep). The 12- and 8-thread `Ion_species.txt` are identical. An evaluate of the evaluate's own output writes the same data (`Hydro_ioniz`, `Ion_species`, `conserved_state`, both `_adv` files, both breakdowns) and the same verdict. Reference EW (`ew_of_state.sh` with the new binary): solution 0.7334, adv 0.7369; with `EW_NONTHERMAL_KERNEL=width_match` 0.7091 and 0.7125, as recorded with 8cc7b546 (`phase5/nonthermal_kernel/ew_HeH2.09`); the solution spectra are bitwise equal to that record, the adv spectra within 4.5e-16 (unbroadened) and 6.4e-14 (broadened) relative. Scratch: `~/.claude/jobs/f24ea156/tmp/evalfix/`.
**Tests** (MEASURED; the gfortran tree build, md5 817048a812c0594155159ee4ba14e480, against the control build of the entry text, 843c97f955b072158ccc8454f2cee6a8).

- `src/tests/certification`: 145 PASS, 0 FAIL before and after.
- `src/tests/grid_and_gates`: 420 PASS, 3 FAIL before; 424 PASS, 3 FAIL after. The three failures are the same rows of `restart_intent_and_metadata.sh` on its stale fixture (`backup/regression/wasp_full_newton`, solved under older physics, which the binary refuses): `stationary_evaluate_exit_status` (2), `stationary_evaluate_certification_unchanged` (writer T, re-evaluated F) and `stationary_evaluate_rebuilt_within_sweep_budget`, whose worst column moved from O III 6.01e-5 (the swept composition) to heat 5.83e-5 (the species now come back as stored; the heat of that fixture was written by an older binary). `stationary_evaluate_products.sh` was rewritten to the rule: four rows added (`header_certificate_is_the_stored_pairs`, `exit_status_is_the_stored_pairs_verdict`, `products_are_made_on_the_stored_pair`, the loaded state bit for bit, `changed_option_swept_reading_reported`), the dump points renamed (`stored_before_products`, `stored_after_products`), the claim rows read the new log lines.
- `src/tests/boundary_state` (the evaluate-route scripts `boundary_of_the_installed_composition`, `interior_composition_held`, `ghost_composition_fixed_point`, `ghost_seed_and_ghost_record`): 27 PASS, 0 FAIL before and after. An intermediate version that entered the reported sweep at the ghosts of the held sweep and rebuilt the boundary unconditionally failed `face_state_gap_of_the_entry_composition_boundary` and `ghost_seed_of_the_installed_rows_is_inert`; both pass with the version described above.
- `src/tests/adv_static_limit`: 64 PASS before and after (the derived column's worst residual 2.98e-11 -> 2.86e-11, now derived from the stored pair); `src/tests/coupled_block_jacobian` (the solve route): 8 PASS before and after, every measured value identical.

**Not verified.** The solve route (`Restart intent: stationary` without `evaluate`) now takes the held sweep before its work-state sweep; the work state is entered at the loaded state, but caches the held sweep filled (the frozen background, the boundary record) can move it at the level of the volume-averaged energy row above; no solve was run with the new

binary. The HeH3.0 k02 closure was not resumed (its branch stays stopped, section 86; its closure was unconverged in any case).

## 89. The column energy balance with absorption by the overlying column; the thin limit flags instead of refusing; phase 5 at He/H 1.5 and 1.2 (2026-10-04)

The column energy balance below the matching level (`src/utils/conducted_heat_column_balance.py`, section 86) treated the band emission as optically thin and refused a column in which the gas above absorbed more than 10 per cent of a bin's emission excess. That refusal stopped He/H 1.5. Records: `LHS1140b/models/.P2/ew_fit/phase5/column_transfer/NOTES.md` (measurements, figure inspected), `phase5/HeH1.5_thick/`, `HeH1.5_thin/`, `HeH1.2_thick/` (`NOTES.md` each), `phase5/compare_column_transfer.py` and `column_transfer_numbers.txt`, `fig_column_transfer_{line,profiles}.png` (inspected). All MEASURED unless marked READ. EW in percent-angstrom.

**Formulation (new option, default off).** The adapter option is `--column-band-transfer cooling_to_space` (default `thin`). The form is the cooling-to-space approximation as generalized to a two-level band by Dickinson (1972, J. Atmos. Sci. 29, 1531), READ in the published version: section 3c Eq. 12 (cooling to space, (1/2) n h nu r T(z,inf)) and section 3g Eq. 33 (net rate [1 + (1/2)(A/lambda) T(z,inf)]ˆ-1 [R_ext + A E(z) - (1/2) A T g eˆ(-h nu/kT)], with E(z) the exchange function of Eq. 30). Rodgers & Walshaw (1966, QJRMS 92, 67) was not read: the publisher's page could not be retrieved, and it is cited only through Dickinson. Lopez-Puertas & Taylor (2001) and Goody & Yung (1989) were not read and are not used.

E(z) is split in two parts:

- Toward the optically thick climate column below, the source function is held at its climate value. This part adds the downward half 1/2 to the escape probability, and the field from below (a = e_LTE(T0)/2, as before) to R_ext.
- Toward the gas above and inside the warmed layer, it is neglected (the cooling-to-space approximation; the gas above the match is the base of the wind, at least as warm as the match).

Escape of node i in bin b: beta = 1/2 + sum_g w_g k_g u_g / sum_g w_g k_g, with u_g = [E_3(tau_t) - E_3(tau_b)]/(2 (tau_b - tau_t)) the layer integral of (1/2) E_2(tau). The optical depths are measured from the top of the profile with the k-distribution at the climate state.

- LTE bands: excess = sum_bins beta (e_bin(T) - e_bin(T0)).
- Two-level bands: beta = sum beta e_bin / e_band and eps_beta = C10/(C10 + beta A10), so a band with beta A10 « C10 thermalizes; excess = eps_beta(T)(beta e(T) - a) - eps_beta(T0)(beta e(T0) - a).

With no absorbing column this is the thin treatment bit for bit.

Validity (it replaces the 10 per cent refusal for this option). The exchange inside the domain that the approximation neglects is computed on each solution:

- exact slab exchange, E_3 of the optical-depth differences between faces, every g-point;

- layers thinner than 1e-6 in optical depth are treated as sheets with E_2 fluxes (the slab form loses them to round-off);
- over the nodes warmed by more than 1e-6 of the match rise.

Its first-order effect on T_match (one solve with the Jacobian of the solution) must not exceed `--column-exchange-shift-max` (default 1 K), else `ColumnTransferRefusal`. The value and the escape probability of each band are written to `conducted_heat_column.txt`; the choice enters the profile header (`column_band_transfer`) and, off the default, the `solution_id`.

**Is the exchange term needed?** `ColumnResponse.solve_with_exchange` solves with the full exchange of the whole 6 H domain. On the He/H 1.8 column it adds +0.493 K (nlte) and +0.605 K (lte) to T_match over cooling to space, whose first-order estimates are 0.395 and 0.480 K; on the test column it adds +0.466 and +0.569, against estimates of 0.373 and 0.452. The estimate is 0.79-0.80 of the full value in all four cases. The exchange deposits the downward emission over 1.5-400 microbar and warms that region by about 2 K. Not needed at these optical depths (0.5 K against the 6 K closure tolerance and the 2.4 K of the transfer itself); measured and bounded on every solution.

**Thin limit: flag, not refusal (user decision 2026-10-04).** Under `thin`, a column past the 10 per cent limit is kept. The 10 per cent stays the stated validity threshold at the code site (`THIN_ESCAPE_MIN`). The adapter log prints a WARNING. `conducted_heat_column.txt` and the profile header carry `thin_layer_limit_exceeded T|F`, `thin_layer_max_absorbed_fraction` and `thin_layer_max_absorbed_bin`. The profile header keys are written by `lower_profile_schema.write_profile`; EXHALE ignores header keys it has no consumer for, READ from `lower_atmosphere_profile.f90`. The closure driver does not repeat the flag. `ThinLayerRefusal` is removed.

Found and documented, not changed: the thin test's escape factor exp(-1.66 tau_bar) is the diffusivity form of the flux transmission 2 E_3. The escape of locally emitted photons is E_2(tau)/2, whose absorbed share is larger at small tau (14.6 against 6.4 per cent at tau = 0.04), so the test understates the absorption. It is kept as the thin treatment's stated criterion (thin unchanged by instruction), noted at the code site. SUPERSEDED (section 90): the check now uses the cooling-to-space escape; every thin absorbed fraction quoted in this section is replaced there.

**Tests** (`src/tests/conducted_heat_column/run.sh`, 66 PASS, 0 FAIL):

- the thin limit of the new transfer is the thin solution bit for bit (lte, nlte);
- energy budget to 6e-11;
- the new transfer is warmer than thin, with escape between 1/2 and 1;
- NH3 x5 is flagged by the thin test and solved within the bound, while NH3 x300 is refused by the bound;
- a grey optically thick emitter (one bin at 2100 cmˆ-1, tau 2 at the match): escape probabilities against independent quadrature of E_2 to 1e-5, tau against quadrature to 1e-5, and T_match against an independent shooting solution (solve_ivp, bisection) of the continuous equations. The discrete solution converges at order 1.95, and its Richardson value agrees to 1.6e-3 K (2.8e-5 of the rise);
- the diagnostic switch below;
- the existing refusal test now expects the flag.

The thin path is bitwise unchanged against the entry text (lte, nlte, nlte_partial, rotational on the test column; the thin adapter runs reproduce the stored T_match 349.876 and 359.546).

**Thin against cooling to space, adapter alone** (k-final chemistry arguments of each closure):

| He/H | thin | cooling to space | difference |
|---|---|---|---|
| 2.09 | 349.876 | 351.922 | +2.05 K |
| 1.8 | 359.546 | 361.949 | +2.40 K |
| 1.2 | 382.735 (flagged) | 386.371 | +3.64 K |

All are below the 6 K closure tolerance, so the converged He/H 1.8 and 2.09 models need not be redone on this ground (decision for the main session).

**The overlying gas.** Both transfers take the column above p_match from the Photochem CH4 and NH3 above the match. In the coupled model that gas belongs to the wind, which carries C, N and O as atoms and is warmer. Diagnostic `include_gas_above_match = False`:

| He/H | thin absorbed fraction, with / without | cooling to space - thin, with / without |
|---|---|---|
| 1.8 | 0.0853 / 0.0099 | +2.42 / +0.12 K |
| 1.5 | 0.1015 / 0.0105 | +3.08 / +0.13 K |
| 1.2 | 0.1130 / 0.0112 | +3.68 / +0.14 K |

SUPERSEDED (section 90): with the correct escape the thin absorbed fractions with / without the gas above are 0.2219 / 0.0293 (1.8), 0.2522 / 0.0307 (1.5), 0.2790 / 0.0325 (1.2), so 87-88 per cent from that gas; the temperature columns are unchanged. The thin flag F at 9.60 per cent of the He/H 1.5 thin state below becomes T at 24.6 per cent, and the 1.2 thin value 0.1129 becomes 0.2788.

About 90 per cent of the absorbed fraction that triggered the thin limit, and 95-96 per cent of the cooling-to-space correction, come from molecules the wind model does not contain. Which overlying column is physical (the Photochem gas, or the wind's atomic gas with no band absorbers) is not settled here.

**Pinning passes.** At He/H 1.5 and 1.2 the Photochem model-top pinning on the warmed column converges by alternating passes and needed 7-11 of them; three first attempts stopped at the 6-pass cap. New adapter option `--top-pressure-passes` (default 6, unchanged); the phase-5 closures at 1.5 and 1.2 pass 12. The 1e-5 agreement is unchanged.

**Phase 5 at He/H 1.5 and 1.2** (continued in He/H from `HeH1.8/k03`, settings of `HeH1.8/closure.json` except He/H, transfer and passes). The EW verdicts all come from binary 99a76cb0 (`written_state certified=T`), EW_NONTHERMAL_KERNEL=width_match:

| He/H, transfer | closure (binary) | T_match | log10 Mdot (N-20) | EW sol / adv | EW width matched sol / adv (f km/s) | Kn_bulk 0.1 / 1 / sonic [R_p] |
|---|---|---|---|---|---|---|
| 1.8, thin (section 86) | k = 3 (8cc7b546) | 359.55 | 7.9359 | 0.7975 / 0.8015 | 0.7710 (section 87) | 6.11 / 18.93 / 26.26 |

| He/H, transfer | closure (binary) | T_match | log10 Mdot (N-20) | EW sol / adv | EW width matched sol / adv (f km/s) | Kn_bulk 0.1 / 1 / sonic [R_p] |
|---|---|---|---|---|---|---|
| 1.5, thin | k = 1, eps 4.3e-4 (8cc7b546); flag F, 0.0960 | 373.54 | 7.9766 | 0.8318 / 0.8364 | 0.8042 / 0.8087 (22.09) | 6.54 / 20.71 / 26.83 |
| 1.5, cooling to space | k = 1, eps 8.2e-3 (99a76cb0) | 375.98 | 7.9770 | 0.8327 / 0.8373 | 0.8050 / 0.8096 (22.09) | 6.55 / 20.72 / 26.83 |
| 1.2, cooling to space | k = 1, eps 0.049, dT 5.7 K (8cc7b546) | 386.37 | 8.0218 | 0.8593 / 0.8648 | 0.8308 / 0.8361 (22.14) | 7.08 / 22.95 / 27.45 |

**Stale (2026-10-07):** the Knudsen numbers, `Kn_bulk = 0.1` radii and exobase radii in this passage were computed by `src/utils/collisional_validity.py` before its resonant H+ + H and He+ + He momentum-transfer coefficients (`md/Update_EXHALE_stage3.md` section 110). On the 102 stored LHS 1140 b states the change moves the `Kn_bulk = 0.1` radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.

- At He/H 1.5 the thin-treatment error is +2.44 K in T_match and 0.0009 (0.1 per cent) in the EW. The converged thin state is inside its limit (9.60 per cent); intermediate iterates passed it (10.15 per cent).
- He/H 1.2 was reached from 1.8 in one jump (no rung). Its closure converged just inside both tolerances.
- Stops, all kept in `attempt*/` or `k*.incomplete.*`: the pinning cap (three times), and the evaluate of binary 8cc7b546 refusing solve-certified states on the mass row at the rounding floor (He/H 1.5 thick k01, 3.197e-12 against 3.0e-12; He/H 1.5 thin k00, 6.853e-12 against 6.7e-12; the defect of section 88). The He/H 1.5 thick closure was restarted on 99a76cb0 from its own k01 state.

**Where the crossing is.** The width-matched EW rises as He/H falls (0.7091, 0.7710, 0.8050, 0.8308 at 2.09, 1.8, 1.5, 1.2), but the step shrinks: +0.034 from 1.8 to 1.5, +0.026 from 1.5 to 1.2. A linear continuation of the last step reaches the observed 1.108 only at a negative He/H. No crossing exists at He/H >= 1.2 in this model family, and the trend gives no sign of one below it (an extrapolation, not a model). The line deepens at a fixed width; the hot peak cools (4795, 4568, 4272 K at 1.8, 1.5, 1.2) as the wind densifies.

**Not verified.** For a two-level band the exchange weighting by eps_beta at emitter and absorber is approximate; it is exact for LTE bands. The first order of the validity bound reads 0.8 of the full exchange. The gas-above-match question is open, as stated. The He/H 1.2 state lies near its closure tolerances.

## 90. Phase 5 closed: the EW has no crossing in He/H; the thin check uses the escape of emitted photons; the model-top pinning averages a 2-cycle (2026-10-04)

User decision (2026-10-04): run He/H 0.8 and 0.5 to measure the upper bound of the EW in He/H and close phase 5 on measured values. Records: `LHS1140b/models/.P2/ew_fit/phase5/NOTES.md` section 8, the `NOTES.md` of `HeH1.0/`, `HeH0.8/`, `HeH0.65/`, `HeH0.5/`, `column_transfer/thin_check_escape/NOTES.md`; `compare_lowHeH.py`, `lowHeH_numbers.txt`, `fig_ew_vs_heh.png`, `fig_lowHeH_line.png` (inspected). All numbers MEASURED. EW in percent-angstrom; the crossing quantity is the width-matched window EW (section 87).

**1. Thin-treatment validity check (fixed).** Physically wrong: `optical_depth_check` of `src/utils/conducted_heat_column_balance.py` took the escape of the band photons emitted in the column as the flux transmission exp(-1.66 tau_bar), the diffusivity form of 2 E_3(tau), i.e. the transmission of a beam leaving the column. Photons emitted isotropically at depth tau escape upward with probability E_2(tau) (Dickinson 1972, Eq. 5 for the transmission to space), and integrated over a layer and the k-distribution g-points this is the escape the `cooling_to_space` transfer already uses (`escape_factors`). The check now takes the absorbed share of the upward emission excess as sum_i ex_i 2 (1 - beta_i) / sum_i ex_i, beta_i the cooling-to-space escape of node i's control layer (one function for both). Only the flag and the recorded fraction change. The docstring paragraph on the overlying gas was re-measured. `DIFFUSIVITY` is removed (no other user).

- Tests (`src/tests/conducted_heat_column/run.sh`, 66 assertions): new `grey_thin_test_absorbed_share_vs_E2_quadrature`, the grey emitter (tau 2 at the match) against an independent 16-point Gauss-Legendre quadrature of E_2 over every control layer: entry text FAIL (0.96661 against 0.96476), corrected PASS (to 1e-6). `thin_column_is_not_flagged` failed on the corrected check (the test column absorbs 20.9 per cent), so it now uses that column with a tenth of the CH4 and NH3 (3.8 per cent). Entry text 65 PASS / 1 FAIL, corrected 66 PASS / 0 FAIL.
- Thin solution bit for bit: on the stored column inputs of He/H 1.8 and 1.2, entry text against corrected module, treatments nlte, lte, nlte_partial, rotational: T and q `array_equal` in all eight; the adapter-alone thin runs reproduce the data rows of `conducted_heat_column.txt` of the entry-text runs (md5) at 2.09, 1.8 and 1.2 and the stored T_match at all five He/H.
- Recomputed absorbed shares (adapter alone, the last chemistry arguments of each closure, transfer thin), CH4 nu4+nu2 dyad / NH3 rotational, old in brackets: 3.0 (k02) 16.1 / 13.1 % (5.76 / 5.60); 2.09 20.3 / 16.5 % (7.68 / 7.48); 1.8 (k03) 22.2 / 18.1 % (8.52 / 8.39); 1.5 (thin k01) 24.6 / 20.1 % (largest 9.60); 1.2 (k01) 27.9 / 22.8 % (largest 11.29). Every thin state of phases 3 and 5 is past the 10 % limit, by 1.6-2.8 times; what that costs is measured by the cooling-to-space transfer (+2.05, +2.40, +2.44, +3.64 K in T_match at 2.09, 1.8, 1.5, 1.2; 0.0009 in the width-matched EW at 1.5; section 89), below the 6 K closure tolerance. Without the gas above p_match the largest share is 2.9-3.7 % at every He/H measured (2.09-0.65), so 87-90 % of it comes from that gas. The numbers of sections 86 and 89 and of `md/NH3_CH4_ISSUES_20261004.md` are marked superseded where they stand.

**2. Model-top pinning: averaged once the passes alternate (fixed).** `steady_state_at_stated_model_top` (`src/utils/photochem_to_lower_profile.py`) iterates the map z -> z'(z) of the Photochem domain top (pin, con-

verge, read the altitude of the stated pressure). At He/H 0.5 on the warmed column the passes alternated without contracting (+1.7e-2, -1.1e-2, ..., +6.2e-3 at pass 12 against 1e-5; `HeH0.5/attempt2_pinning_cycle/`), a 2-cycle of the undamped iteration (slope near -1). Once two successive moves have opposite signs the top is pinned at (z + z')/2: the same fixed point and the same 1e-5 test. On the same arguments the entry text refuses after 12 passes and the corrected adapter converges at pass 9 (1.8e-6); passes that do not alternate are unchanged (the first pinning sequence of that run is identical line for line). Converged closures before this change are not affected (their stored states satisfy the same test); the He/H 1.0 and 0.8 closures and the 0.65 iterates after k00 ran with it. No suite drives this function; the check is the He/H 0.5 run itself (`HeH0.5/attempt4_from_0.65/k00/adapter.log`, `(top averaged)` lines).

**3. Climate starting value.** At He/H 0.8 and 0.5 the clima root solve for the deep temperature does not bracket from the default 400 K. The root does not depend on the starting value (20 bar: 648.4 K at 0.8 from 600 and 650 K; 679.1 K at 0.5 from 650 and 700 K), so the low-He/H closures pass `--climate-t-deep-guess 650`; nothing else changed.

**4. The scan** (cooling to space, settings of `HeH1.2_thick/closure.json`, binary 99a76cb0 for every wind and EW verdict, `written_state certified=T`, evaluate exit 0):

| deep He/H | closure k | T_match [K] | F_cond | log10 Mdot (N-20) | EW sol / adv | EW width matched sol / adv (f km/s) | Kn_bulk 0.1 / exobase / sonic [R_p] | thin flag (largest share) |
|---|---|---|---|---|---|---|---|---|
| 1.2 (section 89) | 1 | 386.37 | 17.16 | 8.0218 | 0.8593 / 0.8648 | 0.8308 / 0.8361 (22.14) | 7.08 / 22.95 / 27.45 | T (0.279) |
| 1.0 (rung) | 1 | 399.39 | 19.82 | 8.0531 | 0.8655 / 0.8714 | 0.8368 / 0.8425 (22.17) | 7.50 / 24.74 / 27.89 | T (0.307) |
| 0.8 | 1 | 410.53 | 22.95 | 8.0930 | 0.8734 / 0.8798 | 0.8444 / 0.8507 (22.21) | 8.04 / 27.08 / 28.40 | T (0.343) |
| 0.65 (rung) | 2 | 412.59 | 25.87 | 8.1325 | 0.8822 / 0.8891 | 0.8529 / 0.8596 (22.24) | 8.57 / 28.63 / 28.79 | T (0.377) |
| 0.5 | not converged | 398.51 (k00, diagnostic) | 29.51 | 8.179 | 0.8773 / 0.8845 | 0.8482 / 0.8552 (22.28), diagnostic | 9.24 / above the domain / none in the domain | |

**Stale (2026-10-07):** the Knudsen numbers, `Kn_bulk = 0.1` radii and exobase radii in this passage were computed by `src/utils/collisional_validity.py` before its resonant H+ + H and He+ + He momentum-transfer coefficients (`md/Update_EXHALE_stage3.md` section 110). On the 102 stored LHS 1140 b states the change moves the `Kn_bulk =`

0.1 radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.
Elemental fluxes at the match (F_H / F_He, g/s, overlap-window spreads 1e-9 to 2.4e-9): 1.0 4.636e7 / 6.665e7; 0.8 5.527e7 / 6.862e7; 0.65 6.479e7 / 7.088e7. Neglected-exchange shift 0.53, 0.59, 0.64 K (bound 1 K); cooling to space - thin in T_match +4.40, +5.36, +6.16 K at 1.0, 0.8, 0.65 (+0.15 to +0.18 K without the gas above p_match).

- He/H 0.8: the jump from 1.2 stopped at k00 (wind info 1; the He 2^3S carrier balance at the first gated cell, r = 1.204 R_p, 4.8e-5 against 1e-5 after the 60 outer passes); converged from the 1.0 rung.
- He/H 0.5: NOT CONVERGED in four attempts, all kept. After the two fixes above, k01 from 1.2 and k00 from the converged 0.65 rung both stopped after the 60 outer passes on the He 2^3S carrier balance at the first gated cell (r = 1.248 R_p; 1.1e-4 and 1.6e-4 against 1e-5). The same row needed 30-40 passes at He/H 1.2 and 1.5. Tolerances and the pass budget were not changed. The k00 diagnostic of the attempt from 1.2 (closure residual 0.27, T_match 398.5 K) has no sonic point in the domain (Mach 0.996 at 29.03 R_p): the sonic point moves outward as He/H falls (26.26, 27.45, 27.89, 28.40, 28.79 R_p at 1.8, 1.2, 1.0, 0.8, 0.65) and leaves the 30 R_p domain between 0.65 and 0.5.

**5. The upper bound.** Below He/H 1.2 the width-matched EW rises by 0.006-0.009 a step, 0.04-0.06 per unit He/H (2.09 to 1.2: 0.07-0.21 per unit). Fits on the four models with He/H <= 1.2 (compare_lowHeH.py): linear, rms 1.1e-3, 0.877 at He/H 0; quadratic, rms 2.6e-4, 0.905 at He/H 0 (EXTRAPOLATIONS beyond the last model, labeled so in fig_ew_vs_heh.png). Since the EW must vanish as He/H -> 0, these are upper estimates of the maximum provided the slope does not steepen below 0.65 (an assumption). Three forms with EW(0) = 0 and a maximum fit with rms 0.008-0.010, five times the scatter of the models: the models show no turnover down to 0.65. The largest model EW, 0.8529 (adv 0.8596) at He/H 0.65, is 8.8 sigma below the observed 1.1084 +- 0.0289, the extrapolated 0.88-0.91 7-8 sigma below; 1.108 would need about 0.39 per unit He/H below 0.65, seven to ten times the measured slope. The width-matched lines peak near 0.98 per cent against the observed 1.25 and converge with each other at low He/H (fig_lowHeH_line.png). **No He/H of this model family reaches the observed EW; phase 5 closes without a crossing.**
**Not verified.** The closure of 0.65 stopped with eps_H 0.045 (tolerance 0.05) and that of 1.2 near both tolerances; at about 1e-3 per K (section 75) and the 6 K T_match tolerance the step-to-step EW differences below 1.2 (0.006-0.009) are of the order of the closure tolerance itself. The He 2^3S carrier-row stall at the first gated cell is not diagnosed. The gas-above-match question (section 89) stays open and now carries 87-90 % of a thin absorbed share that is past its limit at every He/H.

## 91. LHS 1140 b: why the He 10830 EW rises as He/H falls, dEW decomposed by field and by balance term (2026-10-04)

User request: verify numerically the explanation that lower He/H raises the EW through a denser, cooler outer wind with larger n_e and n(He+), n(2^3S) tracking n_e x n(He+). Records: LHS1140b/models/.P2/ew_fit/phase5/dEW_decomposition/NOTES.md, scripts, dEW_table.txt, figures (inspected) fig_dEW_factors.png, fig_n3_balance.png, fig_line_buildup.png, fig_mdot_drivers.png. All

numbers MEASURED; width-matched window EW (section 87), f re-measured for every profile (22.02-22.24 km/s). No source file of the code changed; the stored states were only read.

**Method.** The He I 10830 branch of `EXHALE_transit.py` restated as a function (`he10830_line.py`): on the four stored states (He/H 2.09, 1.8, 1.2, 0.65) its instrument-convolved curve equals the stored `tpm_He10830.txt` to 4.4e-16 and the EW, width-matched EW and f equal the EW.txt records; on three substituted profiles `EXHALE_transit.py` itself agrees to 2e-14. n(2^3S) is written as C times the local root of the code's balance (`tr_triplet_row` plus the H2 sink), rate coefficients from the code's own elemental functions of `Cool_coeff.f90` (a driver linked against a private object directory, removed afterwards); the 2^3S photoionization rate P3 is measured from `carrier_row_terms.txt` and is the same in the four states to 0.3 per cent (6.41-6.54e-6 s^-1). C = n3/n3_loc is 0.9955-1.0072 over 1.2-8 R_p and up to 1.18 above 8 R_p; the local form (C = 1) lowers the width-matched EW by 0.004-0.008. The level is fed by recombination only (the 1^1S collisional feed is below 1e-14 of the gain where n3 matters). Shapley values over nine factor fields (n_H, local y = n_He/n_H, He fractions, H fractions, n_e, T, v, C with P3, R0), 512 syntheses for each pair, order-independent and summing exactly to dEW (residual 3e-16); a radial zone Shapley; a first-order factor by zone split; and a second Shapley over the terms of the balance, n3 = C n_e n_HeII a_eff/(L_P + L_e + L_r).

**Contributions (width-matched EW).**

| factor | 2.09 -> 1.8 | 1.8 -> 1.2 | 1.2 -> 0.65 | 2.09 -> 0.65 |
|---|---|---|---|---|
| n_H (density) | +0.0624 | +0.1353 | +0.1510 | +0.3524 |
| y = n_He/n_H local | -0.0566 | -0.2223 | -0.3428 | -0.6261 |
| He ionization fractions | -0.0257 | -0.0489 | -0.0652 | -0.1406 |
| H fractions | -0.0005 | -0.0016 | -0.0025 | -0.0046 |
| n_e | +0.0645 | +0.1273 | +0.1761 | +0.3687 |
| T | +0.0189 | +0.0708 | +0.1030 | +0.1946 |
| v | -0.0000 | -0.0001 | -0.0001 | -0.0002 |
| transport C, P3 | -0.0019 | -0.0040 | -0.0043 | -0.0112 |
| R0 | +0.0009 | +0.0032 | +0.0069 | +0.0109 |
| dEW | +0.0619 | +0.0598 | +0.0221 | +0.1438 |

The change is a small difference of large terms: density, electrons and the lower temperature add +0.92, the lower helium abundance of the flow (He diffusion: top-of-domain y 0.59 at 2.09, 0.28 at 0.65) and the lower He+ share of the helium remove -0.77. By radius (zone Shapley, 2.09 -> 0.65) the rise comes from beyond 6 R_p (+0.177); inside 3.5 R_p the line weakens (-0.059; the n3 peak falls from 52.5 to 22.5 cm^-3).

**The n_e x n(He+) explanation holds in part.** Balance terms (2.09 -> 0.65): n_e n_HeII +0.703, recombination coefficient a_eff(T) +0.156, He(2^3S)+H ionization loss n_HI Q31 -0.666, electron-collision loss -0.045, A31 + P3 + H2 and transport -0.011, thermal width -0.005, v -0.0002, R0 +0.011. Over 1.5-8 R_p, where most of the line forms, the He(2^3S)+H ionization takes 50-93 per cent of the 2^3S loss rate and A31 takes more than half only above 9.9-13.8 R_p. At fixed r, n_e n_HeII rises x1.2-2.9 from 2.09 to 0.65 but n_HI x3.3-4.6, so n3/(n_e n_HeII) at 0.65 is 0.41-0.51 of the 2.09 value over 2-8 R_p and reaches it only above ~20 R_p. The denser, more neutral hydrogen cancels 95 per cent of the gain of the n_e x n(He+) product; the net comes from the cooler gas (larger recombination coefficient) and

from the outer wind where A31 replaces the Penning loss. Velocity contributes nothing.

**Density and Mdot.** Q_net - L_cond equals the energy the flow carries out to 0.04-0.62 per cent in all four states, at 1.95-1.98e12 erg/g (G M_p/R0 = 1.91-1.92e12). The heat deposited below 1.02 R_p (mostly the molecular-chemistry channel) is close to the conducted loss, which rises x2.7 but does not drive the wind. The heat deposited above 1.02 R_p rises x1.41 (2.09 -> 0.65) and Mdot per unit of it x1.26 (Mdot x1.77); the heating per unit mass in the wind is nearly unchanged, the heat is deposited higher (the radius halving it 1.35 -> 1.53 R_p), mu at T_max falls 2.62 -> 1.91 and the scale height there rises 0.174 -> 0.215 R_p; the mass density beyond 2 R_p is 2.0-2.2 times larger.

**Not verified.** The cause of the Mdot rise: the absorbed stellar XUV power is not written by the code and no solve changing the mean molecular weight alone was run, so "lighter gas, larger scale height, heat deposited over more mass" is an interpretation consistent with the numbers. The rate coefficients were evaluated from the working tree (dd4a591 with uncommitted edits), not from the 99a76cb0 build that solved the states; their agreement is inferred from the smooth, state-independent measured P3. The radial factor split is first order (its zone sums differ from the Shapley values by up to 0.036 on 2.09 -> 0.65).

## 92. Phase 6 of the LHS 1140 b EW plan: numerical and observational validation at He/H 0.65 (2026-10-04)

Decision taken with the user: phase 5 found no crossing in He/H (section 90), so phase 6 validates the numerics at the model with the largest EW, deep He/H 0.65 (`phase5/HeH0.65/`, closure k = 2), to show that the EW deficit is robust. Record: `LHS1140b/models/.P2/ew_fit/phase6/NOTES.md` (runs, attempts, figures). All numbers MEASURED; binary 99a76cb0 for every solve and evaluate; EW = width-matched window EW of the solution product (section 87; f = 22.24 km/s on every state, an added kernel, not a computed physics term). Reference 0.852918 (unbroadened 0.882189), observed 1.1084 +- 0.0289, target 0.0058 for each test.

**Numerical tests.**

| test | EW | change | below 0.0058? |
|---|---|---|---|
| grid 500 -> 1000 cells (all spacings halved, outer face kept), certified | 0.845660 | -0.00726 | NO (1.25 x) |
| grid 1000 -> 2000 cells, diagnostic (not certified, below) | 0.841623 | -0.00404 | yes (0.70 x) |
| Resid tol 4e-6 -> 1e-6, 1e-7, 3e-8, 1e-8, all certified | 0.852918 | 0 | yes |
| closure tol 0.05 -> 0.01, T_match tol 6 -> 2 K, converged | 0.853294 | +0.00038 | yes |
| domain to L1 (`Outer shells: 37.6 15`), certified | 0.854864 | +0.00195 | yes |

- Grid. 1000 cells: `Base grid [dr,cells]: 9.9999997473787516e-5 100`, `Grid cells: 1000`, `Outer radius [R_p]: 29.638956766027057` (the reference's outer face 29.273416 R_p to 1e-15); cell widths 0.499-0.508 of the reference's. The seed needed one correction: the mapper interpolated the first refined cell from the reservoir ghost row (415 K against 576 K in the reference's cell 1), and the JFNK stalled on cell 1; with that cell set to the reference's cell 1 the solve converged, and certified after a continuation (the first run ended on the 60-pass cap with only the He 2^3S carrier row of the first gated cell above tolerance; continued with 150 passes, accepted at pass 116; tolerances unchanged). The refined state has log10 Mdot lower by 0.0034 (-0.8 %), rho and n(2^3S) lower by 0.8 and 0.8-1.5 % from 3 to 28 R_p: the grid error enters through the steep base layer that sets Mdot. The EW falls with refinement, as in the 1000/2000-cell study of 2026-09-24 (first order, -3 % there), so the 500-cell EW is an overestimate and the deficit larger in the continuum limit. 2000 cells (`Base grid 4.999999873689376e-05 200`, same face), seeded from the certified 1000-cell state: not certified after 300 outer passes (He 2^3S carrier row at the first gated cell 4.7e-5 against 1e-5, and the mass row of one cell at its rounding floor), with the EW unchanged to 1e-6 between pass 150 and 300: 0.841623 (diagnostic). Successive differences -0.00726, -0.00404, ratio 0.556, observed order 0.85; Richardson continuum (DERIVED) 0.8366, 0.016 below the 500-cell value (2.8 x the target). Mdot falls 0.8 % and then 0.4 % with each halving.
- Wind tolerance. `Resid tol` is the JFNK target, not a certification tolerance; the certification gates are compile-time (mass row 3e-12 floor or ten times the cell's rounding, momentum 1e-8, energy 1e-6, closure and level 1e-6, carriers and elemental transport 1e-5 at r >= 1.2 R_p). The JFNK reaches its rounding floor (||R|| 1.2-1.6e-8) at every target from 4e-6 to 3e-8, so those states are identical; 1e-8 (below the floor, 21 of 32 solves end info 2) is accepted and moves the state by <= 6.5e-9 (hydro), 1.6e-7 (n(2^3S)). Tightest certifying target 1e-8; EW change 0.
- Closure. Resuming the damped closure from k02 at tol 0.01 stopped twice at k03 on the wind (He 2^3S carrier row at cell 225: 2.1e-5 and 1.15e-4 against 1e-5 after 60 and 150 outer passes; kept). A fresh closure started at the fluxes k03 measured converged at k = 1 (eps 4.7e-5 / 1.2e-4, T_match 415.86 K, moved 0.053 K). Fixed point F_H 6.4803e7, F_He 7.0914e7 g/s, T_match +3.27 K over k02. The k02 eps_H 0.045 was the trial-measured gap; k02's measured flux was within 2e-4 of the fixed point. Settled.
- Domain. L1 extension: sonic point 28.80 (28.79), log10 Mdot unchanged to 2e-5, profile changes <= 1e-4 inside 12.9 R_p. The 30 R_p edge is 0.48 R_p beyond the sonic point and moving it does not change the branch.
- Roche (decision 3, option b'). `Domain mode: Roche` re-solve: certified by the solve (pass 28), but the evaluate of the reloaded state refuses it on the mass row of cell 1 (3.514e-9 against the cell's anchor 3.3e-9; the solve read 3.101e-9 there), so its EWs are diagnostic. Spherical synthesis 0.834533; Roche synthesis (`phase4/roche/roche_he10830.py`, sub 4, 8 azimuths) 0.820499. Geometric interval [Roche-consistent, spherical] = [0.8205, 0.8529] (-3.8 %; -2.4 % at He/H 2.09).

**Observation.** The width-matched model against the released Cherubim et al. (2026) spectrum (three-Gaussian extractor): red depth 0.985 against 1.254 %, blue 0.128 against 0.198 %, red/blue 7.67 against 6.35, FWHM 0.841 (matched) against 0.841 A, shift -0.04 against +2.85 km/s. Residual chi^2 136 for 44 window samples (1962 for a zero model); the deficit sits in the red pair (mean residual 0.224 +- 0.025 % over 10832.9-10833.7 A against 0.051 +- 0.031 % over the blue component). Frame: the model line has no shift; the data prefer +0.081 A (+2.24 km/s) for the model's shape

under independent errors (chi^2 16.9 against 61.2 at zero), the published fit with correlated noise 2.0 (+2.0/-2.2) km/s (READ, dataset README). The window is not symmetric about the red pair, so the shift moves the model EW by +0.0115 (at +0.081 A) to +0.0137 (+0.104 A), 2.0-2.4 times the target; the assumption holds at 1 sigma of the published shift and is carried as a systematic. Figure `phase6/observation/HeH0.65_k02_solution.png` (inspected).

**Uncertainty budget at He/H 0.65** (changes of the EW): grid -0.00726 (1000 -> 2000 -0.00404; to the continuum -0.0164, derived); Resid tol 0; closure +0.00038; domain +0.00195 (numerical, in quadrature 0.0075, or 0.0165 with the grid's continuum estimate); NH3 nu2 + H2 rate: water analog -0.00037 (T_match +7.6 K), no H2 term +0.00105 (+39.0 K), single solves at k02's chemistry arguments, 15-40 times smaller than at He/H 2.09 because the EW is nearly insensitive to T_match here; ion-neutral drift of He I (phase 1 method on this state, which carries He 2^3S: a fixed-background He 2^3S solve with the Stefan-Maxwell drift flux, lambda_3 from 63 stage-channel evaluates) +0.000037 (A) / +0.000034 (B), so no drift treatment is needed (plan section 8); geometry [Roche-consistent] -0.0324 (diagnostic); frame +0.0115 to +0.0137. Every positive item together gives at most 0.870, still 8.2 sigma below the observation: the deficit is robust to every item measured. Validity: 31.4 % of the EW forms beyond Kn_bulk = 0.1 (8.57 R_p), 9.1 % beyond R_*/R_p, 0.01 % beyond the exobase (28.63 R_p, below the sonic point 28.79); max Kn 1.3.

**Stale (2026-10-07):** the Knudsen numbers, `Kn_bulk = 0.1` radii and exobase radii in this passage were computed by `src/utils/collisional_validity.py` before its resonant H+ + H and He+ + He momentum-transfer coefficients (`md/Update_EXHALE_stage3.md` section 110). On the 102 stored LHS 1140 b states the change moves the `Kn_bulk = 0.1` radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.

**Found, not fixed (need a decision).** (1) In Roche mode the evaluate does not reproduce the solve's certification of the written state (cell-1 mass row, above); 99a76cb0 does in spherical mode. (2) The He 2^3S carrier row at the first gated cell (r = 1.2 R_p) decides every hard ending: k03, the first 1000-cell run, the 2000-cell run, He/H 0.5 of phase 5; k02 itself passed it at 9.65e-6. (3) Refining the base block needs the seed correction above (`phase6/grid/base_cell_seed.py`); `map_state_to_grid.py` is unchanged.

**Not verified.** The 2000-cell state is not certified (its EW is diagnostic). The NH3-H2 cases were not re-closed. The drift estimate is a fixed-background linear estimate of a drift that is itself outside its Chapman-Enskog validity where w/v ~ 1.

## 93. Closure acceptance: the conducted heat flux has its own residual; Codex's resume and solve-exit refusals reviewed (2026-10-04)

Review `md/LHS1140b_NH3_CH4_EW_REVIEW_20261004.md` (Codex, 2026-10-04, interrupted by its usage limit). Codex also edited `src/utils/element_flux_closure.py` and added `src/tests/element_flux_closure/` (6 tests): a window whose H or He flux or spread is missing or non-finite is unresolved (`flux_window_resolved`); a `--resume` refuses a last history row without a resolved measurement at the requested tolerance and checks it against the saved wind's own measurement; an EXHALE that exits non-zero is refused before post-processing even when its log says info=0. Read here, compiled, and the suite passes (6/6); kept.

Added here (the review's finding that the energy-coupled acceptance has no explicit conducted-flux test):

conducted_flux_residual = |F_meas - F_triall/|F_meas| of the conducted base heat flux is held to the same tol as the elemental fluxes, both in the iteration and in the resume check, beside the T_match change. Measured on every stored energy-coupled closure (last history row): 0.0001-0.0375 on all converged ones (largest He/H 1.2, 0.0375), so no stored convergence verdict changes; phase5/HeH3.0 (not converged) has 0.218.

Also recorded from the review: the provenance of the phase-1 molecular emission estimate is the He/H 9.0103 state and six molecules (corrected in md/NH3_CH4_ISSUES_20261004.md); sed/scale_sed.py scales the whole SED, not the XUV alone (corrected in md/PLAN_20261004_lhs1140b_heh_high_and_xuv.md); cross-section-weighted photoionization rates at the b orbit, GJ 699 over GJ 1132: H I 38.7, He I 6.44, He 2^3S 2.13 (md/oct4_review_evidence/stellar_photoionization.json); a uniform 1.30x of n(He 2^3S) on the He/H 0.65 reference gives width-matched EW 1.0957 (a scale of the required opacity, not a model).

## 94. The three phase-6 problems fixed: the mapper's base cells, the He 2^3S carrier row, the certificate of a written state (2026-10-04)

User decision of 2026-10-04 (phase 6 of md/PLAN_20261003_lhs1140b_ew.md, section 92 and LHS1140b/models/.P2/ew_fit/phase6/NOTES.md section 4; review md/LHS1140b_NH3_CH4_EW_REVIEW_20261004.md section 6). Each problem was reproduced before it was fixed. Binary of every run below: LHS1140b/models/.P2/EXHALE_c37b3671.x (ifx 2026.1.0, md5 c37b3671aa518ec7ccb75bf2718adaba, manifest .P2/BINARY_MANIFEST_c37b3671aa51.txt, source copy .P2/build_p6fixes_ifx/; against 99a76cb0 exactly three sources differ: EXHALE_main.f90, diffusive_photochemistry.f90, ion_residual_core.f90). Scratch runs: ~/.claude/jobs/f24ea156/tmp/p6fix/. lart4. All numbers MEASURED.

### 94.1 The mapper seeds the first refined cell from the reservoir row

**Reproduced.** map_state_to_grid.py --ic of phase5/HeH0.65/k02 onto the 1000-cell grid writes the data rows of phase6/grid/attempt1_interpolated_seed bit for bit: the first target center (1.0000491 R_p) lies between the source's lower ghost (0.99990 R_p, the reservoir, 415 K) and its first cell (1.0000967 R_p, 576 K) and was interpolated across the base face (536 K); that seed stalled the JFNK (attempt 1).

**Rule adopted** (src/utils/map_state_to_grid.py, map_rows, module header "THE FIRST PHYSICAL CELLS ARE MAPPED FROM PHYSICAL CELLS"): a target physical center that lies between the base face and the source's first physical center is inside the source's first cell and takes that cell's values (its average over the volume that holds the center); every other physical row is interpolated between the two source rows that bracket it, as before. The lower ghost rows are the reservoir, boundary data that load_IC.f90 does not read, never an interpolation partner of a physical cell. The upper ghosts are the free-outflow continuation of the flow and stay interpolation partners (the shifted-grid round trip of the suite needs that; holding the last cell there moved the density round trip from below 2e-5 to 5.5e-4). The # mapped: line counts the rows that took cell 1's values. An unrefined mapping holds no row and is unchanged.

**Verification.** The fixed tool reproduces the hand correction phase6/grid/base_cell_seed.py exactly: the k02 -> 1000-cell seed equals the seed of attempt 4 (Hydro bit for bit, species within 1.2e-16 relative in row 3),

the 1000 -> 2000-cell seed equals that of `grid2000/solve` bit for bit; the workaround is no longer needed (kept as a record). `element_flux_closure.py` maps onto the seed's own centers (a rewritten `# grid` R0 and `--reservoir`), which holds no row: suite row `unrefined_stored_same_grid_reservoir_byte_identical...` below, and `element_flux_closure` 6/6.

**Test** (`src/tests/state_mapper`, section 7 added): a manufactured base layer (cold reservoir ghosts, steep T) refined 2x: the first target cell is the source's cell 1 (1e-14), no base cell carries a value outside the first two source cells, the second cell is interpolated, the mapping line counts the held row; the stored k02 state onto the 1000-cell grid (when the tree carries it): first cell = source cell 1, not colder than it; three unrefined mappings (identity, base-layer identity, the stored state on its own centers with `--reservoir He/H`) byte-identical to the all-rows mapping. RED (entry text): 7 FAIL (4 manufactured, 3 stored, e.g. first cell 536.28 K against 575.80 K); GREEN: 90 PASS, 0 FAIL.

### 94.2 The He 2^3S carrier row at the first gated cell

**Reproduced from the stored runs.** On `phase6/grid2000/continued` the gated measure of `carrier balance He2^3S` sits at cell 894 (1.230 R_p) in every one of 150 passes and wanders between 1.6e-5 and 2.1e-3 without contracting (the 1000-cell continuation and the closure k03 likewise at cells 448 and 225; the accepted states took a pass that happened to fall below 1e-5).

**What the row is there** (row terms of the stored state; the level's gross halves recorded by this section, below): production 41.2515 and loss 41.2515 cm^-3 s^-1, net 8.797e-7, transport 8.796e-7 (the face flux changes sign in this cell: 8.44e-7 in, 3.5e-8 out), residual -8.36e-11. The level is in local balance to 2e-8 of its channels and the row is solved to 2e-12 of them; the measure divided that residual by |J_in| + |J_out| + |net| = 1.76e-6, i.e. by 4e-8 of the row's own terms, and so asked the coupled state for 4e-13 of the channel rates. The JFNK stops at ||R|| ~ 6e-8; the composition and temperature it leaves move the two halves at that order from pass to pass, and the measure moves across 1e-5 with them. Not a property of the equations, not a resolution question (the measure is 2.7e-6 at 1.217 R_p, 4.7e-5 where the flux crosses zero, 2e-7 at 1.37 R_p), not the gate's position, not the Jacobian (the row is linear in n3 at a fixed background; the forward-difference step 1e-12 n_He is exact for it; `helium_metastable_transport` checks d src(He II)/d n3 to 1e-6): the scale of the measure.

**Cause (code defect).** `carrier_residual` (`diffusive_photochemistry.f90`) formed the physical scale `row_terms_phys` from |src|, the NET source, while the declaration of `carrier_accept_tol` states the measure as the row residual over "the transport flux divergence and the reaction production and loss", and the certification prints it as "the sum of the row's own terms". For a row that hands out only its net (helium stages, atomic proton) the two agree; for the level, whose balance is one closed expression of sign-definite channels, the net is the difference of the terms and not a term.

**Fix.**

1. `ion_residual_core.f90`: `tr_triplet_row_channels`, the production n_e (n_HeII a_3 + n_SI q13) and loss n3 (P_3 + n_e (q31g + q31a + q31b + b_3) + A31 + n_HI Q31) of `tr_triplet_row`'s own terms; `carrier_source` records them as `sprod/sloss` of the level (molecular row: + the He(2^3S) + H2 quench), sprod - sloss = src to rounding.

2. `carrier_residual`: the physical scale takes `srcp + srcl` as its reaction terms. Where a row hands out only its net, srcp + srcl = |srcl| bit for bit, so the helium stage rows and the atomic proton row are unchanged; H2, the molecular proton row and the oxygen carriers now read their gross channels, as the stated definition says. The Newton's own scale (`dsc`) keeps |srcl|. The tolerance (1e-5 in the wind) is unchanged.
3. Found on the way and fixed: the carrier relaxation declared "the carrier block closed at this background" after ONE transport step with the level row at 9e-5 of its terms (every pass of grid2000). Its fixed-point test compared the largest change of any carrier with the largest fraction of ALL carriers (x_ref ~ 1, from the helium stages), against which a step that moved the level (5e-11 of helium) by 20 % of itself read 1e-11 < 1e-10. Now (`relax_photochemical_composition`) each carrier is judged against its own largest fraction (`carrier_displacement_on_own_scale`; the old `carrier_composition_displacement`, no longer called, removed), and the fixed point is also asked of the steady carrier residual of the present rows on the certification's measure, at or below `carrier_accept_tol` (1e-8): an implicit step's displacement bounds its residual only by n/dt. MEASURED with item 3 alone (scale unchanged), grid2000: the relaxation closes the block (9.7e-9 after 189 steps) but the measure after the next wind solve still wanders (3.0e-1 on the first pass after the closure, then 4.7e-5, 4.2e-5, 9.5e-5, 1.1e-5, 4.3e-5, 3.9e-5, 4.2e-5, accepted on an 8.7e-6 at pass 14), which is what isolated item 2 as the cause.

**Verification** (binary c37b3671, the phase-5 solve environment).

- `phase6/grid2000` restarted from `continued`: certified at pass 4 (worst gated row 8.0e-8 (He+), then He++ 6.3e-4, 6.8e-5, 2.9e-6, a geometric contraction); the stored-pair evaluate of the written state reproduces the solve (mass row binding cell 843, 3.559e-11 against 3.7e-11; exit 0). Width-matched EW 0.841623, equal to the earlier diagnostic value, now on a certified state; so 1000 -> 2000 cells = -0.004037 stands as measured.
- `phase5/HeH0.5`, the wind of attempt 4 (seed from the 0.65 rung, the k00 inputs): certified at pass 13 (He++ row 1.0 for six passes, then 1.2e-1, 2.3e-2, 4.8e-3, 9.6e-4, 2.0e-4, 3.9e-5, 8.3e-7); stored-pair evaluate certified, exit 0. Only the wind was re-solved; the He/H 0.5 closure was not re-run (its earlier k00 had no sonic point in the 30 R_p domain).
- `phase5/HeH0.65/k02` re-solved from its own seed: certified at pass 1, the state within 1.6e-12 (hydro) and 8.2e-13 (species) of the stored k02, width-matched EW 0.852918 (unbroadened 0.882189), identical to the reference to the printed digits. The equations are unchanged (the scale is the measure's, not the state's), so "unchanged within rounding" applies.
- Gross channels recorded: on the stored 2000-cell state the level row of cell 894 reads production 41.2515, loss 41.2515, measure 1.45e-9 at most over the gated cells (was 4.75e-5); k02 evaluate: He2^3S 2.9e-4 (cell 111, below the gate) -> 1.7e-10; H2 4.0e-9 -> 3.7e-9, H+ 1.1e-7 -> 8.8e-9; He+ and He++ identical (net rows).
- What the gate now asks of the level: its residual against its whole turnover, which in the wind is the transport plus two halves that cancel to 1e-8 at 1.2-1.4 R_p and to 1-4 % in the line-forming region (phase 1); the gate is therefore weaker against a transport defect of the level where the level is in local balance, which is where transport does not set it.

### 94.3 The certificate of a written state and of its reload

**Reproduced.** The stored-pair evaluate of `phase6/roche/solve` with 99a76cb0 reads the cell-1 mass row 3.514150e-9 against its anchor 3.3e-9 (exit 2); the solve's own final certification read 3.101e-9. A re-run of the Roche solve from its seed with a control binary (the 99a76cb0 source plus measurement records in `certify_the_state_to_be_written`) writes the stored files bit for bit and shows the cause.

**Cause.** Not the Roche geometry: the physical cells of the written and the reloaded state are identical (u, f_sp, n_tot, n_e), and the spherical k02 shows the same disagreement at its base cells (cell-2 mass row 6.219e-10 in the solve, 7.407e-10 on reload), only far below that cell's anchor. The two differ in the LOWER GHOST: the solve's final certification stood on the ghost the last sweep of the solve had returned (taken at the temperature of an earlier iterate), the reload on the ghost solved at the state's own temperature. Ghost T differs by 1.8e-12, the base face mass flux F_lower by 4.1e-16 (4e-10 of itself: the base continuity row reads the ghost with a gain of order 1/Mach, Mach ~ 1e-6 there), the cell-1 row by 13 % of its anchor. The Roche state sat 0.93 of its anchor and crossed; the spherical states did not. A certificate taken on quantities that depend on the iteration history is not a statement about the state written.

**Fix** (`EXHALE_main.f90`). `sources_and_ghosts_of_the_held_composition`: the temperature from u through the caloric map of f_sp, the heating and cooling and the lower ghost from one sweep that holds the composition (`ioniz_eq_sources_of_held_composition`, the interior returned bit for bit, asserted), i.e. the stored-pair evaluation of section 88 as one routine. It is called by that evaluation (unchanged in effect), by the joint test of every stationary outer pass before its certification, and by `certify_the_state_to_be_written`, so a pass is accepted, and a file is certified, on the numbers its reload measures. Measurement records added at the final certification (`EXHALE_EVAL_STATE_DUMP`, `EXHALE_BOUNDARY_TRACE`, `EXHALE_GHOST_RECORD`, default off), the same records the evaluation writes.

**Verification.**

- `phase6/roche` re-solved from its seed: certified at pass 8; the stored-pair evaluate of the written state reproduces the final certification entry by entry (mass row max 4.663e-10 at cell 17, binding cell 218 2.654e-11 against 2.9e-11 in both; volume average 7.392e-11 against 7.469e-11), exit 0. Spherical-synthesis EW 0.834533 (as the diagnostic value); the state moved by at most 7.3e-6 (hydro) and 1.7e-7 (species) from the stored one. The Roche synthesis (`roche_he10830.py`) was not re-run on it.
- Spherical evaluates bitwise unchanged: `phase3/he23s_transport/solve` and `phase5/HeH0.65/k02` evaluated with 99a76cb0 and with c37b3671 write `Hydro_ioniz`, `Ion_species`, `conserved_state`, both `_adv` files and both breakdowns with identical data rows; hydrodynamic entries identical; the carrier entries change as in 94.2. (A first version that repeated the derivation of step (1) inside the evaluation moved the two lower ghost rows of k02 by 1e-14; the routine takes the derivation as done there.)

### 94.4 Tests (gfortran tree build, md5 7a48413c1ec482c0305cb69c08a976a9)

state_mapper 90/0 (RED 7 FAIL above); helium_metastable_transport 24/0 (two rows added: the production and loss halves of the level equal the closed form at the local root, 1e-12; RED against the entry text: both FAIL, 0 and 5.7e-11 against 3.85e5); element_flux_closure 6/6; certification 145/0; ionization_stage_flux 45/0; stage_row_balance

10/0; ionization_stage_enthalpy_flux 13/0; carrier_boundary_jacobian 12/0; carrier_constraint_attribution 10/0; carrier_enthalpy_flux 12/0; carrier_helium_inventory 15/0; carrier_outer_boundary 16/0; carrier_reference_scales 16/0; carrier_retry 156/0; carrier_returned_state_acceptance 36/0; coupled_block_handover 33/0; coupled_block_jacobian 8/0; element_operator 65/0; interdiffusion_enthalpy_flux 18/0; ionization_imposed_fractions 48/0; krylov_and_dogleg 347/0; species_masses 9/0; steady_species_rows 197/0; adv_static_limit 64/0; boundary_state 43/0; grid_and_gates 424/3, the same three rows of `restart_intent_and_metadata.sh` on its stale fixture as in section 88 (worst column heat 5.83e-5, exit 2, certification F), unchanged.

**Not verified.** The Roche synthesis on the re-solved Roche state; the He/H 0.5 closure; the solve trajectories of molecular runs other than these four (the relaxation now runs to the closed block, 35-189 transport steps a pass on grid2000); the regression matrix (not run, by rule).

## 95. Route B: the GJ 699 proxy spectrum puts the He 10830 EW three times above the observation; the escape boundary condition's hydrogen carrier split made self-consistent (2026-10-04)

Plan `md/PLAN_20261004_lhs1140b_heh_high_and_xuv.md`, route B step 1 (user decision 2026-10-04: run GJ 699). Records: `LHS1140b/models/.P2/ew_fit/route_xuv/NOTES.md`, `compare_route_xuv.py`, `route_xuv_numbers.txt`, figures (inspected) `fig_route_xuv_line.png`, `fig_route_xuv_profiles_HeH{0.65,2.09,5.0,10.0}.png`, `fig_route_xuv_triplet_balance.png`, `fig_route_xuv_ew_vs_heh.png`. Binary 99a76cb0 for every closure; every final state also evaluated with c37b3671 (section 94): EW, width-matched EW and f agree to all printed digits. All numbers MEASURED; "EW wm" is the width-matched window EW in percent-angstrom (added Gaussian velocity kernel of FWHM f, not a computed physics term). Observed 1.1084 +- 0.0289.

**The GJ 699 product is not LHS 1140's bolometric flux.** Scaled to the LHS 1140 X-ray flux (A_699 = 0.056) the GJ 699 product carries 1.767e6 erg cm^-2 s^-1 at the b orbit, 3.25 times the GJ 1132 product and 3.1 times L_bol/(4 pi a^2) of LHS 1140; the Photochem climate integrates the whole file. With it the climate is another planet (He/H 2.09: deep 790 K against 561 K, tropopause 248 K against 185 K, no water cold trap). Both were run: the pure GJ 699 file in the chemistry and the wind (as instructed), and a split sensitivity construction (`make_split_sed.py`: GJ 699 at lambda <= 2600 A, GJ 1132 above; not a stellar model) in the chemistry with the pure GJ 699 file in the wind (the wind uses only E >= 4.77 eV). The split climate reproduces the GJ 1132 one. The GJ 1132 control at He/H 2.09 with the cooling-to-space transfer was run because no such state existed.

| spectrum | deep He/H | closure | T_match [K] | log10 Mdot | EW wm sol / adv (f km/s) | Kn_bulk 0.1 / sonic [R_p] |
|---|---|---|---|---|---|---|
| GJ 1132 | 2.09 | k = 1 | 350.71 | 7.8864 | 0.7109 / 0.7143 (22.02) | 5.70 / 25.69 |

| spectrum | deep He/H | closure | T_match [K] | log10 Mdot | EW wm sol / adv (f km/s) | Kn_bulk 0.1 / sonic [R_p] |
|---|---|---|---|---|---|---|
| GJ 699 | 2.09 | k00 wind, residuals 2.5e-5 / 1.0e-4 / 0.0054, T_match change 0.20 K (k01 chemistry); k01 wind not run (mapper) | 329.74 | 8.7080 | 3.6977 / 3.6915 (21.52) | 8.46 / 15.01 |
| split | 2.09 | k = 5 | 264.59 | 8.6544 | 3.3876 / 3.3835 (21.52) | 7.88 / 15.18 |
| GJ 1132 | 0.65 | k = 2 (section 90) | 412.59 | 8.1325 | 0.8529 / 0.8596 (22.24) | 8.57 / 28.79 |
| GJ 699 | 0.65 | REFUSED k00 (cooling-to-space exchange bound, 1.72 and 1.60 K > 1 K) | | | | |
| split | 0.65 | k = 4 | 401.53 | 8.8152 | 3.7065 / 3.7114 (21.62) | 11.54 / 14.54 |
| split | 5.0 | k = 3 | 271.78 | 8.6686 | 3.4270 / 3.4202 (21.41) | 7.23 / 15.46 |
| split | 10.0 | k = 3 | 280.31 | 8.6617 | 3.3276 / 3.3203 (21.34) | 6.82 / 15.57 |

**Stale (2026-10-07):** the Knudsen numbers, `Kn_bulk = 0.1` radii and exobase radii in this passage were computed by `src/utils/collisional_validity.py` before its resonant H+ + H and He+ + He momentum-transfer coefficients (`md/Update_EXHALE_stage3.md` section 110). On the 102 stored LHS 1140 b states the change moves the `Kn_bulk = 0.1` radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.

The sonic point lies inside the 30 R_p domain in every state. The GJ 699 lines are 3.0-3.5 times deeper than observed (width-matched red peak 3.8-4.3 per cent against 1.28), at the GJ 1132 line position (10833.27 A, 0.15 A blueward of the observed peak) and red/blue ratio (7.2-7.3). Thin flag T in every state but He/H 10 (largest absorbed share 0.08-0.76, the pure-GJ 699 0.763 from uncondensed water); without the gas above p_match 0.03-0.04.

**No crossing in He/H on GJ 699.** The width-matched EW is 3.71 (0.65), 3.39 (2.09), 3.43 (5), 3.33 (10) on the

split spectrum and 3.70 at 2.09 on the pure file: 2.2-2.6 percent-angstrom above the observation everywhere, weakly dependent on He/H. The observation lies between the two proxy spectra (GJ 1132 0.71-0.85, GJ 699 3.33-3.71); the crossing is in the stellar input, not in He/H.

**Penning loss.** At He/H 2.09, GJ 699 against GJ 1132 at fixed r (1.5-15 R_p): n(2^3S) x2.3-15.6; recombination gain n_e n(He+) a_TR x3.2-26.5; n(H I) x0.61-1.14 (x(H+) rises 5-7 times but Mdot x6.6), so the Penning loss rate falls by at most 40 per cent inside 3 R_p and not beyond 8 R_p; the electron-collision loss rises x2.8-11.9 and takes the Penning share of the loss from 0.5-0.7 to 0.2-0.3. P3 rises x2.12 (1.389e-5 against 6.54e-6 s^-1) and stays below 5 per cent of the loss inside 8 R_p. The 39x H I photoionization does not raise n(2^3S) by cutting the Penning loss; the larger He I photoionization (x6.4) and the hotter, more ionized wind raise its source.

**Fixed: the hydrogen carrier split of the escape boundary condition (`src/utils/photochem_to_lower_profile.py`).** The elemental hydrogen flux is put on H2 when the atomic-H share of the hydrogen nuclei at the model top is below 1 per cent and split between H and H2 otherwise, but the share was measured on the composition the imposition sees, on a fresh start the initial one (1e-40): the split branch could not be reached, and a converged top above 1 per cent was refused (split He/H 2.09 k00: 1.009e-2). Now the converged share is compared with the share the boundary condition carries (`LAST_IMPOSITION`); where they differ by more than 1 per cent the converged share is imposed (`CARRIER_SHARE_STATED`, needed because a re-solve on a new temperature re-initializes the composition from the climate grid) and the chemistry (under conduction, the column iteration) re-solved, up to 8 rounds; from the second round the imposed share is the mean of the imposed and the converged one (the map has slope about -0.75 at He/H 30: 0 -> 0.098 -> 0.025 -> 0.079 -> 0.039). RED: split 2.09 k00 refused; GREEN: one round, imposed 0.0101, converged 0.00914. A case needing no round runs the unchanged path: the GJ 1132 2.09 k00 arguments give a profile byte-identical to the closure's (`cmp`).

**Found here, resolved in section 96.** `map_state_to_grid.py --reservoir` refused a carry whose first physical cell departs from the requested C/H by more than 1e-12 relative; the pure GJ 699 closure at 2.09 was stopped twice by it (1.23e-12, 1.03e-12). The cause is not CO: the k00 state carries no OH, H2O or CO column (`oxychem=F`); its cell-1 C/H stands 1.03e-12 from its own `# reservoir` line, and the factor new/old carries that departure onto the new value (section 96).

He/H 10 and 30 jumped directly from 2.09 stalled in the first wind passes and were stopped (kept); 10 converged from 5, and 30 from 10 was stopped in its first wind pass after 34 min and not pursued.

**Not verified.** The k01 wind of the pure GJ 699 2.09 closure; pure GJ 699 at any other He/H; He/H above 10; the region beyond Kn_bulk = 0.1 (6.8-11.5 R_p).

**Stale (2026-10-07):** the Knudsen numbers, `Kn_bulk = 0.1` radii and exobase radii in this passage were computed by `src/utils/collisional_validity.py` before its resonant H+ + H and He+ + He momentum-transfer coefficients (`md/Update_EXHALE_stage3.md` section 110). On the 102 stored LHS 1140 b states the change moves the `Kn_bulk = 0.1` radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.

## 96. The mapper's base-row check takes the consumer's tolerance, 1e-6 (2026-10-04)

**What refused the route-B carry (MEASURED).** `map_state_to_grid.py --reservoir El/H v` multiplies the ionization-stage columns of El by k = v/old, old the El/H of the source's `# reservoir` line, and then refused the map when the base rows (He: the two inner ghosts and cell 1; a metal: cell 1) came out more than 1e-12 relative from v. The GJ 699 He/H 2.09 k00 state (`LHS1140b/models/.P2/ew_fit/route_xuv/GJ699_HeH2.09/k00/output`) carries no OH, H2O or CO column (`# options ... oxychem=F`), so the CO explanation of section 95 does not apply. Its cell-1 C/H is 2.77551410664327e-4 against the line's 2.7755141066461295e-4, a relative departure of -1.0303e-12 that the solve leaves (C/H varies by 2.5e-10 over the physical cells); k carries it onto the new value unchanged, and the check refused at 1.03e-12. The attempt-3 carry refused at 1.23e-12 the same way. Both reproduced with the script as it stood, on copies.

**Why 1e-6 (READ, `src/modules/files_IO/load_IC.f90`).** The consumer of the mapped state accepts the loaded He/H within `heh_dev_tol = 1.0d-6` (line 76, the check at line 784), and renormalizes every metal column the handoff states by one factor (handoff El/H)/(El/H at the base cell) (the "elements the handoff states" loop, lines 969-1030), so a metal arrives at the run's El/H at the base cell whatever the mapper wrote. A 1e-12 check was a million times stricter than the run that reads the file. The base-row check now uses one named constant, `BASE_ROW_RATIO_TOL = 1.0e-6`, for helium and for the metals, with a comment pointing to `heh_dev_tol` and to the base-cell renormalization; the refusal message states the tolerance and its source. The check still catches the gross failure it exists for: a species holding nuclei of two elements that the element's one factor does not move with it (HeH+; OH, H2O and CO, which the tool does not multiply). The mapping itself is unchanged.

**Tests (`src/tests/state_mapper/`, MEASURED).** New: a metal state whose cell-1 C/H stands 1.03e-12 from its reservoir line maps and its departure comes out unchanged (-1.03040e-12 against the source's -1.03018e-12; bound 1e-15 on the difference); the same state at 2e-6 is refused and the message names `heh_dev_tol`; HeH+ at 0.8 n_H(cell 1) with He/H 1.6 -> 2.13 is refused (the existing 0.05 case is still refused); the route-B k00 -> k01 carry with its closure arguments maps, every requested El/H at cell 1 within 4.4e-12. RED with the previous script: 5 failures (the 1e-12 case refused, the route-B carry refused, three message checks). GREEN: `run.sh` 98 PASS, 0 failures (the stored-state cases included); `element_flux_closure/run.sh` 6 tests OK (it does not call the mapper). Byte identity: the 34 output files of the suite's previously passing maps are identical between the two scripts (`cmp`), and 43 mapper invocations recorded in the `route_xuv/*/closure.log` and `phase5/*/closure.log` files, rerun with both scripts on their recorded sources into scratch directories, give 42 byte-identical pairs of outputs and the route-B carry, refused before, now written. The attempt-3 carry (1.23e-12) also maps now.

Installed at 2026-10-04 15:17:40 KST by an atomic replace (a temporary file in `src/utils/` moved over the script), because the route_heh_high closures invoke the script afresh at each iteration.

**Not verified.** The k01 wind of the route-B closure from the carried state (not run here); a state with oxygen-chemistry carriers through `--reservoir` (no such state exists in the campaign; it is still refused when their share moves an El/H by more than 1e-6).

## 97. Code audit of 2026-10-04: a non-finite restart center is refused; a zero conducted flux can close (2026-10-04)

Audit `../EXHALE_audit_20261004/CODE_AUDIT_20261004.md` (independent copy of the 15:55 source; read in full).
**B1, fixed.** `verify_restart_radii` (`load_IC.f90`) compared |r_file - rl/|rl against a running maximum; a NaN center compares false, so the maximum stayed 0 and a file with a NaN radius passed the grid guard (reproduced by the audit on the real module). A non-finite relative difference is now set to huge() before the comparison, so it is refused as a grid mismatch. Test `src/tests/restart_radius_guard/` (the audit's driver, linked against the tree's objects): finite -> exit 0, different -> exit 1, NaN -> exit 1 (before the fix NaN returned 0). Valid states are untouched: the change acts only on non-finite values.
**B2, fixed.** `conducted_flux_residual` (`element_flux_closure.py`) returned inf whenever the measured conducted flux was zero, so an exact zero/zero fixed point could never close. It now returns 0 for two exact zeros, inf for a nonzero trial against a zero measurement, inf for a missing or non-finite value, and the relative residual otherwise. Tests added to `src/tests/element_flux_closure/` (10 pass). No stored closure has a zero flux, so no verdict changes.
**P1, open (needs a decision).** With `Low-energy electron partition: Furlanetto2010` on, any n(H2) > 0 switches the atomic partition off in that cell (`low_energy_electron_degradation.f90`, the H2 gate), so the deposition is discontinuous between zero and a trace H2 density, and the tables are those of an atomic H/He mixture. The option is off in every LHS 1140 b run, so no stored result is affected. Recorded as TO_BE_DONE (AJ).
**P2, P3** are the documented limitations already recorded: the dissociative-recombination recipients (section 72 range statement at the code site) and the NH3/CH4 overlap (`md/NH3_CH4_ISSUES_20261004.md`, TO_BE_DONE (AI)).
The tree's gfortran build was rebuilt; the running ifx closures use the already-built binaries and are unaffected (the load_IC change concerns only non-finite input).

## 98. Route B step 2: the observed He 10830 EW needs 1.33-1.39 times the GJ 1132 ionizing XUV at He/H 2.09 (2026-10-04)

Plan `md/PLAN_20261004_lhs1140b_heh_high_and_xuv.md`, route B step 2 (user decision 2026-10-04). Records: `LHS1140b/models/.P2/ew_fit/route_xuv_scaled/NOTES.md`, `compare_xuv_scaled.py`, `xuv_scaled_numbers.txt`, figures (inspected) `fig_xuv_scaled_line.png`, `fig_xuv_scaled_profiles.png`, `fig_xuv_scaled_ew_vs_s.png`. All solves and evaluates on lart3, binary c37b3671. All numbers MEASURED; "EW wm" is the width-matched window EW in percent-angstrom (added Gaussian velocity kernel of FWHM f, not a computed physics term). Observed 1.1084 +- 0.0289.
**The spectrum (a sensitivity construction, not a stellar model).** `make_xuv_scaled_sed.py`: the GJ 1132 product with every row at lambda <= 911.75 A (X-ray and hydrogen-ionizing EUV, the band that photoionizes the H I and He I ground states and in which the two proxies differ by 7x in F_EUV) multiplied by s, and every row above by one factor c holding the bolometric flux (trapezoid over the file) at the GJ 1132 value: c - 1 = -1.3e-5 at s = 1.36, -1.07e-4 at s = 4; the written files (17 digits) reproduce the bolometric 5.439384522884e5 erg cm^-2 s^-1 to 0 relative. The 911.75-2600 A band (Lyman alpha, FUV, NUV: the He 2^3S photoionizing band and the H2 Lyman-Werner band) keeps its shape, so the scan raises the He+ recombination source of 2^3S and not its photoionization loss (P_He23S at the orbit +0.6

per cent at s = 1.36, from the part of the 2^3S cross section below 911.75 A; P_HI and P_HeI scale as s). Both the Photochem adapter and the wind read the scaled file: `EXHALE_setup.out` Log EUV / Log X equal the control's plus log10 s on every state, and the profile's `stellar_flux` record names the file with its sha256.

**Settings** identical to the GJ 1132 He/H 2.09 cooling_to_space control of section 95 (`route_xuv/GJ1132_HeH2.09`, k = 1, EW wm 0.7109) except the SED path and the binary; continuation in s from the nearest converged s; no iteration budget raised, no tolerance changed, no climate starting value needed.

| s | F_EUV(100-911) | closure k | T_match [K] | log10 Mdot | T_max [K] | EW wm sol / adv (f km/s) | wm red peak [%] | red/blue | Kn_bulk 0.1 / sonic [R_p] |
|---|---|---|---|---|---|---|---|---|---|
| 1 (control) | 10.05 | 1 | 350.71 | 7.8864 | 4961 | 0.7109 / 0.7143 (22.02) | 0.821 | 7.60 | 5.70 / 25.69 |
| 1.3 | 13.07 | 1 | 350.29 | 8.0399 | 5212 | 1.0452 / 1.0489 (21.98) | 1.207 | 7.52 | 6.06 / 24.06 |
| 1.36 | 13.67 | 1 | 350.39 | 8.0652 | 5250 | 1.1071 / 1.1107 (21.97) | 1.278 | 7.51 | 6.12 / 23.77 |
| 1.4 | 14.07 | 1 | 350.31 | 8.0812 | 5274 | 1.1472 / 1.1509 (21.96) | 1.325 | 7.50 | 6.16 / 23.58 |
| 1.5 | 15.08 | 1 | 349.95 | 8.1188 | 5329 | 1.2444 / 1.2481 (21.95) | 1.437 | 7.49 | 6.25 / 23.12 |
| 2 | 20.10 | 3 | 345.81 | 8.2691 | 5531 | 1.6703 / 1.6735 (21.90) | 1.928 | 7.42 | 6.64 / 21.22 |
| 4 | 40.20 | 3 | 339.95 | 8.6069 | 5894 | 2.7923 / 2.7931 (21.67) | 3.222 | 7.29 | 7.85 / 17.14 |

**Stale (2026-10-07):** the Knudsen numbers, `Kn_bulk = 0.1` radii and exobase radii in this passage were computed by `src/utils/collisional_validity.py` before its resonant H+ + H and He+ + He momentum-transfer coefficients (`md/Update_EXHALE_stage3.md` section 110). On the 102 stored LHS 1140 b states the change moves the `Kn_bulk = 0.1` radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.

Every closure converged (tol 0.05, T_match tol 6 K, conducted-flux residual <= 0.05); every wind certified as written; every evaluate exit 0 with `written_state certified=T`; sonic point inside the 30 R_p domain everywhere; thin flag

T in every state (largest absorbed share 0.203-0.205, CH4 dyad, as the control; 0.028 without the gas above p_match), exchange shift 0.36 K. The s = 1 control re-evaluated on lart3 gives the section-95 EW, EW wm and f to all printed digits.

**The crossing.** EW wm rises monotonically with s. Linear in ln s between the bracketing converged states: the observed 1.1084 at s = 1.361 (solution; 1.358 adv), the 1 sigma band 1.0795-1.1373 at s = 1.333-1.390, i.e. F_EUV(100-911) = 13.4-14.0 erg cmˆ-2 sˆ-1 at the b orbit (crossing 13.68), 0.195 of the GJ 699 value (70.18) and 1.36 times the GJ 1132 value. The converged s = 1.36 state is itself 0.0013 (solution) / 0.0023 (adv) from the observation. At that state the width-matched red peak is 1.278 per cent against the observed 1.275 (expected once EW and FWHM are both matched; not an independent test), the red/blue depth ratio 7.51 against the observed 6.35 (same extractor, section 92; 7.6 was quoted here in error, section 99), and the peak 0.15 A blueward of the observed one, as at every s. The lower column hardly moves over the crossing range (T_match 349.95-350.39 K up to s = 1.5). Interpretation: the EW grows through He+ and n_e (the recombination source of 2ˆ3S) at every radius; n(H I) is nearly unchanged.

**Not verified.** The grid error at these states (about a quarter of the observed 1 sigma at He/H 0.65, section 90); the crossing at other He/H; the Roche-mode evaluate; the region beyond Kn_bulk = 0.1 (6.1 R_p at the crossing). The interval reflects the observed 1 sigma only.

**Stale (2026-10-07):** the Knudsen numbers, `Kn_bulk = 0.1` radii and exobase radii in this passage were computed by `src/utils/collisional_validity.py` before its resonant H+ + H and He+ + He momentum-transfer coefficients (`md/Update_EXHALE_stage3.md` section 110). On the 102 stored LHS 1140 b states the change moves the `Kn_bulk = 0.1` radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.

## 99. Route B step 3: the observed He 10830 EW fixes F_EUV(100-911) at 12.9-13.7 erg cmˆ-2 sˆ-1 over deep He/H 0.65-5 (2026-10-04)

Plan `md/PLAN_20261004_lhs1140b_heh_high_and_xuv.md`, route B (user decision 2026-10-04: repeat the section-98 crossing search at deep He/H 0.65, 1.2 and 5). Records: `LHS1140b/models/.P2/ew_fit/route_xuv_scaled/NOTES.md` section 6, `compare_heh_match.py`, `heh_match_numbers.txt`, figures (inspected) `fig_heh_match_feuv.png`, `fig_heh_match_line.png`, `fig_heh_match_ew_vs_s.png`. All solves and evaluates on lart3, binary c37b3671, the construction and settings of section 98 (cooling_to_space, energy coupling, tol 0.05, T_match tol 6 K); no iteration budget raised, no tolerance changed. All numbers MEASURED; "EW wm" is the width-matched window EW in percent-angstrom (added Gaussian velocity kernel, not a computed physics term). Observed 1.1084 +- 0.0289.

**Seeds.** The s = 1 converged GJ 1132 state at each He/H: `phase5/HeH0.65` k02 (keeps the control's `--climate-t-deep-guess 650`), `phase5/HeH1.2_thick` k01, `route_heh_high/HeH5.0` k01 (read only); later s from the nearest converged s at the same He/H. The two controls solved with older binaries re-evaluate with c37b3671 on lart3 to all printed digits. `setup_case.py` takes `--control`; cases `HeH<x>/xuv_x<s>/`. 14 closures, every one converged at k = 1 and certified, every evaluate exit 0 with `written_state certified=T`, sonic point inside the 30 R_p domain everywhere.

| deep He/H | s for 1.1084 (1 sigma) | F_EUV(100-911) match (1 sigma) | nearest converged state: s, EW wm sol / adv (f km/s) | log10 Mdot (N-20) | T_match / T_max [K] | red/blue (depth) | red peak [A vac] | sonic / Kn_bulk 0.1 / exobase [R_p] |
|---|---|---|---|---|---|---|---|---|
| 0.65 | 1.300 (1.264-1.336) | 13.07 (12.71-13.43) | 1.3, 1.1084 / 1.1159 (22.20) | 8.2408 | 414.9 / 3626 | 7.60 | 10833.276 | 26.20 / 8.78 / above the domain |
| 1.2 | 1.280 (1.249-1.311) | 12.86 (12.55-13.17) | 1.28, 1.1088 / 1.1146 (22.10) | 8.1379 | 390.4 / 4437 | 7.56 | 10833.265 | 25.32 / 7.31 / 24.99 |
| 2.09 (section 98) | 1.361 (1.333-1.390) | 13.68 (13.40-13.97) | 1.36, 1.1071 / 1.1107 (21.97) | 8.0652 | 350.4 / 5250 | 7.51 | 10833.272 | 23.77 / 6.12 / 19.90 |
| 5 | 1.336 (1.309-1.363) | 13.43 (13.16-13.70) | 1.34, 1.1129 / 1.1142 (21.79) | 8.0000 | 326.8 / 6188 | 7.46 | 10833.266 | 22.67 / 5.09 / 15.30 |

**Stale (2026-10-07):** the Knudsen numbers, `Kn_bulk = 0.1` radii and exobase radii in this passage were computed by `src/utils/collisional_validity.py` before its resonant H+ + H and He+ + He momentum-transfer coefficients (`md/Update_EXHALE_stage3.md` section 110). On the 102 stored LHS 1140 b states the change moves the `Kn_bulk = 0.1` radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.

Brackets (linear in ln s): He/H 0.65 s = 1.25-1.29 / 1.3-1.32 / 1.32-1.36 for obs - 1 sigma / obs / obs + 1 sigma (states at 1.15, 1.25, 1.29, 1.3, 1.32, 1.36); He/H 1.2 1.17-1.25 / 1.25-1.28 / 1.28-1.32 (1.17, 1.25, 1.28, 1.32); He/H 5 1.3-1.34 / 1.3-1.34 / 1.34-1.37 (1.3, 1.34, 1.37, 1.42). The adv crossings lie 0.001-0.009 lower in s.

**The degeneracy line.** The F_EUV that reproduces the observed EW is 12.9-13.7 erg cmˆ-2 sˆ-1 at the orbit (s = 1.28-1.36) over He/H 0.65-5, not monotonic (lowest at 1.2, as the s = 1 EW 0.853 / 0.831 / 0.711 / 0.715 is not monotonic). The spread of the four crossings, +-3 per cent about 13.3, is of the order of each one's 1 sigma half-width (+-2.0-2.8 per cent); the union of the 1 sigma intervals is 12.55-13.97. Interpretation: on these equations and this construction the measured EW fixes F_EUV(100-911) at about 13.3 +- 0.7 erg cmˆ-2 sˆ-1 and leaves deep He/H undetermined over 0.65-5; along the line log10 Mdot falls from 8.24 to 8.00 and T_max rises from 3630 to 6190 K.

**Unfitted properties.** The four matched lines are nearly identical. The red/blue depth ratio falls from 7.60 to 7.46 with He/H; the same three-Gaussian extractor on the released spectrum gives 6.35 (16-84 percentiles 5.25-7.89, Monte Carlo

of the tabulated sigma with independent samples, nominal). The model red peak is at 10833.27 A at every He/H, 0.14-0.15 A blueward of the observed 10833.42 A (extractor shift of the data +0.103 A, 0.048-0.142). Neither distinguishes the He/H.

**Correction (documents).** Section 98 and the two route-B NOTES compared the model red/blue with an "observed 7.6"; the observed value from the same extractor is 6.35 (section 92; the published 6.7 +12.7/-3.1, READ from `he_line_metrics.py`). The quantity is the ratio of the fitted red depth to the blue depth, not of equivalent widths. Corrected in `route_xuv/NOTES.md`, `route_xuv_scaled/NOTES.md`, section 98 above and the column labels of `compare_xuv_scaled.py` / `xuv_scaled_numbers.txt`; no number of a model changes.

**Not verified.** Column checks (thin flag, gas above p_match) at the new states; the grid error (about a quarter of the observed 1 sigma at He/H 0.65, section 90); the Roche-mode evaluate; the region beyond Kn_bulk = 0.1 (5.1-8.8 R_p at the matched states). The intervals carry the observed 1 sigma only.

**Stale (2026-10-07):** the Knudsen numbers, `Kn_bulk = 0.1` radii and exobase radii in this passage were computed by `src/utils/collisional_validity.py` before its resonant H+ + H and He+ + He momentum-transfer coefficients (`md/Update_EXHALE_stage3.md` section 110). On the 102 stored LHS 1140 b states the change moves the `Kn_bulk = 0.1` radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.

## 100. LHS 1140 b: heat conduction in the wind makes the mass-loss rate depend on composition (2026-10-04)

Question: log10 Mdot rises from 7.886 (deep He/H 2.09) to 8.133 (0.65) in the present model (section 91) and was flat across composition in the 2026-09 models; which physics does it? Records: `LHS1140b/models/.P2/ew_fit/mdot_attribution/NOTES.md`, `attribution.py`, `conduction_budget.py`, `conduction_ladder.sh`, `make_case.py`, tables `attribution.txt`, `conduction_budget.txt`, `escape_fluxes.txt`, `cooling_channels.txt`, `mach.txt`, figures (inspected) `fig_mdot_ew_vs_heh.png`, `fig_profiles.png`, `fig_dEW_factors.png`. Binary c37b3671, lart4, GJ 1132 SED. All numbers MEASURED; EW is the width-matched window EW (section 87; f 22.02-22.26 km/s, not a computed physics term). No source file changed.

**Configurations** (each at deep He/H 2.09 and 0.65; ionization transport, He 2^3S transport, molecular carriers, element diffusion, grid, domain, outer boundary, Resid tol 4e-6 and closure tolerance 0.05 identical): A, the references (`route_xuv/GJ1132_HeH2.09/k01` and `phase6/closure_direct/k01`, solved with 99a76cb0, re-evaluated with c37b3671, EW unchanged); B, the elemental-flux closure without `energy_coupling` (uncoupled column, conduction in the wind on: the closure before section 76); C, `Conduction: False` (the energy-row conduction source and the base conductive flux off; no flux to pass, so the column is the uncoupled one; element, carrier and stage enthalpy fluxes on, the code ties them to `Interdiffusion enthalpy flux`, not to conduction); D, conduction on, uncoupled column, `Interdiffusion enthalpy flux: False` (all three enthalpy fluxes off). Every closure converged (B k = 0 / 2, C k = 0 / 1, D k = 1 / 0) and every final state was certified. C at 2.09 needed a seed from a conductivity continuation on the fixed B column (`EXHALE_CONDUCTION_SCALE` 0.5, 0.2, 0.05, then off; the direct switch stalled at ||R|| ~ 2 and is kept).

| config | He/H | log10 Mdot | Mdot ratio 0.65/2.09 | EW wm sol / adv | T_match [K] | T_max [K] | mu(T_max) |
|---|---|---|---|---|---|---|---|
| A | 2.09 / 0.65 | 7.8864 / 8.1326 | 1.763 | 0.7109 / 0.7143; 0.8533 / 0.8600 | 350.7 / 415.9 | 4961 / 3501 | 2.621 / 1.906 |
| B | 2.09 / 0.65 | 7.8029 / 8.1276 | 2.112 | 0.5664 / 0.5694; 0.8413 / 0.8479 | 185.0 / 184.6 | 4889 / 3503 | 2.587 / 1.900 |
| C | 2.09 / 0.65 | 8.0495 / 8.1659 | 1.307 | 1.3510 / 1.3540; 1.0764 / 1.0829 | 185.0 / 184.6 | 5353 / 3619 | 2.610 / 1.897 |
| D | 2.09 / 0.65 | 7.8276 / 8.1263 | 1.990 | 0.6860 / 0.6895; 0.8982 / 0.9054 | 185.0 / 184.6 | 4996 / 3527 | 2.626 / 1.917 |

dEW(0.65 - 2.09): A +0.142, B +0.275, **C -0.275**, D +0.212. The energy carried through cell N-20 equals Q_net - L_cond to 0.983-1.005 in all eight states.

**Attribution** (log10 of the Mdot ratio): wind conduction C -> B +0.208 dex (it lowers Mdot by 0.247 dex at 2.09 and 0.038 dex at 0.65); energy coupling B -> A -0.079 dex (it raises Mdot by 0.084 dex at 2.09, 0.005 at 0.65: it reduces the composition dependence); enthalpy fluxes D -> B +0.026 dex; remainder without conduction and coupling (C) 0.116 dex (x1.31), with mu(T_max) 2.61 -> 1.90, consistent with candidate (3) but not isolated. Heat conduction in the wind is what makes Mdot depend strongly on composition. Mdot follows the net heat of the gas above 1.02 R_p (heating - cooling + conduction source - enthalpy-flux divergence): B/C 0.587 at 2.09 (1.21e20 against 2.06e20 erg/s; Mdot 0.567) and 0.930 at 0.65 (Mdot 0.916). At 2.09 conduction keeps the cool layer above 1.02 R_p (1311 K at 1.02 R_p against 2304 K in C), where H3+ infrared cooling takes 2.75e19 erg/s (1.3e15 in C), the heating above 1.02 R_p falls by 0.38e20 and the conduction source removes 0.15e20; at 0.65 heating is unchanged and conduction removes 0.11e20 (interpretation of these budgets: the heavier gas keeps the hot layer close to the cold base).

**Shapley (dEW, 9 factors, `decompose_dEW.py` unchanged)**: n_H +0.353 / +0.340 / +0.442 / +0.362 (A / B / C / D); y -0.628 / -0.470 / -1.142 / -0.560; He fractions -0.140 / -0.161 / -0.128 / -0.175; n_e +0.368 / +0.423 / +0.203 / +0.419; T +0.195 / +0.150 / +0.348 / +0.172; rest below 0.02. No factor changes sign; the net flips in C because the helium-abundance term doubles and the electron term halves: without conduction the 2.09 wind carries more helium (escaping He/H 0.73 against 0.50 in B; F_He 8.3e7 against 4.2e7 g/s) and the hydrogen flux rises less toward 0.65 (x2.36 against x3.04). Synthesis endpoints equal the EW records in all eight states.

**Not verified.** The C winds are subsonic inside the 30 R_p domain (isothermal Mach at most 0.66-0.68; A, B, D become supersonic at 18.5-21.5 R_p), so the C mass-loss rates depend on the outer boundary, which was held fixed; no larger

domain was run. Candidate (3) not isolated by a fixed-opacity mu solve. The mechanism sentence is an interpretation.

## 101. Route A: deep He/H 2.09 to 1e4 on the GJ 1132 spectrum; the He 10830 EW passes a minimum and turns up but stays below the observation (2026-10-04)

Plan `md/PLAN_20261004_lhs1140b_heh_high_and_xuv.md`, route A (user decision 2026-10-04), extended to He/H 1e4 at the user's request of the same day. Records: `LHS1140b/models/.P2/ew_fit/route_heh_high/NOTES.md`, `compare_route_heh_high.py`, `route_heh_high_numbers.txt`, `n3_column_split.txt`, figures (inspected) `fig_heh_high_ew_vs_heh.png`, `fig_heh_high_line.png`, `fig_heh_high_profiles.png`, `fig_heh_high_triplet_shares.png`. Binary c37b3671 (section 94) for every closure and EW, lart4, 12 threads. No source file changed. All numbers MEASURED; "EW wm" is the width-matched window EW (section 87; an added Gaussian velocity kernel, f 21.4-22.0 km/s, not a computed physics term). Observed 1.1084 +- 0.0289.

**Setup.** Every case is the closure of the GJ 1132 control at He/H 2.09 (`route_xuv/GJ1132_HeH2.09/closure.json`: cooling_to_space, top-pressure passes 12, tol 0.05, T_match tol 6 K, energy coupling) with only `--abundances He=<x>`, the binary and the template changed; continuation in He/H (the mapper carries the reservoir; no molecular state rescaled). He/H 3.0 was started fresh from the control, not from the old-treatment `phase5/HeH3.0` k01. No tolerance and no iteration budget was changed (the model-top pinning needed at most 7 of its 12 passes). Jumps of x1.33-x2 solved; three jumps stalled in the first wind pass (JFNK ||R|| 1.5-3 for 400-830 iterations) and are kept: 2.09 -> 5, 400 -> 1000, 1000 -> 3000.

**Scan** (every row: closure converged, written state certified in the wind, EW evaluate certified, exit 0; sonic point inside 30 R_p):

| deep He/H | closure k | T_match [K] | log10 Mdot | T_max [K] | EW sol / adv | EW wm sol / adv (f km/s) | Kn_bulk 0.1 / exobase / sonic [R_p] |
|---|---|---|---|---|---|---|---|
| 2.09 (control, section 95) | 1 | 350.71 | 7.8864 | 4961 | 0.7353 / 0.7388 | 0.7109 / 0.7143 (22.02) | 5.70 / 17.23 / 25.69 |
| 3.0 | 1 | 332.83 | 7.8255 | 5358 | 0.6993 / 0.7020 | 0.6761 / 0.6787 (21.92) | 5.06 / 14.67 / 24.54 |
| 4.0 | 1 | 326.05 | 7.8007 | 5657 | 0.7054 / 0.7076 | 0.6820 / 0.6841 (21.85) | 4.72 / 13.28 / 23.82 |
| 5.0 | 1 | 321.73 | 7.8012 | 5877 | 0.7399 / 0.7418 | 0.7154 / 0.7172 (21.80) | 4.57 / 12.64 / 23.49 |
| 7.0 | 1 | 328.84 | 7.8135 | 6159 | 0.8058 / 0.8072 | 0.7791 / 0.7805 (21.73) | 4.45 / 12.03 / 23.23 |

| deep He/H | closure k | T_match [K] | log10 Mdot | T_max [K] | EW sol / adv | EW wm sol / adv (f km/s) | Kn_bulk 0.1 / exobase / sonic [R_p] |
|---|---|---|---|---|---|---|---|
| 10 | 1 | 334.06 | 7.8136 | 6384 | 0.8398 / 0.8407 | 0.8121 / 0.8129 (21.67) | 4.31 / 11.38 / 22.98 |
| 14 | 1 | 335.13 | 7.8067 | 6541 | 0.8493 / 0.8498 | 0.8212 / 0.8217 (21.62) | 4.18 / 10.81 / 22.76 |
| 20 | 1 | 343.02 | 7.7936 | 6660 | 0.8416 / 0.8418 | 0.8137 / 0.8139 (21.57) | 4.06 / 10.26 / 22.53 |
| 30 | 1 | 357.62 | 7.7749 | 6754 | 0.8212 / 0.8212 | 0.7941 / 0.7941 (21.53) | 3.93 / 9.71 / 22.28 |
| 40 | 2 | 371.07 | 7.7661 | 6803 | 0.8137 / 0.8135 | 0.7868 / 0.7866 (21.51) | 3.87 / 9.44 / 22.16 |
| 60 | 2 | 394.30 | 7.7599 | 6852 | 0.8130 / 0.8126 | 0.7862 / 0.7858 (21.48) | 3.82 / 9.21 / 22.07 |
| 80 | 2 | 412.25 | 7.7579 | 6875 | 0.8160 / 0.8156 | 0.7891 / 0.7887 (21.47) | 3.80 / 9.11 / 22.04 |
| 100 | 2 | 428.94 | 7.7576 | 6888 | 0.8201 / 0.8195 | 0.7931 / 0.7925 (21.46) | 3.79 / 9.06 / 22.03 |
| 200 | 1 | 484.29 | 7.7596 | 6909 | 0.8362 / 0.8354 | 0.8086 / 0.8079 (21.44) | 3.78 / 8.99 / 22.03 |
| 400 | 1 | 546.76 | 7.7657 | 6915 | 0.8573 / 0.8562 | 0.8291 / 0.8281 (21.44) | 3.79 / 9.01 / 22.07 |
| 700 | 1 | 602.09 | 7.7731 | 6914 | 0.8779 / 0.8768 | 0.8491 / 0.8480 (21.44) | 3.81 / 9.08 / 22.13 |
| 1000 | 1 | 635.41 | 7.7781 | 6912 | 0.8913 / 0.8901 | 0.8621 / 0.8609 (21.44) | 3.82 / 9.13 / 22.17 |

| deep He/H | closure k | T_match [K] | log10 Mdot | T_max [K] | EW sol / adv | EW wm sol / adv (f km/s) | Kn_bulk 0.1 / exobase / sonic [R_p] |
|---|---|---|---|---|---|---|---|
| 1400 | 1 | 667.00 | 7.7833 | 6910 | 0.9047 / 0.9034 | 0.8750 / 0.8738 (21.44) | 3.84 / 9.19 / 22.21 |
| 2000 | 1 | 703.55 | 7.7897 | 6906 | 0.9205 / 0.9191 | 0.8904 / 0.8890 (21.44) | 3.86 / 9.27 / 22.26 |
| 3000 | 1 | 748.33 | 7.7979 | 6901 | 0.9406 / 0.9391 | 0.9098 / 0.9084 (21.44) | 3.88 / 9.36 / 22.32 |
| 4000 | 1 | 779.89 | 7.8038 | 6897 | 0.9550 / 0.9535 | 0.9238 / 0.9223 (21.45) | 3.90 / 9.44 / 22.37 |
| 6000 | 1 | 816.20 | 7.8108 | 6892 | 0.9725 / 0.9709 | 0.9407 / 0.9392 (21.45) | 3.93 / 9.52 / 22.42 |
| 7000 | 1 | 826.69 | 7.8131 | 6891 | 0.9781 / 0.9765 | 0.9461 / 0.9446 (21.45) | 3.93 / 9.55 / 22.44 |
| 8000 | 1 | 837.95 | 7.8153 | 6889 | 0.9836 / 0.9820 | 0.9515 / 0.9499 (21.45) | 3.94 / 9.58 / 22.45 |
| 1e4 | 1 | 855.53 | 7.8188 | 6886 | 0.9923 / 0.9907 | 0.9600 / 0.9584 (21.45) | 3.95 / 9.62 / 22.48 |

**Stale (2026-10-07):** the Knudsen numbers, `Kn_bulk = 0.1` radii and exobase radii in this passage were computed by `src/utils/collisional_validity.py` before its resonant H+ + H and He+ + He momentum-transfer coefficients (`md/Update_EXHALE_stage3.md` section 110). On the 102 stored LHS 1140 b states the change moves the `Kn_bulk = 0.1` radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.

**The EW against He/H.** The width-matched EW passes a minimum at He/H 3-4 (0.676), rises to a local maximum at 14 (0.821), falls to 0.786-0.789 at 40-80 and rises again above 100, about 0.10 per decade above 1000, to 0.9600 at 1e4: 0.148 (5.1 nominal sigma) below the observed EW. **No He/H between 2.09 and 1e4 reaches the observed EW on the GJ 1132 spectrum.** Two paths to the same He/H agree (10 from 5 and from 7: 0.8121 / 0.8106; 20 from 10 and from 14: 0.8137 / 0.8141). The lines keep one shape and position (10833.27 A, 0.15 A blueward of the observed peak); the width-matched peak grows to 1.107 per cent at 1e4 against the observed 1.28.

**He 2^3S loss as hydrogen disappears** (the balance terms of `phase5/dEW_decomposition/triplet_balance.py`; P3

6.54e-6 s^-1 measured): Penning ionization by H I, 0.64 of the loss at 1.5-3 R_p at He/H 2.09, gives way to electron collisions between He/H 3 and 4 (0.03 at 100, 3e-4 at 1e4); electron collisions then take 0.97-1.00 inside 3 R_p and radiative decay A31 0.58-0.61 at 8 R_p; P3 stays below 0.05 and the H2 quench below 1e-4. The loss rate at 1.5 R_p falls only to 0.62 of its 2.09 value (the electron-collision floor) while n(He+) rises 1.8 times; n(2^3S) there rises 2.7 times. Above He/H 100 the wind above 1.2 R_p is otherwise unchanged (T_max 6886-6915 K) but Mdot rises 15 per cent (log10 7.758 -> 7.819) with the warming base (T_match 429 -> 856 K: the conducted flux heats a column whose CH4, NH3 and H2O scale with hydrogen), and the He+ front and the 2^3S peak move down to the base (n3 peak 263 cm^-3 at 1.14 R_p at He/H 100, 2.6e3 at 1.03 R_p at 1e4). The dEW was not decomposed on the high side; the mechanism statements are interpretations of the balance terms.

**Validity.** Column: thin flag T up to He/H 5, F from 7 on (largest absorbed share 0.20 at 2.09, 0.004-0.006 at 1000-1e4); exchange shift 0.36 -> 0.02 K. Hydrogen-poor molecular network: the gated carrier rows (H2, H+, He+, He++, He 2^3S) certify in the wind at every He/H (largest He++, 9.6e-6 against 1e-5); match q_H2 0.193 -> 5.0e-5 at 1e4. Climate and Photochem succeeded at every He/H (deep 20-bar temperature 561 K at 2.09, 252 K at 400; tropopause 1.0 -> 9.3 bar). The share of the radial 2^3S column inside the ungated base band r < 1.2 R_p grows from 0.014 (2.09) to 0.231 (100), 0.439 (1000) and 0.598 (1e4); the worst ungated species row of those states is 0.4e-4 to 1.8e-4 (He++ at 1.15 R_p), 4-18 times the wind's 1e-5. The rise above He/H 100 is therefore carried increasingly by cells the certificate does not gate (reported, not changed).

**He/H 8000 refused once, by a hydrogen-specific step.** Seeded from 4000, the k00 wind stopped in its first sweep: "(ioniz_eq) STOP: the base handoff partition of a lower ghost did not close against its own ionization balance" (30 passes, last move of the imposed 2 n(H2)/n_H 5.10e-4 against 1e-10, residual 3.45e-4, pin error 0; EXHALE exit 128). Budget and tolerance are compile-time parameters of `ionization_equilibrium.f90`, unchanged. The same pair closed in at most 4 passes (to 1e-13) at 4000, 6000, 7000, 8000 (seeded from 6000) and 1e4, so the refusal depends on the seed's radiation field at the base, not on He/H; kept in `HeH8000.0/attempt1_from_4000/`. The last good He/H is 1e4; nothing failed for good.

**Not verified.** The base band below 1.2 R_p (above); the region beyond Kn_bulk = 0.1 (3.8-5.7 R_p); a dEW decomposition of the minimum at 3-4 and the maximum at 14; why the 4000-seeded ghost at 8000 did not close.

**Stale (2026-10-07):** the Knudsen numbers, `Kn_bulk = 0.1` radii and exobase radii in this passage were computed by `src/utils/collisional_validity.py` before its resonant H+ + H and He+ + He momentum-transfer coefficients (`md/Update_EXHALE_stage3.md` section 110). On the 102 stored LHS 1140 b states the change moves the `Kn_bulk = 0.1` radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.

## 102. The EUV that matches the observed EW, deep He/H 0.2 to 1e4; the exchange inside the warmed column solved as a transfer option (2026-10-05)

Directory `LHS1140b/models/.P2/ew_fit/route_xuv_scaled/` (NOTES section 7; `setup_lowheh_case.py`, `compare_euv_match_vs_heh.py`, `euv_match_vs_heh.txt`, `fig_euv_match_vs_heh.png`). User request 2026-10-04: lower He/H to 0.1 and find F_EUV matching the EW at each; and (user) the s = 1 route-A states at He/H 14,

100 and 1e4. Same construction and settings as sections 98-99 (GJ 1132 with lambda <= 911.75 A times s, bolometric held; binary c37b3671; cooling_to_space; energy coupling; tol 0.05; T_match tol 6 K; every solve on lart3). Seeds: the s = 1.3 state at the next He/H for the descent 0.65 -> 0.45 -> 0.3 -> 0.2 -> 0.1 (continuation in He/H at fixed s), the converged s = 1 states of `route_heh_high/` for 14, 100 and 1e4. The adapter climate start `--climate-t-deep-guess` 680-750 K at He/H 0.45-0.1 (the deep temperature rises as He/H falls; only a starting value). No tolerance changed.
**Result** (width-matched solution EW interpolated in ln s to 1.1084; 1 sigma interval from 1.0795 and 1.1373; every He/H has a converged state within 0.0058 of the observation except where noted):

| deep He/H | s (1 sigma) | F_EUV(100-911) [erg cmˆ-2 sˆ-1] | nearest converged state: s, EW - obs |
|---|---|---|---|
| 0.1 | 2.532 (2.43-2.64) | 25.45 (24.42-26.50) | 2.53, -0.0005 |
| 0.2 | 1.656 (1.60-1.71) | 16.65 (16.07-17.23) | 1.66, +0.0020 |
| 0.3 | 1.448 (1.40-1.49) | 14.56 (14.09-15.02) | 1.45, +0.0012 |
| 0.45 | 1.339 (1.30-1.38) | 13.46 (13.07-13.86) | 1.34, +0.0006 |
| 0.65 | 1.300 (1.26-1.34) | 13.07 (12.71-13.43) | 1.30, 0.0000 |
| 1.2 | 1.280 (1.25-1.31) | 12.86 (12.55-13.18) | 1.28, +0.0004 |
| 2.09 | 1.361 (1.33-1.39) | 13.68 (13.40-13.97) | 1.36, -0.0013 |
| 5 | 1.336 (1.31-1.36) | 13.43 (13.16-13.70) | 1.34, +0.0045 |
| 14 | 1.271 (1.24-1.30) | 12.77 (12.48-13.08) | 1.27, -0.0008 |
| 100 | 1.331 (1.30-1.37) | 13.38 (13.04-13.72) | 1.33, -0.0008 |
| 1000 | 1.253 (1.22-1.29) | 12.60 (12.29-12.92) | 1.25, -0.0031 |
| 1e4 | 1.143 (1.11-1.17) | 11.49 (11.20-11.79) | 1.14, -0.0031 |

GJ 1132 has F_EUV = 10.05, GJ 699 70.18. The required F_EUV exceeds the GJ 1132 value at every He/H (by at least 14 per cent, at 1e4), is nearly flat (12.8-13.7) over He/H 0.45-100, rises steeply below 0.45 (14.6 at 0.3, 16.7 at 0.2: the s = 1 EW falls below He/H 0.65, so the same EUV gives less line) and falls toward 1e4 (11.5, where the s = 1 EW has turned up, section 101). Interpolated intervals carry the observed 1 sigma only. States at He/H 0.3 and 0.2 have the sonic point inside the domain (26.2-27.7 and 25.4-27.7 R_p), the exobase above it, Kn_bulk = 0.1 at 10.9-12.6 R_p.
**Stale (2026-10-07):** the Knudsen numbers, `Kn_bulk = 0.1` radii and exobase radii in this passage were computed by `src/utils/collisional_validity.py` before its resonant H+ + H and He+ + He momentum-transfer coefficients (`md/Update_EXHALE_stage3.md` section 110). On the 102 stored LHS 1140 b states the change moves the `Kn_bulk = 0.1` radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.
**He/H 0.1.** The first attempt (cooling_to_space) stopped at k01: the exchange of the emission excess inside the warmed layer, neglected by that transfer, moves T_match by 1.003 K to first order against its bound of 1 K (0.89 K at He/H 0.2). The bound was not changed; the transfer was extended instead:
**New option `--column-band-transfer cooling_to_space_exchange`** (`conducted_heat_column_balance.py`, `solve_with_included_exchange`; default unchanged: `thin`). It is `cooling_to_space` with the exchange of the emission excess between all warmed nodes of the domain solved (fixed point on the exchange heating, the exist-

ing solve_with_exchange) instead of neglected, so it has no bound of that kind; the record carries included, the T_match shift from the exchange, heating/F and the first node. Validity as the module docstring states it (exact slab E_3 fractions for every g-point; for a two-level band weighted by eps_beta at both ends, exact for LTE bands). Tests (src/tests/conducted_heat_column, all pass): bit-identical to solve_with_exchange, shift = the history difference, shift >= 0, budget closes to 8.5e-15 (lte and nlte), an NH3 x 5 column's solved exchange is 1.38 times the first-order estimate that cooling_to_space measures (the audit measured 1.25 on the LHS 1140 b columns), the thin limit has no exchange. An NH3 x 300 column's fixed point settles slowly (30 iterations do not suffice: the increments fall like nˆ-1.5); the option's iteration cap is 100 and a non-settling solve raises, it does not return. The second attempt at He/H 0.1 with this transfer passed the column and then stopped in the model-top pinning (12 re-pinned solves left a monotone -8.7e-5 against the 1e-5 test), so --top-pressure-passes was raised to 24 (the test unchanged), as for He/H 0.5 in section 90. Results at He/H 0.1 (cooling_to_space_exchange, top-pressure passes 24, HeH0.1/xuv_x*): the EW (width matched, solution) is 0.7137 at s = 1.3, 0.9829 at 2.1, 1.0422 at 2.3, 1.0995 at 2.5 and 1.1079 at 2.53 (-0.0005 from the observation; closure converged at k = 1-4; sonic point 22.7 R_p, Kn_bulk = 0.1 at 16.4 R_p, exobase above the domain, red depth 1.279 per cent, red/blue 7.70). EW follows sˆ0.643 over 2.1-2.5. The match is s = 2.532 (1 sigma 2.43-2.64), F_EUV = 25.4, i.e. 2.5 times the GJ 1132 value. The s = 2.7 run was stopped by the operator during its first chemistry step to relieve the load of lart3 (its k00 log says so). s = 2.6 (closure converged at k = 2) gives 1.1273 (+0.0189), so the match s = 2.532 is now bracketed by two converged states (2.53 and 2.6); the upper 1-sigma end 2.636 lies just past the bracket.

**Stale (2026-10-07):** the Knudsen numbers, Kn_bulk = 0.1 radii and exobase radii in this passage were computed by src/utils/collisional_validity.py before its resonant H+ + H and He+ + He momentum-transfer coefficients (md/Update_EXHALE_stage3.md section 110). On the 102 stored LHS 1140 b states the change moves the Kn_bulk = 0.1 radius out by 1.0-5.9%, the exobase out by 0.4-5.7% and the largest Kn down by 0-16%; Kn(H2) and the sonic point do not move and no verdict changes. Not re-measured for the state quoted here.

**He/H 1000.** Run on lart4 from the s = 1 state of route_heh_high/: EW wm 0.8621 (s = 1), 1.1053 (s = 1.25, -0.0031), 1.1148 (1.26), 1.1590 (1.31); match s = 1.253, F_EUV = 12.60.

**Spectrum figures.** The overlays of figures 2 and 7 of docs/lhs1140b_exhale_vs_pwinds.tex show the model moved by the C26 fit v_wind = +2.26 km/s, as every earlier overlay of the memo does (make_memo_figures.py); the shift is for the overlay only and every EW, metric and chi^2 is computed from the unshifted spectra (checked: the numbers of phase6/observation/HeH0.65_k02_solution.json are unchanged). Figures 5 and 6 gain a right axis with F_EUV over the GJ 1132 value (= s).

**Not verified:** the column checks (thin flag, gas above p_match) at the states of this section; the grid error; the Roche-mode evaluate; the region beyond Kn = 0.1. The He/H 0.1 closure uses a different column transfer from the others (its T_match differs from the cooling_to_space value by about the exchange shift, 1 K).

## 103. The physics-and-algorithms document audited against the source (2026-10-05)

docs/EXHALE_physics_and_algorithms.tex and every docs/physics_overview/*.tex were checked statement by statement against src/ (and the src/utils/ scripts they describe) and brought to the state of sections 1-102. Corrections of what the document said the code does, each verified in the routine named:

- A Crank-Nicolson conduction stage that fails (floor, non-finite, pivot) stops the run (`conduction_report`, `error stop 1`); it is not retaken at a shorter step as the energy step is. The base conductive flux (`eq:qbase`), its bath at r(0) (`conduction_base_level_T`, the profile's temperature where a profile is in use) and its `EXHALE_resolved.out` record are stated.
- The heating efficiency eta is formed but written to no output file.
- `momentum_row_terms_of_cell` is in `RK_rhs.f90`; `sed_band_fluxes` replaces a routine name that does not exist; the molecular rows no longer carry the retired R20; Penning Q31 is Movre & Meyer (1997) in the closed form of Garcia Munoz (2025); the sweep's XUV field is one block (`xuv_field_block_cells = 0`); the Ly-alpha default is `jlya_mode = 0`.
- The stationary residual's energy row carries the carrier and stage enthalpy divergence; the certification report holds 33 entries; the PTC start is dtau0 = 1 on the marching handoff and the stationary restart, the CFL step only on the direct `EXHALE_PTC=1` route.
- New text: carried ionization stages and He 2^3S transport (rows, fluxes, simplex, certification scales), the 10.2-30 eV partition of Furlanetto & Stoever (2010) with its H2 gate and the (AJ) limitation, the elemental-flux closure and its acceptance rules, the escape boundary condition's hydrogen carrier split, the column energy balance under the conducted heat (two-way energy coupling), the restart mapper rules and `verify_restart_radii`, the element nucleus velocities of the `_adv` step, the stored-pair certificate, the width-matching velocity kernel (post-processing, Sec. 10.5), and the environment variables that select physics or continuation factors. Numbers quoted from earlier sections are labeled with their section; the q31g/(q31a+q31b) ratios were re-measured (0.054 at 1e4 K). Source comments found stale and left for their owners: the header of `viscous_conduction.f90` section 5 and the `conduction_report` message ("B3a will reject and retry"), `diffusive_photochemistry.f90` sections 2 and 6 (ion stages "not transported", "4x4 blocks"), `ionization_stage_transport.f90` "RELATION TO THE PROTON CARRIER", `low_energy_electron_degradation.f90` (parameter name), `species_table.f90` (f_sp columns 7..33), `steady_newton.f90` `set_transported_species_rows`, the `--column-band-transfer` help text of `photochem_to_lower_profile.py` (thin is flagged, not refused), and `EXHALE_transit.py` lines 97-100 (bulk velocity). PDF rebuilt, 102 pages, 0 LaTeX errors, no undefined references.

## 104. The user documents audited against the source (2026-10-05)

`docs/EXHALE_user_manual.tex`, `docs/transmission_spectrum.tex`, `README.md`, `README_HOWTO.md`, `md/input_schema.md`, `docs/lhs1140b_catalog_state.tex`, `docs/Update_EXHALE_appendix.tex` and the workspace `CLAUDE.md` were checked against `src/`, the `src/utils/` scripts and the LHS 1140 b EW tools for what sections 39-102 added or changed; every statement of a key, default, token, output field or option was read in the routine that implements it. Wrong before, corrected:

- The option-token count of the restart contract (24 in `README.md`, 23 in `README_HOWTO.md`, 20 in `docs/composition_restart_and_base_handoff.tex`): `load_IC.f90` has 26 (`opt_name`); the vocabulary is now listed in the manual, the HOWTO and `md/input_schema.md` appendix D.2.
- The conductivity of `Conduction` was still the Watson et al. (1981) atomic-hydrogen value in `README_HOWTO.md`

and md/input_schema.md K34b; it is the mixture value of section 41 (thermal_conductivity), the viscosity alone keeping Watson's (dynamic_viscosity).

- The _adv step was described on the bulk velocity (manual, transit section); it steps each element on its own nuclei velocity (section 82, post_process_adv.f90).
- EXHALE_PTC_DTAU0 also replaces the start 1.0 of the stationary restart (stationary_state_of_the_loaded_restart), not only that of EXHALE_PTC=1.
- README.md called carried ionization "H+ as a fifth carrier"; it is the three stages on their element's nucleus flux, in an atomic gas too.
- The README_HOWTO.md statement of the stationary evaluate (the measured state written back) and "the closure driver, which does not exist yet".
- The manual lacked examples 18 and 19, and docs/transmission_spectrum.tex had a wrong n2_populations signature and notebook path. Added: He 2^3S transport, Interdiffusion enthalpy flux (with the carrier_enth / stage_enth tokens), H2 double ionization, H2 neutral dissociation in the schema; the EXHALE_resolved.out records (base_conduction_*, he23s_transport, low_energy_electron_partition); the cert_reason values (marching_snapshot_s<step>, pass_snapshot_p<k>, projected_seed, option_change_seed, legacy_factor_token, mapped_seed); EXHALE_PTC_SER with its two companions, the row-term, budget and evaluation dump keys; the column energy balance options of photochem_to_lower_profile.py, the profile header keys and which of them enter the solution_id; the energy coupling, acceptance and resume rules of element_flux_closure.py; the mapper's first-cell rule and BASE_ROW_RATIO_TOL; the width-matching kernel, the EW operator and the overlay-only v_wind shift (transmission document section "Comparison with an observed line profile", manual and README for every planet). The appendix was re-measured with codesize.py (core 236 files, 109,539 code lines, 31.2 times ATES's 3,514); the catalog summary gained a section with the energy-coupled He/H and EUV scans (values read from the run files). Dated memos that stated a current state that is no longer true carry a one-line stale note: kompot_comparison, koskinen2022_model_a_comparison, composition_restart_and_base_handoff, lower_atmosphere_coupling.

Found in sources outside these files and left for their owners: section 87 above calls column 3 of tpm_He10830.txt the "instrument- and rotation-convolved" curve, and so does the step-1 comment of LHS1140b/he10830_equivalent_width.py; EXHALE_transit.py writes column 3 as T_instr (instrument only, no planet rotation) and column 4 as T_rot+instr, so the EW and the kernel act on the curve without rotation. Also stale: the cond_on comment of parameters.f90 (Watson conductivity), the docstring of element_flux_closure.py (convergence stated without the conducted-flux residual the code holds to tol), the --column-band-transfer help text (thin is flagged, not refused), the H2 neutral dissociation comment of input_read.f90 (33-41 eV; the ramp is 32-41.5 eV), and the "no element drift" header comment of EXHALE_transit.py and the transit_state_files docstring. Builds: the manual (125 pages), the transmission document (15), the catalog summary (20), the stage 3 log with the appendix and the four noted memos, all without LaTeX errors; the changed pages were rendered and inspected.

## 105. The lower-ghost chemistry of LHS 1140 b at He/H 2e4: the reservoir H2 partition solved with its own ionization, and the molecular rows normalized by the turnover of their element (2026-10-06)

Request of 2026-10-06: the He/H 2e4 rung of the LHS 1140 b He/H ascent at the fixed XUV scale s = 1.14 (GJ 1132 SED) stopped in its first closure iteration (`LHS1140b/models/.P2/ew_fit/route_xuv_scaled/HeH20000/xuv_x1p14/k00`, binary `EXHALE_c37b3671.x`) with `ioniz_eq: persistent non-root chemical equilibrium`: the two lower ghost cells (0 and -1, 900 K, n_H 4.07e8, n_He 8.13e12 cm^-3) were non-roots from the first sweep, 2000 NON-ROOT lines in `run.log`, the stop after 1000 consecutive sweeps, while the interior JFNK reached ||R|| 5e-8. He/H 1e4 had none. Two defects, both in the code, each measured before it was fixed. Scratch runs: `~/.claude/jobs/f24ea156/tmp/heh2e4_diag/`, lart4. All numbers MEASURED unless marked READ.

### 105.1 The reservoir H2 partition was pinned at the ionization of the previous state

The base handoff states x2 = 2q(1 + He/H)/(1 + q) of the NON-IONIZED hydrogen (`base_h2_nuclei_fraction`), so the ghost owes x_H2 = x2 (1 - x_ion), x_ion the hydrogen in H+, H2+, H3+ and HeH+. The sweep pinned row 4 at x2 (1 - x_ion) with the x_ion of the state it was HANDED and closed the pair by passes with a secant step. A pinned x(4) leaves 1 - x(4) of the hydrogen for H I and every ion together, and where the ionization balance asks for more ions than that, no composition satisfies the rows. At He/H 2e4 the H+ of the ghost is made by He+ + H2 -> H+ + H + He at the rate the helium is photoionized, which doubles against the hydrogen from He/H 1e4 to 2e4. At the first sweep the pin was x(4) = 0.9698 (room 0.0302); the root of the same rows with the partition solved together with the ionization has x_ion = 0.03136 (test below). The only root hybrd1 found with the pin (attempt 1, info 1, residual 2e-16) had x(H+) + x(H2) = 1.0011, negative H I, and O+/O = 1.044 (element violation 4.4e-2); every bounded iterate of the three attempts was a non-root (residual 1e-4 to 5e-3), and the retained non-root (x_ion 3.6e-4) fed the next pin x(4) = 0.991, which left less room still.

Fix (`ion_cell_state.f90`, `System_HeH_mol.f90`, `System_HeH_mol_metals.f90`, `ionization_equilibrium.f90`, `constrained_chemical_equilibrium.f90`): the row IS the equation, `fvec(4) = x(4) - x2 (1 - x_ion(x))`, with x_ion an unknown of the same solve (`x_h2_neutral_partition_fixed`, `h2_fraction_of_reservoir`, `ionized_hydrogen_nuclei_fraction`), in the fraction systems and in the constrained continuation, where H2 is now an unknown with that row instead of a held density. The neutral hydrogen left, (1 - x2)(1 - x_ion), is never negative. `impose_transported_fractions` places a starting point and a returned state on the row; the closure passes, the secant and the infeasibility stop are removed (the closure check on the returned state stays). With this alone the ghosts reached roots, but the JFNK refused 275 trial states for ghosts without a root (worst residual 5e-5, just above the 1e-6 tolerance) and stopped with no descent at ||R|| = 1.9, which is the second defect.

### 105.2 The rows were scaled by a bound that places trace ions at the whole of their element

The molecular rows were divided by the sum of the magnitudes their terms reach with every species at the whole of its element (`set_mol_turnover_rates`). Where a trace ion of one element reacts with another element, the bound

exceeds the turnover by the inverse of the ion's fraction. The H+ row carries He+ + H2 (R17) and He(2^3S) + H (Q31) with He+ and the metastable at all of the helium: at the ghost root the bound was 3.9e12 cm^-3 s^-1 against a gross H+ rate P + L of 1.3e3, so the row reached the solve and the acceptance test at 3e-10 of its weight. The Jacobian of the scaled system, its columns times |x|, had its singular value along H+ at 9.8e-11; from the converged ghost at one JFNK perturbation the Newton step asked for 12 per cent of H+, its full length raised the residual from 1.3e-5 to 1.0, and hybrd1 returned info 4 at the starting point. Ten per cent of H+ off the root moved the scaled H+ row by 2.7e-11: the tolerance 1e-6 admitted an H+ imbalance of 3e3 times its gross rate. He+, H2+, H3+ and HeH+ carry the same construction.

Fix: every row of the molecular system is divided by the CHEMICAL TURNOVER OF ITS ELEMENT, G_E = sum over the rows of element E of (P_j + L_j) (hydrogen: H+, H2, H2+, H3+, HeH+; helium: He+, He++, He 2^3S; each metal its two stage rows, oxygen with OH and H2O). The gross rates come from the row routines themselves (`mol_heh_rows`, `oxygen_carrier_rows`, `metal_rows`, `cx_add_to_fvec`, `he_h_cx_fvec`, optional argument `gross`); `element_turnover_reciprocal` and `scale_molecular_rows` (`System_HeH_mol.f90`) and `set_element_turnover_scales` (`System_HeH_mol_metals.f90`) apply them. The scale is taken at the starting point of each solve and HELD through it, and a candidate is judged with the scale taken at the candidate (`normalized_reaction_residual`, which no longer widens the molecular rows by a separate charge-exchange bound: the gross rates contain it). A row with no reaction keeps the old bound (1 for an identity row). The zeros of the rows are unchanged. Two variants were measured and rejected: (i) each row by its own P + L asks a species at 1e-22 of its element (H3+ in the outer wind) to balance to 1e-6 of itself, and 350 cells of the He/H 2e4 run were left without a root (hybrd1 info 1 with H2+, H3+, HeH+ rows at 1e-6 to 5e-1); (ii) the element turnover re-taken at every iterate bounds every row by one and leaves no gradient back from an iterate far outside the simplex: on the cold start of the `mol_metals` regression gate 214 cells were left without a root, attempts went out to fractions of 2e5, and the ghost fixed point stopped the run. The constrained continuation keeps the bound on its own path; its candidate is judged by the same measure as every other. The rate matrix of the decay-rate diagnostic now asks the systems for their rows as volume rates (`mol_rows_as_volume_rates`) instead of undoing a scale.

Also fixed: `cce_probe.x` stopped at once (`keq_H_H_to_H2 called before h2_thermochemistry_init`); the driver now builds the table first. The two charge-exchange reservoir tests that undid the scale by setting `mol_inv_turnover = 1` ask for volume rates instead. The cell-state hash of `attempted_step.f90` covers the two new fields. Documents: the normalization and the reservoir row are written into `docs/physics_overview/ionization.tex` and `molecules_metals.tex`; `docs/EXHALE_physics_and_algorithms.pdf` rebuilt.

### 105.3 Tests

New suite `src/tests/ghost_reservoir_chemistry/` (fixture: ghost 0 of the failing run at its first sweep, from its `EXHALE_CCE_DUMP`): the reservoir row is x(4) - x2 (1 - x_ion) to the bit; from the state the earlier code retained, hybrd1 reaches a root inside every element budget at 4.8e-10 (tolerance 1e-6); that root has x_ion = 0.03136 against the 0.03020 the pin left; ten per cent of H+ off the root moves the H+ row by 5.4e-2 (bound 3.9e12, gross 1.3e3, G_H 1.87e3 cm^-3 s^-1); the hydrogen rows share one divisor; the continuation solves H2 as an unknown, its layout is square, and its H2 row is the reservoir row. GREEN: 9 of 9. RED, the same assertions on the source of `EXHALE_c37b3671.x`

with the pin at 0.9698: hybrd1 info 4, residual 2.8e-3, outside the simplex, x_ion 2.9e-3, and the H+ row 2.7e-11 at ten per cent of H+: 4 FAIL. Suites run with the changed objects: `ghost_reservoir_chemistry` 9/9 (6/6 before the continuation rows were added), `charge_exchange_rows` 257/257 (22 FAIL before the two tests asked for volume rates), `ionization_imposed_fractions` 48/48, `constrained_network_layout` 8/8, `stage_row_balance` 10/10, `charge_exchange_detailed_balance` 298/298, `physics_probe` 2499/2499, `acceptance_classes` 22/22, `boundary_state` 43/43, `molecular_seed` 29/29, `attempted_step` 70/70, `carrier_returned_state_acceptance` 36/36, `carrier_retry` 156/156, `helium_metastable_transport` 24/24, `carrier_constraint_attribution` 10/10, `carrier_reference_scales` 16/16, `carrier_helium_inventory` 15/15, `carrier_boundary_jacobian` 12/12, and `grid_and_gates` 424 PASS, 3 FAIL: `stationary_evaluate_rebuilt_within_sweep_budget` (5.832e-5), `stationary_evaluate_certification_unchanged` and `stationary_evaluate_exit_status` (2) on the `wasp_full_newton` fixture, the same three with the same numbers from the source of `EXHALE_c37b3671.x`, so not of this change (an atomic fixture no longer certified by the current code; reported, not touched).

### 105.4 The actual cases

New frozen binary `LHS1140b/models/.P2/EXHALE_f6945afb.x` (ifx 2026.1.0, md5 f6945afb87281affb90523b0fbb3823d, manifest `.P2/BINARY_MANIFEST_f6945afb8728.txt`, source copy `.P2/build_ghostres_ifx/`). The He/H 2e4 k00 EXHALE stage rerun in scratch with the closure's environment (`closure.json` exhale_env, 12 threads): 0 NON-ROOT lines, 0 constrained-continuation rescues, 0 refused trials, 216 JFNK iterations, `the stationary solve returned info = 0`, CERTIFIED in the wind, 2.3 minutes. The old binary on the same input: 128 NON-ROOT lines in its first 90 s. The converged ghost 0: x(H2) = 0.9394, x(H+) = 0.0534, n_e 2.51e7 cmˆ-3, He++/He 3.0e-14 (the retained non-root had He++/He 2.6e-7 > He+/He 4.2e-8). log10 Mdot from the written profile, median over r > 1.2 R_p: 7.883 (He/H 1e4 k01: 7.875).

He/H 1e4 (`HeH10000/xuv_x1p14/k01` input and IC), `EXHALE_c37b3671.x` against `EXHALE_f6945afb.x`, as written (stationary evaluate) and as a stationary solve (both converge in one JFNK iteration, info = 0, no NON-ROOT): the ghost compositions of every evaluation (`EXHALE_GHOST_RECORD`) agree to 1e-14 relative in n_e and every fraction, `Hydro_ioniz.txt` to 1.9e-14 and `Ion_species.txt` to 8.2e-13. The accepted reaction residual of the ghosts reads 1.4e-17 - 3.8e-17 under the old normalization and 1.6e-15 - 7.2e-15 under the new one, the same roots measured in the new units.

Scoped impact on two regression cases (not the matrix, no golden touched), gfortran builds of the earlier and the new source, 12000 steps, single-threaded: `lower_profile` bit-identical in `Hydro_ioniz`, `Ion_species` and both `_adv` files; `mol_metals` (hot-Uranus gate with trace metals, molecular) within 7.0e-8 (`Hydro_ioniz.txt`), 1.1e-7 (`Ion_species.txt`) and 3.1e-6 (`Hydro_ioniz_adv.txt`) relative, log10 Mdot 10.58 in both, no non-root in either, and the 13 constrained-continuation rescues of the earlier normalization are 0. Marching speed of `mol_metals`, measured over the same four minutes at different steps: 115 (new) against 114 (earlier) steps per minute.

Open: the atomic systems' acceptance scale (`normalized_reaction_residual`, atomic branch) carries the same construction in its H+ row, the He <-> H charge exchange with He+ at all of the helium, (k1 + k2) n_H n_He; it is not changed here (the atomic systems are not solved at He/H 2e4 in this route).

## 106. A failed conduction stage is a refused and retaken step; the handoff-free base-level conflict warns; stale source comments corrected (2026-10-06)

User decisions of 2026-10-05 (items A and B) and the source comments the two documentation audits of sections 103 and 104 left for their owners (item C). Scratch runs: `~/.claude/jobs/f24ea156/tmp/condretry/`, lart4. Numbers MEASURED unless marked READ.

### 106.1 A failed Crank-Nicolson stage refuses the attempted step

Before: `conduction_report` (`viscous_conduction.f90`) ended a failed stage (floor, non-finite input or solution, pivot breakdown) with `error stop 1`; the header (section 5) and the message said "B3a will reject and retry", which the marching loop never did. Now the stage prints its diagnostics, leaves its verdict in `conduction_last_status` and the `status` argument and returns no state; `EXHALE_main` passes `status`, and on a failure refuses the attempt with `as_reject_conduction` at `as_op_conduction` and leaves the trial, exactly as a failed energy update does (both run modes). The controller then restores the checkpoint and retakes the step at half dt (`attempted_step_reduced_dt`) within its budget of `n_step_retry_max` = 8 halvings; at dt/2^8 `stop_on_exhausted_retry_budget` stops the run with exit status 2 and the retry history. The refusal text names the stage's reason and cell. `conduction_stop_on_failure` is removed (only the unit driver set it). Why a shorter step is the remedy (header section 5): a mode of the conduction operator with decay rate lambda/dt is multiplied by (1 - lambda/2)/(1 + lambda/2), negative for lambda > 2, so a step longer than twice the diffusion time of a cell overshoots; halving dt halves lambda. The two remaining `error stop` statements of the module are not stage failures and stay: an `EXHALE_CONDUCTION_SCALE` outside [0, 1] (input) and `h2_conductivity` asked for at a non-positive or non-finite temperature (an argument guard also reached from the stationary residual; the stage's entry test covers every cell its stencil reads).

Test hook, default off: `EXHALE_CONDUCTION_REFUSE_DT_ABOVE = t` [s] makes the stage return the new status `CONDUCTION_INJECTED` (reason `CONDUCTION_REASON_INJECTED`) for every attempt whose longest cell interval exceeds t, and print that interval. New suite `src/tests/conduction_retry` (hot-Uranus gate `mol_carrier` with `Conduction: True`, 1 or 2 steps, 40 s), registered in `executable_identity`: (1) t = 1e-300: 9 refusals at the conduction step, every interval exactly half the one before (worst |ratio - 1/2| = 0), the exhaustion report naming the stage, exit 2; (2) t = 0.3 of the first interval (1.684 s): two refusals, the third attempt (1/4) accepted, exit 0; (3) t = 1e300: no refusal, `Hydro_ioniz.txt` and `Ion_species.txt` byte-identical to the run without the hook. 20 PASS with the tree build; with the control build (the tree before these edits) the rows of (1) and (2) FAIL (no refusal, exit 0). Unit level: `energy_update` `conduction_floor_tests` no longer sets the flag; with the new objects all 69 rows pass, with the control objects the floor case ends in `ERROR STOP 1`.

Byte identity where no stage fails: `mol_carrier` + `Conduction: True`, 300 steps, single thread, control build against the tree build: `Hydro_ioniz`, `Ion_species`, both `_adv` files, the heating and cooling breakdowns and the body of `conserved_state.txt` identical (its `PAIR_CHECKSUM` covers the provenance line). `attempted_step` with its whole-binary rows: 160 PASS, 4 FAIL with both builds, the same rows and values (`exhaustion_on_the_integration_error_*`, `temporal_order_of_the_transported_species_fractions` 1.9978/1.4453, `temporal_order_of_the_electron_fraction_and_T` 3.4109), so they predate this change; not in-

vestigated here.

Docs: hydrodynamics.tex (the stage paragraph), numerics.tex (the refused step statement, now naming the energy and conduction refusals in both modes), src/tests/energy_update/README.md, the EXHALE_main comment on the floor counters.

### 106.2 Base BC: pressure beside a density key without a handoff

input_read.f90: with no base.inp p_base and no profile, Base BC: pressure beside Log10 lower boundary number density is still followed, now with a WARNING naming both levels (n0 and pressure of each) and the one used; the overridden n0 is kept in base_density_key_overridden_n0 and write_setup_report.f90 repeats the warning under "Base level". MEASURED on the LHS 1140 b atomic input of .P1/fiducial_conduction/ew_fid_atomic_nocond, which carries both keys: density key n0 = 3.2063e13 cmˆ-3 (1.0004e-6 bar) against the pressure's 3.2049e13 (1.0000e-6 bar); outputs of a 1-step run byte-identical to the control build, which printed no warning. Docs: introduction.tex (base-level paragraph), the manual's Base BC: entry, md/input_schema.md rows 2 and K32.

### 106.3 Stale source comments (each checked in the code first)

- diffusive_photochemistry.f90 header: the transported set is the run's (carrier_set_init, up to 8 rows: H2, OH, H2O, CO; H+, He+, He++; He 2ˆ3S); the carried stages are charged and their ambipolar drift is neglected, not absent; blocks are 8x8 (n_carrier_max); a handoff's base H2 partition is imposed in the ghosts (section 5 said nothing was).
- ionization_stage_transport.f90: the carrier stage rows ARE formed from its equation (1) (carrier_stage_face_flux).
- low_energy_electron_degradation.f90: the switch is use_low_energy_partition.
- species_table.f90: metal columns 7..33.
- util_ion_eq.f90 HeITR_coeffs: the n >= 3 singlet share is 0.019 of the direct q31a + q31b at 1e4 K and 0.079 at 2e4 K, evaluated from the coded fits (the comment said 0.023 and 0.09); the cascade share of q13 quoted there (0.10 and 0.33) was checked: 0.107 and 0.331.
- ion_residual_core.f90 heh_tr_rows: the rows carry the electron-impact ionization of H I, He I (both levels) and He II.
- steady_newton.f90 set_transported_species_rows: the rows are the elements, then every solved carrier (stages and He 2ˆ3S included), carriers only in a molecular run with carrier transport.
- photochem_to_lower_profile.py: thin is flagged past 10 per cent, not refused.
- parameters.f90 cond_on: the mixture conductivity.
- element_flux_closure.py docstring: convergence with energy coupling also holds the conducted-flux residual to tol.
- The H2 neutral dissociation window is 32-41.5 eV (input_read.f90 comment and its log line, parameters.f90, util_ion_eq.f90, h2_photo_channels.f90 header).

- `EXHALE_transit.py`, `exhale_transit_lib.py`: the `_adv` step marches each element on its own nuclei velocity (section 82); the J_lya estimate of Huang et al. (2017) is the order-of-magnitude estimate in the text after their Eq. (6), Section 2, page 3 (READ in the published PDF, `references/Huang_2017_ApJ_851_150.pdf`; Eq. 6 is the n(2l)/n(1s) ratio), and the recombination assumption is on page 3, not 11.
- `lya_rt.f90`, `excited_hydrogen.f90`: the 1/(1 + tau) attenuation and the column-integrated F_LyC are EXHALE's closures, not the paper's. Also `README_HOWTO.md`: 54 suites (adds `conduction_retry` and `ghost_reservoir_chemistry`).

#### 106.4 The atomic acceptance scale (measured, not changed)

`normalized_reaction_residual`, atomic branch, on the certified LHS 1140 b atomic state `ew_fid_atomic_nocond` (He/H 9.7, He 2^3S on), rows recorded with `EXHALE_STAGE_CHANNELS` at eight cells: the H+ row scale is 3.9e12 (cell 2) to 2.1e2 (cell 495, 26 R_p) times the gross H+ rate, and the term that sets it is the Penning bound f Q31 n_He n_H (He 2^3S at all of the helium), 0.99 to 1.00 of the scale; the He <-> H charge exchange term that section 105 named is 1.6e-6 to 2.4e-6 of it. The He 2^3S row scale is 1.4e14 to 3.9e3 times its gross rate. With `ieq_res_tol` = 1e-6 the test cannot refuse a non-root in these rows inside about 1.35 R_p. The roots on this state are tight (row residual 1e-16 of the gross rate). Recorded as `md/TO_BE_DONE.md` (AM).
Build: bare `make`, tree `EXHALE.x` md5 `3042c8726291e0b02d796c3325fa95cf`. Documents rebuilt without LaTeX errors: the physics document (103 pages; the changed pages 4, 12 and 62 inspected) and the manual (125).

## 107. The helium carry of `map_state_to_grid.py --reservoir` holds the hydrogen-nucleus count of every row (2026-10-06)

The elemental-flux closure of the LHS 1140 b He/H ascent stopped at the He/H 7e4 rung: the seed `route_xuv_scaled/HeH40000/xuv_x1p14/k01/output` (He/H 3.9974e4) carried onto He/H 69936.32, C/H 2.758973e-4, N/H 8.133062e-5, O/H 6.023026e-4 was refused, "C/H ... 2.758969E-04 ... instead of 2.758973E-04 (relative departure 1.52E-06, above the tolerance 1E-06)".

#### 107.1 Cause

Not the file precision: the mapper writes every column at `%24.16E` and the check runs on its own float64 arrays before writing. The departure was a real inconsistency of the helium carry. `--reservoir He/H` multiplies every helium column by k = new/old, HeH+ included (`element_stage_columns`), and HeH+ holds one hydrogen nucleus, so every row gained (k - 1) n(HeH+) hydrogen nuclei and every El/H of the row moved by -(k - 1) n(HeH+)/n_H: the metals that the same call carried by their own factor new/old, and He/H itself. The seed holds HeH+/H = 2.024e-6 at cell 1 (6.0e-7 and 6.2e-7 in the two lower ghosts; MEASURED from its `Ion_species.txt`), and the departure followed (k - 1) x 2.024e-6 exactly (MEASURED with the entry mapper):

| target He/H | k - 1 | departure (C/H, cell 1) |
|---|---|---|
| 4.5e4 | 0.126 | passed |

| target He/H | k - 1 | departure (C/H, cell 1) |
|---|---|---|
| 5e4 | 0.251 | passed |
| 6e4 | 0.501 | 1.01e-6 |
| 69936.32 | 0.750 | 1.52e-6 |
| 1e5 | 1.502 | 3.04e-6 |

Why the 2e4 and 4e4 rungs passed was not measured (their seeds' HeH+/H at cell 1 was not read).

### 107.2 Fix (`src/utils/map_state_to_grid.py`)

The hydrogen-nucleus count of every row is held. After the columns of an element are multiplied, the (k - 1) Hs hydrogen nuclei they brought (Hs the hydrogen nuclei of the multiplied columns: HeH+ for helium, none for a metal) are taken out of the species holding only hydrogen (H I, H II, H2, H2+, H3+), all multiplied by one factor g = 1 - (k - 1) Hs/P, P their hydrogen nuclei, which keeps their ionization and molecular split; their mass (1 per hydrogen nucleus, `bsp_mass`) and particles plus electrons (H I 1, H II 2, H2 1, H2+ 2, H3+ 2) enter the density and the temperature written at fixed pressure. Every row then carries k times its source He/H and its source ratio of every other element exactly, in whatever order the elements are named. The `--uniform` row factor becomes (value n_H - X)/S with n_H held. A HeH+ carry that the hydrogen-only species cannot supply (g <= 0) is refused. The base-row test and its tolerance are unchanged (`BASE_ROW_RATIO_TOL` = 1e-6, `heh_dev_tol` of `load_IC.f90`); it now catches only the oxygen carriers OH, H2O and CO, which hold nuclei of two elements and are not multiplied. Before this, a HeH+-bearing state was refused by design ("no single factor sets both counts"); it is now carried.

### 107.3 Tests (`src/tests/state_mapper/`)

The two assertions that a HeH+ state is refused are replaced by the carry with n_H held (HeH+/H 0.05 and 0.3: n_H equal to the control mapping to 4e-16, HeH+ multiplied by k, base rows at the new He/H to 2e-16, the diffused profile shape and the H II/H I ratio kept, T = p/((n_tot + n_e) k_B) and rho the species mass sum to 3e-15), a refusal when the hydrogen-only species cannot supply the hydrogen (HeH+/H 0.3 carried by 6/1.6), the `--uniform` carry with HeH+, and a synthetic LHS 1140 b state (He/H 3.9974e4, HeH+/H 2.024e-6, C/H, O/H) carried to He/H 69936.32 and 1e5 together with C/H and O/H. RED with the entry mapper: 5 failures (the two LHS carries refused at 1.52e-6 and 3.04e-6, the two HeH+ carries refused, n_H not held under `--uniform`, 0.29). GREEN after: 131 PASS, 0 failures. The suites that call the mapper without `--reservoir` (`boundary_state`, `coupled_block_jacobian`, `grid_and_gates`) were not run: the change lies inside the `--reservoir` branch only, and the suite's own no-option byte-identity assertions pass.

### 107.4 The real seeds (MEASURED)

The closure's command (the seed `output` of `HeH40000/xuv_x1p14/k01` and of `k00`, onto the target grid header of `HeH70000/xuv_x1p14/k00`, C/H, N/H, O/H as in the closure log) for He/H 69936.32 and 1e5 now writes the pair; recomputed from the written `Ion_species_IC.txt` the first-cell departures are, for k01: C/H 1.3e-14, N/H 5e-15, O/H 5e-15, He/H 8.6e-13 (base rows); for k00: C/H 1.9e-14, N/H 2.7e-14, O/H 1.7e-14, He/H 3.2e-12; the same at

1e5. The frozen binary `LHS1140b/models/.P2/EXHALE_f6945afb.x`, run in scratch on the He/H 7e4 pairs of both seeds with the rung's `input_template.inp` and the k00 `lower_atmosphere_profile.dat`, loads them: no restart-contract or He/H refusal, "base cells set to the input He/H = 6.9936E+04", C/H, O/H and N/H "column rescaled by 1.000000E+00". The 1e5 pair was not loaded: no profile at He/H 1e5 exists, and a copy of the 7e4 profile with X_He multiplied is refused by `input_read` ("the base H2 fraction is not attainable at this He/H").
Noticed, not changed: `input_read.f90:3159` and `:3309` print He/H with `F8.5`, which shows `********` at He/H >= 1e3 (the LHS 1140 b ascent) and `q_H2(base) -> 0.00001` for 7.1e-6.
Documents: `docs/physics_overview/boundaries_initial.tex` (the mapper paragraph) and `docs/EXHALE_user_manual.tex` (the mapper paragraph and the closure paragraph) state the held n_H; rebuilt without LaTeX errors, the physics document (103 pages; page 74 inspected) and the manual (125; pages 10 and 83 inspected).

## 108. The EUV that matches the observed EW reaches He/H 1e5 (2026-10-06)

The He/H ascent of section 102 above 1e4 at fixed s = 1.14 (the s that matched at 1e4), each rung seeded from the converged one: 2e4 and 4e4 converged at k = 1, 7e4 and 1e5 at k = 2 (max residual 0.0095, 0.0125, 0.0071, 0.0045; no non-root cell). It could be run only after two code defects were removed: the lower ghost chemistry at 2e4 (section 105) and the mapper's helium carry at 7e4 (section 107); the closures use the binary `EXHALE_f6945afb.x`. The width-matched solution EW at s = 1.14 is 1.1285 (2e4), 1.1607 (4e4), 1.2083 (7e4), 1.2425 (1e5) against 1.1084 observed: the s = 1 EW has turned up above 1e4, so the matching s falls. At He/H 1e5, s = 1.06 gives 1.1575, s = 1.03 gives 1.1247 (EW proportional to s^1.0 here) and the crossing at 1.015 was confirmed by a converged state: s = 1.015 gives 1.1082 (-0.0002; closure k = 1, residual 0.0108). F_EUV(100-911) = 10.20 (1.5 per cent above the unscaled GJ 1132 value 10.05); the 1-sigma interval of s is 0.989-1.042 (lower end extrapolated). The He/H = 0.1 crossing was confirmed too: s = 2.53 gives 1.1079 (-0.0005) and s = 2.6 gives 1.1273, so the table has no extrapolated point left (He/H 1e3 at s = 1.25: 1.1053). Figure 6 of `docs/lhs1140b_exhale_vs_pwinds.tex` now has thirteen He/H, 0.1 to 1e5. The EW of states above He/H 1e4 was evaluated with f6945afb (`measure_case_f6945afb.sh`, `run_ew_f6945afb.sh`, copies of the control scripts with the binary name changed). Record: `NOTES.md` section 7 of `LHS1140b/models/.P2/ew_fit/route_xuv_scaled/`.

## 109. The equivalent width and the width-matching kernel read the planet-rotation column (user decision 2026-10-06)

`tpm_He10830.txt` carries four columns: `T_theo`, `T_instr` (instrument profile only) and `T_rot+instr` (the tidally locked planet's rotation, an azimuthal disk average, and the instrument profile). The EW operator, the width-matching kernel and the memo figures read column 3 (`T_instr`); a literature check (Huang et al. 2023 section 4.3; Taylor et al. 2025 and 2026; Biassoni et al. 2024 section 2, whose transit tool is the ancestor of `EXHALE_transit.py`) shows that published He and metal-line transit models do add the planet's rotation as a computed broadening, and keep any remaining width shortfall as a separate non-thermal term. The convention of CLAUDE.md ("Line width ... every planet") therefore now reads column 4: the rotation is a computed term of the model and the Gaussian velocity kernel

stands only for the velocity field the one-dimensional model does not contain.

Code: `LHS1140b/he10830_equivalent_width.py` defines `TRANSIT_COLUMN = 3` (0-based) and uses it in `transit_file_excess` and `transit_file_air_excess` (and so in the EW, `--width-matched` and `ew_of_state.sh`); `LHS1140b/make_memo_figures.py` `curve_file` reads the same column; `EXHALE_transit.py` fits the `tpm_He10830_metrics.txt` three-Gaussian metrics on the rotation-and-instrument curve as well (it fitted the instrument-only curve). All 1560 stored `tpm_He10830.txt` files have four columns. Tests `transit_state` and `transit_census` pass.

Effect, MEASURED: on the converged He/H 1e5, s = 1.015 state the width-matched EW is 1.10816 (column 3: 1.1082) and f = 21.4743 km/s (21.4803); at He/H 0.1, s = 2.53 the unbroadened EW and the kernel differ by 1e-6 %A and 0.01 km/s (LHS 1140 b: v_rot = 0.03 km/s). Every EW and every figure of this memo series is therefore unchanged at the printed precision; they were computed from column 3 and are not recomputed. For a planet with a fast spin the two columns differ materially and column 4 is the one to use. The documents that described the column (`CLAUDE.md`, README, README_HOWTO, user manual, `transmission_spectrum.tex`, `physics_overview/postprocessing.tex`, the LHS 1140 b memo) are updated.

## 110. Resonant H+/H and He+/He coefficients in the collisional diagnostic; HITRAN band coefficients and two comments of the conducted-heat column corrected (2026-10-07)

Two defects found by the 2026-10-06 audits, each checked against the cited source before the change.

**1. `src/utils/collisional_validity.py`: resonant charge exchange.** The tool computed the ion-parent-atom pairs H+ + H and He+ + He with the non-resonant polarization + rigid-core coefficient of the element operator. That is physically wrong for these pairs: their momentum transfer is set by resonant charge exchange, and the non-resonant form gives 0.52 (H+ + H) and 0.68 (He+ + He) of the published values at 1e4 K (MEASURED from both formulas; 0.70-0.52 and 0.77-0.67 over 500-2e4 K). The published coefficients now REPLACE the non-resonant one for these two pairs (`RESONANT_PAIRS`, `resonant_charge_exchange_rate_coefficient`, dispatched in `momentum_transfer_frequency`), in the tool's convention `nu_st = n_t K`:

- H+ + H: Glassgold, Krstic & Schultz (2005, ApJ 621, 808; publisher PDF via ADS, read), eq. (3.19), `K = 2.13e-9 (v_rms/km s^-1)^0.75 cm^3 s^-1`, with `v_rms = (vbar^2 + w^2)^(1/2)` (their eq. 3.15) and `vbar = (8 a^2/pi)^(1/2)`, `a = (kT/m)^(1/2)` with m the reduced mass m_H/2 (their `vbar(100 K) = 2.05 km/s` confirms it); their drag force `f_d = n_j n_k m_jk w_jk K_jk` (eq. 3.1) is the tool's convention. Accuracy 10-15% for v_rms > 1 km/s (their section 3.3). The drift w is not known to the tool; w = 0 is used, the smallest K.
- He+ + He: Schunk & Nagy (1980, Rev. Geophys. Space Phys. 18, 813; `references/Schunk_1980RGSP_18_813.pdf`), Table 5, `nu_in = 8.73e-11 n(He) T_r^(1/2) (1 - 0.093 log10 T_r)^2 s^-1`, T_r > 50 K, with their eq. (19) convention `n_s m_s nu_st = n_t m_t nu_ts`, so `K = 2 nu_in/n(He)` (conversion of `md/AK_ambipolar_stage_drift_derivation_20261006.md` section 6.1). The K values reproduce that note's Table 6.4 to its digits (H+ + H 6.67e-9, 2.05e-8; He+ + He 2.19e-9, 6.89e-9 at 500 and 1e4 K).

H2+ + H2 keeps the non-resonant coefficient (no published resonant value checked); the header and `md/collisional_validity.md` (section 1 table, the omitted-channel paragraph) say so.

Effect, MEASURED: on `route_xuv_scaled/HeH0.65/xuv_x1p3/k01` (as `measure_case.sh` calls it) the `Kn_bulk = 0.1` radius moves from 8.779 to 8.883 R_p, `Kn_bulk` at the top from 0.993 to 0.963, the largest Kn of the critical region from 0.982 to 0.944 (H I 0.98 to 0.94); the sonic point (26.196 R_p), the exobase (above the 29.03 R_p boundary), the electron-ion energy-coupling ratio (0.0216) and the verdict do not move. Re-run on all 102 stored LHS 1140 b states of `LHS1140b/models/.P2/ew_fit/` that have a `collisional*.txt` and their run directory (output to scratch; the stored files were not regenerated; 9 `compare/` records whose run directory is gone were skipped): `Kn_bulk = 0.1` radius +1.0 to +5.9%, exobase (72 states with one in the domain) +0.4 to +5.7%, largest Kn of the critical region 0 to -16%; no verdict changes (every state stays above 0.1); two exobases at 28.9-29.0 R_p move above the outer boundary (`GJ699split_HeH5.0`, `HeH0.65/xuv_x1p25`). The largest changes are on the most ionized states (GJ 699 spectrum, He/H >= 1e3). On the ionized regression case `backup/regression/wasp_full_newton` (WASP-121 b) the largest Kn falls 31% (2.58e-5 to 1.78e-5).

Documents quoting this tool's numbers are marked stale in place (a "Stale (2026-10-07)" paragraph after each quoting passage; values left as they were, not re-measured for the states quoted): `md/collisional_validity.md` (section 4 header), `md/Update_EXHALE_stage1.md` (3), `md/Update_EXHALE_stage2.md` (2), this log (22 passages between sections 2 and 104), `md/lhs1140b_stationary_L4f`, `_L8` (2), `_L10`, `_L12b_derivation`, `_L22`, `_L36`, `_L36e`, `md/TO_BE_DONE.md`, `md/PLAN_20260923_rev2.md` (2), `md/lhs1140b_p14_outer_closure_20260923.md`, `md/session_handoff_20260930.md` (2), `md/PLAN_20260930_rev1/rev2/rev3.md`, `md/lhs1140b_width_nonthermal_candidates.md`, `md/PLAN_20261003_lhs1140b_ew.md` (5), `md/PLAN_20261004_lhs1140b_heh_high_and_xuv.md`, `md/lhs1140b_ew_codex_review_20261003.md`, `md/LHS1140b_NH3_CH4_EW_REVIEW_20261004.md`, `docs/lhs1140b_catalog_state.tex` (1) and `docs/lhs1140b_exhale_vs_pwinds.tex` (15). The typeset twins `docs/Update_EXHALE_stage<N>.tex` were not edited (generated). Not re-typeset: the two `.tex` documents above.

**2. `src/utils/conducted_heat_column_balance.py`.**

(a) The comment "NH3 relaxes 16 times faster with He than CD4 does" is wrong against the cited tables: Hovis & Moore (1980, J. Chem. Phys. 72, 2397) Table I, NH3(nu2) + He 1.1e-13 (198 K) and 2.8e-13 (293 K), against Siddles, Wilson & Simpson (1994, Chem. Phys. 188, 99) Table 1, CD4(nu2, nu4) + 4He 7.5e-15 (190 K) and 2.8e-14 (295 K) (both read in `references/`): ratios 10.0 (293/295 K) and 14.7 (198/190 K); with the code's Landau-Teller NH3 rate at the CD4 temperatures 13.4, 12.6, 10.1 at 190, 240, 295 K (MEASURED). The comment now says 10 to 15 and gives the four numbers.

(b) The two CH4 A10 values at 185 K (1.96 s^-1 in `NLTE_BANDS`, 2.16 s^-1 in `TWO_LEVEL_CHECK`) are different quantities: the first is the average over the dyad population (nu4 and nu2 levels), the second the nu4 level alone (the check's formula uses g1 = 3 and nu0 = 1306). Recomputing both from the HITRAN line list (hitran.org line-by-line API, retrieved 2026-10-07, main isotopologue; the selected line sets reproduce the code's band intensities, nu4 5.141e-18 and NH3 nu2 2.245e-17) gives A(nu4) 2.25 and A(dyad) 2.04 s^-1 at 185 K, i.e. both code values were 4% low, and the dyad table 4-10% low over 150-700 K. The published value is Yelle (1991, ApJ 383, 380) Table 2, A(nu4) = 2.12 s^-1, 6% below the line sum; the band-intensity relation `A = 8 pi c nu0^2 S/(3 x abundance)` gives 2.23 s^-1 at 296

K, agreeing with the line sum. The same recomputation for NH3 nu2 gives 14.64, 14.69, 14.85, 15.11, 15.36 s^-1 at 150, 185, 296, 500, 700 K against the code's 13.46, 13.21, 12.56, 11.89, 11.37 (9-26% low, with the wrong temperature trend); published 15 s^-1 (Weaver & Mumma 1984, ApJ 276, 782). New values (MEASURED): CH4 dyad 2.11, 2.04, 1.86, 1.68, 1.59 s^-1; NH3 nu2 as above; `TWO_LEVEL_CHECK` A10 2.25 (CH4 nu4, labeled as the nu4 level alone) and 14.69 (NH3). The method and the published comparisons are in the comments. How the earlier table was produced is not recorded, so the origin of its lower values is not identified. The non-LTE efficiency eps = C10/(C10 + A10) falls by (C10 + A_old)/(C10 + A_new): estimated with He 0.35 and H2 0.6 of the particles at 415 and 645 K, 0.1-1.2% at 1e-5 to 2e-6 bar (C10/A10 8.7-97) and 4.5-13% at 1e-7 bar (C10/A10 0.4-1.0). The change to T_match of the stored closure states was NOT measured (it needs the Photochem closure re-run); `conducted_heat_column.txt` files of earlier states carry the old values. `md/AI_stage1_inventory_20261006.md` section 4.1 quotes the old A10 (12.56, 1.77 s^-1 at 296 K, "READ from code") and was not edited (another worker's file).

(c) The rotational-LTE argument compared the NH3 rotation-inversion A values (<= 4.8 s^-1) with the HITRAN pressure-broadening rate (1.4e4 s^-1 at 1 microbar), as if broadening were rotational population relaxation. It is not: broadening counts every phase-interrupting collision. The module header and `ROTATIONAL_LTE_NOTE` now state that rotational LTE is an estimate, give the critical density of about 5e10 cm^-3 from NH3-H2 state-to-state rates of about 1e-10 cm^3 s^-1 (Demes et al. 2023) against about 2e13 cm^-3 at 1 microbar and 400 K, and point to `md/AI_stage1_inventory_20261006.md` section 4 for the excitation data now in hand. `ROTATIONAL_LTE_NOTE` is not written to any output (grep).

**Tests.** `src/tests/conducted_heat_column/run.sh` (the only suite that imports the module; `photochem_to_lower_profile.py` imports it but has no suite here): 77 PASS before and 77 PASS after (MEASURED). Values that move with the A10 tables, before -> after: the `nlte_partial` warming of the test column 195.12 -> 198.56 K (+1.8%); the water-analog T_match 362.00 -> 364.75 K (its reference bracket moves with it); the two-level-with-published-rates margin below LTE 15.37 -> 17.09; the included-exchange record (nlte) 0.4655 -> 0.4599; the `lte` (151.99 K) and `rotational` (370.11 K) warmings are unchanged, as they must be. `collisional_validity.py` has no suite; it was checked by the K values against section 6.4 of the derivation note and by the before/after runs above. No Fortran file was touched; no build.

## 111. The recomputed NH3/CH4 band coefficients barely move the EW-matched state (2026-10-07)

The A10 tables of `conducted_heat_column_balance.py` were recomputed from HITRAN in section 110 (NH3 nu2 9-26% and CH4 dyad 4-10% higher than the old tables). Every energy-coupled state before it was computed with the old tables. MEASURED on the route-B EW-matched state He/H 0.65, s = 1.3 (`LHS1140b/models/.P2/ew_fit/route_xuv_scaled/HeH0.65/xuv_x1p3`), rerun with the same closure configuration and binary (c37b3671), seeded from its converged k01, only the tables changed (`HeH0.65/xuv_x1p3_A10recomputed`, closure converged at k = 1, max residual 9.3e-5): T_match 414.864 -> 416.048 K (+1.18 K), conducted flux into the column 25.654 -> 25.627 erg cm^-2 s^-1 (-0.1%), elemental fluxes F_H +0.010%, F_He +0.020%, log Mdot 8.240 unchanged; width-matched EW 1.1084 -> 1.1086 %A (+0.0002, against the 0.0058 nu-

merical target), unbroadened EW 1.1463 -> 1.1465, kernel f 22.1985 -> 22.1902 km/s, red/blue 7.601 unchanged. The earlier energy-coupled results (the He/H scans, the EUV match of sections 102 and 108) are therefore not recomputed; their numbers stand within the quoted precision.

## 112. The acceptance of the ionization sweep measures every kinetic row against the turnover of its element group; constraint rows are judged in their own units (item (AM), 2026-10-07)

Item (AM) of `md/TO_BE_DONE.md`; plan `md/PLAN_20261006_to_be_done_AI_AM_rev3.md` section 2; inventory `md/AM_row_layout_inventory_20261006.md`; interface proposal `md/AM_interface_proposal_20261006.md`, whose questions Q1-Q6 the user approved on 2026-10-07. Two separately verified changes.

### 112.1 What was wrong

`normalized_reaction_residual` (`ionization_equilibrium.f90`) judged the atomic layouts against an upper bound of each row's rates with every reactant at the whole of its element (the Penning term put He 2^3S at all of the helium): on the LHS 1140 b atomic state `ew_fid_atomic_nocond` the H+ scale exceeded the gross H+ rate by 3.9e12 at the base and 2.1e2 at 26 R_p, so a 1 per cent move of H+ was admitted inside about 1.35 R_p (section 106.4). Four further defects, from the inventory: `res = max(res, a)` drops a NaN row (compiler- and order-dependently), so a state with a nonfinite balance row could be accepted; a zero scale counted as 1, so a row with no turnover was compared with 1e-6 in cm^-3 s^-1; the atomic pinned rows (`x - x_fix`) were divided by a kinetic bound, against the comment of the function; in the molecular layouts the gross rate of a balance that a constraint row replaced (transported H+, carrier H2, the reservoir partition of a lower ghost) still entered the hydrogen turnover the other rows were measured against. `scale_out` was 1 in the molecular branch. And the turnover aggregation `element_turnover_reciprocal` had no layout argument: on the atomic triplet layout it grouped He 2^3S (row 4) with hydrogen.

### 112.2 Change (a): the explicit row layout (no number moves)

New module `src/modules/nonlinear_system_solver/chemical_row_layout.f90` (in `SRC` of the Makefile): the type `chemical_row_layout_of_cell` (kind, units, species slot, acceptance group and constraint tolerance of every row), `ionization_row_layout` (the static layout from the switches; it stops on a row count that does not match them, and checks that every row is grouped with an element its species carries), `mark_constraint_rows_of_cell` (the constraint rows of one cell from the same flags and element densities the systems read), `nuclei_of_species` (the nuclear stoichiometry: HeH+ one H and one He, OH one H and one O, H2O two H and one O, CO one C and one O; kept apart from the grouping), `element_group_turnover_reciprocal` (G_E summed in row order, a status for every row: of group, row without reaction, group without reaction, not finite, constraint row; constraint rows optionally excluded), `constraint_row_tol = 1e-6`, `absent_element_density = 1e-30` (now the one definition read by `metal_rows`, `System_HeH_metals`, `System_HeH_mol_metals`, `constrained_chemical_equilibrium` and `ionization_equilibrium`). `element_turnover_reciprocal` is removed; the molecular solve scale (`set_element_turnover_scales`) is taken through `turnover_or_bound_reciprocal` (`System_HeH_mol.f90`): every row counted, the bound where the status is not "of group", which reproduces the old grouping and fallback exactly

for finite rates (a gross rate of +Inf, which used to give a reciprocal of 0, now gives the bound).

Verification of (a): src/tests/acceptance_scale PIN 36/36 PASS (FUTURE 20 of 30 still failing, the 6 grouping and 1 status assertions now passing); every suite under src/tests as before (55 suites; the pre-existing failures grid_and_gates 3 stationary-evaluate assertions, steady_completion_flag and steady_selfconsistent_residual needing run logs, unchanged). The molecular regression cases run on scratch copies with the control build and the (a) build (gfortran, single thread): mol_base_handoff, mol_metals, mol_lyman_werner, mol_diffusion, mol_ir_bands, mol_sec_ion, mol_carrier, lower_profile, oxygen_chemistry, hp_zero_seed, hp_trace_seed, hp_front: Hydro_ioniz, Ion_species and both _adv files BITWISE identical, and the acceptance record of every cell identical (MEASURED).

### 112.3 Change (b): the acceptance correction

normalized_reaction_residual (signature: the unused n_e_ref argument is removed; optional refusal_reason, turnover_share_out, own_row_out added) evaluates the system the cell solve used at the candidate, the molecular ones as volume rates, and reads the gross rate of every row from the new threadprivate ionization_row_gross (ion_residual_core.f90), which every system now fills from the same factors as its rows: heh_rows and heh_tr_rows gained a gross output (He 2^3S row from tr_triplet_row_channels, as the molecular systems form it), System_H writes its one row's, and the molecular systems' mol_row_gross became this array. The atomic solves stay unscaled and the molecular solve scale is unchanged (user decision Q3). Then, with the layout and the cell's constraint rows:

```
kinetic row i of group E:  m_i = |f_i| / G_E,  G_E = sum of P + L over
                           the KINETIC rows of E at the candidate
constraint row:            m_i = |f_i| ieq_res_tol / constraint_row_tol
group with G_E = 0:        m_i = 0 if f_i = 0, else refused (huge(),
                           refusal_reason 2, class 6)
nonfinite f_i or G_E:      the measure is NaN (refusal_reason 1)
```

and the maximum is taken after both tests. Atomic groups: H+ alone; He rows 2, 3 and He 2^3S; each metal its two rows. scale_out is the group turnover (the constraint tolerance ratio, 1, for a constraint row), which changes the scale column of EXHALE_IEQ_DUMP_CELL (user decision Q6). Certification entries "eliminated-species closure" and "level balance He 2^3S" name the new scale. New measurement switch EXHALE_IEQ_CLASS_RECORD: after every sweep output/ieq_acceptance_classes.txt holds each cell's class, residual, and row by row the turnover share and the row against its own gross rate.

Tests: acceptance_scale PIN 45/45 and FUTURE 47/47 PASS (the 27 FUTURE assertions that failed on the old source, now passing; new assertions: H only, H/He with charge exchange, H/He + metals with an absent and a two-stage element, the H+ row kinetic and pinned at one candidate, a helium group without turnover and without imbalance, the stoichiometry and the slot numbering against the continuation's). The new assertions call the new interface and cannot be compiled against the old one; their failure on the old source is the 27 FUTURE assertions' (same defects). All 56 suites as after (a) except run_mode (see 112.5).

### 112.4 Measurement (control build against the (b) build)

Scratch copies of every case of `DEFAULT_CASES`, the control build (the tree before (a) plus the class record) against the (b) build, gfortran, one thread, `EXHALE_IEQ_CLASS_RECORD=1` (`~/.claude/jobs/f24ea156/tmp/am_eval/`; `iontrans_metals` needs `stage_enth` named on its "Restart option change:" line to start at all, in both builds, see 112.7). Outputs: Hydro_ioniz, Ion_species and the `_adv` pair. Classes: the last sweep cell by cell, and the run-wide ledger (MEASURED).

| Case | Outputs (b) vs control | Last sweep, cell by cell | Run-wide non-roots (class 4), control -> (b) |
|---|---|---|---|
| wasp_full | within 7.8e-10 | all class 1 in both | 0 -> 0 |
| wasp_he23off | within 1.7e-8 | all class 1 in both | 0 -> 0 |
| wasp_full_newton | bitwise | all class 1 in both | 0 -> 0 |
| iontrans_metals | bitwise | all class 1 in both | 0 -> 0 |
| lower_profile | within 2.6e-10 | all class 1 in both | 0 -> 0 |
| hydrostatic_column | CHANGED in values of 1e-87 to 1e-34 (sign) | 166 cells 1 -> 4 | 0 -> 53896 |
| mol_base_handoff | bitwise | all class 1 in both | 16 -> 22 |
| mol_metals | bitwise | all class 1 in both | 0 -> 0 |
| mol_lyman_werner | bitwise | all class 1 in both | 83 -> 85 |
| mol_diffusion | bitwise | all class 1 in both | 16 -> 22 |
| mol_ir_bands | bitwise | all class 1 in both | 0 -> 0 |
| mol_sec_ion | bitwise | all class 1 in both | 36 -> 36 |
| mol_carrier | within 2.3e-9 | all class 1 in both | 1422 -> 1435 |
| oxygen_chemistry | within 7.6e-5 | all class 1 in both | 30 -> 91 (class 5: 2 -> 0) |
| hp_zero_seed | within 1.5e-7 | all class 1 in both | 1188 -> 12020 (class 5: 0 -> 2) |
| hp_trace_seed | within 1.5e-7 | all class 1 in both | 1384 -> 12218 (class 5: 0 -> 3) |
| hp_front | bitwise | all class 1 in both | 184 -> 228 |

LHS 1140 b, stationary evaluate of the stored state (scratch copies): the atomic `ew_fid_atomic_nocond` 504 class-1 cells before and after (one sweep); the stored pair's "eliminated-species closure" measure 2.2e-10 -> 9.6e-7 (tolerance 1e-6, at cell 258), after the sweep 1.1e-15; the molecular `..._HeH2.09_ref_merged01` and `..._cond_iontrans01` 1512 class-1 acceptances (three sweeps) before and after. The certification verdict of these stored pairs is NOT REPRODUCED with the present tree in both builds (the hydrodynamic energy row, 1.48e-6 against 1e-6 on the atomic state), a change of the tree since those states were certified and not of this item (the ifx build of (b) gives the same).

### 112.5 Newly refused states, and what they are

Every newly refused state was checked: its scales are finite (no case logged a class-6 refusal, "CANDIDATE REFUSED", which a NaN or a zero-turnover refusal would end in; every refused state is class 4 with a finite measure at or below 1), its constraint rows are the flags the system imposed, and its measure was compared with the rounding of the row (eps times its gross rate, ~1e-16).

1. **Molecular layouts with a transported carrier** (`hp_*`, `oxygen_chemistry`, the lower ghosts of every handoff case). The refused states are dominated by the H3+ row (MEASURED with the class record taken every sweep: hp_trace_seed, 983 of 986 refused records; oxygen_chemistry, 65 of 91), out of balance by 1e-5 to 3.5e-2 of its OWN gross rate while it carries 1 to 50 per cent of the hydrogen KINETIC turnover once the pinned H+ and H2 balances are left out. These are real non-roots under the stated semantics (the imbalance is 1e11 times the rounding of the row); the old measure hid them by counting the replaced H+/H2 balances, which dominate the hydrogen turnover. Their cause is the solve, not the judgment: the held molecular solve scale (unchanged, Q3) still divides the H3+ row by a turnover that includes the pinned balances, so hybrd1 stops at its step tolerance (info 1) with that row underweighted. Excluding the constraint rows from the solve scale as well is the measured change that would address it (open, 112.7).
2. **A column without ionizing radiation** (`hydrostatic_column`, H/He, T 1140 K). The H+ row of cells above r = 1.0097 R_p is refused with measures 1.4e-6 to 1.0: x(H+) is ~1e-43 of the hydrogen, the only reactions are collisional ionization (b ~ exp(-138)) and recombination, and the state sits above the root, so the row is all recombination and its imbalance is its whole turnover. By the row's own rounding it is a real non-root; in the unknowns it is not resolvable (the Newton step to the root is of the order of x, 1e-43, against the 1e-16 resolution of a fraction of order one). The measure has no absolute floor, by design point 4 of the plan, so such a state is refused. Consequence (MEASURED): the `run_mode` suite's physical run of this column (`Run mode: phys`, which refuses a step with any class-4 cell) now exhausts its step retries at step 0, and 14 of its assertions fail that passed after (a). Whether the acceptance should accept a row whose distance to its root is below the resolution of the unknowns is a decision for the user (112.7); it is not taken here.

### 112.6 Q5: slow rows inside fast groups (measurement and decision)

Rule (proposal section 4, values approved 2026-10-07): for a kinetic row with share s_i = (P_i + L_i)/G_E the group norm admits a relative error of its species up to delta_adm ~ 1e-6/s_i; where delta_adm > delta_req = 1e-3 in a cell in which the species carries at least w_min = 1e-2 of the energy term or observable it feeds, a row criterion |f_il/(P_i + L_i) <= 1e-3 is to be added for that species in those cells. MEASURED on the accepted (last-sweep) states of `mol_base_handoff`, `mol_metals`, `mol_lyman_werner`, `mol_diffusion`, `mol_ir_bands`, `mol_sec_ion`, `mol_carrier`, `oxygen_chemistry`, `hp_front` and the LHS 1140 b states `ref_merged01` and `cond_iontrans01` (script `~/.claude/jobs/f24ea156/tmp/am_eval/ana/q5.py`; weights: H3+ and H2 their IR cooling share, He++ and the molecular ions their share of n_e, He 2^3S the share of its column the cell carries, the metal stages the larger of their cooling and n_e shares):

- slow rows are common: shares down to 4e-12 (He++), 2e-11 (H3+), 6e-9 (He 2^3S), 9e-7 (H2), 1e-9 (HeH+),

1e-8 (O III on LHS);

- at every accepted state the own-row residual of every kinetic row is at or below 7e-13 (in the cells above, 4e-14 or less): no accepted state carries a masked imbalance;
- the trigger of the rule is met for H3+ only: in `oxygen_chemistry`, 44 cells (r = 1.017-1.040 R_p) where H3+ carries 1 to 100 per cent of the cooling, delta_adm up to 0.36; in `mol_lyman_werner`, 5 cells (r = 1.078-1.083) with up to 12 per cent of the cooling, delta_adm up to 3.1e-3. No other species, case or LHS state meets it (He 2^3S: the cells with delta_adm > 1e-3 carry at most 1.6e-7 of the column).

Decision: the group norm alone is adopted for every species except H3+. For H3+ the rule calls for the row criterion in cells where its cooling is material, and it is NOT implemented here: the materiality test needs the H3+ cooling share of the candidate inside the cell acceptance, which the acceptance does not form, and how to form it (which cooling, at which composition, or a materiality proxy such as the H3+ share of the hydrogen nuclei) is a design choice the measurement does not settle. It is recorded as open (112.7) with these numbers; no accepted state of the measured cases is affected (own-row residual of H3+ there <= 4e-14).

### 112.7 Open

1. The molecular solve scale and the transported carriers (112.5 item 1): the measured change Q3 left for later.
2. A resolution floor for rows whose root is below the resolution of the unknowns (112.5 item 2); `run_mode` fails until it is decided.
3. The H3+ row criterion where its cooling is material (112.6).
4. A charge exchange term on the identity row of a two-stage element's X++: `cx_add_to_fvec` adds the X++ charge exchange (a volume rate) to that dimensionless row when x(X++) is nonzero (acceptance_scale RECORD: x = 1e-3 gives f = -2.4e3 for oxygen); the root x = 0 is unchanged, the units of the row are mixed. Reported, not changed.
5. `backup/regression/iontrans_metals` refuses to start with the present tree (its IC predates the stage enthalpy flux token `stage_enth`); its input needs `stage_enth` on the "Restart option change:" line, or a new IC.
6. `CCE` keeps its own species-slot numbering (`is_*`); the test asserts that it agrees with `chemical_row_layout`'s, and moving it onto the module is a separate change.

Frozen binary (ifx 2026.1, a copy of the tree with (a) and (b)): `LHS1140b/models/.P2/EXHALE_15440b7c.x`, manifest `BINARY_MANIFEST_15440b7c0a0b.txt`. Documentation: `docs/physics_overview/ionization.tex` (paragraph "The acceptance measure", Eq. eq:ion:acceptance; the normalization paragraph; class 6 for a NaN), `steady_state.tex` (the chemistry gate), the user manual (`EXHALE_IEQ_CLASS_RECORD`, the scale of `EXHALE_IEQ_DUMP_CELL`).

## 113. The carrier source runs the oxygen cycle only with the oxygen chemistry on (2026-10-07)

Found by the (AN) diagnosis (`md/AN_matching_surface_diagnosis_20261007.md`, summary item 1); fixed as a change separate from section 112 and measured on its own (control: the build of section 112).

**Defect.** `carrier_source` (`src/modules/lower_atmosphere/diffusive_photochemistry.f90`) called `oxygen_carrier_rows` inside `if (thereis_mol)` with no `thereis_oxychem` guard; the two other callers

(System_HeH_mol_metals, constrained_chemical_equilibrium) are guarded. With the oxygen chemistry off, OH and H2O are not carried (zero), so of the oxygen cycle only the loss O + H2 -> OH + H (k2 n_O n_H2) acted, on the H2 row, with no product carried, no reverse and no reaction heat: the free oxygen broke H2 at no energy cost. On the LHS 1140 b energy-coupled states that term was 72-95 per cent of the column H2 destruction, and the H + H + M recombination of the atoms it freed 63-69 per cent of the wind's heating (READ, the (AN) diagnosis).

**Fix.** The call is made only when thereis_oxychem; otherwise rows 9 and 10 (OH, H2O) and the oxygen terms of the H2 row stay at the zero they are initialized to, so src(ic_OH), src(ic_H2O) and pH2ox, lH2ox are zero (no uninitialized read: fv, the p/l terms and src are zeroed before the branch). set_oxygen_coeffs is still called (its coefficients are then not read).

**Test.** src/tests/carrier_constraint_attribution, assertion block (4): with the oxygen chemistry off and no OH/H2O carried, the H2 source of a cell must not depend on the free oxygen, and the OH and H2O sources are zero. RED before the fix (MEASURED): H2 source -6.848e7 cm^-3 s^-1 with the free oxygen 4.8e10 cm^-3 against -50.21 without it, OH + H2O sources 6.85e7; GREEN after: -50.21 both ways, OH + H2O 0. The carrier suites (carrier_boundary_jacobian, carrier_constraint_attribution, carrier_enthalpy_flux, carrier_helium_inventory, carrier_outer_boundary, carrier_reference_scales, carrier_retry, carrier_returned_state_acceptance), stage_row_balance, ionization_stage_flux, helium_metastable_transport, ghost_reservoir_chemistry and acceptance_scale pass.

**Regression cases it reaches.** The carrier operator with molecular chemistry and the oxygen chemistry off: mol_carrier, hp_zero_seed, hp_trace_seed, hp_front (molecular carrier transport). All four carry no metals (no metals.inp), so their free oxygen is zero and the removed term was zero: outputs BITWISE identical to the section-112 build, and the acceptance record identical (MEASURED). oxygen_chemistry (oxygen chemistry on, the guarded call unchanged): bitwise. No other default case runs the molecular carrier operator.

**Where it acts: the LHS 1140 b molecular states** (metals from the lower profile, oxygen chemistry off). Stationary evaluate of the stored ..._HeH2.09_ref_merged01 state: the carrier balance H2 measure 2.2e-7 (within 1e-5) with the section-112 build against 8.4e-3 (ABOVE, at cell 117; 2.3e-5 at the wind cell 217) with this one; on ..._cond_iontrans01, 1.8e-3 -> 8.4e-3. Every LHS molecular state solved with the defect is therefore not a stationary state of the corrected equations and has to be recomputed; the effect on the base temperature and composition is the one the (AN) scratch run estimated (first cell x_H 0.58 -> 0.13, 575 -> 387 K, READ). Not rerun here.

Frozen binary (ifx 2026.1, the tree of section 112 plus this fix): LHS1140b/models/.P2/EXHALE_17c04ea2.x, manifest BINARY_MANIFEST_17c04ea22cea.txt (differs from EXHALE_15440b7c.x in diffusive_photochemistry.f90 alone).

## 114. The oxygen-guard fix on the energy-coupled LHS 1140 b states: the He/H 2.09 line falls by a factor 2.5 (2026-10-07)

Section 113's fix (oxygen_carrier_rows no longer applied to the transported H2 row with the oxygen chemistry off) measured on its own: three certified energy-coupled closures rerun from their converged states with binary 17c04ea2 (sections 112 + 113) and, as control, 15440b7c (section 112 only; the two binaries differ only in

diffusive_photochemistry.f90); dirs <case>_rerun_17c04ea2 and <case>_rerun_15440b7c beside each case.

| Case | Control (15440b7c) |
|---|---|
| He/H 2.09 (phase3/energy_coupling_nlte) | k = 1; log Mdot 7.890; F_H 2.326e7, F_He 5.366e7 g/s; F_cond 9.62; T_match 350.3 K; EW 0.726 |
| He/H 0.65 (phase5/HeH0.65) | k = 1, unchanged from the stored state |
| He/H 0.65, s = 1.3 route B | k = 1, unchanged (width-matched EW 1.1084) |

The control reproduces the stored states (He/H 2.09 stored EW 0.7255), so the whole change is the defect's. At He/H 2.09 Mdot falls by a factor 2.1, the helium flux by 2.7, the conducted heat by 2.0, T_match by 66 K, and the width-matched EW by a factor 2.5 (0.70 -> 0.28 %A against 1.108 observed). Every energy-coupled molecular result of sections 63-111 that rests on these states (the He/H scans, the EUV match of sections 102, 108 and 111) is invalid and is to be recomputed.

The two He/H 0.65 runs end with "the stationary outer iteration spent its budget of 60 passes without an accepted state ... carrier balance He+, measure 4.21e-4 against 1e-5 at cell 217, r = 1.20 R_p" (route B); the (AN) scratch run without the oxygen term met the same refusal. Under diagnosis (md/heplus_carrier_certification_20261007.md).

## 115. The element operator settles on the pressure gradient of the mixture, and its non-resonant ion-neutral pairs are the polarization form alone (2026-10-07)

Both defects were found by the (AK) derivation note (md/AK_ambipolar_stage_drift_derivation_20261006.md, sections 2.1 and 6.3); user decision 2026-10-07: fix both, no option key, the results change.

**1. Defect: settling on gravity instead of the pressure gradient.** Each species obeys n_s m_s Du_s/Dt = -grad p_s + n_s m_s G + n_s e_s E + friction (Schunk 1977, RvGSP 15, 429, eq. 24b). With the drift inertia neglected the sum over all species is the mixture equation rho Du/Dt = -grad p + rho G, and eliminating the bulk acceleration leaves the force on species s against the mixture as -grad p_s + (rho_s/rho) grad p + n_s e_s E (note eq. 2.1): gravity cancels, and the settling acceleration is g_eff = -(1/rho) dp/dr = g + Du/Dt. The code used g (Dphi) alone in settling_coefficient, in the trace-metal coefficients and in the photochemical carrier settling, i.e. it left out the acceleration of the wind (MEASURED on the LHS 1140 b atomic state: v dv/dr up to 0.050 g at 18.8 R_p; O(1) through a sonic point).

**Fix.** New mixture_pressure_acceleration (binary_element_diffusion.f90) forms g_eff = -(p/rho) dln p/dr from the state's own total pressure p = (n_tot + n_e) k T, the particle count being the equation of state's (gas_particle_density, new in composition.f90, which shares one routine, electron_and_particle_densities, with get_species_densities, so the two cannot count particles differently; the existing path is arithmetically unchanged), by the centred logarithmic difference over the two neighbours, the same stencil and form as the ambipolar field eE = -kT dln(n_e T)/dr. An isothermal barometric column returns g exactly, and in a column of uniform composition eE = mbar m_H g_eff, so the ambipolar limits of the relative settling mass (now normalized to g_eff) hold whatever the density profile. Every place the settling force is formed now uses it: settling_coefficient (element step, fixed-wind relaxation, stationary residual element_transport_residual, helium_diffusive_face_flux and with it post_process_adv, the element flux profile of the certification, the base flux record), trace_element_transport_coefficients (one geff argu-

ment, from the same `settling_coefficient` call) and the carrier settling of `diffusive_photochemistry.f90` (`carrier_face_coefficients`; `carrier_geometry` no longer returns a gravity). The JFNK Jacobian of the element rows is a finite difference of the residual, and g_eff reaches the same neighbours j +- 1 as eE, so the stencil is unchanged. `grav_func` is no longer used by either module. Viscous stress is not part of this first-order closure and is left out of the mixture balance (stated in the module header).

**2. Defect: rigid core added to the non-resonant ion-neutral friction.** `ion_neutral_pair_diffusion` combined the induced-dipole coefficient with a rigid sphere of the neutral-neutral diameter, 1/D = 1/D_pol + 1/D_hs. Schunk & Nagy (1980, RvGSP 18, 813) use the polarization form alone for every non-resonant pair (their eq. 17, after Dalgarno et al. 1958, tabulated in Table 6, both checked on the rendered page of the published article), and the quantum-mechanical H+/He coefficient of Kimura et al. (1993, ApJ 405, 801, Table 2) is 1.05-1.17e-9 cm^3/s against the code's 3.2-7.4e-9.

**Fix.** `ion_neutral_pair_diffusion` is removed; `stage_pair_diffusion` calls `polarization_pair_diffusion(TK, ntot, alpha_n, Z_ion, mu_g)`, D = kT/(2.21 pi Z e n (alpha_n mu)^1/2), the ion charge entering the induced-dipole potential as Z^2 e^2 (so He++ + H and the doubly charged metal stages against H carry twice the singly charged friction; before, the charge was absent). Every non-resonant pair of the operator changes: H+/He, H2+/He, H3+/He, He+/H, He++/H, He+/H2, He++/H2 and every metal stage against H and H2. Resonant pairs never enter the element operator (both partners carry one element). The Python twin `src/utils/collisional_validity.py` is changed the same way (its `pair_diffusion` reproduces the Fortran K(H+,He) = 1.3057e-9 cm^3/s, MEASURED). Against Kimura et al. (1993) the polarization form is 1.243 (1000 K), 1.232 (5000 K) and 1.116 (1e4 K) times the published value (MEASURED, test T12e); that 12-24% residual is not explained by the published sources, which do not define the thermal average of the table (note section 6.3), and is stated at the code site.

**Tests** (`src/tests/diffusion_tests.f90`, run against the old operator with only the accessors added, and against the new one; MEASURED):

| test | old operator | new operator |
|---|---|---|
| T1a hydrostatic closed column, logit error at N=400 (g_eff = g, no change) | 4.553e-6 PASS | 4.553e-6 PASS |
| T1c column held by g + a, a = (r-1) v0^2/R0, logit error vs integral of g + a | 1.116 FAIL | 9.3e-6 PASS |
| T12e K(H+,He) vs Schunk & Nagy Table 6 (1.327e-9 with the code masses) | 7.378e-9, rel. 4.56 FAIL | 1.306e-9, rel. 1.6e |
| T12e K(He+,H) vs Table 6 (2.342e-9) | 8.428e-9, rel. 2.60 FAIL | 2.355e-9, rel. 5.8e |
| T12e K(He++,H)/K(He+,H) = 2 (charge of the induced dipole) | rel. 0.5 FAIL | exact PASS |
| T12c half-ionized Blanc average typed with polarization alone | 8.7e-3 FAIL | 0 PASS |
| T15 g_eff of a hydrostatic / an accelerating (a/g up to 3.0) ionized column, against g / g + a | (accessor only) 4.6e-13 / 1.6e-13 | 4.6e-13 / 1.6e-13 |

Fixtures that assumed the old physics were corrected, not loosened: a column whose pressure falls with g = 0 is not at rest, and its settling is now the physical drive, so every "column at rest" or "flat column" of the element suites is now built at uniform pressure (new helper `density_of_uniform_pressure` in `src/tests/test_columns.f90`; T6, T7d, `element_operator` (two rows), `steady_species_rows`, `ionization_stage_flux`, `certification` (`column_background`, rho T = 1)), and the T1a/T1b/T9-front columns are hydrostatic for their own composition (`impose_pressure_profile`). T9 normalizes the relative settling mass to g_eff and types its targets from the col-

umn's own mean particle mass; T7c types G with the column's g_eff; T14 drives the helium outward with a column whose pressure rises outward, because a closed hydrostatic column at this inventory stops near X = 0.8 (MEASURED) and never reaches the bound. The whole driver: 39 passed, 0 failed (old operator: 30 passed, 9 failed).

Suites that link the changed modules (all 56 `src/tests/*/run.sh`), run with the new build and with the control build of the entry text: every suite passes except `carrier_enthalpy_flux` (3 rows), `grid_and_gates` (the stationary-evaluate rows), `run_mode`, `steady_completion_flag` and `steady_selfconsistent_residual`, whose FAIL rows are identical with the control build [CORRECTED 2026-10-10, section 137: the three `carrier_enthalpy_flux` rows were NOT pre-existing; they appeared with the real g_eff of this section, which the test's reference did not follow] (pre-existing; the last two need `EXHALE_STEADY_RUNLOGS`, `run_mode` is the (AM) resolution-floor item). `certification`, `element_operator`, `ionization_stage_flux` and `steady_species_rows` failed on their flat-column fixtures before those were rebuilt at uniform pressure (above) and pass after.

**Regression cases it reaches** (scratch copies, control build of the entry text against the new build, both gfortran, single thread):

The cases with `He_diffusion: True` (the matrix cases `mol_diffusion` and `lower_profile`, the latter with `He_metal_diffusion` too, and the He/H ladder `heh_0p3`, `heh_3`, `heh_10`, `heh_30`, `heh_1_12k`, `heh_1_lw_12k`) and the cases with molecular carrier transport (`mol_carrier`, `hp_front`, `hp_zero_seed`, `hp_trace_seed`, through the carrier settling). The control build reproduces the golden of `lower_profile` to 1e-8. Verdict of `compare_within_tolerance.py` (1e-3) on the four output files, and the largest relative changes of the physical columns (MEASURED):

| case | verdict | rho | T | He/H | n(2^3S) | H2 |
|---|---|---|---|---|---|---|
| mol_diffusion | changed (Hydro: cool col. near zero, 1.3e-2) | 8.4e-5 | 6.6e-5 | 1.1e-4 | 9.2e-5 | 3.5e-5 |
| mol_carrier | Hydro within 1e-3; Ion changed (H3+ at the top, 1.8e-2) | 1.3e-6 | 1.8e-6 | 2e-13 | 1.1e-5 | 1.4e-5 |
| hp_front, hp_zero_seed, hp_trace_seed | Hydro within 1e-3; Ion changed (H3+ at the top, 4-6e-3) | 1.4e-9 | 1.6e-9 | 1e-14 | 7e-9 | 1.9e-8 |
| heh_0p3 | changed (v where it is about -2 cm/s) | 2.3e-4 | 1.7e-4 | 2.7e-4 | 2.9e-4 | 6.0e-5 |
| heh_10 | changed | 1.8e-4 | 1.7e-4 | 5.1e-4 | 2.9e-4 | 4.4e-5 |
| heh_30 | changed | 8.2e-5 | 7.0e-5 | 4.9e-4 | 1.1e-4 | 2.6e-5 |

| case | verdict | rho | T | He/H | n(2^3S) | H2 |
|---|---|---|---|---|---|---|
| heh_1_12k | changed | 2.2e-4 | 2.3e-4 | 2.9e-4 | 4.1e-4 | 1.6e-4 |
| heh_1_lw_12k | changed (v where it crosses zero near the base) | 8.3e-5 | 6.3e-5 | 3.6e-4 | 1.4e-4 | 1.7e-5 |
| heh_3 | changed (trajectory) | 0.18 | 0.14 | 4.5e-4 | 0.16 | 7.6e-4 |
| lower_profile | changed (trajectory) | 0.47 | 0.33 | 5.6e-4 (O/H 3.7e-3) | 0.32 | - |

The composition moves by at most 5.6e-4 (He/H) in every case. In `heh_3` and `lower_profile`, which are relaxation snapshots far from a steady state (final du 0.92 and 2.7), the marching trajectories of the two builds agree to 1e-5 in du up to step 1000 (lower_profile) and 2000 (heh_3) and then separate at once (0.26 and 7e-2), so the hydrodynamic differences are a divergence of two unconverged trajectories set off by the small composition change, not a size of the physics change. No golden was refreshed.

**LHS 1140 b state.** `LHS1140b/models/.P1/fiducial_conduction/ew_fid_atomic_nocond` (He/H 9.7, He_Kzz 1e9, He 2^3S on), stationary re-solve from the stored state in `/home/kiseon/.claude/jobs/f24ea156/tmp/ak_diff/`, info = 0 and certified for both builds. MEASURED: the control build returns the stored state (He/H to 5e-15, Mdot identical); the new build moves He/H by at most +0.63% (at 1.37 R_p; -0.32% at the top), log10 Mdot by -8e-5 (7.87925 -> 7.87917), the radial He 2^3S column by +0.011% and n(2^3S) locally by at most 0.26%, T and v by at most 5e-4 and 6e-4. A third build with the coefficient correction alone attributes nearly all of it to item 2 (He/H +0.67% at 1.38 R_p); g_eff alone moves He/H by at most 4e-4 here, because where g_eff departs from g (5% at 19 R_p) the advection dominates the settling. Near the base the discrete g_eff of the stationary state departs from g by -0.3% to +0.8% in cells 1-5 (the centred log difference against the hydrodynamics' well-balanced balance), below 0.1% from cell 17 (1.003 R_p) out to 3.6 R_p; beyond 1.1 R_p g_eff/g - 1 equals v dv/dr / g of the state to 1.5e-3 (MEASURED), i.e. g_eff is the wind's acceleration plus gravity. Figure `docs/figures/lhs1140b_settling_geff_polarization_resolve.pdf`.

**Stale.** The drift estimates of the derivation note section 10 and the bound quoted in the `ionization_stage_transport.f90` header were computed with the rigid-core inventory; marked stale at both places, not re-measured. Every He-diffusion state and golden computed before this section carries the old settling and friction; no golden was refreshed.

**Documents.** `docs/physics_overview/molecules_metals.tex` (g_eff, the binary coefficient paragraph with Schunk & Nagy eq. 17 and the Kimura residual; carrier settling), bibliography (Schunk 1977, Schunk & Nagy 1980, Kimura et al. 1993), `docs/EXHALE_user_manual.tex`, `README_HOWTO.md`, the derivation note (verdict, sections 2.1, 6.2, 6.3, 10, 15), `md/TO_BE_DONE.md` (AK) (also its broken reference `species_diffusion.f90` -> `binary_element_diffusion.f90`). Physics and manual PDFs rebuilt.

Frozen binary (gfortran 16.2, the source of `EXHALE_17c04ea2.x` plus this section):

LHS1140b/models/.P2/EXHALE_fae54121.x, manifest BINARY_MANIFEST_fae541210744.txt.

## 116. The carried ionization stages are scaled by their gross channels; the molecular solve scale groups as the acceptance does; a resolution floor and an H3+ own-row criterion in the acceptance (item (AO) D1-D4, 2026-10-07)

Item (AO) of md/TO_BE_DONE.md, user decisions of 2026-10-07: D1, D2 and D3 as recommended, D4 a defect fix. Four changes, each measured against the build before it: control = the source of section 115, then D4, D4 + D1, D4 + D1 + D2 and D4 + D1 + D2 + D3 (gfortran 16.2, one thread, scratch copies of the regression cases with EXHALE_IEQ_CLASS_RECORD=1, in ~/.claude/jobs/f24ea156/tmp/ao/). Every number is MEASURED unless marked READ. No golden was refreshed.

### 116.1 D4 (defect): the carried proton and helium stages carry their gross production and loss

**Defect.** The physical scale of a carrier row (row_terms_phys, carrier_residual), which the carrier acceptance and the certification entries "carrier balance ..." divide by, takes the row's production plus loss from carrier_source. For the two ionized helium stages (molecular and atomic gas) and the proton of an atomic gas, carrier_source handed out the positive and negative parts of the NET source, under a comment saying that the two magnitudes "are not available". That was false: mol_heh_rows and heh_tr_rows both return the gross rate of every row (gross, from the same factors as the rows, section 112), and he_h_cx_fvec adds each charge exchange reaction to the gross rate of every row it moves. Where a stage is held by fast chemistry its net is many digits below its channels: He+ in the molecular layer of the LHS 1140 b route-B He/H 0.65 state is made by photoionization and destroyed by He+ + H2 -> H+ + H + He at about 330 cmˆ-3 sˆ-1 each at 1.20 R_p, net 6e-3, so the scale held the transport alone and read a state balanced to 5e-8 of its channels as 4.2e-4 against 1e-5 (md/heplus_carrier_certification_20261007.md). The molecular proton row carried its He <-> H charge exchange as a net on one side as well.

**Fix** (src/modules/lower_atmosphere/diffusive_photochemistry.f90, carrier_source, lines 5254-5760). mol_heh_rows (molecular gas) or heh_tr_rows (atomic gas) and he_h_cx_fvec now fill stage_row_gross. Every term of rows (1)-(3) is sign-definite in the densities, so a row f with gross rate G has production (G + f)/2 and loss (G - f)/2; the metal charge exchange (charge_exchange_stage_sources) is added with its own gross production and loss on its own sides. The atomic rows are in the boundary-flow basis (row (2) the summed He I balance, row (3) the He II -> He III flow, src(He II) = -f2 - f3); each reaction touches He II once and none is in both rows, so the gross rate of He II is G2 + G3. The proton row takes the same form in both gases. src and the Newton's own scale (dsc) are unchanged, and sprod - sloss = src to the rounding of the two sums. The comments of carrier_source, carrier_residual and carrier_row_terms_write are corrected. The He 2ˆ3S row already had this form (section 94).

**Test.** src/tests/helium_metastable_transport, new assertions (4b): the He II density is bisected to the zero of its own net source with the other carried states held (He III and the 2ˆ3S level lowered so that the bracket closes in both gases); there the production of He II must hold the photoionization of the ground singlet, the loss of He II and the production of He III the photoionization of He II (compared on the row's own P + L), and sprod - sloss = src must hold for H II, He II and He III. Control: 6 FAIL (the three bounds in both gases; e.g. the He II production fell short of the

singlet photoionization by 3.0e14 of its own P + L in atomic gas); D4: 40 of 40 PASS.

**The refused state.** The k00 state of `LHS1140b/models/.P2/ew_fit/route_xuv_scaled/HeH0.65/xuv_x1p3_rerun_17c04ea2`, reloaded exactly (code-unit state) in a scratch copy and evaluated (`Restart intent: stationary evaluate`, the run's environment):

| binary | verdict | carrier balance He+ | at cell 217 (r = 1.2006) | He++ | H+ |
|---|---|---|---|---|---|
| 17c04ea2 | NOT CERTIFIED, He+ alone | 8.80e-3 (cell 207, below the gate) | 4.21e-4 | 9.31e-6 | 5.83e-8 |
| source of 17c04ea2 + D4 (ifx, scratch) | CERTIFIED | 6.98e-8 (cell 276) | 5.36e-8 | 2.46e-8 | 5.83e-8 |

The gross-scale measures are the ones the diagnosis predicted (5.4e-8 at the refusing cells, 7.1e-8 worst). With the present source (section 115) the same state is not a stationary state of the equations any more: the hydrodynamic energy row reads 5.3e-2 and the carrier rows 1.6e-3 to 0.24 at the outer cell 500, with and without D4 (an effect of section 115, not examined here); at cell 217 D4 moves the He+ measure from 1.8e-4 to 2.3e-8. A gfortran build of the present source refuses the conserved state the ifx binary wrote on its NORMALIZATION line (the code-unit scales differ in their last digits between the two compilers) and restarts from the dimensional pair; the ifx build of the same source reloads it exactly.

**Regression cases it reaches** (carried ionization stages): `hp_zero_seed`, `hp_trace_seed`, `hp_front` (molecular, relaxation snapshots) and `iontrans_metals` (atomic, stationary): outputs and acceptance record BITWISE identical to the control. The certification of `iontrans_metals` reads the same state with the new scale: H+ 6.47e-5 -> 4.98e-7, He+ 5.48e-6 -> 2.18e-6, He++ 4.70e-4 -> 8.16e-6 (worst over the column; certified in the wind before and after). The atomic proton and stages carried the defect too.

### 116.2 D1: the molecular solve scale sums the kinetic rows of each group

**Change** (`System_HeH_mol_metals.f90`, `set_element_turnover_scales`, lines 138-201; `System_HeH_mol.f90`, `turnover_or_bound_reciprocal`, line 656). The held solve scale marks the cell's constraint rows (`mark_constraint_rows_of_cell`) and sums each acceptance group over its kinetic rows, as the acceptance does (section 112). A constraint row keeps the bound, which is never applied to it (the row is written x - x_fix after the scaling). Before, a pinned H+ or H2 balance, the reservoir partition of a lower ghost and the carried helium and oxygen balances entered the turnover the other rows of their element were divided by.

**Test.** `src/tests/acceptance_scale`: the two assertions that pinned the old solve scale (`PIN_molecular_solve_scale_keeps_pinned_H+_balance`, `..._keeps_reservoir_H2_balance`) became `PIN_molecular_solve_scale_excludes_pinned_H+_balance` and `..._excludes_reservoir_H2_balance` (1/scale(4) with H+ pinned and 1/scale(1) with the reservoir row equal the kinetic hydrogen turnover). D4 build: FAIL (7.16e-7 against 1.42e-6 and 1.40e-6); D1: PASS.

**Measurement** (D4 build -> D1 build; run-wide class-4 records, class-5, outputs; the last sweep of every case is all class 1 in both):

| case | class 4 | class 5 | outputs |
|---|---|---|---|
| hp_zero_seed | 12020 -> 11993 | 2 -> 0 | within 1.0e-9 |
| hp_trace_seed | 12218 -> 12192 | 3 -> 0 | within 7.3e-10 |
| hp_front | 228 -> 228 | 0 | within 7.4e-11 |
| oxygen_chemistry | 91 -> 72 | 0 | within 7.7e-5 |
| mol_base_handoff | 22 -> 24 | 0 | within 1.3e-10 |
| mol_diffusion | 24 -> 24 | 0 | within 3.9e-8 |
| mol_sec_ion | 36 -> 36 | 0 | within 6.6e-9 |
| mol_carrier | 1439 -> 1437 | 0 | within 9.4e-11 |
| mol_metals | 0 -> 0 | 0 | within 2.3e-11 |
| mol_ir_bands | 0 -> 0 | 0 | within 2.5e-11 |
| mol_lyman_werner | 85 -> 85 | 0 | within 3.6e-9 |

**What it does not do.** Section 112.5 attributed the H3+ refusals of the transported-carrier layouts to the solve scale. It is not their cause. With every sweep recorded (scratch build whose class record appends), the 11993 refused records of `hp_zero_seed` after D1 are all H3+ rows (own-row imbalance 2e-6 to 1.0, mean 0.9 per cent) in which H3+ now carries 5 to 56 per cent of the hydrogen kinetic turnover. With the tolerance of the cell solve lowered from sqrt(eps) = 1.5e-8 to 1e-12 (`EXHALE_IEQ_TOL=1e-12`) the same run has no non-root at all, and needs fewer hybrd1 calls (101304 against 126816): hybrd1 stops at its relative step test, taken on the scaled norm of all the unknowns, which the O(1) fractions dominate, before the trace unknown x(H3+) is resolved. The same cause makes two assertions fail after D1 (and not before): `ghost_reservoir_chemistry` `ghost_root_inside_every_element_budget` (the ghost root ends at x(He++) = -1.6e-15, rows within 2.0e-7; with the solve tolerance 1e-12 in a scratch copy of the test, x(He++) = +4.5e-15 and rows of 1.4e-13), and `grid_and_gates` `restart_lower_ghost_is_the_derived_boundary` (the lower ghost rows of two round trips 1.32e-14 apart against 1e-14; 0 with the D4 build; 4.3e-16 with the D1 build and `EXHALE_IEQ_TOL=1e-12`). Not changed here: the solve tolerance is a path of every cell solve of every case (116.6, item 1).

### 116.3 D2: the resolution floor of the acceptance

**Change** (`ionization_equilibrium.f90`, `normalized_reaction_residual`, header and lines 4803-4842, internal `rows_as_volume_rates` 4879-4915; `chemical_row_layout.f90`, `fraction_resolution` line 138). A kinetic row whose measure is above `ieq_res_tol` (in a group with turnover, finite) is accepted when the remaining imbalance is below what the composition resolves: its Newton correction along its own unknown, dx_i = |f_i| / |df_i/dx_i| (forward difference of the same system as volume rates, step sqrt(eps)|x_i|), is at most `fraction_resolution` = epsilon(1) = 2.2e-16, the spacing of the floating-point numbers at the element total (every unknown is a fraction of the nuclei of an element, and a smaller change moves none of the element's sums: the neutral 1 - sum x, the electrons; a correction below eps |x_i|, below the resolution of x_i itself, is a special case of it), or the species and its root both lie below

`absent_element_density` in nuclei, (|x_i| + dx_i) n_E <= 1e-30 cm^-3. The row's measure is then 0, and the new optional output `resolution_floor_out` marks it. The form "below the resolution of x_i itself" alone cannot accept the case the item names: on `hydrostatic_column` the refused H+ rows have dx/x = 1e-6 to 1e-3 (their imbalance over their own gross rate), with x(H+) <= 4e-17 (MEASURED from the class record and the state of the D1 run).

**Test.** `src/tests/acceptance_scale`, section (vii-a): an H-only cell without radiation at 1140 K and n_H 4.7e13 (the column of `hydrostatic_column`), root x* = 2.5e-48: at x = 1e-20 the row is all recombination and its correction is x/2; FUTURE: accepted and floored (D1 build: measure 1.0, FAIL; D2: 0, PASS). At x = 1e-3 the correction is 5e-4: PIN, still refused. An element at n_H = 1e-31 cm^-3: FUTURE, accepted as absent (D1: FAIL; D2: PASS). The new keyword cannot compile against the D1 source, so the red run used the same assertions without it.

**Measurement** (D1 build -> D2 build; `wasp_full` and `wasp_he23off` end with exit status 2, not certified, in every build, as their stored runs do):

| case | last sweep | run-wide class 4 | outputs |
|---|---|---|---|
| hydrostatic_column | 338 c1 + 166 c4 -> 504 c1 | 53896 -> 2695 | changed: n(H+) in 122 cells, \|x(H+)\| <= 5.7e-14, 90 of them in sign (the mirror root, below); He II, He III below 1e-43 of the helium; rho, v, p, T within 5.4e-13 |
| hp_trace_seed | all c1 | 12192 -> 12192 | within 1.3e-9 |
| oxygen_chemistry | all c1 | 72 -> 75 | within 1.1e-5 |
| wasp_full, wasp_he23off, wasp_full_newton, lower_profile, iontrans_metals, hp_zero_seed, hp_front, mol_base_handoff, mol_metals, mol_lyman_werner, mol_diffusion, mol_ir_bands, mol_sec_ion, mol_carrier | all c1 | unchanged | BITWISE |

`hydrostatic_column` against the pre-112 golden: Hydro_ioniz within 7.5e-10; Ion_species identical to 8.8e-11 except n(H+) in 70 cells and n(He++) in 2, which differ in sign (|x(H+)| up to 8.0e-12) or below 1e-243 of the helium. The 2695 class-4 records left are real non-roots of the first eleven steps of the march (199 cells at step 1, falling to 153 at step 11; res 1.0): the initial composition carries x(H+) ~ 1e-14 to 1e-10 where the root is below 1e-40, and the atomic Newton of the cell solve (`newton_dense`) stops at an absolute step of 1e-11 on a row that is quadratic in x near zero, so each sweep halves x only once or twice.

`run_mode`: 16 FAIL lines since section 112 (that section said 14; the log of its (b) build has 16) -> 2. The two left

(an_init_snapshot_loaded_by_a_phys_run_is_the_handoff, the_handoff_clock_starts_at_zero) load a 10-step initialization snapshot of hydrostatic_column into a physical run, whose handoff test refuses 171 cells with x(H+) ~ 3e-14 against a root below 1e-40: real non-roots, the same as above. MEASURED on scratch copies: a 15-step snapshot is refused at 16 cells, a 20- or 30-step snapshot hands off. Not changed (116.6, item 2).

**Found while measuring (not changed).** Seventy accepted cells of hydrostatic_column (142 in the control) carry a NEGATIVE H+ density, n(H+) = -8.9e-2 cmˆ-3 at r = 1.31 R_p, x(H+) = -8.0e-12: the mirror root x = -x* of the H+ row, whose electrons are its own protons (n_e = n(H+)), so that f(-x*) = 0 with a negative electron density. The admissibility test ionization_fractions_physical accepts negative fractions down to -1e-10 (ftol), so this unphysical root passes as class 1 (measure 2.9e-7). The pre-112 golden has none. 116.6, item 3.

### 116.4 D3: H3+ against its own gross rate

**Change** (ionization_equilibrium.f90, normalized_reaction_residual, lines 4790-4801; chemical_row_layout.f90, h3p_row_balance_tol line 148). In the molecular layouts the H3+ row's measure is the larger of its group measure and (|fl|/(P + L)) ieq_res_tol / h3p_row_balance_tol, so it passes only when |fl| <= 1e-6 G_H and |fl| <= 1e-3 (P + L) (section 112.6: the one species the group norm masks where its cooling is material). It applies in every cell: the materiality test of 112.6 needs a cooling share inside the cell acceptance, which it does not form, and a cell where H3+ is too scarce to matter has a correction below the resolution floor of 116.3 or a measure far below 1e-3 at its root.

**Test.** src/tests/acceptance_scale, section (vii-b): at the root of the molecular cell of (iii), where H3+ carries 3.7e-8 of the hydrogen turnover, a 10 per cent move of H3+: FUTURE, refused (D2 build: measure 1.7e-9, admitted, FAIL; D3: 4.8e-5, PASS); a 1e-4 move: PIN, admitted (5.0e-8).

**Measurement** (D2 build -> D3 build): oxygen_chemistry class 4 75 -> 139 (the added records at steps 241-316, cells up to 128, the transient after the cold start, where hybrd1 left H3+ off by more than 1e-3 of its own rate; outputs within 2.8e-6; last sweep all class 1); the H3+ own-row imbalance of the accepted states at r = 1.017-1.040 R_p (the 112.6 cells) 2.4e-16 and 2.2e-16 at most, 1.0e-13 over the column. mol_lyman_werner (the 112.6 cells r = 1.078-1.083): outputs BITWISE, class-4 records 85 in both, H3+ own-row imbalance of the accepted states 2.4e-16 at most there, 4.3e-16 over the column. Every other default case: outputs and classes BITWISE identical.

### 116.5 Suites, frozen binary, documents

All 56 suites under src/tests/*/run.sh with the final source and with the control build of the entry text (run_suites.sh, one thread). Final: 50 pass. Failing in both builds, identical rows (pre-existing, section 115): carrier_enthalpy_flux (3 rows), grid_and_gates (the three stationary-evaluate rows), steady_completion_flag and steady_selfconsistent_residual (need EXHALE_STEADY_RUNLOGS). run_mode: 16 -> 2 rows (116.3). New after D1: ghost_reservoir_chemistry (1 row) and grid_and_gates restart_lower_ghost_is_the_derived_boundary (116.2). The suites the changes touch: acceptance_scale PIN 47/47 FUTURE 50/50, helium_metastable_transport 40/40, every carrier suite but carrier_enthalpy_flux (pre-existing, above), stage_row_balance, ionization_stage_flux, certification pass. Three further suites

(low_energy_electron_degradation, physics_probe, spectrum_type) failed in the control copy only because that copy lacked data files outside src/ (inputdata/, references/, eta_approx.py); they pass in the tree.
Frozen binary (ifx 2026.1, a copy of the final tree in LHS1140b/models/.P2/build_aoacceptance_ifx/): LHS1140b/models/.P2/EXHALE_bd2b93d3.x, manifest BINARY_MANIFEST_bd2b93d30fb0.txt (differs from EXHALE_fae54121.x, a gfortran build, in diffusive_photochemistry.f90, chemical_row_layout.f90, System_HeH_mol.f90, System_HeH_mol_metals.f90 and ionization_equilibrium.f90). On the D4 state above it reloads exactly and reads He+ 2.25e-8 at cell 217.
Documents: docs/physics_overview/ionization.tex (the solve-scale paragraph, Eq. eq:ion:element_turnover over kinetic rows; the acceptance measure with the H3+ criterion, Eq. eq:ion:h3p_own_row, and the resolution floor), steady_state.tex (the carrier scale of the ionized stages; the chemistry gate), docs/EXHALE_user_manual.tex (the certification scale of the carrier rows; EXHALE_CARRIER_ROW_TERMS), md/TO_BE_DONE.md (AO).

### 116.6 Open

1. The tolerance of the cell solve. hybrd1 (molecular) stops at a relative step of sqrt(eps) on the scaled norm of all unknowns, and the atomic newton_dense at an absolute step of 1e-11; both stop before a trace unknown (H3+ at 1e-6, He++ at 1e-15, H+ at 1e-14 of its element) is resolved, and the acceptance then refuses the state (116.2, 116.3). With EXHALE_IEQ_TOL=1e-12 hp_zero_seed has no non-root and the two assertions of 116.2 pass. A change of every cell solve's path; for the user.
2. run_mode rows 2: the 10-step initialization snapshot it hands to a physical run is not a root of the chemistry (116.3); either item 1 or a longer relaxation of the fixture (20 steps hand off) settles it.
3. Negative ionization fractions accepted down to -1e-10 (ionization_fractions_physical, ftol): the mirror root x = -x* of a row whose electrons are its own ions passes as a root (116.3). A negative density is never physical; the admissible bound belongs at the resolution of the fraction.
4. D5 of (AO) (the phase5 He/H 0.65 helium-depleted outer wind) and the LHS 1140 b reruns with the frozen binary.

## 117. The energy-coupled LHS 1140 b states with every correction to date (binary bd2b93d3): the He 10830 line collapses at He/H 0.65 (2026-10-08)

Binary EXHALE_bd2b93d3.x carries sections 112-116 (acceptance repair, the oxygen guard of 113, the element-diffusion corrections of 115, the stage certification scale of 116). The three closures of section 114 rerun from their stored converged states (<case>_rerun_bd2b93d3; Python adapter with the section-110 A10 tables):

| Case | Stored / control (15440b7c) | bd2b93d3 |
|---|---|---|
| He/H 2.09 (phase3/energy_coupling_nlte) | log Mdot 7.890; F_He 5.37e7 g/s; T_match 350.3 K; width-matched EW 0.7025 | converged k = 6 |
| He/H 0.65, s = 1.3 route B | log Mdot 8.240; F_He 9.80e7 g/s; T_match 416.0 K; width-matched EW 1.1084 | converged k = 9 |
| He/H 0.65 (phase5/HeH0.65) | log Mdot 8.130 | STOPPED at k |

At He/H 2.09 the corrections after section 113 move the state by less than 1% (17c04ea2 gave EW 0.2914 / 0.2817). At

He/H 0.65 (route B) the helium escape flux falls by a factor 11.5, Mdot by 4.1 and the width-matched EW by a factor 25, to 0.045 %A against 1.108 observed: the EUV that matched the line at He/H 0.65 (s = 1.3, sections 98-102) now gives 4% of it. The ordering of the line with He/H has reversed (0.045 at He/H 0.65 against 0.28 at 2.09) relative to the defective runs. Every conclusion of sections 63-111 about the EUV and He/H that match the line is void; the search is to be repeated on the corrected model.

## 118. The element simplex of the ionization acceptance is bounded at the resolution of a fraction; the mirror root is refused, a candidate outside the simplex is projected and judged again, and the atomic sweep re-solves a cell without a root (item (AO) (iii), 2026-10-08)

Item (iii) of 116.6. Measured against the source of section 116 (control: gfortran 16.2 build of that source, md5 `b23415a1`, the same binary as the D3 build of section 116, so its regression runs in `~/.claude/jobs/f24ea156/tmp/ao/reg_d3/` are the control; measured runs in `~/.claude/jobs/f24ea156/tmp/neg/`; one thread, `EXHALE_IEQ_CLASS_RECORD=1`). Every number is MEASURED unless marked READ. No golden was refreshed.

### 118.1 The defect

**A negative density was admitted as a root.** `ionization_fractions_physical` (`ionization_equilibrium.f90`), the test that a candidate lies in the element simplex, admitted every fraction down to -1e-10 and every element's tracked species up to 1 + 1e-10 (`ftol`). The band was introduced on 2026-08-11 as a round-off band (READ: `md/hd209_metal_stagnation.md` section 5, where a band at the solver's xtol = 1.5e-8 was tried and measured worse on `examples/15`, the retried root being the better one); its value was not derived. The H+ row of a cell whose electrons are its own protons, n_H g (1 - x) - n_H^2 alpha_B x^2 (+ collisional terms), has beside its root x* the root -x* to O(x*^2). On `hydrostatic_column` (the faint field of `Log10 of EUV luminosity: 10.0`) x* is about 1e-11 above 1.1 R_p, inside the band: 70 cells of the written state carried that root, n(H+) = -8.9e-2 cm^-3 at r = 1.3106 R_p (x = -8.0e-12, the electron density of the cell negative with it), all class 1, and `Ion_species_adv.txt` carried the same 70 negatives. The golden written before section 112 has none (116.3).

**Why the mirror root was reached.** The atomic cell solve (`solve_ieq`: `newton_dense`, `hybrd1` only when the Newton reports failure) stops when its step is below 1e-11 (1 + max |x_i|), an absolute step on fractions of order one, and reports convergence: at x* ~ 1e-11 the attempts from the previous state and from the uncoupled balance stop at a relative error of order one (measure 4e-3 to 1, refused), and the attempt from all nuclei neutral reaches the mirror root, which the measure passes (its imbalance is 1.7e-16 of its gross rate in the fixture below) and the band admitted. The comment of `newton_solver.f90` and the solver paragraph of `docs/physics_overview/ionization.tex` said the fallback made the result "never worse than" hybrd1's; a Newton that stops on its step test is never checked by hybrd1, so that was false (both corrected).

### 118.2 The fix (`src/modules/radiation/ionization_equilibrium.f90`)

1. **The bound** (`ionization_fractions_physical`, lines 4484-4615). The rounding of a number keeps its sign, so a fraction whose root is zero or positive comes out negative only through a sum or difference formed at the scale of the element total, whose spacing is `fraction_resolution` = epsilon(1) = 2.2e-16 (`chemical_row_layout.f90`, the resolution of the floor of 116.3). A candidate is outside the simplex when a fraction is below -epsilon(1), or when the k fractions of one element sum above 1 + k epsilon(1): the neutral remainder 1 - s is a fraction of the same element with the same resolution, and forming s from k terms, each at most one, adds at most (k - 1) epsilon/2 of rounding (a fraction converted into another element's budget, HeH+ into helium, OH and H2O into hydrogen, counts as one term). `ftol` is gone.
2. **A candidate outside the simplex is projected onto it and judged again** (both branches of the sweep; molecular lines 3015-3036 and 3143-3147, atomic lines 3503-3530 and 3549-3553). The projection is `clamp_fractions_to_element_budget` (negative fractions raised to zero, an element's species scaled to its nuclei) and the measure is taken again at the projected state; it is a root when the measure says so (class 3). It is taken only when no attempt gives a root without projection, and before the constrained continuation (molecular), the fourth solve (atomic, item 3) and the bounded non-root. Why: hybrd1 resolves an unknown only to its step test on the scaled norm of all of them, so a root on a face (a stage at a fraction far below that, a neutral remainder of a fully ionized metal) comes back a little outside it; the band admitted those as they were, the bound alone left them non-roots (with items 1 and 3 only, `wasp_full_newton` had 11 non-root records at excursions of 3.6e-14 to 2.1e-13 of the element, and `mol_lyman_werner` 87 against 85). The mirror root projects to x = 0, where the row's imbalance is its whole production: no root.
3. **The atomic cell without a root is re-solved** (atomic branch, lines 3353-3420). The attempt loop of the atomic branch has a fourth attempt, made only when attempts 1-3 gave neither a root nor a projected root: the first bounded non-root of 1-3 is re-solved by hybrd1, whose step test is relative to the scaled norm of the unknowns and resolves a trace stage of that size; the result faces the same judge and takes the class its rank gives. A cell with a root among attempts 1-3 never reaches it.
4. **Admitted states are written on the face** (atomic extraction, lines 3698-3710; the metal neutral of both branches, 3717-3725). A state admitted with a fraction in [-epsilon, 0) or a remainder in [-k epsilon, 0) is on the face to the resolution of the composition; the stage fractions are raised to zero and the neutral remainders H I, He I and X I are taken as max(1 - s, 0), as the molecular branch already did for its species and H I, He I. No negative density leaves the sweep.

Considered and not taken: (a) refusing the candidate and keeping the cell's entry composition (class 6), which leaves the cell on a non-root where the physical root exists and is reachable; (b) changing the stopping test of `newton_dense` for every cell solve (116.6 item (i), still for the user): items 2 and 3 change only the cells that have no root without them.

The H-only branch of the sweep (no helium) has no admissibility test and no acceptance measure. MEASURED on an H-only variant of `hydrostatic_column` (He/H 0, 300 steps): no negative fraction is written, and the control and the final source give bitwise identical outputs. Not changed (118.5).

### 118.3 Tests

**Fixture** (src/tests/acceptance_scale, section (viii), new): an H-only cell at the state of hydrostatic_column at r = 1.31 R_p (T 1125 K, n_H 1.114e10 cm^-3, alpha_B 1.2e-12 cm^3 s^-1, g = 8.55e-25 s^-1), both roots in closed form: x+ = 7.9974e-12, x- = -7.9974e-12; the mirror root's measure 1.66e-16 (PIN: the measure alone cannot refuse it). Control: 3 FAIL (FUTURE_mirror_root_outside_the_simplex, FUTURE_negative_fraction_beyond_resolution_refused at -4 epsilon, FUTURE_neutral_remainder_beyond_resolution_refused at x = 1 + 1e-12); final: PIN 51/51, FUTURE 53/53 pass. The PIN rows (x+ admitted, -epsilon/2 admitted, 1 + epsilon admitted) pass in both.

**The case** (hydrostatic_column, 300 steps):

| | control | item 1 and 4 alone | final |
|---|---|---|---|
| cells with n(H+) < 0, written state and _adv | 70 and 70 | 0 and 0 | 0 and 0 |
| n(H+) at r = 1.3106 R_p [cm^-3] | -8.926128e-2 | +8.926128e-2 | +8.926128e-2 |
| last sweep | 504 class 1 | 504 class 1 | 504 class 1 |
| run-wide class-4 records | 2695 | 3266 | 0 |
| cell solves restarted (run-wide) | 3043 | 3795 | 1032 |

With the bound alone the 70 mirror cells became non-roots under the relaxation amnesty in the first eight steps (267 printed records at step 1 against 199 in the control), and a physical run that admits no non-root (run_mode) refused its first step: run_mode 2 -> 16 FAIL rows. With the fourth attempt every record of the run is a class-1 root (151200), and run_mode has 0 FAIL rows (its two rows of 116.3, the 10-step initialization snapshot handed to a physical run, pass: the snapshot is now a root everywhere). Against the golden written before section 112: n(H+) within 8.8e-11 where it is above 1e-30 of the hydrogen (360 cells), n(He+) within 5.6e-11, rho, v, p, T within 2.6e-10 (the control differed in sign in the 70 cells). The frozen ifx binary gives the same (no negative density, 151200 class-1 records, n(H+) within 6.2e-13 of the gfortran build).

**Suites** (all 56 under src/tests/*/run.sh, one thread, final source, against the control logs of section 116.5): every exit status as in the control except run_mode (2 FAIL rows -> 0); failing rows identical to the control in carrier_enthalpy_flux (3), grid_and_gates (4), steady_completion_flag (1), steady_selfconsistent_residual (1), ghost_reservoir_chemistry (1: the assertion drives hybrd1 directly and checks the strict simplex; its root x(He++) = -1.6e-15 is outside the bound the sweep now applies, -2.2e-16, and would be projected and judged there); acceptance_scale and acceptance_classes pass.

### 118.4 Regression cases (control -> final)

Outputs (Hydro_ioniz, Ion_species and their _adv pairs) and the last sweep's acceptance record; run-wide records of classes 1/3/4 where they move.

The atomic cases ran a build whose molecular branch predates item 2 there and the molecular cases the final build; each runs only its own branch, and `hydrostatic_column` is bitwise between the two builds.

| case | outputs | last-sweep record | run-wide classes (where they move) |
|---|---|---|---|
| `hydrostatic_column` | CHANGED: the 70 negative n(H+) now positive (118.3); rho, v, p, T within 7.5e-10 | 504 class 1 in both | class 1 148505 -> 151200, class 4 2695 -> 0 |
| `wasp_full` (exit 2 in both, as stored) | within 1.9e-9 (Hydro), 5.0e-10 (Ion) | 504 class 1 in both, measure within 6.5e-13 | class 1 68465376 -> 68465375, class 3 0 -> 1 |
| `wasp_full_newton` | within 2.3e-12 (Hydro), 6.2e-12 (Ion), 1.0e-11 (_adv) | 504 class 1 in both, measure within 1.1e-13 | marching ledger: class 1 24993360 -> 24993359, class 3 0 -> 1 |
| `mol_lyman_werner` | BITWISE | BITWISE | class 1 90216419 -> 90216417, class 3 0 -> 2 (class 4 85 in both) |
| `wasp_he23off` (exit 2 in both), `lower_profile`, `iontrans_metals`, `hp_zero_seed`, `hp_trace_seed`, `hp_front`, `mol_base_handoff`, `mol_metals`, `mol_diffusion`, `mol_ir_bands`, `mol_sec_ion`, `mol_carrier`, `oxygen_chemistry` | BITWISE | BITWISE | unchanged |

The class-3 records are the states that the band of 1e-10 admitted as they stood and the bound refuses: projected onto the face, they are roots. With items 1, 3 and 4 alone (no projection) `wasp_full_newton` had 11 non-root records and `mol_lyman_werner` 87 (118.2).

### 118.5 Frozen binary, documents, open

Frozen binary (ifx 2026.1, a copy of the final tree in `LHS1140b/models/.P2/build_simplexbound_ifx/`): `LHS1140b/models/.P2/EXHALE_de993804.x`, manifest `BINARY_MANIFEST_de99380418a5.txt` (differs from `EXHALE_bd2b93d3.x` in `ionization_equilibrium.f90` and a comment of `newton_solver.f90`).
Documents: `docs/physics_overview/ionization.tex` (the solver paragraph corrected; a new paragraph "The element simplex"; the class-3 row of the class table), `md/TO_BE_DONE.md` (AO) (iii) closed, (ii) settled with it.
Open, for the user: (a) the H-only sweep (no helium) has no admissibility test and no acceptance measure; nothing measured goes wrong there (118.2), but no judge stands behind it. (b) 116.6 item (i) in the molecular layouts: hybrd1

stops on a step relative to the norm of all the unknowns, which the O(1) fractions dominate, so a trace molecular ion (H3+) is left unresolved (the 11993 class-4 records of `hp_zero_seed` are unchanged by this section).

## 119. One description of the matching surface: the carrier base face open against the stated ghost partition, the ghost hydrogen census and one ghost temperature from the profile, Photochem's top at p_match, and the wind's H and H2 fluxes handed over separately (item (AN) items 2-4, user decisions 2026-10-07; 2026-10-08)

Design: `md/AN_overlap_structural_design_20261008.md` (the boundary-value problem on each side, the exchanged quantities, the closure variables; no public interface beyond the approved decisions changed: no input.inp key, no restart token, additive keys in EXHALE_resolved.out and the profile header). Diagnosis: `md/AN_matching_surface_diagnosis_20261007.md`. Every number is MEASURED unless marked READ. No golden was refreshed.

### 119.1 What changed

**Wind (Fortran).**

- `diffusive_photochemistry.f90`: where a handoff states the base H2 partition (`carrier_base_composition_imposed`, base.inp `q_H2_base` or the profile), face 0 of the carrier column carries the eddy, molecular and settling flux of H2 against the Dirichlet ghost, formed exactly as an interior face (`carrier_face_coefficients`, `jfirst = 0`; `carrier_residual` forms `Jf(0)`; `solve_carriers` adds only the cell-1 derivative `dJr(0)`, the ghost being data). `carrier_state` takes the ghost value of a stated carrier from the sweep-pinned species vector in every path. The enthalpy of that relative transport at face 0 is opened with it (`molecular_carrier_enthalpy_face_flux`). Without a stated partition the face stays closed (unchanged). Header sections 5 and 6 rewritten.
- The certified state's H2 flux through the base face is recorded (`carrier_steady_residual` optional `base_face_flux`, `carrier_record_base_face_flux`, called by the certification's `carrier_rows_of_state`) and written to EXHALE_resolved.out: `base_ghost_h2_nuclei_fraction`, `base_face_carrier_state`, `base_face_H2_flux_cgs` (molecules cmˆ-2 sˆ-1, outward), its `_adv`/`_dif` halves, `base_face_radius_cm` (`write_setup_report.f90`).
- Ghost hydrogen census (`input_read.f90 apply_lower_atmosphere_profile`, `parameters.f90 h_atomic_share_base`, `composition.f90 base_h2_nuclei_fraction`): with a profile, x2 = 1 - q_H n_tot/n_H, the nuclei per particle from the profile's rho, n_tot and X_El (agrees with the sum over the q_ columns to 2e-5 on the LHS profiles); the hydrogen of CH4, NH3, H2O goes onto H2. base.inp keeps 2q(1 + He/H)/(1 + q).
- One ghost temperature (`base_boundary.f90`): with a profile the two advective ghosts carry the profile's T at their centres (the conduction bath's temperature) and the volume-averaged pressure of the hydrostatic column at that T(r) (`hydrostatic_pressure_on_profile`, Simpson in 1/T, 32 panels); the well-balanced block keeps

chi_j = nhat T_j. The stale comment of `base_reservoir_temperature_at` (profiles "an isotherm at the clima skin temperature") rewritten.

- The profile reader accepts p_top_bar = p_match_bar (`lower_atmosphere_profile.f90`).

**Photochem side (Python).**

- `photochem_to_lower_profile.py`: the top FACE of the chemistry grid is placed at p_match (`steady_state_with_top_face_at_match`, moves of H ln(p_face/p_match), tolerance 1e-3 in ln p, the gas-giant stepper's own re-pinning suspended during the passes); the table ends at p_match with the face state continued from the two top cell centres (`level_at_top_face`); the chemistry runs on the climate column plus the column-balance rise up to the face as a persistent T(p) (`set_chemistry_temperature`); `--trial-flux-H2` splits the hydrogen flux as the wind does (H2 that part, atomic H the rest); an inflow of atoms deposits phi_in D0/2 (D0 = 36118.11 cmˆ-1) in the column energy balance with the shape of the column's own H + H (+M) rate (`recombination_heating_of_inflowing_atoms`); `--toa`/`--p-top-bar` refused. New header keys `trial_flux_H2`, `measured_flux_H2`, `recombination_heat_flux_cgs` (`lower_profile_schema.py`, which also accepts a table whose top is p_match).
- `conducted_heat_column_balance.py`: `chemical_heating` in `solve`; the overlying column is the wind's, no absorber above p_match (`include_gas_above_match` default False; True is the former treatment, a diagnostic).
- `element_flux_closure.py`: fourth variable Phi_H2 from the wind's `base_face_H2_flux_cgs`, damped update, residual |F_H2 - Phi_H2|/|F_H| <= tol; `--phi0-H2`; history columns `trial_F_H2 meas_F_H2 eps_H2`; the empty overlap window (top at the match) read as such, the steady window supplying the flux.

**119.2 Tests (RED on the source of section 118, GREEN now)**

- `carrier_base_face` (new): the base face against the analytic spherical diffusion flux of X = X_inf - C/r (error and order on two widths), the coefficients against their formulas, a closed face without a handoff. RED on the control tree (a copy of `build_simplexbound_ifx/src`): 5 FAIL (face coefficient 0, flux error 1.0). GREEN: 10/10, error 2.50e-5 at dr = 1e-2 and 6.25e-6 at 5e-3, order 2.00003.
- `base_hydrogen_census` (new): a profile with CH4 and NH3 whose top is p_match. RED: the control reader refuses p_top = p_match; with the top above it the ghost atomic share is 6.92e-3 against 1.235e-4. GREEN: 1.2351779e-4, and x2 = 1 - share to 1e-9.
- `matching_surface` (new, Python): the split imposition, the top-face pressure and level, the recombination heat (total and shape), the closure's split variable and history. GREEN 10/10 (new functions, no RED state).
- `conducted_heat_column`: section 9 (a top-node heating equals a top flux to 1e-9 K, the budget closes, the total is recorded, spread heating warms the match less than top heating) added; the transfer tests keep the gas above the match explicitly. All PASS.

All 63 suites (one thread): every exit status and every failing row as in the section-118 control (`carrier_enthalpy_flux` 3, `grid_and_gates` 15 lines, `ghost_reservoir_chemistry` 1, `steady_completion_flag` 1, `steady_selfconsistent_residual` 1, the rows of 118.3), the three new suites pass.

Adapter on the route-B configuration (scratch, trial H2 flux 3.6e7 g/s of 3.44e7, i.e. an inflow of 5.56e10 atoms cmˆ-2 sˆ-1): the face reaches the match in 2-3 passes (final offset 8.5e-4 in ln p); T_match 268.13 K against 271.16 K of the stored iteration under the same F_cond (no absorber above the match, plus F_chem = 0.199 erg cmˆ-2 sˆ-1); 3.9 min.

### 119.3 Regression cases reached (gfortran build, control = the live source before this section, md5 e2e1ca99)

Base.inp handoff without carrier transport (`mol_base_handoff`, `mol_metals`, `mol_lyman_werner`, `mol_diffusion`, `mol_ir_bands`, `mol_sec_ion`): BITWISE, acceptance record bitwise. `mol_carrier` (carrier transport, base.inp q_H2_base 0.84: the face is opened): CHANGED, rho within 7.5e-4 and T within 5.9e-4 over the physical cells. `hp_zero_seed`, `hp_trace_seed`, `hp_front` (the same, 10-step snapshots): CHANGED, rho within 4.6e-4, T within 2.7e-6, base-cell velocity 3 per cent, a trace ion 8.6 per cent. `lower_profile` (profile: ghost temperature and census): CHANGED, rho within 7.9e-3, T within 5.5e-3, ghost-0 T 1455.3 -> 1450.4 K.

### 119.4 Frozen binary

`LHS1140b/models/.P2/EXHALE_a3663329.x` (ifx 2026.1, copy of the tree in `build_matchsurf_ifx/`), manifest `BINARY_MANIFEST_a36633293829.txt` (differs from `EXHALE_de993804.x` in input_read, lower_atmosphere_profile, write_setup_report, composition, parameters, diffusive_photochemistry, base_boundary, certification and viscous_conduction).

### 119.5 The three LHS 1140 b states: the open face does not certify

Closures `<case>_structural_a3663329` beside each case (seeds: route B `xuv_x1p3_rerun_bd2b93d3/k09`, He/H 2.09 `energy_coupling_nlte_rerun_bd2b93d3/k06`, He/H 0.65 the stored `phase5/HeH0.65/k02`; first trials their measured fluxes; adapter without `--toa`; `--kmax 16`).

- **No closure passed k = 0.** He/H 2.09: the stationary route refused at outer pass 37 (the joint distance stalled): the carrier relaxation took no admissible step from pass 10 on, refused on He++ at the outer edge of the H2 front (cell 309, r = 2.00 R_p, |res| over its terms 0.71), carrier balance He+/He++ 0.62/0.89, the element He/H partition 8.6e-3 above its tolerance there (He/H falls from 2.09 at the base to 0.16 at the top and is not monotonic near r = 2). Route B: hydro info = 1 from outer pass 8, He+/He++ refusing at r = 1.2-1.4; stopped by hand at pass 10 (40 min a pass). He/H 0.65: refused at pass 60 (carrier rows refusing on He++ at r = 3.6 R_p from pass 38 on, H2 front x2 = 1e-2 at 27 R_p).
- **Attribution (scratch, the k00 inputs of each closure, the same seed and environment):** a build of this source with face 0 closed again (everything else of this section in place) certifies the He/H 2.09 wind in 5 outer passes and route B in 11; at He/H 0.65 it had not certified after 7 passes either (stopped; the same refusal as the bd2b93d3 rerun of section 117, TO_BE_DONE (AO) D5). The open-face binary seeded from the certified closed-face He/H 2.09 state follows the same path toward the same stall (H2 front x2 = 1e-2 at 1.15 -> 1.54 R_p by pass 7; stopped). The open face is what moves the He/H 2.09 and route-B winds into a regime the stationary route does not certify: with H2 supplied at the eddy rate the H2 layer reaches r = 2 R_p (He/H 2.09; 1.17 R_p with the face closed), and its outer edge, where H2 at 1e-2 of the hydrogen nuclei meets He++ at 1e-5 of the helium, is where the carrier

interval is refused. This appears to be the failure of (AO) D5 and the unresolved trace stage rows of (AP); that attribution is an interpretation and was not tested.

- **The matching surface itself (He/H 2.09, the uncertified k00 state):** ghost T 218.8 / 254.2 K against the profile's 218.1 / 252.6 K at the same pressures (before: 293.7 / 287.6 against 224.8 / 256.4); ghost x_H 6.0e-6 against Photochem's 5.9e-6 (before 4.0e-4); cell 1 x_H 5.4e-6, x_H2 0.1928 against the profile's q_H2 0.1926 (before 0.094 and 0.137, and 0.093 / 0.137 with the face closed and the census and ghost temperature of this section). He/H 0.65 (uncertified k00): ghost x_H 1.03e-5 against 1.01e-5, cell 1 x_H 6.8e-6 and x_H2 0.4344 against 0.4340 (before, the stored state of the oxygen-defect binary: 0.58 and 0.018). On the states the solver returned the jump at the face is gone with the open face and only with it; route B kept no usable state (its last two pass states carry an unconverged hydrodynamic solve). Figure `LHS1140b/models/.P2/ew_fit/an_struct/fig_matching_surface_structural.png`, tables `an_struct_tables.txt` (script `an_structural_compare.py`).
- Not measured, because no state certified: T_match, F_H, F_He, Mdot and the width-matched EW of a converged closure (`route_xuv_scaled/measure_case_a3663329.sh` and `run_ew_a3663329.sh` are in place).

Open, for the user: the open base face is physically required (decision 2) and makes the wind's H2 layer reach 1.5-2 R_p (27 R_p at He/H 0.65 with the unscaled spectrum); certifying those states needs the trace-stage resolution of (AP) and the He++ carrier failure of (AO) D5 first. Items 5 of (AN) (the p_match sensitivity) waits on that.

## 120. The cell solve stops when its Newton correction resolves every unknown relative to its own magnitude; the hydrogen-only sweep faces the same judge (item (AP), user decision 2026-10-08)

Item (AP) of `md/TO_BE_DONE.md` and 116.6 item 1, 118.5 (a) and (b). Every number is MEASURED unless marked READ. Control: the gfortran 16.2 build of the source of section 119 (md5 `52be5b2d`, the source of `EXHALE_a3663329.x`); measured: the same source with this section's change alone (md5 `b0bdc85c`); one thread, scratch copies of the regression cases with `EXHALE_IEQ_CLASS_RECORD=1` in `~/.claude/jobs/f24ea156/tmp/ap_d5/` (`reg_ctrl/`, `reg_ap1/`). No golden was refreshed.

### 120.1 The stopping test

**What was wrong.** The cell solve reported convergence on two tests, neither of which is a statement about each unknown. The atomic `newton_dense` stopped when its step was below 1e-11 (1 + max |x_i|), an absolute step on fractions of order one, or when the residual norm of the cell's network was below sqrt(eps): a trace stage whose root is of that order (H+ at 8e-12 of the hydrogen under a faint field) was returned at a relative error of order one. MINPACK `hybrd1` (the molecular and He 2^3S layouts) stops on a relative step of sqrt(eps) on the SCALED NORM of all the unknowns, which the O(1) fractions dominate, so H3+ and He+ in molecular gas were left out of balance by up to a few per cent of their own gross rates (116.2, 116.6): every H3+ refusal of the hp_* cases had this cause.

**The criterion** (`src/modules/nonlinear_system_solver/newton_solver.f90`, module header and `correction_resolves_every_unknown`). Every unknown is the fraction of the nuclei of one element held by

one species, and a trace species is as much a part of the answer as an O(1) fraction (its density enters its own cooling, the electrons of a shielded layer and the carried stages of the transport solve). The solve has converged when the Newton correction at the returned state resolves every unknown relative to its own magnitude,

```
|dx_i| <= tau |x_i| + epsilon(1)    for every i,
```

with tau the tolerance of the sweep (sqrt(eps) = 1.5e-8, or what `EXHALE_IEQ_TOL` names) and epsilon(1) = 2.2e-16 = `fraction_resolution`, the resolution of a fraction at its element total, below which no change of a fraction moves any sum of its element: the resolution floor of the acceptance measure (116.3). The correction dx = -J^{-1} f is the Newton estimate of the remaining error of each unknown, so a state that meets the test is resolved to about tau^2 of each unknown, far inside the acceptance of every kinetic row.

**The solver** (`damped_newton_to_every_unknown`, with an analytic Jacobian for `newton_dense` and a forward-difference one, the step rule of MINPACK fdjac1, for `newton_fd`; LU with partial pivoting kept for the line search). A trial x + lam dx (lam halved from 1, at most 20 times, 60 iterations) is taken when it lowers the residual norm (the Armijo test that stood) or when the simplified correction at the trial point, J^{-1} f(x + lam dx) with the Jacobian of x and measured relative to each unknown (D_i = 1/(|x_i| + epsilon(1))), falls below (1 - lam/4) of that of the step. The second test is what lets a trace unknown be resolved once the large rows sit at their rounding and the residual norm, which they dominate, can no longer decrease. The residual-norm stop (fnorm < ftol) is gone: a norm over rows of different sizes says nothing about each unknown.

**Where it acts.** `solve_ieq` (H-only, atomic H/He, atomic H/He + metals): `newton_dense` with the new test; on failure `hybrd1` from the same guess and then the Newton from hybrd1's result, kept only when it meets the test. Every `hybrd1` call of the sweep (the molecular attempts, the He 2^3S systems, the fourth attempt of the atomic branch) is now `hybrd1_resolving_every_unknown`: hybrd1, then the forward-difference Newton from its result, kept (info = 1) only when it meets the test; otherwise hybrd1's x and info are returned. The validation switch `EXHALE_FORCE_HYBRD1` skips both Newtons. Not changed: the constrained continuation (class 5; MINPACK hybrd in mode 2 on ln n) and the `_adv` post-processing solves (`post_process_adv.f90`), which are not the cell solve of the sweep.

### 120.2 The hydrogen-only sweep is judged like the others

118.5 (a): the H-only branch had no admissibility test and no acceptance measure. Now (`ionization_equilibrium.f90`, H-only branch): a candidate is a root when it lies in [-epsilon, 1 + epsilon] (`ionization_fractions_physical`, which already carried the H-only layout) and its one row, the H+ row, is within `ieq_res_tol` of its own gross rate or below the resolution floor (`normalized_reaction_residual`, whose layout already carried System_H); two starting points, the guess built from the previous state and the positive root of the quadratic balance in closed form (`System_H.f90`, `hydrogen_ionization_balance_root`: A x^2 + B x - g = 0 with A = n_H (a + b), B = g - n_H b, each sign of B in its cancellation-free form); a candidate outside the simplex projected and judged again (class 3); a cell without a root takes the bounded non-root (class 4) or, above the cap, keeps its entry composition (class 6). The ledger, the class record of each cell and the non-root streak are those of the other branches: their reset and their assembly were moved out of the He branch so that both sweeps fill one ledger.

MEASURED on the H-only variant of `hydrostatic_column` (He/H 0, 300 steps): 151200 class-1 records over the run and none of any other class (the control had no record at all), outputs within 1.0e-6 of the control (Hydro_ioniz column 7 at r = 1.32 R_p, 7.4219e-28 against 7.4219e-28 in its last digits; Ion_species within 5.0e-7).

### 120.3 Tests (RED: the solver of section 119; GREEN: this section)

- `src/tests/acceptance_scale`, section (ix), new: (a) the trace H+ root of the fixture of (viii), x* = 7.9974e-12, from x = 1e-6: `FUTURE_newton_resolves_trace_H+_root` RED x = 7.8125e-9 (976 x* , info 1), GREEN x = x* to 5.5e-15; (b) the same row with its root at 0.38: `PIN_newton_resolves_order_one_H+_root` passes in both; (c) the molecular cell of (iii) solved once as the sweep solves it: `FUTURE_molecular_solve_resolves_every_row`, the largest row against its own gross rate RED 1.25e-8 (H3+; the RED run calls hybrd1 alone, the routine of the control), GREEN 2.1e-16; (d) `FUTURE_closed_form_H_balance_root` (new routine, no RED state) GREEN to 1e-14. PIN 52/52, FUTURE 56/56.
- `src/tests/ghost_reservoir_chemistry`: the driver now solves the ghost as the sweep does (`hybrd1_resolving_every_unknown`). RED (hybrd1 alone): `ghost_root_inside_every_element_budget` FAIL, x(He++) = -1.6e-15, largest row 1.97e-7; GREEN: x(He++) = +4.46e-15, largest row 1.05e-16, every row PASS.
- `src/tests/grid_and_gates restart_lower_ghost_is_the_derived_boundary`: control 1.32e-14 against 1e-14 (FAIL since section 116), this section 3.7e-16 (PASS).
- `src/tests/run_mode`: 32 PASS, 0 FAIL (0 since section 118; the 20-step fixture of (AP) is not needed).
- Every other suite (all 63 under `src/tests/*/run.sh`, on a scratch copy of the tree with this section alone; the suites that read files outside `src/` rerun after those were linked into the copy): exit status and failing rows as in the control of section 119 (`carrier_enthalpy_flux` 3 rows, `grid_and_gates` the three stationary-evaluate rows, `steady_completion_flag`, `steady_selfconsistent_residual`, which need `EXHALE_STEADY_RUNLOGS`), except `physics_probe`, whose Python checks read files the copy lacks (it passes in the tree, 121.2); `krylov_and_dogleg` (which sets `EXHALE_IEQ_TOL`) passes.

### 120.4 Regression cases (control -> this section)

All seventeen default cases. Outputs (`Hydro_ioniz`, `Ion_species` and their `_adv` pairs) compared bitwise, else within 1e-3; run-wide class-4 records (non-roots) from the run's own ledger; the last sweep is all class 1 in every case and build.

| case | outputs | class 4 | hybrd1 calls | wall time |
|---|---|---|---|---|
| `hp_zero_seed` | within 6.2e-7 | 11992 -> 0 | 126814 -> 101304 | 14 -> 15 s |
| `hp_trace_seed` | within 6.2e-7 | 12192 -> 0 | 127722 -> 101304 | 14 -> 15 s |
| `hp_front` | within 9.1e-11 | 228 -> 0 | 151656 -> 151200 | 18 -> 21 s |
| `oxygen_chemistry` | within 1.6e-5 | 139 -> 1 | 7770715 -> 7769613 | 2221 -> 2313 s |
| `mol_base_handoff` | within 3.3e-10 | 24 -> 0 | | 3727 -> 3964 s |

| case | outputs | class 4 | hybrd1 calls | wall time |
|---|---|---|---|---|
| mol_carrier | within 3.3e-7 | 1452 -> 0 | | 3740 -> 3972 s |
| mol_diffusion | within 3.9e-8 | 24 -> 0 | | 3509 -> 3945 s |
| mol_sec_ion | within 2.8e-8 | 36 -> 0 | | 5329 -> 5371 s |
| mol_ir_bands | within 1.7e-9 | 0 -> 0 | | 8832 -> 9953 s |
| mol_metals | within 1.3e-9 | 0 -> 0 | | 8571 -> 9704 s |
| mol_lyman_werner | within 4.2e-9 | 85 -> 0 | 90216689 -> 90216000 | 11385 -> 11740 s |
| hydrostatic_column | rho, v, p, T within 1.1e-6; species above 1e-30 of their column within 5.7e-7 (below it, He III at 1e-232, the trace digits differ) | 0 -> 0 | | 12 -> 13 s |
| iontrans_metals | within 4.0e-8 | 0 -> 0 | | 1784 -> 1921 s |
| lower_profile | within 5.3e-10 | 0 -> 0 | | 6037 -> 7221 s |
| wasp_full_newton | within 1.7e-11 | 0 -> 0 | | 7205 -> 7871 s |
| wasp_full (exit 2 in both, as stored) | within 1.9e-9 | 0 -> 0 | | 15793 -> 17486 s |
| wasp_he23off (exit 2 in both) | within 2.4e-8 | 0 -> 0 | | 11474 -> 11909 s |

The one non-root left in oxygen_chemistry is a cold-start transient: cell 101 at step 203 (r = 1.023 R_p, 762 K), res 0.999, a candidate 1.6 outside the simplex, streak 1 sweep; the largest record of the control was also 0.999. In hydrostatic_column every cell solve is now a Newton root (the control had 1619 of 152711 solves fall back to hybrd1, the fourth attempt of section 118). The wall time rises by 1 to 20 per cent (the forward-difference Newton after every hybrd1 call; measured under a shared load of about 40 processes on lart4, so the figures are indicative).

### 120.5 The three LHS 1140 b k00 stationary solves

The k00 inputs of the three structural closures of section 119.5 (same seeds, same environment, 12 threads, scratch k00_<case>_ap1/ on lart3) with this section's build: He/H 2.09 follows the control's sequence pass for pass until it was stopped at pass 15 (the same worst rows at the same cells, the H2 front at the same radii, 1.54 R_p at pass 7 and 2.06 R_p from pass 9; refused on He++ at cell 309 from pass 10, the control's refusal), and the molecular cell solve at cell 324 (r = 2.30 R_p, the passes 7 and 8) still returns non-roots: 42 class-4 and class-6 records there, against 36 in the control; route B: hydro info = 1 at outer pass 5 (the control at pass 5 too); He/H 0.65: hydro info = 1 at pass 7. The stop of the cell solve is therefore not what refuses these states; section 121 says what does (with the change of section 121 the cell-324 records are gone).

### 120.6 Documents

docs/physics_overview/ionization.tex (the solver paragraph rewritten, Eq. eq:ion:stop; the element-simplex paragraph; the H-only layout under the acceptance classes), docs/EXHALE_user_manual.tex (EXHALE_IEQ_TOL, EXHALE_FORCE_HYBRD1), md/TO_BE_DONE.md (AP) closed, (AO) D5 updated.

## 121. Why the open-face LHS 1140 b winds were refused: an unbounded reconstruction of the trace ionization stages, and an outer alternation that oscillates between a helium-retaining and a helium-depleted wind (item (AO) D5 and section 119.5, 2026-10-08)

Item (AO) D5 of md/TO_BE_DONE.md and the open point of section 119.5. Every number is MEASURED unless marked READ. Builds (gfortran 16.2): control 52be5b2d (source of section 119, that of EXHALE_a3663329.x), section 120 alone b0bdc85c, section 120 + this section 0d817aa3; scratch ~/.claude/jobs/f24ea156/tmp/ap_d5/ (k00_<case>_<build>[_<route>]/: the k00 inputs of the structural closures of 119.5, the closure's environment, 12 threads). No golden was refreshed.

### 121.1 The refusing rows: a trace stage drained through its own outflow face

The control's He/H 2.09 k00 solve (phase3/energy_coupling_nlte_structural_a3663329/k00) kept no carrier step from outer pass 10 to its refusal at pass 37 ("NO ADMISSIBLE ADVANCE"), every interval refused on He++ at cell 309 (r = 2.003 R_p). Its last carrier assembly (output/carrier_row_terms.txt):

| cell | r [R_p] | x(He++) of the helium | He++ face flux out [cm^-3 s^-1] | production | loss | \|res\| / terms |
|---|---|---|---|---|---|---|
| 308 | 1.985 | 3.37e-6 | 6.3e-3 | 1.3e-3 | 1.9e-4 | 0.31 |
| 309 | 2.003 | 6.0e-12 | 4.76e-2 | 2.1e-3 | 3.3e-10 | 0.71 |
| 310 | 2.021 | 5.6e-5 | 2.70e-1 | 9.0e-3 | 3.5e-3 | 0.67 |

A cell holding He++ at 6e-12 of its helium (n = 9e-5 cm^-3) sent 4.8e-2 cm^-3 s^-1 of He++ through its outer face, 23 times its production. He+ had the same pattern (cells 309-312: 4.6e-3, 2.1e-2, 6.0e-2, 2.5e-2 of the helium) and so had the He/H 0.65 closure (He++ 5.5e-19 at cell 364, r = 3.62 R_p, between 5.7e-12 and 6.5e-11). No implicit step of positive density balances such a row, so every substep of the interval was refused.

**The defect.** The stage face fraction is the WENO3 reconstruction of the stage fraction (ionization_stage_face_fractions -> species_face_fraction -> Reconstruct_scalar, src/modules/states/Reconstruction.f90; the stationary route selects WENO3 for the whole run). Its smoothness indicators were b = dq^2 + dr^2: a squared jump of the scalar plus the squared cell width in code units, which do not carry the same units. For every scalar whose jumps are below dr (dr = 1.8e-2 at r = 2 R_p; every trace stage) both indicators are dr^2, the weights are the linear ones, and the face value is the unlimited three-point interpolant, which is not bounded by the two cells of the face. At the minimum of cell 309 it is x309 + 0.33 (x310 - x309) - 0.17 (x309 - x308) = 1.8e-5, three million times the cell's own value (species_face_fraction bounds a face to [0,1] only). The published weights (Yamaleev and Carpenter 2009, J. Comput. Phys. 228, 3025, Eq. 65, READ from the publisher PDF in references/) carry the square of the solution scale in the floor, eps = max(||u0^2||, ||(u0_xi)^2||) dxi^2; docs/physics_overview/numerics.tex already recorded that the code's floor drops that scale.

### 121.2 The fix (`Reconstruct_scalar`, WENO3 branch)

```
eps_j = (dr_j q_s,j)^2 ,   q_s,j = max(|q_j-1|, |q_j|, |q_j+1|) ,
b0 = dq+^2 + eps_j ,       b1 = dq-^2 + eps_j ,
```

with S = 1 where a stencil of zeros gives b = 0. The face values then commute with a scaling of the scalar (q -> c q gives c times the faces). The scale is the local one and not a column norm as in the published prescription, because a stage fraction spans many decades along one column (He++ from 9e-17 to 7e-4 of the helium on the refused He/H 2.09 state) and a column norm would set the floor by its largest value. The primitive reconstruction of the hydrodynamic variables (`Reconstruct`) keeps eps = dr^2 (not changed here: it changes every result of the code; for the user).
**Test** (`src/tests/species_face_flux`, new `test_trace_scalar_reconstruction`): RED = the control's `Reconstruction.f90` (scratch tree `red_tree/`), GREEN = this section.

- `FUTURE_trace_scalar_faces_scale_with_the_scalar` (a smooth O(1) profile and the same times 1e-8): RED 3.9e-11 relative departure, GREEN 2.2e-16.
- `FUTURE_trace_minimum_outflow_face_bounded_by_its_cell` (the triple 3.37e-6, 5.98e-12, 5.56e-5 of cells 308-310, outward mass flux): face fraction RED 1.78e-5, GREEN 0 (at most the cell's 5.98e-12).
- `PIN_order_one_scalar_faces_on_the_profile`: 9.9e-6 in both. Suite: 18/18 (RED 16/18). `src/tests/physics_probe` (`weno3_reconstruction_order.f90`) asserted that `Reconstruct` and `Reconstruct_scalar` put the same face on one profile and that its local copy of the face formula is the scalar routine; with the two floors now different by design both rows read 2.2e-10 against the new source. The local copy takes the scalar floor as an option (`weno3_local_copy_matches_production`, 0 against `Reconstruct_scalar`) and the vector row compares `Reconstruct` with the primitive form (`weno3_vector_matches_primitive_formula`, 0); the order tables are unchanged.

### 121.3 The trajectory: the outer alternation oscillates between two winds

The fix removes the drained minima and the cell-324 non-roots of section 120.5, but not the refusal: with it (build `0d817aa3`) the He/H 2.09 k00 solve fails the hydrodynamic solve at outer pass 8 (hydro info = 1), route B at pass 5 and He/H 0.65 is refused on He++ rows whose physical terms are 1e-12 cm^-3 s^-1 at r = 6.35 R_p (from pass 15). The cause is the order of the alternation itself. Each outer pass solves the wind at a fixed composition, then relaxes the element composition to the fixed point of the element operator AT THAT WIND, then the carriers. On the control's He/H 2.09 k00 run (`stage_state_p<k>_*.txt`, the composition each pass hands on, MEASURED):

| pass | He/H at 2.0 R_p (wind / after the element step) | T at 1.5 R_p [K] after it | x(H2) at 1.5 R_p after it |
|---|---|---|---|
| 1 | 0.50 / 0.30 | 3783 | 5e-6 |
| 3 | 0.19 / 5.5e-4 | 1824 | 7e-6 |
| 4 | 5.5e-4 / 1.30 | 10008 | 1e-8 |
| 5 | 1.30 / 0.91 | 4473 | 5e-6 |
| 6 | 0.91 / 1.21 | 5272 | 3e-6 |

| pass | He/H at 2.0 R_p (wind / after the element step) | T at 1.5 R_p [K] after it | x(H2) at 1.5 R_p after it |
|---|---|---|---|
| 7 | 1.21 / 1.9e-8 | 1321 | 1.4e-2 |
| 8 | 1.9e-8 / 1.35 | 7794 | 8e-7 |

The element fixed point of one wind empties the helium above 1.45 R_p (He/H 1.4 -> 6e-5 at 1.5 R_p, 1.0 -> 1e-11 at 3 R_p: diffusive separation in a wind of 5 m/s at 1.4 R_p, READ from the state); the wind of that composition loses the helium photoionization heating (T 3600 -> 1300-1400 K) and, without He+ to destroy it, H2 survives outward (the front at x2 = 1e-2 walks 1.15 -> 1.54 -> 2.06 R_p, the 2 R_p layer of 119.5); the element fixed point of that cold wind puts the helium back (He/H 1.35), and so on: a period-2 to period-4 oscillation of the Picard alternation between a helium-retaining and a helium-depleted wind, with the inner map distance of the element step 0.13-0.95 over the first ten passes. The carrier refusals of 121.1 are where the oscillation strands the carried stages (steep fronts of trace stages, which the unbounded faces then drained).

**Damping.** Run with `EXHALE_DIFF_OMEGA=0.5` (the starting under-relaxation of the element update; the closures pin 1.0 in their `closure.json`, the code's own start is 0.5) alone, the He/H 2.09 k00 solve still refused at pass 10 (carrier step refused on H+ at r = 3.0 R_p; element map distance 0.2-0.8 at every pass). With the carrier movement bound also held at 0.04 (`EXHALE_CARRIER_TRUST_HOLD=0.04`; by default it doubles from 0.01 to 0.32 over the passes), the alternation contracts: element map distance 0.15 at pass 16, 6.6e-2 at 18, 1.4e-2 at 23, 9.6e-4 at 30, the worst gated row 1.45e-3 at pass 30, and the state is ACCEPTED and CERTIFIED at outer pass 44 (He/H 2.09 -> 0.19 from base to top, smooth; H2 front x2 = 1e-2 at 1.227 R_p, against 1.17 R_p with the face closed; steady Mdot median 2.256e7 g s^-1 with the trial fluxes of k00). The 2 R_p H2 layer of 119.5 was therefore the oscillation, not the open face's solution.

**Attribution** (the same recipe, the same k00 inputs, scratch): the control build (the source of section 119) and the build of section 120 alone are ACCEPTED at outer pass 44 as well, along the same element map distances to the printed digits for the first ten passes. The three certified states agree within 4.8e-4 (Hydro_ioniz, the cooling column; Ion_species 4.4e-4), steady Mdot medians 2.2557662e7, 2.2557662e7 and 2.2557670e7 g/s. So the outer alternation, and not the defects of sections 120 and 121, is what kept the He/H 2.09 open-face wind from certifying; the reconstruction defect is what held the refused control run at no admissible advance once the oscillation had left steep fronts of trace stages.

### 121.4 Regression cases (section 120 -> this section)

Section 120 build -> this section's (gfortran, one thread, the seventeen default cases). BITWISE in sixteen of them: `hp_zero_seed`, `hp_trace_seed`, `hp_front`, `hydrostatic_column`, `oxygen_chemistry`, `mol_base_handoff`, `mol_metals`, `mol_lyman_werner`, `mol_diffusion`, `mol_ir_bands`, `mol_sec_ion`, `mol_carrier`, `lower_profile`, `wasp_full_newton`, `wasp_he23off`, `wasp_full`. CHANGED: `iontrans_metals`, the stationary atomic case whose ionization stages are carried (WENO3 stage faces): rho, v, p, T within 5.0e-4, He II within 7.5e-4, He III within 9.3e-4, He I within 1.0e-2 at 1.2-3 R_p (where the helium is ionized and He I is the minor stage), every species below 1.2 R_p within 5.6e-5; certified in both builds.

### 121.5 What is left, and the proposal

1. **Route B** (He/H 0.65, s = 1.3) is not certified by any route tried: the default and omega 0.5 routes fail the hydrodynamic solve at outer pass 3-5 (info 1, mass row 1.2-1.6; every build, the control included); omega 0.25 with the carrier bound held at 0.02 keeps the hydrodynamic solve but the element map distance stays at 0.3-0.9 through pass 16. The closure with the frozen binary stopped at its k00 (outer pass 4) when the charge-exchange guard of the constrained continuation reported a thread holding no cell (`cx_add_to_turnover`, `error stop`), during the JFNK of the pass after the failed hydrodynamic solve. `cx_set_cell` stores the cell temperature as the tag that guard tests (`cx_cell_T < 0` means "no cell"), so a cell temperature below zero in a failed JFNK iterate would trip it with that message; this was NOT verified (no traceback), and the gfortran runs of the same route did not abort.
2. **He/H 0.65** (`phase5`): the element map distance stays at 3e-2 to 5e-2 per pass and the carried He+ rows at 0.4-0.6 of their terms near r = 1.2 R_p (cells 217-226) at omega 0.5 and 0.25 alike, and the hydrodynamic solve fails at outer pass 11 (closure, 0.5/0.04) or 17 (0.25/0.02). With the face closed (section 117) the same case did not certify either: carrier refusals at r = 3.05-3.9 R_p in a wind at 1600-1700 K whose helium is almost all neutral (He+ 2e-7 and He++ 4e-10 of the helium, production of He++ 6 to 13 times its loss on frozen carriers). The original D5 description (a helium-depleted outer wind) is the depleted branch of the oscillation of 121.3.
3. **Proposal.** The outer route alternates three solves (wind at fixed composition, element composition at a frozen wind to its fixed point, carriers at a frozen wind under a movement bound) whose coupling through the helium (its heating, its mean mass, its settling) is strong enough on these states to make the alternation oscillate. Two remedies, in order of effort: (a) a movement bound on the element update like the one the carriers have, held rather than grown, with omega 0.5 (what converged He/H 2.09; a route setting, not an equation, and the closures of section 122 run with it); (b) the element and stage rows as unknowns of the stationary Newton (`Coupled carrier solve` extended to the carried stages, which it refuses today, and to the element rows), the one route that does not depend on the alternation contracting. Also for the user: the primitive reconstruction (`Reconstruct`) keeps eps = dr^2 (121.2), and the charge-exchange guard of item 1 should not use the cell temperature as its sentinel.

### 121.6 Documents

`docs/physics_overview/numerics.tex` (paragraph "WENO3 of a transported scalar", Eq. eq:weno_scalar_floor; the primitive floor stated with the published Eq. 65), `md/TO_BE_DONE.md` (AO) D5.

## 122. The three LHS 1140 b structural closures with binary a76da494 and the damped outer alternation: He/H 2.09 certifies with the matching surface continuous; route B and He/H 0.65 stop at k = 0 (2026-10-08)

Frozen binary `LHS1140b/models/.P2/EXHALE_a76da494.x` (ifx 2026.1, a copy of the tree in `build_cellsolve_ifx/`), manifest `BINARY_MANIFEST_a76da49445a5.txt` (differs from `EXHALE_a3663329.x` in `newton_solver.f90`, `System_H.f90`, `ionization_equilibrium.f90` and `Reconstruction.f90`: sections

120 and 121). Closures <case>_structural_a76da494 beside the a3663329 ones, the same seeds, first trials, adapter arguments and environment as section 119.5, except the outer alternation: EXHALE_DIFF_OMEGA=0.5 and EXHALE_CARRIER_TRUST_HOLD=0.04 (closure.json exhale_env; section 121.3). Run on lart3. Every number is MEASURED.

### 122.1 He/H 2.09 (phase3/energy_coupling_nlte_structural_a76da494)

Every wind solve certified: k00 at outer pass 44 (from the closed-face seed), k01-k04 in 23, 22, 20 and 18 passes, the element map distance contracting by a factor 0.7 per pass in each. The k = 4 chemistry was refused by the Photochem adapter at its default budget (--blocks 400: the column did not re-reach steady state after the first move of its top face to the match); the closure was resumed at k = 4 with --blocks 1600 (launch_resume.sh, seeded with k03; a first resume took the closed-face seed, which the driver uses as given, and was stopped, launch_k04_seeded_wrong.out). At the writing the k = 5 chemistry was still integrating its column after the same top-face move (more than 85 minutes); the last certified iteration is k04.

| k | trial F_H, F_He [g/s] | measured F_H, F_He [g/s] | eps_H, eps_He | F_cond trial / measured [erg cmˆ-2 sˆ-1] | T_match [K] |
|---|---|---|---|---|---|
| 0 | 1.645e7, 1.987e7 | 1.3028e7, 9.529e6 | 0.263, 1.085 | 4.727 / 2.242 | 283.37 |
| 1 | 1.474e7, 1.470e7 | 1.3016e7, 9.468e6 | 0.133, 0.553 | 3.485 / 2.266 | 262.26 |
| 2 | 1.388e7, 1.208e7 | 1.3009e7, 9.435e6 | 0.067, 0.281 | 2.875 / 2.278 | 251.04 |
| 3 | 1.344e7, 1.076e7 | 1.3006e7, 9.417e6 | 0.034, 0.142 | 2.577 / 2.285 | 245.30 |
| 4 | 1.322e7, 1.009e7 | 1.3004e7, 9.409e6 | 0.017, 0.072 | 2.431 / 2.288 | 242.41 |

The wind's escape hardly moves with the trial (measured F_H 1.30e7, F_He 9.41e6 g/s from k0 to k4); the damped closure approaches it at omega = 0.5, and k4 is within the tolerance 0.05 on H and the temperature (T_match moved 2.9 K < 6 K) but not on He (0.072) or on the hydrogen split (eps_H2 0.074). The wind's H2 flux through the base face is 5.62e11 molecules cmˆ-2 sˆ-1 outward, 2.4 times the hydrogen escape in nuclei, with the atomic H flowing down into the column (the split Phi_H2 = 3.15e7 g/s on H2, -1.83e7 g/s on atomic H at k5): H2 is mixed up by the eddy flux, dissociated in the wind, and its atoms return to the column, where their recombination heat (2.24 erg cmˆ-2 sˆ-1 at k4) is as large as the conducted flux (2.43).

**State k04** (certified, measure_case_a76da494.sh, run_ew_a76da494.sh): log10 Mdot = 7.350 (steady median 2.241e7 g/s), F_H = 1.300e7 g/s, F_He = 9.41e6 g/s at the match, T_match = 242.4 K; H2 front x2 = 0.5 at 1.099 R_p, 1e-2 at 1.227 R_p; He 10830 EW 0.1465 %A unbroadened, width-matched 0.1416 %A (f = 22.06 km/s), red/blue 7.7 (observed 1.108 %A). Collisional validity: exobase 9.5 R_p below the sonic point 22.6 R_p (Kn up to 5.7 in the acceleration region), the hydrodynamic result unvalidated there, as for every LHS 1140 b state so far.

Against the closed-face closure of the same configuration (section 117, bd2b93d3, k06): log Mdot 7.560 -> 7.350, F_H 1.645e7 -> 1.300e7, F_He 1.987e7 -> 9.41e6 g/s, T_match 284.6 -> 242.4 K, width-matched EW 0.2790 -> 0.1416 %A. The open face halves the helium escape and the line.

**The matching surface** (an_struct/fig_matching_surface_structural_a76da494.{png,pdf},

an_struct_tables_a76da494.txt, script an_structural_compare_a76da494.py): ghost T 204.7 / 225.5 K against the profile's 204.4 / 224.8 K at the same pressures; ghost x_H 4.74e-5 against the profile's q_H 4.20e-5 / 4.56e-5; cell 1 x_H 4.70e-5, x_H2 0.1928 against q_H2 0.1926. Before (bd2b93d3, the closed face): cell 1 at 323 K, x_H 0.094, x_H2 0.137 against 284.6 K, 6.8e-6 and 0.193. On a certified closure the temperature and the hydrogen partition are now continuous across the matching surface.

### 122.2 Route B (He/H 0.65, s = 1.3) and He/H 0.65 (phase5)

Both stopped at k = 0, with EXHALE exiting 1 on the charge-exchange guard of the constrained continuation (cx_add_to_turnover: "called on a thread that holds no cell"), route B during the JFNK of outer pass 4 after the hydrodynamic solve of pass 3 had failed (info 1, mass row 1.6), He/H 0.65 at pass 18 after failed hydrodynamic solves from pass 11 (mass row 1.5-1.9). Scratch runs of the same k00 inputs with the same binary at omega 0.25 and the bound held at 0.02 were refused at pass 27 (route B, element map distance 0.3-0.9 throughout) or failed the hydrodynamic solve in 5 of 30 passes (He/H 0.65). Neither state is certified, so T_match, the fluxes, Mdot and the EW of these two are not measured; section 121.5 states what is left.

### 122.3 Open

(AN) item 5 (the p_match sensitivity) can run on He/H 2.09; route B and He/H 0.65 wait on the coupled solve of 121.5. The k = 5 chemistry of the He/H 2.09 closure was running at the writing; its outcome is in phase3/energy_coupling_nlte_structural_a76da494/launch.out.

## 123. Energy continuity at the open matching surface measured; the WENO3 floor of the primitive variables tested (2026-10-08)

Two measurements ordered by the review of 2026-10-08 (user decisions 2 and 3 of md/session_handoff_20261008_rev1.md), no source change.

**Total energy flux through the matching face** (md/AN_energy_continuity_20261008.md, scripts LHS1140b/models/.P2/ew_fit/an_energy/). One sign (outward positive), one reference (sensible enthalpy from T = 0 with the wind's caloric EOS; chemical zero at H2 v = 0 J = 0, He, CH4, NH3, H2O and the free electron, so atomic H carries D0/2), erg cm^-2 s^-1 at R0, He/H 2.09 open face, closure iteration k04: wind side -4.6786, column side -4.6746, difference -0.0041. The difference splits into closure residuals that vanish at convergence (conduction +0.143, atomic-H chemical energy -0.156) and structural omissions of +0.0092. What the column's energy equation does not book is a net heating of +0.018: the relative H2/H sensible enthalpy at the face (-0.0279 on the wind side, 5.20e11 cm^-2 s^-1 of diffusive H2 flux, the code's own formula), the change with the 57 K warming of the advective and interdiffusion terms (-0.0046, -0.0030) and the C/N atomization (-0.0020). That is 0.80% of the conducted heat (2.288) and 0.28% of the wind's XUV heating (6.59); its effect on T_match is at most 0.44 K (0.67 K for the H2/H term alone) against the closure tolerance 6 K and a last-step change of 2.9 K. The recombination heat is counted once, in the column (F_chem); the wind's energy row has no formation-energy density. On the closed face (bd2b93d3, k06) the difference is +0.130. This is an energy-conservation error, not a stated approximation. User decision (2026-10-08): it

is NOT implemented now. The bound stands as the documented limitation of the interface: net +0.018 erg cmˆ-2 sˆ-1, 0.8% of F_cond, T_match shift below 0.7 K; it scales with the H2 face flux and Mdot and is to be re-measured on every other state. The corrections (a new field `base_face_carrier_enthalpy_flux_cgs` carried by the closure and deposited by the adapter, or two perturbation terms built from F_H and F_He in `conducted_heat_column_balance.py`) are specified in the note and will be put in once the closure converges. Two defects found by the measurement, to be fixed after the diagnoses of the day: F_H (wind component 1) includes the C, N and O mass (0.44% on k04) and is converted with the hydrogen mass in the closure and in `photochem_to_lower_profile.py` (lines 286, 301), so the H flux handed to Photochem is 0.44% high (F_chem 0.3%); and the column's conduction is formed from the full T instead of T - T0 (the climate gradient at the 0.85 bar kink injects 0.10 into the bottom flux; not at the match).

**WENO3 floor of the primitive variables** (`md/weno3_primitive_floor_tests_20261008.md`, suite `src/tests/weno3_primitive_floor/`, figure `docs/figures/weno3_primitive_floor_20261008.png`). The test carries a copy of the face formula and equals `Reconstruct` bit for bit with the present floor eps = drˆ2. Against the scale-aware floor eps = (dr q_s)ˆ2 (the one of the transported scalars, section 121): rescaling invariance fails (2.3e-4 to 6.0e-3 on smooth and stored profiles, 1.7e3 on a step) against 4e-16; at outer-layer amplitude (1e-12) a step overshoots by 0.169 of the jump with a negative face from ratio 1e2, repaired only by the positivity limiter; the smooth order (3.00) and the stored k04/k06 profiles (0 faces outside the neighbours, 0 negative) do not discriminate; the face values would change by at most 3.4e-7 (rho), 4.8e-9 (p), 1.4e-6 (v; 1.9e-3 on two base faces of k06). With `Well balanced: True` the pressure faces are second order for every floor (1.99 against 3.00), as in the cited Kaeppeli & Mishra (2016) design (READ: abstract and section 2.1.3). User decision (2026-10-08): change the primitive floor to the scale-aware form, together with the Fortran work of the oscillation/coupling decision, measured on the regression matrix and on restarts of k04 and k06 before any golden refresh.

## 124. The Photochem adapter runs on its stated step budget and retries a stalled steady-state pass once; the He/H 2.09 open-face closure resumed at k = 5 and converged there (user decisions 2026-10-08; `md/k05_chemistry_failure_20261008.md`)

The k = 5 chemistry step of the He/H 2.09 open-face closure (`LHS1140b/models/.P2/ew_fit/phase3/energy_coupling_nlte_stru`) was refused after the top-face pass that followed the warmed-column solve: the BDF integrator locked into a period-2 limit cycle on the condensation switch of the water condensate at level 27 (0.70 bar), and Photochem's own step cap ended the pass (diagnosis in the note above). User decisions (2026-10-08): fix the adapter's two defects and resume the closure at k = 5 with unchanged inputs; do not swap inputs to avoid the stall.

**Adapter** (`src/utils/photochem_to_lower_profile.py`; Python only, no Fortran touched, binary a76da494 unchanged).

1. The step budget. Every steady-state pass now runs through `robust_steps_to_steady_state`, which sets `pc.gdat.max_total_step = 100*blocks` before `initialize_robust_stepper`. Before, the adapter counted `--blocks` x 100 calls of `robust_step` but never set Photochem's cap (`gasgiants.py`, default 100000), so any `--blocks` above 1000 had no effect and the run record stated a budget that was not used. Setting the cap was chosen over refusing `--blocks > 1000` because `--blocks` is documented as the budget (the module docstring,

closure.json with --blocks 1600) and a refusal would have changed the closure's settings.

2. The single stall retry (steady_state_with_stall_retry). A pass that ends without a steady state (budget spent, or Photochem's give_up) is integrated once more from the state the stalled integration holds, at relative tolerance STALL_RETRY_RTOL = 1e-4; the stated rtol is restored afterwards. It is accepted only under the unchanged criteria: Photochem's own steady state, then the top-face tolerance, the measured-flux and the elemental-conservation checks. A second stall refuses. It applies to the first solve, the new-temperature (warmed-column) solve and every pass of steady_state_with_top_face_at_match. A retry is logged as steady_state_retry rtol=1e-04 at pass <label> (labels such as top face 1 after new-temperature solve 1) and written to the profile header as the additive key # steady_state_retry ... (lower_profile_schema.write_profile, written only when a retry happened); reached_steady_state keeps its meaning. The Fortran reader ignores the key (lower_atmosphere_profile.f90, case default) and the Python readers skip # lines. The refusal messages now say that the retry also stalled.

**Tests** (src/tests/matching_surface/, class StallRetryTests, six tests; a stand-in model that follows Photochem's robust_step contract, including the give-up at total_step_counter > max_total_step). RED on the adapter before the change (3 failures, 3 errors: the stall-then-steady pass refused, max_total_step 100000 instead of 160000 at --blocks 1600, no header line); GREEN after: 16/16 in the suite. conducted_heat_column PASSED, element_flux_closure 10/10 OK, physics_probe/jupiter_radius_python_tools.py passed.

**The k05 stall, reproduced with the fixed adapter** (scratch /home/kiseon/.claude/jobs/f24ea156/tmp/k05_fix/, lart4, the k05 command of k05.failed.20261008/adapter.log with its in-memory trial values, the closure's environment). With --blocks 1600 (repro/) the run follows the stored path digit for digit up to the warmed face pass 1 (+5.467e-2 in ln p), which now runs on its stated 160000-step budget; it reached steady state inside that budget without the retry (no steady_state_retry line), so the limit cycle ended somewhere past the old 100000-step cap (the step count of the pass is not logged; about 1.2e5 from the wall time, an estimate). Wall time 8568 s (2 h 23 min). Handed-over state: T_match 240.98 K, q_H(match) 4.883097e-5, measured H / He flux 1.312217e7 / 9.760390e6 g/s (imposed 1.311433e7 / 9.748631e6); against the diagnosis's f_retry_rtol1e-4 profile the largest relative difference over the 202 levels is 1.5e-7 in T and 1.7e-7 in q_H. To exercise the retry on the real stall, the same command with --blocks 100 (repro_blocks100/, a 10000-step budget, the budget of f_retry_rtol1e-4): steady_state_retry rtol=1e-04 at pass top face 1 after new-temperature solve 1, stalled at t = 6.94e6 s after 10000 steps, STEADY after 308 steps of the retry; face passes 2 and 3 at +6.07e-3 and +7.66e-4; the same state at the match (T 240.98 K, q_H 4.883097e-5); header line written; wall time 803 s.

**Closure resumed at k = 5** (launch_resume_k05.sh, lart4, closure.json unchanged: frozen binary a76da494, damped alternation 0.5/0.04, --blocks 1600; seeded from k04/output; the refused k05 kept as k05.failed.20261008; --kmax 11 to bound the resume at six further iterations). On a resume the driver rebuilds the trial values from closure_history.txt, which stores them to 11 significant digits (Phi_H 13114328.6265 against 13114328.62627739 in the first run, a relative difference of 2e-11). That moved the integrator off the path of the first run: the k05 chemistry reached steady state in every pass without a retry (first solve STEADY at block 23; warmed column STEADY at t = 7.9e17 s; face passes +5.52e-2, +6.12e-3, +7.71e-4 in ln p) in 4.7 min. MEASURED, k = 5:

| quantity | k = 4 | k = 5 |
|---|---|---|
| T_match [K] | 242.411 | 240.983 |
| F_H measured / trial [g/s] | 1.300394e7 / 1.322472e7 | 1.300305e7 / 1.311433e7 |
| F_He measured / trial [g/s] | 9.408835e6 / 1.008843e7 | 9.404539e6 / 9.748631e6 |
| Phi_H2 measured / trial [g/s] | 3.194260e7 / 3.098441e7 | 3.194096e7 / 3.146350e7 |
| F_cond measured / trial [erg cm^-2 s^-1] | 2.288208 / 2.430878 | 2.289832 / 2.359543 |
| eps_H | 0.01698 | 0.00856 |
| eps_He | 0.07223 | 0.03659 |
| eps_H2 | 0.07368 | 0.03672 |
| conducted-heat residual | 0.0624 | 0.0304 |
| T_match change [K] | 2.884 | 1.428 |

CONVERGED at k = 5: every residual below tol 0.05 and the T_match change below 6 K; wind solve info = 0 and CERTIFIED in the wind (r >= 1.2 R_p; worst row below it 1.24e-5, He+ carrier balance at r = 1.08 R_p, reported, not gating); log10 Mdot = 7.350, He/H at the match 2.0923. The k05 profile at the match: T 240.98 K, q_H 4.883089e-5, q_H2 0.1926145, measured top fluxes H 1.312137e7 and He 9.753697e6 g/s against the imposed 1.311433e7 and 9.748631e6.

Width-matched He 10830 EW of k05 (`route_xuv_scaled/measure_case_a76da494.sh`, lart4): solution EW 0.1464 per cent-angstrom without the kernel, 0.1416 with it (f = 22.06 km/s, red-pair FWHM 0.841 A); the _adv record 0.1499 / 0.1449 (status conditional). k04: 0.1465 / 0.1416. Collisional record unchanged: exobase 9.495 R_p below the sonic point 22.642 R_p, max Kn 5.75 in the heating region, so the hydrodynamic result remains unvalidated in the sense of (AQ) G2. The gates of (AQ) still apply: G1 (the thin band transfer is outside its validity, k05 max absorbed fraction 0.533 at the NH3 rotational 100-160 cm^-1 bin) and G3 (the interface energy bound of section 123) are not resolved by the convergence.

New item (AS) in `md/TO_BE_DONE.md`: the water-condensate layer of the column (4.4-0.7 bar, 2.9e22 cm^-3 condensed water at 1.7 bar, about 500 times the gas number density) needs a review of Photochem's condensation and sedimentation before the column below 1 bar is used for any result. Also fixed: a forbidden word in (AR) item 1. Docs: `docs/physics_overview/molecules_metals.tex` (step budget and retry) and `docs/EXHALE_user_manual.tex` (the optional header key); both PDFs rebuilt.

## 125. Trial admissibility of the stationary residual, recovery of the outer alternation to the last converged wind, and the WENO3 primitive floor at the domain scale (plan rev2 step 2, user approval 2026-10-09)

Plan: `md/PLAN_20261008_corrected_model_rev2.md` step 2 and section 3 (checkpoint). Background: `md/osc_diagnosis_20261008.md` sections 3-5, `md/weno3_primitive_floor_tests_20261008.md`. Source before: the frozen `EXHALE_a76da494.x` (its copy in `LHS1140b/models/.P2/build_cellsolve_ifx/src/`). Frozen binary after: `LHS1140b/models/.P2/EXHALE_a68121e6.x` (manifest `BINARY_MANIFEST_a68121e60694.txt`); it

supersedes EXHALE_7fe0b827.x, which carried the local WENO3 floor refused in 125.3 and was used only for the reproductions of 125.5 marked as such. Every number below is MEASURED on lart4 unless marked READ.

### 125.1 Trial admissibility (steady_newton.f90, UW_conversions.f90, charge_exchange.f90, constrained_chemical_equilibrium.f90)

- eval_residual screens every cell of the unpacked conserved state with admissible_conserved_cell right after unpack_U, before any equation of state, boundary or rate: a new named refusal refuse_conserved_state_inadmissible (code 8), with its counter in n_eval_refusal(0:8) and its text in write_last_evaluation_refusal. No clipping of T. The early return writes every output (Fvec a quiet NaN, heat = cool = 0, n_part = 0, f_sp = the seed, the judged measures sqrt(huge) so that a division by a tolerance stays finite, cells_judged = the refused cell) and puts back the products a discarded evaluation saves. An evaluation that does not ask for admissible is an evaluation of the iterate: meeting a state that is not a gas there stops the run with the cell named (error stop).
- Both solves screen their entry state (conserved_state_of_the_entry_is_a_gas) and return info = 1 with the cell named, so the iterate is admissible by construction.
- The merit of a refused trial is not formed: the line searches of PTC and JFNK, the Levenberg-Marquardt curve and escapes, and the trust-region ray probes set the merit from the flag (the ray probes take the iterate's merit, which makes no slope, and ray_slope_verdict reads the flag first).
- The finite-difference banded Jacobian (build_banded_jac_full) samples a color on its backward side, [F(Y) - F(Y - h e)]/h, when every forward length (sqrt(eps) halved six times) is refused, as jv_product already did; counted as n_jac_backward_color and reported. A forward column the forward probe resolves cannot move.
- The refusal census is printed for every solve ((JFNK) residual samples: admitted ..., refused for a conserved state that is not a gas ...), also on the route without species rows, where it was not printed before.
- admissible_conserved_cell also requires the internal energy formed exactly as U_to_W_comp forms it, E - 0.5 m m/rho, to be positive where it cannot overflow: a cell with E set to its kinetic energy passed the scaled test while the conversion gave rho e = 0 and the sweep was handed T = 0 (MEASURED, kind rho_e_zero of the new suite).
- Charge exchange: the "thread holds a cell" state is its own threadprivate flag cx_cell_loaded, set with cx_cell_T in cx_set_cell; cx_require_cell checks the flag first; cx_add_to_turnover requires T (the caller in set_turnover_scales_at_field passes ieq_cell%T_K); cx_set_cell stops on a temperature that is not finite and positive (a rate coefficient is a Maxwellian average).

Tests (RED: a76da494 source plus the test hooks only, ctl_hooks; GREEN: this source). New suite src/tests/trial_admissibility/run.sh on the iontrans_metals state (metals on), 16 rows: five kinds of trial (rho e < 0 in cell 2, rho e = 0, rho = 0, NaN energy, infinite momentum), each as an adoptable trial and as a probe: GREEN admissible = F with no radiative pair, no particle count and no finite row (10/10); RED admissible = F but 1499 finite rows from a sweep at T = -2.4e-3 K for the first two kinds, and an abort in charge exchange "at T = NaN" at rho

= 0. Iterate after the refused trials: 0 of 1500 entries differ (GREEN). Jacobian at a cell whose rho e is 1e-12 of its kinetic energy: forward probe at the shortest length refused, 0 unresolved colors, the momentum column equal to the backward difference of that column alone to 2.9e-4 of its largest entry. An iterate that is not a gas stops the run named (RED: evaluated, "REACHED"). A solve entered at a state that is not a gas returns info = 1 at once, named (RED: a full JFNK run). `src/tests/charge_exchange_rows/run.sh cell_tag`: a worker thread that never loaded a cell stops with "holds no cell" (RED: "last filled at T = -1 K"), a cell at -1.4356e-5 K stops in `cx_set_cell` (RED: given rate coefficients, "REACHED").

### 125.2 Failed-wind recovery (`EXHALE_main.f90`, `steady_wind_with_element_diffusion`)

Definition (written at the declaration of `u_wind_conv`): the last converged wind is the conserved state u (ghosts included) and the species fractions f_sp of the most recent pass whose solve returned 0, taken where its joint test takes them (after `sources_and_ghosts_of_the_held_composition`), with the carrier module state and the acceptance record of that solve (`gate_rnorm_accepted`, `gate_fspread_accepted`) beside them; every other quantity the loop reads (primitive state, T, excited hydrogen, heat, cool, eta, the sweep ledger, the ghosts) is rebuilt from the pair at held composition. The wind is a solution for its composition, so the two are restored together. The boundary inputs do not change inside one call, and the record does not outlive it.

- After a failed solve (info 1, 2 or 3) the pair is restored, the pass keeps its flag, and the update is retried from the restored wind with a shorter step: omega halves (floor 0.125) when an element update was consumed; the carrier bound halves (floor 1e-3) whether or not the update ended on it, unless held. Where nothing can be shortened the alternation ends with the new named ending `outer_failed_update_unshortenable` (the next pass would repeat the failed one; the state handed back is the restored wind, not accepted).
- First pass (no converged wind): the solve is repeated once from its own iterate at the same composition (nothing is relabeled; no update has been consumed); if that fails too the alternation ends, `outer_no_converged_wind`, with the iterate and its flag.
- The acceptance record is restored with the wind: before, the written restored state was compared with the flux spread of the discarded iterate and the run printed "WARNING: the face-flux spread of the written state differs from the value the solve accepted" (route B 8.06e-2 accepted against 2.17e-12 written); after, 1.785e-12 against 1.893e-12, no warning.

Tests: new suite `src/tests/failed_wind_recovery/run.sh` (iontrans_metals, hook `EXHALE_TEST_FAILED_SOLVE_PASS` reports the listed passes failed and raises their returned internal energy by 1e-3), 8 rows GREEN: the retried update is taken from the converged wind (distance 0) and not from the failed iterate (1.43e-2); the bound shortened 1e-2 -> 5e-3; the failed pass keeps info 1; the restored state stands at the joint distance the converged pass recorded (1.94072e3 both); the retry converges; a second failure returns to the same wind; a first-pass failure is solved again once and then ends with no update. RED (`ctl_hooks`): the update is taken from the failed iterate (distance to the converged wind 1.45e-2, to the iterate 0), a second failure enters from the next iterate (2.1e-2), and a first-pass failure continues. `grid_and_gates/output_state_consistency.sh`: its stagnation fixture now ends on `outer_failed_update_unshortenable` (pass 12 of the fixture ends info = 2 under the new floor, the

bound is held, nothing can be shortened); the row accepts that ending, which hands back the certified state as the stagnation ending does (the next row, `outer_iteration_ending_hands_back_one_state`, 7.4e-16, PASS); reason written at the row.

### 125.3 The WENO3 primitive floor at the domain scale (`Reconstruction.f90`, `hydrodynamic_rows_body.inc`)

The approved floor eps_k = (dr_j q_s,k)^2, q_s,k the largest |W_k| of cells j-1, j, j+1, was installed first and REFUSED by measurement: on the regression matrix the marching WENO3 stage of `wasp_full` and `wasp_he23off` no longer reached their du < 1e-3 stop (wasp_full: smallest du in each 2000 steps 3.8e-3, 3.7e-3, 3.6e-3, 2.7e-3, 3.2e-3, 3.7e-3, 3.4e-3, 3.6e-3, 3.7e-3, 3.7e-3, 4.0e-3, 4.4e-3, 4.1e-3 up to step 26 803; wasp_he23off du 3.77e-3 at step 34 451; both killed by PID, outputs not compared), while the control stops at step 12 627 (du 9.989e-4, log10 Mdot 13.32). The cause, by test: the new binary with the dr_j^2 floor (`diag/floorB`) is bitwise the control through the stop at step 12 627 (outputs identical), so changes 125.1-125.2 do not touch the marching path; one component at a time (smallest du in steps 2500-3000; control 1.24e-3, local floor on every component 3.81e-3): local floor on rho alone 3.38e-3, on p alone 1.26e-3, local p and v with the sound speed in the v scale 3.81e-3, local p and v with the domain scale on rho 1.51e-3, the domain scale on every component 1.48e-3. The density component of the local floor is what keeps the marching state unsteady; the Newton-finished state is the same under either (wasp_full_newton log10 Mdot 13.29). Adopted: eps_k = (dr_j W_s,k)^2 with W_s,k the largest |W_k| over the physical cells, the solution-scaled floor of Yamaleev and Carpenter (2009, Eq. 65) with its norm over the domain as that equation writes it; S = 1 where b = 0. It is invariant under W -> cW (suite `weno3_primitive_floor`: domain_scale rescaling rows 2.2e-16 to 4.3e-16 on smooth, step and stored states, PASS, where the retired dr_j^2 floor FAILs at 2.2e-4 to 1.7e3), third order on smooth data without the well-balanced option (3.00-3.01), and identical to the state-scale rows on every test of the suite (the suite scales whole profiles). Its cost, stated at the code: where a component is far below its domain maximum (the outer layer) the weights are linear and a jump there overshoots by up to 1/6 of the jump, which the positivity limiter bounds. With `Well balanced: True` the pressure face values are second order (1.99-2.06) for every floor: the second-order construction of Kaeppeli and Mishra (2016); now stated in `docs/physics_overview/numerics.tex` and at the code. NEEDS CONFIRMATION by the user: this replaces the approved local form by the published domain form. The suite driver keeps the retired, state-scale and data-scale floors beside the production one (`KIND_DOMAIN`), and its CHECK rows compare production with the domain copy (8/8 PASS, difference 0). `weno3_vector_matches_primitive_formula` (physics_probe) uses the domain scale: 0 (PASS).

### 125.4 Checkpoint: regression matrix (gfortran, OMP_NUM_THREADS = 1, no golden refresh)

Control: gfortran build of the a76da494 source (`EXHALE_gfortran` md5 540dff43...). The full matrix ran once with the local-floor build (bf98bd71); the four cases that run WENO3 or the alternation were run again with the step-2 final build (2215c911, `regm/s2f`). The other 13 cases are not reached by the floor (their capped marching never leaves the PLM stage) nor by 125.1-125.2 (no stationary solve), and they were bitwise the control with bf98bd71; the diff from bf98bd71 to 2215c911 is the floor, the acceptance-record restore inside the alternation and a log line, none on their path. Table (max relative difference of Hydro_ioniz.txt / Ion_species.txt data rows; this run's outputs only):

| case | build compared | Hydro_ioniz new vs control | Ion_species new vs control | Hydro_ioniz control vs golden | verdict |
|---|---|---|---|---|---|
| wasp_full | 2215c911 (step-2 final) | running (not stopped at du < 1e-3) | running (not stopped at du < 1e-3) | 2.96e-03 (row 217 col 2) | pending |
| wasp_he23off | 2215c911 (step-2 final) | running (not stopped at du < 1e-3) | running (not stopped at du < 1e-3) | 1.44e-02 (row 259 col 2) | pending |
| wasp_full_newton | 2215c911 (step-2 final) | 3.95e-03 (row 501 col 6) | 1.92e-03 (row 501 col 7) | 3.80e-05 (row 357 col 6) | changed |
| mol_base_handoff | bf98bd71 (local floor; path not reached by the floor) | bitwise | bitwise | 5.61e-01 (row 379 col 6) | bitwise |
| mol_metals | bf98bd71 (local floor; path not reached by the floor) | bitwise | bitwise | 1.22e-01 (row 364 col 6) | bitwise |
| mol_lyman_werner | bf98bd71 (local floor; path not reached by the floor) | bitwise | bitwise | 5.09e-01 (row 385 col 6) | bitwise |
| mol_diffusion | bf98bd71 (local floor; path not reached by the floor) | bitwise | bitwise | 4.65e-01 (row 379 col 6) | bitwise |
| mol_ir_bands | bf98bd71 (local floor; path not reached by the floor) | bitwise | bitwise | 1.22e-01 (row 364 col 6) | bitwise |
| mol_sec_ion | bf98bd71 (local floor; path not reached by the floor) | bitwise | bitwise | 1.83e+00 (row 370 col 6) | bitwise |
| mol_carrier | bf98bd71 (local floor; path not reached by the floor) | bitwise | bitwise | 7.04e-01 (row 387 col 6) | bitwise |

| case | build compared | Hydro_ioniz new vs control | Ion_species new vs control | Hydro_ioniz control vs golden | verdict |
|---|---|---|---|---|---|
| lower_profile | bf98bd71 (local floor; path not reached by the floor) | bitwise | bitwise | 1.69e+00 (row 429 col 6) | bitwise |
| hydrostatic_column | bf98bd71 (local floor; path not reached by the floor) | bitwise | bitwise | 1.14e-06 (row 280 col 6) | bitwise |
| oxygen_chemistry | bf98bd71 (local floor; path not reached by the floor) | bitwise | bitwise | 9.72e-01 (row 5 col 6) | bitwise |
| hp_zero_seed | bf98bd71 (local floor; path not reached by the floor) | bitwise | bitwise | 3.04e-02 (row 1 col 2) | bitwise |
| hp_trace_seed | bf98bd71 (local floor; path not reached by the floor) | bitwise | bitwise | 3.04e-02 (row 1 col 2) | bitwise |
| hp_front | bf98bd71 (local floor; path not reached by the floor) | bitwise | bitwise | 3.15e-02 (row 1 col 2) | bitwise |
| iontrans_metals | 2215c911 (step-2 final) | 3.95e-03 (row 501 col 6) | 1.99e-03 (row 501 col 7) | 1.69e-03 (row 501 col 6) | changed |

The two changed cases differ from the control by at most 1.5e-4 (Hydro_ioniz) and 9.9e-5 (Ion_species) over cells 1-497 (medians 1e-6) and by 2e-3 to 4e-3 only in the outermost rows; both are accepted states in both builds (wasp_full_newton log10 Mdot 13.29 in both; iontrans_metals ACCEPTED at outer pass 15 in both). wasp_full and wasp_he23off with the step-2 final build had not stopped when this was written (step 23 595, du 1.73e-3; step 29 090, du 1.72e-3, against the control's stop at 12 627); a discriminating run (the 9ac38309 source with only the floor set back to dr_j^2, wasp_full, one thread, capped at 14 000 steps) was started; the earlier test of the same kind (the 125.1-125.2 source with dr_j^2, `diag/floorB`) was bitwise the control through its stop.

Invariants: mass row, momentum row, energy row of the accepted states of iontrans_metals (outer pass 15 ACCEPTED in both) and wasp_full_newton (JFNK info = 0, ‖R‖ 9.6e-9 in both) are each within the certification tolerances of the run; the nuclei and charge closures printed by the certification (element census, `max |sum f_i A_i - 1|` at round-off) unchanged in kind. The goldens of the molecular cases differ from the control itself by up to 1.8 (stale since earlier

series, READ: section 122 and before); no golden was refreshed.

Suites (all 62, new tree against the a76da494 source as control, `run_all_suites.sh`): failing in both and unchanged: `carrier_enthalpy_flux` (3 rows, `the_counterflow_face_flux_is_Phi_H2_times_k_u_rv_minus_5T_over_2` and two proton recoil rows), `steady_completion_flag` and `steady_selfconsistent_residual` (need `EXHALE_STEADY_RUNLOGS`), `grid_and_gates` (3 `stationary_evaluate_*` rows). Failing only in the control and only by its scratch environment (no `references/`, `inputdata/` beside it): `low_energy_electron_degradation`, `h2_level_ladder` (skipped), `spectrum_type`, `grid_and_gates  sed_coverage_sed_file`; and in the a76da494 source `physics_probe weno3_local_copy_matches_production` (2.2e-10, its own older probe text). Changed by this step and green: `charge_exchange_rows` (+2 cell_tag rows), `krylov_and_dogleg` (347/347 after the kind-generic copy took the floor; 7 rows red while it did not), `weno3_primitive_floor`, `physics_probe` (2499/2499), new `trial_admissibility` (16/16), `failed_wind_recovery` (8/8).

### 125.5 The route-B and He/H 0.65 k00 entries (closure environment, omega 0.5, bound 0.04 held, 9-12 threads)

Inputs: `LHS1140b/models/.P2/ew_fit/{route_xuv_scaled/HeH0.65/xuv_x1p3,phase5/HeH0.65}_structural_a76da494/k00`, copied to scratch. The charge-exchange abort does not occur; no trial was refused for a conserved state that is not a gas on any pass (census "not a gas 0" on every solve). What limits them now is the wind-carrier half: the hydrodynamic solve after the second (route B) and eighth (He/H 0.65) composition update fails (info 1, ~8500 samples admitted, none refused), the converged wind is restored, and the retries fail too. Per pass (binary a68121e6; hydro info; T of cells 1-3 [K] at the certification; He/H at 2 R_p):

| case | pass | hydro info | T cells 1/2/3 at certification [K] | He/H at 2 R_p | samples admitted / refused (not a gas) |
|---|---|---|---|---|---|
| route B | 1 | 0 | 57/229/295 | 0.1463 | 515 / 0 |
| route B | 2 | 0 | 1488/2866/48 | 0.06041 | 1768 / 0 |
| route B | 3 | 1 | 1488/2866/48 | 0.06041 | 8527 / 0 |
| route B | 4 | 1 | 1488/2866/48 | 0.06041 | 8580 / 0 |
| route B | 5 | 1 | 1488/2866/48 | 0.06041 | 8564 / 0 |
| He/H 0.65 | 1 | 0 | 86/486/634 | 0.3813 | 586 / 0 |
| He/H 0.65 | 2 | 0 | 85/199/269 | 0.3689 | 439 / 0 |
| He/H 0.65 | 3 | 0 | 85/165/213 | 0.3524 | 377 / 0 |
| He/H 0.65 | 4 | 0 | 84/153/183 | 0.3398 | 377 / 0 |
| He/H 0.65 | 5 | 0 | 82/149/164 | 0.3279 | 378 / 0 |
| He/H 0.65 | 6 | 0 | 80/150/152 | 0.3163 | 376 / 0 |
| He/H 0.65 | 7 | 0 | 76/157/144 | 0.3037 | 377 / 0 |
| He/H 0.65 | 8 | 0 | 73/174/136 | 0.291 | 376 / 0 |
| He/H 0.65 | 9 | 1 | 73/174/136 | 0.291 | 8052 / 0 |
| He/H 0.65 | 10 | 1 | 73/174/136 | 0.291 | 8056 / 0 |
| He/H 0.65 | 11 | 1 | 73/174/136 | 0.291 | 8057 / 0 |

On a failed pass the certification columns are those of the restored wind.
Both endings are the restored wind, not accepted (route B: no progress in 3 passes at pass 5; He/H 0.65: no step can be shortened at pass 11, bound held and omega at 0.125). The cold base cell is unchanged by this step (route B cell 1 at 57 K on pass 1, as in section 122); it is the subject of section 126. With `EXHALE_7fe0b827.x` (local floor) the records were the same in kind (route B: passes 3-5 failed, ending no progress; He/H 0.65: passes 6, 10-12 failed, ending at pass 12).

### 125.6 Also corrected

- The base conductive flux formula in `viscous_conduction.f90` (comment of `heat_conducted_into_lower_atmosphere`) and in the header line `write_setup_report.f90` prints into every `EXHALE_resolved.out` read q = -kappa (T_1 - T_bath)/(r_1 - r_0) for a value labelled positive into the lower atmosphere; the value is kappa (T_1 - T_bath)/(r_1 - r_0) (READ: `lower_atmosphere_heat_flux_cgs = -q_into_wind`, q_into_wind = blo(1)(T_b - T_1)... > 0 into the wind). Corrected there and in `docs/physics_overview/hydrodynamics.tex`, `molecules_metals.tex`, `postprocessing.tex`, `docs/EXHALE_user_manual.tex` and `md/input_schema.md`. The record `md/interface_definitions_step3_20261009.md` 7.2 that found it is left as written.
- `backup/regression/iontrans_metals/input.inp`: `Restart option change: iontrans, wellbal, stage_enth`. Without `stage_enth` the case refused to start ("metadata field options differs ... stage_enth: F -> T") with both binaries, so the matrix could not run it.

Not changed, reported: a stationary restart whose solve returns info = 1 is written with "the state written is a relaxation snapshot: the run made no stationary claim ... exits 0" (`certify_the_state_to_be_written(info_jfnk .eq. 0, ...)` in `stationary_state_of_the_loaded_restart`, READ; the same in a76da494). The pair is consistent with the code's rule (a claim is made only by a converged solve; status 2 belongs to a claimed and refused state), but the wording "relaxation snapshot" is wrong for a stationary restart whose solve failed, and the exit status 0 lets a script read a failed stationary solve as a run without a claim; changing the status is an interface change of the closure and is left for decision.
Docs: `docs/physics_overview/steady_state.tex` (trial refusal; failed solve and recovery; progress control), `docs/physics_overview/numerics.tex` (the floor, the well-balanced order), `docs/EXHALE_user_manual.tex` (endings, recovery, refusal census), `md/TO_BE_DONE.md` (AR) 1 and (AQ).

## 126. An open base face starts from a seed that agrees with its ghost: the base cells take the ghost hydrogen partition, and the carriers are relaxed at the entry wind before the first solve (change 4 of step 2, user approval 2026-10-09; `md/cold_base_cell_diagnosis_20261009.md`)

Measured separately from section 125, after its checkpoint. Frozen binary `LHS1140b/models/.P2/EXHALE_9ac38309.x` (manifest `BINARY_MANIFEST_9ac3830993b9.txt`; source = that of `EXHALE_a68121e6.x` plus the two blocks below, in `EXHALE_main.f90`). Every number MEASURED on lart4 unless marked READ.

### 126.1 What was implemented

(a) `stationary_state_of_the_loaded_restart`, step (2b): on an open face (`base_h2_composition_imposed`) the cells of the uniform base region, 1..`N_low_cells`, take the n(H I)/n(H2) of ghost 0 as the boundary sweep of step 2 leaves it, each cell keeping its hydrogen nuclei and its mass (f(H I) + 2 f(H2) held; f is a number per unit mass). On the stationary restart route only: a marching run evolves the seed in time. The diagnosis's control `match_base_partition.py` did the same on the IC file (READ). (b) `steady_wind_with_element_diffusion`, before pass 1: on an open face with transported carriers alternated, the carrier rows are relaxed at the entry wind (`relax_photochemical_composition` at the largest bound `carrier_trust_max` = 0.5, repeated up to 20 times while it ends on the bound and keeps steps), and the dependent state is rebuilt, so the energy row is not converged against an unbalanced carrier composition. The rule is stated at the code.

### 126.2 Tests

New suite `src/tests/open_face_seed/run.sh` (route B k00, one pass): GREEN (this source) base partition of cells 1..50 at the hydrodynamic entry equal to the ghost's to 7.6e-5 (tolerance 1e-3: the entry is read after the composition sweep that re-solves the eliminated H I), hydrogen nuclei as loaded to 3.9e-10 (tolerance 1e-9: the build without (a) departs by 3.7e-10 over the same load-to-entry path), Sidf of cell 1 -9.0e-4 (19.67 before), the entry relaxation attempted; RED (the source of section 125): partition 1.3e4 from the ghost's, Sidf 19.67, no relaxation; the nuclei row passes in both.

### 126.3 Pass 1, cells 1-60 (the diagnosis's control repeated)

Route B k00 (before: `EXHALE_a76da494.x`, the diagnosis run RB; after: `EXHALE_9ac38309.x`), certification of pass 1, heat and Sidf in code units:

| cell | T before [K] | heat before | Sidf before | T after [K] | heat after | Sidf after |
|---|---|---|---|---|---|---|
| 1 | 57.2 | 1.931e+01 | 1.967e+01 | 334.8 | 1.252e-03 | -8.950e-04 |
| 2 | 229.0 | 6.583e-02 | -2.885e-03 | 416.9 | 8.660e-04 | -1.212e-03 |
| 3 | 294.8 | 2.259e-02 | -6.299e-04 | 487.9 | 6.998e-04 | 7.127e-05 |
| 10 | 495.8 | 2.160e-03 | 6.370e-06 | 858.3 | 6.961e-04 | 4.140e-05 |
| 20 | 643.4 | 7.375e-04 | 1.071e-05 | 1227.6 | 8.228e-04 | 3.268e-05 |
| 30 | 748.5 | 4.719e-04 | 1.119e-05 | 1514.9 | 8.780e-04 | 2.988e-05 |
| 40 | 832.5 | 3.764e-04 | 1.084e-05 | 1784.3 | 1.845e-03 | 4.250e-05 |
| 45 | 869.2 | 3.491e-04 | 1.059e-05 | 1985.5 | 3.744e-03 | 1.046e-04 |
| 48 | 889.8 | 3.361e-04 | 1.042e-05 | 2231.2 | -3.726e-04 | 3.901e-04 |
| 49 | 896.5 | 3.321e-04 | 1.036e-05 | 2381.5 | -2.677e-02 | -3.369e-02 |
| 50 | 903.0 | 3.284e-04 | 1.029e-05 | 2618.9 | -2.277e-01 | -1.707e+01 |
| 51 | 909.5 | 3.249e-04 | 1.023e-05 | 54.9 | 4.392e+00 | 1.695e+01 |
| 52 | 916.0 | 3.214e-04 | 1.020e-05 | 184.0 | 3.788e-02 | 2.637e-03 |
| 53 | 922.5 | 3.181e-04 | 1.015e-05 | 242.2 | 1.296e-02 | -1.792e-04 |
| 55 | 935.5 | 3.116e-04 | 1.003e-05 | 319.2 | 4.406e-03 | 1.477e-04 |

| cell | T before [K] | heat before | Sidf before | T after [K] | heat after | Sidf after |
|---|---|---|---|---|---|---|
| 60 | 968.3 | 2.969e-04 | 9.715e-06 | 448.2 | 1.290e-03 | 8.097e-05 |

He/H 2.09 k00 (seed rebuilt as in the diagnosis; before: `EXHALE_a68121e6.x`; after: `EXHALE_9ac38309.x`):

| cell | T before [K] | heat before | Sidf before | T after [K] | heat after | Sidf after |
|---|---|---|---|---|---|---|
| 1 | 57.4 | 1.648e+01 | 1.689e+01 | 354.5 | 1.611e-03 | -1.562e-03 |
| 2 | 209.8 | 6.375e-02 | -3.354e-03 | 436.6 | 1.016e-03 | -1.200e-03 |
| 3 | 269.1 | 2.076e-02 | -4.588e-04 | 508.2 | 7.799e-04 | 2.988e-05 |
| 10 | 459.6 | 1.618e-03 | 1.216e-05 | 885.5 | 1.145e-03 | 1.773e-05 |
| 20 | 613.5 | 5.288e-04 | 1.307e-05 | 1260.5 | 1.650e-03 | 1.378e-05 |
| 30 | 729.9 | 3.483e-04 | 1.192e-05 | 1545.6 | 1.663e-03 | 1.263e-05 |
| 40 | 826.9 | 2.837e-04 | 1.078e-05 | 1809.1 | 2.914e-03 | 2.096e-05 |
| 45 | 870.3 | 2.647e-04 | 1.029e-05 | 1998.5 | 5.691e-03 | 5.543e-05 |
| 48 | 895.0 | 2.556e-04 | 1.002e-05 | 2205.1 | 9.769e-03 | 1.856e-04 |
| 49 | 903.1 | 2.528e-04 | 9.926e-06 | 2317.1 | 9.986e-03 | -5.197e-02 |
| 50 | 911.0 | 2.502e-04 | 9.826e-06 | 2470.7 | -2.175e-04 | -1.415e+01 |
| 51 | 918.9 | 2.477e-04 | 9.742e-06 | 60.6 | 2.222e+00 | 1.408e+01 |
| 52 | 926.8 | 2.452e-04 | 9.682e-06 | 170.2 | 3.338e-02 | 2.523e-03 |
| 53 | 934.7 | 2.429e-04 | 9.614e-06 | 223.7 | 1.087e-02 | -1.391e-05 |
| 55 | 950.7 | 2.383e-04 | 9.451e-06 | 297.6 | 3.368e-03 | 1.518e-04 |
| 60 | 991.3 | 2.278e-04 | 9.052e-06 | 429.2 | 8.885e-04 | 8.225e-05 |

Cells 1-60 in full: `LHS1140b/models/.P2/ew_fit/cold_cell/change4_cells_1_60_{routeB,HeH209k00}.txt`. (a) removes the cold cell 1 (route B 57.2 K -> 334.8 K, Sidf 19.67 -> -9.0e-4; He/H 2.09 57.4 K -> 354.5 K, 16.89 -> -1.6e-3), and the partition jump moves to the top of the base region, where the same cold cell reappears on pass 1 (cell 51: 54.9 K, Sidf 16.95 against -17.07 in cell 50; He/H 2.09 cell 51 60.6 K), as the diagnosis's control RB_ghost50 had (READ: 54.9 K). (b) does not act on these entries: the relaxation is refused at its first trial, "the shortest admissible trial was not covered", because the carrier substep acceptance refuses the H2 row of cell 2 (route B; relative residual 5.6 at full length falling to 4.7e-3 at 2^-10 of it, the solve itself stopping at the round-off of its row terms, `carrier_solve_unreachable_at_substep`; diagnosed with an instrumented scratch build, not in the tree), so the composition enters pass 1 unchanged. Why the carrier substep cannot close that row at the entry state is not identified.

**126.4 Full alternations from k00 (closure environment, omega 0.5, bound 0.04 held, 9 threads, budget 60 passes)**

- Route B: every hydrodynamic solve converges (49 passes, info 0 on each, against failures from pass 3 on without change 4); the base stays smooth (cells 1-3 at 270/302/331 K from pass 6 on; the cold cell 51 of pass 1 is gone by the end: cells 48-53 932-969 K at pass 49); the helium of the wind leaves (He/H at 2 R_p 0.146 -> 8.3e-4), and the alternation ends at pass 49 with no progress in 3 passes: the He+ and He++ carrier balances of cell 354 stand at 0.96 and 1.00 of their tolerance scale (1e-5), He 2^3S 0.13, the elemental He/H row 8.8e-2. Not certified.

- He/H 0.65: 60 passes, all but passes 33-34 (info 2, restored and retried) converged; cells 1-3 at 420-435 K; He/H at 2 R_p 0.38 -> 0.17-0.20 oscillating with a period of 6-7 passes; budget ended at pass 60 with the He+ balance of cell 217 at 0.88 (H+ 0.23, H2 0.16). Not certified. A shallow cool region remains at the old jump (cells 49-53 at 173-195 K against 245 K at cell 48, pass 60).

Neither k00 certifies with change 4. The wind-carrier failure of section 125.5 is gone on both; what refuses them now are the ionization-stage carrier balances in the wind (route B: at the helium separation front of a wind that loses its helium, the regime of `md/osc_diagnosis_20261008.md` section 1; He/H 0.65: the He+ row at 1.2 R_p of section 4 there).

### 126.5 Matrix

No case of the regression matrix runs a stationary restart on an open face or alternates carriers on one (the only stationary case, iontrans_metals, is atomic; the molecular cases are marching snapshots), so (a) and (b) do not reach it (code path, READ); the matrix was not rerun for this change.
Docs: `docs/physics_overview/steady_state.tex` (paragraph "An open base face starts from a seed that agrees with its ghost").

## 127. The marching stops on the face mass flux and the state change, not on the cell-centered du; the local WENO3 floor is restored; secondary ionization is on from step 0 (2026-10-09/10, user decisions)

**Why.** An independent diagnosis (`md/weno3_local_floor_marching_stall_diagnosis_20261009.md`) showed that the local WENO3 primitive floor eps_k = (dr_j q_s,k)^2 (approved 2026-10-08, replaced in section 125 by a domain-scale floor because wasp_full never reached du < 1e-3 with it) is not defective. Its certified stationary wasp_full state (500 cells, secondary ionization off) has a face mass-flux spread of 5.3e-13 but a cell-centered du = (max - min)/mean of rho_j v_j r_j^2 of 3.57e-3 (1.67e-3 on 1000 cells), set by cells 499/500 next to the outer ghost; the domain-floor stationary state has du 1.76e-3. du is a resolution-dependent property of the cell averages, not a measure of stationarity. The secondary-ionization switch was gated on the same du: in the local-floor march it never switched on. User decisions (2026-10-09): keep the local floor; stop the marching on the numerical face mass flux and the state change, du kept as a diagnostic; decouple the secondary ionization from du; review the outer ghost condition (design note only, below). Threshold of the face flux for the marching stop: 4e-4 (user decision 2026-10-10, from the measurement in 127.2).

### 127.1 Changes (working tree; frozen binary `LHS1140b/models/.P2/EXHALE_5ed2b073.x`, manifest `BINARY_MANIFEST_5ed2b073943e.txt`, ifx; gfortran tree build md5 7a1ff33f)

1. `Reconstruction.f90`, `hydrodynamic_rows_body.inc`: eps_k = (dr_j q_s,k)^2, q_s,k = max |W_k| over cells j-1, j, j+1 (the form of `Reconstruct_scalar`); comments no longer state that the local floor was refused.
2. `EXHALE_main.f90`, `parameters.f90`: the marching stop is fspread_gate < `march_flux_spread_th` (4e-4; the face spread of r_{j+1/2}^2 F_rho over r >= r_flux, the steady solvers' gate quantity) AND chg_win

< `march_change_th` (1e-4), chg_win = max over cells of |rho/rho_win - 1|, |E/E_win - 1|, |(rho v) - (rho v)_win|/sqrt(rho E)_win over one monitor window of N_resid = 500 steps. A `[stop]` line prints both every window. The cell-centered du is computed and printed as a diagnostic and still drives the PLM -> WENO3 switch and the JFNK hand-off (both unchanged). The stall net (`relaxation_stalled`, formerly `du_plateaued`) counts settled windows only while the face spread has stopped falling (less than 1% drop since the previous window, `flux_still_falling`; added by the advisor after 127.2). Overrides for measurement: `EXHALE_MARCH_FLUX_TH`, `EXHALE_MARCH_CHANGE_TH`, `EXHALE_MONITOR_STEPS`.

3. `input_read.f90`, `parameters.f90`, `write_setup_report.f90`: `Secondary_ionization: True` (the default) now applies the coupling from step 0; `Staged` keeps the earlier behavior, its switch now made when chg_win < march_change_th (out of the PLM stage), not on a du stop flag. The cold-start runaway that staging was made for did not occur (MEASURED: wasp_full from its cold IC, coupling on from step 0, 20 000 steps, no NaN; wasp_full_newton from its cold IC, Newton finish at log10 Mdot 13.29, as with staging; `tmp/step2/diag/imm_*`).
4. No consumer parses the old stop text (searched `src/utils`, `LHS1140b`, `backup/regression`, `examples`, `src/tests`); `LHS1140b/models/run_case.sh` reads only the `-> stopped:` prefix.

**127.2 Measurements behind the thresholds (wasp_full, local floor unless stated, secondary ionization off, continued from a loaded state)**

- Both floors give the same face-spread history (`tmp/step2/diag/cont_loc`, `cont_dr2`): 7.5e-3 at step 5500 to 4.0e-4 at step 31 500, falling 4-5% per 500-step window. The first implementation of the stall net ended both runs there as "stall, target not met" (chg_win 8.5e-5 < 1e-4 for 2000 steps) while they were still converging: a defect, fixed by item 2's `flux_still_falling`.
- Continued with the fix (`cont2_loc`, 8 threads): the face spread fell smoothly (e-folding about 9000 steps) to 1.94e-5 at a further 27 500 steps, chg_win 4.0e-6, where the stop at the steady gate 2e-5 was reached.
- Between the states at face spread 4.0e-4 and 1.9e-5: rho changes by at most 1.45e-3 (r = 1.415 R_p), v by 1.24e-3, T by 3.3e-4 (medians 2e-4 to 5e-4); log10 Mdot 13.45 in both. Hence the marching threshold 4e-4 (user decision): the remaining change is below the proposed profile budget of `md/acceptance_by_observable_precision_20261009.md` (1%).

**127.3 Regression matrix (gfortran 7a1ff33f, one thread, `tmp/step2/regm/fin`), against the step-2 matrix of 7fe0b827 (`regm/new`: same local floor, staged secondary ionization)**

| Case | log10 Mdot | Largest relative change (rho, v, T) |
|---|---|---|
| wasp_full_newton | 13.29 -> 13.29 | 4e-13 (identical to round-off) |
| iontrans_metals | - | BITWISE |
| mol_sec_ion | 10.51 -> 10.51 | BITWISE (already `Immediate`) |
| mol_base_handoff, mol_diffusion | 10.57 -> 10.51 | 0.30, 1.2 (v near the base), 0.19 |
| mol_metals, mol_ir_bands | 10.58 -> 10.52 | 0.34, 1.1, 0.22 |

| Case | log10 Mdot | Largest relative change (rho, v, T) |
|---|---|---|
| mol_lyman_werner | 10.54 -> 10.49 | 0.24, 0.92, 0.15 |
| mol_carrier | 10.56 -> 10.51 | 0.36, 1.2, 0.24 |
| lower_profile | 9.56 -> 9.57 | 0.41, 1.5, 0.35 |
| oxygen_chemistry | 9.40 -> 9.40 | 0.022, 1.7 (v near zero), 0.18 |
| hp_zero_seed, hp_trace_seed | 10.56 -> 10.56 | 7.8e-5, 0.43 (base v), 4.1e-3 |
| hp_front | 10.56 -> 10.56 | 6.4e-5, 0.31, 3.1e-3 |
| hydrostatic_column | 8.27 -> 8.27 | 3e-11 |
| wasp_full, wasp_he23off | stopped unconverged (see below) | |

wasp_full and wasp_he23off (cold start, coupling on from step 0, the L1 outer face at Mach 0.96) were stopped on 2026-10-10 when the user decided to move the WASP outer face past the sonic point (section 127.6): at that time the face spread stood at 1.7e-2 (wasp_full, step 28 500, one thread), 1.3e-2 (wasp_he23off, step 35 000) and 4.3e-3 (wasp_full on 8 threads, step 46 500), still above 4e-4. Every capped case still ends on its step cap (the new stop does not fire earlier). The moved cases are the relaxation snapshots in which the secondary ionization was previously never switched on (staged, never settled within the cap); the molecular snapshots move toward `mol_sec_ion`'s Mdot 10.51. No golden refreshed (end of series).

### 127.4 Outer ghost condition: design note only

`md/outer_boundary_subsonic_review_20261009.md`: the hydrostatic ghost extension with copied velocity (`Apply_BC.f90:458-472`) is applied at a subsonic outer face (Mach 0.85 on wasp_full) where v dv/dr is 5-9 times the extension's gravity, outside the validity its own comment states. Options from the published sources (Thompson 1987, 1990; Poinsot & Lele 1992; Murray-Clay et al. 2009; Salz et al. 2016; Caldiroli et al. 2021): an outer face beyond the sonic point for transonic winds, a relaxed characteristic condition (whose Mdot is then conditional on an exterior pressure) for edges that must stay subsonic. Not implemented; awaits the user. DECIDED 2026-10-12 (section 144): no subsonic-edge treatment; the Mdot of such a run is labeled not quantitative.

### 127.5 Exit status of a failed stationary solve; change 4 (b) diagnosed (2026-10-10, user decisions)

**Exit status.** `certification.f90` (`stationary_solve_failed`, `certification_note_stationary_solve_failed`), called by the stationary restart and the direct steady route of `EXHALE_main.f90` with info /= 0: the state is written with the reason `stationary_solve_failed` and the run exits 3 (it was written as a "relaxation snapshot", exit 0). Exit 2 (converged, not certified) is unchanged in the binary and is now a flag for its callers: `src/utils/element_flux_closure.py` (`WRITTEN_STATE_STATUSES = (0, 2, 3)`; a failed solve still stops the closure; exit 2 without the certification's closing line is a gfortran runtime error and stops it), `LHS1140b/models/run_case.sh` (accepts 3 beside 0 and 2 before classifying the ending), `LHS1140b/models/.P1/fiducial_conduction/ew_of_state.sh` (evaluate exit 2 makes the products, labeled uncertified; exit 2 without the closing line is fatal). TEST (gfortran 106de97f): `iontrans_metals` as a

stationary restart with EXHALE_JFNK_MAXIT=1 EXHALE_OUTER_PASSES=1 returns info 1, writes certified=F cert_reason=stationary_solve_failed and exits 3. A converged route is untouched (code path).

**Change 4 (b): the refusal of the entry relaxation (diagnosis corrected the same day).** A first reading (a scratch build printing the worst row when a substep ended "unreachable at substep") found the H2 row of base cells 4-11 at 1e-8 to 4e-8 of its physical terms, with the time term 2e8-1e9 times those terms at substeps of 1.6e-3 to 6.2e-3 s; a step floor built on it (step 2e-1 s) changed nothing, and a second scratch build printing EVERY rejected substep showed why: those substeps are the END of the halving chain (step fraction 4e-3 of a 50 s interval), and such round-off-limited rows are already accepted with a flag by carrier_returned_state_verdict. The refusals that decide the outcome are earlier and different (MEASURED, tmp/c4b_run2/run.log, route B k00, first entry relaxation, grow 1 to 1/16): at every substep length from the full interval down, the Newton itself ends converged on its own scale (raw status 0) or stagnated (3), yet the RETURNED state has 6-8 H2 rows at 0.2 to 5.6 of their physical terms, in cells 2-9 at the base and 48-51 at the top of the seeded region (cells 1-50 take the ghost partition under change 4 (a)). The step floor was therefore removed again (the tree is back to the binary of the exit-status change, md5 116c26a8). What moves the returned state off the Newton's root (MEASURED, a third scratch build printing the hydrogen budget of every cell before the element limiter, tmp/c4b_run5/run.log): in the seeded cells 1-50 the H2 of the entry state already holds ALL the hydrogen the cell has (2 n(H2) = nH_free to 1e-5: the ghost partition x(H I) ~ 1e-5 that change 4 (a) writes), while above the seed (cell 51 on) 2 n(H2) ~ 0.86 nH_free. The carrier relaxation moves the carriers at FROZEN element densities, so any net gain of H2 in a saturated cell asks for hydrogen the cell does not have: at the full interval cell 1 asks for 2 n(H2) = 1.31 nH_free (inflow through the open base face), cells 2-12 for 1.13 down to 1.000002, and cells 41-50, at the top edge of the seed, for up to 1.08 (H2 brought up from below while the seed edge cannot pass it on). limit_to_element_budget scales the H carriers back into the budget, and the scaled state no longer solves its rows (0.2 to 5.6 of the physical terms). Cause: change 4 (b) relaxes carriers at frozen elements on a seed that saturates the hydrogen budget, so a stationary carrier profile there needs a redistribution of the hydrogen ELEMENT that only the element relaxation can make. Measured beside it (section 126.4): with (a) alone, (b) refused at every entry, route B converged 49 solves in a row and the cold cell left the grid, so (b) has contributed nothing. REMOVED (user decision 2026-10-10): the entry relaxation block and its variables in steady_wind_with_element_diffusion (EXHALE_main.f90, a comment states why), the row entry_carrier_relaxation_attempted of src/tests/open_face_seed, and the (b) sentence of docs/physics_overview/steady_state.tex; change 4 (a) is kept. TO_BE_DONE (AX) 1 closed.

### 127.6 The WASP regression cases move their outer face past the sonic point (user decision 2026-10-10)

Measured on wasp_full_newton (md/outer_boundary_subsonic_review_20261009.md section 5, figure docs/figures/outer_radius_wasp_full_newton_20261010.png): with the default Roche L1 outer face (1.574 R_p, Mach 0.95, sonic point near 1.6 R_p) rho and v differ from the 2 R_p solution by up to 15-17% near 1.31 R_p and log10 Mdot by 0.01, five times the grid effect seen between the 2 and 3 R_p domains (both supersonic). Decision: the WASP cases whose L1 face is subsonic (wasp_full, wasp_he23off, wasp_full_newton; iontrans_metals is a stationary restart of a stored L1 state and needs a new stored state) take Outer radius [R_p]: 2.0. Test runs at 2 R_p from the cold IC: tmp/r2cases/ (running).

## 128. The column conduction as an exact excess over the climate column; the latent and eddy heat of the column measured (2026-10-10; TO_BE_DONE (AR) 3(ii), (AU))

Record: `md/column_conduction_and_latent_heat_20261010.md`.

- `conducted_heat_column_balance.py` solves the total equation (option T of `md/interface_definitions_step3_20261009.md`) in the excess unknown dT = T - T0 with the exact discrete perturbation flux (both terms) and the reference source S0 explicit (`balance_terms`); the zero-input shortcut is removed (it returned a non-solution where T0 has a gradient), the depth test is on the response's bottom flux, the Newton scale is the largest term of the balance. New suite `src/tests/conducted_heat_column/reference_conduction_tests.py` (RED on the entry text, GREEN now).
- k05 (scratch rerun): T_match unchanged, 240.982642 K with either text; S0 moves T by at most 3.1e-5 K (0.85 bar); the held geometry costs -6.4e-3 K on T_match (MEASURED).
- (AU): the change of the latent deposit after the coupling moves T_match by -1.5e-12 K (not booked). The eddy heat flux of Photochem's K_zz = 1e9, booked as a change over T0, would lower T_match by 55.4 K; not implemented, user decision on K_zz near the match needed (note section 5).

## 129. One stellar geometry at the matching face, set by the wind's input (user decision 2026-10-10)

`md/matching_face_radiation_python_20261010.md` found three conventions for the same stellar beam: the wind takes xi_w F at normal incidence (xi_w = `dayside_dilution()` of the input's `2D approximate method`: 1, 1/2 or 1/4), Photochem 0.5 F at 48 deg (0.335 F on the horizontal), the climate column F/4. Decision: every model at the face takes the energy the wind's input states.

- `photochem_to_lower_profile.py`: `--face-geometry wind` is the default (`per_model` reproduces earlier runs); it needs the wind's xi_w, from `--wind-dilution` or the face file.
- `element_flux_closure.py`: the chemistry call adds `--face-geometry wind --wind-dilution xi_w` (xi_w from the template's `2D approximate method`, the rule of `dayside_dilution`) unless the configuration names a geometry, and every wind run's `EXHALE_resolved.out` value is compared with it (`check_wind_dilution`, refusal on a mismatch).
- `write_setup_report.f90`: `dayside_dilution` written to `EXHALE_resolved.out`. What it changes for LHS 1140 b: its runs use `2D approximate method: Mdot` (xi_w = 1), so the column now takes the substellar beam; on k05 (worker's scratch run) the climate deep boundary moves from 561 to 878 K, the tropopause from 1.004 bar / 185 K to 0.48 bar / 220 K, the cold-trap water from 1.35e-7 to 5.85e-5 and T_match from 240.98 to 248.99 K (with the face file). The `2D approximate method` of the LHS inputs is a physical choice of the model and is reported with every result. Tests: `matching_face_radiation` 16, `matching_surface` 16, `element_flux_closure` 10 (one row updated: a status 3 is now refused as "stationary solve failed", and a status 2 without the certification's closing line as a runtime error, both before the post-processing pass; that suite had

not been run after the exit-status change of section 127.5 and its old row failed until updated), all pass. Fortran: tree build md5 7cc33e0b.

## 130. Eddy heat transport in the reference climate and the column; with K_zz = 1e9 no reference climate exists (user decision 2026-10-10, option (b))

Record: `md/eddy_heat_transport_column_20261010.md`.

- Physics: q_e = rho c_p K (dT/dz + g/c_p) (Johnstone et al. 2018, Eq. 44, read; the potential-temperature form), with the chemistry's own K_zz (`lower_profile_schema.eddy_diffusion_coefficient`, one definition). Hunten (1974), on the dissipation heating that may balance the eddy cooling, was read in its ADS abstract only (paywalled; listed for the user).
- `radiative_convective_column.py`: `stratosphere_energy_balance` replaces clima's isothermal stratosphere by the solution of radiation (clima's own `TOA_fluxes_dry`), molecular and eddy conduction on the column's own nodes (dlnp 0.01 anchored at the match, no non-radiative flux through it), troposphere held; Newton with an analytic conduction and a finite-difference radiative Jacobian; clima's altitude rtol tightened to 1e-12 (at 1e-9 its fluxes are a step function of T, MEASURED); superadiabatic or sub-50 K solutions refused.
- `conducted_heat_column_balance.py`: eddy face coefficient and the offset rho K g in the exact excess form, the reference's held radiative heating in S0 (S0 is now the reference's own imbalance), JANAF heat capacities, flux parts and imbalance records.
- `photochem_to_lower_profile.py`: `--eddy-heat-transport {on,off}`, default on; `off` reproduces the earlier runs bit for bit (k05 command: T_match 240.98267612 K, identical data rows, profile and solution_id); the reference nodes are added to the column's levels.
- k05 (MEASURED): clima's isotherm carries a net radiative heating of 1.78e4 erg cm^-2 s^-1 above the tropopause (7.51e4 with the wind geometry), so it is not a steady state; the eddy-consistent reference exists only up to K_zz = 1.78e5 (per_model) and 1.42e6 (wind) cm^2 s^-1, where it reaches the 50 K end of the opacity tables; K_zz = 1e9 is refused. With the K_zz = 1e5 chemistry (composition held) T_match goes from 408.87 K (off) to 411.93 K (on): +7.18 K from the new T0, -4.13 K from the eddy conduction, 0.10 K from the reference imbalance. Photochem's equilibrium initializer fails below about 180-184 K, so the eddy-on adapter cannot run the chemistry on such references.
- User decision needed: every LHS 1140 b closure with `--kzz-const 1.0e9` stops in its first chemistry call under the default; add `"--eddy-heat-transport", "off"` to the closure's `adapter_args` to run as before, or decide on the dissipation heating (papers needed) or a K_zz for heat (note section 6). Tests: `eddy_heat_tests.py` (16) and `stratosphere_balance_tests.py` (9), new, RED on the entry texts and GREEN; `conducted_heat_column` run.sh 126 PASS 0 FAIL, `matching_face_radiation` 16 and `matching_surface` 16 pass. No Fortran.

**130, addendum (user decision 2026-10-10, after the measurement above).** The default of `--eddy-heat-transport` is OFF: the chemistry's K_zz is used as an effective tracer-mixing coefficient that carries no heat, since as a heat diffusivity no reference climate exists above K_zz ~ 1.4e6 cm^2 s^-1 (xi_w = 1) and the LHS 1140 b closures use 1e9.

The approximation and its size are stated at the option in photochem_to_lower_profile.py. on stays for the review of turbulent dissipation heating (Hunten 1974, JGR 79, 2533; Osborn 1980, J. Phys. Oceanogr. 10, 83, both supplied by the user) and for heat-diffusivity sensitivities.

## 131. The JFNK hand-off reads the face mass-flux spread (user decision 2026-10-10)

**Why.** The hand-off of a Solver: Newton run was keyed on the cell-centered du < 1e-2. With the outer face of wasp_full_newton at 2 R_p (section 127.6) du never falls below 1.04e-2 (its resolution floor next to the outer ghost, section 127), and the hand-off came only through its stall clause, after 45 118 marching steps.

**Change** (EXHALE_main.f90, parameters.f90, input_read.f90, write_setup_report.f90; gfortran tree build a8ad24c4): the hand-off is armed the first time a monitor step sees the face mass-flux spread (fspread_gate) at or above newton_flux_switch, and fired at a monitor step below it; a state whose spread has settled (the marching stall net) within 5 times the switch is handed over too. Solver: Newton [value] now sets this FACE-SPREAD threshold (it set a du threshold before); the variable is newton_flux_switch (EXHALE_resolved.out key of the same name), default 0.1. du stays a printed diagnostic and still drives the PLM -> WENO3 switch.

**Measured** (wasp_full_newton, 8 threads, tmp/handoff/):

| outer face | switch | hand-off at step | JFNK | log10 Mdot |
|---|---|---|---|---|
| 2 R_p | 0.2 | 11 500 | info 0 (5 non-monotone accepts) | 13.28 |
| 2 R_p | 0.1 | 16 500 | info 0 (2) | 13.28 |
| 2 R_p | 0.03 | 21 500 | info 0 (1) | 13.28 |
| 2 R_p | du < 1e-2 (before) | 45 118 (stall clause) | info 0 | 13.28 |
| L1 | 0.1 | 7 000 | info 0 | 13.29 |

Default 0.1: converged at both outer faces with a margin (0.2 also converged but needed more non-monotone accepts), a third of the steps of the du switch.

iontrans_metals keeps its L1 domain: it is a stationary restart of a stored L1 state whose purpose is the transported ionization stages, not the outer boundary; a new stored state on a 2 R_p domain would change what it guards without guarding anything new (rev4 plan A2). Its outer face is subsonic (Mach 0.95), which its output now reports.

## 132. The H3+ LTE cooling function is MiZATeP (Mizus et al. 2017); the Miller et al. 2013 fit is removed (plan rev4 step B)

**Why** (md/h3p_cooling_cause_20261010.md): the Miller et al. 2013 LTE fit cannot be reproduced from the line list and partition function its authors state; the NMT and MiZATeP line lists agree with each other to 1%, and the fit lies above both by 1.14 (1000 K), 1.48 (1500 K), 1.89 (2000 K), 2.11 (3000 K) (MEASURED). A correction of existing physics against its published source.

**Change** (worker diff tmp/h3p_impl/h3p_mizatep.diff, applied unchanged; note

md/h3p_cooling_mizatep_implementation_20261010.md): new module h3p_mizatep_cooling_table.f90 (4981 nodes, 20-5000 K, verbatim from the ExoMol file 1H3_p__MiZATeP.cf, kept as references/ExoMol_1H3_p__MiZATeP_20170330.cf; generator src/utils/h3p_mizatep_cooling_table.py); h3p_lte_emission_mizatep replaces h3p_emission_lte for the emission and for the W(T_rad) absorption term, ln W linear in 1/T between nodes; Miller's non-LTE factor s kept (a ratio of its own vibrational sums); renames in certification.f90; comments in util_ion_eq.f90 and Cool_coeff.f90. Units unchanged (W molecule^-1 sr^-1). **Verified by the advisor**: the built module returns 2.92879e-20, 1.29793e-19, 3.19812e-19, 9.54546e-19 W sr^-1 at 1000/1500/2000/3000 K, the ExoMol values; suites physics_probe 2519 PASS / 0 FAIL (51 new H3+ assertions, 31 of them RED on the old module) and certification 145 / 0. gfortran tree build f69f7371. Measured by the worker on 12 000-step molecular snapshots: log10 Mdot unchanged (10.51, 10.52), total cooling -19% and -7% at 1.02 R_p, T within 0.5 K. Findings recorded in the note: the ExoMol .cf is wrong below ~16 K (rounded partition function; the table starts at 20 K), and Table 7 lies 0.5-4% above the .cf at 3000-5000 K (the code follows the .cf, which a full line sum reproduces). Living documents that still name the Miller fit are updated in step D. Every H3+-bearing golden moves; refreshed at the end.

## 133. Turbulent dissipation heating does not make the chemistry's K_zz = 1e9 a heat diffusivity; recorded, not implemented (review, user directive 2026-10-10)

Record: md/eddy_dissipation_heating_review_20261010.md. No file in src/ changed (user directive: physics that published models do not treat is recorded, not forced into the code).

- Papers read in full (both supplied by the user): Hunten (1974, JGR 79, 2533) and Osborn (1980, J. Phys. Oceanogr. 10, 83; a scan, quotations checked on the page images).
  - Turbulence that mixes a stable gas with K draws the mechanical power K N^2/R_f per unit mass, and dissipates eps = K N^2/Gamma of it into heat. Here Gamma = R_f/(1 - R_f) <= 0.2 (Osborn Eqs. 7-10); Hunten's Eqs. 7-9 have Ri_c ~ 0.2 in its place.
  - Hunten: eddy mixing "probably heats the atmosphere and almost certainly does not cool it"; "It is better to leave out two compensating effects than to include only one of them."
  - Derived here: the mixing of theta also adds the work against buoyancy, rho K N^2, to the enthalpy of a hydrostatic column. The eddy flux of section 130 (Johnstone et al. 2018, Eq. 44) has neither term.
- Measured on k05 (xi_w = 1, scratch copy of stratosphere_energy_balance with the heating, clima re-solved before every solve):
  - The eddy flux alone ends between 1.0e6 and 1.38e6 cm^2 s^-1 (below 50 K); with the work against buoyancy alone it ends before 2.79e6.
  - With dissipation a steady T(p) exists to 1e9: T at the match is 345 K (dissipation) or 383 K (both terms), and the stratosphere reaches 650-720 K.
  - At 1e9 the turbulent heating is 4.2e8-5.3e8 erg cm^-2 s^-1: 790-1000 times the absorbed stellar flux (5.36e5). It is conducted into the held troposphere, and OLR - ASR = 3.5e6-6.4e6.

  - The heating passes the absorbed stellar flux at K ~ 1.2e6.
  - Verdict: K_zz = 1e9 cannot be a local stably stratified turbulence (no energy source of that size). It is an effective tracer coefficient, and `--eddy-heat-transport off` stays.

- Published models (READ):
  - Johnstone et al. 2018 carry eddy conduction (free K_E <= 6e8) without the dissipation heating.
  - Koskinen et al. 2013a/2022, Huang et al. 2023 and Yelle 2004 carry no eddy term in the energy equation; in CETIMB K_zz enters the species diffusion only.
  - Zhang & Showman 2018 (ApJ 866, 1): K_zz is a tracer-dependent effective coefficient.
- Proposal, not merged: if a heat diffusivity is ever switched on, the work against buoyancy and the dissipation go with it (scratch `turbulent_heating_functions`, off by default, 7 scratch tests PASS).
- Defect found, reported, not fixed: clima's `TOA_fluxes_dry` leaves its last profile in `c.T`, so a second `stratosphere_energy_balance` call on one clima object holds a drifted troposphere (bottom node 234.5 K or 289.4 K instead of 222.66 K).
  - The adapter (one call per fresh object) is not affected.
  - The K_zz continuation tables of section 130 are affected beyond their first entry.
- Not measured: the column T_match on the heated references (the analysis stopped at the verdict).
- Figure: `docs/figures/eddy_dissipation_reference_k05_20261010.png`.

## 134. The chemistry is handed the hydrogen-NUCLEUS flux, not the component-1 mass flux (plan rev4 step C)

**Defect.** The elemental flux windows of `write_element_flux_profile` (`binary_element_diffusion.f90`) give F_H = 4 pi r^2 (F_rho (1 - Y_He) - J), the mass flux of component 1 of the binary split: the hydrogen carriers TOGETHER with the trace metals and the C, O bound in molecules (`mixture_mass_split`). The closure handed it to the chemistry as `--trial-flux-H`, which the adapter divides by m_H as a flux of hydrogen nuclei, so the metal mass was counted as hydrogen (0.44% on LHS 1140 b).

**Fix.** The windows also reduce F_Hnuc = s_H F_H, s_H = m_H n_H / m_1 the hydrogen-nucleus share of component 1's mass in the upwind cell (the metals of component 1 share its advective and diffusive flux in this operator), written as `lower_profile_F_Hnuc_{median,spread}` and `steady_F_Hnuc_{median,spread}` in `EXHALE_resolved.out` (`lower_atmosphere_profile.f90`, `write_setup_report.f90`). `element_flux_closure.py` reads F_Hnuc (`hydrogen_nucleus_flux`), falling back to F_H with a stated reason for an older EXHALE. F_H and F_He keep their meaning (F_H + F_He = Mdot). A resumed closure whose history holds component-1 fluxes no longer matches its saved measurement and is refused, as intended for a changed meaning. MEASURED (`lower_profile`, HD 209458 b with solar metals, 8 threads, tree build d10b3634): F_Hnuc/F_H = 0.98820 in both windows (3.3144e12 / 3.3539e12 g/s overlap; 6.7128e9 / 6.7926e9 steady); F_H, F_He, Mdot unchanged. Tests: `element_flux_closure` 10 and `matching_surface` 16 pass.

**Also fixed (found by the dissipation-heating review, section 133).** clima's `TOA_fluxes_dry` leaves its last trial profile in `c.T`; the stratosphere solve (eddy heat transport on only) now restores the climate object's solution on exit (`stratosphere_energy_balance`, a wrapper around `stratosphere_energy_balance_of_held_troposphere`); the module docstrings no longer say the eddy option is on by default.

## 135. Step D: documents brought to the code; four defects the documentation review found (2026-10-10)

The user-facing documents were updated to sections 125-134 (README.md, README_HOWTO.md, docs/EXHALE_user_manual, docs/EXHALE_physics_and_algorithms and its physics_overview sections, docs/lower_atmosphere_coupling, docs/molecular_hydrogen_treatment, docs/methodology_comparison, md/input_schema, md/code_status_20260910 banner, examples/README, examples/15_molecular), and three run recipes were added (examples/recipe_hd209458b_atomic, examples/recipe_koskinen2022_model_a, examples/recipe_lhs1140b_photochem_closure; their keys checked with 5-step runs). Defects the review found in the code, fixed here (gfortran tree build af223ef9):

1. `dayside_dilution` and `newton_flux_switch` were written only to the parse dump, not to `EXHALE_resolved.out`, so the closure's geometry cross-check (section 129) never compared; both are now resolved-configuration keys.
2. The marching stop counted as a stationarity claim, so a march stopped at face spread 4e-4 exited 2 unless it certified; the stop is twenty times looser than the steady gate and its state is a relaxation snapshot, so it no longer claims stationarity (exit 0, reason no_stationary_claim). A Newton finish and the residual gate still claim.
3. Stale comments and messages: the stop's threshold named as flux_spread_th; section numbers 130 for the hand-off (131); the startup line "then WENO3 until du < du_th"; the open-face seed comment that still named the removed entry relaxation.
4. `md/acceptance_by_observable_precision_20261009.md` A2 still named du < 1e-3 as the marching stop: annotated in place. Reported by the review and left as stated: the recipes give run times READ from earlier logs (no recipe was run to completion with the current binary); the example-15 hand-off switch default (5.0e-2 dropped) is not re-measured on a molecular run.

## 136. Checkpoint of plan rev4 steps A-D: frozen binary and regression matrix (2026-10-10)

Frozen: `LHS1140b/models/.P2/EXHALE_f8a38866.x` (ifx, manifest `BINARY_MANIFEST_f8a38866*.txt`), the tree of sections 125-135. Matrix: the gfortran build af223ef9 of the same source, one thread, `tmp/step2/regm/stepAD`. Against the goldens every case moves (the goldens are stale since before section 125 and are refreshed once at step G). Against the matrix of section 127.3 (build 7a1ff33f: secondary ionization from step 0 and the new stop, before the H3+ cooling, F_Hnuc, the hand-off and the 2 R_p WASP faces):

| Case | log10 Mdot | Largest change (rho, v, T) | Cause |
|---|---|---|---|
| lower_profile, hydrostatic_column, iontrans_metals | unchanged | BITWISE | none reaches them |
| mol_* (7 snapshots) | unchanged (10.49-10.52) | rho, T <= 0.9%; v near the base where v ~ 0 | H3+ cooling (132) |
| oxygen_chemistry (1000-step snapshot) | 9.40 | rho 6%, T 38% at 1.77 R_p | H3+ cooling of a transient snapshot |
| hp_zero_seed, hp_trace_seed, hp_front | 10.56 | rho 2e-5, T 1e-4 | H3+ cooling |
| wasp_full, wasp_he23off (15 000-step snapshots, 2 R_p) | 13.24 | new domain, not comparable row by row | 127.6 |
| wasp_full_newton (2 R_p, hand-off at face spread 0.1, step 16 500) | 13.29 -> 13.28 | new domain | 127.6, 131; JFNK info 0, outer Mach 1.50 |

The capped WASP snapshots end on their cap with no stationarity claim (exit 0, section 135). Steps A-D of plan rev4 are complete; the scope of E, F and G is the user's next decision.

**136, addendum (2026-10-10).** After the freeze, comments only were corrected in `EXHALE_main.f90` (which stops claim stationarity; the du-stop wording), `parameters.f90` (du_th, du_th_plm) and `input_read.f90` (the du_th key); no executable statement changed, but the line shifts move the md5 of a build (tree gfortran build 169a0f2b). The frozen `EXHALE_f8a38866.x` stays the A-D binary; its manifest differs from the tree by these comments only.

## 137. The six long-failing test rows were stale expectations, not code defects; two rows made stale by section 127.5 (2026-10-10)

Diagnosis: `md/failing_rows_diagnosis_20261010.md` (worker; scratch `tmp/diag6/`).

- `carrier_enthalpy_flux`, 3 rows: the test's reference settling drift used the gravity Dphi v0^2/R0, while the operator uses g_eff = -(1/rho) dp/dr (`mixture_pressure_acceleration`) since section 115; on the test's isothermal, outward-dissociating column the two differ at O(1), sign included. The code is right; the test reference now calls the same function (test file only). Section 115's statement that these rows were "pre-existing" was wrong and is annotated there. Now 12 PASS / 0 FAIL.
- `grid_and_gates` `stationary_evaluate_*`, 3 rows: the fixture `backup/regression/wasp_full_newton/IC/` (2026-09-27) was no longer a solution of the current equations (changes of sections 66-72, 112, 125, bisected by frozen binaries), and since section 127.6 (outer face 2 R_p) it no longer matched the case grid at all. Refreshed: the 2 R_p state re-solved with the tree binary 169a0f2b (certified as loaded, JFNK 1 iteration, info 0, ||R|| 9.1e-10); old pair kept in `IC_pre_outer2_20261010/`, record in `IC/README.md`.
- Two further rows (`outer_iteration_ending_run`, `direct_steady_setup_report`) had become stale with section 127.5: their bounded stationary solves end with info /= 0 and now exit 3; both accept a written state with

status 3. Result on the tree: `grid_and_gates` 427 PASS / 0 FAIL, `carrier_enthalpy_flux` 12 / 0. The two other suites that check exit codes (`species_face_flux`, `coupled_source_step`) run marching routes, which cannot exit 3 (code path). Also: the header of `restart_intent_and_metadata.sh` corrected (since section 88 the evaluate route writes the stored pair and rebuilds only heat and cool). Not touched: `iontrans_metals/IC/` (a copy of the old pair, run on its L1 domain as decided in section 131).

## 138. A Roche domain extended past the potential ridge: the cold IC, the outer ghost and the absorbing columns assumed a potential rising outward (plan rev4 step E, 2026-10-11)

**Found by** the step-E benchmarks (Salz et al. 2016 HD 209458 b / HD 189733 b, Huang et al. 2023 WASP-121 b Cases B and D, Yan et al. 2022 WASP-52 b), every one a Roche domain with `Outer radius [R_p]:` well past L1. Diagnosis note: `md/roche_outer_domain_diagnosis_20261011.md`. Frozen binary f8a38866 unless stated; MEASURED.
**Defects.** Past the ridge of the Roche potential (L1) phi falls outward, toward the star; three parts of the code assumed it rises.

1. Cold hydrostatic IC (`set_IC.f90`, rho = rho_bc exp(-(phi - phi(1)))): the density rises without bound past the ridge (HD 209458 b, 15 R_p: 5.5e10 cm^-3 at 10.5 R_p, 5.1e43 at 15 R_p). Both Salz cold starts failed at step 0.
2. Outer ghost (`Apply_BC.f90`, hydrostatic continuation exp(-(phi_g - phi_N)/c^2)): past the ridge the factor exceeds one. Huang B at 12 R_p, step 0: cell N 1.7e6, ghosts 3.4e34 and 4.9e63 cm^-3; Yan WASP-52 b at step 6780: 3.3e3 against 3.6e22 and 8.1e41. Live runs at 15 / 8.85 / 8.2 R_p carried ghost/cell-N ratios of 1e5 to 1e14.
3. The absorbing columns started at N+Ng, so the ghosts were gas on the line of sight (`calc_column_dens` and its two variants in `utilities.f90`, the sweep in `util_ion_eq.f90`, the Ly-alpha line-center depth in `lya_rt.f90`, the fine-structure escape column in `Cool_coeff.f90`). An exploding ghost shielded the whole domain: zero heating in every cell of the Yan atomic runs, T to 13 K, energy update failure (exit 2). In a domain with sane ghosts it added a two-cell column of no physical meaning. The regression case `wasp_full` (outer face 2.0 R_p, ridge 1.539 R_p) carried defect 1: in 200 steps from the cold IC the density rose 2000-fold toward the outer face with inflow at 1.96 R_p.

**Fix** (physics stated at the code):

- IC: stratified on the running maximum of phi from the base outward; bitwise the old IC inside the ridge, held at the ridge value (in practice the 1e-8 floor) past it.
- Outer ghost of a supersonic cell N (v_N above the adiabatic sound speed): the steady isothermal wind through cell N, v_g from the supersonic Bernoulli root, rho_g from rho v r^2 constant, p_g/rho_g = p_N/rho_N; valid in any potential. The subsonic edge keeps the hydrostatic continuation (open user decision, sections 127.4 and 127.6). The Bernoulli root solver has one definition, `isothermal_wind_speed` in `grav_func`, shared with the transonic IC.
- Every stellar-beam and escape column starts at the outer face r_edg(N); the outer ghosts carry no depth.
- Setup report: a Roche domain with `Outer radius` reads "extended to the user outer radius X R_p; Hill radius Y R_p" (it said "truncated at the Hill/L1 radius").

- Test drivers `xuv_cell_mean_attenuation.f90`, `lya_beam_cell_mean.f90` (new assertion `lya_outer_ghosts_carry_no_depth`), `lya_escape_probability.f90` on the outer-face convention.

**Tests** (gfortran scratch build of this source): the five failing setups (Salz HD 209458 b and HD 189733 b cold IC at 15 R_p, Huang B at 14.11 R0, Huang D at 13.28 R0, Yan atomic 92/8 at 10 R_p) ran 3000-8000 steps with exit 0 and no energy-update failure; outer cells supersonic (Mach 9.4, 12.9, 181, 53, 50), ghosts at or below cell N, rho v r^2 constant through the ghosts (Salz HD 209458 b: 3.49e14 in cell 502 and ghost 504). `wasp_full` 200 steps: unchanged below 1.56 R_p (median 2e-5), the cold-IC rise removed, outer Mach 0.067 -> 0.589. `mol_base_handoff` 200 steps (Hill domain): outermost cell only, at most 3.1e-3. Suites grid_and_gates free_outflow_boundary and hydrostatic_residual 82 PASS; physics_probe no failing assertion. Not done: the column change alone on wasp_full. Every golden moves (column change); refreshed at step G.

## 139. The internal energy of cold, highly supersonic expanding cells from the advected adiabat (dual-energy method) (plan rev4 step E, 2026-10-11)

**Found by** the Yan et al. 2022 WASP-52 b atomic runs after section 138 (Roche, 10 R_p): h98he2_f1, h92he8_f1, h98he2_f0p5 failed at steps 7290, 8795 and ~8000, exit 2, "incoming temperature already below the floor" (T_floor = max(0.01, 1 K/T0) = 13.04 K, `energy_semi_implicit.f90`). Note: `md/cold_supersonic_internal_energy_20261011.md`.

**Cause** (MEASURED, last accepted state of h98he2_f1): the scheme recovers rho e = E - rho v^2/2; at 9-10 R_p rho e/(rho v^2/2) = 5.7e-4 to 8.2e-4 (M ~ 50), and the difference carries the truncation error of the kinetic-energy flux. Between 9.03 and 9.14 R_p the adiabat p/rho^(5/3) drops 7.2-fold (1.00e-15 to 1.38e-16 cgs) where the net source is positive (heating 1e-16, cooling 2e-19 erg cm^-3 s^-1), which an adiabatic flow with heating cannot do. Not round-off, not the outer ghost (the collapse starts ~15 cells inside the face). Yan Fig. 23a (READ, digitized) gives 377 K at 9.15 R_p and 334 K at 9.73 R_p.

**Fix** (new module `adiabatic_entropy_transport.f90`; Ryu et al. 1993, ApJ 414, 1; Bryan et al. 1995, CoPhC 89, 149; full texts not read, their threshold values not compared): the adiabat K = p/rho^gamma is carried with the face mass fluxes in every SSP-RK3 stage (donor-cell face value), taken fresh from the state at the start of each step; after an accepted step, rho e is taken from K where rho e < 0.01 E (M >~ 13), div v >= 0 (no shock) and the gas has the constant gamma (atomic). Heating, cooling, conduction and viscosity are applied after it as before (operator split). Total energy is not conserved in the reset cells by less than 1% of their E; the run log prints the number of reset cell-steps.

**Tests**: h98he2_f1 8000 steps exit 0 (256 080 cell-steps reset; outer T 820-990 K, adiabat 1.05e-15 to 1.12e-15); h92he8_f1 9000 steps exit 0 (294 429; T 427-527 K at 9-10 R_p, outer Mach 22); `wasp_full` 200 steps bitwise identical to section 138 (0 cells reset). Reached by: every Yan run, Huang B and D (75 and 78 cells, 2.8-13.8 R_p; Huang B showed the same adiabat fall at its outer face), the Salz production states (22 and 261 cells). Open: in the 98/2 test the base cells flowed at -290 cm/s at step 8000 against +675 cm/s in the control at step 7271 (base breathing or this change, not separated).

**Benchmark binary**: `EXHALE_dualE_f9047247` (gfortran 16.2, source of sections 136 + 138 + 139, manifest beside it); all step-E runs were restarted on it from their last states (`Restart intent: relaxation`) on 2026-10-11 10:15 KST.

## 140. Step-E inputs moved into the tree; documents brought to sections 138-139 (plan rev4 step E, 2026-10-11)

**SEDs.** `inputdata/sed/hd209458b_solar_salz2016.txt` and its builder `build/build_hd209458b_solar_salz.py` moved in from the session scratch; the builder run in the tree reproduces the file byte for byte (MEASURED, `cmp`). The two Huang Lyman-continuum files were already in the tree; `build/build_wasp121b_composite_lyc_huang.py` with and without `--xuv-multiplier 0.74` reproduces both byte for byte (MEASURED). `build/sed_coverage_and_order_check.py` now checks the three files (file rules, band fluxes) and the three Huang configurations, and asserts the normalizations: Salz F(5-100 A) = 4.0041e1 and F(100-912 A) = 1.1028e3 against log L_X = 26.40 and log L_EUV = 27.84 at 0.04723 au; Huang F(1-912 A) = 2.69e5 and 0.74 x 2.69e5 (all PASS, exit 0). `inputdata/sed/README.md`: bin convention, inventory, band-flux table and build commands for the three files.

**A finding stated in the inventory.** The WASP-121 b files normalized to F_XUV = 1.6e6 (`wasp121b_solar_huang2023.txt`, `_wasp17_`, `_composite_`) carry the number Huang et al. 2023 quote, not the flux their model was driven by (2.69e5; benchmark README of `benchmarks/huang2023_wasp121b/`): about 6x brighter below 912 A. `WASP-121b/` and `benchmarks/wasp121/` still use the 1.6e6 composite; whether they switch is the user's decision.

**Case D checked against the paper.** Section 6.1 (READ): "we reduce the stellar flux at wavelengths shorter than 1700 A to 74% of the reference spectrum ... and weaken the stellar Lya radiation intensity to 35% of the canonical value", the second being "the Lya flux used in the Lya radiative transfer calculation". The case-D SED (everything below 1700 A x 0.74, Lya line included) with `Stellar Lya flux: 3.5e4` follows that; no change.

**Comparison script.** `benchmarks/benchmark_profile_comparison.py` (from the scratch `compare/`), paths now relative to its own directory: runs under `benchmarks/<benchmark>/`, output under `benchmarks/comparison/`. The scratch copy keeps the scratch layout until the runs move.

**Documents** (sections 138-139 were in the code only):

- `docs/physics_overview/radiation.tex`: the column starts at r_edg(N), sum over j < j' <= N (it said r(N+N_g)); the ghosts carry no depth.
- `physics_overview/hydrodynamics.tex`: either potential can be carried past L1 (it said only the spherical one could).
- `physics_overview/boundaries_initial.tex`: the steady-wind ghost of a supersonic cell N; the cold IC on the running maximum of phi; the columns do not read the ghosts.
- `physics_overview/numerics.tex`: the dual-energy method in the operator-split step; Ryu et al. 1993, Bryan et al. 1995 and Yan et al. 2022 added to the bibliography (bibcodes and author lists confirmed on ADS).
- `docs/EXHALE_BC_and_IC.tex`: four stale statements fixed (no incoming characteristic at any outflowing outer face; the outermost ghost given its own column; Roche always truncated at the Hill radius; cold-IC line numbers), plus the steady-wind ghost and the running-maximum IC.
- `docs/EXHALE_user_manual.tex`: Roche past L1 paragraph (IC, ghosts, columns), a paragraph on supersonic outer cells and the reset count of the run log; the `wasp_full_newton` outer-face numbers marked as measured

before section 138.

- README.md (features), README_HOWTO.md (the same outer-face numbers marked).
- src/modules/states/Apply_BC.f90: the comment of free_outflow_ghost listed the column integral among the readers of the ghost; fixed (comment only). All three PDFs built with no undefined reference.

## 141. Stale folders deleted (user decision, 2026-10-11)

A read-only survey of EXHALE_v1.00/ (three workers, references re-checked by the advisor) sorted every leftover into delete / ask / keep; the user approved the delete list and added build/tests/. 233 paths (list and log: the session scratch delete_list.txt, delete_stale_20261011.log), about 10.6 GB, none tracked by git, none in use by a process on lart2/3/4:

- LHS1140b/models/.P1/: 64 audit, diagnostic and test directories of plan rev2 (binaries before sections 112-139). Kept: fiducial_conduction, v5_test, audit_F1 (live), atomic97, s2_continuation, newphys, ghost_trial_refusal, publish_rows, cx_round4, s1_remeasure, connectivity (cited by living documents or open items), and iontr_diag, progress_control, omp_scaling_lart2 (cited as the measurement behind constants in EXHALE_main.f90 and init.f90).
- LHS1140b/models/.L*: 14 stage-2 solver diagnostic trees and .L8 except .L8/r45 (cited by src/utils/collisional_validity.py); .L14, .L22, .L25, .L26, .L34 kept (catalog provenance, CLAIMS_59bfdb3fc4d0.md). LHS1140b/lower_profile_vulcan/ pruned to vulcan_work/STOPPED.txt.
- backup/regression/: 28 golden_* snapshots of closed series, baseline_post170/2a/2b_20260905, parse_golden_blockE_20260903, parse_golden_retired_20260904; backup/EXHALE_pre_handoff_restart_rescale.x. Kept: golden/, golden_pre_final_20260927, golden_pre_F1_20260929, every case and fixture.
- scratchpad/; the agent worktree .claude/worktrees/agent-a213596977999e418 and its branch (its uncommitted diff and untracked tests saved in the session scratch, worktree_agent-a2135_record.tgz; every test file exists in the main tree); build_f6/, build_ifx/, build_w1/, build_w1g/, build_w2/, build_w3/, EXHALE_ifx.x, EXHALE_w*.x, build_ifx.log, tp_mod.mod, pid; build/tests/ (1.1 GB of test products, rebuilt by src/tests/*/run.sh).
- Planet, benchmark and example folders: the output_pre_* snapshots of the August fixes, the tpm_* directories of those states, output_jun7, output_caseA2058_jun7, output_nan_20260810, output_march1e6_20260811, WASP-121b/solver_validation/, the WASP-52 b *_pre_kb_20260815 copies and the variants fxuv0p25_he98_L1_noLya, fxuv0p25_he98_L1trunc, fxuv1p0_he98_rbase; __pycache__/ and .ipynb_checkpoints/ throughout.

Text fixed with it: the inventory tables of HD189733b/, HD209458b/, WASP-121b/, benchmarks/hd189/, hd209/, wasp121/ README files (rows removed, one line naming what was deleted), WASP-121b/STALE_P48.md, examples/README.md, four places in md/TO_BE_DONE.md, a header note in the three dated memos that cited scratchpad/. Update logs and dated memos are records and otherwise unchanged. The Makefile comment and the backup/regression/run_check.sh header that name baseline_post170_* and golden_block* are edited after the regression run that was executing the script ends.

Executed 2026-10-11: 229 paths deleted, 4 entries absent because their parent had gone first; the worktree carried a lock of a dead agent process (pid 61637, lock dated 2026-10-03, no such process on lart2/3/4) and was removed with `git worktree remove -f -f`, its branch deleted. Every kept item checked present afterwards; no git-tracked file deleted.

Second round, 2026-10-11 (user decision): deleted `LHS1140b/models/.P2/build_*_ifx` (20 build trees; the frozen binaries `EXHALE_<md5>.x` and their `BINARY_MANIFEST_*.txt` stay), `vulcan_work/hd209_wind_response/` and `vulcan_work/pc_compare_p1/exhale_hd209/` (the 30 runs invalid by item P35), `backup/HD209458b_test/` (closed June study, retired keys), `lart_runs/*_pre_kb_20260815/` (8, untracked, unreferenced); compressed `benchmarks/hd189/rerun_lyafix.log` and `run_20260815_kb.log` (175 MB to 39 MB, `.gz`); removed the empty tracked `input.inp` at the root (re-added by accident in 460ea67; it made the existence check of `run_EXHALE.sh` pass on an empty file), which git now shows as deleted. `LHS1140b/archive_20260830/` was NOT split: 88 of its 119 run directories are named in the update logs, memos or `md/TO_BE_DONE.md`, and the other 31 are 113 MB. Text fixed: `src/utils/pin_base_grid.py` (FROZEN_ROOTS), `backup/regression/run_parse_corpus.sh` (header and exclusion list), three places in `md/TO_BE_DONE.md`, a header note in the eight memos that cite the deleted runs. Log: session scratch `delete_stale_20261011_round2.log`.

## 142. The regression matrix measured across sections 138-139 (2026-10-11/12)

Both binaries gfortran 16.2, `-O3 -fopenmp`, the 17 default cases single-threaded: the tree build (md5 41af91c7, sections 136-139) against the same source before section 138 (`T/rochefix/ctrl/EXHALE.x`, md5 72ccde1b; its source differs from the tree only in the files of the 138 and 139 diffs, checked with `diff -rq`). A first control with the frozen ifx binary f8a38866 gave the same numbers to the digits below (compiler effect negligible), except that it refused to load `iontrans_metals`: r_max of the L1 domain differs in the last bit between ifx and gfortran and the case's `IC/` (2026-09-27) was written by gfortran.

- Every case ran to completion; `iontrans_metals` CERTIFIED in both. The printed log Mdot is the same in every case (13.24, 13.24, 13.28, 10.49-10.52, 9.57, 8.27, 9.40, 10.56).
- The dual-energy reset of section 139 acted in no case (0 cell-steps), so every difference is section 138 (columns from the outer face, steady-wind ghost of a supersonic outer cell, IC on the running maximum of phi).
- Largest relative differences in the physical cells, MEASURED (`T/cmp_phys.py`; cells where both values are below 1e-6 of the column maximum skipped): `wasp_*` rho 1.1e-2, heating 2.0e-2, |dv|/c_s 4.5e-3, all at 1.98 R_p next to the 2.0 R_p outer face, median rho 1.3e-4; molecular cases T 3e-3, heating 1.8e-2, |dv|/c_s 8e-3 at 4.62 R_p next to the outer face, median rho 7e-7; `lower_profile`, `oxygen_chemistry`, `hp_*` T and heating below 2e-3; `hydrostatic_column` round-off (1e-13); `iontrans_metals` rho 1.1e-2 at 1.32 R_p, median 6.4e-4. Relative cooling differences up to 0.4 occur only where the cooling is 1e-4 of the local heating and changes sign (`mol_base_handoff`, 1.74-2.36 R_p).
- Verdict: the change acts where it should (the outermost cells, where the two ghost cells used to add column and where the ghost rule changed), and moves no Mdot at the printed precision. Every golden moves (the column change touches every cell above round-off), so the refresh belongs to plan rev4 step G as planned. Outputs:

T/reg_new_41af91c7/, T/reg_ctl_72ccde1b/, T/reg_ctl_f8a38866/.

## 143. WASP-121 b run of record: the spectrum Huang et al. computed with (user decision, 2026-10-12)

WASP-121b/input.inp and benchmarks/wasp121/input.inp now load inputdata/sed/wasp121b_composite_lyc_huang2023.tx (F(1-912 A) = 2.69e5 at the orbit, the Lyman-continuum flux Huang et al. 2023 section 2.2 states for their spectrum) in place of wasp121b_composite_huang2023.txt (normalized to the F_XUV = 1.6e6 the same section quotes from Salz et al. 2019; about 6 x brighter below 912 A; the evidence is in section 140 and the benchmark README). These were the only inputs using a 1.6e6 file. Checked: sed_coverage_and_order_check.py exit 0, no FAIL (both configurations name the new file); a one-step run of the WASP-121b input (scratch copy, tree build 41af91c7) reads it and prints Log XUV flux at the planet distance: 5.430. No stored output was regenerated; both READMEs carry a 2026-10-12 section saying a run with this input is needed.

Found and fixed with it: both READMEs named the authors "Huang, Koskinen, Lavvas and Ben-Jaffel" (ADS: Huang, Koskinen, Lavvas, Fossati) and said Table 1 supplies R_p = 2.2075 R_J and R_star = 1.458 R_sun; Table 1 (READ, Delrez et al. 2016 values) gives the optical R_p = 1.766 R_J and R_star = 1.4572 R_sun. The input's Planet radius 2.2075 R_J (1.25 x 1.766) has no identified source; it is a model input and is left for the user (Huang's Case A base, from log g = 2.84 at 1 microbar, is 2.058 R_J, the value of benchmarks/huang2023_wasp121b/caseA).

Follow-up the same day (user decision): Planet radius [R_J] of both inputs 2.2075 -> 2.058 (Huang case A, 1 microbar level from Table 3 log g = 2.84). Only the radius changed: these inputs keep a fixed base density (log n = 12.49) and Equilibrium temperature 2358 K, whereas benchmarks/huang2023_wasp121b/caseA sets the base by Base BC: pressure 1.0 (1 microbar) and 3960 K (the first point of Huang's published T profile); the two base conditions differ and are reported to the user.

## 144. A subsonic outer face: no further treatment; its Mdot labeled not quantitative (user decision, 2026-10-12)

Decision on the open item of section 127.4 (md/outer_boundary_subsonic_review_20261009.md section 6): transonic winds keep the outer face past the sonic point; no relaxed characteristic condition is implemented for a subsonic face, because the escape models the code is compared with (Salz et al. 2016, Murray-Clay et al. 2009, Caldiroli et al. 2021) put the face past the sonic point and state no incoming condition, and that condition would make Mdot depend on an exterior pressure none of them states (project rule on physics the published models do not treat).

Code (steady_mass_loss_rate, EXHALE_main.f90): below Mach 1.1 at the outermost cell the run log prints "NOT A QUANTITATIVE MDOT" with the reason, and EXHALE_setup.out now carries the same line (it carried only the Mach number). Output text only. Also fixed: the comment of the beam ledger in util_ion_eq.f90 said the column N(N+Ng) is the outermost cell's own content; since section 138 the columns start at r_edg(N) and are zero above cell N. TEST (gfortran, md5 e6f7fcb7): hydrostatic_column 300 steps in a scratch copy, outer Mach 0.79: both lines written; data rows of Hydro_ioniz.txt and Ion_species.txt bitwise identical to the 41af91c7 build (only the provenance time stamp differs). Documents: user manual (new paragraph), physics document (upper boundary),

README, README_HOWTO.

Stored states of record ending subsonic at the default L1 face (estimate from the stored profile, gamma 5/3): `WASP-121b/` 1.02, `examples/18_oxygen_chemistry` 0.96, regression `iontrans_metals` 0.95; `benchmarks/wasp121` 1.14 just above. Their inputs are unchanged (an input change is the user's).

Second follow-up the same day (user decision): both WASP-121 b inputs now take the base and outer face of Huang's case A: `Equilibrium temperature 3960 K`, `Base BC: pressure 1.0` (1 microbar; the density key removed), `Outer radius [R_p]: 16.403` (Roche potential kept past L1). In `benchmarks/wasp121/` the VULCAN handoff `base.inp` (T_base 2135.9 K, r_base 1.98187 R_J, HeH_base 0.0909, which override T0, R0 and He/H) was renamed `base_vulcan_photochemistry.inp`, so it is no longer read. One-step runs of both inputs (binary e6f7fcb7, scratch copies): p_base 1 microbar, n0 1.8275e12 / 1.8290e12 cm^-3 (case A benchmark: 1.83e12), planet radius 2.058 R_J, upper boundary 16.40 R_p, log F_XUV 5.430. Stored outputs are not regenerated; both READMEs say so.

## 145. Diagnosis: the base-cell mass row refusal is the known split between the solver merit and the certification scale (2026-10-12)

Note: `md/base_mass_row_refusal_diagnosis_20261012.md`. The Yan WASP-52 b molecular benchmark (cold IC) ended with the wind stationary (face mass-flux spread 1.9e-4, JFNK ||R|| 4.4e-10) and was refused by the mass row of base cell 1, 9.3e-3 against 2.2e-11. A stationary evaluation with the boundary trace and the mass-floor scan (lart4) shows face mass fluxes of cells 1-12 differing by 1.4 percent of their mean, 9-10 decades above rounding. Cause, as stated at `residual_row_scale`: the certification divides the mass row by its flux, the Newton merit by the state scale, on which the row reads its fractional flux error times the local Mach number (1.2e-4 here), so JFNK stops at a base layer it cannot see; the marching stop reads only r >= r_flux. Not a rounding-anchor problem, and not the Salz HD 209458 b JFNK stall (merit stuck at 1.9). Under the acceptance rule of 2026-10-09 the benchmark Mdot (log 11.76) stands, the base-layer flux differences bounding the base supply's remaining effect. No code changed: a solver merit on the flux scale is a solver item for the user to decide.

## 146. Section 145 corrected after the Codex review: the refused Yan state is the marching fallback (2026-10-12)

Review: `md/REVIEW_20261012_base_layer_mass_flux_solver.md` (Codex). Its seven corrections were checked against the code and the publisher PDFs and all hold (table in `md/PLAN_20261012_base_layer_mass_flux_solver.md` section 0.1): the Newton mass column scale is rho (`state_scales_of_cell`), the printed `||R||` is the gate measure `rnorm` over the whole column and `||Fs||2` the merit; Koskinen et al. 2013a sect. 2.1.1 set only the lower-boundary velocity from the flux constant (READ); Murray-Clay et al. 2009's six dependent variables do not include Mdot (READ); the 1.4 percent spread is not an Mdot bound. Re-reading the raw log as the review asked: the Yan molecular stationary route converged its hydrodynamic rows (pass 20: base mass row 3.5e-11 against 1.7e-11, `rnorm` 4.4e-10) and failed on the carrier transport rows (H2 0.41, H+ 0.45, He+ 0.80 against 1e-5); the run fell back to marching and the 9.7e-3 refusal of section 145 is the verdict on that marching state. Section 145's cause ("the merit cannot see the base mass row") is withdrawn; its benchmark

statement for Yan (log Mdot 11.76 accepted by observable precision) stands, without the "1.4 percent bound". Huang Case B and Salz HD 209458 b stalled with interior rows near the measure's maximum (1.9), far from a root. The review's redesign (a common-flow constraint) is not started; the plan is rescoped to F1 (carrier rows in the partitioned outer iteration), F2 (the marching stop does not read the base layer), F3 (JFNK stalls far from a root).

## 147. Second Codex review: the F1-F3 plan corrected (rev2) (2026-10-12)

Review: `md/REVIEW_20261012_stationary_failures_F1_F3.md`. Each finding was checked against the logs and the source before it was taken in; all held. Plan rev2: `md/PLAN_20261012_stationary_failures_F1_F3.md` (its section 6 lists every correction with its evidence). The corrections that change what section 146 said:

- The Yan written state is a marching state that fails mass (9.68e-3), momentum (5.65e-4), energy (0.202) and the H2 and H+ carrier rows, not only the base mass row; the marching stop's state-change part reads all cells (`EXHALE_main.f90:4116-4133`).
- A stationary evaluation of the same byte-identical restart files does not reproduce the run's own residual (cell-1 energy flux divergence 4.89e-2 against 3.41e-2; energy row 0.444 against 0.202): reproducing evaluations is now prerequisite P0. The "1.4 percent" base-layer flux difference came from that reload's trace of raw F without r^2 and is withdrawn.
- F1: the carrier relaxation does close its own block (2.39e-10 at pass 18 against `carrier_accept_tol` 1e-8); the alternation with the hydrodynamic solve is what fails. The worst gated cell at r = 1.2002 R_p is the first cell of the certification gate, which is not a transport boundary; the H2 front is at 1.057 / 1.106 R_p. The coupled block is refused with ionization transport (`input_read.f90:2454-2520`). Pass 20 is not retained (p18, p19 only).
- F3: the energy row is multi-term and not bounded by 2 (Salz HD 209458 b starts its JFNK at 2.206); the statement of section 146 that rows near 1.9 mean "terms of opposite sign" is withdrawn. Huang B's energy row is at r = 5.51 R0 (divergence -1.02e-4 and net heating +9.81e-5 add), its mass row at 1.64 R0.

## 148. Error-testing material deleted from the source and the test set (2026-10-16)

User decision 2026-10-16 (`EXHALE_v1.10/md/environment_variable_options_survey_20261016.md`, D1 and D2): every piece of material created for error testing is deleted (fault injection, forced or expected failures, error reproduction, invalid-input and error-path tests, and the routines, variables, drivers, scripts, fixtures and environment controls only they use). Production failure detection, retry, rollback and recovery stay. No replacement error test was written.

**Deleted from the production source.**

- Environment controls and their injected behavior (19 names): `EXHALE_REJECT_STEP`, `EXHALE_ELEMENT_REFUSE_STEPS`, `EXHALE_TEST_FAILED_SOLVE_PASS`, `EXHALE_REJECT_AFTER_OP`, `EXHALE_REJECT_AT_STEP`, `EXHALE_REJECT_ATTEMPTS`, `EXHALE_AS_REJECT_IN_PASS`, `EXHALE_INJECT_SOURCE_ENERGY`, `EXHALE_AS_ERR_INJECT`, `EXHALE_AS_ERR_INJECT_COUNT`, `EXHALE_AS_FORCE_UNRESOLVED`, `EXHALE_CONDUCTION_REFUSE_DT_ABOVE`, `EXHALE_ENERGY_FAIL_AT_STEP`,

EXHALE_ENERGY_FAIL_CALLS, EXHALE_ADV_TEST_REJECT, EXHALE_TRIAL_ADMISSIBILITY_TEST (the whole trial-admissibility block of EXHALE_main.f90), EXHALE_PASS_STATE_STOP_AT, and the two found by the source search, EXHALE_CSM_GEOM_REFUSE and EXHALE_CSM_EXTRAP_REFUSE (the coupled source step refused every error estimate or every extrapolation candidate).

- Routines and variables only these used: attempted_step_injected_refusal, the reason code as_reject_injected (11 is now unassigned), err_pass_index, inject_source_energy_offset, the error-estimate multiplier and the forced unresolved verdict, attempted_step_perturb_checkpointed_state_for_test and its bump_* helpers, element_refuse_leading_steps_for_test, element_refuse_after_accepted_for_test, conduction_refuse_dt_above_s with CONDUCTION_INJECTED/CONDUCTION_REASON_INJECTED, energy_update_read_environment, species_face_identity_injection, stop_if_pass_state_step, carrier_perturb_checkpointed_state_for_test and its bump_* helpers, carrier_reject_leading_attempts_for_test with the reason carrier_reject_forced_for_test, carrier_history_reset_for_test, the test-only counter carrier_transport_stops_suppressed, lin_test_dense_preconditioner_fails, and the public export of build_banded_jac_full.
- D2: put_back_2d_logical, same_2d_logical; the always-true flag resid_at_own_composition (the sweep to the fixed point now runs unconditionally; the comment in parameters.f90 states it as fact, with its measurements); the never-set stop step_change_converged with its loop term, its report branch and the parameter dtu_th (dtu stays a printed diagnostic). The if (.false.) line of element_census.f90 went with the dead snapshot argument it silenced, and the commented-out Norcross block of cross_sec.f90 was removed.

**Kept at the boundaries, renamed for what they carry.** The negative as_inject_op that carried the real exhaustion of the nested positivity retry is now the logical hydro_bisections_exhausted (named hydro_stage_retry_exhausted until 2026-10-17) (verdict as_reject_hydro_stage, operation as_op_hydro). read_test_reject_reason is now read_column_energy_solve_controls, still reading EXHALE_ADV_COLUMN_FLOOR_IT and EXHALE_ADV_COLUMN_EXCLUDE_UNROOTED once, with the logical adv_column_controls_read in place of the -1 marker. chem_cycles_cap_for_test stays as a measurement control (the single closure increments of closure_returned_state.f90); carrier_initial_substep_for_test stays for the step-doubling rows; EXHALE_AS_FIXED_DT, EXHALE_ERR_EVERY, EXHALE_RESID_SC_MAX and n_eq_sweeps_fixed stay.

**Deleted from the test set** (src/tests/, not tracked by git). Whole suites: failed_wind_recovery, trial_admissibility, conduction_retry, restart_radius_guard; files certification/certification_capacity.f90, charge_exchange_rows/{stale_cell_rates_refused,stale_cell_stage_sou
grid_and_gates/{restart_grid_guard,sed_coverage_stop}.sh. Rows in 27 further suites: the injected refusals and exhaustions of attempted_step (including order_by_class, the temporal-order measurement that needed the error injection), the checkpoint perturbation rows of attempted_step and carrier_retry, the forced refusals of carrier_retry (sections 2, 4, 8, 12, 13) and of closure_returned_state (section 4), the face-identity injection of species_face_flux, the injected rejection reasons and the no-record transit refusal of adv_static_limit, the refused-key, missing-clock and inadmissible-handoff rows of run_mode, the conflicting-name

rows of executable_identity, the NaN/Inf/negative-input rows of the unit drivers, the expected-failure controls of steady_species_rows, and the refused-input rows of grid_and_gates, state_mapper, transit_state, molecular_seed, spectrum_type, energy_update, element_flux_closure, matching_face_radiation, matching_surface, conducted_heat_column, krylov_and_dogleg, coupled_block_handover, certification and physics_probe. The executable_identity SUITES list, README.md, README_HOWTO.md (suite list, 61 suites) and docs/EXHALE_user_manual.tex (PDF rebuilt) were updated; the Makefile discovers suites by directory and needed no change.

**Verification** (all after the edits, once).

- make (gfortran 16.2): no warnings. -Wunused-variable -Wunused-parameter on the 13 edited production sources against the same flags on the pristine source: no new warning (the six as_op_* imports and build_banded_jac_full it found were removed). Binaries: control (pristine source) md5 51092371..., measured md5 bdd9fe4d....
- Regression, 17 default cases single-threaded, control binary in backup/regression/<case> against the measured binary in sibling copies of the same case directories: the data lines of Hydro_ioniz.txt, Ion_species.txt and both _adv files are bitwise identical in all 17 cases (iontrans_metals writes no _adv pair), and the stop and final lines of the 17 logs are identical. golden/ was not written.
- make test, control on a pristine copy of the source (git archive plus the untracked src/tests and links to the untracked fixtures) and measured on the tree: no row fails in the measured tree that passes in the control. Measured: 60 PASS, 4 FAIL entries (grid_and_gates 4 rows of stationary evaluation and outer-iteration ending, residual_determinism seed family, steady_completion_flag, steady_selfconsistent_residual), each failing identically in the control. The two residual_determinism branch runs (atomic_elem_newton) were still marching after 5.5 h in both trees and were stopped by hand. Control-only failures: low_energy_electron_degradation (the copy cannot reach ../references/) and the deleted run_mode rows.
- Whole-binary rows of attempted_step (run separately, they need EXHALE_ATTEMPTED_STEP_EXE): control 164 rows (2 FAIL, the deleted temporal-order rows), measured 64 rows, all PASS, each with the verdict and value it has in the control. carrier_retry: control 156 rows, measured 84, all PASS and present in the control. run_closure_returned_state.sh: PASSED.
- grep of every deleted name in src/, Makefile, README*.md and the user manual: 0.

**Coverage given up.** The recovery paths are no longer exercised on demand: a refusal after each operation of the trial, retry exhaustion in both run modes, the energy-update, conduction and element-step failures, the failed-solve recovery of the outer alternation, the checkpoint completeness (perturbation) rows, the interrupted publication, and the invalid-input refusals. The temporal order of each row class (RK3 p+1 = 4, species split p+1 = 2), from which err_order_p was set, has no direct measurement in the suite any more. Production detection, retry and rollback are unchanged (regression above).

**Files.** Three __pycache__ directories under src/ (765 377 bytes) and nine .ipynb_checkpoints copies byte-identical to their originals (745 623 bytes; in docs/, docs/figures/k22_model_a/, LHS1140b/, cooling_data/;

none tracked) were deleted. The other byte-identical groups found by `md5sum` (audit run states and blocks under `docs/audit_20260905/`, the Cloudy decks of `src/utils/h2_shielding_*`, `inputdata/windae_grid/`, `benchmarks/koskinen2022_model_a/` restarts, the `docs/version_compare/` binaries) are run records or inputs whose identity is itself a recorded result, and were kept. The public verdict variables `conduction_last_nfail`, `conduction_last_T_floor` and `conduction_last_energy_deficit` of `viscous_conduction.f90`, written and never read once the conduction tests were gone, were deleted on 2026-10-17 (below).

**Follow-up, 2026-10-17: the two trees made to agree, the write-only conduction verdicts removed, the residual_determinism branch bounded.** The deletion was made separately in v1.00 and v1.10 (`EXHALE_v1.10/md/environment_variable_options_survey_20261016.md`, "Executed 2026-10-16"), and the two judgments differed. Every suite present in both trees was compared row by row (Fortran `check*/call test_*`, shell `PASS|FAIL <name>` lines, Python test methods) against the pre-deletion copies of both test sets, and both trees now carry the same rows under one rule: a row that constructs a failing input or state and asserts the failure report, or reproduces a past defect ("red" rows), is deleted; a row that asserts normal behavior stands, even where it was first written as the control of an error row. The test files of the suites present in both trees are now byte-identical, apart from `carrier_enthalpy_flux`, `interdiffusion_enthalpy_flux` and `steady_species_rows`, which differed before the deletion (v1.10 physics) and carry the same rows. Rows deleted in v1.10 to match v1.00: the expected-failure rows of `energy_update` (`test_no_bracket`, `test_temperature_floor`, `conduction_floor_tests.f90` `test_floor`, with `cool_constant` of `cooling_models.f90`) and its three defect reproductions (`falling_branch_two_iteration_residual_red`, `floor_clamp_would_have_moved_the_state_red`, `h2_conductivity_8p57K_exp_form_nan_red`); `inadmissible_probe_is_refused_at_any_ratio` (`krylov_and_dogleg`); the defect rows `live_token_on_the_stalling_pass_is_not_the_bound` and `old_conjunction_cannot_hold` (`coupled_block_handover`); `nh3_x300_exchange_beyond_its_bound_is_refused` (`conducted_heat_column`). Rows deleted in v1.00 to match v1.10: `kzz_1e9_refused_out_of_the_table_range` (`conducted_heat_column`), `test_nonzero_trial_against_zero_measured_is_refused` (`element_flux_closure`), the `hydrogen_split_residual(None, ...)` assertion and the `SystemExit` expectation of `test_blocks_is_the_step_budget` (`matching_surface`; the step budget is now asserted on a stall followed by a steady retry), and `reader_reports_incomplete_evidence_on_truncated_final_solve` with its fixture (`steady_selfconsistent_residual`). Rows restored in v1.00 from v1.10: `a_zero_residual_norm_is_admissible`, `a_finite_positive_residual_norm_is_admissible`, `the_largest_finite_residual_norm_is_admissible` (`carrier_retry`); `an_unread_healthy_step_is_accepted`, `an_unread_healthy_step_moves_the_composition` (`element_operator`); `the_block_was_still_entered_once` (`coupled_block_handover`); `the_increment_test_is_the_fallback` on the run with no estimate (`coupled_source_step`); and the two covered-SED runs of `sed_coverage` (`sed_coverage_covered_floor_runs/_output`, `sed_coverage_balmer_table_runs/_output`, new file `grid_and_gates/sed_coverage_covered_runs.sh`, README section 8). One name each: the `restart_continuation_factor` row is `factor_zero_is_a_factor` (its "not T" half was a refusal and is gone), and the `charge_exchange_rows` selector is `transport_with_metals`. Comments, dead imports and unused

variables were made the same in both trees; the reference-run row of attempted_step/run.sh, which in the v1.10 copy printed PASS before its check, now prints one verdict.

Production. The public variables conduction_last_nfail, conduction_last_T_floor and conduction_last_energy_deficit of viscous_conduction.f90 were deleted (declaration, export, assignment); nfail, T_flr and deficit are still computed because conduction_report prints them in the failure report. conduction_last_step is likewise written and never read, and conduction_last_T_solution is read only inside the module; both were left (reported, not deleted). The two names the two deletions gave to one object are now one: the logical carrying the exhaustion of the nested positivity retry is hydro_bisections_exhausted (what is exhausted is the bisection of the hydro dt), and the reader of the two column measurement keys is read_column_energy_solve_controls; post_process_adv.f90 is now identical in the two trees.

residual_determinism. The branch runs of atomic_elem_newton never ended because run_residual_determinism.sh loaded the fixture under the default relaxation intent: the run marched until the face mass-flux spread, read every 500 steps, fell below the hand-off switch 0.1, and the fixture (mapped onto the face-level grid on 2026-09-27) enters at a face spread of 1.82 (measured at step 500; still marching at step 662 after 300 s). The branch run now appends Restart intent: stationary, which measures the loaded state and enters the stationary solve with no CFL step, which is what the family is about; the report then takes 6 s. Both runs of the script carry EXHALE_MAXSTEPS=1 (no measurement takes a marching step, so 1 is the smallest cap under which both are measured) and timeout 600; a run that ends on either limit prints ==> LIMIT step_cap or ==> LIMIT wall_clock, and run.sh fails the row with measured=step_cap or measured=wall_clock (the step-cap path was exercised on a copy of the script without the intent: it ends in 4 s with that row). Measured 2026-10-17: the suite takes 20 s (v1.00) and 28 s (v1.10), all nine rows PASS in both, the seed family finding one root with a closure spread of 0.1635 of the row tolerance.

Verification (once, after all edits). make in both trees: no warnings, make -q clean; binaries v1.00 md5 6d7e381a..., v1.10 07d6cef1.... backup/regression/hp_zero_seed (EXHALE_MAXSTEPS=100 OMP_NUM_THREADS=1), binary of each tree before and after these edits: the data lines of Hydro_ioniz.txt, Ion_species.txt and both _adv files are identical in both trees. The touched suites (those above plus adv_static_limit, attempted_step with its whole-binary rows, carrier_retry with run_closure_returned_state.sh, certification, charge_exchange_rows, executable_identity, grid_and_gates, molecular_seed, physics_probe, run_mode, species_face_flux, spectrum_type, state_mapper) were run in both trees: every row that existed before keeps its verdict, and every restored row passes. No row went from PASS to FAIL. The failures are the ones that stood before: in grid_and_gates the three stationary_evaluate_* rows (certification unchanged, exit status, rebuilt within the sweep budget) in both trees and outer_iteration_ending_is_the_stagnation_one in v1.00; steady_selfconsistent_residual without a run log (its fixture rows pass in both trees when it is given one); and in v1.10 the rows whose inputs are not in that tree (examples/14_diffusion, examples/15_molecular, examples/lya_insitu_emissivity.py, the LHS 1140 b model states and SED, the WASP-52b SED: rows of adv_static_limit, charge_exchange_rows, conducted_heat_column, grid_and_gates and physics_probe) and carrier_retry the_full_term_scale_accepts_the_imbalance_at_a_short_step, which fails identically in the v1.10 control. Wall time of the slowest suites: grid_and_gates 26 min (v1.00) and 50 min (v1.10),

conducted_heat_column 20 min; coupled_source_step with its whole-binary rows (EXHALE_EXE set) 5.5 min, 44 rows PASS in both trees, measuring 5.05 passes per coupled step against 6.03 for the run with no estimate in both.

## 149. Uncalled routines, unbuilt sources and work-item option names removed from the source (2026-10-17)

User decision 2026-10-17: what md/leftover_test_code_survey_20261016.md (sections 2.3 and 5) and the two environment-variable surveys (item-named category) found unneeded is removed, in both trees (EXHALE_v1.00, EXHALE_v1.10; in v1.10 only src/, Makefile and docs/, never workflows/, runs/ or provenance/). Each routine was checked first with grep -rniw over src/ (tests, Python, Makefile included) of both trees: no caller outside the files deleted with it.

**Routines deleted (23).** h2_vibrational_critical_density (h2_vibrational_relaxation.f90), hep_h2_three_body_domain (mol_rates.f90), molecular_infrared_ready (molecular_infrared_cooling.f90), formation_energy_report (molecular_reaction_heat.f90), the internal gp of published_rate_coefficient (charge_exchange.f90), free_opacity_tables (opacity_models.f90), reset_coupled_block_entry_count (coupled_block_handover.f90), wae_get_ion_pot (wae_soe.f90); the ten Wind-AE input readers reached only from the unbuilt comparison programs, wae_load_all_params, val_after_colon, read_n_reals, read_n_ints, read_planet, read_phys, read_bcs, read_term, read_tech (wae_params.f90) and wae_load_params (wae_spectrum.f90); and the five parse_csv / parse_csv_after_colon copies inside those programs. The survey counted eleven readers: wae_load_spectrum, after_colon and next_csv of wae_spectrum.f90 are called by wind_ae_ic.f90 and wae_exhale_bridge.f90 and stay, so the readers are ten and the total is 23.

**What became unused with them, also deleted.** The run counters n_hep_h2_evaluations, n_hep_h2_three_body and hep_h2_worst_ratio of mol_rates.f90, read only by hep_h2_three_body_domain (the one-time warning of the He+ + H2 three-body domain guard stays); the module wae_ion_pots.f90 (its table was read only by wae_get_ion_pot) with its use and ion_pots_init call in wind_ae_ic.f90 and wae_exhale_bridge.f90; the wae_config import of wae_params.f90; and in oxygen_rates.f90 the rate functions rk_O8_O_H_assoc, rk_O9_H_OH_assoc, rk_O10_OH_OH_water, rk_O12_O1D_H2O_hydroxyl and lindemann_rate, whose only caller was the hand-compiled a2_m1_rate_check.f90. Their published values, ranges, sources and verdicts stay in the comments of oxygen_rates.f90 (O8-O12 are not in the minimal set: 7.8e-9, 9.7e-8, 5.5e-7 and 2.8e-5 of O1 at the HD 189733 b base), so the audit still records a value and a verdict; the header and the THIRD BODY M note now say that no code evaluates them.

**Unbuilt sources deleted (15) and what referred to them.** src/modules/wind_ae/{base_bcs_test,eqn_trace_f,glq_trace_f,ic
src/tests/a2_m1/a2_m1_rate_check.f90, src/tests/a2_m2/{a2_m2_g3_run_check,a2_m2_kinetics_check}.f90,
src/tests/e1_h2/e1_h2_channel_check.f90 with their saved .out; src/tests/grid_and_gates/base_closure_candidates_p
(named by no script). backup/regression/windae_oracle/relax_gate.py (v1.00 only) was read: it neither builds nor runs relax_ae_f, it only compares a CSV that program wrote, so it went with it. a2_m1_thermo_xcheck.py and its .out went with the rate check because the script reads the check's saved output; the three directories a2_m1, a2_m2, e1_h2 are therefore gone (none held a run.sh). Comments that named the deleted files were corrected:

element_census_tests.f90 now points to physics_probe/h2_channel_detector_ratio.f90, which gates the H2 channel sum and stoichiometry; wae_params.f90 and wae_spectrum.f90 describe what they now hold. No Makefile, README or docs/*.tex named any of the 15.

**Options and identifiers renamed for what they compute.** Each of the eleven item-named environment variables was read in the code. Ten have a current user and were kept: the v1.10 workflows set EXHALE_DIAG_BASE, EXHALE_P54_TS, EXHALE_L15_TRACE and EXHALE_L15_DUMP_AT; README_HOWTO.md documents the four L22B keys; the two mask keys change a result when set and were kept with the default (unset) unchanged. EXHALE_L15_REPEAT had no user in src/tests, a README, the Makefile or the workflows and was deleted with the repeat check it switched on (l15_repeat_check, l15_first_difference, the l15_repeat_at switch). The v1.10 workflow scripts that set the old names run snapshot executables built before the change and were left as they are; a run of the current source needs the new names.

| old |
|---|
| EXHALE_DIAG_BASE, diag_base, output/base_diag.txt |
| EXHALE_P54_TS, output/p54_flux_ts.txt |
| p54_ts_every/unit/open/face/r, p54_flux_timeseries_append, p54_cell_spread |
| EXHALE_L15_TRACE, EXHALE_L15_DUMP_AT |
| l15_setup, l15_iteration_record, l15_krylov_record, l15_dump_state |
| l15_hash_r8/logical/scalar, l15_fnv_basis, l15_fnv_prime, l15_number_of_active_bounds |
| l15_ready/trace_on/unit/dump_at/iter_now/path |
| EXHALE_L22_MASK, EXHALE_L22_MASK_MODE |
| l22_interval_mask_setup, l22_particle_count_change_by_set, l22_safety_tripped, l22_mask_*, l22_last_{j,d}_{select,omit} |
| EXHALE_L22B_JAC_RECON |
| EXHALE_L22B_JAC_ACTION, EXHALE_L22B_JAC_CELLS |
| EXHALE_L22B_DISPLACEMENT |
| l22b_setup, l22b_advective_band_derivative, l22b_jacobian_action_report, l22b_jac_recon, l22b_jac_action(_done), l22b_displa |
| upmap_* (EXHALE_main, the EXHALE_UPDATE_MAP diagnostic) |

Log lines that named the items ((L15), (L22 interval mask), (L22b), # L22 step 2b, # L22 step 3, increment I1) and the comments that cited P54/(AD) and P158 now say what they report. The tests carrier_base_face, carrier_enthalpy_flux and carrier_outer_boundary call the renamed carrier_jacobian_options_setup and carrier_advective_band_derivative; README_HOWTO.md, src/tests/coupled_block_jacobian/{README.md,run.sh} and the user manual (item L22 increment I3) were updated and the manual PDF rebuilt in both trees. The two environment-variable surveys were recounted from the source (v1.00 187 names, v1.10 196) and their item-named category is empty.

**Verification** (once, after all edits).

- make in both trees: no warnings, make -q clean; make wind_ae_ic builds without warnings in both. Binaries v1.00 md5 737270bf..., v1.10 e17134e1.... -Wall -Wextra on the 16 edited production sources: no warning names a deleted or renamed entity or an entity left unused by the deletions.

- Default behavior, binary of each tree before (`pre18_v100.x`, `pre18_v110.x`) and after, on scratch copies of `backup/regression` cases, `OMP_NUM_THREADS=1`, data lines (non-#) of `Hydro_ioniz.txt`, `Ion_species.txt` and both `_adv` files compared with `cmp`: identical in every file for `hp_zero_seed` (v1.00 at its `maxsteps` 100; v1.10 at `EXHALE_MAXSTEPS=100`), `lower_profile` and `mol_carrier` (v1.00, at their `maxsteps` 12000), 504 data lines each, with identical `final:` lines.
- Renamed diagnostics switched on once (post binaries): `hp_zero_seed` with `EXHALE_FACE_MASS_FLUX_SERIES=5` `EXHALE_BASE_CELL_STARTUP=1` `EXHALE_CARRIER_JACOBIAN_ACTION=jac_action.txt` `EXHALE_CARRIER_JACOBIAN_ACTION_CELLS=1,20` writes `output/face_mass_flux_series.txt`, `output/base_cell_startup.txt` and `jac_action.txt` in both trees; a stationary restart of the `wasp_full` snapshot with `EXHALE_NEWTON_STATE_HASH_TRACE=hash_trace.txt` `EXHALE_NEWTON_STATE_DUMP_AT=1` `EXHALE_JFNK_MAXIT=2` writes `hash_trace.txt` (header `# Newton state hash trace dump_at 1`, iteration and Krylov records) and `hash_trace.txt.dump1` in both trees; the `carrier_retry` driver (it calls `relax_photochemical_composition`) with `EXHALE_CARRIER_INTERVAL_MASK` (cells 2, 3), `..._MODE=veto` and `EXHALE_CARRIER_STATE_DISPLACEMENT=1` prints the `(carrier interval mask)` and `(carrier state)` reports and keeps its 86 PASS rows, as without them.
- Suites that use a touched module directly (35 in v1.00 counting `element_census_tests`, `closure_returned_state` and `grid_and_gates`; 38 in v1.10): every v1.00 suite passes. The rows of the suites run earlier the same day (`physics_probe`, `carrier_retry`, `charge_exchange_rows`, `certification`, `krylov_and_dogleg`, `coupled_block_handover`, `coupled_source_step`, `element_operator`, `attempted_step`, `spectrum_type`, `closure_returned_state`) have the same verdicts in both trees. v1.10 failures, each present before these edits: `carrier_retry` `the_full_term_scale_accepts_the_imbalance_at_a_short_step` and the missing-input rows of `charge_exchange_rows`, `coupled_block_jacobian` and `physics_probe` (section 148); `carrier_helium_inventory` (2 rows), `carrier_returned_state_acceptance` (1), `helium_metastable_transport` (2) and `steady_species_rows` (9), which give the same verdicts on a copy of the v1.10 source with these edits reversed. `grid_and_gates` (run separately: 28 min in v1.00, 52 min in v1.10) gives the same verdicts as the run made earlier the same day, 384 rows in v1.00 and 349 in v1.10, with the failures recorded in section 148 (the three `stationary_evaluate_*` rows in both trees, `outer_iteration_ending_is_the_stagnation_one` in v1.00, and the v1.10 rows whose inputs are not in that tree).
- `grep` of the old names (`EXHALE_L15_*`, `EXHALE_L22*`, `EXHALE_P54_TS`, `EXHALE_DIAG_BASE`, `l15_`, `l22_`, `l22b_`, `p54_`, `upmap_`, `p54_flux_ts`, `base_diag.txt`) and of every deleted routine and file in `src/`, `Makefile`, `README*` and `docs/*.tex`: none, apart from the typeset update logs `docs/Update_EXHALE_stage{1,2,3}.tex` (records of their time) and a file name `lhs1140b_stationary_L22_step3_design_20260916.md` cited by `coupled_block_jacobian`.

**Found, not fixed.** `make wind_ae_ic` writes `build/wind_ae_ic.o`, a main program, next to the production objects; 18 suite scripts take `build/*.o` less `(EXHALE_main|_tests|_probe)\.o` and then fail to link (`multiple definition`

of main). Measured here: after make wind_ae_ic, the suites with that filter that were run failed to build until the object was removed (it was removed in both trees before the verification above; wind_ae_ic.x stays). The filter is in grid_and_gates/run.sh among others, which this change did not touch; the one-rule fix (exclude wind_ae_ic.o in all 18, or build the IC generator's main object elsewhere) is left for the next change to src/tests.

# Code-size summary: EXHALE vs. the original ATES (2026-06-07; updated 2026-10-05)

A line-by-line comparison of the Fortran source against the pristine upstream code (ATES/ATES-Code-main) quantifies the extent of the in-house extension. *Pure code lines (SLOC)* exclude blank lines and comments (the comment stripper is string-aware, so a ! inside a quoted string is not treated as a comment); the line-by-line diff additionally normalizes whitespace and case, so reformatting and comment-only edits are *not* counted as changes. The measurement script is src/utils/codesize.py, which reprints the totals, the diff counts, every row of the two module tables and the byte-identical list of Table 77; the split of the core into the simulation build and the test drivers and the figures of the ten heaviest modified files are formed with its own counting functions. All numbers of this appendix are MEASURED on the working tree of 2026-10-05, after section 102 of the stage 3 log (md/Update_EXHALE_stage3.md). Up to 2026-09-22 (the previous edition, after section 16 of the stage 2 log) the extension comprised the acceptance and certification contracts, the attempted-step controller, the partitioned stationary solver, the oxygen and carbon chemistry, the molecular infrared bands, the ionization-stage transport and the charge-exchange rows, the base contact boundary and its trace, the molecular seed taken from an atomic state, the transition into the coupled carrier solve, the discrete conservation ledger, and the test suites those series brought with them. Stage 3 added the atomic-data audit and its repairs, the exact code-unit restart file (conserved_state_restart.f90), heat conduction with the conductivity of the mixture, the interdiffusion enthalpy flux, the enthalpy flux of the carried ionization stages and molecular carriers, the 10.2–30 eV photoelectron partition of Furlanetto & Stoever (2010), the He $2^3$S transport, the advection-corrected systems stepped on the velocity of each element's nuclei, and 25 new test drivers.

**What the script counts as *core*.** Every .f90 file under src/ except those in src/modules/wind_ae/, which is the ported Wind-AE IC generator and is listed separately. That rule puts the standalone drivers under src/tests/ in the core figure, and they are now the larger part of it by file count, so the table splits the core row into the simulation build and those drivers. Of the 105 files outside src/tests/, 104 are in the SRC list of the Makefile and are compiled into EXHALE.x; the one exception is nonlinear_system_solver/constrained_equilibrium_probe.f90, which is built on its own. Each test suite likewise builds its own driver from src/tests/<suite>/run.sh.

| | Fortran files | code lines (SLOC) |
|---|---|---|
| Original ATES (`ATES-Code-main`) | 47 | 3 514 |
| EXHALE core (excl. Wind-AE port) | 236 | 109 539 |
| of which the simulation build | 105 | 70 592 ($\approx 20.1\times$) |
| of which the `src/tests/` drivers | 131 | 38 947 |
| net change (core) | +189 | +106 025 ($\approx 31.2\times$) |
| Wind-AE IC-generator port (`wind_ae/`) | +36 | +4 756 |
| EXHALE total | 272 | 114 295 ($\approx 32.5\times$) |

**Line-by-line diff, core** (comments + blanks excluded; whitespace/case normalized): **107 432** code lines added in EXHALE (new modules + the new side of modified lines) and **1 407** removed from the original (deleted + the old side of modified lines), for **108 839** total changed lines. At the file level, of the 47 original files **38 were modified**, **8 are unchanged** (all eight byte-for-byte; the list is Table 77) and **one was removed**: `flux/speed_estimate_HLLC.f90`, Toro's adaptive $p_\star$ estimate, was used by `Num_Fluxes.f90` and never called, because the HLLC branch builds its own Davis/Einfeldt speeds inline. **190 new core modules** were added. Of those, 59 are part of the simulation build:

| New module | Purpose | SLOC |
|---|---|---|
| `time_step/steady_newton.f90` | stationary solve of the hydrodynamic rows: residual $F(Y) = 0$ on the physical cells, Newton/Krylov with pseudo-transient continuation, line search, best-iterate ledger | 11122 |
| `lower_atmosphere/diffusive_photochemistry.f90` | vertical transport of the molecular carriers ($H_2$, OH, $H_2O$, CO) and of the transported ionization stages ($H^+$, He II, He III) and He $2^3$S level, solved together with the chemistry that makes and destroys them, with the enthalpy fluxes they carry | 4587 |
| `functions/binary_element_diffusion.f90` | conservative binary H/He element transport (stage-resolved pair coefficients, donor-cell drift, metals slaved to H) and the interdiffusion enthalpy flux it carries | 2610 |
| `lower_atmosphere/molecular_infrared_data.f90` | band and line data of the molecular infrared coolant (generated by `cooling_data/molecular_infrared_bands.py`) | 2182 |
| `time_step/certification.f90` | one evaluator of the equations a run actually solves, and the contexts that read its verdict | 1991 |
| `time_step/attempted_step.f90` | the attempted-step controller: one checkpoint, one trial, one adoption boundary, and the rollback contract | 1630 |
| `radiation/low_energy_electron_tables.f90` | the electronic tables of Furlanetto & Stoever (2010), energies 10.0–32.2 eV at their 14 ionized fractions (generated by `src/utils/furlanetto2010_tables_to_fortran.py`) | 1502 |
| `radiation/charge_exchange.f90` | charge-exchange network (Huang et al. 2023, Table 4), as residual rows and as the stoichiometric sources of the transported stages | 1498 |
| `nonlinear_system_solver/constrained_chemical_equilibrium.f90` | chemical equilibrium of the molecular H/He network in logarithmic number densities, element conservation as explicit residual rows | 1238 |
| `lower_atmosphere/h2_self_shielding_table.f90` | $H_2$ Lyman–Werner dissociation cross section with overlapping lines, and its two branching ratios | 952 |
| `states/base_boundary.f90` | the lower boundary as a characteristic condition at the face, the number of conditions set by the Mach number there | 723 |

| New module | Purpose | SLOC |
|---|---|---|
| `time_step/viscous_conduction.f90` | Navier–Stokes viscous momentum force, its dissipation, and thermal conduction with the conductivity of the mixture (Crank–Nicolson) | 671 |
| `init/molecular_seed_from_atomic_state.f90` | the molecular state a solved atomic state of the same wind implies, written as an initialization file: an imposed $H_2$ partition capped by each cell's own neutral hydrogen, closed on the caloric equation of state at fixed pressure or at fixed temperature | 595 |
| `time_step/steady_residual.f90` | finite-volume stationary residual $R = \mathrm{d}u/\mathrm{d}t$ and its norms, shared by the diagnostic, the convergence monitor and the stationary solver | 580 |
| `time_step/energy_semi_implicit.f90` | implicit source step for the thermal energy of a cell, at fixed composition and heating | 535 |
| `radiation/excited_hydrogen.f90` | non-LTE H($n$=2) 2s/2p equilibrium (Christie, Arras & Li 2013) and the Balmer feedback | 508 |
| `time_step/conservation_budget.f90` | every term of the three finite-volume conservation rows of one assembly, with the geometry, the potential and the gravitational work the energy flux difference carries inside it | 493 |
| `files_IO/conserved_state_restart.f90` | the exact code-unit state $(\rho, \rho v, E, f_{\mathrm{sp}})$ written beside the dimensional pair, and read back only when every recorded condition matches, so that a restart starts at the bits the run ended on | 466 |
| `lower_atmosphere/element_inventory.f90` | one stoichiometric map of the H, He, O and C nuclei of a cell, and the three constraints a composition state is judged by | 448 |
| `nonlinear_system_solver/System_HeH_mol.f90` | coupled H/He + molecular ($H_2/H_2^+/H_3^+/HeH^+$) system, turnover-scaled rows, with the gross production and loss of each row on request | 448 |
| `lower_atmosphere/molecular_reaction_heat.f90` | net chemical heat of the collisional reactions of the $H_2$/He network, with the dissociative recombinations that leave a product excited charged to that level and not to the gas | 416 |
| `files_IO/lower_atmosphere_profile.f90` | lower-atmosphere profile reader: log-$p$ interpolation, elemental reservoirs, $K_{zz}(p)$ | 396 |
| `nonlinear_system_solver/ion_residual_core.f90` | shared ionization-residual rows and analytic-Jacobian pieces of the `System_*` family | 387 |
| `states/caloric_eos.f90` | caloric equation of state: internal energy density, pressure and temperature, with the $H_2$ level ladder | 367 |
| `lower_atmosphere/oxygen_rates.f90` | rate coefficients and thermodynamic data of the oxygen-hydrogen chemistry, with their validity ranges | 316 |
| `functions/element_census.f90` | one definition of the elemental and charge content of a composition state, and the invariant checks built on it | 307 |
| `radiation/electron_energy_degradation.f90` | degradation of a fast photoelectron in a partly ionized H/$H_2$/He gas: the heat fraction and the ionizations of each target | 305 |
| `time_step/coupled_block_handover.f90` | the state at which the wind-composition alternation stops on its movement bound, and the one recorded pair $(u, f_{\mathrm{sp}})$ the coupled carrier solve is entered from | 298 |
| `radiation/lya_rt.f90` | Ly$\alpha$ transfer by the escape-probability closure, evaluated in line each step | 261 |
| `functions/ionization_stage_transport.f90` | transport of the ionization stages of one element on that element's own nucleus flux, the neutral stage closing the simplex, so that moving charge between stages moves no nucleus | 235 |

| New module | Purpose | SLOC |
|---|---|---|
| `radiation/low_energy_electron_degradation.f90` | partition of the 10.2–30 eV photoelectrons into heat, H I and He I ionizations and escaping line radiation (Furlanetto & Stoever 2010), the option `Low-energy electron partition` | 234 |
| `lower_atmosphere/molecular_infrared_cooling.f90` | infrared exchange of the molecular layer: $H_2O$ and CO vibration-rotation bands, $H_2$ quadrupole and magnetic-dipole lines | 231 |
| `lower_atmosphere/h3p_cooling.f90` | optically thin $H_3^+$ infrared cooling (Miller et al. 2013) | 228 |
| `functions/h2_photo_channels.f90` | $H_2$ photoabsorption resolved into mutually exclusive final-state channels, with one stoichiometric table | 224 |
| `lower_atmosphere/mol_rates.f90` | molecular reaction rates (Koskinen et al. 2022, Table 1) | 206 |
| `files_IO/metals_input_read.f90` | parse `metals.inp`: trace-metal abundances at runtime | 191 |
| `radiation/opacity_models.f90` | opacity-model dispatcher for the photoionization cross sections (A/C/P/T) | 180 |
| `states/boundary_state_trace.f90` | the distance between the cached base face state and the one the installed composition would give now, measured and never decided on | 175 |
| `nonlinear_system_solver/System_HeH_metals.f90` | coupled H/He+metals ionization (+ analytic Jacobian) | 166 |
| `flux/species_face_flux.f90` | advection of the species mass fractions on the same faces, with the same mass flux, as the density | 163 |
| `functions/composition.f90` | single point turning $(\rho, f_{\mathrm{sp}})$ into every species density, $n_e$ and $n_{\mathrm{tot}}$, and converting pressure and temperature | 162 |
| `time_step/hydrodynamic_rows.f90` | the hydrodynamic rows of the stationary residual in double precision, instantiated from the kind-generic text | 161 |
| `lower_atmosphere/h2_vibrational_relaxation.f90` | the fate and the size of the vibrational energy left in an $H_2$ molecule, for both of its drivers | 155 |
| `nonlinear_system_solver/System_HeH_mol_metals.f90` | molecular + metals merged ionization system | 147 |
| `lower_atmosphere/lyman_werner.f90` | $H_2$ photodissociation in the Lyman and Werner bands (the Solomon process) | 142 |
| `lower_atmosphere/co_photodissociation.f90` | CO photodissociation on the 912–1201 Å band | 127 |
| `nonlinear_system_solver/newton_solver.f90` | damped Newton with an analytic Jacobian for the ionization systems, with the MINPACK `hybrd1` retry | 127 |
| `init/species_table.f90` | metadata table of the metal stages and of the base, molecular and oxygen species | 126 |
| `lower_atmosphere/water_photolysis.f90` | $H_2O$ and OH photodissociation in the four far-ultraviolet bands of the oxygen option | 111 |
| `lower_atmosphere/co_self_shielding_table.f90` | CO self-shielding function $\Theta[N(\mathrm{CO}), N(\mathrm{H_2})]$ (Visser, van Dishoeck & Black 2009) | 110 |
| `nonlinear_system_solver/ion_cell_state.f90` | named-field cell state of the ionization residuals, in place of the position-packed `params` | 110 |
| `states/stationary_operator.f90` | the spatial operator a stationary state is a state of, and the face mass flux $r^2(\rho v)_f$ it puts through every face | 107 |
| `radiation/hydrogen_n2_rates.f90` | atomic data and rate coefficients of the H $n$=2 manifold and of the Ly$\alpha$ line, in one place | 99 |
| `flux/low_mach_dissipation.f90` | gated fourth-difference stress damping the two-cell velocity mode a contact-resolving flux leaves undamped | 87 |
| `lower_atmosphere/lower_column.f90` | analytic lower-atmosphere column from the 1-bar level to the base pressure (Koskinen et al. 2022) | 87 |

| New module | Purpose | SLOC |
|---|---|---|
| files_IO/opacity_input_read.f90 | parse opacity.inp | 85 |
| init/blas_thread_policy.f90 | the thread count of the LAPACK library the binary links, resolved at run time and set to one unless the run states it | 73 |
| nonlinear_system_solver/System_HeH_TR_metals.f90 | merged He $2^3$S+metals system | 72 |
| nonlinear_system_solver/constrained_equilibrium_probe.f90 | standalone re-evaluation of one saved constrained-equilibrium state, without the hydrodynamics | 15 |
| | simulation-build new-module total | 42 168 |

The other 131 are the standalone test drivers under src/tests/, which did not exist in ATES at all. They state what a routine is allowed to return and are run by suite, not by the production binary:

| Test driver | What it asserts | SLOC |
|---|---|---|
| tests/krylov_and_dogleg/krylov_and_dogleg_tests.f90 | what the linear solve and the step geometry of the stationary solver are allowed to return | 2755 |
| tests/steady_species_rows/steady_species_rows_tests.f90 | the unknown space of the stationary system, one row per transported balance | 1826 |
| tests/ionization_stage_flux/ionization_stage_flux_tests.f90 | the stage fluxes sum to the element nucleus flux at every face, against the formula of the derivation and not against the module itself | 1536 |
| tests/carrier_retry/carrier_retry.f90 | the carrier-local checkpoint and the rejection-and-retry controller, on a real molecular column | 1463 |
| tests/certification/certification_contexts.f90 | the certification evaluator and its contexts, against the predicates they replace | 1156 |
| tests/diffusion_tests.f90 | the binary H/He element-diffusion operator: analytic limits and conservation | 1043 |
| tests/grid_and_gates/hydrostatic_residual.f90 | the momentum row returns zero on a hydrostatic state | 1031 |
| tests/charge_exchange_detailed_balance/charge_exchange_detailed_balance.f90 | the forward and reverse rate coefficients of every charge-exchange pair obey microscopic reversibility with the partition functions of the four species | 950 |
| tests/element_operator/element_operator_tests.f90 | the boundary the element transport operator is measured under | 945 |
| tests/stage_row_balance/stage_row_balance_tests.f90 | the signed sum of the gross channels of an ion-stage row is that stage's source, in molecular and in atomic gas and with the He $2^3$S level tracked and untracked | 898 |
| tests/carrier_retry/closure_returned_state.f90 | what the thermochemical closure hands back on a molecular column: the composition, its chemical residual, the conditioning, and the state a refused trial puts back | 890 |
| tests/physics_probe/molecular_energy_recipients.f90 | the enthalpy of a reaction is untouched by where its energy goes: heat, radiation and internal energy close on the one formation table, and the R12/R15 pair stays in detailed balance collider by collider | 750 |
| tests/grid_and_gates/base_boundary_continuity_probe.f90 | the base face density does not move along a perturbation of the wind window once the branch is decided, and the two limits coincide at a vanishing window | 592 |

| Test driver | What it asserts | SLOC |
|---|---|---|
| tests/h2_level_ladder/h2_ladder_model.f90 | a reduced statistical equilibrium of the $H_2$ ground-state ladder from the published level, radiative and collisional data, which brackets the scalar thermalized fraction | 561 |
| tests/species_face_flux/species_face_flux_tests.f90 | species advection on the face mass fluxes, and the element rows of the diffusion operator | 549 |
| tests/low_mach_stress_energy/low_mach_stress_energy_form.f90 | the discrete kinetic-energy quadratic form of the gated fourth-difference stress, and the sign a coefficient carried outside both differences does not have | 519 |
| tests/energy_update/conduction_floor_tests.f90 | what the Crank–Nicolson transport stage returns, refuses and conserves | 482 |
| tests/physics_probe/weno3_reconstruction_order.f90 | convergence order of the face reconstruction against the exact face value | 482 |
| tests/helium_metastable_transport/helium_metastable_transport.f90 | the He $2^3$S population carried as a fourth helium state: the carried simplex, the write-back, the local limit, the source ledger and a manufactured three-state helium column | 438 |
| tests/charge_exchange_rows/reaction_source_orientation.f90 | one electron transfer moves $-R$, $+R$, $-R$, $+R$ over four stages, creates no nucleus and makes no net charge, in each of the three equation bases | 428 |
| tests/grid_and_gates/base_reservoir_discrete_equilibrium.f90 | whether the base boundary shares one discrete hydrostatic equilibrium with the well-balanced reconstruction, on static columns above the reservoir | 420 |
| tests/charge_exchange_rows/transported_stage_sources.f90 | the charge-exchange source of a transported ionization stage is the source the local systems' rows carry, in each of the three bases the metal systems assemble | 419 |
| tests/spectrum_type/loaded_sed_table_semantics.f90 | what a row of a loaded SED table means, and what the production consumers make of it | 417 |
| tests/attempted_step/attempted_step_tests.f90 | the checkpoint and the step-size policy of the attempted-step controller | 415 |
| tests/a2_m1/a2_m1_rate_check.f90 | the oxygen rate coefficients against their published sources | 406 |
| tests/spectrum_type/balmer_band_quadrature.f90 | quadrature of the Balmer-continuum band integrals of `excited_hydrogen` | 367 |
| tests/coupled_block_jacobian/coupled_block_linear_system.f90 | the directional action of the coupled block on one frozen iterate: its repeatability, and its perturbation ladder row group by row group | 353 |
| tests/physics_probe/metal_photoionization_fits.f90 | every metal photoionization cross section against the published fit, and only where it is valid | 343 |
| tests/element_census_tests.f90 | the elemental and charge invariants of a composition state | 340 |
| tests/interdiffusion_enthalpy_flux/interdiffusion_enthalpy_flux_tests.f90 | the interdiffusion enthalpy flux $\sum_s h_s J_s$ (Cook 2009): conservative, zero where the element fluxes are, equal to the hand-evaluated face value, and absent with its key off | 336 |
| tests/energy_update/energy_update_tests.f90 | the residual-controlled temperature solve of the implicit energy source step | 334 |
| tests/grid_and_gates/base_closure_candidates_probe.f90 | two discriminants for the same characteristic base condition, read on one state under the production stationary operator | 328 |
| tests/physics_probe/recombination_photoelectron_partition.f90 | the photoelectrons of the recombination radiation absorbed on the spot pass through the partition of the stellar photoelectrons | 326 |

| Test driver | What it asserts | SLOC |
|---|---|---|
| tests/boundary_state/contact_law_at_the_base.f90 | the lower boundary as a contact: a column on the reservoir isentrope stays at rest, a pressure mismatch launches the acoustic response, and the entropy is upwinded on the matched face velocity | 316 |
| tests/carrier_helium_inventory/carrier_helium_inventory.f90 | the helium a cell holds is the helium it had: the admissible stage partition leaves out the nucleus inside $HeH^+$ and the population in He $2^3S$ | 310 |
| tests/charge_exchange_rows/he_h_pair_reservoir.f90 | the He $\leftrightarrow$ H charge-exchange pair reacts from the ground singlet, the same one in every system carrying helium and a metastable | 304 |
| tests/spectrum_type/loaded_sed_threshold_edges.f90 | the photon grid built from a loaded SED, and the rates formed on it | 304 |
| tests/grid_and_gates/low_mach_face_mass_flux.f90 | the face mass flux stays accurate relative to the wind as the Mach number of the lower layer falls to $10^{-9}$ | 303 |
| tests/ionization_imposed_fractions/imposed_ionization_rows.f90 | an imposed ionization row is the identity $x - x_{\mathrm{fix}}$ and is one number in all seven systems, and nothing depends on the carried values where nothing is imposed | 289 |
| tests/coupled_block_handover/coupled_block_handover_contract.f90 | the transition into the coupled carrier solve is reached once and from the recorded entry state bit for bit, never from a live token the stalling pass cannot carry | 285 |
| tests/h2_level_ladder/h2_ladder_statistical_equilibrium.f90 | the solved ladder against the first-event branching fraction: detailed balance and the thermal limit, and the four ways an injected formation energy leaves | 275 |
| tests/physics_probe/species_formation_energy_table.f90 | the formation-energy table of the energy ledger, and the reaction energies that are its differences | 275 |
| tests/spectrum_type/wasp121b_sed_sensitivity.f90 | the production band quantities of the three WASP-121 b spectrum candidates | 273 |
| tests/physics_probe/lya_band_transmission.f90 | the stellar Ly$\alpha$ band reaching a cell through the atomic hydrogen column above it | 272 |
| tests/physics_probe/one_minus_exp_over_x_sites.f90 | $(1-e^{-x})/x$ of the whole code and the four quantities formed from it besides the absorbed fraction | 263 |
| tests/coupled_source_step/coupled_source_tests.f90 | the coupled temperature-composition source step | 260 |
| tests/grid_and_gates/base_equilibrium_columns.f90 | the static and the wind-carrying columns, and the one production evaluation, shared by the two measurements of the discrete equilibrium of the base | 256 |
| tests/physics_probe/inner_shell_photoabsorption_ledger.f90 | the relaxation of an inner-shell metal vacancy (Kaastra & Mewe 1993), the energy ledger of the event, and the two-stage jump in the metal rows | 246 |
| tests/charge_exchange_rows/metal_helium_reservoir.f90 | the metal + He reactions of group C react from the ground singlet, the same one in the three systems that assemble them | 235 |
| tests/physics_probe/lya_beam_cell_mean.f90 | the stellar Ly$\alpha$ field of a cell is the beam mean over the cell's own optical depth | 234 |
| tests/physics_probe/lyman_werner_cell_mean.f90 | the Lyman–Werner rate a cell carries is the mean of the local rate over that cell | 234 |
| tests/spectrum_type/balmer_continuum_field.f90 | the two scalar Balmer-continuum rates of H($n$=2) | 227 |
| tests/carrier_returned_state_acceptance/carrier_returned_state_acceptance.f90 | what makes a returned carrier state a solution, and what does not | 225 |

| Test driver | What it asserts | SLOC |
|---|---|---|
| tests/physics_probe/positivity_limiter_scaling.f90 | the positivity scaling of the reconstructed face states (Zhang & Shu 2010) | 223 |
| tests/physics_probe/ground_capture_and_cascade_data.f90 | the atomic data of the recombination and $n$=2 physics against the published tables they come from | 222 |
| tests/low_energy_electron_degradation/low_energy_electron_partition.f90 | the 10.2–30 eV photoelectron partition against the authorsówn interpolator of the same tables, and the energy budget the tables close | 216 |
| tests/grid_and_gates/photon_grid_threshold_edges.f90 | every active absorber's threshold against the bin partition of the photon grid | 213 |
| tests/physics_probe/photoionization_field_substitution.f90 | the order in which the field and the composition that attenuates it are brought into agreement | 209 |
| tests/adv_static_limit/enthalpy_flux_of_mass_divergence.f90 | the third term of the advected energy equation | 202 |
| tests/adv_static_limit/stationarity_by_the_mass_row.f90 | which test decides that the state the post-process was handed is stationary in a cell | 202 |
| tests/charge_exchange_rows/advection_pair_reservoir.f90 | the advection-corrected system hands the He $\leftrightarrow$ H pair the same ground singlet the equilibrium system does | 199 |
| tests/carrier_enthalpy_flux/carrier_enthalpy_flux.f90 | the sensible enthalpy flux of the transported molecular carriers: zero at the column ends, conservative, the carrier operatorś own face flux times the $H_2$ enthalpy, and absent with its key off | 198 |
| tests/carrier_outer_boundary/carrier_outer_boundary.f90 | the carrier column is closed to diffusion at both ends, one outflow ghost rule serves every evaluation, and the advective band entries are the operator's own derivative | 196 |
| tests/physics_probe/co_destruction.f90 | the CO destruction rates, the shielding table, the deposited energies and the cell mean | 195 |
| tests/carrier_reference_scales/carrier_reference_scales.f90 | the element a transported carrier is made of, and whether its chemistry rows can be differentiated | 191 |
| tests/ionization_stage_enthalpy_flux/ionization_stage_enthalpy_flux.f90 | the sensible enthalpy the carried ionization stages move relative to their element, against a manufactured column | 186 |
| tests/spectrum_type/photon_grid_n2_floor.f90 | the photon grid built when the excited-hydrogen coupling is armed | 184 |
| tests/carrier_constraint_attribution/carrier_constraint_attribution.f90 | which residuals belong to the solver and which to a constraint | 179 |
| tests/physics_probe/recombination_radiation_on_the_spot.f90 | the escape weights of the ground captures and the photon and energy accounting of the recombination radiation absorbed where it is emitted | 176 |
| tests/grid_and_gates/base_branch_discriminant.f90 | what the base contact reads its direction off: the wind window wherever it carries one flux, the matched face velocity elsewhere, never the first cell's own product | 174 |
| tests/grid_and_gates/free_outflow_boundary.f90 | the outer free-outflow boundary on a supersonic wind, and its ghost state | 174 |
| tests/spectrum_type/planck_photon_field.f90 | the incident flux put on the photon grid for the photospheric blackbody type | 173 |
| tests/physics_probe/lya_escape_probability.f90 | the escape probability of the two-level closure against the published slab solution | 169 |
| tests/physics_probe/recombination_coefficient_fits.f90 | every recombination coefficient against the published fit | 169 |

| Test driver | What it asserts | SLOC |
|---|---|---|
| `tests/carrier_boundary_jacobian/carrier_boundary_jacobian.f90` | the boundary derivatives of the carrier advection, a copied ghost carrying one and a stated one carrying none | 165 |
| `tests/a2_m2/a2_m2_kinetics_check.f90` | the kinetics quantities the oxygen option adds on top of the rate audit | 163 |
| `tests/e1_h2/e1_h2_channel_check.f90` | the mutually exclusive $H_2$ photoabsorption channels | 161 |
| `tests/physics_probe/roe_low_mach_velocity_jump.f90` | the low-Mach scaling of the normal velocity jump on the ROE branch, against the analytic three-wave flux (Rieper 2011) | 152 |
| `tests/physics_probe/photoionization_field_self_consistency.f90` | the field one cell is solved in, against the composition that cell returns | 148 |
| `tests/physics_probe/co_helium_ion_sink.f90` | $He^+ + CO$ as a species sink, not only as a heat source | 147 |
| `tests/physics_probe/h3p_cooling_limits.f90` | the limits, the continuities and the published values of the $H_3^+$ cooling rate | 147 |
| `tests/grid_and_gates/photon_grid_quadrature.f90` | the photoionization rates of one unattenuated cell against the band quadrature | 146 |
| `tests/physics_probe/riemann_wave_speeds.f90` | signal speeds at a face, and the state one conservative update with that flux leaves behind | 143 |
| `tests/physics_probe/helium_level_cooling_ledger.f90` | which helium level each cooling channel of `eval_cool` is charged to | 140 |
| `tests/physics_probe/molecular_third_body_in_the_chemistry.f90` | the three-body collider sum of the R12/R15 pair is one number for the composition and for the energy, and is not the total density | 140 |
| `tests/physics_probe/heating_channel_closure.f90` | one assembly of the volumetric heating rate, and the closure of its channels | 138 |
| `tests/adv_static_limit/upstream_caloric_energy.f90` | the upstream energy of the advected energy equation | 137 |
| `tests/acceptance_classes/acceptance_class_contract.f90` | one meaning of the ionization acceptance classes for every consumer | 132 |
| `tests/physics_probe/metal_ionization_energy_ledger.f90` | the energy metal collisional ionization and recombination take from the electron gas, and the ledger of electron gas, ionization energy and radiation they close | 128 |
| `tests/physics_probe/absorbed_fraction_switch.f90` | $(1-e^{-d})/d$, the fraction of the photons entering a cell that the cell absorbs | 126 |
| `tests/physics_probe/opacity_model_p_ledger.f90` | the radiative ledger of a cell whose cross sections are broadened by opacity model P | 124 |
| `tests/physics_probe/helium_584_resonance_line.f90` | the fate of the He I 584 Å recombination photons in a cell with He I (Wood, Mathis & Ercolano 2004) | 123 |
| `tests/grid_and_gates/outer_shells_base_identity.f90` | the outer shells leave the constructed grid unchanged bit for bit, close on the stated face and widen in geometric progression | 122 |
| `tests/physics_probe/cooling_assembly_identity.f90` | the cooling the post-process temperature root balances is the cooling `eval_cool` reports, channel for channel | 118 |
| `tests/physics_probe/photoevent_energy_ledger.f90` | the energy an isolated photoabsorption event has to account for | 117 |
| `tests/physics_probe/xuv_cell_mean_attenuation.f90` | the beam attenuation the photoionization rate of a cell is evaluated with | 115 |
| `tests/adv_element_velocity/element_velocity_rate_weights.f90` | each elementś rows of the advection-corrected systems are stepped on the velocity of that elementś nuclei | 112 |
| `tests/physics_probe/helium_bray_collision_strengths.f90` | the electron-impact collision strengths of the He I metastable network against Bray et al. (2000), Tables 2 and 3 | 105 |

| Test driver | What it asserts | SLOC |
|---|---|---|
| `tests/grid_and_gates/grid_window_indices.f90` | the two index statements that decide which cells a quantity is read from | 101 |
| `tests/grid_and_gates/base_cell_width_provenance.f90` | the base cell width is written as a decimal that reads back to the same bits, and the default and the pinned line state the widths their runs were made on | 98 |
| `tests/physics_probe/h2_rovibrational_identity.f90` | the identities tying the $H_2$ partition function of the chemistry to the internal energy of the caloric EOS | 96 |
| `tests/adv_static_limit/stationary_flux_refusal.f90` | the sensitivity of the advected energy correction to a non-stationary mass flux | 91 |
| `tests/physics_probe/ionization_threshold_turn_on.f90` | one definition of each ionization threshold, and every cross section turning on exactly there | 90 |
| `tests/grid_and_gates/base_grid_pinned_width.f90` | which grid each of the four base cell widths builds, compared on the bits of every cell center | 86 |
| `tests/physics_probe/voronov_table_rows.f90` | every Voronov (1997) electron-impact ionization fit of the code against the published row | 84 |
| `tests/grid_and_gates/grid_width_identity.f90` | the cell-width and total-width identities of the radial grid | 83 |
| `tests/physics_probe/h2_channel_detector_ratio.f90` | the measured proton-to-$H_2^+$ ratio of Chung et al. (1993) against the four channels | 83 |
| `tests/physics_probe/helium_metastable_detailed_balance.f90` | the He I $1^1\mathrm{S} \leftrightarrow 2^3\mathrm{S}$ electron collisions in detailed balance, finite where the excitation underflows | 83 |
| `tests/physics_probe/fine_structure_escape_probability.f90` | the escape probability of the ground-term fine-structure lines, and the depth it is fed | 82 |
| `tests/physics_probe/hydrogen_helium_excitation_data.f90` | the collisional-excitation cooling of H I, He I and He II against the coronal sums it is fitted to, and the shared H($n$=2) rate set | 81 |
| `tests/physics_probe/atomic_mass_and_radius_constants.f90` | the helium mass and the astronomical constants of the input readers, one definition each | 79 |
| `tests/physics_probe/recombination_energy_loss.f90` | the kinetic energy the recombination cooling removes in each capture onto H II, He II and He III, $kT\alpha(3/2 + \mathrm{d}\ln\alpha/\mathrm{d}\ln T)$ | 79 |
| `tests/physics_probe/assertion_report.f90` | one printed verdict line per assertion, and the failure count the driver exits on | 74 |
| `tests/adv_static_limit/thermal_damkohler_limit.f90` | the validity condition of the advection-corrected energy solve, as a function of the flow speed | 72 |
| `tests/a2_m2/a2_m2_g3_run_check.f90` | with every photolysis rate at zero, the solved O/OH/$H_2O$ partition is the chemical equilibrium | 71 |
| `tests/constrained_network_layout/transported_proton_constraint_layout.f90` | which species are unknowns of the continuation, and whether the system built is square | 71 |
| `tests/test_columns.f90` | synthetic columns whose species carry the mass density they are measured against | 67 |
| `tests/physics_probe/h2_infrared_line_populations.f90` | the LTE populations weighting the $H_2$ quadrupole lines are those of the caloric EOS ladder | 65 |
| `tests/charge_exchange_rows/turnover_cell_rates.f90` | the charge-exchange turnover bound is the loaded cell's, carries no history, and is each thread's own | 63 |
| `tests/physics_probe/wind_ae_bridge_composition.f90` | the composition an EXHALE `input.inp` hands the Wind-AE solver | 54 |

| Test driver | What it asserts | SLOC |
|---|---|---|
| `tests/spectrum_type/fuv_band_quadrature.f90` | the band integral of a loaded spectrum against closed-form answers, with the band edges inside segments, on nodes and at the table ends | 53 |
| `tests/physics_probe/voronov_carbon_ionization.f90` | the electron-impact ionization of $C^0$ and $C^+$ against the published Voronov (1997) rows | 52 |
| `tests/physics_probe/water_photolysis_lyman_alpha_yields.f90` | the $H_2O$ branching on the band that contains Ly$\alpha$, against the published yields | 49 |
| `tests/physics_probe/h3p_recombination_branching.f90` | the two product channels of $H_3^+$ dissociative recombination | 44 |
| `tests/energy_update/cooling_models.f90` | analytic cooling laws for the tests of the implicit energy update | 43 |
| `tests/species_masses/carrier_background_masses.f90` | the mass the carrier transport gives each of its background collision partners | 36 |
| `tests/charge_exchange_rows/stale_cell_rates_refused.f90` | a residual assembled at a temperature the rate coefficients were not filled at is refused | 31 |
| `tests/charge_exchange_rows/stale_cell_turnover_refused.f90` | a turnover bound taken at a temperature the rate coefficients were not filled at is refused | 27 |
| `tests/certification/certification_capacity.f90` | one equation past a full certification report is refused by name, before any field is written | 26 |
| `tests/restart_radius_guard/restart_radius_guard_probe.f90` | a restart whose recorded cell centers are not finite or differ from the runś grid is refused | 26 |
| `tests/charge_exchange_rows/stale_cell_stage_sources_refused.f90` | a stage source evaluated at a temperature the rate coefficients were not filled at is refused | 24 |
| | test-driver total | 38 947 |
| | new-module total | 81 115 |

So the 190 new core modules contribute 81 115 of the 107 432 added code lines; the remaining 26 317 added (and 1 338 of the 1 407 removed, the other 69 being the deleted HLLC estimator) are edits spread across the 38 modified files. The ten heaviest, by added code lines, are the main driver (`EXHALE_main.f90`, 5 110 added, 246 → 5 225 SLOC: the marching loop, the attempted-step adoption boundary, the run-mode contracts and the written-state assertions), the cooling module (`Cool_coeff.f90`, 3 837 added, 228 → 3 882 SLOC: CHIANTI metal-line fits, Fe tables, fine-structure statistical equilibrium, Penning and molecular rate coefficients), the ionization equilibrium and its diagnostics (`ionization_equilibrium.f90` 3 243 and `util_ion_eq.f90` 2 139: the metal, molecular and oxygen networks, the acceptance classes, the photoionization field and the heating channels), input parsing (`input_read.f90`, 1 986), the state reader and writer (`load_IC.f90` 1 774, `write_output.f90` 1 562, `write_setup_report.f90` 989), the advection-corrected post-processor (`post_process_adv.f90`, 1 533) and the cross sections (`cross_sec.f90`, 596). Those ten carry 22 769 of the 26 317. The rest of the edits are in the reconstruction, the fluxes, the boundary conditions, the grid, the time step, the energy equation and the `System_*` ionization systems, which wire the metals, the molecules, the opacity dispatcher, the Roche potential, the transonic/auto/Wind-AE IC, the excited-H and Ly$\alpha$ physics, the element diffusion and the Brent/Newton solver upgrades into the original solver.

In total the EXHALE core is now ≈31.2× the original code base (3 514 → 109 539 SLOC), or ≈20.1× counting only the simulation build (70 592 SLOC), touching almost all of the original files (38 of 47 modified, one deleted, plus 190 new); with the ported Wind-AE IC generator the full Fortran tree is 272 files / 114 295 SLOC (≈32.5×). Beyond

Fortran, the tree also carries the Python tooling (GUI, the `EXHALE_transit.py` transmission post-processor, plotting/-analysis, the lower-atmosphere adapters and the elemental-flux closure driver) and 351 309 lines of documentation (460 md/tex files, notebook checkpoints excluded: the memos, the manual, the dated audit directories and the update logs) under `docs/` and `md/` (the Markdown memos moved to `md/` on 2026-09-25).

## Fortran source inherited unchanged from ATES

Of the 47 upstream Fortran files (`ATES/ATES-Code-main`), **8 are byte-for-byte identical** in EXHALE (re-measured 2026-10-05, and the same eight as on 2026-09-11 and 2026-09-22: nothing entered or left the set in between): the numerics EXHALE reuses without touching, which are now the linear-algebra part of the included MINPACK nonlinear solver plus the gravity pre-evaluation. Table 77 lists them with their role. No file is identical after whitespace and case normalization but not byte-identical, so the normalized-unchanged set and the byte-identical set are the same eight.
Five files left the byte-identical set since the 2026-08-30 edition of this table, each for a documented reason rather than drift.

- `nonlinear_system_solver/hybrd.f90`, `hybrd1.f90` and `fdjac1.f90` carry a new `unit_step_floor` argument threaded from `hybrd` to `fdjac1`, which selects the forward-difference step of the Jacobian: `.false.` is MINPACK's own $h = \varepsilon|x_j|$ with the exact-zero fallback, `.true.` floors it at unit magnitude, $h = \varepsilon \max(1, |x_j|)$, which is the rule for unknowns already scaled so that unity is their natural size (the logarithmic unknowns of the constrained chemical equilibrium). The `params` dummy also became assumed-size, so one transport array serves systems of different length. `hybrd1` passes `.false.` and is therefore unchanged in behavior.
- `states/Source.f90` gained the well-balanced option of 2026-09-10 (item N37 of the stage-2 solver program): under `Well balanced: True` the gravitational source and the geometric pressure term are not evaluated here at all, because they cancel the equilibrium part of the momentum flux difference identically and `RK_rhs` assembles the row from what is left (Käppeli & Mishra 2014, J. Comput. Phys. 259, 199, eq. 2.26). The file also reads its pressure through `caloric_eos` instead of a constant $\gamma$.
- `radiation/J_inc.f90` carries the single-spectrum-type decision: `Spectrum type:` now states the spectrum of the whole photon grid, the XUV and the part below 13.6 eV alike, and no band is filled from a type the input did not select. The file also became the one definition of the eV-to-Hz conversion and of the Balmer-continuum head $e_{\mathrm{th,HI}}/4$.

The two wave-speed estimators are no longer in the "effectively untouched" category either. `flux/speed_estimate_HLLC.f90` was deleted as dead code, as recorded above. `flux/speed_estimate_ROE.f90` was rewritten (99 code lines added, 43 removed) into a star-state estimate that returns a status: the primitive-variable, two-rarefaction and two-shock branches of Toro (2009) chapter 9 are selected by the pressure ratio, the vacuum condition of his eq. 4.76 is tested first, and a face with no admissible star state is reported rather than given a square root of a negative ratio.

Table 77: The 8 Fortran files byte-identical between the original ATES (`ATES-Code-main`) and EXHALE (measured 2026-10-05), grouped by role.

| File | Purpose |
|---|---|
| *MINPACK linear algebra and constants* (Argonne; dir `nonlinear_system_solver/`) | |
| `qrfac.f90` | QR factorization via Householder transformations |
| `qform.f90` | Accumulate the orthogonal factor $Q$ from the QR factorization |
| `r1updt.f90` | Rank-one update of the QR factorization |
| `r1mpyq.f90` | Apply the Givens-rotation sequence of the rank-one update |
| `dogleg.f90` | Dogleg (trust-region) step combining Gauss–Newton and steepest descent |
| `enorm.f90` | Euclidean norm of a vector (overflow/underflow safe) |
| `dpmpar.f90` | Machine floating-point precision / range constants |
| *Grid setup* | |
| `init/set_gravity_grid.f90` | Pre-evaluate gravity at cell centers and interfaces |